Differential Topology

— ***EYNTKA*** **Series** —

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Introduction

A topological space and a Euclidean space differ in how much structure they impose. On a topological space almost nothing is fixed: a homeomorphism may bend, stretch, and crumple it, and the properties that survive (connectedness, compactness, the number of holes, etc.) are exactly those indifferent to such distortion. A Euclidean space fixes a great deal: it measures lengths and angles, and its theorems, the ones about triangles and circles and curvature, are precisely the ones a distortion would destroy. Rigidity, in this sense, is imposed, and the more a space imposes, the more its maps must preserve. Between the two lies the smooth manifold, the object of this book, and where it falls is the first surprise the subject has to offer.

A smooth manifold carries just enough structure to do calculus: tangent vectors, derivatives, the chain rule, and so forth. It is tempting to conclude that calculus makes it geometric, that once a space is smooth, lengths and curvature come along for the ride. They do not. A smooth manifold has no notion of length, no angle between two tangent vectors, no curvature; a diffeomorphism may stretch it as freely as a homeomorphism bends a topological space, only more smoothly. The emblem of how little smoothness adds is Whitney's approximation theorem (chapter 7): every continuous map between smooth manifolds is a uniform limit of smooth ones [18], so the smooth world sits against the topological one as closely as the Weierstrass theorem lets polynomials sit against continuous functions. Smoothness gives *calculus*, not geometry.

Geometry, then, is not smoothness but something imposed on it: a rigidifying structure the smooth manifold does not itself possess. A Riemannian metric gives lengths and angles, and with them curvature, the content of Riemannian geometry [6]. A symplectic form, a complex structure, a reduction of the frame bundle's structure group to a subgroup of the general linear group (section 11.5) each rigidify in their own way; and some objects, an algebraic variety among them, are so rigid from the start that they induce such structures intrinsically. The key is that none is forced by the smooth structure alone. This book studies the smooth manifold before any such choice is made. That is why the title is not *differential geometry*, the study of the rigid structures one may impose, but *differential topology*, the study of the bare substrate they are imposed on and of the properties intrinsic to it, the ones no diffeomorphism can alter (it is fair to say that that smooth manifolds are *geometrizable* objects just like some topological spaces are *metrizable*).

Differential Data, Topological Invariants

The guiding question is which properties of a smooth manifold are intrinsic: unchanged by diffeomorphism, stable under small perturbation, insensitive to homotopy or to the coarser equivalence of cobordism. Each of the quantities the calculus produces (the critical points of a function, the signed count of transverse intersections, the integral of a differential form, etc.) is read off an arbitrary choice, and one might expect the answer to depend on that choice, and so to be no invariant at all. The recurring discovery of differential topology is the opposite: such quantities compute strong invariants exactly, the same for every choice. For example, the number of solutions of an equation, counted with sign, is a homotopy invariant, the degree (chapter 8). The critical points of a generic function reconstruct the manifold up to homotopy, one cell for each (chapter 9). The failure of a closed differential form to be exact measures a hole, and the de Rham cohomology it defines is isomorphic to the singular cohomology that algebraic topology builds from purely continuous data (chapter 14). The reason is always the same in shape: each quantity is read from a choice the manifold does not give, genericity (chapter 7) makes almost every choice a legitimate one, and any two legitimate choices are joined by a deformation along which the answer cannot jump (a cobordism for the degree, a chain homotopy for de Rham, a cancellation of handles for Morse, etc.) so it depends on the manifold alone.

The subject grew from exactly this discovery. Poincaré, seeking to organize the qualitative behaviour of differential equations, founded *analysis situs* on the manifolds their flows inhabit [15], reading topology from analytic data. Whitney, in the 1930s, gave the manifold its modern definition [18] and proved the two theorems that pin the smooth category to the topological one: every manifold embeds in a Euclidean space [19], and every continuous map is a limit of smooth ones [18]. Morse showed that a single function's critical points determine a manifold's shape [10]; Pontryagin and Thom turned the counting of framed submanifolds into the homotopy groups of spheres and the algebra of cobordism [17]; de Rham proved that differential forms compute the same cohomology as chains [9]. By mid-century the pattern had become a method, and this book is an account of it.

What is a Manifold: Gluing and Base Space

A manifold is built the way many spaces in mathematics are built: locally it is a copy of a fixed model, and globally it is those copies glued together, the regularity of the gluing fixing the kind of object obtained.[1] The model here is Euclidean space $\mathbb{R}^n$, and the gluing maps are diffeomorphisms between its open subsets; demanding only homeomorphisms would give a topological manifold, while linear isomorphisms on fibres would give a vector bundle (chapter 11). The same recipe with a different model gives different objects; for example the zero sets of polynomials or $\mathbb{C}^n$ begins algebraic and complex geometry, and the general study of spaces locally modelled on a fixed base is sheaf theory [1]. What distinguishes the objects is the model and the glue; the choice made here, $\mathbb{R}^n$ glued smoothly and with no "infinitesimally close" points (Hausdorffness), is the focus a book. Because a point of a manifold has no coordinates of its own, only those a chart assigns it, a persistent aim of the subject is to describe what is intrinsic: the properties of a manifold and its maps that no change of chart can alter.

[1] This local-to-global assembly is a completion: the circle is the colimit, in the category of open intervals and translations, of a diagram with images of morphisms representing overlap. Replacing the model or the maps completes a different category, and the pattern recurs across geometry.

The Shape of the Book

The book follows the smooth substrate up to its topology, and then to the threshold where geometry would enter. Part I builds the smooth manifold: charts and smooth maps, the tangent space and the differential that linearizes a map at a point, the rank theorems that classify maps locally, and vector fields with their flows. Part II is the differential topology proper: Sard's theorem and transversality make "generic" precise and prove Whitney's theorems; the intersection of submanifolds and the degree of a map turn geometric incidence into integer invariants; Morse theory reads a manifold's cells off a single function; and cobordism, through the Pontryagin-Thom construction, identifies framed manifolds with the homotopy of spheres. Part III returns to the calculus: vector and cotangent bundles and the constructions on them, among which the reduction of the structure group is the exact point at which a geometric structure would first appear; then differential forms, their integration and Stokes' theorem; and finally de Rham cohomology, where the method culminates in an isomorphism between the analytic and the topological.

A reader who reaches the end will compute the de Rham cohomology of a manifold, wield the degree and the Morse inequalities, and see a geometric structure for what it is: a rigidification of a substrate this book has studied bare. Those rigidifications, the metric of Riemannian geometry, the symplectic and complex structures, and the connections that measure their curvature, are the subject of the companion volumes; the reduction of the structure group in part III is the door to them.

A quick review

In this preliminary chapter, we quickly go over elementary knowledge of differentation and integration. Some notions from Topology that may be uncommon in a usual topology shall class shall quickly be defined.

0.1 Topology

I am assuming the reader is familiar with Topology, compactness, connectedness, component, local connectedness, in $\mathbb{R}^n$, secound-countable/seperable.

Definition 0.1.1: Refinement

Let $\mathcal{O}$ be an open cover for U. Then a *refinement* of $\mathcal{O}$ is another open cover $\mathcal{V}$ such that for every set $V \in \mathcal{V}$, there is a set $U \in \mathcal{O}$ such that $V \subseteq U$. if $\mathcal{V}$ is a refinement of $\mathcal{O}$, we say that $\mathcal{V}$ refines $\mathcal{O}$.
If B indexes the sets of $\mathcal{V}$ and A indexes the sets of $\mathcal{O}$, we will say there is a *refiniement map* $\varphi : \mathcal{V} \to \mathcal{O}$ satisfying $V_\beta \subseteq U_{\varphi(\beta)}$

Note that every subcover is trivially a refinement, but not all refinements are open covers.

Definition 0.1.2: Locally Finite Set

Let $\mathcal{O}$ be an open cover for a set U. Then $\mathcal{O}$ is said to be locally finite if for any point $x \in U$, only finitely many sets of $\mathcal{O}$ contain x.

Definition 0.1.3: Paracompact

Let X be a topological space. Then X is said to be *paracompact* if for any open cover $\mathcal{O}$ of X, there exists a refinement that is locally finite.

Definition 0.1.4: Component

Let X be a topological space. Then the component containing x, denoted $C(x)$ is defined as the union of all connected subsets of X containing x. A subset of a topological space X is called a [connected] component if it is equal to $C(x)$ for some $x \in X$.

Definition 0.1.5: Locally Connected

A topological space X is *locally connected* if it has a basis consisting of connected sets.

Proposition 0.1.6: Equivalence Of Paracompact

Let X be a locally compact Hausdorff space that is locally connected and in which every point has a second-countable open neighbourhood. Then the following are equivalent:

1. Every component of X is σ-compact (countable union of compact sets)
2. Every component is 2nd countable
3. X is metrizable
4. X is paracompact
5. There exists a C^0 partition of unity subordinate to any open cover

***Proof*:**
These equivalences are point-set topology. The cycle through (1) to (4) is [7, Paracompactness for Locally Compact Spaces], and the equivalence of (4) with (5) is, for a Hausdorff space, [7, Paracompactness from Partitions of Unity]. They are recorded here in the combined form the manifold theory uses, as we sought to show.

(equivalence between connected and path connected components in the right situation. On a manifold, connected if and only if path connected)

(Show that there is only one norm on finite dimensional vector space)

(In general, if a function takes in a function and spits out another function, it is called an *operator*)

0.2 Differentiation

Being differentiable at a point x essentially means that if we zoom in enough at the point x, we can arbitrarily well approximate an open neighborhood of that point with some linear function, i.e., the function is locally linear. In $\mathbb{R}$, we can capture this intuition with the following: the function $f : \mathbb{R} \to \mathbb{R}$ is differentiable at a point x if

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

exists, or equivalently if we substitute $a = x + h$, we get that $h \to 0$ become $a \to x$ and

$$\lim_{a \to x} \frac{f(a) - f(x)}{a - x}$$

exists. If it exists, we usually denote the result of this limit as $f'(x)$. In higher dimensions, this formulation of differentiability doesn't quite work (for example, the composition of differentiable functions need not be differentiable). Hence, we re-formulate this the definition to be that the function f is differentiable at x if there exists a $m = f'(x)$ such that

$$\lim_{h \to 0} \frac{f(x+h) - f(x) - mh}{h} = 0$$

In this way, $f'(x)$ will in fact give us back linear maps, which in this case can always be represented as 1×1 matrices. In higher dimensions $\mathbb{R}^n$, we generalize this notion by taking the norm: a function $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at a point $x \in \mathbb{R}^n$ if there exists a linear transformation $A_x : \mathbb{R}^n \to \mathbb{R}^m$ such that:

$$\lim_{h \to 0} \frac{\|f(x+h) - f(x) - A_x(h)\|}{\|h\|} = 0$$

with the usual $\mathbb{R}^m$ norm in the numerator and $\mathbb{R}^n$ norm in the denominator. Recall (or re-prove) that the derivative is unique, so if two linear transformations A_x and B_x satisfy this same limit, they are in fact equal: $A_x = B_x$. If we choose the standard basis $\beta = \{e_1, ..., e_n\}$ for $\mathbb{R}^n$, then the matrix representation of any derivative at a, $Df(a)$, is will be an $m \times n$ matrix and is called the *Jacobian Matrix*. If $U \subseteq \mathbb{R}^n$ is an open set and if for every $x \in U$, f is differentiable at x, then we say that $f : U \to \mathbb{R}^m$ is a differentiable function. Usually, A_x is written as $Df(x) : \mathbb{R}^n \to \mathbb{R}^m$. If f is differentiable on U, then we can say that $Df : U \to \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is a map that takes in a point $x \in U \subseteq \mathbb{R}^n$ and gives a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$. Since $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m) \cong \mathbb{R}^{nm}$ as a vector-space, then it is equivalent to consider $Df : \mathbb{R}^n \to \mathbb{R}^m$, meaning we can consider the differeniability of Df. If Df is differentiable at a in $\mathbb{R}^n$, then we would write $D(Df)(a)$, or more succiently $D^2 f(a)$. If $D^2 f$ is differentiable at every $x \in \mathbb{R}^n$, then we would get a map $D^2 f : \mathbb{R}^n \to \mathbb{R}^{n^2 m}$, or in terms of linear transformation $D^2 f : \mathbb{R}^n \to \mathcal{L}(\mathbb{R}^n, \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m))$. It is a useful fact to remember that $\mathcal{L}(\mathbb{R}^n, \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m))$ is isomorphic to $\mathcal{L}^2(\mathbb{R}^n, \mathbb{R}^m)$ which is the set of all bilinear linear transfomrations from $\mathbb{R}^n$ to $\mathbb{R}^m$, that is all functions $f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^m$ such that $f(rx, y) = f(xr, y) = rf(x, y)$ and f is linear in the first and second commponent. We can repeat this process to define $D^n f$, and if $D^n f$ exists for all $a \in \mathbb{R}^n$, then $Df^n f : \mathbb{R}^n \to \mathcal{L}^n(\mathbb{R}^n, \mathbb{R}^m)$ exists. The linear function in the codomain of DF can be "realised" as mappings between tangent spaces from U to $\mathbb{R}^m$. Recall that $\mathbb{R}^n_a$ is the vector-space centered at a. Then $Df(a)$ can be thought of having domain and codomain

$$Df(a) : \mathbb{R}^n_a \to \mathbb{R}^n_{f(a)}$$

This will become very important once we start trying to define a notion of integrating manifolds.

The composition of two differnetiable functions is differentiable: If $f : \mathbb{R}^n \to \mathbb{R}^m$ and $g : \mathbb{R}^m \to \mathbb{R}^p$ are two functions such that f is differentiable at a and g is differentiale at $f(a)$, then $g \circ f(a)$ is also differentiable and

$$D(g \circ f)(a) = Dg(f(a)) \circ Df(a)$$

given the jacobian matrix representation of these linear transofmrations, we can see this as multiplying matrices $[Dg(f(a))]_\beta \cdot [Df(a)]_\beta$. Using the chain rule, we can see that addition of two differentiable functions is differentiable (since $f + g = s(f, g)$ where $s(x, y) = x + y$ and $(f, g)(a) = (f(a), g(a))$), and $D(fg)(a) = Df(a)g(a) + f(a)Dg(a)$.

Finding the jacobian matrix $[Df(a)]_\beta$ is made easy using patial derivatives. If β is the standard basis (in particular, it is ordered), $f : \mathbb{R}^n \to \mathbb{R}$ is a function, and $a \in \mathbb{R}^n$, then f has a *ith partial derivaitve at a* if

$$\lim_{h \to 0} \frac{f(a_1, a_2, ..., a_i + h, a_{i+1}, ..., a_n)}{h}$$

exists and is denoted $D_i f(a)$ of $\frac{\partial f(a)}{\partial x_i}$. In particular, notice that this is just a function from $\mathbb{R}$ to $\mathbb{R}$ in disguise: letting $g(x) = f(a_1, a_2, ..., a_{i-1}, x, a_{i+1}, ..., a_n)$, we see that $g'(a_i) = D_i f(a)$. Hence, all standard notions and tecniques of differentiation are already available to us. As with differentiation of $f : \mathbb{R}^n \to \mathbb{R}^m$, if $D_i f(a)$ is defined for all $x \in \mathbb{R}^n$, then we have a function $D_i f : \mathbb{R}^n \to \mathbb{R}$. Hence, we can take the partial derivative of $D_i f$ in any direction: $D_j(D_i f)$, which we usually denote $D_{i,j} f$. We will soon cover how $D_{i,j} f = D_{j,i} f$ (If $D_{i,j} f$ and $D_{j,i} f$ are continuous) after we cover how we can use partial derivative information to find $[Df]_\beta$.

If Df exists, then each $D_i f$ exists. However, the converse is not true: every $D_i f$ can exist but Df might not exist. In fact, if we generalize our notion form partial derivative to directional derivative (where instead of $a_i + h$ we take $a_i + \mathbf{u}h$ for some unit vector $\mathbf{u}$), then it is possible that *every* directional derivative exists but Df still doesn't exist. However, if each $D_i f$ is *continuous*, then this in fact implies that Df exists. It would be nice if Df existing implied that each $D_i f$ was continuous, however that is not the case. Thus, since we are comfortable with working with $D_i f$, we usually don't look at differentiable functions, but the stronger class of functions with continuous partial derivatives, denoted $C^1(U, \mathbb{R}^n)$. If $n = 1$, we usually denote this as $C^1(U)$. If the partial derivatives are r-times differentiable with the partial of $D^r f$ being continuous, we denote this as $C^r(U, \mathbb{R}^n)$. If f is in C^r for all $r \geq 0$, then we say that f is *smooth* and is in C^∞. Furthermore, if f is bijective (hence invertible), then if f^{-1} is also smooth, we say that f is a *diffeomorphism*. Theorem 0.2.1 shows that if f is C^∞ and invertible, then f^{-1} is automatically C^∞, giving us a simpler criterion for finding the "isomorphisms" in the appropriate category (which we'll define soon).

One important trick to remember is that:

$$D_x f(p) = \left.\frac{d}{dt}\right|_{t=0} f(p + tx)$$

With partial derivatives defined, we can finally find $[Df(a)]_\beta$ using them. If $f : \mathbb{R}^n \to \mathbb{R}^m$ is a functions where $f = (f^1, ..., f^m)$ where $f^i : \mathbb{R}^n \to \mathbb{R}$, then

$$[Df(a)]_\beta = \begin{pmatrix} D_1 f^1(a) & D_2 f^1(a) & \cdots & D_n f^1(a) \\ D_1 f^2(a) & D_2 f^2(a) & \cdots & D_n f^2(a) \\ \vdots & \vdots & \ddots & \vdots \\ D_1 f^m(a) & D_2 f^m(a) & \cdots & D_n f^m(a) \end{pmatrix}$$

i.e., if $M = [Df(a)]_\beta$, then $M_{ij} = D_j f^i$. We see that this matrix is indeed the derivative by first taking $m = 1$, then taking $h(x) = (a_1, ..., x, ..., a_n)$ with x in the jth place, and then noticing that

$D_j f(a) = (f \circ h)'(a_j)$ so that

$$\begin{aligned}(f \circ h)'(a_j) &= f'(a) \cdot h'(a_j) \\ &= f'(a) \cdot \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}\end{aligned}$$

We end this review with a reminder of the differentiability of f^{-1} whenever it is defined for a differentiable function f and a theorem which let's us convert "relations" into functions, in particular it is used to easily go back and forth between different coordinates:

Theorem 0.2.1: Inverse Function Theorem

Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be a C^1 function. $a \in U$ and $\det Df(a) \neq 0$. Then there exists an open set $V \subseteq \mathbb{R}^n$ and $W \subseteq \mathbb{R}^n$ where $a \in V$, $f(a) \in W$, such that $f : V \to W$ has a differentiable inverse $f^{-1} : W \to V$, and for every $y \in W$:

$$D(f^{-1})(y) = [Df(f^{-1}(y))]^{-1}$$

One of the great results of the inverse function theorem is that it suffices that f is C^∞ and invertible to be a diffeomorphism. First, we prove a lemma:

Lemma 0.2.2: Bounding On Closed Rectangles

Let $A \subseteq \mathbb{R}^n$ be a closed rectangle and let $f : A \to \mathbb{R}^n$ be continuously differentiable. If M exists such that for all $x \in \operatorname{int} A$, $|D_i f^i(x)| \leq M$, then:

$$|f(x) - f(y)| \leq n^2 M |x - y|$$

for all $x, y \in A$

***Proof*:**
We first create a tellescoping sum:

$$|f^i(y) - f^i(x)| = \sum_{j=1}^{n} f(y^1, y^2, ..., y^j, x^{j+1}, ..., x^n) - f(y^1, y^2, ..., y^{j-1}, x^j, ..., x^n)$$

Applying the man value to each term on the right hand side, we get:

$$\begin{aligned}f(y^1, y^2, ..., y^j, x^{j+1}, ..., x^n) - f(y^1, y^2, ..., y^{j-1}, x^j, ..., x^n) &= (y^j - x^j) D_j f^i(z_{ij}) \\ &\leq |y^j - x^j| M\end{aligned}$$

for appropraite z_{ij}, thus since $|y^i - x^i| \leq |x-y|$ by the reverse triangle inequality, $|f^i(y) - f^i(x)| \leq nM|y-x|$. Finally, with some norm manipulation and the triangle inequality, we get:

$$|f(y) - f(x)| \leq \sum_{i=1}^{n} |f^i(y) - f^i(x)| \leq n^2 M|y - x$$

completing the proof

We know proceed with the proof for the Inverse Function Theorem:

***Proof*:**
We will start by showing that an inverse exists, which will use the fact that $\det Df(a) \neq 0$ to find a neigbhorhood around a that is invertible. Showing that it's continuous will almost immediately come out of the proof of bijectivity. Then,

For notational simplicity, let λ be the linear transformation $Df(a)$. Then λ is non-singular since $\det Df(a) \neq 0$. Now, consider

$$D(\lambda^{-1} \circ f)(a) = D(\lambda^{-1})(f(a)) \circ Df(a) = \lambda^{-1} \circ Df(a) = \text{id}$$

Notice that if we prove the theorem for $\text{id} = \lambda^{-1} \circ f$, it is certainly true for f (since λ is non-singular, it would be the composition of two invertible functions). Thus, without loss of generality, we may assume λ is the identity.

This will now force injectivity on a local level. Let's say $f(a+h) = f(a)$. Then:

$$\frac{|f(a+h) - f(a) - \lambda(h)}{|h|} = \frac{|h|}{|h|} = 1$$

However,

$$\lim_{h \to 0} \frac{|f(a+h) - f(a) - \lambda(h)}{|h|} = \frac{|h|}{|h|} = 0$$

Thus, there must exist a clsoed rectangle, U, containing a in its interior such that $f(x) \neq f(a)$, $x \in U$ when $x \in A$. Since det is a continuous function and f is continuously differentiable, we can also assume that:

$$\det Df(x) \neq 0 \quad x \in U \tag{1}$$

$$|D_j f^i(x) - D_j f^i(a)| \leq \frac{1}{2n^2} \quad \text{for all } i, j \text{ and } x \in U \tag{2}$$

The partiular choice of bound on the partial derivaties is to employ the following trick; we see to make sure that f grows faster than the distance between two points so that we can prove that we can find an inverse. To that end, we made the choice in the 2nd bullent point so that if we combine the last point with lemma 0.2.2 and apply it to $g(x) = f(x) - x$, we get:

$$|f(x_1) - x_1 - (f(x_2) - x_2)| \leq \frac{1}{2}|x_1 - x_2|$$

Using the reverse triangle inequality, we get:

$$|x_1 - x_2| + |f(x_1) - f(x-2)| \leq |f(x_1) - x_1 - (f(x_2) - x_2)| \leq \frac{1}{2}|x_1 - x_2|$$

Thus we also get a bound of:

$$|x_1 - x_2| \leq 2|f(x_1) - f(x_2)| \tag{3}$$

for all $x_1, x_2 \in U$, showing us that f must grow faster than the linear distance between two points.

Now, since ∂U is compact, so is $f(\partial U)$. By the the fact that $f(x) \neq f(a)$ for all $a \neq x \in U$, $f(a) \notin f(\partial U)$. Therefore, there exists a $d > 0$ such that $|f(x) - f(a)| \geq d$ for all $x \in \partial U$. Let $W = \{y \mid |y - f(a)| \geq d/2\}$. Note that for any $y \in W$, the distnce between y and $f(x)$ where $xin\partial U$ must always be greater than the distance between y and $f(a)$:

$$|y - f(a)| < |y - f(x)| \tag{4}$$

With this, we can show that for all $y \in W$, there exists a unique $t \in \int(U)$ such that $f(t) = Y$ (showing the existence of an inverse on W). To show existence, we will take advantage of the determinant information we established in equation (1) and (2). In particular, define $g : U \to \mathbb{R}$,

$$g(x) = |y - f(x)^2|^2 = \sum_{i=1}^{n} (y^i - f^i(x))^2$$

Since g is continuous on U and U is bounded, it's minimum must be achieved somehwere. Let's say the minimum as at t. Since g is differentiable, we can find the minimum using the derivative. We know that the minimum of g is achieved. Importantely, by equation (4), $g(a) < g(x)$ for all $x \in \partial U$, so the minimum is not achieved on the border (which will be importance for f^{-1} being differentiable). Then:

$$\sum_{i=1}^{n} 2(y^1 - f^i(t)) \cdot D_i f^i(t) = 0$$

however, $\det Df(x) \neq = 0$ for all $x \in \int U$! Therefore, the only possibility left for making this equation is if $y^i - f^i(t) = 0$ for all i, i.e. $y = f(t)$. For uniqueness, if two points $x_1 \neq x_2$ had $f(x_1) = f(x_2)$, then this would violate equation (3).

Thus, let $V = \int U \cap f^{-1}(W)$, so that $f : V \to W$ has inverse $f^{-1} : W \to V$. To show that f^{-1} is continuous, notice that equation (3) also implies:

$$|f^{-1}(y_1) - f^{-1}(y_2)| \leq 2|x_1 - x_2| \tag{5}$$

which immediately implies continuity of f^{-1}

Finally, we show that f^{-1} is differentiable., in particular if $\mu = Df(x)$ ($x \in V$), then $D(f^{-1})(x) = \mu^{-1}$. Similarly to how we proved the chain rule, let

$$f(x_1) = f(x) + \mu(x_1 - x) + \varphi(x_1 - x)$$

where the error term φ tends to zero faster than linearly

$$\lim_{x_1 \to x} \frac{|\varphi(x_1 - x)|}{x_1 - x} = 0$$

Manipulating the equation around, we get the equivalent equation of:

$$\mu^{-1}(f(x_1) - f(x)) = x_1 - x + \mu^{-1}(\varphi(x_1 - x))$$

$$x_1 = x + \mu^{-1}(f(x_1) - f(x)) - \mu^{-1}(\varphi(x_1 - x))$$

Since f is invertible, we can re-write $x_1 = f^{-1}(y_1)$ and $x = f^{-1}(y)$, and re-shuffle to get:

$$f^{-1}(y_1) = f^{-1}(y) + \mu^{-1}(y_1 - y) - \mu^{-1}(\varphi(f^{-1}(x_1) - f^{-1}(x)))$$

and so it suffices to show that the error term goes to zero faster than linearly:

$$\lim_{y_1 \to y} \frac{|\mu^{-1}(\varphi(f^{-1}(y_1) - f^{-1}(y)))|}{y_1 - y} = 0$$

Using the operator norm on μ^{-1}, we get that if this limit exists, then

$$\begin{aligned}\lim_{y_1 \to y} \frac{|\mu^{-1}(\varphi(f^{-1}(y_1) - f^{-1}(y)))|}{y_1 - y} &\leq \lim_{y_1 \to y} \frac{M|\varphi(f^{-1}(y_1) - f^{-1}(y))|}{y_1 - y} \\ &= M \lim_{y_1 \to y} \frac{|\varphi(f^{-1}(y_1) - f^{-1}(y))|}{y_1 - y}\end{aligned}$$

which will go to zero if and only if $\lim_{y_1 \to y} \frac{|\varphi(f^{-1}(y_1) - f^{-1}(y))|}{y_1 - y}$ goes to zero. With this, we can do the same trick as we've done with the chain-rule:

$$\begin{aligned}&\frac{|\varphi(f^{-1}(y_1) - f^{-1}(y))|}{y_1 - y} \\ &\qquad = \frac{|\varphi(f^{-1}(y_1) - f^{-1}(y))|}{f^{-1}(y_1) - f^{-1}(y)} \cdot \frac{|f^{-1}(y_1) - f^{-1}(y)|}{y_1 - y} \\ &\qquad \leq \frac{|\varphi(f^{-1}(y_1) - f^{-1}(y))|}{f^{-1}(y_1) - f^{-1}(y)} \cdot 2 \qquad \text{by equation(5)}\end{aligned}$$

Since f^{-1} is continuous, as $y_1 \to y$, $f^{-1}(y_1) \to f^{-1}(y)$, in particular limits are preserved when taking composing with continuous functions, and hence the first factor goes to 0, and so the whole limit goes to zero. Since this is true for any $y \in W$, f^{-1} is indeed differentiable with derivative μ^{-1}, as we sought to show.

Note that it is possible that f^{-1} exists even if $\det Df(a) = 0$, (ex. x^3), howver f^{-1} cannot be differentiable at the point where $\det Df(a) = 0$

The next theorem is useful in identifying when we can write a set of equations in terms of different parameters. In particular, if we have a function $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$, and take the collection of points where $F(x^1, ..., x^n, y^1, ..., y^m) = 0$ (i.e. $F^{-1}(0)$), then under appropriate conditions, there exists an $f : \mathbb{R}^n \to \mathbb{R}^m$ such that $F(x^1, ..., x^n, f(x^1, ..., x^n)) = 0$

This theorem is particularly useful if you recall that a manifold can be defned as the zero locus of a C^1 function with Df having constant rank.

Theorem 0.2.3: Implicit Function Theorem

Let $f : U \subseteq \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ be a continuously differentiable function and $(a, b) \in U$, $f(a, b) = 0$. Let M be an $m \times m$ matrix such that

$$M_{ij} = D_{n+j}f^i(a, b) \qquad 1 \leq i, j \leq m$$

If $\det M \neq 0$, then there exists an open set $A \subseteq \mathbb{R}^n$ containing a and an open set $B \subseteq \mathbb{R}^m$ containing b such that for each $x \in A$, there is a unique $g(x) \in B$ such that $f(x, g(x)) = 0$. Furthermore, the function g is differentiable.

If you are visual, then M is

$$\begin{pmatrix} D_{n+1}f^1(a,b) & D_{n+2}f^1(a,b) & \cdots & D_{n+m}f^1(a,b) \\ D_{n+1}f^2(a,b) & D_{n+2}f^2(a,b) & \cdots & D_{n+m}f^2(a,b) \\ \vdots & \vdots & \ddots & \vdots \\ D_{n+1}f^m(a,b) & D_{n+2}f^m(a,b) & \cdots & D_{n+m}f^m(a,b) \end{pmatrix}$$

***Proof*:**
We will be using the Inverse Function theorem to prove this result. Let $F : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n \times \mathbb{R}^m$, $F(x, y) = (x, f(x, y))$. Since $\det M \neq 0$, then by construction of F, if $0 = f(a, b)$ $\det DF(a, b) = \det M \neq 0$. Thus, by the Inverse Function theroem, there exists an $W \subseteq \mathbb{R}^n \times \mathbb{R}^m$ containing $F(a, b) = (a, 0)$ and an open set $V \subseteq \mathbb{R}^n \times \mathbb{R}^m$ containing (a, b), which we see can be easily written as $A \times B$, such that $F : A \times B \to W$ has a differentiable inverse function $F^{-1} : W \to A \times B$. By the form of F, $F^{-1}(x, y) = (x, k(x, y))$ for appropriate differentiable function k.

With this function defined, letting $\pi : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ be the natural projection function $\pi(x, y) = y$, we get the following chain of equalities:

$$\begin{aligned} f(x, k(x, y)) &= f \circ F^{-1}(x, y) \\ &= ((\pi \circ F) \circ F^{-1})(x, y) \\ &= (\pi \circ (F \circ F^{-1}))(x, y) \\ &= \pi(x, y) \\ &= y \end{aligned}$$

Thus, $f(x, k(x, y)) = y$, in particular $f(x, k(x, 0)) = 0$. Letting $g(x) = k(x, 0)$, we get $f(x, g(x)) = 0$, completing the proof.

In my mind, the key use of the Implicit function theorem is that it let's you relate a set of "inputs" to some desired output

Example 0.2.4: Implicit Function Theorem

Let $p : \mathbb{C} \to \mathbb{C}$, $p(z) = a_0 + \cdots + a_{n-1}z^{n-1}z$. Let's say all the roots of p are distinct and r is a root, $p(r) = 0$. If $\mathbb{C}^n$ is the set of all possible coefficients, is there a smooh way of relating the coefficents to the root? That is, is there a function: $f_r : \mathbb{C}^n \to \mathbb{C}$ such that given the coefficients, we can find a smooth function from $\mathbb{C}^n$ to the roots.

Going one way is easy: if $p(z) = (z - \alpha_1) \cdots (z - \alpha_n)$ are the roots, then multiplying up we get

$p(z) = z^n + \sigma_{n-1} z^{n-1} + \cdots + \sigma_1 z = \sigma_0$, where σ_i are the symmetric functions. Findin an inverse of this function is really hard. However, we can find it "implicitely" by defining:

$$F(a_0, ..., a_{n-1}, z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n$$

Then $f(a_0, a_1, ..., a_{n-1}, r) = 0$, and since all the roots are distinct we have that $\partial F(a, r)/\partial z \neq 0$, and hence we can find a C^1 inverse!

The key here is that there was some "input" (i.e. coefficients) that we wanted to relate to some output (in this case a root).

(A word on why it has to one root, namely the need for bijectivity).

Finding the derivative of $g(x)$ is feasible as well. Given that $f(x, g(x)) = 0$, $f^i(x, g(x)) = 0$. Taking the derivative D_j on both sides, we get:

$$D_j f^i(x, g(x))) = D_j f^i(x, g(x)) = \sum_{\alpha=1}^{m} D_{n_\alpha} f^i(x, g(x)) \cdot D_j g^\alpha(x)$$

and since $\det M \neq 0$, these equations can be solved fof $D_j g^\alpha(x)$. The answer will depend on $D_j f^i(x, g(x))$, and hence ultimately on $g(x)$. Unfortuantely, there is no more real simplification we can do here, namely since $g(x)$ is not necessarily a unique answer. For example, if we consider $f(x, y) = x^2 + y^2 - 1$, then we get $g(x) = \pm\sqrt{1 - x^2}$ as possible solutions to $f(x, g(x)) = 0$. If we differentiate f at 0, at $f(x, g(x)) = 0$, we get:

$$(D_1 f(x, g(x)) + D_2 f(x, g(x))) \cdot g'(x) = 0$$

i.e.

$$2x + 2g(x) \cdot g'(x) = 0 \quad \Leftrightarrow \quad g'(x) = \frac{-x}{g(x)}$$

which works for both $\pm$.

(when you get to Clairuts theorem show how it is a simple task to generalize form when we can swap any two indices).

0.3 Integration

Integrating over $\mathbb{R}$ and $\mathbb{R}^n$ will be taken as known (in particular, integrable functions, zero measures sets, and Fubini a). We will be reviewing what is important for understanding integrating manifolds. The first important concept to remember is that integrals are invariant under diffeomorphism as lon as the domain of integration changes properly as well:

Theorem 0.3.1: 1 Dimensional Change Of Variables

Let g be C^1 on $[a, b]$ and let f be continuous on $g([a, b])$. Then, with $u = g(x)$:

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

***Proof*:**
This is the substitution rule of elementary calculus [4]: if $F' = f$, then $(F \circ g)' = (f \circ g)g'$ by the chain rule, and the fundamental theorem of calculus evaluates both sides to $F(g(b)) - F(g(a))$.

Theorem 0.3.2: Change Of Variables

Let $\Omega \subseteq \mathbb{R}^n$ be open, let $g\colon \Omega \to g(\Omega)$ be a diffeomorphism, and let f be a Lebesgue measurable function on $g(\Omega)$. If $f \geq 0$ or f is integrable, then:

$$\int_{g(\Omega)} f \, d\lambda = \int_{\Omega} (f \circ g) \, |\det Dg| \, d\lambda$$

***Proof*:**
This is proved in *Real Analysis* [14, Change of Variables for C^1 Diffeomorphisms], where g is only required to be a C^1 diffeomorphism (so the smooth diffeomorphisms of this book qualify). Note the absolute value on the Jacobian determinant: the integral against Lebesgue measure carries no orientation, and $|\det Dg|$ is exactly the factor that keeps the identity true when g reverses orientation.

Example 0.3.3: Change Of Variables
put here all the common change of var's.

Theorem 0.3.4: Subordinate Partitions Of Unity

Let $A \subseteq \mathbb{R}^n$ and let $\mathcal{O}$ be an open cover for A. Then there exists a collection Φ of C^∞ functions φ defined defined in an open set containing A such that

1. For each $a \in A$, $0 \leq \varphi(x) \leq 1$ for all $\varphi \in \Phi$
2. For each $a \in A$, there exists an open neighborhood V_a such that $a \in V_a$ and only finitely many $\varphi \in \Phi$ are nonzero at a
3. For each $a \in A$, $\sum_{\varphi \in \Phi} \varphi(a) = 1$ (by the previous condition, this is always a finite sum).
4. For each $\varphi \in \Phi$, there is an open set $U \in \mathcal{O}$ such that $\varphi = 0$ outside some closed set V containing U.

If we just have the first 3 conditions, we call Φ a partition of unity of A, and if we add the 4th condition we say that the partition of unity is subordinate to $\mathcal{O}$. The proof follows a similar strategy to many of its kind: first we prove it on compact sets. Let U be an open set contianing A. Since A is compact, a we can choose a finite open cover automatically, satisfying the 2nd condition. Then we can take the closure of these sets, or try to refine our choice after applying compactness so that each $U_i \subseteq A$ (such an open cover is caleld admisible) and repeat a slightly more elaborate but elementary construction (Spivak calculus on Manifolds p.63). Let's label the compact sets whose interior cover A in an admissible $\{D_i\}_i$. way Then we can choose some C^∞ functions ψ_i so that they are 0 outside of D_i. Then we can define $\varphi_i = \frac{\psi_i}{\sum_i \psi_i}$. Finally, choosing some C^∞ function $f : U \to [0,1]$ that is zero outisde of A. Then $\Phi = \{f \circ \varphi_i\}_{i=1}^n$, completing the compact case. Then we expand this in a similar way we would do for a σ-algebra, and then to general open sets by taking advantage of the fact that any open set has an "open set exaustion", and finally for arbitrary sets by taking the union of all open sets of the cover of A to be the open set containing A, then applying the previous step.

0.4 Category Theory

(If these notes expand into something publishable, I will add some elementary category theory so that I may use the terminology throughout the book. Important concecpts are universal properties, limits and colimits, isomorphisms, natural transfomrations)

0.5 Linear and Multilinear Algebra

(put dual spaces, tensor algebras, and their further derivations here like the alternating and symmetric algebra)

0.5.1 Dual Space

Recall that $V^* = V^\vee = \mathrm{Hom}_V(V, k)$ is the *dual vector space* of V, and is isomorphic to V.

Proposition 0.5.1: Basis For Dual Space

Let V be a vector-space with basis $(e_1, e_2, ..., e_n)$. Then the collection $(f_1, f_2, ..., f_n)$ defined by:

$$f^i(e_j) = \delta^i_j$$

where δ^j_i is the kronecker delta is a basis for V^*

***Proof*:**
exercise

Thus, if $v \in V$, then $v = \sum_i f^i(v_i)$. Since a linear functional $f : \mathbb{R}^n \to \mathbb{R}$ would have a matrix representation as a row, we sometimes write f^i in a matrix representation as:

$$f^1 = (1 \ 0 \ \cdots \ 0) \quad f^2 = (0 \ 1 \ 0 \ \cdots \ 0) \quad \cdots \quad f^n = (0 \ 0 \ \cdots \ 1)$$

If $\omega \in V^*$, we can write it in terms of the basis as:

$$\omega = \omega_i f^i \qquad \omega(e_i) = \omega_i$$

If $f : V \to W$, then $f^* : W^* \to V^*$ is defined by taking the function $x \in W^*$ to the function $f^*(x) \in V^*$ defined by:

$$f^*(x) = x \circ f$$

Furthermore:

$$(g \circ f)^* = f^* \circ g^* \qquad (\mathrm{id}_V)^* = \mathrm{id}_{V^*}$$

showing that $(\_\_\_)^*$ is a functor from $\mathbb{R}$-**Vect** to itself (more generally, any field k will work). Furthermore, the map $(\_\_\_)^{**}$ is a natural isomorphism.

(If you want, add a word on angle bracket notation like John Lee doesp. 274)

0.5.2 Multilinear Algebra

Recall the following in the context of vector-spaces:

Definition 0.5.2: Multilinear Map

Let $V_1, V_2, ..., V_k$ and W be vector-spaces. Then $f : V_1 \times V_2 \times \cdots \times V_k \to W$ is called *multilinear* if it is linear in each component.

Example 0.5.3: Multi-Linear Maps

1. the dot product from $\langle \cdot,\ \cdot \rangle : \mathbb{R}^n \to \mathbb{R}$ is bilinear map that is interpreted as the "length" of vectors along with some orientation them
2. the cross product on $\cdot : \mathbb{R}^3 \to \mathbb{R}$ is a bilinear map interpreted as computing the area of a parallelogram, or finding an orthogonal vector to two vectors which with priority given to "right-handedne".
3. The determinant $\det(\_\_\_) : \mathbb{R}^{n^2} \to \mathbb{R}$ as a n-linear function giving us a notion of volume
4. The lie bracket for a lie algebra $\mathfrak{g}$ is a bilinear map (not sure how to interpret it yet)
5. The Euler form on root lattices from Lie Algebra representation is another example.

We see how multilinear maps are pretty common in geometry, and many represent geometric notions like volume or length (idk what lie brackets yet represent TBD). Recall from geometry that given a R-module M, we can define a new R-module

$$\mathcal{T}(M) = \bigoplus_{i \in \mathbb{N}} \mathcal{T}^k(M)$$

where $\mathcal{T}^k(M) = \bigotimes_i^k M$. In our case, we will let $M = V^*$ be the dual of a vector-space. Then for any $f, g \in V^*$, we define their tensor-product to be:

$$f \otimes g(a, b) = f(a)g(b)$$

notice that $f \otimes g$ is bilinear, and so by the universal property of tensors we are jusitifed in writing $f \otimes g$. We will say that $f \otimes g \in \mathcal{L}^2(V, V, R)$, or since the vector-spaces are the same, we will write $\mathcal{L}^2(V, \mathbb{R})$. Furthermore, $\mathcal{L}^2(V, \mathbb{R})$ is still a vector-space.

This process easily generalized: Let $F \in \mathcal{L}^n(V_1, V_2, ..., V_n, \mathbb{R})$ and $G \in \mathcal{L}^m(W_1, W_2, ..., W_m, \mathbb{R})$ be an n-linear and m-linear map respectively. Then:

$$F \otimes G(v_1, ..., v_n, w_1, .., w_m) = F(v_1, ..., v_n, w_1, ..., w_n)$$

is an $n + m$-linear map and belongs in $\mathcal{L}^{n+m}(V_1, V_2, ..., V_n, W_1, W_2, ..., W_m, \mathbb{R})$ (recall that tensor-products are associative, and hence this is indeed well-defined).

Recall that a basis for $\mathcal{L}^n(V_1, V_2, ..., V_n, \mathbb{R})$ is simply the k-tensor product of the basis for each V_i^*. Due to this, we have:

$$V_1^* \otimes \cdots V_k^* \cong_{k\text{-}\mathbf{Vect}} \mathcal{L}^k(V_1, ..., V_k, \mathbb{R})$$

and so since $V_1 \cong V_1^{**}$ canonically, we also have:

$$V_1 \otimes \cdots V_k \cong_{k\text{-}\mathbf{Vect}} \mathcal{L}^k(V_1^*, ..., V_k^*, \mathbb{R})$$

giving us a characterization of tensor products in terms of a k-linear map on *dual spaces*.

0.5.3 Symmetric And Alternating Tensors

The most important tensors products for us are those that have more geometric meaning. We present the two important ones here:

Symmetric Tensors

Definition 0.5.4: Symmetric Tensor

Let V be a finite dimensional vector space. Then a covariant k-tensor on V is said to be *symmetric* if it's value is unchanged if we interchange any pairs of arguments

The collection of all symmetric k-tensors is denoted $\Sigma^k(V)$

(eventually will fill)

Alternating Tensors

Definition 0.5.5: Alternating Tensor

Let V be a finite dimensional vector space. Then a covariant k-tensor on V is said to be *alternating* if it's sign changes if we interchange any pairs of arguments

The collection of all alternating k-tensors is denoted $\Lambda^k(V)$

Proposition 0.5.6: Properties Of Alternating Tensors

Let α be a covariant k-tensor on a finite dimensional vector-space V. Then the following are equivalent:

1. α is alternating;
2. $\alpha(v_1, \ldots, v_k) = 0$ whenever $(v_1, \ldots, v_k)$ is linearly *dependent*;
3. $\alpha(v_1, \ldots, v_k) = 0$ whenever two of the arguments are equal.

***Proof*:**
(1)⇒(3): interchanging two equal arguments both fixes α and negates it, so $\alpha = -\alpha = 0$ there. (3)⇒(1): for the transposition of arguments i and j, expand $0 = \alpha(\dots, v_i + v_j, \dots, v_i + v_j, \dots)$ by multilinearity; the two repeated-argument terms vanish by (3), leaving $\alpha(\dots, v_i, \dots, v_j, \dots) + \alpha(\dots, v_j, \dots, v_i, \dots) = 0$, and every permutation is a product of transpositions. (3)⇒(2): if $(v_1, \dots, v_k)$ is dependent, some v_j is a combination of the others; multilinearity expands $\alpha(v_1, \dots, v_k)$ into terms each with a repeated argument, all zero by (3). (2)⇒(3): equal arguments make the tuple dependent.

0.5.4 The Wedge Product and the Exterior Algebra

The alternating tensors of every degree assemble into a single graded algebra once equipped with a product that keeps the result alternating. That product is the wedge, and the resulting exterior algebra is the pointwise model for the differential forms of chapter 12.

Definition 0.5.7: Wedge Product

For $\alpha \in \Lambda^k(V)$ and $\beta \in \Lambda^l(V)$, the *wedge product* $\alpha \wedge \beta \in \Lambda^{k+l}(V)$ is

$$(\alpha \wedge \beta)(v_1, \dots, v_{k+l}) = \sum_{\sigma} \operatorname{sgn}(\sigma)\, \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)})\, \beta(v_{\sigma(k+1)}, \dots, v_{\sigma(k+l)}),$$

the sum over all (k, l)-*shuffles*: permutations $\sigma \in S_{k+l}$ with $\sigma(1) < \cdots < \sigma(k)$ and $\sigma(k+1) < \cdots < \sigma(k+l)$.

The result is again alternating: interchanging two of the v's either keeps both inside a single block or crosses the blocks, and in each case pairs the shuffles into terms of opposite sign.

Proposition 0.5.8: Properties of the Wedge Product

The wedge product is bilinear and associative, and *graded anti-commutative*: for $\alpha \in \Lambda^k(V)$ and $\beta \in \Lambda^l(V)$,

$$\alpha \wedge \beta = (-1)^{kl}\, \beta \wedge \alpha.$$

In particular $\alpha \wedge \alpha = 0$ when α has odd degree, and for covectors $e^i \wedge e^j = -\, e^j \wedge e^i$.

***Proof*:**
Bilinearity is immediate from the formula. For graded anti-commutativity, the permutation that moves the last l arguments in front of the first k has sign $(-1)^{kl}$ and carries the (k, l)-shuffles for $\alpha \wedge \beta$ bijectively onto the (l, k)-shuffles for $\beta \wedge \alpha$, so the two sums agree up to $(-1)^{kl}$. Associativity holds because both $(\alpha \wedge \beta) \wedge \gamma$ and $\alpha \wedge (\beta \wedge \gamma)$ equal the sum over (k, l, m)-shuffles of $\operatorname{sgn}(\sigma)$ times the product of the three factors.

The wedge writes every alternating tensor in a basis of decomposable elements and pins the dimension of $\Lambda^k(V)$.

Theorem 0.5.9: Basis of the Alternating Tensors

Let $(e_1, \dots, e_n)$ be a basis of V with dual basis $(e^1, \dots, e^n)$. Then

$$\{\, e^{i_1} \wedge \cdots \wedge e^{i_k} : 1 \le i_1 < \cdots < i_k \le n \,\}$$

is a basis of $\Lambda^k(V)$, so $\dim \Lambda^k(V) = \binom{n}{k}$ (and $\Lambda^k(V) = 0$ for $k > n$). The pairing with basis vectors is the determinant,

$$(e^{i_1} \wedge \cdots \wedge e^{i_k})(e_{j_1}, \dots, e_{j_k}) = \det\big(e^{i_a}(e_{j_b})\big)_{a,b},$$

and every $\alpha \in \Lambda^k(V)$ expands as $\alpha = \sum_{i_1 < \cdots < i_k} \alpha(e_{i_1}, \dots, e_{i_k})\, e^{i_1} \wedge \cdots \wedge e^{i_k}$.

***Proof*:**
Applying the wedge formula to basis vectors, the general entry is the permutation expansion of the determinant $\det(e^{i_a}(e_{j_b}))$; for increasing multi-indices $I = (i_1 < \cdots < i_k)$ and $J = (j_1 < \cdots < j_k)$ this is δ^I_J (that is, 1 if $I = J$ and 0 otherwise). Hence pairing α against $(e_{j_1}, \dots, e_{j_k})$ for increasing J returns the coefficient of e^J, so the e^I are linearly independent. An alternating tensor is determined by its values on increasing tuples (it is multilinear and alternating, proposition 0.5.6), so the stated expansion holds and the e^I span. There are $\binom{n}{k}$ increasing multi-indices, and none when $k > n$.

Example 0.5.10: The Exterior Powers of $\mathbb{R}^3$
For $V = \mathbb{R}^3$ with dual basis (e^1, e^2, e^3): $\Lambda^1(V) = V^*$ has dimension 3; $\Lambda^2(V)$ has basis $e^1 \wedge e^2$, $e^1 \wedge e^3$, $e^2 \wedge e^3$ (dimension $\binom{3}{2} = 3$, matching the cross product); and $\Lambda^3(V)$ has the single basis element $e^1 \wedge e^2 \wedge e^3$, the determinant, with $\Lambda^k(V) = 0$ for $k \ge 4$.

Definition 0.5.11: Exterior Algebra

The *exterior algebra* of V is the graded algebra

$$\Lambda^\bullet(V) = \bigoplus_{k=0}^{n} \Lambda^k(V), \qquad \Lambda^0(V) = \mathbb{R}, \quad \Lambda^1(V) = V^*,$$

with multiplication the wedge product; it is associative and graded anti-commutative (proposition 0.5.8), with total dimension $\sum_k \binom{n}{k} = 2^n$.

0.5.5 Covariant and Contravariant tensors on Vector Spaces

The last thing we showed is that we can take the tensor product of V_i's or V_i^*'s. These behave differently, and so we will give these a name:

Definition 0.5.12: Covariant And Contravariant Tensor

Let V be a finite dimensional vector space and $k \in \mathbb{N}_{>0}$. Then a *covariant* k-tensor on V is an element of

$$V^* \otimes \cdots \otimes V^* = \mathcal{T}^k(V^*)$$

and a *contravariant* k-tensor on V is an element of

$$V \otimes \cdots \otimes V = \mathcal{T}^k(V)$$

Example 0.5.13: Covariant And Contravariant Tensors

1. Any linear functional $\omega : V \to \mathbb{R}$ is linear, and so a *covariant 1-tensor*. We also call such a function a covector, an element of $T^1(V^*)$.
2. The dot product, or more generally the inner product, is a *covariant 2-tensor*. Covariant 2-tensors are also called *bilinear forms*.
3. The determinant is a covariant n-tensor.
4. I will see if I can find a good way of adding the contravariant tensors here TBD

it is sometimes useful to mix these two:

Definition 0.5.14: Mixed Tensors

Let V be a finite dimensional vector space, and $k, l \in \mathbb{N}_{>0}$. Then ω is a *mixed tensor* of type (k, l) if it is in:

$$T^{(k,l)}(V) := \underbrace{V \otimes \cdots \otimes V}_{k \text{ times}} \otimes \underbrace{V^* \otimes \cdots \otimes V^*}_{l \text{ times}}$$

Part I

Smooth Manifolds

On a high level, a manifold is a Hausdorff topological space which is nice enough to do some useful analysis (is σ-compact) that, crucially, *locally* resembles Euclidean space. At the beginning of this book, manifolds were said to be spaces X that are copies of a base space B glued together in appropriate ways. Though this is the case, we shall be taking a more axiomatic approach to manifolds since there are multiple ways of "concretely interpreting" a manifold. This is similar to how in group theory, groups are first defined using axioms, and only some concrete "representations" of groups are introduced (ex. Caley's theorem shows that all groups embedd into $\mathrm{Sym}(G)$, Frucht's theorem which states that every finite group is the group of symmetries of a finite undirected graph, and the universal property of free groups which states that every group is the quotient of a free group).

Manifolds will be the focus of this part of the book, however to contextualize them, a quick general overview of thinking of a space as being locally another space (i.e. locally like the "base space") paints a better pictures of where manifold theory fits into mathematics. In mathematics, there are many spaces which will be defined by the "type" of locality (how subsets of a the space is isomorphic to a subset of a base space), and the "regularity" of the locality. The type of locality will depend on the choice of *base space* (ex. $\mathbb{R}^n$, $L^2(\mathbb{R})$, zero sets of multi-variate polynomials); for example a circle will be seen to be locally $\mathbb{R}$ since for any $x \in S^1$, there is a neighborhood of x, $U \subseteq S^1$, that maps isomorphically to $V \subseteq \mathbb{R}$. In the case of manifolds, they will always be locally $\mathbb{R}^n$ for some $n \in \mathbb{N}$, that is if M is a manifold, then for each $x \in M$, there exists an open neighborhood $U \subseteq M$ of x and an isomorphism $f : U \to V \subseteq \mathbb{R}^n$. For the readers awareness, if we choose a different base space, then we get different mathematical objects, for example in algebraic geometry, spaces that are locally the zero set of polynomials, are called [affine] variety. Later in this part of the book, we shall also see a slight generlization of manifolds called *orbifolds* which (for reasons that shall be clear later) shall have as base space a finite group quotient of Euclidean space.

There are two other types of space we shall study in this book that is very close to being a manifold, but will be distinct since it will have a different base space associated to it. The other two spaces we will see have the word manfold inside their name, but should not be confused with manifolds since they shall have different base spaces; these are *manifold with borders* and *manifold with corners*[2], which have respective local spaces:

$$\mathbb{H}^n = \{x \in \mathbb{R}^n \mid x_1 \geq 0\} \qquad \text{and} \qquad \mathbb{R}^n_{\geq 0} = \{x \in \mathbb{R}^n \mid x_i \geq 0,\ 1 \leq i \leq n\}$$

The type of isomorphism from U to a part of the base space will define the type of regularity; for example homeomorphic, C^k, smooth, analytic, etc. The choice of the type of map will define the type of manifold we choose, in particular for the four type of maps just mentioned we would get:

1. a Topological Manifold
2. a Differentiable Manifolds (C^1 manifold)
3. a Smooth Manifold (C^∞-manifolds)
4. an Analytic Manifold (C^ω-manifolds)

Topological manifolds can be seen as the "basic" manifold from which we may obtain other manifolds by adding other strucutres. In this book, differentiable manifolds shall be the main focus, (which is why the title of the book specifies that we are studying *differential* geometry) which shall be

[2]Though as we shall see, all manifolds are (isomorphic to) manifolds with boundary/corners, but the converse is not the case

topological manifolds with a differentiable sructure that shall be added. Given a differential structure on a manifold, we shall be able to define the the *derivative* of a map between two differentiable manifolds $f : M \to N$. Recall that a derivative is a map that is locally linear. The derivative of a map between smooth manifolds shall be a linear map at each point that will "vary smoothly"; this is a technical endeavour that shall introduce the notion of a tangent bundle to be defined and will be the focus of chapter 2.

After having defined differential manifolds and their associated tangent bundle, we shall look at the behavior of maps between differential manifolds. The key property of maps between differentaible manifolds is that they "locally" will have the property of the manifold. This will differ from the property functions usually preserve in algebra (the image $f(G)$ is another algebraic obect) or topology (we can study the local properties of codomain N via the local propeties of the domain M). The advantage of this appraoch will be evident in chapter 3 where a lot of information about a map $f : M \to N$ will be deduced by the local information of f,and lead to the generalization of theroems like the implicit and inverse function theorem, and allow for the fruitful study of the intersection of manifolds.

We shall conclude this part of the book with a chapter on vector-fields on Manifolds. Vector-fields are very useful to Physicists for studying many natural phenomena; to a mathematician, a vector-field on a manifold is a generalizaiton of the notion of solving a system of ODE's on a manifold. We shalls see when a solution to a system of ODE's exist, and give show how being able to define certina vector-fields on a manifold gives property of a manifold;for example, we shall be able to detect when a 2-dimensional manifold has a "hole"; or if a sphere has even or odd dimension. This exploration will naturally lead to part III

1

Manifolds

In this chapter, we define topological, differentiable, and smooth manifolds. We give many examples of each type of manifold, and give some ways of constructing and comparing these manifolds.

After having defined a nice range of manifolds, we will define the maps between them. These maps will have many nice properties we expect from morphisms in categories, however they will in other ways be lacking in important ways due to their "local" nature. The introduction of maps will allow us to ask questions like what are the subobjects of our category, the product objects, and the quotient objects. All of these will be explored in this chapter.

We will end the chapter by introducing partitions of unity for manifolds. Partitions of unity will be invaluable for finding the existence of many extensions of functions and to develop the theory of integration later on.

1.1 Topological Manifolds

A manifold is, before anything else, a space that looks locally like $\mathbb{R}^n$. Making "looks like" precise is the work of a single definition, after which two further global conditions rule out the pathologies a purely local require cannot see.

Definition 1.1.1: Locally Euclidean Space

A topological space M is *locally Euclidean of dimension* n if every point $x \in M$ has an open neighbourhood $U \subseteq M$ homeomorphic to an open subset of $\mathbb{R}^n$.

The homeomorphisms witnessing this local structure are the coordinate systems of the theory.

Definition 1.1.2: (Topological) Chart

A *chart* on a topological space M is a homeomorphism $\varphi\colon U \to \varphi(U)$ from an open set $U \subseteq M$, the *chart domain*, onto an open subset $\varphi(U) \subseteq \mathbb{R}^n$. The chart is *centred at* $p \in U$ when $\varphi(p) = 0$.

Definition 1.1.3: Coordinate Functions

Let $\varphi\colon U \to \mathbb{R}^n$ be a chart and $\pi_i\colon \mathbb{R}^n \to \mathbb{R}$ the projection $\pi_i(a^1, \ldots, a^n) = a^i$. The i-th *coordinate function* of the chart is $x^i = \pi_i \circ \varphi$, so that $\varphi = (x^1, \ldots, x^n)$.

Definition 1.1.4: (Topological) Atlas

An *atlas* on a locally Euclidean space M is a collection of charts whose domains cover M.

That single local require is by itself far too weak: it constrains nothing global, and spaces satisfying it can fail to separate their points or can be too large to be described by countably many pieces. Two global hypotheses repair this, and they are exactly the ones that place every manifold inside the class governed by the point-set criterion of *Topology* [7, Second-Countable Locally Compact Hausdorff Spaces are Paracompact]: a second countable, locally compact Hausdorff space is paracompact.

Each of the two hypotheses is justified below by a space that satisfies the other two conditions and fails it, so that neither can be dropped without changing the subject.

Definition 1.1.5: Topological Manifold

Let $n \geq 0$ be an integer. A topological space M is a *topological n-manifold* if

1. M is *locally Euclidean of dimension n*: every point of M has an open neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$;
2. M is Hausdorff ([7, Hausdorff Space]);
3. M is second countable ([7, Second-Countability]).

A homeomorphism $\varphi\colon U \to \varphi(U)$ from an open set $U \subseteq M$ onto an open subset $\varphi(U) \subseteq \mathbb{R}^n$ is a *chart* on M, and U is its *chart domain*. A family of charts whose domains cover M is an *atlas*. When $M \neq \emptyset$, the integer n is the *dimension* of M; the empty space satisfies (1), (2) and (3) vacuously for every n and is assigned no dimension.

Condition (1) has an equivalent form. Requiring a neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$ is the same as requiring one homeomorphic to $\mathbb{R}^n$ itself. Indeed, if $\varphi\colon U \to \varphi(U)$ is a chart around p, then $\varphi(U)$ contains an open ball $B(\varphi(p), \rho)$, and the map

$$B(0,1) \to \mathbb{R}^n, \qquad x \mapsto \frac{x}{1 - \|x\|}$$

is a homeomorphism with inverse $y \mapsto y/(1 + \|y\|)$: both formulas are continuous where written, and substituting one into the other gives

$$\frac{x/(1-\|x\|)}{1+\|x\|/(1-\|x\|)} = \frac{x/(1-\|x\|)}{1/(1-\|x\|)} = x$$

while the same manipulation with $1+\|y\|$ in place of $1-\|x\|$ returns y, so the two maps are mutually inverse and each is continuous. Composing with the translation and dilation carrying $B(0,1)$ onto $B(\varphi(p),\rho)$ produces a chart domain around p homeomorphic to all of $\mathbb{R}^n$. For nonempty M the dimension n is not an arbitrary label attached to M but is determined by the space.[1]

The sphere is the standard object satisfying all three conditions, and its charts can be written down. Let $S^n = \{x \in \mathbb{R}^{n+1} \big| \|x\| = 1\}$ carry the subspace topology from $\mathbb{R}^{n+1}$ ([7, Subspace Topology]), and write $N = (0,\dots,0,1)$, so that N and $-N$ are the two poles. Stereographic projection from N is

$$\sigma_+ : S^n \setminus \{N\} \to \mathbb{R}^n, \qquad \sigma_+(x_1,\dots,x_{n+1}) = \frac{1}{1-x_{n+1}}(x_1,\dots,x_n)$$

which is continuous because $x_{n+1} < 1$ on its domain, so the denominator never vanishes. Its inverse is

$$\sigma_+^{-1}(u) = \frac{1}{\|u\|^2+1}\big(2u,\ \|u\|^2-1\big)$$

and three short computations confirm this. First the image lies on the sphere, since the sum of the squares of its coordinates is

$$\frac{4\|u\|^2 + (\|u\|^2-1)^2}{(\|u\|^2+1)^2} = \frac{(\|u\|^2+1)^2}{(\|u\|^2+1)^2} = 1$$

and its last coordinate $(\|u\|^2-1)/(\|u\|^2+1)$ is strictly smaller than 1, so the image avoids N. Second, writing $x = \sigma_+^{-1}(u)$ gives $1 - x_{n+1} = 2/(\|u\|^2+1)$, so $\sigma_+(x) = \frac{\|u\|^2+1}{2} \cdot \frac{2u}{\|u\|^2+1} = u$. Third, the composite in the other order is also the identity, which is what makes σ_+ injective. For $x \in S^n \setminus \{N\}$ put $x' = (x_1,\dots,x_n)$ and $u = \sigma_+(x) = x'/(1-x_{n+1})$. Then $\|x'\|^2 = 1 - x_{n+1}^2$, so

$$\|u\|^2 = \frac{1-x_{n+1}^2}{(1-x_{n+1})^2} = \frac{1+x_{n+1}}{1-x_{n+1}}$$

whence $\|u\|^2+1 = 2/(1-x_{n+1})$ and $\|u\|^2-1 = 2x_{n+1}/(1-x_{n+1})$. Therefore $2u/(\|u\|^2+1) = (1-x_{n+1})u = x'$ and $(\|u\|^2-1)/(\|u\|^2+1) = x_{n+1}$, that is, $\sigma_+^{-1}(\sigma_+(x)) = x$. The two maps are mutually inverse and both are continuous, so σ_+ is a chart with domain $S^n \setminus \{N\}$, which is open in S^n. Projection from the opposite pole, $\sigma_-(x) = (1+x_{n+1})^{-1}(x_1,\dots,x_n)$, is a chart on $S^n \setminus \{-N\}$ by the same computation with the sign of the last coordinate reversed, and the two domains cover S^n. The remaining two conditions are inherited: a subspace of a Hausdorff space is Hausdorff, since the separating open sets may be intersected with the subspace, and if $\mathcal{B}$ is a countable basis for a space Z then $\{B \cap A \big| B \in \mathcal{B}\}$ is a countable basis for a subspace $A \subseteq Z$ directly from the definition of the subspace topology. So S^n is a topological n-manifold, with an atlas of two charts.

Dropping the Hausdorff condition loses more than it might appear to, and the cheapest space that shows this is obtained by gluing two lines together along everything except their origins.

[1] What determines n is Brouwer's invariance of domain, in the form: a nonempty open subset of $\mathbb{R}^n$ is homeomorphic to an open subset of $\mathbb{R}^m$ only when $n = m$. Charts land in open subsets, so the weaker statement that $\mathbb{R}^n$ and $\mathbb{R}^m$ are homeomorphic only when $n = m$, which is a corollary of it, does not by itself settle the matter. The proof belongs to algebraic topology, where the theorem is read off from the homology of spheres, and is not available at this point. The case $n = 1$ against $n \geq 2$ is elementary, since deleting a point disconnects an open interval but leaves an open ball in $\mathbb{R}^n$ connected for $n \geq 2$ ([7, Connected]), and connectedness is preserved by homeomorphism.

Example 1.1.6: The Line with Two Origins

Let $Y = \mathbb{R} \sqcup \mathbb{R}$, realized as $\mathbb{R} \times \{0,1\}$ with $\{0,1\}$ discrete, and let $\sim$ be the equivalence relation generated by $(x,0) \sim (x,1)$ for every $x \neq 0$. Let $X = Y/\sim$ carry the quotient topology ([7, Quotient Topology]) and let $q\colon Y \to X$ be the quotient map. Write $0_0 = q(0,0)$ and $0_1 = q(0,1)$ for the two origins, and write x for the common image $q(x,0) = q(x,1)$ when $x \neq 0$.

1. *The map q is open.* Let $s\colon Y \to Y$ be the homeomorphism $s(x,i) = (x, 1-i)$ exchanging the two copies. For $U \subseteq Y$ open, a point (x,i) lies in $q^{-1}(q(U))$ exactly when $(x,i) \in U$ or both $x \neq 0$ and $s(x,i) \in U$, so
$$q^{-1}(q(U)) = U \cup s\big(U \setminus (\{0\} \times \{0,1\})\big)$$
Both sets on the right are open, hence so is $q^{-1}(q(U))$, and therefore $q(U)$ is open in X.

2. *X is locally Euclidean of dimension* 1. Put $A_i = q(\mathbb{R} \times \{i\})$ for $i \in \{0,1\}$. Each A_i is open, since q is open and $\mathbb{R} \times \{i\}$ is open in Y, and $A_0 \cup A_1 = X$ because q is surjective. The restriction $q|_{\mathbb{R}\times\{i\}}\colon \mathbb{R} \times \{i\} \to A_i$ is injective (the relation $\sim$ never identifies two points of the same copy), it is surjective onto A_i by definition, it is continuous, and it is open because an open subset of $\mathbb{R} \times \{i\}$ is open in Y and q is open. A continuous open bijection is a homeomorphism, so $A_i \cong \mathbb{R}$. Every point of X therefore lies in an open set homeomorphic to $\mathbb{R}$.

3. *X is second countable.* The family $\mathcal{B} = \{(a,b) \times \{i\} \big| a < b \text{ rational},\ i \in \{0,1\}\}$ is a countable basis for Y. Each $q(B)$ with $B \in \mathcal{B}$ is open. Given $W \subseteq X$ open and $p \in W$, choose $y \in q^{-1}(p)$; then $q^{-1}(W)$ is an open set containing y, so there is $B \in \mathcal{B}$ with $y \in B \subseteq q^{-1}(W)$, and applying q gives $p \in q(B) \subseteq W$. So $\{q(B) \big| B \in \mathcal{B}\}$ is a countable basis for X.

4. *X is not Hausdorff.* Let U and V be open sets with $0_0 \in U$ and $0_1 \in V$. Then $q^{-1}(U)$ is open in Y and contains $(0,0)$, so it contains $(-\epsilon, \epsilon) \times \{0\}$ for some $\epsilon > 0$; likewise $q^{-1}(V)$ contains $(-\delta, \delta) \times \{1\}$ for some $\delta > 0$. Choosing t with $0 < t < \min(\epsilon, \delta)$ gives $q(t,0) = q(t,1) \in U \cap V$, so the two origins admit no disjoint neighbourhoods.

The failure is confined to a single pair of points, and it is not a failure of the weaker separation axioms: points of X are closed, since $q^{-1}(X \setminus \{0_0\})$ is $Y \setminus \{(0,0)\}$ and $q^{-1}(X \setminus \{x\})$ is $Y \setminus \{(x,0),(x,1)\}$ for $x \neq 0$, both open. What the failure destroys is the interaction between compactness and closedness. The set $K = q([-1,1] \times \{0\})$ is compact, being the image of a compact interval under a continuous map, but it is not closed: every open neighbourhood of 0_1 contains points $q(t,1) = q(t,0)$ with $0 < t < 1$, so 0_1 lies in $\overline{K} \setminus K$. In a Hausdorff space this cannot happen ([7, compact subspaces of Hausdorff Spaces are closed]), and a space in which compact sets need not be closed is one where local compactness carries no useful global consequence. That is what the Hausdorff clause in definition 1.1.5 excludes.

Second countability is the second global require, and it too is independent of the others. The witness is the long line built, in the discussion following [7, The Minimal Uncountable Well-Ordered Set], from the minimal uncountable well-ordered set Ω: a linearly ordered space that is locally indistinguishable from $\mathbb{R}$ and yet is uncountably long, and therefore admits no countable basis.

Example 1.1.7: The Long Line Is Not a Manifold

Let Ω be the minimal uncountable well-ordered set of [7, The Minimal Uncountable Well-Ordered Set], with least element 0, so that every proper initial segment $S_\alpha = \{x \in \Omega \big| x < \alpha\}$ is countable, and let $\Omega \times [0,1)$ carry the lexicographic order, in which $(\alpha, s) < (\beta, t)$ when $\alpha < \beta$, and also when $\alpha = \beta$ and $s < t$. The long line is the ordered set

$$L = \big(\Omega \times [0,1)\big) \setminus \{(0,0)\}$$

introduced after [7, The Minimal Uncountable Well-Ordered Set], with the order topology. Deleting the least element $(0,0)$ leaves a linearly ordered set with neither a least nor a greatest element, and for $\alpha \in \Omega$ with $\alpha > 0$ the set

$$L_\alpha = \{x \in L \big| x < (\alpha, 0)\}$$

is the open initial ray cut off by the α-th block.

1. *L is Hausdorff.* In any linearly ordered set with its order topology, two points $x < y$ are separated as follows: if some z satisfies $x < z < y$, the open rays $\{t \big| t < z\}$ and $\{t \big| t > z\}$ are disjoint and contain x and y respectively; if no such z exists, the rays $\{t \big| t < y\}$ and $\{t \big| t > x\}$ are disjoint, since a point in both would lie strictly between x and y, and they contain x and y. One of the two cases applies to every pair, so L is Hausdorff.

2. *L is locally Euclidean of dimension* 1. For $\beta < \alpha$ in Ω write

   $$J(\beta, \alpha) = \{x \in \Omega \times [0,1) \big| (\beta, 0) \leq x < (\alpha, 0)\} = \{\gamma \in \Omega \big| \beta \leq \gamma < \alpha\} \times [0,1)$$

   The claim is that $J(\beta, \alpha)$ is order-isomorphic to $[0,1)$. Two observations drive the proof. First, a single block $\{\gamma\} \times [0,1)$ is order-isomorphic to $[0,1)$ by $(\gamma, t) \mapsto t$. Second, if a linearly ordered set J is the concatenation of blocks $B_0 < B_1 < B_2 < \cdots$ indexed by $\mathbb{N}$ (each element of J lies in exactly one B_n, and every element of B_n is below every element of B_{n+1}) and each B_n is order-isomorphic to $[0,1)$, then J is order-isomorphic to $[0,1)$: compose the isomorphism of B_n with the increasing affine bijection of $[0,1)$ onto $[1-2^{-n}, 1-2^{-n-1})$ and take the union of the resulting maps. The union is a bijection onto $\bigcup_{n \in \mathbb{N}} [1-2^{-n}, 1-2^{-n-1}) = [0,1)$, and it is increasing because the blocks and their target intervals occur in the same order.

   Now argue by transfinite induction on $\alpha \in \Omega$, the inductive statement being that $J(\beta, \alpha) \cong [0,1)$ for every $\beta < \alpha$. For $\alpha = 0$ there is no such β and nothing to prove. Let $\alpha > 0$, assume the statement for every $\alpha' < \alpha$, and fix $\beta < \alpha$.

   Suppose first that α has an immediate predecessor α^-. If $\beta = \alpha^-$ then $J(\beta, \alpha) = \{\beta\} \times [0,1)$, which is order-isomorphic to $[0,1)$. If $\beta < \alpha^-$ then $J(\beta, \alpha)$ is the concatenation of $J(\beta, \alpha^-)$ and $\{\alpha^-\} \times [0,1)$; the first is order-isomorphic to $[0,1)$ by the inductive statement at $\alpha^- < \alpha$ and the second is by the first observation, so sending the first onto $[0, 1/2)$ and the second onto $[1/2, 1)$ by increasing affine maps gives $J(\beta, \alpha) \cong [0,1)$.

   Suppose instead that α has no immediate predecessor. The set $\{\gamma \in \Omega \big| \beta \leq \gamma < \alpha\}$ lies inside S_α and is therefore countable; enumerate it as $\delta_0, \delta_1, \ldots$, repeating terms if it is finite. Put $\gamma_0 = \beta$ and, given $\gamma_n < \alpha$, set $\mu = \max(\gamma_n, \delta_n) < \alpha$; the set $\{x \in \Omega \big| \mu < x < \alpha\}$ is nonempty, since otherwise α would be the immediate successor of μ, so let γ_{n+1} be its least element. Then $\gamma_n < \gamma_{n+1} < \alpha$ and $\delta_n < \gamma_{n+1}$ for every n. Consequently each γ with $\beta \leq \gamma < \alpha$

satisfies $\gamma_n \leq \gamma < \gamma_{n+1}$ for exactly one n, namely the least n with $\gamma < \gamma_{n+1}$, which exists because $\gamma = \delta_m$ for some m and hence $\gamma < \gamma_{m+1}$. So $J(\beta, \alpha)$ is the concatenation of the blocks $J(\gamma_n, \gamma_{n+1})$, each order-isomorphic to $[0, 1)$ by the inductive statement at $\gamma_{n+1} < \alpha$, and the second observation applies. The induction is complete.

Taking $\beta = 0$ gives, for each $\alpha > 0$, an order isomorphism

$$h\colon J(0, \alpha) = \{x \in \Omega \times [0, 1) \big| x < (\alpha, 0)\} \to [0, 1)$$

An order isomorphism carries a least element to a least element, so h sends $(0, 0)$ to 0. Since L_α is exactly $J(0, \alpha) \setminus \{(0, 0)\}$, restricting h gives an order isomorphism of L_α onto $[0, 1) \setminus \{0\} = (0, 1)$.

An order isomorphism between linearly ordered sets carries open rays onto open rays and is therefore a homeomorphism for the order topologies. What remains is that the order topology of L_α is its subspace topology from L, and this holds because L_α is an interval: the trace on L_α of a ray of L is empty, or all of L_α, or a ray of L_α, and conversely every ray of L_α is such a trace. Hence L_α is homeomorphic to $(0, 1)$, an open subset of $\mathbb{R}$. Each L_α is open in L, being an open ray, and every point of L lies in one of them: a point (β, t) lies in L_α for any $\alpha > \beta$, and such an α exists because Ω has no greatest element, a greatest element g making $\Omega = S_g \cup \{g\}$ countable.

3. *L is not second countable.* The sets $\{L_\alpha\}$, for $\alpha \in \Omega$ with $\alpha > 0$, form an open cover of L by the previous item. Let $\{L_{\alpha_n}\}_{n\in\mathbb{N}}$ be a countable subfamily. The set $\{\alpha_n \big| n \in \mathbb{N}\}$ has a strict upper bound γ in Ω: otherwise every element of Ω would satisfy $x \leq \alpha_n$ for some n, giving $\Omega = \bigcup_n (S_{\alpha_n} \cup \{\alpha_n\})$, a countable union of countable sets and hence countable. Then $\gamma > 0$, so $(\gamma, 0)$ is a point of L, and $(\gamma, 0) > (\alpha_n, 0)$ for every n, so $(\gamma, 0)$ lies in no L_{α_n}. The subfamily does not cover L, and the cover has no countable subcover. A second countable space admits no such cover: if $\mathcal{B}$ is a countable basis and $\mathcal{A}$ an open cover, let $\mathcal{B}'$ consist of those $B \in \mathcal{B}$ contained in some member of $\mathcal{A}$ and choose one such $U_B \in \mathcal{A}$ for each; every x lies in some $U \in \mathcal{A}$ and hence in some $B \in \mathcal{B}$ with $x \in B \subseteq U$, so $B \in \mathcal{B}'$ and $x \in U_B$, and $\{U_B \big| B \in \mathcal{B}'\}$ is a countable subcover. Therefore L is not second countable.

The long line satisfies (1) and (2) of definition 1.1.5 and fails (3), so it is not a manifold. The failure is not a technicality about bases. The cover $\{L_\alpha\}$ exhibited above has no countable subcover, and L is not paracompact either: the transfer of the argument of [7, The Minimal Uncountable Well-Ordered Set] from Ω to L is carried out where L is constructed, immediately after that example. So L is a space that looks locally like $\mathbb{R}$ and yet lies outside the paracompact class, which is what the third condition is there to exclude. The exclusion is not the only possible one: some treatments replace second countability by paracompactness in definition 1.1.5, which admits, for instance, the disjoint union of uncountably many copies of $\mathbb{R}$, a space that is locally Euclidean, Hausdorff and paracompact but has no countable basis. The two conventions differ only in how many components are allowed.

With the three conditions in place, the promised inclusion is a matter of producing the three hypotheses that [7, Second-Countable Locally Compact Hausdorff Spaces are Paracompact] asks for. Second countability and the Hausdorff property are two of them outright; the third, local compactness, is not assumed but follows, because a chart transports the compact closed balls of $\mathbb{R}^n$ into M.

Theorem 1.1.8: Manifolds Are Paracompact

Every topological manifold is paracompact.

***Proof*:**
Let M be a topological n-manifold.

Step 1: M is locally compact. Fix $p \in M$. By definition 1.1.5 there is a chart $\varphi\colon U \to \varphi(U)$ with $p \in U$ and $\varphi(U) \subseteq \mathbb{R}^n$ open. Since $\varphi(U)$ is open and contains $\varphi(p)$, there is $\rho > 0$ with $B(\varphi(p), \rho) \subseteq \varphi(U)$; put $r = \rho/2$, so that the closed ball $\overline{B}(\varphi(p), r)$ is contained in $B(\varphi(p), \rho)$ and hence in $\varphi(U)$. That closed ball is closed and bounded in $\mathbb{R}^n$, hence compact by the Heine-Borel theorem. Set

$$C = \varphi^{-1}\big(\overline{B}(\varphi(p), r)\big)$$

Then C is compact, being the image of a compact space under the continuous map φ^{-1}, and compactness of C does not depend on whether it is viewed inside U or inside M. Moreover C contains $\varphi^{-1}(B(\varphi(p), r))$, which is open in U and therefore open in M because U is open in M, and which contains p. So C is a compact subspace of M containing a neighbourhood of p, which is what local compactness ([7, Local Compactness]) requires. As p was arbitrary, M is locally compact.

Step 2: the remaining hypotheses. Clauses (2) and (3) of definition 1.1.5 say that M is Hausdorff and second countable. Combined with Step 1, M is a second countable, locally compact Hausdorff space.

Step 3: conclusion. Every second countable locally compact Hausdorff space is paracompact ([7, Second-Countable Locally Compact Hausdorff Spaces are Paracompact]). Applying this to M gives the claim, completing the proof.

Each of the three clauses does identifiable work in the argument. The locally Euclidean clause gives compactness locally; the Hausdorff clause is what makes that local compactness usable, since without it compact sets need not be closed, as the line with two origins shows; and second countability gives the countable exhaustion, without which the conclusion fails outright, as the long line shows. Two consequences follow at once. First, since M is locally compact Hausdorff, the local-compactness criterion in Hausdorff spaces ([7, Locally Compact in Hausdorff Space]) upgrades Step 1: every point of M has, inside any prescribed neighbourhood, a neighbourhood whose closure is compact, so manifolds admit neighbourhood bases of relatively compact open sets. Second, a paracompact Hausdorff space is normal ([7, Paracompact Hausdorff Spaces Are Normal]), so every manifold is normal and Urysohn's lemma ([7, Urysohn]) is available on it: disjoint closed subsets of a manifold are separated by continuous real-valued functions.

Normality separates two closed sets at a time, which is the input to the construction of a single function. The cover-indexed family of functions is what global constructions require, and theorem 1.1.8 delivers exactly the hypothesis under which such a family exists.

Corollary 1.1.9: Partitions of Unity on a Manifold

Let M be a topological n-manifold and let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of M. Then M admits a partition of unity subordinate to $\{U_\alpha\}_{\alpha \in A}$.

***Proof*:**
By definition 1.1.5 the space M is Hausdorff, and by theorem 1.1.8 it is paracompact. So M satisfies the hypotheses of the partition-of-unity existence theorem ([7, Existence of Subordinate Partitions of Unity]), which produces for any open cover of a paracompact Hausdorff space a partition of unity subordinate to it in the sense of the definition of a partition of unity ([7, Partition of Unity]) and subordination to a cover ([7, Subordinate to a Cover]). Applying that theorem to the cover $\{U_\alpha\}_{\alpha \in A}$ gives the assertion, completing the proof.

The corollary is the form in which the chapter's machinery is used elsewhere. Suppose $\{U_\alpha\}$ is an open cover of M and a continuous function $f_\alpha \colon U_\alpha \to \mathbb{R}$ is given on each piece, with no compatibility assumed on the overlaps. Let $\{\psi_\alpha\}$ be a partition of unity subordinate to the cover and define $g_\alpha \colon M \to \mathbb{R}$ by $g_\alpha = \psi_\alpha f_\alpha$ on U_α and $g_\alpha = 0$ on $M \setminus \operatorname{supp}\psi_\alpha$. Those two sets are open and cover M, and the two formulas agree on the overlap, so each g_α is continuous on all of M. Subordination puts $\operatorname{supp}\psi_\alpha$ inside U_α, and local finiteness of the supports means every point has a neighbourhood on which all but finitely many g_α vanish, so the sum

$$f = \sum_{\alpha \in A} g_\alpha$$

is a finite sum near each point and hence a continuous function on M. Since $\sum_\alpha \psi_\alpha = 1$, the value $f(p)$ is a convex combination of the values $f_\alpha(p)$ over those α with $p \in U_\alpha$; in particular, if every f_α takes values in a fixed interval, so does f. Locally defined continuous data on a manifold can therefore be averaged into global continuous data, with no agreement on overlaps required. The partition of unity itself is written down explicitly for $M = \mathbb{R}$ in [7, A Partition of Unity on the Line].

What corollary 1.1.9 asserts is a C^0 statement, and it is a statement about the underlying space alone. A smooth structure on M is extra data, an atlas whose charts overlap by transition maps that are C^∞ with C^∞ inverses. It does not alter the topology, so the corollary applies verbatim to every smooth manifold and already yields continuous ψ_α there. What the smooth theory has to add is regularity of the functions themselves, namely that the ψ_α can be chosen C^∞, which requires replacing the Urysohn functions used here by the explicit bump functions on $\mathbb{R}^n$ constructed in [4], and then transporting those through charts. The point-set half of that upgrade is not redone in the differential topology companion, which takes it as given.

Note that it suffices to show that M can be covered by countably many charts to show that it is second-countable. Every manifold has an atlas, and most of the time, an atlas will only need finitely many charts. *Any atlas covering a manifold will henceforth assumed to be countable.* Notice that the M in example 1.1.6 is not a topological manifold. if we remove the hasudroff condition, the resulting class of sets is usually called *non-hausdorff manifolds*; these will not be studied in these notes. Note that a manifold, being second countable and regular (Hausdorff and locally Euclidean), is *metrizable* by the Urysohn metrization theorem. Putting a metric on M will be explored in the companion Riemannian Geometry volume [6, Riemannian Metric], where the Riemann metric is defined.

Since the three properties of a manifold are all topological properties the collection of topological manifolds along with continuous maps forms a category:

Definition 1.1.10: The Category of Topological Manifolds

The topological manifolds, with continuous maps between them, form a category **TMan**: composition and identities are those of the underlying topological spaces.

Using Atlases, we may check for continuity of maps locally:

Lemma 1.1.11: Continuity Is Local in Charts

Let M and N be topological manifolds and $f\colon M \to N$ a map. Then f is continuous if and only if for every $p \in M$ there are a chart (U, φ) of M with $p \in U$ and a chart (V, ψ) of N with $f(p) \in V$ such that $f(U) \subseteq V$ and $\psi \circ f \circ \varphi^{-1}\colon \varphi(U) \to \psi(V)$ is continuous.

***Proof*:**
Continuity is tested pointwise, and at each point on any neighbourhood, since charts are homeomorphisms onto open sets.

If f is continuous, fix $p \in M$ and any chart (V, ψ) of N with $f(p) \in V$. Then $f^{-1}(V)$ is open and contains p, so it contains the domain of a chart (U, φ) with $p \in U$ and $f(U) \subseteq V$, and $\psi \circ f \circ \varphi^{-1}$ is continuous as a composite of the continuous maps φ^{-1}, f, ψ.

Conversely, suppose the condition holds and fix $p \in M$ with charts (U, φ), (V, ψ) as stated. On U, $f = \psi^{-1} \circ (\psi \circ f \circ \varphi^{-1}) \circ \varphi$, a composite of continuous maps, so f is continuous at p. As p was arbitrary, f is continuous, as we sought to show.

Continuity of a map between manifolds is thus a chart-local question.

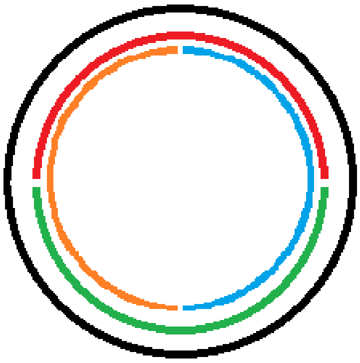
Example 1.1.12: Building Topological Manifolds
The following examples include the most common manifolds and the standard ways of building new manifolds from old ones.

1. The empty space $\emptyset$ is a topological manifold of every dimension, vacuously, and is the only manifold not assigned a single dimension. A one-point space is a 0-manifold.
2. $\mathbb{R}^n$ is the n-manifold underlying all the others: the identity map is a global chart, so a single chart already forms an atlas.
3. An open subset U of a topological n-manifold X is again a topological n-manifold (exercise 1.1.1 (2)). Many manifolds arise this way, as a complement $X \setminus C$ of a closed set. For instance $\mathrm{GL}_n(\mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\})$ is open in $\mathbb{R}^{n^2}$, since the determinant is continuous, and is therefore an n^2-manifold.
4. If M_1 and M_2 are topological manifolds of dimensions n_1 and n_2, then $M_1 \times M_2$ with the product topology is a topological manifold of dimension $n_1 + n_2$ (exercise 1.1.1 (1)): a product $\varphi_p \times \psi_q$ of charts is a chart, and the Hausdorff and second countable properties pass to products.
5. The disjoint union $X \sqcup Y$ of two n-manifolds is an n-manifold, its atlas the union of the two atlases; the dimensions must agree. A manifold therefore need not be connected.

6. The first genuinely nontrivial example is the circle

$$S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$

with the subspace topology. It is Hausdorff and second countable as a subspace of $\mathbb{R}^2$, and compact. Covering S^1 by four open arcs and projecting each onto a coordinate axis



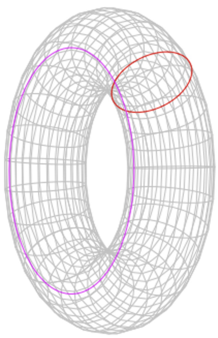
gives four charts, hence an atlas, so S^1 is a 1-manifold. It is not homeomorphic to $\mathbb{R}$: removing a point leaves S^1 connected but disconnects $\mathbb{R}$, and connectedness is a homeomorphism invariant.

7. Since a product of manifolds is a manifold, the n-fold product

$$\underbrace{S^1 \times S^1 \times \cdots \times S^1}_{n} = T^n$$

is a manifold, the *n-torus*; for $n = 2$ one says simply the torus. The 2-torus embeds in $\mathbb{R}^4$, and famously in $\mathbb{R}^3$ through the parameterization

$$x = (R + r\cos\psi)\cos\theta, \qquad y = (R + r\cos\psi)\sin\theta, \qquad z = r\sin\psi$$



Checking that this image, in the subspace topology, is homeomorphic to T^2 is a genuine computation, made routine once a smooth structure is available.

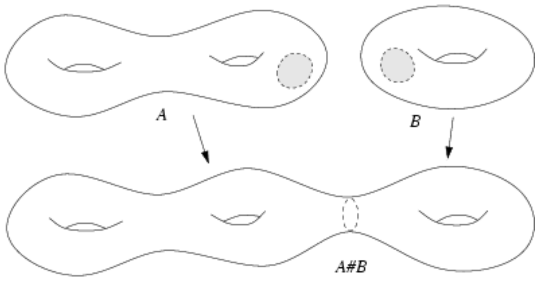
8. The triangle, being homeomorphic to the circle, is a 1-manifold in the topology induced by that homeomorphism.

9. Let $U \subseteq \mathbb{R}^n$ be open and $f\colon U \to \mathbb{R}^k$ continuous. The graph

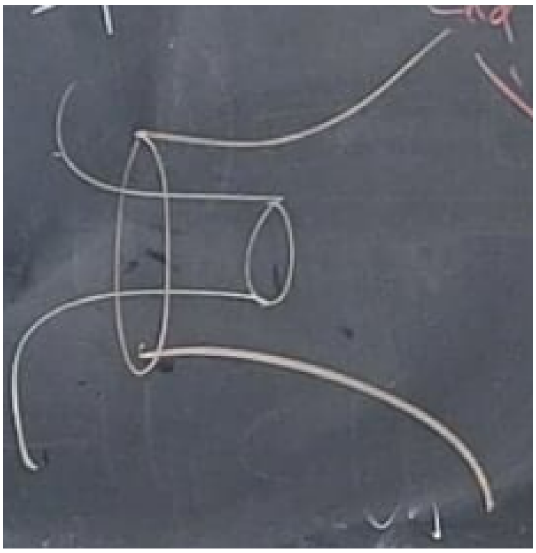
$$\Gamma(f) = \left\{(x,y) \in \mathbb{R}^n \times \mathbb{R}^k \;\middle|\; y = f(x)\right\}$$

   with the subspace topology is a topological n-manifold. The projection $\pi\colon \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ restricts to a continuous $\varphi\colon \Gamma(f) \to U$ with continuous inverse $x \mapsto (x, f(x))$, so φ is a homeomorphism exhibiting $\Gamma(f) \cong U$ as a single chart. A space that is locally a graph in this sense is thus a manifold; the converse, that every manifold is locally a graph, is settled once smooth structures and the implicit function theorem are available.

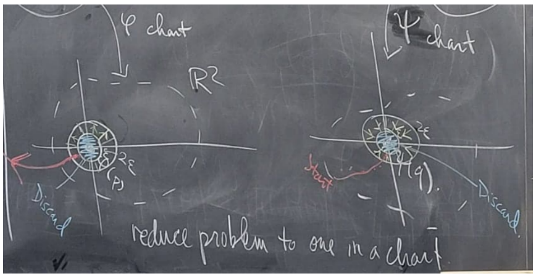
10. Two n-manifolds M_1 and M_2 can be joined at chosen points $p \in M_1$, $q \in M_2$ into their *connected sum* $M_1 \# M_2$.



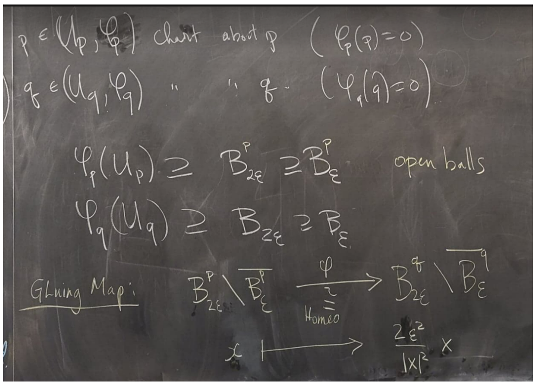
   A small chart ball is deleted around each point and the two resulting collars are glued together.



   The gluing is carried out through the charts,



by an explicit identification,



giving the definition

$$X\#Y = \Big(X \setminus \overline{\varphi_p^{-1}(B_\epsilon)}\Big) \coprod \Big(Y \setminus \overline{\varphi_q^{-1}(B_\epsilon)}\Big) \Big/ \big(\varphi_p^{-1}(B_{2\epsilon}) \ni x \sim \varphi_q^{-1}(\psi(\varphi_p(x))) \in \varphi_q^{-1}(B_{2\epsilon})\big)$$

The result is independent of the chosen points and charts, though it can depend on the orientation of the gluing; independence rests on the annulus theorem. Connected sum is the operation behind the classification of surfaces: every closed connected surface is the sphere S^2, a connected sum of tori T^2, or a connected sum of projective planes $\mathbb{R}\mathrm{P}^2$.

11. The figure below is not a topological manifold: the crossing point has no neighbourhood homeomorphic to any $\mathbb{R}^n$.

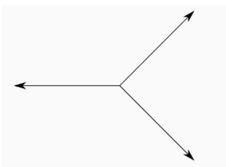


Figure 1.1: The crossing point is not locally Euclidean.

The dimension of a nonempty topological manifold is well-defined: an n-manifold cannot be homeomorphic to an m-manifold for $m \neq n$, which is why the definition fixes no particular Euclidean model, and on a connected manifold the dimension is constant. This is a theorem of Brouwer, proved through the homology of spheres in algebraic topology; it is recorded and cited here, not proved.

Theorem 1.1.13: Invariance of Domain

Let $U \subseteq \mathbb{R}^n$ be open and $f\colon U \to \mathbb{R}^n$ continuous and injective. Then $f(U)$ is open in $\mathbb{R}^n$ and $f\colon U \to f(U)$ is a homeomorphism. In particular, if a nonempty open subset of $\mathbb{R}^n$ is homeomorphic to an open subset of $\mathbb{R}^m$, then $n = m$; and a homeomorphism between nonempty manifolds $M^m \to N^n$ forces $m = n$ on each connected component.

***Proof*:**
Invariance of domain is a theorem of algebraic topology, read off from the homology of spheres; its proof is given in *Algebraic Topology* [11, Invariance Of Domain]. The consequences are formal. Suppose $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ are nonempty, open, and homeomorphic, with $n \leq m$. Let $h\colon U \to V$ be the homeomorphism. Composing its inverse with the inclusion $\mathbb{R}^n \hookrightarrow \mathbb{R}^m$ onto the first n coordinates gives an injective continuous map $V \to \mathbb{R}^m$ on the open set $V \subseteq \mathbb{R}^m$; by invariance of domain its image is open in $\mathbb{R}^m$. But that image is $U \times \{0\} \subseteq \mathbb{R}^n \times \{0\}$, which has empty interior in $\mathbb{R}^m$ unless $n = m$, so $n = m$. For manifolds, a homeomorphism carries a chart around p to a chart around $f(p)$, matching an open subset of $\mathbb{R}^m$ with one of $\mathbb{R}^n$; the local dimensions therefore agree, and on a connected manifold the dimension is constant, completing the proof.

If the manifold has a differentiable structure (as we'll define in section 1.2.1), it is much easier to show that M must a constant dimension n using this structure (as we'll soon show).

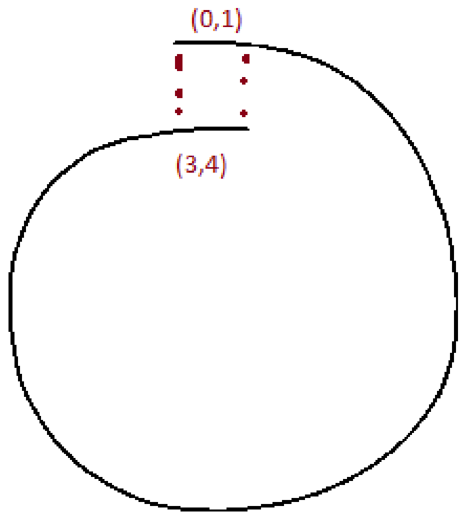
Exercise 1.1.1

1. Let M_1 and M_2 be topological manifolds of dimensions n_1 and n_2. Show that $M_1 \times M_2$ with the product topology is a topological manifold of dimension $n_1 + n_2$, verifying each clause of definition 1.1.5.
2. Show that an open subset U of a topological n-manifold is a topological n-manifold. Deduce that $\mathrm{GL}_n(\mathbb{R})$ is a topological manifold, and give its dimension.
3. Show that a topological manifold is locally path-connected, and conclude that its connected components are open and are themselves path-connected manifolds.
4. For the line with two origins (example 1.1.6), exhibit a compact subset that is not closed, and explain why this alone shows the space is not Hausdorff.

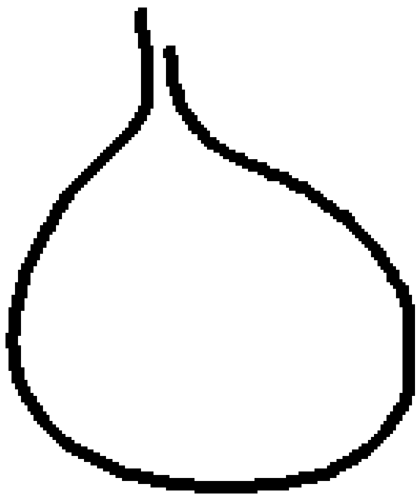
1.1.1 Constructing Manifolds Using Quotient

A manifold can be assembled from open pieces of $\mathbb{R}^n$ glued along open subsets, exactly as an atlas presents an existing one. The construction is worth isolating, since it produces manifolds that are not handed over as subspaces of a Euclidean space.

Consider first a single gluing. Let $B = (0, 4)$ and identify $(0, 1)$ with $(3, 4)$ through $\varphi(x) = x + 3$, forming $M = B/(x \sim \varphi(x))$ with the quotient topology ([7, Quotient Topology]).



The result is homeomorphic to S^1. The homeomorphism condition on φ is not by itself enough: gluing instead by $x \mapsto 4 - x$ brings the points 1 and 3 arbitrarily close without identifying them,



and the quotient fails to be Hausdorff, in the manner of the line with two origins (example 1.1.6). A gluing that remains a manifold must guarantee that leaving one chart's domain means entering another's; the general construction enforces this by a closedness condition on the gluing graphs, alongside the cocycle conditions that make the identifications consistent.

Proposition 1.1.14: Gluing Construction of a Manifold

Let $\{U_i\}_{i\in I}$ be a countable family of open subsets of $\mathbb{R}^n$; for each pair (i,j) let $U_{ij} \subseteq U_i$ be open and $\varphi_{ij}\colon U_{ij} \to U_{ji}$ a homeomorphism, satisfying

1. $U_{ii} = U_i$ and $\varphi_{ii} = \mathrm{id}$;
2. $\varphi_{ji} = \varphi_{ij}^{-1}$;
3. $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ on $U_{ij} \cap \varphi_{ij}^{-1}(U_{jk})$;

and suppose each graph $\{(x, \varphi_{ij}(x)) : x \in U_{ij}\}$ is closed in $U_i \times U_j$. Then

$$M = \left(\coprod_{i\in I} U_i\right) \Big/ (U_{ij} \ni x \sim \varphi_{ij}(x) \in U_{ji})$$

with the quotient topology is a topological n-manifold, and the maps $U_i \to M$ are charts forming an atlas.

Readers who know some algebraic geometry will recognize this as the gluing of a sheaf; the same data assembles schemes from affine pieces [1].

***Proof*:**
Write $q\colon \coprod_i U_i \to M$ for the quotient map and $q_i = q|_{U_i}$.

Step 1: each q_i is an open embedding. Conditions (1) and the cocycle condition (3) make the generated relation restrict to the identity on each U_i, so q_i is injective. It is open: for $V \subseteq U_i$ open, $q^{-1}(q(V)) = \bigsqcup_j \varphi_{ij}(V \cap U_{ij})$, a union of open sets by (2), so $q(V)$ is open. A continuous open injection is an embedding; the images $q_i(U_i)$ are open and cover M, so M is locally Euclidean of dimension n with the q_i as charts.

Step 2: second countable. The index set is countable and each U_i is second countable, so M is covered by countably many charts and is second countable.

Step 3: Hausdorff. Take distinct points of M. If they share a chart they separate as in $\mathbb{R}^n$. Otherwise represent them by $x \in U_i$, $y \in U_j$ with $q(x) \neq q(y)$, so (x, y) is not on the graph of φ_{ij}; as that graph is closed, some basic box $V \times W$ about (x, y) misses it, and then $q_i(V)$ and $q_j(W)$ are disjoint neighbourhoods of the two points. So M is Hausdorff, completing the proof.

Example 1.1.15: The Circle by Gluing Two Lines
The circle is built by gluing two copies of $\mathbb{R}$ along $\mathbb{R}^\times$. The stereographic charts of S^1 have transition map $x \mapsto x^{-1}$ on $\mathbb{R}^\times$, so

$$S^1 = (\mathbb{R} \sqcup \mathbb{R})/(\mathbb{R}^\times \ni x_1 \sim x_2^{-1} \in \mathbb{R}^\times)$$

realizes S^1 through the construction, with two charts and the single transition $x \mapsto x^{-1}$. The same pattern with $\mathbb{R}^n$ and inversion $u \mapsto u/\|u\|^2$ builds S^n from two charts.

Exercise 1.1.2

1. Following the circle example, present S^n as a gluing of two copies of $\mathbb{R}^n$ along $\mathbb{R}^n \setminus \{0\}$ by the inversion $u \mapsto u/\|u\|^2$, and verify the closed-graph condition of proposition 1.1.14.
2. For the gluing of $B = (0,4)$ by $\varphi(x) = 4 - x$ on $(0,1) \to (3,4)$, show that the graph of φ is not closed in $(0,4) \times (0,4)$, and identify the pair of points of the quotient with no disjoint neighbourhoods.
3. Show that conditions (1)-(3) of proposition 1.1.14 are exactly what make the relation generated by $x \sim \varphi_{ij}(x)$ an equivalence relation.

1.1.2 Topological Manifolds with Boundary

Many familiar shapes are locally Euclidean everywhere except along an edge, where a neighbourhood looks like a half-space rather than all of $\mathbb{R}^n$. The closed disk $D^2 = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ is the model case: its interior points have $\mathbb{R}^2$-neighbourhoods, but a point of the bounding circle does not. Admitting such an edge widens the theory to manifolds with boundary.

Definition 1.1.16: Topological Manifold with Boundary

Let $\mathbb{H}^n = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$ be the closed upper half-space, with $\partial\mathbb{H}^n = \{x \in \mathbb{H}^n \mid x_n = 0\} \cong \mathbb{R}^{n-1}$. A Hausdorff, second countable space M is a *topological n-manifold with boundary* if every point has an open neighbourhood homeomorphic to an open subset of $\mathbb{H}^n$. A point of M is a *boundary point* if some chart carries it into $\partial\mathbb{H}^n$, and an *interior point* otherwise; the boundary points form ∂M and the interior points $\operatorname{Int} M$.

Constraining any single coordinate on either side gives a homeomorphic model, so the choice $x_n \geq 0$ is only a convention. The classification of a point as interior or boundary is, crucially, independent of the chart used to test it.

Proposition 1.1.17: The Boundary Is a Manifold

Let M be a topological n-manifold with boundary. Then ∂M and $\operatorname{Int} M$ partition M; $\operatorname{Int} M$ is an n-manifold, and ∂M is an $(n-1)$-manifold without boundary.

***Proof*:**
An interior point of $\mathbb{H}^n$ (one with $x_n > 0$) has a neighbourhood inside the open set $\{x \mid x_n > 0\}$, hence a neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$; a point with $x_n = 0$ has none, since every neighbourhood in $\mathbb{H}^n$ meets $\partial\mathbb{H}^n$, which is not open in $\mathbb{R}^n$. Invariance of domain (theorem 1.1.13) makes this intrinsic: a homeomorphism between open subsets of $\mathbb{H}^n$ must send interior points to interior points, as an interior point's $\mathbb{R}^n$-neighbourhood has open image. So membership in ∂M or $\operatorname{Int} M$ does not depend on the chart, and the two sets partition M. Charts with $x_n > 0$ present $\operatorname{Int} M$ as an n-manifold; restricting boundary charts to $\partial\mathbb{H}^n \cong \mathbb{R}^{n-1}$ presents ∂M as an $(n-1)$-manifold, and it has empty boundary because $\partial\mathbb{H}^n$ does, completing the proof.

Example 1.1.18: Manifolds with Boundary

1. The closed disk $D^n = \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}$ is an n-manifold with boundary $\partial D^n = S^{n-1}$.
2. The interval $[0, 1]$, with the subspace topology, is a 1-manifold with boundary $\{0, 1\}$.
3. A product of two manifolds with boundary need not be one: the square $[0, 1] \times [0, 1]$ has corners where a neighbourhood looks like a quadrant, not a half-space. But the product of a boundaryless manifold X with a manifold-with-boundary Y is again a manifold with boundary, and $\partial(X \times Y) = X \times \partial Y$.

The manifold boundary ∂M is not the topological boundary of M inside an ambient space: the two coincide for the disk in $\mathbb{R}^2$, but the topological boundary depends on the embedding, whereas ∂M is intrinsic to M.

Exercise 1.1.3

1. Show that $\{x \in \mathbb{H}^n \mid x_n > 0\}$ is open in $\mathbb{R}^n$ while no neighbourhood of a point with $x_n = 0$ is homeomorphic to an open subset of $\mathbb{R}^n$; deduce from invariance of domain that a homeomorphism of open subsets of $\mathbb{H}^n$ preserves $\partial\mathbb{H}^n$.
2. Identify the boundary of the cylinder $S^1 \times [0, 1]$ and of the solid torus $S^1 \times D^2$, with their dimensions.
3. Show that gluing two copies of a manifold-with-boundary M along ∂M by the identity yields a manifold without boundary, the *double* of M, and identify the double of D^n.

1.2 Differentiable Manifolds

Topological manifolds are simply a topological space with some extra conditions. We shall now work with manifolds where we add an additional *differential structure* It shall be shown that the same manifold may have different differential structures, hence the differential structure will be considered as part of the information of a differentiable manifold[2]. In this section, we define a differential structure and define the types of maths between differential manifolds that are needed to form a category.

In most of this section, we shall be working with infinitely differentiable structures, i.e. *smooth structures*. This shall make no difference since in this section they are essentially the same, where the smooth structure is in fact simpler since there is no worry about keeping track how many times we "differentiate".

1.2.1 Smooth Charts and Smooth Structures

A topological manifold looks locally like $\mathbb{R}^n$; a *differentiable* manifold adds the data needed to do calculus on it. That data is an atlas whose charts overlap smoothly, so that differentiability transported from $\mathbb{R}^n$ through one chart agrees with the notion transported through any other. The

[2]Though as we shall see, many common manifolds shall have a unique differential structure up to isomorphism

same space can carry different such atlases, so the atlas is part of the data, not optional as it was in the topological case. The construction is set up on a bare set: the atlas *induces* a topology, on which the two global manifold conditions are then imposed.

Definition 1.2.1: (Set) Chart

A *chart* on a set M is a bijection $\varphi\colon U \to \varphi(U)$ from a subset $U \subseteq M$ onto an open subset of $\mathbb{R}^n$, written (U, φ); the integer n is its dimension. The domain U is the *patch* and φ the *coordinate map*, with components $\varphi = (x^1, \dots, x^n)$.

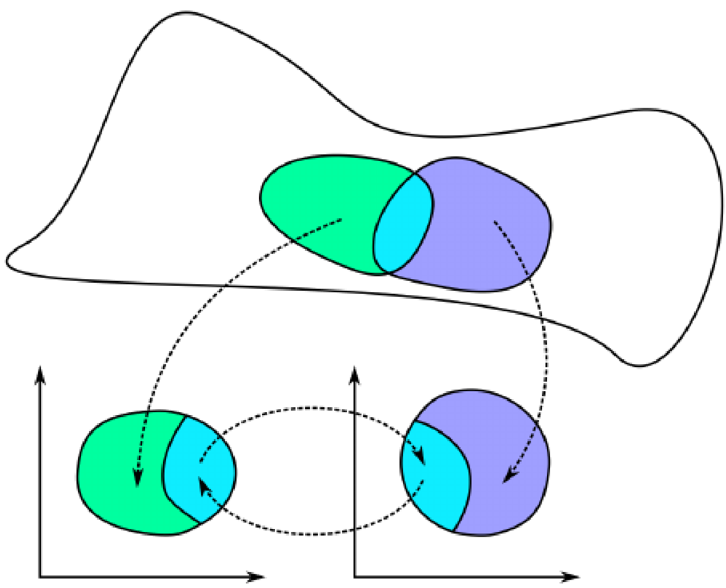
Definition 1.2.2: C^r Atlas

A C^r *atlas* of dimension n on a set M is a collection of n-charts $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ such that

1. the patches cover M, $\bigcup_\alpha U_\alpha = M$;
2. $\varphi_\alpha(U_\alpha \cap U_\beta)$ is open in $\mathbb{R}^n$ for all α, β;
3. for all α, β with $U_\alpha \cap U_\beta \neq \emptyset$, the *transition map* $\varphi_\beta \circ \varphi_\alpha^{-1}\colon \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is a C^r diffeomorphism.

When $r = \infty$ the atlas is *smooth* and its charts are smooth charts.

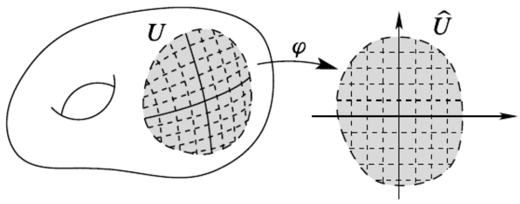
Constraining the transition maps differently selects a different category: real analytic (C^ω) if they are required analytic, and, when $n = 2m$ so that $\mathbb{R}^n \cong \mathbb{C}^m$, complex analytic if they are required holomorphic. The transition maps carry all the compatibility information and are what one computes with.



For charts (U, φ) with $\varphi = (x^1, \ldots, x^n)$ and (V, ψ) with $\psi = (y^1, \ldots, y^n)$, expressing $y^i \circ \varphi^{-1}$ as a function of $(x^1, \ldots, x^n)$ and dropping the composition gives

$$y^i = y^i(x^1, \ldots, x^n) \tag{1.1}$$

leaving deliberately ambiguous whether the x^j are numbers or coordinate functions; the same double reading appeared for the torus parameterization in example 1.1.12. A chart thus temporarily identifies its patch with a piece of $\mathbb{R}^n$, laying a coordinate grid over the set.



Example 1.2.3: Atlases on the Sphere

Regard $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ as a bare set. Split it into the four open arcs U, D, L, R (up, down, left, right)

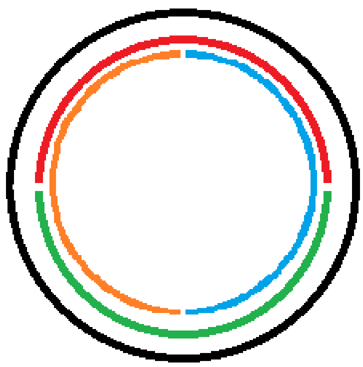


Figure 1.2: Four arcs covering the circle.

and project each to a coordinate axis: $\varphi_U(x, y) = \varphi_D(x, y) = x$ and $\varphi_L(x, y) = \varphi_R(x, y) = y$. These four injective maps are charts. Since $U \cap D = L \cap R = \emptyset$, the only overlaps are of the form $U \cap L$, where $\varphi_L^{-1}(y) = (-\sqrt{1 - y^2}, y)$ gives the transition $\varphi_U \circ \varphi_L^{-1}(y) = -\sqrt{1 - y^2}$, smooth on the open interval $\varphi_L(U \cap L) = (0, 1)$; the remaining overlaps differ only by signs. So the four charts form a smooth atlas.

The same sphere carries many atlases. The most economical uses the two stereographic charts from the poles N and S,

$$\varphi_N\colon S^n \setminus \{N\} \to \mathbb{R}^n, \quad \varphi_N(\vec{x}, t) = \frac{\vec{x}}{1 - t}, \qquad \varphi_S\colon S^n \setminus \{S\} \to \mathbb{R}^n, \quad \varphi_S(\vec{x}, t) = \frac{\vec{x}}{1 + t}$$

whose domains cover S^n. On the overlap, which φ_N maps onto $\mathbb{R}^n \setminus \{0\}$, the identity $\|\vec{x}\|^2 = 1 - t^2$

gives the transition

$$\varphi_S \circ \varphi_N^{-1}(u) = \frac{u}{\|u\|^2}$$

inversion in the unit sphere, smooth away from 0. For $n = 1$ this is $x \mapsto x^{-1}$, the transition already met in example 1.1.15.

Different atlases on a set can define the same differentiable structure, precisely when their charts overlap smoothly with one another.

Definition 1.2.4: Compatible Atlases

A chart (U, φ) is *compatible* with a C^r atlas $\mathcal{A}$ if $\mathcal{A} \cup \{(U, \varphi)\}$ is again a C^r atlas. Two atlases $\mathcal{A}$, $\mathcal{B}$ are compatible if $\mathcal{A} \cup \mathcal{B}$ is a C^r atlas.

Compatibility is an equivalence relation on atlases (exercise 1.2.1 (1)), so the atlases on a set fall into classes, each defining one differentiable structure.

Example 1.2.5: Compatible and Incompatible Atlases

1. The projection and stereographic atlases on S^1 from example 1.2.3 are compatible: their mutual transition maps are smooth (exercise 1.2.1 (2)).
2. On the set $M = \mathbb{R}$, the one-chart atlases $(\mathbb{R}, \mathrm{id})$ and $(\mathbb{R}, x \mapsto x^3)$ are *not* compatible: the transition $\mathrm{id} \circ (x \mapsto x^3)^{-1} = x^{1/3}$ is not differentiable at 0. They define two different smooth structures on the set $\mathbb{R}$.

Because compatible charts can always be adjoined, each class has a largest representative.

Definition 1.2.6: Maximal Atlas

The *maximal atlas* $\overline{\mathcal{A}}$ of a C^r atlas $\mathcal{A}$ is the collection of all charts compatible with $\mathcal{A}$. No atlas properly contains it, and it is the union of the compatibility class of $\mathcal{A}$.

Definition 1.2.7: Differentiable Structure

A C^r *differentiable structure* on a set M is a maximal C^r atlas $\mathcal{A}$, equivalently a compatibility class $[\mathcal{A}]$; when $r = \infty$ it is a *smooth structure*. Working with the class rather than a fixed atlas is what lets a structure be checked on a small, convenient atlas: for S^1 the two stereographic charts suffice.

A differentiable structure carries no topology a priori. The atlas gives one.

Lemma 1.2.8: An Atlas Induces a Topology

Let M carry a C^r atlas $\mathcal{A}$ with maximal atlas $\overline{\mathcal{A}}$. The patches of $\overline{\mathcal{A}}$ form a basis for a topology on M, the *topology induced by the C^r structure*, in which every chart is a homeomorphism onto its open image. Any subatlas of $\overline{\mathcal{A}}$ induces the same topology.

***Proof*:**
A family of subsets is a basis for a topology when it covers M and, for members B_1, B_2 and $p \in B_1 \cap B_2$, some member B_3 satisfies $p \in B_3 \subseteq B_1 \cap B_2$. The patches cover M by the atlas axiom. Given charts $(U, \varphi), (V, \psi) \in \overline{\mathcal{A}}$ and $p \in U \cap V$, the atlas axiom makes $\varphi(U \cap V)$ open in $\mathbb{R}^n$, so $(U \cap V, \varphi|_{U \cap V})$ is a chart; its transition maps against any chart of $\overline{\mathcal{A}}$ are restrictions of transition maps of $\overline{\mathcal{A}}$, hence C^r, so it is compatible and $U \cap V$ is itself a patch of $\overline{\mathcal{A}}$, giving $p \in U \cap V \subseteq U \cap V$. The patches are therefore a basis. In the induced topology a set is open exactly when its image under each chart is open, so every chart is a homeomorphism onto its image; and a subatlas has the same patches up to this basis relation, hence the same topology, completing the proof.

With a topology in hand, the two global manifold conditions become conditions on the atlas.

Proposition 1.2.9: When the Induced Topology Is a Manifold

Let $(M, \mathcal{A})$ be a C^r structure with its induced topology.

1. If any two points of M lie in disjoint patches, M is Hausdorff.
2. If $\mathcal{A}$ has a countable subatlas, M is second countable.

When both hold, M is a topological manifold.

***Proof*:**

1. Disjoint patches are disjoint open sets, so two points lying in such patches are separated.
2. A countable subatlas covers M by countably many patches, each homeomorphic to an open subset of $\mathbb{R}^n$ and so second countable; a countable union of second countable open subspaces is second countable.

Under both hypotheses M is locally Euclidean by construction, Hausdorff and second countable by (1) and (2), and locally compact because a chart carries the compact closed balls of $\mathbb{R}^n$ into M. A second countable, locally compact Hausdorff space is paracompact ([7, Second-Countable Locally Compact Hausdorff Spaces are Paracompact]), so M meets every clause of the definition of a topological manifold, completing the proof.

Definition 1.2.10: Differentiable Manifold

A C^r *manifold* of dimension n is a set M with a C^r structure whose induced topology is Hausdorff and second countable; by proposition 1.2.9 its underlying space is then a topological n-manifold. When $r = \infty$, M is a *smooth manifold*, and $\dim M$ denotes its dimension.

Unless stated otherwise, "manifold" means smooth manifold: Whitney showed that every maximal C^r atlas with $r \geq 1$ contains a compatible C^∞ atlas, so the smooth theory loses no generality. Requiring C^ω or holomorphic transitions instead gives a real-analytic manifold or a complex manifold.

Whether the topology alone fixes the smooth structure is a deep question. In low dimensions it does: manifolds of dimension at most 3 have a unique smooth structure, as does $\mathbb{R}^n$ for every $n \neq 4$. It fails in general, though: Kervaire and Milnor found that S^7 carries 28 distinct smooth structures, and Freedman's E_8 manifold is a topological 4-manifold with no smooth structure at all. Smoothness is genuinely additional data.

Definition 1.2.11: Product Manifold

For manifolds M, M' of dimensions n, n' with atlases $\{(U_\alpha, \varphi_\alpha)\}$ and $\{(V_\beta, \psi_\beta)\}$, the *product* $M \times M'$ is the $(n+n')$-manifold with atlas $\{(U_\alpha \times V_\beta,\ \varphi_\alpha \times \psi_\beta)\}$, where $(\varphi_\alpha \times \psi_\beta)(p, q) = (\varphi_\alpha(p), \psi_\beta(q))$.

A set may already carry a topology; the induced one agrees with it under the expected compatibility.

Proposition 1.2.12: The Induced Topology Matches a Given One

Let $(M, \mathcal{T})$ be a topological space carrying a C^r atlas $\mathcal{A}$. If every patch is $\mathcal{T}$-open and every chart is a $\mathcal{T}$-homeomorphism onto its image, then $\mathcal{T}$ is the topology induced by $\mathcal{A}$.

***Proof*:**
Write $\mathcal{T}'$ for the induced topology, with the patches as a basis. Each patch is $\mathcal{T}$-open by hypothesis, so $\mathcal{T}' \subseteq \mathcal{T}$. Conversely, let $O \in \mathcal{T}$ and $p \in O$, and choose a chart (U, φ) with $p \in U$. Then $O \cap U$ is $\mathcal{T}$-open, and since φ is a $\mathcal{T}$-homeomorphism $\varphi(O \cap U)$ is open in $\mathbb{R}^n$, so $(O \cap U, \varphi|_{O\cap U})$ is a chart and $O \cap U$ a patch, with $p \in O \cap U \subseteq O$. Thus O is a union of patches, hence $\mathcal{T}'$-open, giving $\mathcal{T} \subseteq \mathcal{T}'$. So $\mathcal{T} = \mathcal{T}'$, completing the proof.

Example 1.2.13: Smooth Manifolds
The topological manifolds of example 1.1.12, with the atlases given there, are smooth manifolds.

1. Any open subset U of a manifold M is a manifold, a *smooth open submanifold*, with the restricted charts. These are a rich source of examples: $\mathrm{GL}_n(\mathbb{R})$, open in the matrix space $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$, is one.

2. A product of smooth manifolds is smooth (definition 1.2.11).

3. $\mathbb{R}^n$ is smooth with its single identity chart; the structures $(\mathbb{R}, [\mathrm{id}])$ and $(\mathbb{R}, [x^3])$ are distinct, though they will turn out isomorphic.

4. S^n is smooth by the atlas of example 1.2.3.

5. A group G that is also a smooth manifold, on which multiplication $G \times G \to G$ and inversion $G \to G$ are smooth, is a *Lie group*; the model cases are $\mathrm{GL}_n(\mathbb{R})$, the circle $S^1 \subseteq \mathbb{C}$ under multiplication, and $\mathbb{R}^n$ under addition. Lie theory proper is developed in [16].

6. For smooth $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$, the graph $\Gamma(f)$ is a smooth n-manifold, covered by the single chart $\pi : \Gamma(f) \to U$, $(x, f(x)) \mapsto x$. That a space is locally the graph of a smooth function characterizes smooth manifolds, as chapter 3 makes precise.

Exercise 1.2.1

1. Show that compatibility of C^r atlases (definition 1.2.4) is an equivalence relation.
2. Verify that the projection and stereographic atlases on S^1 (example 1.2.3) are compatible, hence define the same smooth structure.
3. Show that $x \mapsto x^3$ makes $\mathbb{R}$ a smooth manifold whose induced topology is the usual one, yet whose smooth structure differs from $(\mathbb{R}, [\mathrm{id}])$.
4. Let V be an n-dimensional real vector space. Show that a linear isomorphism $\mathbb{R}^n \to V$ is a global chart, and that the resulting smooth structure is independent of the isomorphism chosen.

1.2.2 Pertinent Examples of Manifolds

Points and Vector Spaces

The extremes are quick. A one-point space is a 0-manifold: its single chart maps it to $\mathbb{R}^0 = \{0\}$, and every clause holds vacuously. At the other end, any finite-dimensional real vector space V is a manifold. A linear isomorphism $E\colon \mathbb{R}^n \to V$ is a global chart (exercise 1.2.1 (4)), and the transition between the charts of two bases is the change-of-basis matrix, a linear isomorphism and hence a diffeomorphism; so the smooth structure is independent of the basis.

This already gives many manifolds: the matrix space $M_{m\times n}(\mathbb{R})$, the space $\mathrm{Hom}_{\mathbb{R}}(V, W)$ of linear maps, and their open submanifolds. Chief among the latter is $\mathrm{GL}_n(\mathbb{R})$, open in $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$. The subgroups $\mathrm{SL}_n(\mathbb{R})$ and $\mathrm{O}(n)$ are *not* open, but will appear as regular submanifolds in chapter 3.

1.2.3 Level Sets

Let $\varphi\colon U \subseteq \mathbb{R}^N \to \mathbb{R}$ be smooth and $c \in \mathbb{R}$. The level set $M = \varphi^{-1}(c)$ is a manifold wherever φ is regular: if $D\varphi(a) \neq 0$ for each $a \in M$, the implicit function theorem writes M near a as a graph

$$x^i = f(x^1, \dots, x^{i-1}, x^{i+1}, \dots, x^N)$$

of a smooth f on an open subset of $\mathbb{R}^{N-1}$. As for the sphere, the graph charts are smooth and mutually compatible, so M is a smooth $(N-1)$-manifold. This is the special case of the regular value theorem developed in section 1.4.

Real Projective Space

The lines through the origin of a vector space form a manifold of their own, whose geometry is that of perspective, where parallel lines meet at infinity. The one-dimensional case is real projective space.

Definition 1.2.14: Real Projective Space

Real projective space is the quotient

$$\mathbb{R}\mathrm{P}^n := (\mathbb{R}^{n+1} \setminus \{0\})/\sim, \qquad x \sim y \iff x = \lambda y \text{ for some } \lambda \in \mathbb{R}^\times$$

with the quotient topology, equivalently the orbit space of the scaling action $\mathbb{R}^\times \curvearrowright (\mathbb{R}^{n+1} \setminus \{0\})$.

Every class has a representative on the unit sphere S^n, so $\mathbb{R}\mathrm{P}^n = q(S^n)$ is a continuous image of the compact S^n, hence compact and in particular paracompact. It is also Hausdorff, because the graph $\Gamma_\sim = \{(x,y) \mid x \sim y\}$ is closed: $x \sim y$ exactly when all the 2×2 minors $x_i y_j - x_j y_i$ vanish, so $\Gamma_\sim$ is the common zero set of finitely many continuous functions. A class is written in *homogeneous coordinates* $(x^0 : x^1 : \cdots : x^n) = [x]$, the notation recording that only the ratios are determined.

Proposition 1.2.15: Real Projective Space Is a Manifold

$\mathbb{R}\mathrm{P}^n$ is a smooth n-manifold.

***Proof*:**
Hausdorffness and paracompactness are shown above, so it remains to give a smooth atlas. For $0 \le i \le n$ let $U_i = \{(x^0 : \cdots : x^n) \mid x^i \neq 0\}$ and

$$\varphi_i \colon U_i \to \mathbb{R}^n, \qquad (x^0 : \cdots : x^n) \mapsto \left(\frac{x^0}{x^i}, \ldots, \frac{x^{i-1}}{x^i}, \frac{x^{i+1}}{x^i}, \ldots, \frac{x^n}{x^i}\right)$$

which rescales the representative so the i-th coordinate is 1 and reads off the rest; for instance $\varphi_1(7:3:2) = (7/3, 2/3)$. Each φ_i is a bijection onto $\mathbb{R}^n$, and a homeomorphism for the quotient topology, and the U_i cover $\mathbb{R}\mathrm{P}^n$. The transition $\varphi_i \circ \varphi_j^{-1}$, defined on the set where the i-th coordinate is nonzero, again rescales to make that coordinate 1: a rational map whose only denominators are the nonvanishing i-th coordinate, hence smooth. So $(\mathbb{R}\mathrm{P}^n, \{(U_i, \varphi_i)\})$ is a smooth n-manifold, completing the proof.

Each U_i consists of the lines meeting the affine hyperplane $\{x \mid x^i = 1\} \cong \mathbb{R}^n$, and its complement is the projective space of the remaining hyperplane. This gives the cell decomposition

$$\mathbb{R}\mathrm{P}^n = \mathbb{R}^n \sqcup \mathbb{R}\mathrm{P}^{n-1} = \mathbb{R}^n \sqcup \mathbb{R}^{n-1} \sqcup \cdots \sqcup \mathbb{R}^0$$

as a set: an affine $\mathbb{R}^n$ together with the $\mathbb{R}\mathrm{P}^{n-1}$ "at infinity". The dimension of $\mathbb{R}\mathrm{P}^n$ is nonetheless the constant n.

Complex Projective Space

The same construction over $\mathbb{C}$ gives a manifold of twice the real dimension.

Definition 1.2.16: Complex Projective Space

Complex projective space is

$$\mathcal{P}^n = (\mathbb{C}^{n+1} \setminus \{0\})/\sim, \qquad x \sim y \iff x = \lambda y \text{ for some } \lambda \in \mathbb{C}^\times$$

with the quotient topology.

Identifying $\mathbb{C}^{n+1} = \mathbb{R}^{2n+2}$, every class has a representative on the unit sphere S^{2n+1}, unique up to a unit scalar, so $\mathcal{P}^n = S^{2n+1}/\sim$ with $\|\lambda\| = 1$; Hausdorffness and paracompactness follow as for $\mathbb{R}\mathrm{P}^n$. The charts $U_i = \{(z^0 : \cdots : z^n) \mid z^i \neq 0\}$ with

$$\varphi_i \colon U_i \to \mathbb{C}^n \cong \mathbb{R}^{2n}, \qquad (z^0 : \cdots : z^n) \mapsto \left(\frac{z^0}{z^i}, \ldots, \frac{z^{i-1}}{z^i}, \frac{z^{i+1}}{z^i}, \ldots, \frac{z^n}{z^i}\right)$$

have holomorphic, hence smooth, transition maps exactly as in the real case. So $\mathcal{P}^n$ is a smooth manifold of real dimension $2n$, with the analogous cell decomposition $\mathcal{P}^n = \mathbb{C}^n \sqcup \mathbb{C}^{n-1} \sqcup \cdots \sqcup \{\mathrm{pt}\}$.

Grassmannians

Projective spaces are the lines through the origin; taking k-planes instead gives the Grassmannian.

Definition 1.2.17: Grassmannian

For an n-dimensional vector space V and $0 \le k \le n$, the *Grassmannian* $G_k(V)$ is the set of k-dimensional subspaces of V. When $V = \mathbb{R}^n$ one writes $G_k(\mathbb{R}^n) = G(k,n)$.

$G_k(V)$ is a smooth manifold of dimension $k(n-k)$, generalizing $G_1(\mathbb{R}^{n+1}) = \mathbb{R}\mathrm{P}^n$ and $G_1(\mathbb{C}^{n+1}) = \mathcal{P}^n$. To build charts, fix a splitting $V = P \oplus Q$ with $\dim P = k$, $\dim Q = n-k$. The graph of a linear map $T\colon P \to Q$,

$$\Gamma(T) = \{v + T(v) \mid v \in P\}$$

is a k-subspace meeting Q trivially; conversely a k-subspace S with $S \cap Q = \{0\}$ is the graph of the unique $T = \pi_Q \circ (\pi_P|_S)^{-1}$, where π_P, π_Q are the projections of V. So, writing $U_Q \subseteq G_k(V)$ for the subspaces meeting Q trivially, the map $\varphi\colon U_Q \to \mathrm{Hom}(P,Q) \cong \mathbb{R}^{k(n-k)}$, $\Gamma(T) \mapsto T$, is a bijection onto an open set.

Given a second splitting $V = P' \oplus Q'$ with chart φ', the overlap $\varphi(U_Q \cap U_{Q'})$ is the set of T whose graph meets Q' trivially. Writing $I_T(v) = v + T(v)$, so $\Gamma(T) = \mathrm{im}(I_T)$, this happens exactly when $\pi_{P'} \circ I_T$ is invertible; the entries of $\pi_{P'} \circ I_T$ are linear in T, so the condition $\det(\pi_{P'} \circ I_T) \neq 0$ is open, and $\varphi(U_Q \cap U_{Q'})$ is open. On it, the transition sends T to the T' with $\Gamma(T) = \Gamma(T')$, namely

$$T' = (\pi_{Q'} \circ I_T) \circ (\pi_{P'} \circ I_T)^{-1}$$

whose entries are rational in T with nonvanishing denominator, hence smooth. So $G_k(V)$ is a smooth $k(n-k)$-manifold.

The chart is $\mathrm{Hom}(P,Q)$ on the nose, which makes the tangent space of a Grassmannian readable off the construction with no further work. The identification below is the one every deformation-theoretic use of a Grassmannian runs on: a tangent vector at W is a recipe for moving each vector of W, and only the component transverse to W matters.

Proposition 1.2.18: The Tangent Space of a Grassmannian

Let V be an n-dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$ and $W \in G_k(V)$. There is a canonical isomorphism

$$T_W G_k(V) \cong \mathrm{Hom}(W, V/W)$$

characterized as follows: if M is a manifold, $g\colon M \to G_k(V)$ is smooth, $x \in M$, $\xi \in T_x M$, and σ is a smooth map from a neighbourhood of x into V with $\sigma(y) \in g(y)$ for every y, then

$$dg_x(\xi)\big(\sigma(x)\big) = \partial_\xi \sigma \bmod g(x)$$

It is $\mathrm{GL}(V)$-equivariant: the differential of $W \mapsto hW$ carries $\theta \in \mathrm{Hom}(W, V/W)$ to $\bar{h} \circ \theta \circ (h|_W)^{-1}$, where $\bar{h}$ is the map induced by h on quotients. Over $\mathbb{C}$ the charts are holomorphic, $G_k(V)$ is a complex manifold, and the isomorphism is $\mathbb{C}$-linear.

***Proof*:**
Choose Q with $V = W \oplus Q$ and use the chart $\varphi\colon U_Q \to \mathrm{Hom}(W, Q)$ built above, which sends W itself to 0. As $\mathrm{Hom}(W, Q)$ is a vector space, $d\varphi_W$ identifies $T_W G_k(V)$ with $\mathrm{Hom}(W, Q)$, and the projection $V \to V/W$ restricts to an isomorphism $Q \to V/W$; composing gives $T_W G_k(V) \cong \mathrm{Hom}(W, V/W)$.

The characterization, and independence of Q. Let g and σ be as stated and write $\sigma(y) = w(y) + T_y(w(y))$ in the chart, where $T_y = \varphi(g(y))$ and $w(y) \in W$. Differentiate at x, where $T_x = 0$ and hence $w(x) = \sigma(x)$:

$$\partial_\xi \sigma = \partial_\xi w + (\partial_\xi T)(\sigma(x))$$

The first term lies in W, so modulo W the derivative is $(\partial_\xi T)(\sigma(x))$, which is $d\varphi_W(dg_x(\xi))$ evaluated at $\sigma(x)$. That is the stated formula. It mentions no splitting, so the isomorphism does not depend on Q; and through each $v \in W$ there is such a σ with $\sigma(x) = v$, so the formula determines the tangent vector uniquely.

Equivariance. Apply the characterization to $h \circ \sigma$, which is a section of $y \mapsto h\, g(y)$. Its derivative is $h(\partial_\xi \sigma)$, and reducing modulo $h\, g(x)$ gives $\bar{h}$ applied to the reduction of $\partial_\xi \sigma$; reading the source through $h|_W$ gives the stated formula. Over $\mathbb{C}$ the transition maps computed above are rational with nonvanishing denominator, hence holomorphic, and every step here is $\mathbb{C}$-linear.

Complex Manifolds

Requiring the charts to map to $\mathbb{C}^n$ with holomorphic transitions defines a *complex n-manifold.* Every complex manifold is in particular a real $2n$-manifold, since $\mathbb{C} \cong \mathbb{R}^2$ smoothly; $\mathcal{P}^n$ is the running example.

1.2.4 Smooth Manifolds with Boundary

The smooth counterpart of section 1.1.2 needs a notion of differentiability on the half-space $\mathbb{H}^n$, whose boundary points are not interior to $\mathbb{R}^n$.

Definition 1.2.19: Differentiable over the Half-Space

A map f on $\mathbb{H}^n$ is *smooth* if it is smooth on the interior $\operatorname{int} \mathbb{H}^n$ and all partial derivatives extend continuously to $\partial\mathbb{H}^n$; equivalently, if f extends to a smooth map on an open neighbourhood in $\mathbb{R}^n$.

A *smooth manifold with boundary* is then a manifold with boundary (definition 1.1.16) whose transition maps are diffeomorphisms in this sense. Boundary points are those a chart sends to $\partial\mathbb{H}^n$, and ∂M is a smooth $(n-1)$-manifold, as in the topological case. A collar gives the boundary a standard neighbourhood.

Theorem 1.2.20: Collar Neighbourhood Theorem

The boundary of a smooth manifold with boundary has a neighbourhood diffeomorphic to $\partial M \times [0, 1)$, with ∂M corresponding to $\partial M \times \{0\}$.

The collar is what makes gluing along boundaries produce a smooth manifold: two copies of a bounded surface, such as a torus with an open disk removed, glue along their boundary circles into a closed surface. The proof integrates an inward-pointing vector field and so must wait for flows; it is given in chapter 5.

1.2.5 Orientation

Definition 1.2.21: Oriented Manifold

A smooth manifold M is *oriented* by an atlas whose transition maps all have positive Jacobian determinant,

$$\det D(\psi \circ \varphi^{-1}) > 0$$

on every overlap; a maximal such atlas is an *orientation*. A manifold is *orientable* if it admits one.

The sign is a convention: composing every chart with a fixed orientation-reversing diffeomorphism turns a positive atlas into a negative one. What matters is that the charts be mutually consistent. A connected orientable manifold has exactly two orientations. The sphere S^2 is orientable; the Möbius band is not, since going around it reverses the sign of the Jacobian.

Exercise 1.2.2

1. Let M be a nonempty smooth manifold of dimension ≥ 1. Show that M carries uncountably many distinct smooth structures. (They need not be pairwise non-diffeomorphic; only the maximal atlases must differ.)

1.3 Smooth Maps

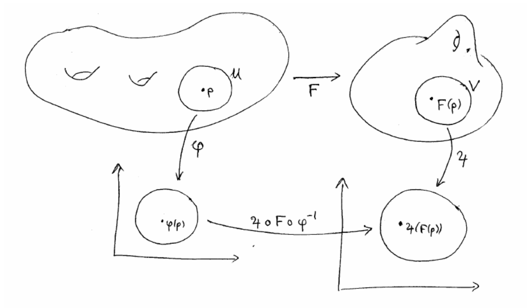
A differentiable structure lets a map be tested for differentiability through charts: read a map $f\colon M \to N$ in coordinates on both sides, and ask whether the resulting map of open subsets of Euclidean space is smooth. Because the charts overlap by diffeomorphisms, the answer does not depend on which charts are used.

Definition 1.3.1: Smooth Map

Let M, N be manifolds with C^r atlases, $f\colon M \to N$ a map, and $p \in M$. Then f is a C^r *map at* p (*smooth* when $r = \infty$) if there are a chart (U, φ) of M with $p \in U$ and a chart (V, ψ) of N with $f(p) \in V$ such that $f(U) \subseteq V$ and the *coordinate representative*

$$\psi \circ f \circ \varphi^{-1}\colon \varphi(U) \to \psi(V)$$

is C^r. It is a C^r *map* if it is one at every point.



The condition is chart-independent. If (U', φ') and (V', ψ') are other charts around p and $f(p)$ with $f(U') \subseteq V'$, then on the relevant overlap

$$\psi' \circ f \circ \varphi'^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1}) \circ (\varphi \circ \varphi'^{-1})$$

is a composite of the smooth representative with two transition diffeomorphisms, hence smooth. In particular a chart $\varphi\colon U \to \mathbb{R}^n$ is itself a smooth map, with smooth inverse – its representative $\mathrm{id} \circ \varphi \circ \varphi^{-1}$ is the identity – so a smooth manifold is *locally diffeomorphic* to $\mathbb{R}^n$, the differentiable analogue of local homeomorphism.

Definition 1.3.2: The Maps $C^r(M, N)$

For C^r manifolds M, N, write $C^r(M, N)$ for the set of C^r maps $M \to N$, and $C^r(M) = C^r(M, \mathbb{R})$.

Under pointwise operations $C^r(M)$ is an $\mathbb{R}$-algebra, and $C^r(M, \mathbb{C})$ a $\mathbb{C}$-algebra. When a map is read in coordinates on both sides, one writes $f(x^1, \ldots, x^n) = (y^1, \ldots, y^m)$ for the representative, thinking of f as a map of coordinates.

The first structural fact is that smoothness refines continuity.

Proposition 1.3.3: A Smooth Map Is Continuous

Every smooth map $f\colon M \to N$ is continuous.

***Proof*:**
Fix $p \in M$ and charts $(U, \varphi), (V, \psi)$ as in definition 1.3.1, so $f(U) \subseteq V$ and $\psi \circ f \circ \varphi^{-1}$ is smooth, hence continuous. Then $f|_U = \psi^{-1} \circ (\psi \circ f \circ \varphi^{-1}) \circ \varphi$ is a composite of continuous maps, so f is continuous at p. As p was arbitrary, f is continuous, as we sought to show.

The clause $f(U) \subseteq V$ is what forces continuity; dropping it admits discontinuous maps that are smooth chart by chart.

Example 1.3.4: An "Almost Smooth" Discontinuous Map
The step function $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = 1$ for $x \geq 0$ and 0 for $x < 0$, is discontinuous, hence not smooth by proposition 1.3.3. Take the identity charts $U = (-1, 1)$ on the domain and $V = (\frac{1}{2}, \frac{3}{2})$ on the codomain. Here $f(U) = \{0, 1\} \not\subseteq V$, so the clause $f(U) \subseteq V$ fails; but on the set $U \cap f^{-1}(V) = [0, 1)$ where a representative is defined, $\psi \circ f \circ \varphi^{-1} \equiv 1$ is constant, hence smooth.

Dropping the $f(U) \subseteq V$ requirement would thus admit this discontinuous f as "smooth chart by chart"; the clause is exactly what excludes it.

If instead one requires that *every* coordinate representative be smooth, the resulting notion agrees with definition 1.3.1 exactly for continuous maps.

Proposition 1.3.5: Equivalent Tests for Smoothness

Let $f\colon M \to N$ be continuous. Then f is smooth if and only if, for every chart (U,φ) of M and (V,ψ) of N, the representative

$$\psi \circ f \circ \varphi^{-1}\colon \varphi(U \cap f^{-1}(V)) \to \psi(V)$$

(defined on the open set $U \cap f^{-1}(V)$) is smooth.

***Proof*:**
Suppose f is smooth and fix charts $(U,\varphi), (V,\psi)$. Near any $q \in U \cap f^{-1}(V)$ there are charts as in definition 1.3.1 on which f is smooth; the given representative differs from that one by transition diffeomorphisms, so is smooth near q, hence smooth. Conversely, assume the representative condition and fix p; choose (V,ψ) with $f(p) \in V$ and any (U,φ) with $p \in U$. By continuity $U \cap f^{-1}(V)$ is open and contains p; restricting φ to it gives a chart with image in V on which the representative is smooth. So f is smooth, completing the proof.

Smoothness is a local property, which is what lets maps be built piece by piece.

Proposition 1.3.6: Smoothness Is Local

A map $f\colon M \to N$ is smooth if and only if every point of M has a neighbourhood on which f restricts to a smooth map.

Corollary 1.3.7: Gluing Smooth Maps

Let $\{U_\alpha\}$ be an open cover of M and $f_\alpha\colon U_\alpha \to N$ smooth maps agreeing on overlaps, $f_\alpha|_{U_\alpha \cap U_\beta} = f_\beta|_{U_\alpha \cap U_\beta}$. Then there is a unique smooth $f\colon M \to N$ with $f|_{U_\alpha} = f_\alpha$.

Requiring the pieces to agree on overlaps is restrictive; partitions of unity (chapter 3) give a more flexible tool for assembling maps.

Proposition 1.3.8: Basic Smooth Maps

Let M, N, P be smooth manifolds.

1. Every constant map $M \to N$ is smooth.
2. The identity $\mathrm{id}\colon M \to M$ is smooth.
3. The inclusion $\iota\colon U \hookrightarrow M$ of an open submanifold is smooth.
4. If $f\colon M \to N$ and $g\colon N \to P$ are smooth, so is $g \circ f$.

***Proof*:**

1. In any charts the representative of a constant map is constant, hence smooth.
2. The representative of id in a chart and itself is the identity of an open subset of $\mathbb{R}^n$.
3. For U open in M, a chart of M restricts to a chart of U, and the representative of ι is the identity.
4. Given p, choose charts (U, φ) at p, (V, ψ) at $f(p)$ with $f(U) \subseteq V$ and $\psi \circ f \circ \varphi^{-1}$ smooth, and (W, χ) at $g(f(p))$ with $g(V) \subseteq W$ and $\chi \circ g \circ \psi^{-1}$ smooth. Shrinking U so $f(U) \subseteq V \cap g^{-1}(W)$, the representative of $g \circ f$ is $(\chi \circ g \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1})$, a composite of smooth maps.

This completes the proof.

Note that smoothness of $g \circ f$ implies neither factor is smooth: a non-smooth bijection f with $f \circ f^{-1} = \mathrm{id}$ is a counterexample. Because composition preserves smoothness, the smooth manifolds and smooth maps form a category, written **Man**.

Proposition 1.3.9: Projections Are Smooth

Let $M_1, \ldots, M_k$ be manifolds. Each projection $\pi_i \colon M_1 \times \cdots \times M_k \to M_i$ is smooth, and a map $f \colon M \to M_1 \times \cdots \times M_k$ is smooth if and only if every $\pi_i \circ f$ is smooth.

***Proof*:**
In product charts $\varphi_1 \times \cdots \times \varphi_k$, the representative of π_i is the coordinate projection $\mathbb{R}^{n_1 + \cdots + n_k} \to \mathbb{R}^{n_i}$, which is smooth; so each π_i is smooth, and $\pi_i \circ f$ is smooth whenever f is. Conversely, in the same charts the representative of f has the representatives of the $\pi_i \circ f$ as its component blocks, and a map into a product of Euclidean spaces is smooth iff each component is; so f is smooth when all $\pi_i \circ f$ are, completing the proof.

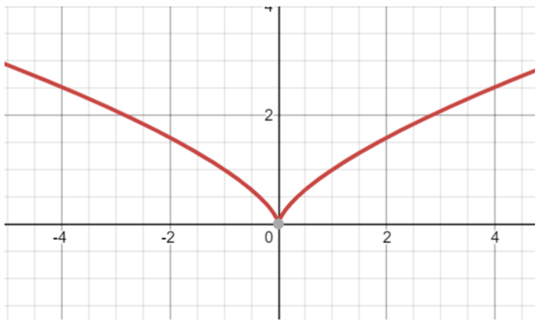
Example 1.3.10: Smooth Maps

1. Any map from a 0-manifold to a manifold is smooth, being locally constant.
2. The winding map $f \colon \mathbb{R} \to S^1$, $x \mapsto e^{2\pi i x}$, is smooth: in an angle chart its representative is $x \mapsto 2\pi x + c$. More generally $\mathbb{R}^n \to T^n$, $(x_1, \ldots, x_n) \mapsto (e^{2\pi i x_1}, \ldots, e^{2\pi i x_n})$, is smooth.
3. Complex conjugation $S^1 \to S^1$, $z \mapsto \bar{z}$, is smooth: in the two stereographic charts its representative is $a \mapsto 1/a$ from one pole and the identity between opposite poles.
4. The inclusion $S^n \hookrightarrow \mathbb{R}^{n+1}$ is smooth, as the stereographic representatives are rational and smooth.

An image of a manifold under a smooth map need not be a manifold – unlike the subobjects in many categories.

Example 1.3.11: The Image of a Smooth Map Need Not Be a Submanifold

The cuspidal cubic $f\colon \mathbb{R} \to \mathbb{R}^2$, $t \mapsto (t^2, t^3)$, is smooth (each component is a polynomial).



Its image is a topological manifold – f is a proper injection, hence a homeomorphism onto its image – but not a *smooth* submanifold of $\mathbb{R}^2$: f fails to be an immersion at 0 (as $f'(0) = 0$), and the image has a cusp there, so it is not locally a smooth graph. Requiring f to be an immersion does not by itself fix this: the line of irrational slope $g\colon \mathbb{R} \to T^2 = \mathbb{R}^2/\mathbb{Z}^2$, $t \mapsto [(t, \alpha t)]$ with α irrational, is an injective immersion whose image is dense in the torus, hence not a submanifold. The failure is repaired for injective maps out of *compact* manifolds, and is analyzed in general through the rank in chapter 3.

Derivations and Germs

Smooth maps deserve a notion of derivative. The full derivative of $f\colon M \to N$ is a linear map of tangent spaces, built in chapter 3; for a real-valued $f\colon M \to \mathbb{R}$ the coordinate partial derivatives are already available, read off through a chart.

Definition 1.3.12: Coordinate Derivative

Let (U, φ) be a chart of an n-manifold M with $\varphi = (x^1, \dots, x^n)$, and $f \in C^1(M)$. The *i-th coordinate derivative* of f is the function on U

$$\frac{\partial f}{\partial x^i}(p) := \partial_i (f \circ \varphi^{-1})(\varphi(p))$$

the ordinary i-th partial derivative of the representative $f \circ \varphi^{-1}$ at $\varphi(p)$.

If f is C^r then $\partial f/\partial x^i$ is C^{r-1}, and these derivatives inherit linearity and the Leibniz rule from their Euclidean counterparts. They depend on the chart: for a second chart (V, ψ) with $\psi = (y^1, \dots, y^n)$ the chain rule gives the change of variables

$$\frac{\partial f}{\partial y^i}(p) = \sum_{j=1}^{n} \frac{\partial f}{\partial x^j}(p)\, \frac{\partial x^j}{\partial y^i}(p) \tag{1.2}$$

This chart-dependence is exactly what the tangent space will resolve, by packaging the derivatives into a single intrinsic object.

Differentiability is local, so it is convenient to speak of maps defined only near a point, or on an arbitrary subset.

Definition 1.3.13: Locally C^r

Let $S \subseteq M$ be any subset of a smooth manifold and $f\colon S \to N$ continuous. Then f is *locally* C^r if every $s \in S$ has an open neighbourhood $O \subseteq M$ and a C^r map $\tilde{f}$ on O with $\tilde{f}|_{S\cap O} = f$.

Definition 1.3.14: Germs

For $p \in M$, call two smooth maps defined on neighbourhoods of p *equivalent* if they agree on some neighbourhood of p; an equivalence class is a *germ* at p, and the germs of maps into N form $C^\infty_p(M, N) = \big(\bigcup_{U\ni p} C^\infty(U, N)\big)/\sim$.

A germ has a well-defined value $[f](p) = f(p)$, and $C^\infty_p(M) = C^\infty_p(M, \mathbb{R})$ is an $\mathbb{R}$-algebra.

Diffeomorphisms

The isomorphisms of **Man** are the diffeomorphisms.

Definition 1.3.15: Diffeomorphism

A *diffeomorphism* is a smooth bijection $f\colon M \to N$ whose inverse is also smooth; then M and N are *diffeomorphic*, written $M \cong N$. The diffeomorphisms of M to itself form the group $\mathrm{Diff}(M)$.

Proposition 1.3.16: Characterizations of Diffeomorphisms

Let M, N be smooth manifolds with maximal atlases and $f\colon M \to N$ a bijection. The following are equivalent:

1. f is a diffeomorphism;
2. a chart (V, ψ) lies in the atlas of N iff $(f^{-1}(V), \psi \circ f)$ lies in that of M;
3. for every g, g is smooth on N iff $g \circ f$ is smooth on M.

***Proof*:**
(1) $\Rightarrow$ (3): if f and f^{-1} are smooth, then $g \circ f$ is smooth when g is (composition), and $g = (g \circ f) \circ f^{-1}$ is smooth when $g \circ f$ is. (3) $\Rightarrow$ (2): apply (3) to the coordinate maps, which are the smooth maps to $\mathbb{R}^n$ that are charts. (2) $\Rightarrow$ (1): condition (2) says f carries charts to charts, so both f and f^{-1} have identity representatives and are smooth. This completes the proof.

The content of (3) is that precomposing with a diffeomorphism f detects smoothness: $g \circ f$ is smooth exactly when g is. This holds only because f is invertible with smooth inverse; for a general smooth f, smoothness of $g \circ f$ says nothing about g (as proposition 1.3.8 already noted).

Example 1.3.17: Diffeomorphisms

1. The open ball and Euclidean space are diffeomorphic: $B(0,1) \to \mathbb{R}^n$, $x \mapsto x/\sqrt{1-\|x\|^2}$, is smooth with smooth inverse $y \mapsto y/\sqrt{1+\|y\|^2}$.

2. The two structures $(\mathbb{R}, [\mathrm{id}])$ and $(\mathbb{R}, [x^3])$ are distinct but *diffeomorphic*: the map $x \mapsto x^{1/3}$, read from the first to the second, has identity representative (the target chart x^3 composed with it is $x \mapsto x$), so it is a diffeomorphism. (Two copies of $(\mathbb{R}, [\mathrm{id}])$ joined by $x \mapsto x^3$ would not be, since $x^{1/3}$ is not differentiable at 0.) Distinct smooth structures can thus be isomorphic.

3. $\mathcal{P}^1$ is the Riemann sphere S^2. Both are covered by two charts glued by a single transition in a complex coordinate z: $z \mapsto 1/z$ for the projective charts of $\mathcal{P}^1$, while the stereographic charts of S^2 (example 1.2.3), under $\mathbb{C} \cong \mathbb{R}^2$, have transition $z \mapsto 1/\bar{z}$. The two agree after conjugating one chart, so matching them – identity on one, conjugation $z \mapsto \bar{z}$ on the other – is a diffeomorphism $\mathcal{P}^1 \cong S^2$. For $n \geq 1$, however, $\mathcal{P}^n \not\cong \mathbb{R}\mathrm{P}^{2n}$: the transition Jacobians of $\mathcal{P}^n$ all have positive determinant, so $\mathcal{P}^n$ is orientable, whereas $\mathbb{R}\mathrm{P}^{2n}$ is not.

Whether a homeomorphism can always be improved to a diffeomorphism is dimension dependent. Up to dimension 3 it can, so those manifolds have a unique smooth structure. It fails in dimension 4: Donaldson and Taubes showed $\mathbb{R}^4$ carries uncountably many pairwise non-diffeomorphic smooth structures, while $\mathbb{R}^n$ for $n \neq 4$ has a unique one. In higher dimensions homeomorphic manifolds may still fail to be diffeomorphic, as the 28 smooth structures on S^7 show.

Proposition 1.3.18: Properties of Diffeomorphisms

1. A composite of diffeomorphisms is a diffeomorphism.
2. A product of diffeomorphisms $f_i \colon M_i \to N_i$ is a diffeomorphism of the product manifolds.
3. Every diffeomorphism is a homeomorphism.
4. A diffeomorphism restricts on an open submanifold to a diffeomorphism onto its image.
5. Being diffeomorphic is an equivalence relation.

***Proof*:**
(1), (4) follow from proposition 1.3.8 (composition and restriction of smooth maps are smooth), applied to f and f^{-1}; (2) from proposition 1.3.9, since a product map is smooth iff its components are; (3) because a diffeomorphism and its inverse are continuous bijections; (5) from (1), the identity being a diffeomorphism, and symmetry of the definition, completing the proof.

Invariance of dimension, deferred in the topological setting to algebraic topology (theorem 1.1.13), is elementary once smoothness is available.

Theorem 1.3.19: Invariance of Dimension for Diffeomorphisms

If $f \colon M \to N$ is a diffeomorphism of nonempty manifolds, then $\dim M = \dim N$.

***Proof*:**
In charts, f has a representative $\hat{f}\colon U \subseteq \mathbb{R}^n \to V \subseteq \mathbb{R}^m$ that is a diffeomorphism of open sets. Differentiating $\hat{f}^{-1} \circ \hat{f} = \mathrm{id}$ by the chain rule gives $D\hat{f}^{-1}(\hat{f}(a)) \circ D\hat{f}(a) = \mathrm{id}$, and likewise in the other order, so $D\hat{f}(a)\colon \mathbb{R}^n \to \mathbb{R}^m$ is a linear isomorphism, forcing $n = m$, completing the proof.

Corollary 1.3.20: Invariance of the Boundary

A diffeomorphism $f\colon M \to N$ of manifolds with boundary satisfies $f(\partial M) = \partial N$ and restricts to a diffeomorphism $\operatorname{Int} M \to \operatorname{Int} N$.

***Proof*:**
In charts f is a diffeomorphism between open subsets of $\mathbb{H}^n$. An interior point has a neighbourhood mapping to an open subset of $\mathbb{R}^n$, whose image under the diffeomorphism is again open, so an interior point maps to an interior point; applying the same to f^{-1} shows boundary maps to boundary. Hence $f(\partial M) = \partial N$ and f restricts to a diffeomorphism of interiors, completing the proof.

Finally, a bijection can carry a smooth structure across, the smooth analogue of transporting a topology.

Proposition 1.3.21: Transfer of Structure

Let M be a smooth manifold and $\varphi\colon M \to X$ a bijection of sets. There is a unique smooth structure on X making φ a diffeomorphism, obtained by declaring the charts of X to be $\varphi \circ \varphi^{-1}$ for charts φ of M.

If X already carries a topology and φ is a homeomorphism, the transferred structure induces that topology. This is how spaces with no evident smooth structure – a triangle, say, homeomorphic to S^1 – acquire one.

Exercise 1.3.1

1. Show that the quotient map $\mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{R}\mathrm{P}^n$ is smooth.
2. Show that $\mathbb{R}\mathrm{P}^1 \cong S^1$, and locate the homeomorphism explicitly through the standard charts.
3. Show that the rotation $r_\theta(x, y, z) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta,\ z)$ is a diffeomorphism of S^2.
4. Give smooth manifolds and maps f, g with $g \circ f$ smooth but g not smooth, and explain why proposition 1.3.16 does not apply.

1.4 Regular Submanifolds

Manifolds should be closed under passing to well-behaved subsets. Restricting an atlas does not work directly – the charts of $\mathbb{R}^2$ do not restrict to charts of S^1, since no single chart covers the circle – so a submanifold is instead defined by requiring that, in suitable charts, it sits inside M the way $\mathbb{R}^k$ sits inside $\mathbb{R}^n$.

Definition 1.4.1: Regular Submanifold

A subset $S \subseteq M$ of an n-manifold is a *k-dimensional (regular) submanifold* if every $p \in S$ lies in a chart (U, φ) of M with

$$\varphi(U \cap S) = \varphi(U) \cap (\mathbb{R}^k \times \{0\})$$

Such a chart is a *submanifold chart* (or *slice chart*) for S.

Proposition 1.4.2: A Submanifold Is a Manifold

A k-dimensional submanifold $S \subseteq M$ is a smooth k-manifold in the subspace topology, with charts $(U \cap S,\ \pi \circ \varphi|_{U \cap S})$ obtained from the submanifold charts by the projection $\pi \colon \mathbb{R}^n \to \mathbb{R}^k$ onto the first k coordinates; the inclusion $\iota \colon S \hookrightarrow M$ is smooth.

***Proof*:**
For submanifold charts $(U, \varphi), (V, \psi)$, the restricted charts of S have transition $\pi \circ \psi \circ \varphi^{-1} \circ \iota_k$, where $\iota_k(x) = (x, 0)$: the restriction of the smooth $\psi \circ \varphi^{-1}$ to a slice, hence smooth. So the restricted charts form a smooth atlas, and each is a homeomorphism onto an open piece of a slice, so the induced topology is the subspace topology. In a submanifold chart the representative of ι is $x \mapsto (x, 0)$, smooth, so ι is smooth, completing the proof.

Regular submanifolds are the subobjects of **Man**; henceforth "submanifold" means regular submanifold. A smooth map restricts to one.

Corollary 1.4.3: Restriction to a Submanifold

If $f \colon M \to N$ is smooth and $S \subseteq M$ a submanifold, then $f|_S = f \circ \iota \colon S \to N$ is smooth.

***Proof*:**
$f|_S = f \circ \iota$ is a composite of the smooth maps ι (proposition 1.4.2) and f, hence smooth, completing the proof.

The converse fails: a smooth map out of S need not extend to M (exercise 1.4.1 (3)).

Example 1.4.4: Submanifolds

1. The n-dimensional submanifolds of $\mathbb{R}^n$ are exactly its open subsets.
2. $\mathbb{R}^k \times \{0\} \subseteq \mathbb{R}^n$ is a k-submanifold, the standard coordinates giving a global submanifold chart.
3. $S^k \subseteq S^n$, the vanishing of the last $n - k$ coordinates, is a k-submanifold.
4. Every open subset of M is an n-submanifold.

Several equivalent tests recognize a subset of $\mathbb{R}^n$ as a submanifold, each convenient in a different situation.

Proposition 1.4.5: Characterizations of a Submanifold of $\mathbb{R}^n$

For $S \subseteq \mathbb{R}^n$, the following local conditions at each point are equivalent, and each exhibits S as a k-dimensional submanifold:

1. (*regular level set*) near the point $S = f^{-1}(0)$ for a C^r map f into $\mathbb{R}^{n-k}$ with Df of rank $n-k$;
2. (*graph*) near the point S is the graph of a C^r map $\mathbb{R}^k \to \mathbb{R}^{n-k}$, after permuting coordinates;
3. (*slice*) a C^r diffeomorphism carries S near the point onto an open piece of $\mathbb{R}^k \times \{0\}$;
4. (*embedding*) near the point S is the image of a C^r embedding of an open subset of $\mathbb{R}^k$.

1.4.6 Key Ideas The cycle (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1) and the equivalence (3) $\Leftrightarrow$ (4) run through the implicit and inverse function theorems: a regular level set is a graph by the implicit function theorem; a graph $y = g(x)$ is flattened by $(x, y) \mapsto (x, y - g(x))$; a slice is the zero set of the complementary projection; and the slice and embedding descriptions are mutually inverse. Replacing $\mathbb{R}^n$ by an arbitrary manifold, these become the rank theorem, proved in full in chapter 3, which is where the general submanifold theory is developed.

Quotients are the construction still missing. Some manifolds do arise as quotients – the Möbius band, for one – but a quotient of a manifold need not even be Hausdorff (example 1.1.6), and smooth structures behave badly under quotients in general. The conditions under which a quotient is a smooth manifold are taken up, through group actions, in chapter 4.

Exercise 1.4.1

1. Show that being a submanifold is transitive: if S is a submanifold of T and T of M, then S is a submanifold of M.
2. Show that every open subset of M is a regular submanifold of the same dimension.
3. Give a submanifold $S \subseteq M$ and a smooth $f : S \to N$ that does not extend to any smooth map $M \to N$.

1.5 Bump Functions and Partitions of Unity

Partitions of unity turn local data on a manifold into global data, and their existence in the smooth category is what sets smooth manifolds apart from complex or real-analytic ones. It rests on two inputs cited from upstream: the smooth bump function on $\mathbb{R}^n$, from [4], and the topological existence of partitions of unity on a paracompact Hausdorff space, from [7, Existence of Subordinate Partitions of Unity].

Definition 1.5.1: Support

The *support* of a function $f\colon X \to \mathbb{R}$ is the closure of the set on which it is nonzero,

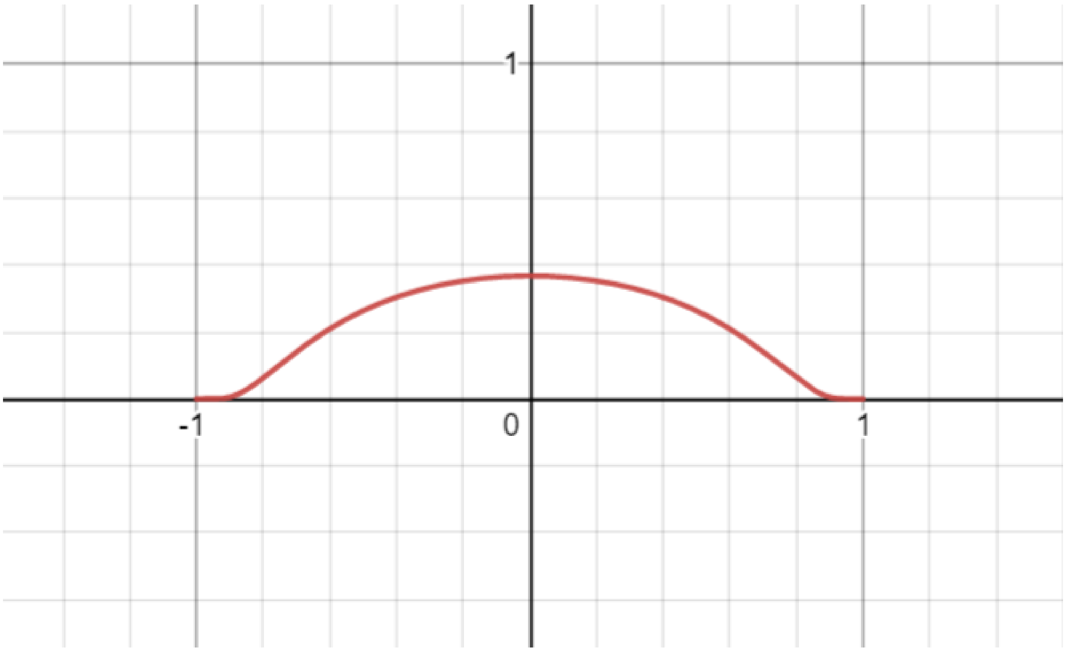
$$\operatorname{supp} f = \overline{\{x \in X \mid f(x) \neq 0\}}$$

Example 1.5.2: A Smooth Bump on $\mathbb{R}$

The function

$$\Phi(x) = \begin{cases} e^{-1/(1-x^2)} & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}$$

is smooth with support the closed unit interval, hence compactly supported.



This function and the general smooth bump on $\mathbb{R}^n$ are constructed in [4].

Definition 1.5.3: Partition of Unity Subordinate to a Cover

For an open cover $\{U_\alpha\}_{\alpha\in A}$ of a space M, a *partition of unity subordinate to it* is a family of continuous functions $\psi_\alpha\colon M \to [0,1]$ such that

1. $\mathrm{supp}\psi_\alpha \subseteq U_\alpha$ for each α;
2. the supports are locally finite: every point has a neighbourhood meeting $\mathrm{supp}\psi_\alpha$ for only finitely many α;
3. $\sum_{\alpha\in A} \psi_\alpha(x) = 1$ for every $x \in M$ (a finite sum near each point, by (2)).

The partition is *smooth* if every ψ_α is smooth.

On a paracompact Hausdorff space a continuous partition of unity subordinate to any open cover exists ([7, Existence of Subordinate Partitions of Unity]), and every manifold is such a space (theorem 1.1.8). The smooth refinement replaces those continuous functions by smooth ones built from bump functions.

Theorem 1.5.4: Existence of Smooth Bump Functions

Let M be a smooth manifold, $K \subseteq M$ compact, and $O \subseteq M$ open with $K \subseteq O$. There is a smooth $\beta\colon M \to [0,1]$ identically 1 on K and with compact support in O.

***Proof*:**
On $\mathbb{R}^n$, for concentric balls $B(0,r) \subseteq B(0,R)$ the smooth bump of [4] gives such a β, equal to 1 on $B(0,r)$ with support in $B(0,R)$; translating handles a ball about any point. For general M, cover the compact K by finitely many chart balls (U_i, φ_i) with $\overline{U_i} \subseteq O$, each carrying a bump β_i (through φ_i, extended by 0 off U_i) equal to 1 on a smaller compact K_i, with the interiors of the K_i still covering K. Then

$$\beta = 1 - \prod_i (1-\beta_i)$$

is smooth, takes values in $[0,1]$, has support in $\bigcup_i U_i \subseteq O$, and equals 1 wherever some $\beta_i = 1$, in particular on K, completing the proof.

The bump functions upgrade the topological partition of unity to a smooth one; this is the result the rest of the geometry track uses.

Theorem 1.5.5: Existence of Smooth Partitions of Unity

Every open cover of a smooth manifold admits a subordinate smooth partition of unity.

***Proof*:**
Let $\{U_\alpha\}$ cover M. By paracompactness (theorem 1.1.8) refine it to a locally finite cover $\{V_i\}$ of relatively compact chart balls with $\overline{V_i} \subseteq U_{\alpha(i)}$, and by the shrinking lemma ([7, Shrinking Lemma]) choose an open cover $\{W_i\}$ with $\overline{W_i} \subseteq V_i$. A smooth bump β_i (theorem 1.5.4) is 1 on $\overline{W_i}$ with support in V_i. Local finiteness makes $\sigma = \sum_i \beta_i$ smooth, and $\sigma > 0$ since the W_i cover

M. Then $\psi_i = \beta_i/\sigma$ are smooth, sum to 1, and, gathered by α, form a smooth partition of unity subordinate to $\{U_\alpha\}$, completing the proof.

The existence of Riemannian metrics, of connections, and of global sections of vector bundles all reduce to this theorem downstream ([6, Existence of Riemannian Metrics], [13]). It is a genuinely C^∞ phenomenon: a nonzero real-analytic function cannot have compact support, so no analytic partition of unity subordinate to a proper cover exists (exercise 1.5.1 (3)).

The remaining corollaries separate and extend across closed sets.

Corollary 1.5.6: Smooth Bump on a Closed Set

For a closed $A \subseteq M$ and open $U \supseteq A$, there is a smooth $\psi\colon M \to [0,1]$ with $\psi \equiv 1$ on A and $\operatorname{supp}\psi \subseteq U$.

***Proof*:**
Take a smooth partition of unity $\{\psi_0, \psi_1\}$ subordinate to the cover $\{U, M \setminus A\}$ (theorem 1.5.5). Then $\operatorname{supp}\psi_0 \subseteq U$, and on A the term ψ_1 vanishes, its support avoiding A, so $\psi_0 = 1 - \psi_1 = 1$ there. Thus $\psi = \psi_0$ works, completing the proof.

Corollary 1.5.7: Smooth Urysohn Lemma

Disjoint closed sets A, B in a smooth manifold are separated by a smooth $f\colon M \to [0,1]$ with $f|_A = 0$ and $f|_B = 1$.

***Proof*:**
Apply corollary 1.5.6 to the closed set B inside the open set $M \setminus A$: the resulting f is 1 on B and, having support in $M \setminus A$, is 0 on A, completing the proof.

Corollary 1.5.8: Extending a Smooth Function off a Closed Set

Let $A \subseteq M$ be closed, $f\colon A \to \mathbb{R}^k$ smooth (definition 1.3.13), and $U \supseteq A$ open. There is a smooth $\tilde{f}\colon M \to \mathbb{R}^k$ with $\tilde{f}|_A = f$ and $\operatorname{supp}\tilde{f} \subseteq U$.

***Proof*:**
Smoothness of f on A gives a smooth extension F to an open $O \supseteq A$; shrink O into U. Let ψ be a smooth bump equal to 1 on A with $\operatorname{supp}\psi \subseteq O \cap U$ (corollary 1.5.6). Then $\tilde{f} = \psi F$, set to 0 off $\operatorname{supp}\psi$, is smooth on M, equals $F = f$ on A, and has support in U, completing the proof.

Corollary 1.5.9: A Closed Set as a Zero Set

Every closed $K \subseteq M$ is the zero set $f^{-1}(0)$ of a smooth $f\colon M \to [0, \infty)$.

***Proof*:**
The open complement $M \setminus K$ is a countable union of chart balls B_i; on each take a smooth $\beta_i \geq 0$ positive on B_i with support $\overline{B_i}$, scaled so that it and its derivatives up to order i are below 2^{-i}, so that $f = \sum_i \beta_i$ converges to a smooth function. Then $f > 0$ exactly on $\bigcup_i B_i = M \setminus K$, so $f^{-1}(0) = K$, completing the proof.

Exercise 1.5.1

1. Verify corollary 1.5.8 in the case $M = \mathbb{R}$, $A = \{0\}$, $f(0) = 1$: write down an explicit smooth extension supported in $(-1, 1)$.
2. Show the extension in corollary 1.5.8 can fail without closedness: exhibit a non-closed $A \subseteq \mathbb{R}$ and a smooth f on it with no smooth extension to $\mathbb{R}$.
3. Explain why no real-analytic partition of unity is subordinate to a proper open cover of $\mathbb{R}$.

2

Derivative and Tangent Structures

On $\mathbb{R}^n$ the derivative of a smooth map $f\colon U \to V$ at a point p is a linear map $Df(p)\colon \mathbb{R}^n \to \mathbb{R}^m$ best approximating f near p, and Df is itself a smooth map into the space of linear maps. This chapter builds the analogue on a manifold: to each $p \in M$ a vector space T_pM, the *tangent space*, and to each smooth $f\colon M \to N$ a linear $T_pf\colon T_pM \to T_{f(p)}N$ approximating f at p and varying smoothly with p. Two things must be constructed: the vector space T_pM itself – there is no ambient $\mathbb{R}^n$ whose affine subspaces could serve – and the object over which T_pf varies smoothly, the tangent bundle. The linear map also records dimensional information; it is what lets one control the rank of a smooth map, and so recognize when an image fails to be a submanifold (example 1.3.11), the theme of chapter 3.

2.1 Tangent Space

There is no single natural definition of T_pM; three constructions, each illuminating, turn out to agree. The *kinematic* one takes tangent vectors to be velocities of curves through p. The *chart* one takes them to be $\mathbb{R}^n$-vectors, one per chart, glued by the derivatives of the transition maps. The *algebraic* one takes them to be derivations of germs of functions at p – the definition that generalizes to algebraic geometry and is easiest to handle without coordinates. The three are canonically isomorphic (proposition 2.1.17); which to use is a matter of convenience.

A single mechanism underlies all three: each chart identifies a neighbourhood of p with a piece of $\mathbb{R}^n$, and the tangent space is $\mathbb{R}^n$ transported through that identification, the transition-map derivatives keeping the transport consistent. The rule

$$\overline{a}^i = \sum_{j=1}^{n} \frac{\partial \overline{x}^i}{\partial x^j}\, a^j$$

relating the components of one tangent vector in two charts is the classical *transformation law.*

The Kinematic Tangent Space

Two curves through p should determine the same tangent vector when they have the same velocity there, tested against every smooth function.

Definition 2.1.1: Tangent Curves

Curves $c_0, c_1 \colon I \to M$ with $c_0(0) = c_1(0) = p$ are *tangent at* p if $(f \circ c_0)'(0) = (f \circ c_1)'(0)$ for every smooth real-valued f defined near p.

This is an equivalence relation, and a class collects all curves through p with a common velocity.

Definition 2.1.2: Tangent Vector (via Curves)

The *tangent space* T_pM is the set of tangency classes at p; its elements are *tangent vectors*. A class is written $v_p = [c]$.

The test against real-valued functions is the same as the test against vector-valued ones.

Lemma 2.1.3: Scalar and Vector Tests Agree

Curves c_0, c_1 are tangent at p if and only if $(f \circ c_0)'(0) = (f \circ c_1)'(0)$ for every $\mathbb{R}^k$-valued smooth f near p.

***Proof*:**
If the curves are tangent, then for $f = (f^1, \ldots, f^k)$ each component satisfies $(f^i \circ c_0)'(0) = (f^i \circ c_1)'(0)$, so $(f \circ c_0)'(0) = (f \circ c_1)'(0)$. Conversely, given the vector-valued equality, apply it to $f = (g, 0, \ldots, 0)$ for an arbitrary real-valued g to recover $(g \circ c_0)'(0) = (g \circ c_1)'(0)$, completing the proof.

The class structure is geometric but gives no vector-space structure directly. That comes by transporting the structure of $\mathbb{R}^n$ through a chart, consistently across charts.

Proposition 2.1.4: Consistent Transfer of a Linear Structure

Let S be a set and $\{b_\alpha \colon V_\alpha \to S\}_{\alpha \in A}$ a family of bijections from n-dimensional vector spaces. If every $b_\beta^{-1} \circ b_\alpha \colon V_\alpha \to V_\beta$ is a linear isomorphism, then S carries a unique vector-space structure making each b_α a linear isomorphism.

***Proof*:**
Transport the operations from one index: $s_1 + s_2 := b_\alpha(b_\alpha^{-1} s_1 + b_\alpha^{-1} s_2)$ and $a \cdot s := b_\alpha(a\, b_\alpha^{-1} s)$. The sum is independent of α: inserting $b_\beta b_\beta^{-1} = \mathrm{id}$ and using that $b_\alpha^{-1} b_\beta$ is linear,

$$b_\alpha(b_\alpha^{-1} s_1 + b_\alpha^{-1} s_2) = b_\alpha b_\alpha^{-1} b_\beta(b_\beta^{-1} s_1 + b_\beta^{-1} s_2) = b_\beta(b_\beta^{-1} s_1 + b_\beta^{-1} s_2)$$

and likewise for scalars. With these operations each b_α is a linear isomorphism, and any structure making some b_α linear must be this one, giving uniqueness and completing the proof.

For a chart (U, φ_α) at p, define $b_\alpha \colon \mathbb{R}^n \to T_pM$ by $v \mapsto [\gamma_v]$, where $\gamma_v(t) = \varphi_\alpha^{-1}(\varphi_\alpha(p) + tv)$ on a small interval about 0.

Lemma 2.1.5: Charts Transfer the Structure Consistently

Each $b_\alpha \colon \mathbb{R}^n \to T_pM$ is a bijection, and for a second chart (V, φ_β),

$$b_\beta^{-1} \circ b_\alpha = D(\varphi_\beta \circ \varphi_\alpha^{-1})(\varphi_\alpha(p))$$

a linear isomorphism.

***Proof*:**
A chart recovers the defining vector: $(\varphi_\alpha \circ \gamma_v)'(0) = \frac{d}{dt}\big|_0(\varphi_\alpha(p) + tv) = v$.

Injectivity. If $b_\alpha(v) = b_\alpha(w)$ then $[\gamma_v] = [\gamma_w]$, so lemma 2.1.3 applied to the $\mathbb{R}^n$-valued φ_α gives $v = (\varphi_\alpha \circ \gamma_v)'(0) = (\varphi_\alpha \circ \gamma_w)'(0) = w$.

Surjectivity. Given $[c] \in T_pM$, set $v = (\varphi_\alpha \circ c)'(0)$. For any smooth f near p, the chain rule applied to $f \circ c = (f \circ \varphi_\alpha^{-1}) \circ (\varphi_\alpha \circ c)$ gives

$$(f \circ c)'(0) = D(f \circ \varphi_\alpha^{-1})(\varphi_\alpha(p))\, v = (f \circ \gamma_v)'(0)$$

so c is tangent to γ_v and $b_\alpha(v) = [c]$.

Change of chart. Taking $f = \varphi_\beta$ in the surjectivity computation, $b_\beta^{-1}(b_\alpha(v)) = (\varphi_\beta \circ \gamma_v)'(0) = D(\varphi_\beta \circ \varphi_\alpha^{-1})(\varphi_\alpha(p))\, v$, a linear isomorphism since the transition map is a diffeomorphism, completing the proof.

By proposition 2.1.4 the maps b_α endow T_pM with a canonical $\mathbb{R}^n$-vector-space structure. This construction is the *kinematic tangent space*, $(T_pM)_{\mathrm{kin}}$.

The Tangent Space via Charts

The chart construction records the same data – a vector per chart, glued by transition derivatives – without reference to curves.

Definition 2.1.6: Tangent Space via Charts

For a manifold M with maximal atlas $\mathcal{A}$ and $p \in M$, let Γ_p be the set of triples $(p, v, (U, \varphi))$ with $v \in \mathbb{R}^n$ and $(U, \varphi) \in \mathcal{A}$, $p \in U$. Declare

$$(p, v, (U, \varphi)) \sim (p, w, (V, \psi)) \iff w = D(\psi \circ \varphi^{-1})(\varphi(p))\, v$$

The quotient $(T_pM)_{\mathrm{phys}} = \Gamma_p/\sim$ is the *tangent space via charts.*

Each chart gives a bijection $b_{(U,\varphi)} \colon \mathbb{R}^n \to \Gamma_p/\sim$, $v \mapsto [(p, v, (U, \varphi))]$: it is injective because $D(\varphi \circ \varphi^{-1}) = \mathrm{id}$ forces $v = w$, and surjective by definition. The change of chart is again $D(\psi \circ \varphi^{-1})(\varphi(p))$, so proposition 2.1.4 transfers the vector-space structure of $\mathbb{R}^n$. This is the *physical tangent space*, the description favoured in physics, where one says $v \in \mathbb{R}^n$ represents the tangent vector in the chart (U, φ).

The Algebraic Tangent Space

The third construction takes tangent vectors to be operators on functions rather than geometric objects. On $\mathbb{R}^n$ the directional derivative $v \cdot \nabla$ in a direction v acts on smooth functions and satisfies two rules: it is linear, and it obeys the product rule. These two rules turn out to characterize directional derivatives completely, and they refer to no coordinates. Taking them as the definition gives a tangent space built from the algebra of functions alone.

The motivation is the gradient. For smooth $f : \mathbb{R}^n \to \mathbb{R}$ the partial derivatives $\partial f / \partial x^i(p)$ assemble into $\nabla f(p)$, one distinguished vector in the tangent space at p – the direction of steepest ascent. A tangent vector should be an *arbitrary* vector there, not this single one; what the gradient exposes is that the operators $\partial/\partial x^i|_p$ span the possible directions. Tracking those operators, rather than the values they produce, is what frees the construction from any chart.

Definition 2.1.7: Tangent Vector as Derivation

Let M be an n-manifold and $p \in M$. A *tangent vector at p* is a linear map $v_p : C^\infty(M) \to \mathbb{R}$ satisfying the *Leibniz rule*

$$v_p(fg) = v_p(f)\, g(p) + f(p)\, v_p(g), \qquad f, g \in C^\infty(M)$$

Such a map is a *derivation at p*: a derivation of the algebra $C^\infty(M)$ relative to the evaluation homomorphism $\mathrm{ev}_p(f) = f(p)$.

One writes $v_p f$ for $v_p(f)$. A derivation at p is in particular an element of the dual space $C^\infty(M)^*$, and the derivations at p form a linear subspace: a linear combination of maps obeying the Leibniz rule at p obeys it again. This subspace is the *algebraic tangent space* $(T_pM)_{\mathrm{Alg}}$. Smoothness is essential to finite-dimensionality: for finite r the derivations of $C^r(M)$ form an infinite-dimensional space, so C^∞ is the right regularity for a tangent space of dimension n.[1]

2.1.8 Remark The derivations carry more than linear structure. Assembling a derivation at every point into a single operator on $C^\infty(M)$ yields a *vector field*, and the commutator of two such operators is again one; this bracket makes the vector fields a Lie algebra, developed in chapter 5.

The prototype derivation is the coordinate partial. Recall (definition 1.3.12) that for a chart (U, φ) at p with $\varphi = (x^1, \dots, x^n)$,

$$\frac{\partial f}{\partial x^i}(p) = D_i\big(f \circ \varphi^{-1}\big)(\varphi(p))$$

[1] For every finite r the space of point derivations of $C^r(\mathbb{R})$ at the origin is infinite-dimensional: functions like $x|x|^\alpha$ with $0 < \alpha < 1$ lie in C^1 yet not in $\mathfrak{m}_0^2$, giving a continuum of independent classes in $\mathfrak{m}_0/\mathfrak{m}_0^2$. Only at $r = \infty$ does it collapse to dimension 1.

Definition 2.1.9: Coordinate Derivations

For a chart (U, φ) of an n-manifold with $p \in U$, the *coordinate derivation* $\partial/\partial x^i|_p$ is

$$\left.\frac{\partial}{\partial x^i}\right|_p : C^\infty(M) \to \mathbb{R}, \qquad \left.\frac{\partial}{\partial x^i}\right|_p f = \frac{\partial f}{\partial x^i}(p)$$

Each coordinate derivation depends on the chart. Setting $c_i(t) = \varphi^{-1}(\varphi(p)+te_i)$ for the i-th standard basis vector e_i, the chain rule gives

$$\left.\frac{\partial}{\partial x^i}\right|_p f = \lim_{h\to 0} \frac{f(c_i(h)) - f(p)}{h}$$

so $\partial/\partial x^i|_p$ is the velocity of the i-th coordinate curve and is a derivation at p, an element of $(T_pM)_{\text{Alg}}$. The goal of this subsection is that these n operators form a basis: for any chart at p,

$$v_p = \sum_{i=1}^{n} v_p(x^i) \left.\frac{\partial}{\partial x^i}\right|_p \qquad \text{for every } v_p \in (T_pM)_{\text{Alg}}$$

where $x^i \in C^\infty(U)$ is the i-th coordinate function. In particular $\dim(T_pM)_{\text{Alg}} = n$ independently of the chart. The first step is that a derivation sees only local behaviour.

Lemma 2.1.10: Derivations Are Local

Let $v_p \in (T_pM)_{\text{Alg}}$ and $f, g, h \in C^\infty(M)$.

1. If $f = g$ on a neighbourhood of p, then $v_p(f) = v_p(g)$.
2. If h is constant on a neighbourhood of p, then $v_p(h) = 0$.

***Proof*:**
For (1), linearity reduces the claim to: if f vanishes on a neighbourhood U of p, then $v_p(f) = 0$. Choose a bump function β with $\operatorname{supp}\beta \subseteq U$ and $\beta(p) = 1$ (theorem 1.5.4). Then $\beta f \equiv 0$ on M: it vanishes on U because f does, and off U because β does. Since $f(p) = 0$, the Leibniz rule gives

$$0 = v_p(\beta f) = \beta(p)\, v_p(f) + f(p)\, v_p(\beta) = v_p(f)$$

For (2), a global constant $c = c \cdot 1$ has $v_p(1) = v_p(1 \cdot 1) = 2v_p(1)$, so $v_p(1) = 0$ and $v_p(c) = c\, v_p(1) = 0$. A locally constant h agrees with a global constant near p, so $v_p(h) = 0$ by (1), completing the proof.

Locality says v_p is determined by germs, yet the definition feeds it global functions. The next result reconciles the two: a derivation on an open $U \ni p$ carries the same information as one on all of M.

Proposition 2.1.11: Restriction to an Open Set is an Isomorphism

For open $U \subseteq M$ with $p \in U$, the map $\Phi\colon (T_pU)_{\text{Alg}} \to (T_pM)_{\text{Alg}}$, $\Phi(w_p)(f) = w_p(f|_U)$, is a linear isomorphism.

***Proof*:**
Each $\Phi(w_p)$ is a derivation on M because restriction $f \mapsto f|_U$ is an algebra homomorphism and evaluation at p factors through it, and Φ is linear. As neither space is yet known to be finite-dimensional, injectivity and surjectivity are shown directly.

Injectivity. Suppose $\Phi(w_p) = 0$ and let $h \in C^\infty(U)$. Choose a bump β with $\mathrm{supp}\beta \subseteq U$ and $\beta \equiv 1$ near p; then βh, extended by 0, is a global $f \in C^\infty(M)$ agreeing with h near p, so $f|_U$ agrees with h near p. Lemma 2.1.10 gives $w_p(h) = w_p(f|_U) = \Phi(w_p)(f) = 0$. As h was arbitrary, $w_p = 0$.

Surjectivity. Given $v_p \in (T_pM)_{\mathrm{Alg}}$, fix such a bump β and set $w_p(h) = v_p(\beta h)$, with βh extended by 0 to M. This is independent of β: two such bumps make βh and $\beta' h$ agree near p, so v_p gives the same value by lemma 2.1.10. It is a derivation, since $(\beta h_1)(\beta h_2) = \beta^2 h_1 h_2$ agrees with $\beta(h_1 h_2)$ near p and the Leibniz rule for v_p then transfers to w_p. Finally, $\beta \cdot f|_U$ agrees with f near p, so

$$\Phi(w_p)(f) = w_p(f|_U) = v_p\big(\beta \cdot f|_U\big) = v_p(f)$$

whence $\Phi(w_p) = v_p$, completing the proof.

The isomorphism justifies treating $\partial/\partial x^i|_p$ as living in both $(T_pU)_{\mathrm{Alg}}$ and $(T_pM)_{\mathrm{Alg}}$, and reduces the basis theorem to a local computation on $\mathbb{R}^n$. That computation rests on a first-order Taylor expansion with a smooth remainder.

Lemma 2.1.12: Hadamard's Lemma

Let $U \subseteq \mathbb{R}^n$ be a star-convex open neighbourhood of a and $f \in C^\infty(U)$. There are smooth $f_i \in C^\infty(U)$ with

$$f(x) = f(a) + \sum_{i=1}^n (x^i - a^i)\, f_i(x), \qquad f_i(a) = \frac{\partial f}{\partial x^i}(a)$$

***Proof*:**
With $h(t) = f(a + t(x-a))$ on $[0,1]$, the chain rule gives $h'(t) = \sum_i \frac{\partial f}{\partial x^i}(a + t(x-a))\,(x^i - a^i)$. Since $h(1) = f(x)$ and $h(0) = f(a)$,

$$f(x) - f(a) = \int_0^1 h'(t)\,dt = \sum_{i=1}^n (x^i - a^i) \int_0^1 \frac{\partial f}{\partial x^i}(a + t(x-a))\,dt$$

so $f_i(x) = \int_0^1 \frac{\partial f}{\partial x^i}(a + t(x-a))\,dt$ is smooth in x (differentiation under the integral) and gives the stated formula. Setting $x = a$, $f_i(a) = \int_0^1 \frac{\partial f}{\partial x^i}(a)\,dt = \frac{\partial f}{\partial x^i}(a)$, completing the proof.

Theorem 2.1.13: Coordinate Derivations Form a Basis

Let (U,φ) be a chart of an n-manifold with $p \in U$ and $\varphi = (x^1,\dots,x^n)$. Then $\big(\partial/\partial x^1|_p,\dots,\partial/\partial x^n|_p\big)$ is a basis of $(T_pM)_{\text{Alg}}$, and every v_p expands as

$$v_p = \sum_{i=1}^n v_p(x^i)\, \frac{\partial}{\partial x^i}\bigg|_p$$

In particular $\dim(T_pM)_{\text{Alg}} = n$.

***Proof*:**
By proposition 2.1.11 work on U, and shrinking U assume $\varphi(U)$ is star-convex; composing φ with a translation makes $\varphi(p) = 0$, so $x^i(p) = 0$. For $f \in C^\infty(U)$, apply lemma 2.1.12 to $f \circ \varphi^{-1}$ at $a = 0$ and pull back by φ: there are $g_i \in C^\infty(U)$ with

$$f = f(p) + \sum_{i=1}^n x^i\, g_i, \qquad g_i(p) = \frac{\partial f}{\partial x^i}(p)$$

Applying a derivation v_p, using $v_p(f(p)) = 0$ and $x^i(p) = 0$,

$$v_p(f) = \sum_{i=1}^n \big(v_p(x^i)\, g_i(p) + x^i(p)\, v_p(g_i)\big) = \sum_{i=1}^n v_p(x^i)\, \frac{\partial f}{\partial x^i}(p) = \sum_{i=1}^n v_p(x^i)\, \frac{\partial}{\partial x^i}\bigg|_p f$$

so the coordinate derivations span. For independence, suppose $\sum_i a_i\, \partial/\partial x^i|_p = 0$ and apply it to x^j: since $\partial x^j/\partial x^i = \delta_i^j$,

$$0 = \sum_{i=1}^n a_i\, \frac{\partial x^j}{\partial x^i}\bigg|_p = \sum_{i=1}^n a_i\, \delta_i^j = a_j$$

for every j, so the operators are independent and form a basis, completing the proof.

Coordinate charts thus give a working basis, with one notational hazard: for $\mathbb{R}^n$ under the identity chart the coordinate functions are $x^1,\dots,x^n$, yet a point is also written $(x_1,\dots,x_n)$, so the tangent vector $\sum_i x_i\, \partial/\partial x^i|_p$ mixes subscripts (components) with superscripts (coordinate functions). On $\mathbb{R}$ the single coordinate is usually named t, with basis vector $\partial/\partial t|_{t_0}$ at t_0.

*The Tangent Space via Germs

The derivation construction used only local behaviour, so it descends to germs. Let $\mathcal{F}_p = C_p^\infty(M)$ be the algebra of germs at p: equivalence classes of smooth functions defined near p, two agreeing when they coincide on some neighbourhood. The value $[f](p) = f(p)$ is well defined, giving an evaluation homomorphism $\mathrm{ev}_p\colon \mathcal{F}_p \to \mathbb{R}$.

Definition 2.1.14: Germ Derivation

A *derivation* of $\mathcal{F}_p$ (relative to ev_p) is a linear $\mathcal{D}_p \colon \mathcal{F}_p \to \mathbb{R}$ with

$$\mathcal{D}_p([f][g]) = f(p)\,\mathcal{D}_p[g] + g(p)\,\mathcal{D}_p[f], \qquad [f],[g] \in \mathcal{F}_p$$

The derivations of $\mathcal{F}_p$ form a real vector space $\mathrm{Der}(\mathcal{F}_p)$. By lemma 2.1.10 and proposition 2.1.11 a derivation of $C^\infty(M)$ is determined by, and extends to, germ data, so $\mathrm{Der}(\mathcal{F}_p)$ is another model of the tangent space. Its value is that it exposes the tangent space as a purely algebraic object attached to a local ring.

Lemma 2.1.15: The Local Ring of Germs

Let $\mathcal{F}_a = C^\infty_a(M)$ be the ring of germs at a in an n-manifold. Then

1. $\mathcal{F}_a$ is a local ring with maximal ideal $\mathfrak{m}_a = \{[f] \in \mathcal{F}_a \mid f(a) = 0\}$;
2. for $M = \mathbb{R}^n$, the ideal $\mathfrak{m}_a$ is generated by $x^1 - a^1, \dots, x^n - a^n$;
3. $\mathcal{F}_a/\mathfrak{m}_a \cong \mathbb{R}$, and $\mathfrak{m}_a/\mathfrak{m}_a^2$ is an n-dimensional real vector space.

***Proof*:**
(1) The set $\mathfrak{m}_a$ is the kernel of the homomorphism ev_a, hence an ideal. If $[f] \notin \mathfrak{m}_a$ then $f(a) \neq 0$, so f is nonzero on a neighbourhood of a and $1/f$ defines a germ inverse to $[f]$. Thus every germ outside $\mathfrak{m}_a$ is a unit, so $\mathfrak{m}_a$ contains every proper ideal and is the unique maximal ideal.

(2) For $[f] \in \mathfrak{m}_a$, Hadamard's lemma (lemma 2.1.12) on a star-convex neighbourhood of a gives smooth f_i with $f = f(a) + \sum_i (x^i - a^i) f_i = \sum_i (x^i - a^i) f_i$, since $f(a) = 0$. So the germs $x^i - a^i$ generate $\mathfrak{m}_a$.

(3) Evaluation ev_a is surjective onto $\mathbb{R}$ with kernel $\mathfrak{m}_a$, so $\mathcal{F}_a/\mathfrak{m}_a \cong \mathbb{R}$. For the second, map

$$\varphi \colon \mathfrak{m}_a \to \mathbb{R}^n, \qquad [f] \mapsto \left(\tfrac{\partial f}{\partial x^1}(a), \dots, \tfrac{\partial f}{\partial x^n}(a)\right)$$

which is well defined (representatives agree near a, hence in all derivatives), linear, and surjective (the coordinate germs realize any target). The product rule gives, for $[f],[g] \in \mathfrak{m}_a$,

$$\varphi([f][g])_i = \tfrac{\partial f}{\partial x^i}(a)\,g(a) + f(a)\,\tfrac{\partial g}{\partial x^i}(a) = 0$$

so $\mathfrak{m}_a^2 \subseteq \ker\varphi$. Conversely, if $\varphi([h]) = 0$ then in a chart centred at a the germ h has vanishing value and first derivatives, so its Taylor expansion begins at order two; writing each second-order term via Hadamard as a product of two elements of $\mathfrak{m}_a$ shows $[h] \in \mathfrak{m}_a^2$. Hence $\ker\varphi = \mathfrak{m}_a^2$ and $\mathfrak{m}_a/\mathfrak{m}_a^2 \cong \mathbb{R}^n$, completing the proof.

The quotient $\mathfrak{m}_a/\mathfrak{m}_a^2$ is the *cotangent space*; a derivation kills constants and, by the Leibniz rule, kills $\mathfrak{m}_a^2$, so it factors through $\mathfrak{m}_a/\mathfrak{m}_a^2$ and $\mathrm{Der}(\mathcal{F}_a) \cong (\mathfrak{m}_a/\mathfrak{m}_a^2)^*$.

Proposition 2.1.16: Algebraic Tangent Space

The *algebraic tangent space* of M at a is $\tilde{T}_a M = (\mathfrak{m}_a/\mathfrak{m}_a^2)^*$. A smooth $f\colon M \to N$ induces a linear $\partial_a f\colon \tilde{T}_a M \to \tilde{T}_{f(a)} N$ with $\partial_a(g \circ f) = \partial_{f(a)} g \circ \partial_a f$.

***Proof*:**
Pullback $[h] \mapsto [h \circ f]$ carries $\mathfrak{m}_{f(a)}$ into $\mathfrak{m}_a$ and $\mathfrak{m}_{f(a)}^2$ into $\mathfrak{m}_a^2$: if $[h] = [h'] + [g_1 g_2]$ with $[g_1], [g_2] \in \mathfrak{m}_{f(a)}$, then $(g_1 \circ f)(g_2 \circ f) \in \mathfrak{m}_a^2$, so the composite agrees modulo $\mathfrak{m}_a^2$. Dualizing the induced map $\mathfrak{m}_{f(a)}/\mathfrak{m}_{f(a)}^2 \to \mathfrak{m}_a/\mathfrak{m}_a^2$ gives $\partial_a f$, and functoriality of pullback under composition yields the chain rule, completing the proof.

Relating the Definitions

The three constructions – kinematic $(T_pM)_{\mathrm{kin}}$, physical $(T_pM)_{\mathrm{phys}}$, algebraic $(T_pM)_{\mathrm{Alg}}$ – are the same space seen three ways. Each comes equipped, for every chart (U, φ) at p, with a linear bijection $b_\varphi\colon \mathbb{R}^n \to (T_pM)_\bullet$, and in all three the change of chart is the transition derivative $D(\psi \circ \varphi^{-1})(\varphi(p))$ (lemma 2.1.5, definition 2.1.6, and theorem 2.1.13). Maps intertwining these parametrizations are therefore canonical isomorphisms.

Proposition 2.1.17: The Three Tangent Spaces Agree

There are canonical vector-space isomorphisms

$$(T_pM)_{\mathrm{kin}} \cong (T_pM)_{\mathrm{phys}} \cong (T_pM)_{\mathrm{Alg}}$$

independent of the chart used to define them.

***Proof*:**
Fix a chart (U, φ) at p with coordinates x^i. Define $\Lambda\colon (T_pM)_{\mathrm{kin}} \to (T_pM)_{\mathrm{Alg}}$ by $\Lambda[c](f) = (f \circ c)'(0)$; this is a derivation at p by the Leibniz rule for one-variable derivatives, and depends only on the tangency class of c (definition 2.1.1). If $[c] = b_\varphi(v)$ in the kinematic space, so $v = (\varphi \circ c)'(0)$, then the chain rule gives

$$\Lambda[c](f) = (f \circ c)'(0) = D(f \circ \varphi^{-1})(\varphi(p))\, v = \sum_{i=1}^{n} v^i \left.\frac{\partial}{\partial x^i}\right|_p f$$

so $\Lambda \circ b_\varphi = b_\varphi$ as maps $\mathbb{R}^n \to (T_pM)_{\mathrm{Alg}}$, the right-hand b_φ being the basis parametrization of theorem 2.1.13. Both parametrizations are linear bijections, so $\Lambda = b_\varphi \circ b_\varphi^{-1}$ is a linear isomorphism. It is chart-independent because in a second chart both parametrizations transform by the same transition derivative, which therefore cancels.

Identically, $[(p, v, (U, \varphi))] \mapsto \sum_i v^i\, \partial/\partial x^i|_p$ intertwines the chart parametrizations of $(T_pM)_{\mathrm{phys}}$ and $(T_pM)_{\mathrm{Alg}}$, giving a canonical isomorphism between them; composing yields the isomorphism with $(T_pM)_{\mathrm{kin}}$, completing the proof.

Henceforth T_pM denotes the tangent space in any of the three descriptions, whichever is convenient, and its elements are tangent vectors. A curve carries a distinguished one.

Definition 2.1.18: Velocity Vector of a Curve

A *curve* in M is a smooth map $\gamma\colon I \to M$ on an interval $I \subseteq \mathbb{R}$. Its *velocity* at t_0 is the tangent vector $\gamma'(t_0) \in T_{\gamma(t_0)}M$ acting by

$$\gamma'(t_0)\, f = (f \circ \gamma)'(t_0), \qquad f \in C^\infty(M)$$

The map γ, not its image $\gamma(I)$ alone, is part of the data; γ together with $\gamma(I)$ is a *parametrized curve*. Common notations for the velocity are $\dot\gamma(t_0)$, $\frac{d\gamma}{dt}(t_0)$, and $\frac{\partial\gamma}{\partial t}|_{t_0}$. Once the tangent map is available (section 2.2), the velocity is $\gamma'(t_0) = T_{t_0}\gamma\big(\partial/\partial t|_{t_0}\big)$, the image of the standard basis vector of $T_{t_0}\mathbb{R}$.

The Tangent Space of a Regular Submanifold

For a submanifold $M \subseteq \mathbb{R}^n$ the tangent space at a can be computed concretely, and each of the four characterizations of proposition 1.4.5 produces the same linear subspace of $T_a\mathbb{R}^n \cong \mathbb{R}^n$. Throughout, T_aM is realized inside $T_a\mathbb{R}^n$ as the velocities of curves in M through a.

Proposition 2.1.19: Tangent Space of a Regular Submanifold

Let $M \subseteq \mathbb{R}^n$ be a C^r submanifold of dimension k and $a \in M$. Under the identification $T_a\mathbb{R}^n \cong \mathbb{R}^n$, the tangent space T_aM equals each of the following, taken from the local descriptions of proposition 1.4.5:

1. $\ker Df(a)$, where $M = f^{-1}(0)$ near a for a submersion f into $\mathbb{R}^{n-k}$;
2. $\big\{(Y, Z) \in \mathbb{R}^k \times \mathbb{R}^{n-k} \;\big|\; Z = Dg(b)Y\big\}$, where M is the graph $z = g(y)$ near $a = (b, g(b))$;
3. $Dh(a)^{-1}\big(\mathbb{R}^k \times \{0\}\big)$, where h is a diffeomorphism flattening M onto a slice;
4. $D\varphi(b)(\mathbb{R}^k)$, where φ is an embedding with $\varphi(b) = a$ parametrizing M near a.

***Proof*:**
The argument computes T_aM from the embedding (4), then matches the other three by linear algebra.

$T_aM = D\varphi(b)(\mathbb{R}^k) = \ker Df(a)$. Three inclusions close the loop, with no need to invert φ. For $v \in \mathbb{R}^k$ the curve $t \mapsto \varphi(b + tv)$ lies in M, passes through a, and has velocity $D\varphi(b)v$, so $D\varphi(b)(\mathbb{R}^k) \subseteq T_aM$. Next, any smooth curve γ in M through a satisfies $f\circ\gamma \equiv 0$ (as $M = f^{-1}(0)$), so differentiating gives $Df(a)\,\gamma'(0) = 0$ and $T_aM \subseteq \ker Df(a)$. Finally $f \circ \varphi$ vanishes, so $Df(a)\, D\varphi(b) = 0$ and $D\varphi(b)(\mathbb{R}^k) \subseteq \ker Df(a)$; since $D\varphi(b)$ is injective and $Df(a)$ has rank $n - k$, both spaces have dimension k. The chain $D\varphi(b)(\mathbb{R}^k) \subseteq T_aM \subseteq \ker Df(a) = D\varphi(b)(\mathbb{R}^k)$ forces all three equal.

$= \{(Y, Z) \mid Z = Dg(b)Y\}$. The graph is the level set of $f(y, z) = z - g(y)$, a submersion with $Df(a) = (-Dg(b),\, I)$. Hence $\ker Df(a) = \{(Y, Z) \mid Z - Dg(b)Y = 0\}$, the stated subspace.

$= Dh(a)^{-1}(\mathbb{R}^k \times \{0\})$. The flattening h satisfies $M \cap U = h^{-1}(\mathbb{R}^k \times \{0\})$, so $f = \pi \circ h$ with $\pi\colon \mathbb{R}^n \to \mathbb{R}^{n-k}$ the last-coordinate projection is a defining submersion. Then $\ker Df(a) = \ker(\pi \circ Dh(a)) = Dh(a)^{-1}(\ker \pi) = Dh(a)^{-1}(\mathbb{R}^k \times \{0\})$, completing the proof.

For the sphere $S^{n-1} = f^{-1}(1)$ with $f(x) = |x|^2$ one has $Df(a) = 2a^\top$, so $T_a S^{n-1} = \ker Df(a) = a^\perp$, the hyperplane orthogonal to a – the familiar picture of the tangent plane.

The Tangent Space of a Vector Space

A finite-dimensional real vector space V is its own tangent space at every point. Concretely, define $\iota_p\colon V \to T_pV$ by sending v to the velocity at 0 of the line $t \mapsto p + tv$; in the three descriptions this is

$$\iota_p(v) = [\,t \mapsto p + tv\,] = \big[(p,\, E(v),\, (V, E))\big] = \Big(f \mapsto \tfrac{d}{dt}\big|_0 f(p + tv)\Big)$$

for any linear chart $E\colon V \to \mathbb{R}^n$. The map ι_p is a linear isomorphism, so $T_pV \cong V$ canonically, independently of p and of the chart. This identification is used silently whenever the tangent space of $\mathbb{R}^n$ is treated as $\mathbb{R}^n$ again.

Exercise 2.1.1

1. Let (U, φ) and (V, ψ) be charts at p with coordinates x^j and $\bar{x}^i$. For a tangent vector $v_p = \sum_j a^j\, \partial/\partial x^j|_p$, use theorem 2.1.13 to show its components in the barred chart are $\bar{a}^i = \sum_j \frac{\partial \bar{x}^i}{\partial x^j}(p)\, a^j$, the transformation law.
2. Show that $T_A\, \mathrm{GL}_n(\mathbb{R}) \cong M_n(\mathbb{R})$ for every A, and that the curve $t \mapsto A + tB$ realizes $B \in M_n(\mathbb{R})$ as a velocity at A.
3. Show that $v_p(f)$ depends only on the first-order Taylor data of f at p: if f and g have equal value and equal first partial derivatives at p in some chart, then $v_p(f) = v_p(g)$ for every $v_p \in T_pM$.

2.2 Tangent Map

Each description of the tangent space produces a description of the map a smooth f induces on tangent spaces, and the three agree. Some authors write f_{*p}, but that notation is reserved here for the *pushforward* of vector fields, so the tangent map is written T_pf.

The geometric version pushes velocities forward along f.

Definition 2.2.1: Tangent Map (via Curves)

For a smooth $f\colon M \to N$ and $p \in M$ with $q = f(p)$, the *tangent map* $T_pf\colon T_pM \to T_qN$ sends $v_p = [c]$ to

$$T_pf(v_p) = [f \circ c]$$

the tangency class of the image curve $f \circ c$.

This is well defined: if $c \sim c'$ then for every $g \in C^\infty(N)$ one has $g \circ f \in C^\infty(M)$, so $(g \circ f \circ c)'(0) = (g \circ f \circ c')'(0)$ and $f \circ c \sim f \circ c'$. Geometrically $T_p f$ carries the velocity $c'(0)$ to the velocity $(f \circ c)'(0)$, the infinitesimal image of the curve.

In coordinates this is the Jacobian acting on components.

Definition 2.2.2: Tangent Map (via Charts)

For a smooth $f\colon M \to N$ and $p \in M$ with $q = f(p)$, choose charts (U, φ) at p and (V, ψ) at q. If v_p is represented by $(p, v, (U, \varphi))$, then $T_p f(v_p)$ is represented in (V, ψ) by $(q, w, (V, \psi))$ with

$$w = D(\psi \circ f \circ \varphi^{-1})(\varphi(p))\, v$$

The chain rule makes this independent of the charts. This description is best when the coordinate representation of f is explicit; for instance the inclusion $\iota\colon M \to M \times N$ at a fixed second point has $T_p\iota(v) = (v, 0)$, read off from the product charts.

Example 2.2.3: Nondegenerate Fixed Points

Let $f\colon M \to M$ fix p, so $f(p) = p$ and $T_p f\colon T_pM \to T_pM$. Call p *nondegenerate* if $T_p f - \mathrm{id}$ is invertible, i.e. 1 is not an eigenvalue of $T_p f$. In a chart centred at p the coordinate map $\hat{f} = \varphi \circ f \circ \varphi^{-1}$ fixes 0 with $D\hat{f}(0)$ representing $T_p f$ (definition 2.2.2), so $D\hat{f}(0) - I$ is invertible. Then $x \mapsto \hat{f}(x) - x$ has invertible derivative at 0, hence is a local diffeomorphism (theorem 0.2.1), so its only zero near 0 is 0 itself: p is an *isolated* fixed point. These nondegenerate fixed points are the ones counted by the Lefschetz and Poincaré-Hopf theorems (section 8.6).

The algebraic version is coordinate-free and the most convenient for computation: $T_p f$ precomposes a derivation with f.

Definition 2.2.4: Tangent Map (via Derivations)

For a smooth $f\colon M \to N$ and $p \in M$ with $q = f(p)$, the *tangent map* $T_p f\colon T_pM \to T_qN$ sends v_p to the derivation acting by

$$(T_p f\, v_p)(g) = v_p(g \circ f), \qquad g \in C^\infty(N)$$

Since $g \circ f \in C^\infty(M)$, the value $v_p(g \circ f)$ makes sense, and $T_p f\, v_p$ is a derivation at q: it is linear in g, and the Leibniz rule for v_p gives $(T_p f\, v_p)(gh) = (T_p f\, v_p)(g)\, h(q) + g(q)\, (T_p f\, v_p)(h)$, so $T_p f\, v_p \in T_qN$. One notational trap: although written with subscript p, the vector $T_p f\, v_p$ lives in T_qN. When a construction needs the tangent map distinguished in notation from the tangent space, it is written $D_p f$ or df_p.

Proposition 2.2.5: Properties of the Tangent Map

Let $f\colon M \to N$ and $g\colon N \to P$ be smooth. Then

1. $T_p f\colon T_pM \to T_{f(p)}N$ is linear;
2. $T_p(g \circ f) = T_{f(p)}g \circ T_p f$;
3. $T_p\mathrm{id} = \mathrm{id}_{T_pM}$;
4. if f is a diffeomorphism, $T_p f$ is a linear isomorphism with $(T_p f)^{-1} = T_{f(p)}(f^{-1})$.

***Proof*:**
(1) For $g \in C^\infty(N)$ and scalars a, b, $T_p f(av_p + bw_p)(g) = (av_p + bw_p)(g \circ f) = a\, T_p f(v_p)(g) + b\, T_p f(w_p)(g)$, so $T_p f$ is linear.

(2) For $h \in C^\infty(P)$, both sides act by $v_p \mapsto v_p(h \circ g \circ f)$: on the left, $T_p(g \circ f)(v_p)(h) = v_p(h \circ (g \circ f))$; on the right, $\big(T_{f(p)}g \circ T_p f\big)(v_p)(h) = T_p f(v_p)(h \circ g) = v_p((h \circ g) \circ f)$. These agree.

(3) $T_p\mathrm{id}(v_p)(h) = v_p(h \circ \mathrm{id}) = v_p(h)$.

(4) By (1) $T_p f$ is linear, and f^{-1} is smooth. Applying (2) and (3) to $f^{-1} \circ f = \mathrm{id}_M$ and $f \circ f^{-1} = \mathrm{id}_N$ gives $T_{f(p)}f^{-1} \circ T_p f = \mathrm{id}_{T_pM}$ and $T_p f \circ T_{f(p)}f^{-1} = \mathrm{id}_{T_{f(p)}N}$, so $T_p f$ is invertible with the stated inverse, completing the proof.

The converse of (4) fails: $f\colon \mathbb{R} \to S^1$, $x \mapsto e^{2\pi i x}$, is not a diffeomorphism (it is not injective) yet $T_p f$ is an isomorphism at every point. It is a *local* diffeomorphism, a notion central to chapter 3. The following partial converse, stated now to show how $T_p f$ captures f locally, is the inverse function theorem for manifolds.

Theorem 2.2.6: Inverse Mapping Theorem for Manifolds

Let $f\colon M \to N$ be smooth with $T_p f$ an isomorphism. Then f restricts to a diffeomorphism of an open neighbourhood of p onto an open neighbourhood of $f(p)$. If $T_p f$ is an isomorphism for every p, then f is a local diffeomorphism.

***Proof*:**
In charts $T_p f$ is represented by $D(\psi \circ f \circ \varphi^{-1})(\varphi(p))$ (definition 2.2.2), which is therefore invertible; the Euclidean inverse function theorem (theorem 0.2.1) then makes $\psi \circ f \circ \varphi^{-1}$ a local diffeomorphism, and composing with the charts gives the claim. The full argument is theorem 3.1.10 in chapter 3, completing the proof.

Properties (1)–(3) say T is a functor.

Definition 2.2.7: Pointed Tangent Functor

On the category Man_* of pointed manifolds, the *pointed tangent functor* T sends (M, p) to T_pM and a smooth $f\colon (M,p) \to (N,q)$ to the linear map $T_pf\colon T_pM \to T_qN$. Its basepoint-free counterpart on whole manifolds is the tangent functor of corollary 2.4.7.

Property (4) gives the dimension of a tangent space at once.

Corollary 2.2.8: Dimension of the Tangent Space

For an m-manifold M, $\dim T_pM = m$ at every p.

***Proof*:**
A chart (U, φ) at p is a diffeomorphism onto $\varphi(U) \subseteq \mathbb{R}^m$, so $T_p\varphi\colon T_pU \to T_{\varphi(p)}\varphi(U)$ is a linear isomorphism by proposition 2.2.5. With $T_pU \cong T_pM$ and $T_{\varphi(p)}\varphi(U) \cong T_{\varphi(p)}\mathbb{R}^m \cong \mathbb{R}^m$ (proposition 2.1.11), transitivity gives $T_pM \cong \mathbb{R}^m$, completing the proof.

Every tangent space of M is thus isomorphic to $\mathbb{R}^m$, hence to every other; the vector-space structure is the same at each point. The isomorphisms are not canonical, though – the tangent spaces sit differently in space, as the tangent lines around a circle already show. Section 2.4 assembles them into a single object that records this placement, and chapter 11 settles when they can be globally identified by translation, the question of parallelizability.

The open-submanifold identification $T_pU \cong T_pM$ (proposition 2.1.11) will be used silently from now on. In particular $T_p\mathbb{R}^n \cong \mathbb{R}^n$ and, for any finite-dimensional V, $T_pV \cong V$, compatibly with linear maps.

Proposition 2.2.9: Tangent Space of a Vector Space

Let V be a finite-dimensional vector space with its standard smooth structure. The canonical isomorphism $\iota_p\colon V \to T_pV$ sending v to the velocity of $t \mapsto p + tv$ is natural in V: for a linear $L\colon V \to W$ the square

$$\begin{tikzcd} T_pV \arrow[r, "T_pL"] & T_{Lp}W \\ V \arrow[u, "\iota_p"] \arrow[r, "L"] & W \arrow[u, "\iota_{Lp}"'] \end{tikzcd}$$

commutes.

***Proof*:**
For $v \in V$, $\iota_p(v)$ is the velocity of $t \mapsto p + tv$, so $T_pL(\iota_p v)$ is the velocity of $t \mapsto L(p + tv) = Lp + t\,Lv$, which is exactly $\iota_{Lp}(Lv)$. Thus $T_pL \circ \iota_p = \iota_{Lp} \circ L$, completing the proof.

The tangent map between manifolds with boundary needs the tangent spaces at boundary points handled with care, through the constant rank theorem; that discussion is deferred to chapter 3, where the theorem is proved.

2.2.1 Computing the Tangent Map

Since $T_p f\colon T_pM \to T_{f(p)}N$ is linear, it has a matrix once bases are chosen, and the coordinate basis makes that matrix the Jacobian. Fix a chart (U,φ) at p. By the proof of corollary 2.2.8, $T_p\varphi\colon T_pM \to T_{\varphi(p)}\mathbb{R}^n$ is an isomorphism, and $T_{\varphi(p)}\mathbb{R}^n$ has the coordinate basis $\partial/\partial x^i|_{\varphi(p)}$. Its preimage is a basis of T_pM,

$$\left.\frac{\partial}{\partial x^i}\right|_p = T_{\varphi(p)}(\varphi^{-1})\left(\left.\frac{\partial}{\partial x^i}\right|_{\varphi(p)}\right)$$

written with the same symbol to stress the identification with the coordinate representation. By definition 2.2.4, for $f \in C^\infty(M)$ with $\hat{f} = f \circ \varphi^{-1}$ and $\hat{p} = \varphi(p)$,

$$\left.\frac{\partial}{\partial x^i}\right|_p (f) = \left.\frac{\partial}{\partial x^i}\right|_{\hat{p}} (\hat{f}) = \frac{\partial \hat{f}}{\partial x^i}(\hat{p})$$

so $\partial/\partial x^i|_p$ differentiates the coordinate representation of f. These are the *coordinate vectors* at p; on $\mathbb{R}^n$ they are the partial derivative operators.

To confirm T_pf generalizes the derivative, take $f\colon U \subseteq \mathbb{R}^n \to V \subseteq \mathbb{R}^m$ with standard coordinates x^i on $\mathbb{R}^n$ and y^j on $\mathbb{R}^m$. For $h \in C^\infty(V)$, the chain rule gives

$$T_pf\left(\left.\frac{\partial}{\partial x^i}\right|_p\right)(h) = \left.\frac{\partial}{\partial x^i}\right|_p (h \circ f) = \sum_{j=1}^m \frac{\partial h}{\partial y^j}(f(p))\,\frac{\partial f^j}{\partial x^i}(p) = \left(\sum_{j=1}^m \frac{\partial f^j}{\partial x^i}(p)\, \left.\frac{\partial}{\partial y^j}\right|_{f(p)}\right)(h)$$

so, as a map of coordinate bases,

$$T_pf\left(\left.\frac{\partial}{\partial x^i}\right|_p\right) = \sum_{j=1}^m \frac{\partial f^j}{\partial x^i}(p)\, \left.\frac{\partial}{\partial y^j}\right|_{f(p)} \tag{2.1}$$

whose matrix is the Jacobian

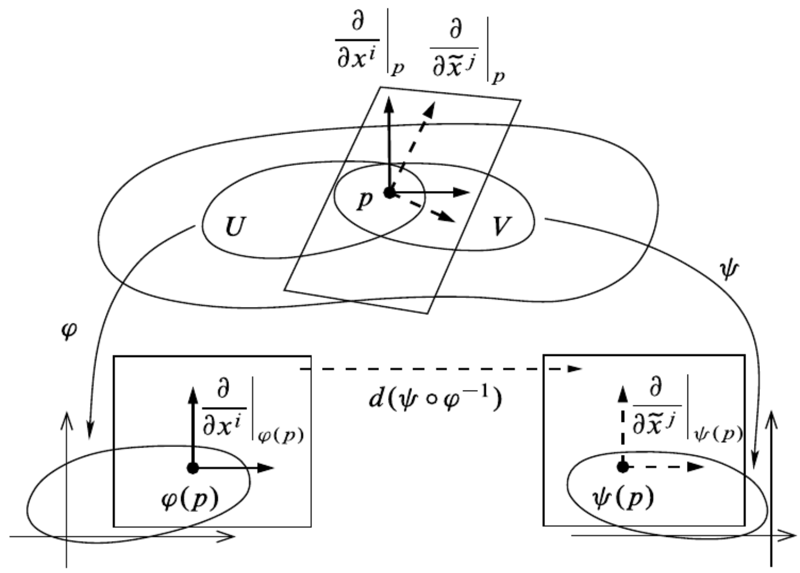
$$\begin{pmatrix} \frac{\partial f^1}{\partial x^1}(p) & \cdots & \frac{\partial f^1}{\partial x^n}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial x^1}(p) & \cdots & \frac{\partial f^m}{\partial x^n}(p) \end{pmatrix}$$

Thus for $T_pf\colon T_p\mathbb{R}^n \to T_{f(p)}\mathbb{R}^m$ one recovers the total derivative $Df(p)$. For general M and N, choosing charts at p and $f(p)$ reduces T_pf to this Jacobian of the coordinate representation, so the tangent map genuinely extends the derivative – built without reference to a basis, but recovering the classical one whenever a basis is chosen.

2.2.2 Change of Coordinates

It remains to record how the coordinate basis transforms between two charts (U,φ) and (V,ψ) with $U \cap V \neq \emptyset$, coordinates x^i and y^j. At $p \in U \cap V$ a tangent vector expands in either $\partial/\partial x^i|_p$ or $\partial/\partial y^j|_p$. The transition map is $\psi \circ \varphi^{-1}$, whose j-th component at $\varphi(p)$ is y^j read in the x-coordinates. By eq. (2.1),

$$T_{\varphi(p)}(\psi \circ \varphi^{-1})\left(\left.\frac{\partial}{\partial x^i}\right|_{\varphi(p)}\right) = \sum_{j=1}^n \frac{\partial y^j}{\partial x^i}(\varphi(p))\, \left.\frac{\partial}{\partial y^j}\right|_{\psi(p)}$$



Transporting through $T(\psi^{-1})$, the coordinate basis transforms as

$$\frac{\partial}{\partial x^i}\bigg|_p = \sum_{j=1}^{n} \frac{\partial y^j}{\partial x^i}(\hat{p})\, \frac{\partial}{\partial y^j}\bigg|_p, \qquad \hat{p} = \varphi(p)$$

and applying this to $v = \sum_i v^i\, \partial/\partial x^i|_p$ gives its components in the y-chart,

$$v = \sum_j \tilde{v}^j\, \frac{\partial}{\partial y^j}\bigg|_p, \qquad \tilde{v}^j = \sum_{i=1}^{n} \frac{\partial y^j}{\partial x^i}(\hat{p})\, v^i$$

the transformation law again, now derived from the tangent map.

Exercise 2.2.1

1. Using definition 2.2.2, verify that the inclusion $\iota\colon M \to M \times N$, $x \mapsto (x, q_0)$ at a fixed q_0, has $T_p\iota(v) = (v, 0)$ under the product-chart identification $T_{(p,q_0)}(M \times N) \cong T_pM \times T_{q_0}N$.
2. For the projection $\pi\colon \mathbb{R}^2 \to \mathbb{R}$, $\pi(x, y) = x$, and a curve $c(t) = (a(t), b(t))$, compute $T_{c(0)}\pi(c'(0))$ directly from definition 2.2.1 and check it against the Jacobian (eq. (2.1)).
3. Let $f\colon M \to N$ be a diffeomorphism and (U, φ) a chart at p. Show that $\big(T_pf(\partial/\partial x^i|_p)\big)_i$ is a basis of $T_{f(p)}N$, and identify the chart on N for which it is the coordinate basis.

2.3 Tangents of Products

Having found the tangent space of a regular submanifold and of an open submanifold, one case remains: a product. Its tangent space is the product of the factors' tangent spaces, matched by the projections.

Proposition 2.3.1: Tangent Space of a Product

Let $M_1, \dots, M_k$ be smooth manifolds with projections $\pi_i\colon M_1 \times \dots \times M_k \to M_i$. For $p = (p_1, \dots, p_k)$ the map

$$\alpha\colon T_p(M_1 \times \dots \times M_k) \to T_{p_1}M_1 \times \dots \times T_{p_k}M_k, \qquad \alpha(v) = (T_p\pi_1(v), \dots, T_p\pi_k(v))$$

is a linear isomorphism. The same holds if one factor is a manifold with boundary.

***Proof*:**
Each $T_p\pi_i$ is linear (proposition 2.2.5), so α is linear. Both sides have dimension $\sum_i \dim M_i$: the product manifold has that dimension, and so does the product of the tangent spaces. It therefore suffices that α is injective. A product chart $\varphi_1 \times \dots \times \varphi_k$ gives $T_p(M_1 \times \dots \times M_k)$ a coordinate basis split into k blocks, the i-th being the coordinates from M_i; the coordinate representation of π_i is the projection onto that block, so by eq. (2.1) $T_p\pi_i$ returns the i-th block of a vector. If $\alpha(v) = 0$ then every block of v vanishes, so $v = 0$, completing the proof.

The identification is natural in the factors.

Corollary 2.3.2: Tangent Map of a Product Map

For smooth $f_i\colon M_i \to N_i$, the product $f_1 \times \dots \times f_k$ is smooth, and under α

$$T_p(f_1 \times \dots \times f_k) = T_{p_1}f_1 \times \dots \times T_{p_k}f_k$$

***Proof*:**
Smoothness is checked in product charts, where the coordinate representation of $f_1 \times \dots \times f_k$ is the product of those of the f_i. For the tangent map, $\pi_i^N \circ (f_1 \times \dots \times f_k) = f_i \circ \pi_i^M$; applying T and proposition 2.2.5 gives $T\pi_i^N \circ T(f_1 \times \dots \times f_k) = Tf_i \circ T\pi_i^M$, which is the i-th component of the block-diagonal form under α, completing the proof.

2.4 Tangent Bundle

The tangent spaces have been treated one at a time. Yet the change-of-coordinate computation showed that the coordinate bases of nearby tangent spaces are related smoothly, so the collection of all tangent spaces should itself carry a smooth structure. Assembling them gives the tangent bundle, the object over which the tangent map varies smoothly and the setting for higher derivatives.

Definition 2.4.1: Tangent Bundle

The *tangent bundle* of a manifold M is the disjoint union

$$TM = \coprod_{p \in M} T_pM$$

with the *projection* $\pi\colon TM \to M$ sending $v_p \in T_pM$ to p.

The projection is part of the data: it is the glue that will make TM more than a disjoint set of spaces.

Example 2.4.2: The Tangent Bundle of $\mathbb{R}^n$

For $M = \mathbb{R}^n$ each $T_p\mathbb{R}^n$ is identified with $\{p\} \times \mathbb{R}^n$, so

$$T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$$

an element being a pair (p, v). This product form is special: TS^1 admits a product description but no canonical one, and TS^2 is not a product at all (chapter 11).

A smooth map lifts to the bundles fibrewise.

Definition 2.4.3: Tangent Lift

For smooth $f\colon M \to N$, the *tangent lift* $Tf\colon TM \to TN$ is the map restricting to $T_pf\colon T_pM \to T_{f(p)}N$ on each fibre.

As a bare set TM is only a disjoint union, and Tf only a set map that is linear on each fibre. The projection π is what promotes it: much as an initial topology is pulled back along a map, the smooth structure on TM is the one making the chart data of M lift. It is *not* the disjoint-union topology, which would record no closeness between different tangent spaces.

Example 2.4.4: Tangent Lift on $\mathbb{R}^n$

For open $U_1 \subseteq \mathbb{R}^n$, $U_2 \subseteq \mathbb{R}^m$ and smooth $f\colon U_1 \to U_2$, the identifications $TU_1 \cong U_1 \times \mathbb{R}^n$ and $TU_2 \cong U_2 \times \mathbb{R}^m$ (example 2.4.2) turn Tf into

$$(p, v) \mapsto (f(p), Df(p)\,v)$$

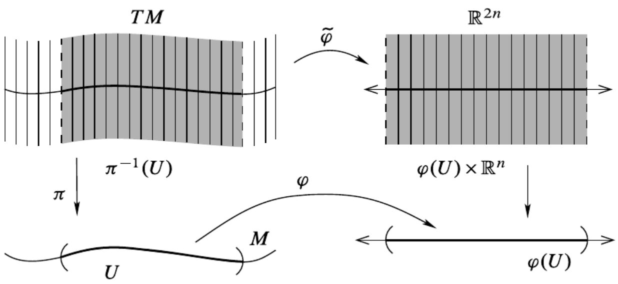
Not every TM is a global product $M \times \mathbb{R}^n$, but over a small chart domain it is, so Tf looks like $(p, v) \mapsto (f(p), Df(p)v)$ in local coordinates. The precise statement is that TM is itself a manifold.

Proposition 2.4.5: The Tangent Bundle is a Manifold

For an m-manifold M, the tangent bundle TM carries a natural topology and smooth structure making it a $2m$-manifold, and the projection $\pi\colon TM \to M$ is smooth.

***Proof*:**
A chart (U, φ) of M with $\varphi = (x^1, \ldots, x^m)$ gives a map $\tilde{\varphi} \colon \pi^{-1}(U) \to \mathbb{R}^{2m}$,

$$\tilde{\varphi}\left(\sum_i v^i \left.\frac{\partial}{\partial x^i}\right|_p\right) = (x^1(p), \ldots, x^m(p), v^1, \ldots, v^m)$$



Its image is $\varphi(U)\times\mathbb{R}^m$, open in $\mathbb{R}^{2m}$, and it is a bijection with inverse $(x, v) \mapsto \sum_i v^i\, \partial/\partial x^i|_{\varphi^{-1}(x)}$. For a second chart (V, ψ), $\psi = (y^1, \ldots, y^m)$, the change-of-coordinates rule $\partial/\partial x^i|_p = \sum_j \frac{\partial y^j}{\partial x^i}(\varphi(p))\, \partial/\partial y^j|_p$ gives the transition

$$\tilde{\psi} \circ \tilde{\varphi}^{-1}(x, v) = \left(\psi \circ \varphi^{-1}(x),\ \sum_{i=1}^m \frac{\partial y^1}{\partial x^i}(x)\, v^i, \ldots, \sum_{i=1}^m \frac{\partial y^m}{\partial x^i}(x)\, v^i\right)$$

on $\varphi(U \cap V) \times \mathbb{R}^m$. This is smooth, its components being the transition map $\psi \circ \varphi^{-1}$ and its partial derivatives, so the charts $\tilde{\varphi}$ form a smooth atlas.

The induced topology – $W \subseteq TM$ open iff $\tilde{\varphi}(W \cap \pi^{-1}(U))$ is open for every chart – is second countable, since a countable atlas $\{U_i\}$ of M gives the countable cover $\{\pi^{-1}(U_i)\}$ of TM. It is Hausdorff: two vectors over distinct $p \neq q$ are separated by $\pi^{-1}(U), \pi^{-1}(V)$ for disjoint $U \ni p$, $V \ni q$; two distinct vectors over the same p lie in the one chart $\pi^{-1}(U)$ and are separated there. Finally π reads $(x, v) \mapsto x$ in these charts, hence is smooth, completing the proof.

Corollary 2.4.6: Single-Chart Bundles are Trivial

If a smooth m-manifold M is covered by a single chart, then $TM \cong M \times \mathbb{R}^m$.

***Proof*:**
For the global chart (M, φ), the map $\tilde{\varphi}$ of proposition 2.4.5 is a diffeomorphism $TM \to \varphi(M) \times \mathbb{R}^m \cong M \times \mathbb{R}^m$, completing the proof.

The converse fails: $TS^1 \cong S^1 \times \mathbb{R}$ although S^1 needs more than one chart (exercise set 2.4.1). As with T_pf detecting only local diffeomorphism, triviality of TM carries only local information – being locally a product is automatic, being globally one is not.

Corollary 2.4.7: The Tangent Functor

For smooth $f\colon M \to N$, the tangent lift $Tf\colon TM \to TN$ is smooth, with coordinate representation

$$Tf(x,v) = \left(\hat{f}(x),\ \sum_i \frac{\partial \hat{f}^1}{\partial x^i}(x)\, v^i, \dots, \sum_i \frac{\partial \hat{f}^n}{\partial x^i}(x)\, v^i\right), \qquad \hat{f} = \psi \circ f \circ \varphi^{-1}$$

Hence $T\colon \mathrm{Man} \to \mathrm{Man}$, $M \mapsto TM$, $f \mapsto Tf$, is a functor.

***Proof*:**
In the bundle charts of proposition 2.4.5, Tf is the stated map, smooth because $\hat{f}$ and its partials are. Functoriality $T(g \circ f) = Tg \circ Tf$ and $T\mathrm{id} = \mathrm{id}$ hold fibrewise by proposition 2.2.5, completing the proof.

Proposition 2.4.8: Properties of the Tangent Lift

Let $f\colon M \to N$ and $g\colon N \to P$ be smooth. Then

1. $T(g \circ f) = Tg \circ Tf$;
2. $T\mathrm{id} = \mathrm{id}_{TM}$;
3. if f is a diffeomorphism, Tf is a diffeomorphism, linear on each fibre, with $(Tf)^{-1} = T(f^{-1})$.

***Proof*:**
Statements (1) and (2) hold on each fibre by proposition 2.2.5 and, both sides being smooth (corollary 2.4.7), hold as maps of bundles. For (3), f^{-1} is smooth, so Tf and $T(f^{-1})$ are smooth and, by (1)–(2), mutually inverse; thus Tf is a diffeomorphism, linear on each fibre by proposition 2.2.5, completing the proof.

Property (3) makes Tf a *bundle isomorphism*. In general Tf carries the fibre $\pi_M^{-1}(p)$ into $\pi_N^{-1}(f(p))$, so the square

$$\begin{tikzcd} TM \arrow[r, "Tf"] \arrow[d, "\pi_M"'] & TN \arrow[d, "\pi_N"] \\ M \arrow[r, "f"] & N \end{tikzcd}$$

commutes; a fibre-respecting map of this kind is a *bundle homomorphism*, the notion generalized to arbitrary fibre bundles in chapter 11.

The bundle can also be built by gluing, as manifolds themselves were. For a countable, locally finite atlas $\{(U_i, \varphi_i)\}$ of an m-manifold, set $\coprod_i \varphi_i(U_i) \times \mathbb{R}^m$ (Euclidean opens) and identify

$$(x,v) \sim (y,w) \iff y = \varphi_j \varphi_i^{-1}(x), \quad w = D(\varphi_j \varphi_i^{-1})(x)\, v$$

All atlases give canonically isomorphic results, recovering TM.

Exercise 2.4.1

1. Show that $TS^1 \cong S^1 \times \mathbb{R}$, and conclude that the converse of corollary 2.4.6 fails, since S^1 is not covered by a single chart.

3

Smooth Maps and Submanifolds

The tangent map Tf linearizes a smooth f at every point, and its one diffeomorphism-invariant – the rank – controls the local shape of f. This chapter extracts that local structure. The constant rank theorem puts a map of locally constant rank into a linear normal form, and the maximal-rank cases (immersions, submersions, embeddings) are read off from it. That machinery then settles the questions left open about submanifolds in section 1.4: which subsets $S \subseteq M$ are submanifolds, when the image of a smooth map is one, what the tangent space of a submanifold is, and how level sets cut them out. The measure-zero refinement of the rank – Sard's theorem – and its consequences (embeddings into $\mathbb{R}^N$, smooth approximation, transversality, intersection theory) open Part II.

3.1 Rank

The rank is the one feature of a linear map invariant under change of basis, so it passes to $T_p f$ without reference to coordinates.

Definition 3.1.1: Rank

The *rank* of a smooth $f \colon M \to N$ at p is the rank of the linear map $T_p f$. If it is the same at every point, f has *constant rank*. If it is the largest possible, $\operatorname{rank} T_p f = \min(\dim M, \dim N)$, then f has *full rank at p*; if f has full rank at every point, f has *full rank*.

Rank is a diffeomorphism invariant, so it can be computed from any coordinate representative. The two full-rank cases behave very differently and are named separately.

Definition 3.1.2: Immersion and Submersion

Let $f\colon M \to N$ be smooth.

1. f is an *immersion at p* if T_pf is injective; an *immersion* if it is one at every point (so $\operatorname{rank} f \equiv \dim M$).

2. f is a *submersion at p* if T_pf is surjective; a *submersion* if it is one at every point (so $\operatorname{rank} f \equiv \dim N$).

An immersion may still self-intersect, retrace itself, or accumulate onto itself, since injectivity of T_pf is purely local; controlling the global image needs the extra hypothesis that f is a homeomorphism onto its image, taken up in section 3.1.2. The full-rank locus is open, a fact packaged in the following matrix lemma.

Lemma 3.1.3: Full-Rank Matrices

Let $M(m,n)$ be the space of $m \times n$ real matrices and $M(m,n;k)$ the set of those of rank k. Then $M(m,n;k)$ is a submanifold of $M(m,n)$ of dimension $k(m+n-k)$. In particular, for $k = \min(m,n)$ it is an open submanifold.

***Proof*:**
Fix $X_0 \in M(m,n;k)$. Since X_0 has rank k, some $k \times k$ minor is nonsingular; permutation matrices P, Q move it to the top left, so

$$PX_0Q = \begin{pmatrix} A_0 & B_0 \\ C_0 & D_0 \end{pmatrix}, \qquad A_0 \in M(k,k) \text{ nonsingular}$$

As det is continuous and $\det A_0 \neq 0$, the top-left block A stays nonsingular on a neighbourhood Ω of X_0 (after the fixed permutation); on Ω write $Y = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$.

Rank criterion. For such Y with A nonsingular, the row operation

$$\begin{pmatrix} I_k & 0 \\ -CA^{-1} & I_{m-k} \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}$$

preserves rank, and the right-hand side has rank $k + \operatorname{rank}(D - CA^{-1}B)$ since A is nonsingular. Hence $\operatorname{rank} Y = k \iff D = CA^{-1}B$.

Submanifold. On Ω define the smooth map (its entries are rational with nonvanishing denominator $\det A$)

$$g\colon \Omega \to M(m-k, n-k) \cong \mathbb{R}^{(m-k)(n-k)}, \qquad g(Y) = D - CA^{-1}B$$

By the criterion, $M(m,n;k) \cap \Omega = g^{-1}(0)$. Moreover g is a submersion: holding A, B, C fixed, $D \mapsto D - CA^{-1}B$ is an affine isomorphism onto $M(m-k, n-k)$, so Dg is surjective everywhere. Thus 0 is a regular value, and by the regular-level-set characterization (proposition 1.4.5) $g^{-1}(0)$ is a submanifold of codimension $(m-k)(n-k)$, i.e. of dimension $mn - (m-k)(n-k) = k(m+n-k)$. For $k = \min(m,n)$ one factor of $(m-k)(n-k)$ vanishes, so g has target $\mathbb{R}^0$ and $M(m,n;k) \cap \Omega = \Omega$ is open, completing the proof.

The lemma turns pointwise full rank into an open condition, which is what makes the immersion and submersion properties locally stable.

Proposition 3.1.4: Immersion or Submersion is Locally Stable

Let $f\colon M \to N$ be smooth and $p \in M$. If T_pf is injective (resp. surjective), then f is an immersion (resp. submersion) on a neighbourhood of p.

***Proof*:**
Injectivity or surjectivity of T_pf means f has full rank at p. In coordinate charts the Jacobian of f is a matrix-valued smooth function, and by lemma 3.1.3 the full-rank matrices form an open set; by continuity the Jacobian remains of full rank on a neighbourhood of p, so f is an immersion (resp. submersion) there, completing the proof.

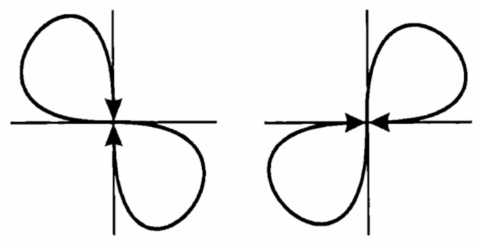
Example 3.1.5: Immersions and Submersions

1. The standard immersion of the torus $T^2 = S^1 \times S^1$ into $\mathbb{R}^3$,

$$(e^{i\theta_1}, e^{i\theta_2}) \mapsto \big((a + b\cos\theta_1)\cos\theta_2,\ (a + b\cos\theta_1)\sin\theta_2,\ b\sin\theta_1\big)$$

is an injective immersion, and hence – as T^2 is compact – an embedding, exactly when $a > b > 0$, when the tube of radius b clears the axis; for $a \le b$ the tube reaches the axis and the parametrization fails to be an immersion where $a + b\cos\theta_1 = 0$.

2. The figure-eight $\gamma\colon (-\pi, \pi) \to \mathbb{R}^2$, $t \mapsto (\sin 2t, \sin t)$, is an injective immersion whose image is a figure-eight. It is *not* an embedding: the image with the subspace topology is not homeomorphic to an interval, as the crossing point has no interval neighbourhood.



3. The projection $\pi\colon \mathbb{R}^n \to \mathbb{R}^k$ onto the first k coordinates is a submersion, as is any surjective linear map; the quotient $\mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{RP}^n$ is a submersion onto a manifold.

The case where T_pf is a linear isomorphism deserves its own name; the next section proves that it is exactly the infinitesimal form of a local diffeomorphism.

Definition 3.1.6: Local Diffeomorphism

A smooth map $f\colon M \to N$ is a *local diffeomorphism* if every $p \in M$ has an open neighbourhood U with $f(U)$ open and $f|_U\colon U \to f(U)$ a diffeomorphism (and similarly with C^r in place of smooth).

Example 3.1.7: Local Diffeomorphism

The quotient $\pi\colon S^2 \to \mathbb{R}\mathrm{P}^2$ sending a point to the line through it and the origin is a local diffeomorphism but not a global one: it is two-to-one, while a diffeomorphism is injective.

Proposition 3.1.8: Properties of Local Diffeomorphisms

Let $f\colon M \to N$ and $g\colon N \to P$ be smooth.

1. A composition of local diffeomorphisms is a local diffeomorphism.
2. A finite product of local diffeomorphisms is a local diffeomorphism.
3. Every local diffeomorphism is an open map and a local homeomorphism.
4. The restriction of a local diffeomorphism to an open submanifold is a local diffeomorphism.
5. Every diffeomorphism is a local diffeomorphism, and a bijective local diffeomorphism is a diffeomorphism.
6. f is a local diffeomorphism if and only if each point has a coordinate representative that is a diffeomorphism onto an open set.

***Proof*:**
(1)–(2): near a point, each factor restricts to a diffeomorphism of open sets, and compositions and finite products of diffeomorphisms of open sets are again such. (3): f is locally a diffeomorphism onto an open set, hence locally open, hence open, and a local homeomorphism. (4): an open submanifold meets each neighbourhood in an open set, on which f still restricts to a diffeomorphism. (6): restated locally, "$f|_U$ a diffeomorphism onto an open set" is exactly the existence of a diffeomorphism coordinate representative.

(5) A diffeomorphism restricts to one on any open set, so it is a local diffeomorphism. Conversely let f be a bijective local diffeomorphism; it is a continuous open bijection, hence a homeomorphism, and its inverse is smooth because on each $f(U)$ it agrees with the smooth $(f|_U)^{-1}$ (proposition 1.3.6). Thus f is a diffeomorphism, completing the proof.

Immersion and submersion together pin down the isomorphism case.

Proposition 3.1.9: Local Diffeomorphism via Rank

For smooth $f\colon M \to N$:

1. f is a local diffeomorphism if and only if it is both an immersion and a submersion;
2. if $\dim M = \dim N$ and f is an immersion or a submersion, then f is a local diffeomorphism.

***Proof*:**
$T_p f$ is a linear isomorphism if and only if it is both injective and surjective, and (proved in the next subsection) f is a local diffeomorphism near p exactly when $T_p f$ is an isomorphism. This gives (1). When $\dim M = \dim N$ a linear map between the tangent spaces is injective if and only if surjective, so either hypothesis in (2) forces $T_p f$ to be an isomorphism at every point, completing the proof.

3.1.1 Rank Theorem

A linear map of rank r can be row- and column-reduced to the block $\left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$. The rank theorem is the nonlinear analogue: a smooth map of constant rank r becomes, in suitable charts, exactly the linear map $(x^1, \ldots, x^m) \mapsto (x^1, \ldots, x^r, 0, \ldots, 0)$. The bijective case is the inverse function theorem, cheap and worth stating first.

Theorem 3.1.10: Inverse Function Theorem for Manifolds

Let $f\colon M \to N$ be smooth and $p \in M$ with $T_p f$ invertible. Then f restricts to a diffeomorphism of an open neighbourhood of p onto an open neighbourhood of $f(p)$.

***Proof*:**
Choose charts (U, φ) at p and (V, ψ) at $f(p)$ with $f(U) \subseteq V$, and set $\hat{f} = \psi \circ f \circ \varphi^{-1}$. By definition 2.2.2 the Jacobian $D\hat{f}(\varphi(p))$ represents $T_p f$ in the coordinate bases, so it is invertible. The Euclidean inverse function theorem (theorem 0.2.1) gives an open $\hat{U} \ni \varphi(p)$ on which $\hat{f}$ is a diffeomorphism onto the open set $\hat{f}(\hat{U})$. Then $U_0 = \varphi^{-1}(\hat{U})$ and $V_0 = \psi^{-1}(\hat{f}(\hat{U}))$ are open, and

$$f|_{U_0} = \psi^{-1} \circ \hat{f} \circ \varphi$$

is a composition of diffeomorphisms, hence a diffeomorphism $U_0 \to V_0$, completing the proof.

The general statement drops invertibility for constant rank; the inverse function theorem is the case $r = m = n$.

Theorem 3.1.11: Constant Rank Theorem

Let $f\colon M \to N$ have constant rank r, with $\dim M = m$, $\dim N = n$. Then each $p \in M$ has charts (U, φ) centred at p and (V, ψ) centred at $f(p)$ with $f(U) \subseteq V$ in which

$$\psi \circ f \circ \varphi^{-1}(x^1, \ldots, x^r, x^{r+1}, \ldots, x^m) = (x^1, \ldots, x^r, 0, \ldots, 0)$$

In particular a submersion is locally $(x^1, \ldots, x^m) \mapsto (x^1, \ldots, x^n)$, and an immersion is locally $(x^1, \ldots, x^m) \mapsto (x^1, \ldots, x^m, 0, \ldots, 0)$.

***Proof*:**
Working in charts, take M open in $\mathbb{R}^m$ and N open in $\mathbb{R}^n$, and after permuting coordinates and translating assume $p = 0$, $f(0) = 0$, with the top-left $r \times r$ block of $Df(0)$ nonsingular. Split coordinates as (x, y) on the domain ($x \in \mathbb{R}^r$, $y \in \mathbb{R}^{m-r}$) and (u, v) on the codomain ($u \in \mathbb{R}^r$, $v \in \mathbb{R}^{n-r}$), and write $f(x, y) = (Q(x, y), R(x, y))$ with $Q \in \mathbb{R}^r$, so $\partial Q/\partial x$ is nonsingular at 0.

The map $\varphi(x, y) = (Q(x, y), y)$ has derivative $\begin{pmatrix} \partial Q/\partial x & \partial Q/\partial y \\ 0 & I \end{pmatrix}$, nonsingular at 0, so by theorem 0.2.1 it has a local smooth inverse $\varphi^{-1}(x, y) = (A(x, y), B(x, y))$. From $\varphi \circ \varphi^{-1} = \mathrm{id}$ comes $B(x, y) = y$, hence

$$f \circ \varphi^{-1}(x, y) = (x,\ S(x, y)), \qquad S(x, y) = R(A(x, y), y)$$

Its derivative is $\begin{pmatrix} I_r & 0 \\ \partial S/\partial x & \partial S/\partial y \end{pmatrix}$. Composing with a diffeomorphism does not change rank, so $f \circ \varphi^{-1}$ still has rank r; the first r columns already contribute r, forcing $\partial S/\partial y = 0$. Thus $S = S(x)$ is independent of y, and $f \circ \varphi^{-1}(x, y) = (x, S(x))$.

Finally $\sigma(u, v) = (u, v - S(u))$ is a diffeomorphism of the codomain, and $\sigma \circ f \circ \varphi^{-1}(x, y) = (x, 0)$. Taking $\varphi = \varphi$ and $\psi = \sigma$ (composed with the initial charts) gives the stated normal form, completing the proof.

Restated invariantly: constant rank is exactly the condition for f to look linear in suitable coordinates.

Corollary 3.1.12: Constant Rank Means Locally Linear

For a smooth $f\colon M \to N$ with M connected, the following are equivalent:

1. every point has charts in which f has a linear coordinate representative;
2. f has constant rank.

***Proof*:**
If f has constant rank, the constant rank theorem gives the linear (indeed projection) representative near each point. Conversely a linear representative has a constant rank on its chart, and the rank, being locally constant, is constant on the connected M, completing the proof.

The theorem localizes injectivity and surjectivity of f from those of $T_p f$.

Theorem 3.1.13: Local Injectivity and Surjectivity

Let $f\colon M^m \to N^n$ be smooth and $p \in M$.

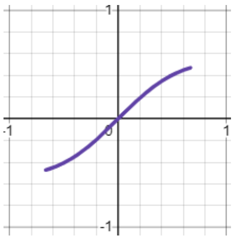
1. If T_pf is injective, then f is injective on a neighbourhood of p.
2. If T_pf is surjective, then f is surjective onto a neighbourhood of $f(p)$.
3. If T_pf is bijective, then f restricts to a bijection near p.

***Proof*:**
If T_pf is injective, f is an immersion near p (proposition 3.1.4), hence of constant rank m there; the constant rank theorem represents it as $x \mapsto (x, 0)$, injective, so f is injective near p. If T_pf is surjective, the same theorem represents f as the projection $(x, y) \mapsto x$, which is surjective onto a neighbourhood of $f(p)$. Bijectivity follows from both, completing the proof.

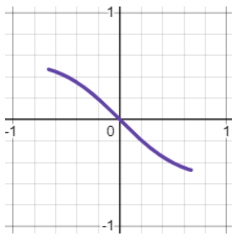
Local injectivity is not global injectivity. The map $f\colon S^1 \to \mathbb{R}^2$ tracing the lemniscate,

$$f(t) = \left(\frac{a\sqrt{2}\cos t}{1+\sin^2 t}, \frac{a\sqrt{2}\cos t\sin t}{1+\sin^2 t}\right)$$

is an immersion and is injective on each of the arcs $(\frac{\pi}{4}, \frac{3\pi}{4})$ and $(\frac{5\pi}{4}, \frac{7\pi}{4})$,



(a) $\hat{f}$ on $(\pi/4, 3\pi/4)$



(b) $\hat{f}$ on $(5\pi/4, 7\pi/4)$

yet not globally injective, since $f(\frac{\pi}{2}) = f(\frac{3\pi}{2})$ at the crossing. Surjectivity has no such defect: if $f|_U$ is surjective onto $f(U)$ then f is too. This asymmetry – injectivity needing a global hypothesis, surjectivity not – is pursued in section 3.1.2.

Global constant rank upgrades these local statements to global ones.

Theorem 3.1.14: Global Rank Theorem

Let $f\colon M \to N$ have constant rank. Then:

1. if f is injective, it is an immersion;
2. if f is surjective, it is a submersion;
3. if f is bijective, it is a diffeomorphism.

***Proof*:**
Let $\operatorname{rank} f \equiv r$, $\dim M = m$, $\dim N = n$.

(1) If f were not an immersion then $r < m$, and the constant rank theorem represents f near a point as $(x^1, \dots, x^r, x^{r+1}, \dots, x^m) \mapsto (x^1, \dots, x^r, 0, \dots, 0)$, which identifies $(0, \dots, 0, \varepsilon)$ with $(0, \dots, 0)$ for small ε – contradicting injectivity. So $r = m$.

(2) If f were not a submersion then $r < n$. Near each p the constant rank theorem represents f as $(x^1, \dots, x^m) \mapsto (x^1, \dots, x^r, 0, \dots, 0)$; shrinking to a coordinate ball U with $f(\overline{U}) \subseteq V$, the image $f(\overline{U})$ is a compact subset of the slice $\left\{y \in V \;\middle|\; y^{r+1} = \cdots = y^n = 0\right\}$, hence closed with empty interior, i.e. nowhere dense in N. Covering M by countably many such U_i writes $f(M)$ as a countable union of the nowhere-dense $f(\overline{U}_i)$, so by the Baire category theorem $f(M)$ has empty interior and cannot be all of N – contradicting surjectivity. So $r = n$.

(3) A bijection is by (1) and (2) an immersion and a submersion, hence a local diffeomorphism (proposition 3.1.9); a bijective local diffeomorphism is a diffeomorphism (proposition 3.1.8), completing the proof.

This answers the question left open earlier: a bijective smooth map is a diffeomorphism provided it has constant rank. Without that hypothesis it can fail.

Example 3.1.15: A Bijective Non-Diffeomorphism
The map $x \mapsto x^3$ of $\mathbb{R}$ (with the standard structure) is a smooth bijection but not a diffeomorphism: its inverse $x^{1/3}$ is not differentiable at 0, where $T_0 f$ drops rank. The rank is 1 everywhere except at the single point 0, so constant rank fails on a measure-zero set – exactly enough to break the conclusion of theorem 3.1.14.

3.1.2 Embeddings

As example 3.1.5 shows, an immersion may fail to be a submanifold: injectivity of $T_p f$ is local, and even a globally injective immersion need not carry the subspace topology onto its image. Demanding that f be a homeomorphism onto its image removes both defects.

Definition 3.1.16: Smooth Embedding

A *smooth embedding* is a smooth immersion $f\colon M \to N$ that is a homeomorphism onto its image $f(M)$ with the subspace topology.

A smooth embedding is in particular a topological embedding, and its image is a topological manifold; the next section shows it is a smooth submanifold. Several common hypotheses upgrade an injective immersion to an embedding.

Proposition 3.1.17: Sufficient Conditions for an Embedding

An injective smooth immersion $f\colon M \to N$ is a smooth embedding if any of the following holds:

1. f is an open or a closed map;
2. f is proper;
3. M is compact;
4. M has empty boundary and $\dim M = \dim N$.

***Proof*:**
It suffices in each case that f is a homeomorphism onto its image. (1) An injective continuous map that is open or closed onto its image is a homeomorphism onto it. (2) A proper map into a locally compact Hausdorff space is closed, so (1) applies. (3) A continuous map from a compact space to a Hausdorff space is closed, so (1) applies. (4) With $\dim M = \dim N$ and empty boundary, the immersion f is a local diffeomorphism (proposition 3.1.9), hence an open map, so (1) applies, completing the proof.

Immersions are exactly the maps that are embeddings locally.

Theorem 3.1.18: Immersions are Local Embeddings

A smooth map $f\colon M \to N$ is an immersion if and only if every $p \in M$ has a neighbourhood U on which $f|_U$ is a smooth embedding.

***Proof*:**
If $f|_U$ is an embedding for a neighbourhood of each point, then f is an immersion at each point, hence an immersion. Conversely let f be an immersion. By the constant rank theorem (theorem 3.1.11) with rank $m = \dim M$, in suitable charts near p the map is the linear inclusion $x \mapsto (x, 0)$. Restrict to a coordinate ball U with compact closure carried into the chart: on it f is a continuous injection from a set with compact closure into a Hausdorff space, hence a homeomorphism onto its image, and an immersion, so $f|_U$ is a smooth embedding, completing the proof.

3.1.3 Submersions

A submersion is the smooth counterpart of a topological quotient map. Recall that a surjective continuous $\pi\colon X \to Y$ presents Y as a quotient of X when it carries saturated open sets to open sets – equivalently, when a set in Y is open iff its preimage is; being an open or closed surjection suffices. Quotient maps satisfy a universal factorization property, and open ones additionally admit continuous local right inverses; both features reappear in the smooth category. A surjective submersion realizes both smoothly, with one caveat – a smooth submersion need not admit a *global* smooth section, only local ones.

Definition 3.1.19: Local Section

A *section* of a smooth $\pi\colon M \to N$ is a smooth right inverse, a smooth $\sigma\colon N \to M$ with $\pi \circ \sigma = \mathrm{id}_N$; a *local section* at p is one defined on a neighbourhood of $\pi(p)$ with $\sigma(\pi(p)) = p$.

Theorem 3.1.20: Submersions and Local Sections

A smooth $\pi\colon M \to N$ is a submersion if and only if every $p \in M$ lies in the image of a smooth local section.

***Proof*:**
If $p = \sigma(\pi(p))$ for a local section σ, then differentiating $\pi \circ \sigma = \mathrm{id}$ gives $T_p\pi \circ T_{\pi(p)}\sigma = \mathrm{id}$, so $T_p\pi$ is surjective; as p was arbitrary, π is a submersion. Conversely let π be a submersion. By the constant rank theorem, near p it is the projection $(x, y) \mapsto x$ in suitable charts; the map $x \mapsto (x, 0)$ is then a smooth local section through p, completing the proof.

Proposition 3.1.21: A Submersion is an Open Map

Every smooth submersion $f\colon M \to N$ is an open map; if it is also surjective, it is a topological quotient map.

***Proof*:**
By the constant rank theorem f is, near each point and in charts, the projection $(x, y) \mapsto x$, which is open; since the charts are diffeomorphisms (open maps), f is locally open, hence open. A surjective open continuous map is a quotient map, completing the proof.

Corollary 3.1.22: Submersions off a Compact Manifold

A smooth submersion $f\colon M \to N$ with M compact and N connected is surjective.

***Proof*:**
The image $f(M)$ is open (proposition 3.1.21) and, being the continuous image of a compact set, closed; a nonempty clopen subset of the connected N is all of N, completing the proof.

Local sections give surjective submersions their defining feature: smoothness of a map out of the base is detected upstairs.

Proposition 3.1.23: Characteristic Property of Submersions

Let $\pi\colon M \to N$ be a surjective smooth submersion and P a smooth manifold (with or without boundary). A map $f\colon N \to P$ is smooth if and only if $f \circ \pi$ is smooth.

***Proof*:**
If f is smooth, so is $f \circ \pi$. Conversely suppose $f \circ \pi$ is smooth and fix $q \in N$. Choose p with $\pi(p) = q$ (surjectivity) and a smooth local section σ through p (theorem 3.1.20), so $\pi \circ \sigma = \mathrm{id}$ near q. Then $f = (f \circ \pi) \circ \sigma$ near q is a composition of smooth maps, hence smooth near q; as q was arbitrary, f is smooth, completing the proof.

Theorem 3.1.24: Universal Property of Submersions

Let $\pi\colon M \to N$ be a surjective smooth submersion and P a smooth manifold (with or without boundary). If $f\colon M \to P$ is smooth and constant on the fibres of π, then there is a unique smooth $\tilde{f}\colon N \to P$ with $\tilde{f} \circ \pi = f$:

$$\begin{tikzcd} M \arrow[d, "\pi"'] \arrow[dr, "f"] & \\ N \arrow[r, dashed, "\tilde{f}"] & P \end{tikzcd}$$

***Proof*:**
Since f is constant on fibres and π is surjective, setting $\tilde{f}(q) = f(p)$ for any $p \in \pi^{-1}(q)$ is well defined and is the unique map with $\tilde{f} \circ \pi = f$. It is smooth: $\tilde{f} \circ \pi = f$ is smooth, so proposition 3.1.23 makes $\tilde{f}$ smooth, completing the proof.

Exercise 3.1.1

1. For $f\colon \mathbb{R}^2 \to \mathbb{R}^2$, $f(x, y) = (x^2 - y^2, 2xy)$, compute $\operatorname{rank} T_p f$ at each p and determine where f is an immersion, a submersion, and a local diffeomorphism.
2. Show that the figure-eight $\gamma\colon (-\pi, \pi) \to \mathbb{R}^2$, $t \mapsto (\sin 2t, \sin t)$, is an injective immersion but not a smooth embedding.
3. Reprove that a smooth submersion $\pi\colon M \to N$ is an open map using local sections (theorem 3.1.20) rather than the constant rank theorem.

3.2 Submanifolds

Section 1.4 defined a regular submanifold by the slice condition and left two things unproved: that the slice condition really makes S a smooth manifold with the inclusion an embedding (and conversely), and that the four descriptions of proposition 1.4.5 agree. The rank theorem settles both, and then delivers the level-set constructions and the tangent space of a submanifold. Recall (definition 1.4.1) that $S \subseteq M$ in an m-manifold is a *regular k-submanifold* if every $p \in S$ lies in a chart (U, φ) of M with $\varphi(U \cap S) = \varphi(U) \cap (\mathbb{R}^k \times \{0\})$; such charts are *submanifold* (or *slice*) charts, and a submanifold of codimension one is a *hypersurface*. The precise slice vocabulary is as follows.

Definition 3.2.1: Local Slice

Let $U \subseteq \mathbb{R}^n$ be open.

1. A *k-slice* of U is a subset $\{(x^1, \ldots, x^n) \in U \mid x^{k+1} = c^{k+1}, \ldots, x^n = c^n\}$ for constants c^i (usually 0; the positions of the constants may be permuted).

2. For a smooth chart (U, φ) and $S \subseteq U$ with $\varphi(S)$ a k-slice of $\varphi(U)$, one says S is a k-slice of U.

3. A subset $S \subseteq M$ satisfies the *local k-slice condition* if each of its points lies in a smooth chart (U, φ) of M with $S \cap U$ a single k-slice of U. Such a chart is a *slice chart*, its coordinates *slice coordinates*, and a covering family a *slice atlas*.

A submanifold atlas is a slice atlas. The next lemma proves that the slice condition is equivalent to being a smoothly embedded subset – discharging the slice $\Leftrightarrow$ embedding half of proposition 1.4.5.

Lemma 3.2.2: Slice Condition Equals Embedding

Let M be a smooth m-manifold and $S \subseteq M$. If S satisfies the local k-slice condition, then with the subspace topology S is a topological k-manifold carrying a smooth structure that makes it a k-dimensional regular submanifold and the inclusion $\iota\colon S \to M$ a smooth embedding. Conversely, if $\iota\colon S \to M$ is a smooth embedding, then S satisfies the local k-slice condition.

***Proof*:**
Suppose S satisfies the local k-slice condition, and give it the subspace topology, inheriting Hausdorff and second countable. Let $\pi\colon \mathbb{R}^m \to \mathbb{R}^k$ be projection onto the first k coordinates. For a slice chart (U, φ) set $V = U \cap S$, $\hat{V} = \pi(\varphi(V))$, and $\psi = \pi \circ \varphi|_V \colon V \to \hat{V}$. Since $\varphi(V)$ is $\varphi(U)$ intersected with a k-slice A (setting $x^{k+1} = c^{k+1}, \ldots$), it is open in A, and $\pi|_A \colon A \to \mathbb{R}^k$ is a diffeomorphism, so $\hat{V}$ is open in $\mathbb{R}^k$. The map ψ is a homeomorphism, with continuous inverse $\varphi^{-1} \circ j$, $j(x^1, \ldots, x^k) = (x^1, \ldots, x^k, c^{k+1}, \ldots, c^m)$. Thus S is a topological k-manifold.

For smoothness, two slice charts $(U, \varphi), (U', \varphi')$ give charts $(V, \psi), (V', \psi')$ of S with transition $\psi' \circ \psi^{-1} = \pi \circ \varphi' \circ \varphi^{-1} \circ j$, a composition of smooth maps; so the induced atlas is smooth. In a slice chart the inclusion is $(x^1, \ldots, x^k) \mapsto (x^1, \ldots, x^k, c^{k+1}, \ldots, c^n)$, a smooth immersion, and it is a topological embedding, hence a smooth embedding.

Conversely let $\iota\colon S \to M$ be a smooth embedding. As an immersion, by the constant rank theorem it has, near each $p \in S$, charts (U, φ) of S and (V, ψ) of M in which $\iota|_U$ is $(x^1, \ldots, x^k) \mapsto (x^1, \ldots, x^k, 0, \ldots, 0)$. Shrinking to a ball $B_\varepsilon(p) \subseteq U$ with $B_\varepsilon(\iota(p)) \subseteq V$, the image $\iota(B_\varepsilon(p))$ is a single slice of $B_\varepsilon(\iota(p))$. Since S carries the subspace topology, $B_\varepsilon(p) = W \cap S$ for some open $W \subseteq M$; then $V' = B_\varepsilon(\iota(p)) \cap W$ is a slice chart with $V' \cap S = B_\varepsilon(p)$, so S meets the slice condition, completing the proof.

Uniqueness of the smooth structure induced on S is taken up with the general uniqueness of subspace structures in chapter 4. A second equivalent description cuts S out as a level set, the form closest to algebraic and complex geometry.

Proposition 3.2.3: Submanifold as a Local Zero Set

A subset $S \subseteq M$ with a submanifold chart (U, φ) at p (so $\varphi(U \cap S) = \varphi(U) \cap \mathbb{R}^k$) is, near p, the zero set $U \cap S = f^{-1}(0)$ of the submersion $f = \pi \circ \varphi \colon U \to \mathbb{R}^{m-k}$, where π projects onto the last $m - k$ coordinates. Conversely such a zero set is a submanifold chart. Thus S has codimension $m - k$.

***Proof*:**
With $\varphi(U \cap S) = \varphi(U) \cap \mathbb{R}^k$ (the slice $x^{k+1} = \cdots = x^m = 0$),

$$f^{-1}(0) = \varphi^{-1}(\pi^{-1}(0)) = \varphi^{-1}(\varphi(U) \cap \mathbb{R}^k) = \varphi^{-1}(\varphi(U \cap S)) = U \cap S$$

Conversely, if $U \cap S = f^{-1}(0)$ for $f = \pi \circ \varphi$, then $\varphi(U \cap S) = \varphi(U) \cap \mathbb{R}^k$, a submanifold chart, completing the proof.

Together with lemma 3.2.2 this discharges the equivalences of proposition 1.4.5: slice charts, smooth embeddings, and regular level sets describe the same submanifolds, and the graph description joins them – a graph is an embedding (example 3.2.6), and a regular level set is a graph by the implicit function theorem. Restriction of a smooth map to a submanifold is smooth, a fact already recorded in corollary 1.4.3.

With embeddings and the rank theorem in hand, the image of an embedding is a submanifold.

Theorem 3.2.4: The Image of an Embedding is a Submanifold

If $f \colon M \to N$ is a smooth embedding, then $f(M)$ is a regular submanifold of N of dimension $\dim M$.

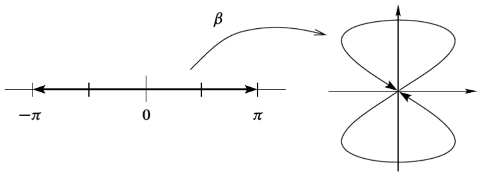
***Proof*:**
Let $\dim M = k$, $\dim N = n$, and $q = f(p) \in f(M)$; injectivity gives a unique such p. By the constant rank theorem (theorem 3.1.11) there are charts (U, φ) at p and (V, ψ) at q with $\psi \circ f \circ \varphi^{-1}(x^1, \ldots, x^k) = (x^1, \ldots, x^k, 0, \ldots, 0)$. As f is a homeomorphism onto its image, $f(U)$ is open in $f(M)$, so $f(U) = f(M) \cap O_0$ for some open $O_0 \subseteq V$. Shrinking to $O = O_0 \cap \psi^{-1}(\varphi(U) \times \mathbb{R}^{n-k})$ preserves $f(M) \cap O = f(U)$ and gives $\psi(O) \cap (\mathbb{R}^k \times \{0\}) = \varphi(U) \times \{0\} = \psi(f(U))$, so $(O, \psi|_O)$ is a submanifold chart for $f(M)$ at q. Since q was arbitrary, $f(M)$ is a regular k-submanifold, completing the proof.

Dropping either hypothesis – immersion or homeomorphism onto the image – breaks the conclusion.

Example 3.2.5: Non-Embeddings

1. The cuspidal cubic $t \mapsto (t^2, t^3)$ (example 1.3.11) is a topological embedding and a smooth map, but not an immersion (rank 0 at $t = 0$), hence not a smooth embedding: its image is not a smooth submanifold. So a smooth embedding is more than a topological embedding that happens to be smooth.
2. The figure-eight $\beta \colon (-\pi, \pi) \to \mathbb{R}^2$, $t \mapsto (\sin 2t, \sin t)$, is an injective immersion but not an embedding: its image is compact (the ends $t \to \pm\pi$ return to $\beta(0) = 0$), while the domain is

not, so β cannot be a homeomorphism onto its image.



3. The irrational-slope line $\gamma\colon \mathbb{R} \to T^2$, $\gamma(t) = (e^{2\pi i t}, e^{2\pi i \alpha t})$ with α irrational, is an injective immersion whose image is dense in the torus; it is not an embedding and its image is not a submanifold.

Conversely, many familiar constructions are embeddings.

Example 3.2.6: Smooth Embeddings

1. An open subset $U \subseteq M$ is a regular submanifold of the same dimension.
2. For a fixed $q \in N$, the slice $x \mapsto (x, q)$ embeds M as the regular submanifold $M \times \{q\}$ of $M \times N$.
3. (Graphs) For open $U \subseteq M$ and smooth $f\colon U \to N$, the graph map $\gamma_f\colon U \to M \times N$, $x \mapsto (x, f(x))$, is a smooth embedding, so the graph $\Gamma(f) = \{(x, f(x)) \mid x \in U\}$ is a regular m-submanifold of $M \times N$. Indeed γ_f is an immersion (its coordinate Jacobian $\left(\begin{smallmatrix} I \\ Df \end{smallmatrix}\right)$ has injective top block) and a homeomorphism onto its image, with continuous inverse the projection $M \times N \to M$ restricted to $\Gamma(f)$.

A smooth embedding is *proper* when it is a proper map; a submanifold is properly embedded if and only if it is a closed subset, so every compact submanifold is properly embedded, as is every graph $\Gamma(f)$ in $U \times N$ (though not in $M \times N$ unless $U = M$).

Level sets generalize the $\mathbb{R}^n$ construction of section 1.2.3 to maps between manifolds. Not every level set is a submanifold – $f_1(x, y) = x^2 - y^2$ has for $f_1^{-1}(0)$ a crossed pair of lines, and $f_2(x, y) = x^2 - y^3$ a cusp – and indeed every closed subset of M is the zero set of a smooth function (corollary 1.5.9), so an unconstrained level set is arbitrary. A rank hypothesis is what forces a submanifold.

Proposition 3.2.7: Constant-Rank Level Set Theorem

If $f\colon M \to N$ has constant rank r, then for each $q \in f(M)$ the level set $f^{-1}(q)$ is a regular submanifold of M of codimension r.

***Proof*:**
Fix $x \in f^{-1}(q)$. By the constant rank theorem there are charts (U, φ) at x and (V, ψ) at q, the latter recentred so $\psi(q) = 0$, in which

$$\psi \circ f \circ \varphi^{-1}(x^1, \ldots, x^m) = (x^1, \ldots, x^r, 0, \ldots, 0)$$

Then $f(y) = q$ for $y \in U$ exactly when the first r coordinates of $\varphi(y)$ vanish, so $\varphi(U \cap f^{-1}(q)) = \varphi(U) \cap \{x^1 = \cdots = x^r = 0\}$, an $(m - r)$-slice. Thus (U, φ) is a submanifold chart, and $f^{-1}(q)$ is a regular submanifold of codimension r, completing the proof.

Corollary 3.2.8: Submersion Level Set Theorem

If Df has full rank at every point of $f^{-1}(q)$, then $f^{-1}(q)$ is a regular submanifold of M.

***Proof*:**
The full-rank locus is open (proposition 3.1.4), so f has constant full rank on an open $U \supseteq f^{-1}(q)$; applying proposition 3.2.7 to $f|_U$ gives the claim, completing the proof.

This is the manifold form of the linear-algebra fact that the kernel of a surjective linear map $\mathbb{R}^m \to \mathbb{R}^r$ is a subspace of codimension r. It needs the rank hypothesis only on the level set itself, which the following vocabulary isolates.

Definition 3.2.9: Critical and Regular Points and Values

For a smooth $f\colon M \to N$, a point $p \in M$ is a *regular point* if T_pf is surjective, and a *critical* (or *singular*) point otherwise. A point $q \in N$ is a *regular value* if every point of $f^{-1}(q)$ is regular (vacuously so when $f^{-1}(q) = \emptyset$), and a *critical value* otherwise; $f^{-1}(q)$ is then a *regular level set*.

A submersion has only regular points, and when $\dim M < \dim N$ every point is critical. The set of regular points is open (proposition 3.1.4).

Corollary 3.2.10: Regular Level Set Theorem

Every regular level set of a smooth $f\colon M \to N$ is a properly embedded submanifold of codimension $\dim N$.

***Proof*:**
Let c be a regular value. Every $x \in f^{-1}(c)$ is regular, so Df has full rank on $f^{-1}(c)$, and corollary 3.2.8 makes $f^{-1}(c)$ an embedded submanifold of codimension $\dim N$. It is closed in M by continuity, hence properly embedded, completing the proof.

Not every submanifold is a global regular level set ($\mathcal{P}^1$ is not, and a torus in S^3 is only awkwardly one), but every submanifold is *locally* one – the converse of proposition 3.2.3 without assuming the cutting function has the special form $\pi \circ \varphi$.

Theorem 3.2.11: Submanifolds are Locally Level Sets

A subset $S \subseteq M$ is an embedded k-submanifold if and only if every $x \in S$ has a neighbourhood U on which $U \cap S$ is a level set of a smooth submersion $f\colon U \to \mathbb{R}^{m-k}$.

***Proof*:**
If S is an embedded submanifold, then by proposition 3.2.3 one may take $f = \pi \circ \varphi$ in a submanifold chart, a submersion with $U \cap S = f^{-1}(0)$. Conversely, if $U \cap S$ is a level set of a

submersion $f\colon U \to \mathbb{R}^{m-k}$, then corollary 3.2.8 makes $U \cap S$ an embedded submanifold of U, so it meets the local slice condition (lemma 3.2.2); as this holds near each point, S is an embedded submanifold, completing the proof.

Definition 3.2.12: Defining Map

A *defining map* for an embedded submanifold $S \subseteq M$ is a smooth $\varphi\colon M \to N$ having S as a regular level set; when $N = \mathbb{R}^{m-k}$ it is a *defining function*. A *local defining map* is one defined on an open U with $S \cap U$ a regular level set.

The sphere S^n has defining function $f(x) = |x|^2$, and by theorem 3.2.11 every embedded submanifold admits a local defining map near each of its points.

3.2.1 Immersed Submanifolds

An embedded submanifold carries the subspace topology, but the image of an injective immersion need not: the figure-eight and the dense torus line (example 3.2.5) have images that, with the subspace topology, are not manifolds. Remembering the parametrizing map rather than only the image repairs this.

Definition 3.2.13: Immersed Submanifold

An *immersed submanifold* of M is a subset $S \subseteq M$ with a manifold topology (possibly finer than the subspace topology) and a smooth structure making the inclusion $\iota\colon S \hookrightarrow M$ an injective immersion.

Every embedded submanifold is immersed, with the subspace topology; the converse fails, as the figure-eight shows. Conversely, the image of any injective immersion $f\colon P \to M$ is an immersed submanifold when given the topology and smooth structure transported from P (so $f\colon P \to f(P)$ is a diffeomorphism), and it is embedded exactly when f is an embedding. Immersed submanifolds are the natural home of Lie subgroups and of the integral manifolds of the Frobenius theorem (chapter 6), whose leaf topology is genuinely finer than the subspace one.

3.2.2 Tangent Space of a Submanifold

For an immersed or embedded submanifold $S \subseteq M$ the inclusion $\iota\colon S \hookrightarrow M$ is an injective immersion, so $T_p\iota\colon T_pS \to T_pM$ is injective and identifies T_pS with a subspace of T_pM. On derivations $T_p\iota(v)$ acts by $T_p\iota(v)(f) = v(f \circ \iota) = v(f|_S)$: a tangent vector to S sees only the restriction of f to S. The three descriptions of the Euclidean case (proposition 2.1.19) now take their general form.

Proposition 3.2.14: Tangent Space via Curves

Let $S \subseteq M$ be a submanifold and $p \in S$. A vector $v \in T_pM$ lies in T_pS if and only if $v = \gamma'(0)$ for a smooth curve γ into M whose image lies in S, smooth as a map into S, with $\gamma(0) = p$.

***Proof*:**
If $\gamma = \iota \circ \gamma_S$ for a smooth curve γ_S into S, then $v = \gamma'(0) = T_p\iota(\gamma_S'(0)) \in T_pS$. Conversely, if $v \in T_pS$, write $v = T_p\iota(w)$; realize $w = \gamma_S'(0)$ for a smooth curve γ_S in S through p (definition 2.1.18), and set $\gamma = \iota \circ \gamma_S$, completing the proof.

Proposition 3.2.15: Tangent Space via Vanishing Functions

For a regular submanifold $S \subseteq M$ and $p \in S$,

$$T_pS = \{v \in T_pM \mid v(f) = 0 \text{ whenever } f \in C^\infty(M),\ f|_S = 0\}$$

***Proof*:**
If $v = T_p\iota(w) \in T_pS$ and $f|_S = 0$, then $v(f) = w(f \circ \iota) = w(f|_S) = 0$. Conversely, suppose v kills every f vanishing on S. In a slice chart at p, S is $\{x^{k+1} = \cdots = x^m = 0\}$; for $i > k$ the function βx^i (with β a bump equal to 1 near p, supported in the chart) is a global smooth function vanishing on S and agreeing with x^i near p, so $v(x^i) = v(\beta x^i) = 0$. By the coordinate expansion $v = \sum_i v(x^i)\,\partial/\partial x^i|_p = \sum_{i \le k} v(x^i)\,\partial/\partial x^i|_p$, which lies in $T_pS = \operatorname{span}\{\partial/\partial x^i|_p : i \le k\}$, completing the proof.

Proposition 3.2.16: Tangent Space via a Defining Map

If $\varphi\colon U \to N$ is a local defining map for a regular submanifold $S \subseteq M$, then $T_pS = \ker T_p\varphi$ for each $p \in S \cap U$.

***Proof*:**
Identify T_pS with $T_p\iota(T_pS) \subseteq T_pM$. Since $\varphi \circ \iota$ is constant on $S \cap U$, $T_p\varphi \circ T_p\iota = 0$, so $\operatorname{im} T_p\iota \subseteq \ker T_p\varphi$. As φ is a submersion, $T_p\varphi$ is surjective, so by rank–nullity $\dim\ker T_p\varphi = \dim T_pM - \dim T_{\varphi(p)}N = k = \dim T_pS = \dim \operatorname{im} T_p\iota$. Equal dimensions force equality, completing the proof.

Corollary 3.2.17: Tangent Space via a Defining Function

If $S = \varphi^{-1}(0)$ for a submersion $\varphi = (\varphi^1, \ldots, \varphi^k)\colon M \to \mathbb{R}^k$, then $v \in T_pM$ is tangent to S if and only if $v(\varphi^1) = \cdots = v(\varphi^k) = 0$.

***Proof*:**
By proposition 3.2.16, $v \in T_pS$ iff $T_p\varphi(v) = 0$. The j-th component of $T_p\varphi(v)$ is $T_p\varphi(v)(y^j) = v(y^j \circ \varphi) = v(\varphi^j)$, so $T_p\varphi(v) = 0$ iff every $v(\varphi^j) = 0$, completing the proof.

3.2.3 Maps and Submanifolds

A smooth $f\colon M \to N$ restricts to a smooth map $f|_S = f \circ \iota\colon S \to N$ on any submanifold $S \subseteq M$, with $T_p(f|_S) = T_pf \circ T_p\iota$ on tangent spaces. Corestriction runs the other way and deserves a statement of record, since it is what makes a factored map smooth through a submanifold.

Proposition 3.2.18: Corestriction to a Regular Submanifold

Let $f\colon M \to N$ be smooth and let $S \subseteq N$ be a regular k-submanifold with $f(M) \subseteq S$. Then the corestriction $f\colon M \to S$ is smooth for the submanifold structure of S.

***Proof*:**
Since S carries the subspace topology and f is continuous into N, the corestriction is continuous into S. For smoothness fix $p \in M$ and a submanifold chart (V, ψ) of N at $f(p)$ (definition 1.4.1), so that $\psi(V \cap S) = \psi(V) \cap (\mathbb{R}^k \times \{0\})$ and the first k components of ψ form a chart of S on $V \cap S$. On the open set $f^{-1}(V)$ the composite $\psi \circ f$ is smooth with values in $\psi(V) \cap (\mathbb{R}^k \times \{0\})$, so its first k components are smooth, and these are the coordinate representation of the corestriction in the chosen charts, completing the proof.

Corestriction to an *immersed* submanifold is subtler, since its topology may be finer than the subspace topology; the criterion is taken up with the Frobenius theorem (chapter 6).

3.2.4 Submanifolds with Boundary

The level-set constructions extend to manifolds with boundary once the boundary is required to be regular as well.

Proposition 3.2.19: Regular Preimage with Boundary

Let $f\colon M \to N$ be smooth from a manifold with boundary, and $q \in N$ a regular value of both f and $f|_{\partial M}$. Then $f^{-1}(q)$ is a k-manifold with boundary, where $k = \dim M - \dim N$, and $\partial(f^{-1}(q)) = f^{-1}(q) \cap \partial M$.

***Proof*:**
On the interior, q is a regular value of $f|_{\operatorname{int} M}$, so $f^{-1}(q) \cap \operatorname{int} M$ is a k-manifold (corollary 3.2.10). At a boundary point $p \in f^{-1}(q) \cap \partial M$, take a boundary chart identifying a neighbourhood with $\mathbb{H}^m$. Regularity of f at p and of $f|_{\partial M}$ at p make f a submersion whose restriction to $\partial\mathbb{H}^m$ is still a submersion, so in the submersion normal form $f^{-1}(q)$ is a k-slice of $\mathbb{H}^m$ meeting $\partial\mathbb{H}^m$ in a $(k-1)$-slice. Thus $f^{-1}(q)$ is a manifold with boundary $f^{-1}(q) \cap \partial M$, completing the proof.

Proposition 3.2.20: Preimage of a Regular Interval

Let $f\colon M \to \mathbb{R}$ be smooth and proper on a boundaryless manifold, and $a < b$ regular values. Then $f^{-1}([a, b])$ is a compact manifold with boundary $f^{-1}(a) \sqcup f^{-1}(b)$.

***Proof*:**
As $[a, b]$ is compact and f proper, $f^{-1}([a, b])$ is compact, and $f^{-1}((a, b))$ is an open submanifold. At $p \in f^{-1}(a)$, regularity makes f a submersion, so by the constant rank theorem f is the coordinate x^m in suitable charts (after recentering); there $f^{-1}([a, b])$ is $\{x^m \geq 0\}$, a boundary chart into $\mathbb{H}^m$, and similarly $\{x^m \leq 0\}$ at $p \in f^{-1}(b)$. Hence $f^{-1}([a, b])$ is a manifold with boundary $f^{-1}(a) \sqcup f^{-1}(b)$, completing the proof.

This is the basic building block of cobordism theory (chapter 10): the level sets $f^{-1}(a)$ and $f^{-1}(b)$ together bound the compact region between them.

Exercise 3.2.1

1. Using corollary 3.2.17 and the defining function $f(x) = |x|^2$, show that the tangent space to the sphere is $T_pS^{n-1} = p^\perp = \{v \in \mathbb{R}^n \mid \langle v, p\rangle = 0\}$.
2. Show that $\mathrm{O}(n) = \left\{A \in M(n,n) \;\middle|\; A^\top A = I\right\}$ is a regular submanifold of $M(n,n)$ and that $T_I\,\mathrm{O}(n)$ is the space of skew-symmetric matrices, using the defining map $F(A) = A^\top A$ into the symmetric matrices.
3. For a smooth $f\colon \mathbb{R}^n \to \mathbb{R}^m$, show that the graph $\Gamma(f) = \{(x, f(x)) \mid x \in \mathbb{R}^n\}$ is a regular submanifold of $\mathbb{R}^{n+m}$ diffeomorphic to $\mathbb{R}^n$.

4

Quotient Manifolds and Homogeneous Spaces

Quotients are the third way of building manifolds, after charts (chapter 1) and level sets (chapter 3). Collapsing a manifold by an equivalence relation, identifying the points of each orbit of a group action or gluing along a symmetry, is how projective spaces, Grassmannians, and the homogeneous spaces of geometry arise. But a quotient of a manifold need not be a manifold, or even Hausdorff, so the construction requires hypotheses. This chapter isolates them: an action that is *free* and *proper* yields a smooth quotient with the projection a submersion, and the characteristic property of submersions (theorem 3.1.24) then gives the smooth maps out of it.

Section 4.1 sets up group actions, orbits, and the orbit space; section 4.2 isolates proper actions and treats the discrete case, where the quotient is a covering manifold; section 4.3 proves the quotient manifold theorem; and section 4.4 specializes to homogeneous spaces G/H and builds the examples of projective spaces, Grassmannians, Stiefel and flag manifolds, and the classical matrix groups as regular-value examples. Throughout, a Lie group means a smooth manifold with smooth group operations; their structure theory is developed in [16], and only the manifold structure is used here.

4.1 Group Actions

Every quotient construction in this chapter begins with a group moving the points of a manifold. This section fixes the language for that movement, the action map and the maps it induces, and reads off the two invariants a point carries under it: its orbit and its stabilizer. The orbit space M/G is then assembled with the quotient topology, and its one structural feature available before any smoothness is proved, the universal property, is the tool used throughout to identify a quotient with a manifold one already knows.

Throughout, G is a Lie group or a discrete group Γ, and a discrete group is given the discrete topology.

Definition 4.1.1: Lie Group

A *Lie group* is a smooth manifold G that is also a group and whose multiplication $G \times G \to G$ and inversion $G \to G$ are smooth. Only the underlying manifold and the smoothness of these operations are used in this book; the structure theory of Lie groups (the exponential map, the Lie algebra, the closed-subgroup theorem) is developed in [16]. The general linear group $\mathrm{GL}_n(\mathbb{R})$, an open submanifold of $\mathbb{R}^{n^2}$ with smooth matrix multiplication and inversion, is the running example.

Definition 4.1.2: Group Action

Let G be a Lie group and M a smooth manifold. A (left) *action* of G on M, written $G \curvearrowright M$, is a map

$$\theta\colon G \times M \to M, \qquad (g,p) \mapsto g \cdot p$$

satisfying $e \cdot p = p$ for all $p \in M$ and $g \cdot (h \cdot p) = (gh) \cdot p$ for all $g, h \in G$ and $p \in M$. The action is *continuous* if θ is continuous (G a topological group), and *smooth* if θ is smooth (G a Lie group). For a discrete group Γ the product $\Gamma \times M$ is the disjoint union $\coprod_{g \in \Gamma} M$, so a smooth action is precisely an action of Γ by diffeomorphisms.[a]

[a] A *right* action is a map $M \times G \to M$, $(p, g) \mapsto p \cdot g$, with $p \cdot (gh) = (p \cdot g) \cdot h$; it becomes a left action on setting $g \cdot p := p \cdot g^{-1}$, so no generality is lost by working with left actions.

Fixing $g \in G$ and letting the point vary produces the map $\theta_g\colon M \to M$, $p \mapsto g \cdot p$. The two axioms say exactly that $\theta_e = \mathrm{id}_M$ and $\theta_{gh} = \theta_g \circ \theta_h$, so $\theta_g \circ \theta_{g^{-1}} = \theta_e = \mathrm{id}_M$ and likewise in the other order: each θ_g is a bijection with inverse $\theta_{g^{-1}}$. When the action is smooth, θ_g is smooth with smooth inverse, hence a diffeomorphism, and $g \mapsto \theta_g$ is a group homomorphism

$$G \to \mathrm{Diff}(M)$$

into the diffeomorphism group of M. An action is thus the same data as such a homomorphism: a rule assigning to each g a symmetry of M, compatibly with the group law. The simplest example is the defining action of $\mathrm{GL}_n(\mathbb{R})$ on $\mathbb{R}^n$ by $(A, x) \mapsto Ax$, which is smooth because matrix-vector multiplication is polynomial in the entries; another is the group $\mathbb{Z}$ acting on $\mathbb{R}$ by translation, $n \cdot x = x + n$, each θ_n being the diffeomorphism $x \mapsto x + n$.

These maps describe the action globally; fixing instead a single point p and asking how the group meets it isolates two pieces of data, the points p can be moved to and the elements that hold it fixed.

Definition 4.1.3: Orbit and Stabilizer

Let $G \curvearrowright M$ and $p \in M$. The *orbit* of p is the set of points reachable from p,

$$G \cdot p = \{g \cdot p \mid g \in G\} \subseteq M$$

and the *stabilizer* (or isotropy group) of p is the subgroup of elements fixing p,

$$\mathrm{Stab}(p) = G_p = \{g \in G \mid g \cdot p = p\} \leq G$$

The orbits partition M: they are the equivalence classes of the relation $p \sim q$ iff $q = g \cdot p$ for some g, which is reflexive (e), symmetric (g^{-1}), and transitive (the second axiom). A point's orbit and stabilizer are not independent. Sending a group element to where it moves p collapses along cosets of G_p and gives the *orbit-stabilizer correspondence*

$$G/G_p \to G \cdot p, \qquad g\,G_p \mapsto g \cdot p \tag{4.1}$$

The map (4.1) is well defined and injective, since $g\,G_p = g'\,G_p$ holds exactly when $g^{-1}g' \in G_p$, that is when $g^{-1}g' \cdot p = p$, that is when $g \cdot p = g' \cdot p$; it is surjective onto the orbit by definition; and it intertwines the left G-actions, $h \cdot (g \cdot p) = (hg) \cdot p$. So an orbit is a homogeneous copy of the coset space G/G_p. When G is a Lie group acting smoothly this set bijection is upgraded to a diffeomorphism, the content of the homogeneous-space theorem (theorem 4.4.2) developed in section 4.4. For a concrete reading, take $\mathrm{SO}(n)$ acting on $\mathbb{R}^n$: the orbits are the spheres $\{|x| = r\}$ of each radius $r \geq 0$ together with the fixed origin, and the stabilizer of a point $p \neq 0$ consists of the rotations of the hyperplane $p^\perp$, a copy of $\mathrm{SO}(n-1)$.

The action's global shape is captured by three adjectives, according to how large the orbits and stabilizers are.

Definition 4.1.4: Free, Transitive, and Effective Actions

An action $G \curvearrowright M$ is

1. *free* if every stabilizer is trivial: $G_p = \{e\}$ for all p, equivalently $g \cdot p = p$ for some p forces $g = e$;
2. *transitive* if there is a single orbit: $G \cdot p = M$ for one (hence every) p; and
3. *effective* (or faithful) if only the identity acts as the identity map: $g \cdot p = p$ for all p forces $g = e$, that is, $g \mapsto \theta_g$ is injective.

Free is the strongest of the three fixed-point conditions and effective the weakest: a free action is effective, since a trivial stabilizer at even one point already rules out a nonidentity g fixing everything, but not conversely. The kernel of the homomorphism $g \mapsto \theta_g$ is $\bigcap_p G_p$, so effectiveness is the statement that this kernel is trivial and G embeds in $\mathrm{Diff}(M)$. Transitivity is a condition of a different kind, on the orbits rather than the stabilizers: it says M is a single orbit, so by (4.1) a transitive action exhibits M as a coset space G/G_p. The three run independently. The translation action of $\mathbb{Z}$ on $\mathbb{R}$ is free but far from transitive (its orbits are the countable cosets $x + \mathbb{Z}$); the action of $\mathrm{GL}_n(\mathbb{R})$ on $\mathbb{R}^n$ is effective but neither free (every matrix fixes 0) nor transitive (no matrix moves 0); and $\mathrm{SO}(n)$ restricted to the sphere S^{n-1} is transitive, with each stabilizer a copy of $\mathrm{SO}(n-1)$, so it is not free (for $n \geq 3$); it is the transitive but non-free case, in contrast to the two-orbit $\mathrm{GL}_n(\mathbb{R})$ action above.

With orbits understood, the object of interest is the space of orbits itself. Collapsing each orbit to a point produces the set M/G, and the topology it should carry is forced by the requirement that the collapsing map be continuous and carry as many open sets as possible.

Definition 4.1.5: Orbit Space

Let $G \curvearrowright M$. The *orbit space* M/G is the set of orbits, and the *quotient map* $\pi\colon M \to M/G$ sends p to its orbit $G \cdot p$. The orbit space carries the *quotient topology*: a set $V \subseteq M/G$ is open exactly when its preimage $\pi^{-1}(V) \subseteq M$ is open. For a discrete group Γ the orbit space is also written M/Γ.

The quotient topology is the finest topology on M/G making π continuous, and by construction the open sets of M/G are the images of the π-saturated open sets of M. Nothing here uses smoothness or even continuity of the action, only that the orbits partition M; this is the same construction the reader met assembling $\mathbb{RP}^n$ and other quotients in chapter 1. The orbit space of $\mathbb{Z}$ acting on $\mathbb{R}$ by translation is the circle, computed below in example 4.1.7. What makes the quotient topology usable is not its definition but the property it was built to have: maps out of M/G are detected upstairs, on M.

Proposition 4.1.6: Universal Property of the Orbit Space

Let G act on a topological space M, give the orbit space M/G the quotient topology, and let $\pi\colon M \to M/G$ be the quotient map. Let N be any topological space.

1. A map $h\colon M/G \to N$ is continuous if and only if $h \circ \pi\colon M \to N$ is continuous.

2. If $f\colon M \to N$ is continuous and constant on orbits, that is $f(g \cdot p) = f(p)$ for all $g \in G$ and $p \in M$, then there is a unique continuous map $\overline{f}\colon M/G \to N$ with $\overline{f} \circ \pi = f$:

$$\begin{tikzcd} M \arrow[d, "\pi"'] \arrow[dr, "f"] & \\ M/G \arrow[r, dashed, "\overline{f}"'] & N \end{tikzcd}$$

***Proof*:**

1. Since π is continuous, if h is continuous then so is $h \circ \pi$. Conversely suppose $h \circ \pi$ is continuous and let $V \subseteq N$ be open. Then

$$\pi^{-1}\big(h^{-1}(V)\big) = (h \circ \pi)^{-1}(V)$$

is open in M, so by the definition of the quotient topology $h^{-1}(V)$ is open in M/G. As V was arbitrary, h is continuous.

2. The orbits are exactly the fibres of π, so f is constant on orbits if and only if it is constant on the fibres of π. Since π is surjective, setting $\overline{f}(G \cdot p) = f(p)$ is well defined and is the only map with $\overline{f} \circ \pi = f$. This map is continuous: $\overline{f} \circ \pi = f$ is continuous, so part (1) applies to $h = \overline{f}$, completing the proof.

The universal property is the device by which a quotient is recognized. To identify M/G with a familiar space N, one produces a continuous map $f\colon M \to N$ constant on orbits, and proposition 4.1.6 descends it to $\overline{f}\colon M/G \to N$ with no further work; the descended map is automatically continuous.

Whether $\overline{f}$ is a homeomorphism, however, is a separate question the universal property does not settle, and in practice it is closed under a compactness argument: a continuous bijection from a compact space to a Hausdorff space is a homeomorphism ([7, making continuous function homeomorphisms]). This division of labour, descent for free and homeomorphism by compactness, recurs in every identification below.

Example 4.1.7: Orbit Spaces of Group Actions

1. *The circle as* $\mathbb{R}/\mathbb{Z}$. The group $\mathbb{Z}$ acts on $\mathbb{R}$ by translation, $n \cdot x = x + n$. Each stabilizer is trivial, so the action is free; it is not transitive, its orbits being the cosets $x + \mathbb{Z}$. The map $x \mapsto e^{2\pi i x}$ from $\mathbb{R}$ to the unit circle $S^1 \subseteq \mathbb{C}$ is continuous, surjective, and constant on orbits since $e^{2\pi i(x+n)} = e^{2\pi i x}$, so proposition 4.1.6 descends it to a continuous bijection $\mathbb{R}/\mathbb{Z} \to S^1$ (injective because $e^{2\pi i x} = e^{2\pi i y}$ forces $x - y \in \mathbb{Z}$). Because π maps the compact interval $[0, 1]$ onto $\mathbb{R}/\mathbb{Z}$, the orbit space is compact, and a continuous bijection from a compact space onto the Hausdorff S^1 is a homeomorphism; hence $\mathbb{R}/\mathbb{Z} \cong S^1$.

2. *Projective space as* $S^n/(\mathbb{Z}/2)$. The group $\mathbb{Z}/2 = \{\pm 1\}$ acts on the sphere $S^n \subseteq \mathbb{R}^{n+1}$ by the antipodal map, $(-1) \cdot x = -x$. Since $0 \notin S^n$, no point equals its antipode, so the stabilizers are trivial and the action is free; the orbits are the antipodal pairs $\{x, -x\}$, so it is not transitive. The orbit space is the real projective space $\mathbb{RP}^n$ of chapter 1, and $\pi\colon S^n \to \mathbb{RP}^n$ is its standard two-to-one quotient. This action is properly discontinuous, the hypothesis put to work in section 4.2.

3. *The Hopf action on* S^3. The circle $G = S^1 = \{\lambda \in \mathbb{C} \mid |\lambda| = 1\}$ acts on $S^3 \subseteq \mathbb{C}^2$ by $\lambda \cdot (z_1, z_2) = (\lambda z_1, \lambda z_2)$, a smooth action since scalar multiplication is smooth. It is free: if $(\lambda z_1, \lambda z_2) = (z_1, z_2)$ with $(z_1, z_2) \neq (0, 0)$, then $\lambda z_i = z_i$ for some nonzero z_i, forcing $\lambda = 1$. The orbits are the circles $\{(\lambda z_1, \lambda z_2) \mid |\lambda| = 1\}$, so the action is not transitive. Two points of S^3 share an orbit exactly when they differ by a unit complex scalar, and rescaling any nonzero $(z_1, z_2) \in \mathbb{C}^2$ to unit norm shows every complex line meets S^3 in one such orbit; hence the orbit space is the complex projective line $\mathcal{P}^1$ of chapter 1, diffeomorphic to S^2. The quotient map $\pi\colon S^3 \to S^2$ is the Hopf map.

4. *A linear group on* $\mathbb{R}^n$. The group $\mathrm{GL}_n(\mathbb{R})$ acts on $\mathbb{R}^n$ by $(A, x) \mapsto Ax$. The action is smooth and effective, since $Ax = x$ for all x forces $A = I$, but it is not free (every A fixes 0) and not transitive on all of $\mathbb{R}^n$ (no A moves 0); it has exactly two orbits, $\{0\}$ and $\mathbb{R}^n \setminus \{0\}$, and is transitive on the latter. Restricting to the orthogonal group $\mathrm{O}(n)$ refines the picture: the orbits are the spheres $\{|x| = r\}$, and on a single sphere S^{n-1} the action is transitive with the stabilizer of a point a copy of $\mathrm{O}(n-1)$. By (4.1) this presents S^{n-1} as a coset space $\mathrm{O}(n)/\mathrm{O}(n-1)$, an identification made smooth in section 4.4.

Three of these four actions are free but not transitive, and they are the ones that will yield smooth quotients in the sections to follow: translation, the antipodal action, and the Hopf action each collapse a manifold onto a manifold of strictly lower dimension, the fibres of π being the orbits. The linear action fails both conditions, and the failure shows what they exclude. It is not free at the origin, and the origin is exactly the point whose orbit is a single point rather than a copy of the others, so the orbit space $\mathbb{R}^n/\mathrm{GL}_n(\mathbb{R})$ is the two-point set $\{[0], [\text{rest}]\}$ with a non-Hausdorff topology. Freeness excludes such collapsed orbits, and a second condition, properness, will exclude the topological pathologies that freeness alone leaves open; both are taken up in section 4.2.

Exercise 4.1.1

1. Let $G \curvearrowright M$ be a smooth action. Show that each $\theta_g \colon M \to M$ is a diffeomorphism and that $g \mapsto \theta_g$ is a group homomorphism $G \to \operatorname{Diff}(M)$. Deduce that the action is effective if and only if this homomorphism is injective, and that its kernel is $\bigcap_{p \in M} G_p$.

2. Let $\mathbb{Z}$ act on $\mathbb{R}^2$ by $n \cdot (x, y) = (x + n, y)$. Determine the orbits and stabilizers, decide whether the action is free and whether it is transitive, and identify the orbit space $\mathbb{R}^2/\mathbb{Z}$ with the cylinder $S^1 \times \mathbb{R}$.

3. Let $\mathrm{SO}(2)$ act on $\mathbb{R}^2$ by rotation about the origin. Describe the orbits and the stabilizer of each point, and decide whether the action is free, effective, or transitive. Using proposition 4.1.6, identify the orbit space $\mathbb{R}^2/\mathrm{SO}(2)$ with the half-line $[0, \infty)$.

4. Let a group G act on itself by conjugation, $g \cdot x = gxg^{-1}$. Identify the orbits and stabilizers in group-theoretic terms, and identify the kernel of the homomorphism $G \to \operatorname{Diff}(G)$. For which groups G is this action free, and for which is it effective?

5. Give a complete proof, via proposition 4.1.6, that $\mathbb{R}/\mathbb{Z}$ is homeomorphic to S^1, making explicit where compactness of the source and the Hausdorff property of the target are used.

4.2 Proper Actions and Covering Quotients

An action furnishes a candidate quotient, the orbit space M/G of definition 4.1.5, but the quotient topology alone controls nothing: M/G can fail to be Hausdorff, and even when the underlying set is a manifold the projection π need not be smooth in any usable sense. This section isolates the two conditions that tame the orbit space. *Properness* of the action is what forces M/G to be Hausdorff, and in the discrete case it specializes to a local separation condition under which $\pi \colon M \to M/\Gamma$ is a covering map. The general quotient manifold theorem, where G is a positive-dimensional Lie group, is taken up in section 4.3; here the payoff is the self-contained discrete case, which already builds the tori, the projective spaces, and the mapping tori.

The obstruction to a Hausdorff quotient is that two distinct orbits can be arbitrarily close: a sequence lying in one orbit may converge to a point of another. Properness rules this out by constraining how the group can move a compact set.

Definition 4.2.1: Proper Action

Let G be a topological group acting continuously on a manifold M. The action is *proper* if the map

$$\Theta \colon G \times M \to M \times M, \qquad (g, p) \mapsto (g \cdot p,\, p)$$

is a proper map, that is, $\Theta^{-1}(K)$ is compact for every compact $K \subseteq M \times M$.

The image of Θ is the *orbit relation*

$$R = \{(g \cdot p,\, p) \mid g \in G,\ p \in M\} \subseteq M \times M$$

the set of pairs sharing an orbit; R is exactly the equivalence relation whose quotient is M/G. Properness is a compactness bound on how far G can carry a compact set back onto itself: unwinding the definition, the action is proper if and only if for every compact $K \subseteq M$ the return set $G_K = \{g \in G \mid (g \cdot K) \cap K \neq \emptyset\}$ has compact closure, which for a discrete group says G_K is finite. Two families of proper actions recur below. Every action of a compact group G is proper, since $\Theta^{-1}(K)$ is a closed subset of $G \times \mathrm{pr}_2(K)$, a product of compact sets; in particular every finite group acts properly. The translation action of $\mathbb{Z}^n$ on $\mathbb{R}^n$, $k \cdot x = x + k$, is proper: a compact K lies in a ball of radius ρ, and $(k \cdot K) \cap K \neq \emptyset$ forces $|k| \leq 2\rho$, leaving finitely many k.

Whether M/G is Hausdorff is decided entirely by the orbit relation, because the projection π is an open map. This is the tool that converts the geometric condition of properness into the topological one of separation.

Proposition 4.2.2: Hausdorff Criterion for Orbit Spaces

Let G act continuously on a manifold M, with orbit relation $R = \{(g \cdot p,\, p) \mid g \in G,\ p \in M\}$.

1. M/G is Hausdorff if and only if R is closed in $M \times M$.
2. If the action is proper, then R is closed, so a proper action has Hausdorff orbit space.

***Proof*:**
First, π is open. For open $U \subseteq M$,

$$\pi^{-1}(\pi(U)) = \bigcup_{g \in G} g \cdot U$$

because $\pi(x) = \pi(y)$ holds iff $y = g \cdot x$ for some g. Each $g \cdot U$ is open, as $p \mapsto g \cdot p$ is a homeomorphism with inverse $p \mapsto g^{-1} \cdot p$; so the union is open, and by definition of the quotient topology $\pi(U)$ is open.

1. Suppose R is closed, and take $[p] \neq [q]$ in M/G. Then p and q lie in different orbits, so $(p, q) \notin R$. As R is closed there is a basic open box $U \times V \ni (p, q)$ with $(U \times V) \cap R = \emptyset$. The sets $\pi(U)$ and $\pi(V)$ are open (since π is open) and contain $[p]$ and $[q]$; they are disjoint, for if $[u] = [v]$ with $u \in U$, $v \in V$ then $(u, v) \in R \cap (U \times V) = \emptyset$. Thus M/G is Hausdorff. Conversely, suppose M/G is Hausdorff, so its diagonal Δ is closed in $(M/G) \times (M/G)$. Since $(a, b) \in R$ iff $\pi(a) = \pi(b)$, one has $R = (\pi \times \pi)^{-1}(\Delta)$, the preimage of a closed set under a continuous map, hence closed.
2. The product $M \times M$ is locally compact Hausdorff, and a proper continuous map into a locally compact Hausdorff space is closed: if A is closed in the domain and $y \in \overline{\Theta(A)}$, choose a compact neighbourhood K of y. Every neighbourhood of y may be shrunk into the interior of K and still meets $\Theta(A)$, so $y \in \overline{\Theta(A) \cap K}$, and $\Theta(A) \cap K \subseteq \Theta(A \cap \Theta^{-1}(K))$, which is compact, being the continuous image of a closed subset of the compact set $\Theta^{-1}(K)$, and hence closed. Therefore $y \in \Theta(A)$. Thus Θ is a closed map, and $R = \Theta(G \times M)$ is closed. Part (1) then gives that M/G is Hausdorff, completing the proof.

Properness therefore does more than give a Hausdorff quotient; it delivers closedness of R through one compactness argument, the mechanism theorem 4.3.1 will exploit. For the separation question only the closedness of R matters, and the sharpest way to see how it can fail is an action whose orbit relation is not closed.

Example 4.2.3: A Non-Hausdorff Orbit Space
Let $M = \mathbb{R}^2 \setminus \{0\}$ and let $\mathbb{Z}$ act smoothly by

$$n \cdot (x, y) = (2^n x,\ 2^{-n} y)$$

each n acting by the linear diffeomorphism $\operatorname{diag}(2^n, 2^{-n})$. The action is free: $2^n x = x$ and $2^{-n} y = y$ with $(x, y) \neq 0$ force $n = 0$. Its orbit relation is not closed. For $k \geq 1$ set $p_k = (2^{-k}, 1)$; then $k \cdot p_k = (2^{-k} \cdot 2^k,\ 2^{-k}) = (1, 2^{-k})$, so

$$(k \cdot p_k,\ p_k) = \big((1, 2^{-k}),\ (2^{-k}, 1)\big) \in R$$

As $k \to \infty$ one has $p_k \to (0, 1)$ and $k \cdot p_k \to (1, 0)$, both points of M, so $\big((1, 0), (0, 1)\big) \in \overline{R}$. Yet $(1, 0)$ and $(0, 1)$ share no orbit: $g \cdot (0, 1) = (0, 2^{-g})$ has vanishing first coordinate and so never equals $(1, 0)$. Hence $\big((1, 0), (0, 1)\big) \notin R$, the relation R is not closed, and by proposition 4.2.2 the orbit space $M/\mathbb{Z}$ is not Hausdorff: the points $[(1, 0)]$ and $[(0, 1)]$ have no disjoint neighbourhoods. This $\mathbb{Z}$-action is free and, as definition 4.2.4 makes checkable, properly discontinuous, while it is not proper. Proper discontinuity alone is thus not enough for a manifold quotient; Hausdorffness has to be secured separately.

For the discrete quotients the properness condition takes a local form that can be tested one point at a time, by asking that the group carry a small neighbourhood entirely off itself.

Definition 4.2.4: Properly Discontinuous Action

An action of a discrete group Γ on a manifold M is *properly discontinuous* if every $p \in M$ has a neighbourhood U such that

$$(g \cdot U) \cap U = \emptyset \quad \text{for all } g \in \Gamma,\ g \neq e$$

The condition forces the action to be free: if $g \cdot p = p$ for some $g \neq e$ then $p \in (g \cdot U) \cap U$, against the definition. So a properly discontinuous action is automatically free, and "free and properly discontinuous" names one hypothesis twice. Such a U is disjoint from all its nontrivial translates, and this is precisely the local picture of a covering: the translates $\{g \cdot U\}_{g \in \Gamma}$ are pairwise disjoint, since $(g \cdot U) \cap (h \cdot U) \neq \emptyset$ gives $(h^{-1} g) \cdot U \cap U \neq \emptyset$, hence $g = h$, and Γ permutes them freely. Proper discontinuity is the local part of properness: it is implied by a free proper action but is strictly weaker, and the action of example 4.2.3 is properly discontinuous yet not proper, which is exactly why its quotient loses Hausdorffness. What proper discontinuity does control unaided is the projection π. The translation action of $\mathbb{Z}$ on $\mathbb{R}$ is properly discontinuous: for $U = (p - \frac{1}{2}, p + \frac{1}{2})$ the translate $U + n$ misses U whenever $n \neq 0$; balls of radius $\frac{1}{2}$ give the same for $\mathbb{Z}^n$ on $\mathbb{R}^n$.

The quotient map of a properly discontinuous action is the differential-topological form of a covering. Recording the target notion first turns the theorem into an identification.

Definition 4.2.5: Smooth Covering Map

A *smooth covering map* is a smooth surjection $\pi\colon E \to M$ of smooth manifolds such that every $p \in M$ has a connected open neighbourhood U that is *evenly covered*: the preimage $\pi^{-1}(U)$ is a disjoint union of open subsets of E, the *sheets*, each carried diffeomorphically onto U by π. Then E is a *covering manifold* of the *base* M; when E is connected and simply connected it is the *universal cover*.

Restricted to a sheet, π is a diffeomorphism, so a smooth covering map is a surjective local diffeomorphism, in particular a surjective submersion. It therefore carries the characteristic property of proposition 3.1.23: a map out of the base is smooth iff its composite with π is. The topological theory of coverings, path lifting and the classification of covers by subgroups of the fundamental group, is developed in [7, paths can be lifted, Homotopies Can Be Lifted, The Lifting Criterion, Classification of Coverings]; what is added here is that all of it runs in the smooth category. The exponential $\pi\colon \mathbb{R} \to S^1$, $t \mapsto e^{2\pi it}$, is a smooth covering: any proper open arc U has $\pi^{-1}(U)$ a disjoint union of intervals, each mapped diffeomorphically onto U. The map $z \mapsto z^k$ on S^1 is a k-sheeted smooth covering.

The topological theory just cited also settles the largest cover, the simply connected one, and its smooth upgrade is immediate: a covering map is a local homeomorphism, so the smooth structure of the base transports across it.

Theorem 4.2.6: Existence of the Smooth Universal Cover

Let M be a connected smooth manifold. Then M has a smooth universal cover: a simply connected smooth manifold $\widetilde{M}$ together with a smooth covering map $\pi\colon \widetilde{M} \to M$. It is unique up to isomorphism, in that for any other smooth universal cover $\pi'\colon \widetilde{M}' \to M$ there is a diffeomorphism $\Phi\colon \widetilde{M} \to \widetilde{M}'$ with $\pi' \circ \Phi = \pi$.

***Proof*:**
Being connected, M is path-connected, and being locally Euclidean it is locally path-connected and semilocally simply connected: each point has a chart neighbourhood diffeomorphic to a ball, in which every loop contracts, hence a fortiori contracts in M. The topological existence theorem ([7, Existence of a Universal Cover]) therefore produces a covering map $p\colon \widetilde{M} \to M$ with $\widetilde{M}$ simply connected. As a covering space of a manifold it is again a topological manifold: it is Hausdorff and locally Euclidean like M, and it is second countable because a manifold has countable fundamental group, so the cover has countably many sheets over a second-countable base.

The smooth structure of M transports across p. Each point $\widetilde{q}$ has an open neighbourhood $\widetilde{U}$ carried by p homeomorphically onto an open set inside a chart (W, φ) of M, and then $\varphi \circ p|_{\widetilde{U}}$ is a homeomorphism of $\widetilde{U}$ onto an open subset of $\mathbb{R}^n$, hence a chart on $\widetilde{M}$. Two such charts have transition $(\varphi' \circ p) \circ (\varphi \circ p)^{-1} = \varphi' \circ \varphi^{-1}$ on their overlap, since the two restrictions of p cancel there, a transition of M and hence smooth. These charts form a smooth atlas, in which p reads locally as id, a local diffeomorphism. Every evenly covered neighbourhood of the topological cover thus has its sheets carried diffeomorphically onto it, so $\pi = p$ is a smooth covering map and $\widetilde{M}$, still simply connected, is a smooth universal cover.

For uniqueness, the topological uniqueness theorem ([7, Uniqueness of the Universal Cover]) gives a homeomorphism $\Phi\colon \widetilde{M} \to \widetilde{M}'$ with $\pi' \circ \Phi = \pi$. Near each point Φ equals $(\pi'|_S)^{-1} \circ \pi$ for the sheet S of π' containing its image, the inverse of a diffeomorphism onto an open set composed with the smooth π, hence smooth. The same argument applied to Φ^{-1}, which satisfies $\pi \circ \Phi^{-1} = \pi'$, makes Φ a diffeomorphism, completing the proof.

These pieces assemble into the discrete quotient theorem: a properly discontinuous action produces a covering, and the covering is smooth because π is a local diffeomorphism onto whose base the smooth structure of M descends.

Theorem 4.2.7: Covering Quotient Theorem

Let Γ be a discrete group acting smoothly and properly discontinuously on a smooth n-manifold M, and suppose the orbit space M/Γ is Hausdorff (by proposition 4.2.2, this holds whenever the action is proper, in particular whenever Γ is finite). Then M/Γ carries a unique smooth structure for which $\pi\colon M \to M/\Gamma$ is a smooth covering map, and with it M/Γ is a smooth n-manifold.

***Proof*:**
Write $\theta_g\colon M \to M$ for the diffeomorphism $p \mapsto g \cdot p$, with inverse $\theta_{g^{-1}}$; smoothness of the action makes each θ_g a diffeomorphism (definition 4.1.2).

Step 1: π is open. For open $U \subseteq M$,

$$\pi^{-1}(\pi(U)) = \bigcup_{g\in\Gamma} \theta_g(U)$$

since $\pi(x) = \pi(y)$ iff $y = \theta_g(x)$ for some g. Each $\theta_g(U)$ is open, so $\pi^{-1}(\pi(U))$ is open and $\pi(U)$ is open. Thus π is an open continuous surjection.

Step 2: even covering. Fix $p \in M$ and, by proper discontinuity, a connected neighbourhood $U \ni p$ with $\theta_g(U) \cap U = \emptyset$ for $g \neq e$. The translates $\{\theta_g(U)\}_{g\in\Gamma}$ are pairwise disjoint: $\theta_g(U) \cap \theta_h(U) \neq \emptyset$ yields $\theta_{h^{-1}g}(U) \cap U \neq \emptyset$, forcing $h^{-1}g = e$. The restriction $\pi|_U\colon U \to \pi(U)$ is a continuous open bijection (surjective by construction, and injective because $\pi(x) = \pi(y)$ with $x, y \in U$ gives $y = \theta_g(x) \in \theta_g(U) \cap U$, so $g = e$ and $x = y$), hence a homeomorphism onto the open set $\pi(U)$, which is connected as a continuous image of U. For each g the map $\pi|_{\theta_g(U)} = \pi|_U \circ \theta_{g^{-1}}$ is likewise a homeomorphism onto $\pi(U)$, and $\pi^{-1}(\pi(U)) = \coprod_{g\in\Gamma} \theta_g(U)$. So $\pi(U)$ is evenly covered and π is a local homeomorphism.

Step 3: M/Γ is a topological n-manifold. It is Hausdorff by hypothesis. It is second countable: for a countable basis $\{B_i\}$ of M, the open sets $\{\pi(B_i)\}$ form a countable basis, because any open $V \subseteq M/\Gamma$ has $\pi^{-1}(V) = \bigcup\{B_i \mid B_i \subseteq \pi^{-1}(V)\}$ and therefore $V = \pi(\pi^{-1}(V)) = \bigcup\{\pi(B_i) \mid B_i \subseteq \pi^{-1}(V)\}$. It is locally Euclidean: with U as in Step 2 chosen inside a chart domain, $(\pi|_U)^{-1}$ carries $\pi(U)$ homeomorphically onto the locally Euclidean U.

Step 4: descent of the smooth structure. For each p take U as in Step 2 inside a smooth chart (U, φ) of M (intersecting with a chart domain preserves the disjointness) and set

$$\widetilde{\varphi} = \varphi \circ (\pi|_U)^{-1}\colon \pi(U) \to \varphi(U) \subseteq \mathbb{R}^n$$

a homeomorphism onto an open set, hence a chart on M/Γ. Let (U, φ) and (U', φ') be two such with $\pi(U) \cap \pi(U') \neq \emptyset$. Fix $[q]$ in the overlap, and let $x = (\pi|_U)^{-1}([q]) \in U$ and

$x' = (\pi|_{U'})^{-1}([q]) \in U'$; since $[x] = [x']$ there is a unique $g \in \Gamma$ with $x' = \theta_g(x)$, the uniqueness by freeness. For $[r]$ near $[q]$ the points $(\pi|_{U'})^{-1}([r])$ and $\theta_g\big((\pi|_U)^{-1}([r])\big)$ both lie in $\theta_g(U)$ and map to $[r]$; as $\pi|_{\theta_g(U)}$ is injective they agree. Hence, on a neighbourhood of $[q]$,

$$\widetilde{\varphi}' \circ \widetilde{\varphi}^{-1} = \varphi' \circ (\pi|_{U'})^{-1} \circ \pi|_U \circ \varphi^{-1} = \varphi' \circ \theta_g \circ \varphi^{-1}$$

a composition of smooth maps, so the transition is smooth and the charts $\{(\pi(U), \widetilde{\varphi})\}$ form a smooth atlas.

Step 5: π is a smooth covering. In the charts (U, φ) and $(\pi(U), \widetilde{\varphi})$ the projection reads as $\widetilde{\varphi} \circ \pi \circ \varphi^{-1} = \varphi \circ (\pi|_U)^{-1} \circ \pi \circ \varphi^{-1} = \mathrm{id}$ on $\varphi(U)$, so π is a local diffeomorphism. By Step 2 each $\pi(U)$ is evenly covered, its sheets $\theta_g(U)$ each carried onto $\pi(U)$ by the diffeomorphism $\pi|_U \circ \theta_{g^{-1}}$. Thus π is a smooth covering map, and M/Γ, Hausdorff and second countable and locally Euclidean with this structure, is a smooth n-manifold. For uniqueness, any smooth structure making π a local diffeomorphism has each $\widetilde{\varphi} = \varphi \circ (\pi|_U)^{-1}$ as a smooth chart, so its atlas is compatible with the one built here and defines the same structure, completing the proof.

The theorem reduces a range of manifold constructions to a checklist: produce a discrete group acting smoothly and freely, verify the local disjointness of proper discontinuity, and confirm Hausdorffness, most often by proving the action proper or by noting that Γ is finite. The quotient is then a manifold of the *same* dimension as M, and π is a covering, so every local computation downstairs lifts to M. The standard flat and elliptic quotients are all instances.

Example 4.2.8: Tori, Projective Spaces, and Mapping Tori

1. The lattice $\mathbb{Z}^n$ acts on $\mathbb{R}^n$ by $k \cdot x = x + k$. The action is smooth, free, and properly discontinuous (balls of radius $\frac{1}{2}$), and proper, since for compact $K \subseteq \mathbb{R}^n$ only finitely many k satisfy $(K + k) \cap K \neq \emptyset$; so $\mathbb{R}^n/\mathbb{Z}^n$ is Hausdorff. By theorem 4.2.7 it is a smooth n-manifold, the torus T^n of chapter 1, and $\pi\colon \mathbb{R}^n \to T^n$ is a smooth universal covering. The map $x \mapsto (e^{2\pi i x^1}, \ldots, e^{2\pi i x^n})$ descends to a diffeomorphism $T^n \cong (S^1)^n$.

2. The group $\mathbb{Z}/2\mathbb{Z} = \{e, a\}$ acts on S^n by the antipodal map $a \cdot x = -x$. It is smooth and free (no $x \in S^n$ has $x = -x$), and properly discontinuous: the open hemisphere $U_x = \{y \in S^n \mid \langle y, x\rangle > 0\}$ contains x, while $a \cdot U_x = \{y \mid \langle y, x\rangle < 0\}$ is disjoint from it. Being finite the action is proper, so $S^n/(\mathbb{Z}/2\mathbb{Z})$ is Hausdorff; by theorem 4.2.7 it is a smooth n-manifold, real projective space $\mathbb{RP}^n$, with $\pi\colon S^n \to \mathbb{RP}^n$ a smooth double cover. This recovers the smooth structure on $\mathbb{RP}^n$ from chapter 1, now presented as a covering quotient.

3. For a diffeomorphism $\varphi\colon N \to N$ of a smooth manifold, let $\mathbb{Z}$ act on $N \times \mathbb{R}$ by $k \cdot (p, t) = (\varphi^k(p),\ t+k)$. The action is smooth, free (the coordinate t is shifted by k, so $k \neq 0$ moves every point), properly discontinuous (take $U = N \times (t - \frac{1}{2}, t + \frac{1}{2})$, whose translates shift the interval off itself), and proper (a compact K lies in $N \times [-\rho, \rho]$, and $(k \cdot K) \cap K \neq \emptyset$ forces $|k| \leq 2\rho$). By theorem 4.2.7 the *mapping torus* $N_\varphi = (N \times \mathbb{R})/\mathbb{Z}$ is a smooth manifold of dimension $\dim N + 1$, and π is a smooth covering. The $\mathbb{Z}$-invariant map $(p, t) \mapsto [t] \in \mathbb{R}/\mathbb{Z} \cong S^1$ descends (proposition 4.1.6) to a smooth surjection $N_\varphi \to S^1$ realizing N_φ as fibred over the circle with fibre N. With $N = \mathbb{R}$ and $\varphi(x) = -x$ this produces the open Möbius band; with $\varphi = \mathrm{id}$, the trivial cylinder $N \times S^1$.

Exercise 4.2.1

1. Let a finite group Γ act smoothly and freely on a smooth n-manifold M. Show that the action is proper and properly discontinuous, and conclude that M/Γ is a smooth n-manifold and π an $|\Gamma|$-sheeted smooth covering.

2. For the antipodal action of $\mathbb{Z}/2\mathbb{Z}$ on S^n, write down an explicit evenly covered neighbourhood of an arbitrary point of $\mathbb{RP}^n$ and verify that $\pi\colon S^n \to \mathbb{RP}^n$ restricts to a diffeomorphism on each sheet.

3. Show that if G acts properly on M, then for every compact $K \subseteq M$ the return set $G_K = \{g \in G \mid (g \cdot K) \cap K \neq \emptyset\}$ is compact. (This is the compact-return form of properness quoted after definition 4.2.1.)

4. Let $\varphi\colon N \to N$ be a diffeomorphism. Verify that the $\mathbb{Z}$-action $k \cdot (p, t) = (\varphi^k(p), t + k)$ on $N \times \mathbb{R}$ is free, properly discontinuous, and proper, and deduce that the mapping torus N_φ is a smooth manifold. Show that the second-coordinate projection descends to a smooth surjection $N_\varphi \to S^1$.

5. Let $\pi\colon E \to M$ be a smooth covering map with M connected. Show that π is a submersion and that the number of sheets $|\pi^{-1}(q)|$ is the same for every $q \in M$.

4.3 The Quotient Manifold Theorem

Section 4.2 settled the case of a discrete group, where a free proper action makes the quotient map a covering and the orbit space a manifold of the same dimension. Dropping discreteness is the task of this section. When a Lie group G acts freely and properly on M, the orbit space M/G is again a smooth manifold, now of dimension $\dim M - \dim G$, and $\pi\colon M \to M/G$ is a submersion. The general construction rests on two facts of Lie-group structure theory, stated and cited to [16]; the descent of smooth data across π is the submersion machinery of chapter 3, applied without change.

The discrete case built charts on M/Γ from evenly covered neighbourhoods, one for each point upstairs. For a positive-dimensional G the orbits are themselves positive-dimensional, so a chart on M/G can no longer come from an open set of M mapped bijectively down; it must instead be cut out *transverse* to the orbits, meeting each nearby orbit in a single point. Freeness forces every orbit to have the same dimension $\dim G$, so a transverse slice has a well-defined codimension; properness makes M/G Hausdorff and lets the slices patch. The theorem packages these two hypotheses into a single statement.

Theorem 4.3.1: Quotient Manifold Theorem

Let G be a Lie group acting smoothly, freely, and properly on a smooth manifold M. Then the orbit space M/G carries a smooth manifold structure of dimension $\dim M - \dim G$ for which the quotient map $\pi\colon M \to M/G$ is a smooth submersion.

***Proof*:**
The discrete case is proved in full; the positive-dimensional case is reduced to two inputs from Lie-group structure theory, cited to [16], followed by the submersion descent of chapter 3.

The discrete case ($\dim G = 0$*).* Here G is a discrete group Γ acting smoothly and freely on M, and properness is the assertion that the map $\Gamma \times M \to M \times M$, $(g, p) \mapsto (g \cdot p, p)$, is proper.

Two consequences follow.

First, the action is properly discontinuous. Fix $p \in M$. Since M is locally compact Hausdorff, choose a compact neighbourhood K of p. Properness makes $\{g \in \Gamma \mid gK \cap K \neq \emptyset\}$ finite; call its nonidentity members $g_1, \ldots, g_k$. Freeness gives $g_i \cdot p \neq p$ for each i, so, using the Hausdorff property and continuity of $x \mapsto g_i \cdot x$, pick an open $V_i \ni p$ with $g_i V_i \cap V_i = \emptyset$. Set $U = \operatorname{int} K \cap \bigcap_i V_i$, an open neighbourhood of p. For any $g \neq e$: if $gU \cap U \neq \emptyset$ then $gK \cap K \neq \emptyset$, so $g = g_i$ for some i, contradicting $g_i U \cap U \subseteq g_i V_i \cap V_i = \emptyset$. Thus $gU \cap U = \emptyset$ for all $g \neq e$, which is the properly discontinuous condition (definition 4.2.4).

Second, M/Γ is Hausdorff. A proper map into a locally compact Hausdorff space is closed, so the image of $(g, p) \mapsto (g \cdot p, p)$, which is the orbit relation $\{(g \cdot p, p) \mid g \in \Gamma,\ p \in M\}$, is closed in $M \times M$. By proposition 4.2.2, M/Γ is Hausdorff.

The action is now free and properly discontinuous, so theorem 4.2.7 makes $\pi \colon M \to M/\Gamma$ a smooth covering map and equips M/Γ with an atlas pulled back through the evenly covered charts; the Hausdorff property just established upgrades this locally Euclidean space to a manifold. A covering map is a local diffeomorphism (definition 4.2.5), hence a submersion, and preserves dimension, so $\dim M/\Gamma = \dim M = \dim M - \dim G$. This settles the discrete case.

The positive-dimensional case. Two facts about the action of a positive-dimensional Lie group are the input from [16], where the structure theory of Lie groups and their actions is developed; only the smooth-manifold consequences are used here.

1. Because G is a Lie group and the action is smooth, free, and proper, each orbit $G \cdot p$ is a properly embedded submanifold of M and the orbit map $g \mapsto g \cdot p$ is a diffeomorphism of G onto it; in particular every orbit has dimension $\dim G$.
2. *(Slice.)* Each p has an embedded submanifold $S \ni p$ of dimension $k = \dim M - \dim G$, transverse to the orbit, together with a G-invariant open neighbourhood of $G \cdot p$ and a diffeomorphism $G \times S \to G \cdot S$ under which the action becomes translation in the first factor, $g \cdot (h, s) = (gh, s)$.

Granting these, the remaining work is the submersion descent, which is in scope. Read off the local model $G \times S \cong G \cdot S$: the restriction of π to this set is the projection $(h, s) \mapsto s$ followed by $S \hookrightarrow M/G$, so π carries the open set $G \cdot S$ onto the image of S and is an open map near $G \cdot p$; as p was arbitrary, π is open. The slice S meets each orbit of $G \cdot S$ exactly once, so $\pi|_S \colon S \to M/G$ is a continuous open injection, hence a homeomorphism onto the open set $\pi(S)$; composing $(\pi|_S)^{-1}$ with a chart of S gives a chart on M/G, and these cover M/G with k-dimensional charts. On an overlap $\pi(S) \cap \pi(S')$ the transition sends $s \in S$ to the unique point of S' on its orbit, namely $(\pi|_{S'})^{-1}(\pi(s))$; in the two local models this is a composition of the smooth trivializations, hence smooth, so the charts form a smooth atlas. Properness makes the orbit relation closed exactly as in the discrete case, so M/G is Hausdorff by proposition 4.2.2, and M/G is second countable as the image of the second-countable M under the open surjection π. Thus M/G is a smooth k-manifold. In the slice charts π is the projection $(h, s) \mapsto s$, a smooth submersion, so $\pi \colon M \to M/G$ is a smooth submersion of the asserted dimension $k = \dim M - \dim G$, completing the proof.

The theorem adds the third and last of the book's manifold-building constructions to charts (chapter 1) and level sets (chapter 3). Its output is a submersion, not a bare topological space, and that is what makes it usable: every statement proved about surjective smooth submersions in chapter 3 now applies to π. In particular a map out of M/G is smooth if and only if its composition with π is (proposition 3.1.23), so smooth functions on the orbit space are exactly the G-invariant smooth functions on M; and a G-invariant smooth map $M \to P$ descends uniquely to a smooth map $M/G \to P$ (theorem 3.1.24). The quotient is thereby characterized by how maps leave it, not by any chart used to build it.

The discrete case already produces the two most familiar quotients. The antipodal action of $\mathbb{Z}/2$ on S^n is free and, being an action of a finite group, proper, so $\mathbb{RP}^n = S^n/(\mathbb{Z}/2)$ is a smooth n-manifold with π a double cover; and the translation action of $\mathbb{Z}^n$ on $\mathbb{R}^n$ is free and proper, so the torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ is a smooth n-manifold with π the universal covering. Both sit inside the theorem as the case $\dim G = 0$.

The atlas built in the proof depended on choices: a slice S at each point and a chart of S. A different choice of slices and charts would build a different atlas. That the resulting smooth structure does not depend on these choices is what justifies speaking of *the* quotient manifold, and it follows from the submersion requirement alone.

Proposition 4.3.2: Uniqueness of the Quotient Smooth Structure

Let G act smoothly, freely, and properly on M. There is at most one smooth manifold structure on M/G for which $\pi\colon M \to M/G$ is a smooth submersion.

***Proof*:**
Suppose two such structures, giving smooth manifolds N_1 and N_2 with the same underlying topological space M/G, and write $\pi_i\colon M \to N_i$ for the common map π regarded as landing in N_i; each π_i is a surjective smooth submersion. Consider the set-theoretic identity $\mathrm{id}\colon N_1 \to N_2$. Its composition with π_1 is $\mathrm{id} \circ \pi_1 = \pi_2$, which is smooth. Since π_1 is a surjective smooth submersion, proposition 3.1.23 makes $\mathrm{id}\colon N_1 \to N_2$ smooth. By the symmetric argument $\mathrm{id}\colon N_2 \to N_1$ is smooth, so id is a diffeomorphism and the two smooth structures coincide, completing the proof.

Uniqueness converts the theorem into a recognition principle. Because the quotient smooth structure is pinned down by the single requirement that π be a submersion, any smooth manifold N equipped with a surjective smooth submersion $M \to N$ whose fibres are exactly the G-orbits must be M/G: the universal property (theorem 3.1.24) produces mutually inverse smooth maps between N and M/G. To identify an orbit space with a familiar manifold, then, it suffices to exhibit one submersion off M that separates the orbits and lands on that manifold. The Hopf fibration (example 4.3.3) is identified with S^2 in exactly this way.

Both hypotheses are necessary, and each fails in a recognizable manner. Freeness controls the dimension of the fibres. Without it the orbit through p has dimension $\dim G - \dim \mathrm{Stab}(p)$, which jumps as the stabilizer grows: the rotation action of $\mathrm{SO}(2)$ on $\mathbb{R}^2$ fixes the origin, where $\mathrm{Stab}(0) = \mathrm{SO}(2)$ and the orbit is a point, while every other orbit is a circle, so the orbit dimension is not constant. The orbit space $\mathbb{R}^2/\mathrm{SO}(2) \cong [0, \infty)$, coordinatized by $r = |x|$, is a manifold with boundary rather than a manifold; the image of the origin is a boundary point with no Euclidean neighbourhood. Properness controls separation. Its failure is precisely the failure of M/G to be Hausdorff, by proposition 4.2.2: a non-proper action can have orbits that accumulate on one another, so no two of them can be prised apart in the quotient. The non-Hausdorff quotient of example 4.2.3 in section 4.2 is the discrete instance, and the dense-orbit example of exercise 4.3.1 (5) below is the continuous one.

One example repays a full computation: the action producing the Hopf fibration, the first fibre bundle whose total space, base, and fibre are all spheres.

Example 4.3.3: The Hopf Fibration $S^3 \to S^2$

Regard $S^3 \subseteq \mathbb{C}^2$ as the unit sphere, $S^3 = \{(w_1, w_2) \in \mathbb{C}^2 \mid |w_1|^2 + |w_2|^2 = 1\}$, and let the circle $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ act by scalar multiplication,

$$z \cdot (w_1, w_2) = (zw_1, zw_2)$$

This is smooth, being the restriction of a smooth action on $\mathbb{C}^2$. It is free: if $(zw_1, zw_2) = (w_1, w_2)$ with $(w_1, w_2) \neq 0$, then $z = 1$. It is proper because S^1 is compact: the action map $S^1 \times S^3 \to S^3 \times S^3$ has compact domain and Hausdorff codomain, so it is closed, and the preimage of a compact set is a closed subset of the compact $S^1 \times S^3$, hence compact. By theorem 4.3.1 the orbit space S^3/S^1 is a smooth manifold of dimension $3 - 1 = 2$ and $\pi: S^3 \to S^3/S^1$ is a smooth submersion.

To name the quotient, note that the orbit of (w_1, w_2) is $S^3 \cap \mathbb{C}(w_1, w_2)$, the intersection of S^3 with the complex line through (w_1, w_2). The orbit space is therefore the set of complex lines through the origin, which is $\mathcal{P}^1$. Precisely, the quotient map $q: \mathbb{C}^2 \setminus \{0\} \to \mathcal{P}^1$ is a surjective smooth submersion (chapter 1) whose fibres are the punctured complex lines. Its restriction $q|_{S^3}$ is constant on the S^1-orbits and has those orbits as its fibres, and it is a submersion: writing $T_w S^3 = \{v \in \mathbb{C}^2 \mid \operatorname{Re}\langle v, w\rangle = 0\}$ for the real orthogonal complement, the kernel of $T_w q$ is the real 2-plane $\mathbb{R}w \oplus \mathbb{R}(iw)$ tangent to the complex line, of which only the line $\mathbb{R}(iw)$ lies in $T_w S^3$; hence $T_w q$ maps the 3-dimensional $T_w S^3$ onto the 2-dimensional $T_{q(w)}\mathcal{P}^1$. By the universal property (theorem 3.1.24) $q|_{S^3}$ descends to a smooth bijection $S^3/S^1 \to \mathcal{P}^1$, and being a bijective submersion between 2-manifolds it is a local diffeomorphism, hence a diffeomorphism. Since $\mathcal{P}^1 \cong S^2$ (chapter 1),

$$\pi: S^3 \to S^2$$

is the *Hopf fibration*. Under the identification $\mathcal{P}^1 \cong S^2 \subseteq \mathbb{C} \times \mathbb{R} \cong \mathbb{R}^3$ it is the *Hopf map*

$$h(w_1, w_2) = \left(2w_1\overline{w_2},\ |w_1|^2 - |w_2|^2\right)$$

whose image lies in S^2 because $|h(w)|^2 = 4|w_1|^2|w_2|^2 + (|w_1|^2 - |w_2|^2)^2 = (|w_1|^2 + |w_2|^2)^2 = 1$, and which is constant on orbits since $zw_1\,\overline{zw_2} = |z|^2 w_1\overline{w_2} = w_1\overline{w_2}$.

The fibres of the Hopf fibration are the great circles $\pi^{-1}(\mathrm{pt})$, and any two of them are linked once in S^3, which is why the bundle is nontrivial: were S^3 diffeomorphic to $S^2 \times S^1$ over S^2, the fibres would be unlinked. The number recording this linking is the first invariant this book computes that distinguishes a bundle from a product, and it returns later in the guise of a degree and of a de Rham class. For now the example shows the theorem: a free proper action of a positive-dimensional group has produced the 2-sphere as the base of a submersion off the 3-sphere.

Exercise 4.3.1

1. Let a compact Lie group G act smoothly on a smooth manifold M. Show that the action is proper. Conclude that if the action is also free, then M/G is a smooth manifold with π a submersion.
2. The antipodal action of $\mathbb{Z}/2 = \{\pm 1\}$ on S^n sends x to $-x$. Show it is free and proper, and deduce from the discrete case of theorem 4.3.1 that $\mathbb{RP}^n = S^n/(\mathbb{Z}/2)$ is a smooth n-manifold with $\pi: S^n \to \mathbb{RP}^n$ a smooth covering map. For which n is it the universal cover?

3. Let G act smoothly, freely, and properly on M, and let $F\colon M \to N$ be a surjective smooth submersion onto a smooth manifold N whose fibres are exactly the G-orbits. Show that N is diffeomorphic to M/G.

4. Fix coprime integers $p \geq 1$ and q, and let $\zeta = e^{2\pi i/p}$. Let $\mathbb{Z}/p$ act on $S^3 \subseteq \mathbb{C}^2$ by $m \cdot (w_1, w_2) = (\zeta^m w_1, \zeta^{qm} w_2)$. Show that the action is free and proper, and conclude that the *lens space* $L(p;q) = S^3/(\mathbb{Z}/p)$ is a smooth 3-manifold with $\pi\colon S^3 \to L(p;q)$ a smooth covering map with p sheets.

5. Fix an irrational α and let $\mathbb{R}$ act on the torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ by $t \cdot [x, y] = [x + t, y + \alpha t]$. Show that the action is free but not proper, and that $T^2/\mathbb{R}$ is not Hausdorff. Which hypothesis of theorem 4.3.1 does this violate?

4.4 Homogeneous Spaces and Examples

The quotient manifold theorem (theorem 4.3.1) turns a free proper action into a smooth quotient. This section takes up the opposite extreme of the orbit structure: an action with a *single* orbit. Such an action recovers its manifold as a coset space, and the coset spaces of the classical groups $\mathrm{O}(n)$ and $\mathrm{U}(n)$ furnish many examples: projective spaces, Grassmannians, Stiefel and flag manifolds.

A transitive action makes every point look like every other, so a manifold carrying one has no distinguished point and its global shape is pinned down by the symmetry group together with a single stabilizer. This is the situation to name.

Definition 4.4.1: Homogeneous Space

Let G be a Lie group. A *homogeneous space* (or *homogeneous G-space*) is a smooth manifold M with a smooth transitive action $G \curvearrowright M$. Transitivity (definition 4.1.4) means $M = G \cdot p$ is a single orbit for any $p \in M$.

The sphere is the first instance. The orthogonal group $\mathrm{O}(n)$ acts on $S^{n-1} \subseteq \mathbb{R}^n$ by $A \cdot x = Ax$; since an orthogonal transformation carries any unit vector to any other, the action is transitive and S^{n-1} is a homogeneous $\mathrm{O}(n)$-space. Fixing the pole e_n, an orthogonal A with $Ae_n = e_n$ also preserves $e_n^\perp = \mathbb{R}^{n-1}$ and acts on it by some $C \in \mathrm{O}(n-1)$, so the stabilizer $\mathrm{Stab}(e_n)$ is the copy $\{\mathrm{diag}(C, 1) \mid C \in \mathrm{O}(n-1)\}$ of $\mathrm{O}(n-1)$. Theorem 4.4.2 reads this off as a diffeomorphism $S^{n-1} \cong \mathrm{O}(n)/\mathrm{O}(n-1)$.

To make the identification $M \cong G/G_p$ precise, two things are needed: that the coset space G/H is a manifold for a closed subgroup H, and that the orbit map through p descends from G to a diffeomorphism of G/G_p. Both are given at once. The first is a descent along the submersion machinery of chapter 3; the Lie-theoretic inputs (that G is a manifold, that closed subgroups are embedded Lie subgroups, and that the relevant action is proper) are cited to [16].

Theorem 4.4.2: Homogeneous Space Theorem

Let G be a Lie group.

1. *(Coset spaces.)* For a closed subgroup $H \leq G$, the coset space G/H carries a unique smooth manifold structure of dimension $\dim G - \dim H$ for which the quotient $\pi\colon G \to G/H$ is a smooth submersion. The left action $G \curvearrowright G/H$, $g \cdot (g'H) = (gg')H$, is smooth and transitive, making G/H a homogeneous space.

2. *(Orbit-stabilizer.)* If $G \curvearrowright M$ is a smooth transitive action and $p \in M$, then the stabilizer $G_p = \mathrm{Stab}(p)$ is a closed subgroup, and the orbit map $g \mapsto g \cdot p$ descends to a G-equivariant diffeomorphism

$$F\colon G/G_p \to M, \qquad F(gG_p) = g \cdot p$$

***Proof*:**

1. Because H is closed, it is an embedded Lie subgroup of G, and the right-translation action $G \times H \to G$, $(g, h) \mapsto gh$, is smooth, free, and proper ([16]). Freeness is immediate, since $gh = g$ forces $h = e$; the orbits of this action are exactly the left cosets gH, so its orbit space is G/H. The quotient manifold theorem (theorem 4.3.1) then makes G/H a smooth manifold of dimension $\dim G - \dim H$ with $\pi\colon G \to G/H$ a smooth submersion, and proposition 4.3.2 makes this structure the unique one with π a submersion.

   For the left action, consider the smooth map $G \times G \to G/H$, $(g, g') \mapsto \pi(gg')$. It is constant on the fibres of the surjective submersion $\mathrm{id}_G \times \pi\colon G \times G \to G \times G/H$, so by the universal property of submersions (theorem 3.1.24) it descends to a smooth map $G \times G/H \to G/H$, which is the action $g \cdot (g'H) = (gg')H$. Any two cosets $g'H$ and $g''H$ are exchanged by left translation by $g''(g')^{-1}$, so the action is transitive, completing the proof of the first part.

2. The orbit map $\theta_p\colon G \to M$, $g \mapsto g \cdot p$, is the restriction of the smooth action to $G \times \{p\}$, hence smooth. Since M is Hausdorff, $\{p\}$ is closed, so $G_p = \theta_p^{-1}(p)$ is a closed subgroup, and part (1) makes G/G_p a smooth manifold with $\pi\colon G \to G/G_p$ a submersion. As $\theta_p(gh) = (gh) \cdot p = g \cdot p = \theta_p(g)$ for $h \in G_p$, the orbit map is constant on the fibres of π, so theorem 3.1.24 descends it to a smooth map $F\colon G/G_p \to M$ with $F(gG_p) = g \cdot p$.

   The map F is a bijection: it is surjective because the action is transitive, and injective because $g \cdot p = g' \cdot p$ gives $g^{-1}g' \in G_p$, hence $gG_p = g'G_p$. It is G-equivariant, since $F(g \cdot (g'G_p)) = F(gg'G_p) = (gg') \cdot p = g \cdot F(g'G_p)$.

   Equivariance forces F to have constant rank. Write ℓ_g for left translation by g on G/G_p and τ_g for the action of g on M, both diffeomorphisms (with inverses $\ell_{g^{-1}}$ and $\tau_{g^{-1}}$). Equivariance is the identity $F \circ \ell_g = \tau_g \circ F$; differentiating at the base point $o = eG_p$ gives $T_{gG_p}F \circ T_o\ell_g = T_{F(o)}\tau_g \circ T_oF$. The outer maps $T_o\ell_g$ and $T_{F(o)}\tau_g$ are isomorphisms, so $\mathrm{rank}\, T_{gG_p}F = \mathrm{rank}\, T_oF$; as every point of G/G_p has the form gG_p, the rank is constant. A smooth bijection of constant rank is a diffeomorphism (chapter 3): injectivity forces the rank to equal $\dim(G/G_p)$, so F is an immersion, surjectivity forces it to equal $\dim M$, so F is a submersion, and a bijective local diffeomorphism is a diffeomorphism. Thus F is a G-equivariant diffeomorphism, as we sought to show.

The theorem is a dictionary between symmetry and shape, read in two directions. Read forward, it manufactures manifolds: any closed subgroup H of a Lie group produces the smooth quotient G/H, whose dimension is the bookkeeping difference $\dim G - \dim H$. Read backward, it recognizes a manifold already in hand: a transitive action exhibits M as G/G_p, so computing $\dim M$ reduces to computing a single stabilizer. The base point p is a choice, and different choices give conjugate stabilizers $G_{g\cdot p} = gG_pg^{-1}$, so the coset space is well defined up to the induced diffeomorphism. Everything below is one application of the backward reading, with G a classical group.

The Grassmannian was built by hand in chapter 1 as an atlas of graph charts; the homogeneous-space description recovers it in a line and, as a bonus, exhibits its compactness.

Example 4.4.3: The Grassmannian as $\mathrm{O}(n)/(\mathrm{O}(k) \times \mathrm{O}(n-k))$

Let $\mathrm{Gr}(k,n) = G_k(\mathbb{R}^n)$ be the Grassmannian of k-planes in $\mathbb{R}^n$ (definition 1.2.17). The orthogonal group $\mathrm{O}(n)$ acts on it by $A \cdot W = A(W)$; in the graph charts this action is given by rational formulas in the entries of A, so it is smooth. Here $\mathrm{O}(n)$ is used as a Lie group: its manifold structure is the regular-value computation of exercise 3.2.1 (2) (revisited in example 4.4.5 below), and its full group structure is developed in [16].

The action is transitive: given a k-plane W, an orthonormal basis of W extends to an orthonormal basis of $\mathbb{R}^n$, and the matrix sending the standard basis to it is orthogonal and carries the standard plane $\mathbb{R}^k = \mathrm{Span}(e_1, \dots, e_k)$ to W. To compute the stabilizer of $\mathbb{R}^k$, note that an orthogonal A with $A(\mathbb{R}^k) = \mathbb{R}^k$ also preserves the orthogonal complement $(\mathbb{R}^k)^\perp = \mathbb{R}^{n-k}$, so in the splitting $\mathbb{R}^n = \mathbb{R}^k \oplus \mathbb{R}^{n-k}$ it is block-diagonal $A = \mathrm{diag}(B, C)$ with $B \in \mathrm{O}(k)$ and $C \in \mathrm{O}(n-k)$. Hence

$$\mathrm{Stab}(\mathbb{R}^k) = \mathrm{O}(k) \times \mathrm{O}(n-k)$$

embedded block-diagonally. By the orbit-stabilizer diffeomorphism (theorem 4.4.2),

$$\mathrm{Gr}(k,n) \cong \mathrm{O}(n)/\big(\mathrm{O}(k) \times \mathrm{O}(n-k)\big)$$

The dimension count reproduces the chart computation of chapter 1:

$$\dim \mathrm{Gr}(k,n) = \frac{n(n-1)}{2} - \frac{k(k-1)}{2} - \frac{(n-k)(n-k-1)}{2} = k(n-k)$$

Because $\mathrm{Gr}(k,n)$ is the continuous image of the compact group $\mathrm{O}(n)$, it is compact. The special case $k = 1$ is real projective space: the 1-planes in $\mathbb{R}^n$ are the lines through the origin, so

$$\mathbb{R}\mathrm{P}^{n-1} = \mathrm{Gr}(1,n) \cong \mathrm{O}(n)/\big(\mathrm{O}(1) \times \mathrm{O}(n-1)\big)$$

with $\mathrm{O}(1) = \{\pm 1\}$ and dimension $1 \cdot (n-1) = n-1$.

The presentation of $\mathrm{Gr}(k,n)$ as a compact homogeneous space is the one that later chapters use: compactness is what makes the Grassmannian a legitimate target for degree and intersection arguments, and the symmetry group acts by diffeomorphisms, so no point of the Grassmannian is special. Remembering a little more than the k-plane (an ordered orthonormal basis of it) gives the next family.

Example 4.4.4: Stiefel and Flag Manifolds

The *Stiefel manifold* $V_k(\mathbb{R}^n)$ is the set of orthonormal k-frames $(v_1, \dots, v_k)$ in $\mathbb{R}^n$. Regarding a frame as the $n \times k$ matrix with columns v_i, one has $V_k(\mathbb{R}^n) = \{A \in M(n,k) \mid A^\top A = I_k\}$, and the map $G(A) = A^\top A$ into the symmetric $k \times k$ matrices $\mathrm{Sym}(k)$ has derivative $DG(A)H = H^\top A + A^\top H$; at $A \in V_k(\mathbb{R}^n)$ this is onto $\mathrm{Sym}(k)$, since $H = \frac{1}{2}AS$ yields $H^\top A + A^\top H = S$ for any $S = S^\top$. So I_k is a regular value and $V_k(\mathbb{R}^n)$ is a properly embedded submanifold of $M(n,k)$ (corollary 3.2.10) of dimension $nk - \frac{k(k+1)}{2}$.

The group $\mathrm{O}(n)$ acts smoothly on $V_k(\mathbb{R}^n)$ by $A \cdot (v_1, \dots, v_k) = (Av_1, \dots, Av_k)$, the restriction of the linear action. It is transitive, because any orthonormal k-frame extends to an orthonormal basis of $\mathbb{R}^n$, giving an orthogonal matrix carrying the standard frame $(e_1, \dots, e_k)$ to it. The stabilizer of $(e_1, \dots, e_k)$ consists of the orthogonal A fixing $e_1, \dots, e_k$; such an A fixes $\mathbb{R}^k$ pointwise and preserves $\mathbb{R}^{n-k}$, so $A = \mathrm{diag}(I_k, C)$ with $C \in \mathrm{O}(n-k)$. Hence $\mathrm{Stab}(e_1, \dots, e_k) = \mathrm{O}(n-k)$ and, by theorem 4.4.2,

$$V_k(\mathbb{R}^n) \cong \mathrm{O}(n)/\mathrm{O}(n-k)$$

of dimension $\frac{n(n-1)}{2} - \frac{(n-k)(n-k-1)}{2} = nk - \frac{k(k+1)}{2}$, in agreement with the regular-value count. The extremes are $V_1(\mathbb{R}^n) = S^{n-1}$ and $V_n(\mathbb{R}^n) = \mathrm{O}(n)$. Forgetting the frame and keeping only its span sends $V_k(\mathbb{R}^n) \to \mathrm{Gr}(k,n)$; in coset terms this is $\mathrm{O}(n)/\mathrm{O}(n-k) \to \mathrm{O}(n)/(\mathrm{O}(k) \times \mathrm{O}(n-k))$, whose fibre $(\mathrm{O}(k) \times \mathrm{O}(n-k))/\mathrm{O}(n-k) \cong \mathrm{O}(k)$ is the set of orthonormal frames of a fixed k-plane.

Iterating the block decomposition gives the *flag manifolds*. Fix a composition $n = n_1 + \dots + n_r$ into positive integers, and let $\mathrm{Fl}(n_1, \dots, n_r)$ be the set of orthogonal decompositions $\mathbb{R}^n = W_1 \oplus \dots \oplus W_r$ with $\dim W_i = n_i$. The group $\mathrm{O}(n)$ acts transitively, and the stabilizer of the standard decomposition into coordinate blocks is the block-diagonal subgroup $\mathrm{O}(n_1) \times \dots \times \mathrm{O}(n_r)$, so

$$\mathrm{Fl}(n_1, \dots, n_r) \cong \mathrm{O}(n)/\big(\mathrm{O}(n_1) \times \dots \times \mathrm{O}(n_r)\big)$$

The Grassmannian is the two-block case $\mathrm{Fl}(k, n-k) = \mathrm{Gr}(k,n)$.

Each of these spaces is compact, being a continuous image of $\mathrm{O}(n)$, and each is manufactured from the same group by choosing which subgroup to quotient by. The acting group has so far been treated as a Lie group on faith; the last example makes its manifold structure explicit and records the classical groups as regular-value examples, the form in which they appear throughout the book.

Example 4.4.5: The Classical Matrix Groups as Regular Level Sets

Each of the classical groups is cut out of a matrix space as the regular level set of a symmetry condition, which is what makes it a manifold; the Lie-group structure (the bracket, the exponential map, and the correspondence with a Lie algebra) is developed in [16].

1. *Orthogonal group.* As in exercise 3.2.1 (2), the map $F(A) = A^\top A$ from $M(n,n)$ to the symmetric matrices $\mathrm{Sym}(n)$ has I as a regular value, so $\mathrm{O}(n) = F^{-1}(I)$ is a submanifold of dimension $n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$. By proposition 3.2.16 its tangent space at the identity is

   $$T_I\,\mathrm{O}(n) = \ker DF(I) = \{H \in M(n,n) \mid H^\top + H = 0\}$$

   the space of skew-symmetric matrices, of dimension $\frac{n(n-1)}{2}$.

2. *Unitary group.* Over $\mathbb{C}$, view $M(n,n;\mathbb{C}) \cong \mathbb{R}^{2n^2}$ and let $G(A) = A^*A$ land in the Hermitian matrices $\mathrm{Herm}(n)$, a real vector space of dimension n^2, where A^* is the conjugate transpose.

Its derivative $DG(A)H = H^*A + A^*H$ is onto $\mathrm{Herm}(n)$ at any $A \in \mathrm{U}(n)$: for a Hermitian S, taking $H = \frac{1}{2}AS$ gives $H^*A + A^*H = \frac{1}{2}(SA^*A + A^*AS) = S$. So I is a regular value and $\mathrm{U}(n) = G^{-1}(I)$ is a submanifold of dimension $2n^2 - n^2 = n^2$, with

$$T_I\,\mathrm{U}(n) = \{H \in M(n, n; \mathbb{C}) \mid H^* + H = 0\}$$

the skew-Hermitian matrices, of real dimension n^2.

3. *Special orthogonal group.* On $\mathrm{O}(n)$ the determinant takes the values ± 1 and is continuous, hence locally constant, so $\mathrm{SO}(n) = \mathrm{O}(n) \cap \{\det = 1\}$ is open and closed in $\mathrm{O}(n)$: it is the identity component, an open submanifold of the same dimension $\frac{n(n-1)}{2}$, with the same tangent space $T_I\,\mathrm{SO}(n) = T_I\,\mathrm{O}(n)$ of skew-symmetric matrices.

The tangent space at the identity of each of these groups is its Lie algebra: skew-symmetric matrices for $\mathrm{O}(n)$ and $\mathrm{SO}(n)$, skew-Hermitian matrices for $\mathrm{U}(n)$. That the identity component alone carries this infinitesimal data, and that the group is reconstructed from it by the exponential map, is the content of the Lie theory picked up in [16]. What this book needs is only what has been proved: these are smooth manifolds, cut out by regular values, and they act smoothly on the projective spaces, Grassmannians, and frame manifolds that populate the rest of the geometry track.

Every space above arose from a transitive action. A general smooth action need not be transitive, but it partitions M into orbits, each a homogeneous space in its own right. The orbit map exhibits an orbit as an injective immersion of a coset space, and whether that immersion is an embedding is decided by a single topological condition on the orbit.

Theorem 4.4.6: Orbits with Locally Closed Image Are Embedded

Let G be a Lie group acting smoothly on a manifold M, and let $p \in M$ have orbit $\mathcal{O} = G \cdot p$. The orbit map descends to an injective immersion $F\colon G/G_p \to M$ with image $\mathcal{O}$. If $\mathcal{O}$ is locally closed in M, then F is a homeomorphism onto $\mathcal{O}$, so $\mathcal{O}$ is an embedded submanifold and $F\colon G/G_p \to \mathcal{O}$ is a diffeomorphism.

***Proof*:**
The stabilizer G_p is a closed subgroup (theorem 4.4.2), so G/G_p is a manifold and the orbit map $\theta_p\colon G \to M$, $g \mapsto g \cdot p$, descends to a smooth injection $F\colon G/G_p \to M$ with image $\mathcal{O}$, injective because $g \cdot p = g' \cdot p$ forces $gG_p = g'G_p$. Equivariance forces F to have constant rank, exactly as in theorem 4.4.2, and injectivity then makes that rank full, so F is an immersion. It remains to show F is a homeomorphism onto $\mathcal{O}$ when $\mathcal{O}$ is locally closed, for which it suffices that the orbit map $G \to \mathcal{O}$ be open.

The key point is that for every compact symmetric neighbourhood K of the identity, $K \cdot p$ has nonempty interior in $\mathcal{O}$. A locally closed subset of a manifold is locally compact Hausdorff, hence a Baire space, so $\mathcal{O}$ in its subspace topology is a nonempty Baire space. Being second countable and locally compact, G is σ-compact, so countably many left translates g_nK cover G. Then $\mathcal{O} = \bigcup_n g_n \cdot (K \cdot p)$, each $g_n \cdot (K \cdot p)$ compact as the continuous image of the compact K, hence closed in $\mathcal{O}$. By the Baire property some $g_n \cdot (K \cdot p)$ has nonempty interior, and translation by the homeomorphism g_n^{-1} carries that interior into $K \cdot p$: some $h \cdot p$ with $h \in K$ is interior to $K \cdot p$.

Openness of the orbit map now follows by homogeneity. Fix a neighbourhood W of the identity, and by continuity of multiplication a compact symmetric neighbourhood K of the identity with $KK \subseteq W$. The previous paragraph gives $h \in K$ with $h \cdot p$ interior to $K \cdot p$. Translation by h^{-1} is a homeomorphism of $\mathcal{O}$ carrying $h \cdot p$ to p and $K \cdot p$ into $h^{-1}K \cdot p \subseteq KK \cdot p \subseteq W \cdot p$ (using $h^{-1} \in K$), so p is interior to $W \cdot p$. Applying the diffeomorphism $q \mapsto g \cdot q$ shows $g \cdot p$ is interior to $gW \cdot p$ for every g, and as gW runs over a neighbourhood basis of g the orbit map $G \to \mathcal{O}$ is open. It descends to a continuous open bijection $F\colon G/G_p \to \mathcal{O}$, hence a homeomorphism. An injective immersion that is a homeomorphism onto its image is an embedding, so $\mathcal{O}$ is embedded and F is a diffeomorphism onto it, completing the proof.

Exercise 4.4.1

1. For $n \geq 2$, show that $\mathrm{SO}(n)$ acts transitively on S^{n-1} with stabilizer $\mathrm{SO}(n-1)$, so that $S^{n-1} \cong \mathrm{SO}(n)/\mathrm{SO}(n-1)$, and check that the dimension count gives $n-1$.
2. Show that the unitary group acts transitively on complex projective space $\mathcal{P}^n$ with stabilizer $\mathrm{U}(1) \times \mathrm{U}(n)$, giving $\mathcal{P}^n \cong \mathrm{U}(n+1)/(\mathrm{U}(1) \times \mathrm{U}(n))$, and verify that the real dimension is $2n$.
3. Show that orthogonal complementation $W \mapsto W^{\perp}$ is a diffeomorphism $\mathrm{Gr}(k,n) \to \mathrm{Gr}(n-k,n)$, and explain how this is visible in the coset description of example 4.4.3.
4. Real projective space has two descriptions: the coset space $\mathrm{O}(n)/(\mathrm{O}(1) \times \mathrm{O}(n-1))$ of example 4.4.3, and the quotient $S^{n-1}/\{\pm 1\}$ of the sphere by the antipodal action. Verify that the antipodal action is free and properly discontinuous, so that theorem 4.2.7 makes $S^{n-1} \to \mathbb{R}\mathrm{P}^{n-1}$ a smooth double cover, and reconcile the two descriptions.
5. Let $\mathrm{SU}(n) = \mathrm{U}(n) \cap \{\det = 1\}$. Using that $\det\colon \mathrm{U}(n) \to \mathrm{U}(1)$ is a submersion, show that $\mathrm{SU}(n)$ is a regular submanifold of dimension $n^2 - 1$ and that $T_I \,\mathrm{SU}(n)$ is the space of traceless skew-Hermitian matrices.

5

Vector Fields and Flows

A vector field assigns a tangent vector to each point of a manifold, smoothly. Two readings run through the chapter. Analytically, a vector field is an ordinary differential equation without coordinates: its integral curves are the trajectories of the equation, and their assembly into a flow is how a manifold moves under an infinitesimal prescription, the setting for geometric evolution equations like the Ricci flow. Structurally, a vector field is an infinitesimal symmetry, the derivative of a one-parameter family of diffeomorphisms; the vector fields form a Lie algebra under a bracket, and on a Lie group this is the Lie algebra whose exponential recovers the group.

Section 5.2 bases vector fields locally through frames; section 5.3 identifies them with derivations of $C^\infty(M)$, section 5.4 tracks them under smooth maps, and section 5.5 builds their bracket; sections 5.6 to 5.8 develop integral curves, flows, and completeness, with the Lie derivative (section 5.10) tying the bracket to the flow.

5.1 Vector Fields

A vector field is a section of the tangent bundle. The section language is set up in general, so that it transfers unchanged to the vector bundles of chapter 11.

Definition 5.1.1: Section

A *section* of a smooth map $\pi\colon E \to M$ is a map $\sigma\colon M \to E$ with $\pi \circ \sigma = \mathrm{id}_M$; a *local section* over an open $U \subseteq M$ satisfies $\pi \circ \sigma = \mathrm{id}_U$. It is a *smooth* (or C^r) section when σ is smooth (or C^r).

$$\begin{tikzcd} E \arrow[r, "\pi"] & M \arrow[l, bend right, dashed, "\sigma"'] \end{tikzcd}$$

Definition 5.1.2: Vector Field

A *vector field* on a manifold M is a smooth section $X\colon M \to TM$ of the tangent bundle $\pi\colon TM \to M$, so $X_p := X(p) \in T_pM$ for each p. A section that is not required to be smooth is a *rough vector field*.

The section formulation pays off in chapter 11, where TM is replaced by a general vector bundle and smoothness by other regularity. The *support* of X is the closure of $\{p \in M \mid X_p \neq 0\}$; X is *compactly supported* when its support is compact. Vector fields are read in coordinates through the coordinate basis.

Definition 5.1.3: Component Functions

For a chart (U, φ) with coordinates x^i, a rough vector field X is written at each $p \in U$ as

$$X_p = \sum_{i=1}^{n} X^i(p)\ \frac{\partial}{\partial x^i}\bigg|_p$$

The functions $X^i\colon U \to \mathbb{R}$ are the *component functions* of X in the chart, and one writes $X = \sum_i X^i\, \partial/\partial x^i$.

Proposition 5.1.4: Smoothness in Coordinates

A rough vector field X is smooth on a chart (U, φ) if and only if its component functions X^i are smooth on U.

***Proof*:**
In the natural coordinates (x^i, v^i) on $\pi^{-1}(U) \subseteq TM$ (proposition 2.4.5) the coordinate representation of X is

$$\widehat{X}(x) = (x^1, \ldots, x^n, X^1(x), \ldots, X^n(x))$$

whose smoothness is equivalent to that of the X^i, completing the proof.

A vector field given on one chart transports to another by the transition derivative, the transformation law of tangent vectors applied pointwise.

Lemma 5.1.5: Change of Chart

Let (U, φ) and (V, ψ) be charts with $U \cap V \neq \emptyset$ and X a smooth vector field on $U \cap V$. In the two charts $X = \sum_k a^k\, \partial/\partial x^k = \sum_k b^k\, \partial/\partial y^k$, with the components related by

$$b^k = \sum_l \frac{\partial y^k}{\partial x^l}\, a^l, \qquad \text{equivalently} \qquad X_{\psi\text{-rep}} = D(\psi \circ \varphi^{-1})\, X_{\varphi\text{-rep}}$$

***Proof*:**
Apply the change-of-coordinates law for the coordinate basis (section 2.2): $\partial/\partial x^l = \sum_k (\partial y^k/\partial x^l)\,\partial/\partial y^k$, so $\sum_l a^l\,\partial/\partial x^l = \sum_k (\sum_l (\partial y^k/\partial x^l) a^l)\,\partial/\partial y^k$, giving $b^k = \sum_l (\partial y^k/\partial x^l) a^l$. This is $D(\psi \circ \varphi^{-1})$ applied to the component vector, and being a smooth function of a smooth field it is again smooth, completing the proof.

The lemma is what lets a field defined chart by chart be glued into a global field when the local pieces agree on overlaps, and it is the pointwise form of the pushforward studied in section 5.4.

Example 5.1.6: Vector Fields

1. *(Coordinate fields.)* On a chart $(U, (x^i))$, $p \mapsto \partial/\partial x^i|_p$ is a smooth local vector field $\partial/\partial x^i$ (its components are constant), defined on U, not on all of M.
2. *(Euler field.)* On $\mathbb{R}^n$, $E = \sum_i x^i\,\partial/\partial x^i$ is smooth (linear components), vanishes only at the origin, and points radially outward; it appears in Euler's homogeneous-function theorem.
3. *(Angle field on S^1.)* In the angle coordinate θ, the field $\partial/\partial\theta$ is a nowhere-vanishing smooth global field on S^1 (the velocity of the standard parametrization), the global frame of corollary 2.4.6.
4. *(Angle fields on T^n.)* On $T^n = (S^1)^n$ the fields $\partial/\partial\theta^1, \ldots, \partial/\partial\theta^n$ are nowhere-zero and independent at each point, a global frame; T^n is parallelizable.
5. *(Gluing on S^2.)* Take the stereographic charts φ_N, φ_S with transition $\varphi_{NS}(s,t) = (s,t)/(s^2+t^2)$. Setting $X_N = \partial/\partial s$ on the N-chart and pushing forward by φ_{NS}, the derivative
$$D\varphi_{NS}(s,t) = \frac{1}{(s^2+t^2)^2}\begin{pmatrix} t^2 - s^2 & -2st \\ -2st & s^2 - t^2 \end{pmatrix}$$
evaluated at $(s,t) = \varphi_{NS}^{-1}(u,v)$ (the involution gives $s^2 + t^2 = (u^2+v^2)^{-1}$) becomes $\begin{pmatrix} v^2-u^2 & -2uv \\ -2uv & u^2-v^2 \end{pmatrix}$, so
$$X_S = (v^2 - u^2)\,\frac{\partial}{\partial u} - 2uv\,\frac{\partial}{\partial v}$$
which extends smoothly across $(u,v) = 0$; the two pieces agree on the overlap and define a smooth field on S^2 vanishing only at the north pole, a single zero of index 2.

Every such field on S^2 vanishes somewhere: no smooth vector field on S^2 is nowhere zero, the *hairy ball theorem* (section 8.6).

Because T_pU is identified with T_pM, a vector field on U is read as a map $U \to TU$ or $U \to TM$ interchangeably, and $X|_U$ is a smooth field on U whenever X is smooth on M. Compactly supported fields extend from closed sets, exactly as smooth functions do.

Proposition 5.1.7: Extension of Vector Fields

Let $A \subseteq M$ be closed, $U \supseteq A$ open, and X a smooth vector field on a neighbourhood of A. There is a smooth vector field $\tilde{X}$ on all of M with $\tilde{X} = X$ on A and $\text{supp}\tilde{X} \subseteq U$. In particular a tangent vector at a point extends to a compactly supported field.

***Proof*:**
Take a smooth bump β equal to 1 on A with $\text{supp}\beta \subseteq U$ contained in the domain of X (corollary 1.5.6). Then $\tilde{X} = \beta X$, extended by 0 off $\text{supp}\beta$, is smooth, agrees with X on A, and has support in U, completing the proof.

Definition 5.1.8: The Space of Vector Fields

$\mathfrak{X}(M)$ denotes the set of all smooth vector fields on M, with the zero field $p \mapsto 0 \in T_pM$.

Proposition 5.1.9: Vector Fields as a Module

$\mathfrak{X}(M)$ is a module over $C^\infty(M)$ under pointwise operations, $(fX + gY)_p = f(p)X_p + g(p)Y_p$.

***Proof*:**
Pointwise sum and scaling by functions produce rough vector fields, and in any chart the components of $fX + gY$ are $fX^i + gY^i$, smooth when X, Y are smooth and $f, g \in C^\infty(M)$; so $\mathfrak{X}(M)$ is closed under these operations, and the module axioms hold pointwise from those of each T_pM, completing the proof.

5.2 Local and Global Frames

The coordinate vector fields $(\partial/\partial x^i)$ give a basis of T_pM at each point of a chart. A collection of vector fields with this property carries a name.

Definition 5.2.1: Frame

An ordered tuple $(X_1, \ldots, X_n)$ of vector fields on an open $U \subseteq M$ is a *local frame* on U if $(X_1|_p, \ldots, X_n|_p)$ is a basis of T_pM for every $p \in U$; it is a *global frame* when $U = M$, and a *smooth frame* when each X_i is smooth. A smooth local frame is classically a *moving frame*.

A frame is abbreviated (X_i) with the index range understood.

Example 5.2.2: Frames

1. The standard coordinate fields form a smooth global frame on $\mathbb{R}^n$.
2. On any chart $(U, (x^i))$ the coordinate fields $(\partial/\partial x^i)$ form a smooth local frame, the *coordinate frame*; so every point of M lies in the domain of a smooth local frame.

3. The angle field $\partial/\partial\theta$ is a smooth global frame on S^1, and $(\partial/\partial\theta^1, \ldots, \partial/\partial\theta^n)$ one on T^n (example 5.1.6).

Independent vector fields can always be completed to a frame nearby, because independence is an open condition.

Proposition 5.2.3: Completion of Frames

Let $(Y_1, \ldots, Y_k)$ be smooth vector fields on an open set, linearly independent at each point. Then every point p has a neighbourhood on which there are smooth vector fields $Y_{k+1}, \ldots, Y_n$ making $(Y_1, \ldots, Y_n)$ a smooth local frame.

***Proof*:**
At p the vectors $Y_1|_p, \ldots, Y_k|_p$ are independent in T_pM; complete them to a basis by adjoining coordinate vectors $\partial/\partial x^{j_{k+1}}|_p, \ldots, \partial/\partial x^{j_n}|_p$ from a chart at p. The n smooth fields $Y_1, \ldots, Y_k, \partial/\partial x^{j_{k+1}}, \ldots, \partial/\partial x^{j_n}$ are then independent at p. In the coordinate frame their components form an $n \times n$ matrix whose determinant is a smooth function, nonzero at p, hence nonzero on a neighbourhood; there the fields remain a basis, giving a smooth local frame, completing the proof.

Local frames are thus abundant, but global ones are not: a smooth global frame gives M a strong triviality.

Definition 5.2.4: Parallelizable

A smooth manifold is *parallelizable* if it admits a smooth global frame.

A parallelizable manifold has trivial tangent bundle, $TM \cong M \times \mathbb{R}^n$ (chapter 11), the frame giving the trivialization. Among the examples, $\mathbb{R}^n$, S^1, and T^n are parallelizable, as are S^3 and S^7; but S^2 is not, by the hairy ball theorem (section 8.6), and in fact S^1, S^3, S^7 are the only parallelizable spheres, a theorem of algebraic topology tied to the real division algebras. Every Lie group is parallelizable, its frame built from a basis of the tangent space at the identity by left translation, so among the spheres a Lie-group structure can occur only for S^1, S^3, S^7; in fact only S^1 and S^3 are Lie groups, as S^7 is parallelizable through octonion multiplication but non-associative. Parallelizability is not the same as being covered by one chart: the torus T^n is parallelizable but needs several charts.

5.3 Vector Fields as Derivations

A vector field acts on functions: for $X \in \mathfrak{X}(M)$ and $f \in C^\infty(M)$, define Xf pointwise by $(Xf)(p) = X_p f$, using that $X_p \in T_pM$ is a derivation at p. The action is local, $(Xf)|_V = X(f|_V)$ for open V, since a tangent vector reads only the germ of f. Smoothness of X is detected through this action.

Proposition 5.3.1: Smoothness via the Action on Functions

For a rough vector field X on M, the following are equivalent:

1. X is smooth;
2. $Xf \in C^\infty(M)$ for every $f \in C^\infty(M)$;
3. $Xf \in C^\infty(U)$ for every open $U \subseteq M$ and every $f \in C^\infty(U)$.

***Proof*:**
(1) ⇒ (2). In a chart $Xf = \sum_i X^i\, \partial f/\partial x^i$; the X^i are smooth (proposition 5.1.4) and so is each $\partial f/\partial x^i$, so Xf is smooth there, hence on M.

(2) ⇒ (3). For $f \in C^\infty(U)$ and $p \in U$, a bump β equal to 1 near p with $\operatorname{supp}\beta \subseteq U$ makes βf a global smooth function agreeing with f near p, so $Xf = X(\beta f)$ near p by locality, smooth near p; as p was arbitrary, $Xf \in C^\infty(U)$.

(3) ⇒ (1). On a chart with coordinates x^j, apply (3) to $f = x^j$: then $Xx^j = X^j$, the j-th component of X, is smooth. As all components are smooth, X is smooth by proposition 5.1.4, completing the proof.

A smooth X thus gives a map $C^\infty(M) \to C^\infty(M)$ that is $\mathbb{R}$-linear and, by the Leibniz rule at each point, satisfies $X(fg) = (Xf)g + f(Xg)$: it is a *derivation* of the algebra $C^\infty(M)$. The correspondence is exact.

Proposition 5.3.2: Vector Fields are the Derivations of $C^\infty(M)$

A map $D\colon C^\infty(M) \to C^\infty(M)$ is a derivation if and only if $Df = Xf$ for a unique smooth vector field $X \in \mathfrak{X}(M)$.

***Proof*:**
A smooth X gives such a derivation, as above. Conversely let D be a derivation. It is local: if f vanishes on a neighbourhood of p, take a bump β with $\beta(p) = 1$ and support in that neighbourhood, so $\beta f \equiv 0$; then $0 = D(\beta f)(p) = D\beta(p)\, f(p) + \beta(p)\, Df(p) = Df(p)$ by the Leibniz rule and $f(p) = 0$. Hence $X_p\colon C^\infty(M) \to \mathbb{R}$, $X_p f = (Df)(p)$, is well defined and, inheriting linearity and the Leibniz rule from D, is a derivation at p, so $X_p \in T_pM$. This defines a rough vector field X with $Xf = Df$; since $Df \in C^\infty(M)$ for all f, X is smooth by proposition 5.3.1. Uniqueness holds because $Xf = X'f$ for all f forces $X_p = X'_p$ at each p, completing the proof.

The identification $\mathfrak{X}(M) \cong \operatorname{Der}(C^\infty(M))$ is the algebraic home of vector fields, and the source of the Lie bracket (section 5.5): the commutator of two derivations is again one.

5.4 Vector Fields and Smooth Maps

A smooth $f\colon M \to N$ sends X_p to $T_pf(X_p) \in T_{f(p)}N$, but these need not assemble into a vector field on N: a non-injective f can send different X_p to the same point, and a non-surjective f leaves points uncovered. The vector fields that do correspond are named.

Definition 5.4.1: f-Related Vector Fields

For a smooth $f\colon M \to N$, fields $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ are *f-related* if $T_pf(X_p) = Y_{f(p)}$ for every p, that is $Tf \circ X = Y \circ f$.

Relatedness is exactly compatibility of the two derivations through pullback.

Proposition 5.4.2: f-Relatedness on Functions

X and Y are f-related if and only if $X(g \circ f) = (Yg) \circ f$ for every $g \in C^\infty(N)$.

***Proof*:**
For $p \in M$ and $g \in C^\infty(N)$, $X(g \circ f)(p) = X_p(g \circ f) = T_pf(X_p)(g)$ by the definition of the tangent map (definition 2.2.4), while $((Yg) \circ f)(p) = Y_{f(p)}(g)$. These agree for all g exactly when $T_pf(X_p) = Y_{f(p)}$, completing the proof.

For an arbitrary f a given X may have no f-related field on N; a diffeomorphism removes the obstruction.

Proposition 5.4.3: Pushforward along a Diffeomorphism

If $f\colon M \to N$ is a diffeomorphism, then each $X \in \mathfrak{X}(M)$ has a unique f-related field on N.

***Proof*:**
Relatedness $T_pf(X_p) = Y_{f(p)}$ determines Y at $q = f(p)$ as $Y_q = T_{f^{-1}(q)}f\big(X_{f^{-1}(q)}\big)$, so Y is unique. This $Y = Tf \circ X \circ f^{-1}$ is a composition of smooth maps, hence a smooth vector field, and is f-related to X by construction, completing the proof.

The two directions of transport have names. Functions pull back along any smooth map.

Definition 5.4.4: Pullback

The *pullback* along a smooth $f\colon M \to N$ is $f^*\colon C^\infty(N) \to C^\infty(M)$, $f^*h = h \circ f$.

Pullback is an algebra homomorphism, making $C^\infty(-)$ a contravariant functor from manifolds to $\mathbb{R}$-algebras. Vector fields push forward, but only along a diffeomorphism.

Definition 5.4.5: Pushforward

The *pushforward* along a diffeomorphism $f\colon M \to N$ is $f_*\colon \mathfrak{X}(M) \to \mathfrak{X}(N)$, $f_*X = Tf \circ X \circ f^{-1}$, the unique field f-related to X.

By construction f_*X is f-related to X, so proposition 5.4.2 gives $(f_*X)(h) = \big(X(h \circ f)\big) \circ f^{-1}$ for $h \in C^\infty(N)$; and pushforward is functorial, $(f \circ g)_* = f_* \circ g_*$ and $(\mathrm{id})_* = \mathrm{id}$, from the chain rule. These are the transformations under which the Lie bracket is natural (section 5.5).

5.5 The Lie Bracket

The composition XY of two vector fields, read as derivations, is a second-order operator, not a derivation: applied to a product it produces the extra term $(Xf)(Yg)+(Yf)(Xg)$. The antisymmetric combination removes it.

Definition 5.5.1: Lie Bracket

The *Lie bracket* of $X, Y \in \mathfrak{X}(M)$ is the operator $[X,Y] = XY - YX$ on $C^\infty(M)$, $[X,Y]f = X(Yf) - Y(Xf)$.

Proposition 5.5.2: The Bracket is a Vector Field

For $X, Y \in \mathfrak{X}(M)$ the operator $[X,Y]$ is a derivation of $C^\infty(M)$, hence a smooth vector field. In a chart its components are

$$[X,Y]^j = \sum_i \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i} \right)$$

***Proof*:**
$[X,Y]$ is $\mathbb{R}$-linear, and the Leibniz rule holds because the two second-order terms cancel: $X(Y(fg)) - Y(X(fg))$ expands, and the pieces $(Xf)(Yg) + (Yf)(Xg)$ appear symmetrically in both $X(Y(fg))$ and $Y(X(fg))$, so they subtract to zero, leaving $[X,Y](fg) = ([X,Y]f)g + f([X,Y]g)$. Thus $[X,Y]$ is a derivation and, by proposition 5.3.2, a smooth vector field. For the components, with $X = \sum_i X^i \partial/\partial x^i$ and $Y = \sum_j Y^j \partial/\partial x^j$, the second derivatives $\partial^2 f/\partial x^i \partial x^j$ cancel by symmetry of mixed partials, and collecting the first-order terms gives the stated formula, completing the proof.

The bracket makes $\mathfrak{X}(M)$ a Lie algebra, the infinitesimal counterpart of the diffeomorphism group.

Proposition 5.5.3: The Lie Algebra of Vector Fields

The bracket on $\mathfrak{X}(M)$ is $\mathbb{R}$-bilinear and antisymmetric, $[X, Y] = -[Y, X]$, and satisfies the Jacobi identity

$$[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0$$

Over functions it is not bilinear: for $f, g \in C^\infty(M)$,

$$[fX, gY] = fg\,[X, Y] + f(Xg)\,Y - g(Yf)\,X$$

***Proof*:**
Bilinearity and antisymmetry are immediate from $[X, Y] = XY - YX$. Writing each bracket as a commutator of operators, the Jacobi sum expands into twelve terms $XYZ, XZY, \ldots$ that cancel in pairs, the standard commutator identity. For the last, the Leibniz rule gives $(fX)(gYh) = f(Xg)(Yh) + fg\,X(Yh)$, and subtracting the symmetric expression collects into the stated three terms, completing the proof.

The bracket is natural under f-relatedness, which is what lets it descend along maps and, on a Lie group, single out the left-invariant fields.

Proposition 5.5.4: Naturality of the Bracket

If X is f-related to X' and Y to Y' under a smooth $f\colon M \to N$, then $[X, Y]$ is f-related to $[X', Y']$.

***Proof*:**
By proposition 5.4.2, for $g \in C^\infty(N)$, $X(g \circ f) = (X'g) \circ f$ and likewise for Y. Then

$$[X, Y](g \circ f) = X\big((Y'g) \circ f\big) - Y\big((X'g) \circ f\big) = (X'Y'g) \circ f - (Y'X'g) \circ f = ([X', Y']g) \circ f$$

so $[X, Y]$ is f-related to $[X', Y']$ by proposition 5.4.2, completing the proof.

In particular a diffeomorphism intertwines brackets, $f_*[X, Y] = [f_*X, f_*Y]$, so pushforward is a Lie-algebra isomorphism. Two brackets vanish for a reason worth naming: $[\partial/\partial x^i, \partial/\partial x^j] = 0$ for coordinate fields, since their components are constant, so a coordinate frame commutes. The converse, that a commuting frame is locally a coordinate frame, is the flow-box theorem of section 5.10, and vanishing brackets are exactly the integrability condition of the Frobenius theorem (chapter 6).

Exercise 5.5.1

1. On $\mathbb{R}^2$ compute $[X, Y]$ for $X = \partial/\partial x$ and $Y = x\,\partial/\partial y$, and for $X = y\,\partial/\partial x$, $Y = x\,\partial/\partial y$.
2. Show that $[\partial/\partial x^i, \partial/\partial x^j] = 0$ for all i, j, and deduce that any two constant-coefficient fields on $\mathbb{R}^n$ commute.
3. Deduce the Jacobi identity for the Lie bracket from the associativity of composition of operators, by expanding the three double commutators.

5.6 Integral Curves

An integral curve of a vector field follows the field: its velocity at each point is the field's value there. Finding integral curves is solving a system of ordinary differential equations in a chart, and the standard existence theory transfers to manifolds unchanged.

Definition 5.6.1: Integral Curve

An *integral curve* of a vector field X on M is a smooth curve $\gamma\colon I \to M$ with $\gamma'(t) = X_{\gamma(t)}$ for all $t \in I$. If $0 \in I$, the point $\gamma(0)$ is the *starting point*.

Example 5.6.2: Integral Curves

1. For $V = \partial/\partial x$ on $\mathbb{R}^2$, the integral curves are the horizontal lines $\gamma(t) = (a + t, b)$.
2. For $W = x\,\partial/\partial y - y\,\partial/\partial x$ on $\mathbb{R}^2$, writing $\gamma(t) = (x(t), y(t))$ the condition $\gamma'(t) = W_{\gamma(t)}$ is the system $x' = -y$, $y' = x$, with solution

$$\gamma(t) = (a \cos t - b \sin t,\ a \sin t + b \cos t)$$

Starting at (a, b), this is the constant curve when $(a, b) = 0$ and a circle traversed counter-clockwise otherwise; the integral curves are the origin and the circles about it.

In a chart, writing $\gamma(t) = (\gamma^1(t), \dots, \gamma^n(t))$, the condition $\gamma'(t) = X_{\gamma(t)}$ is the autonomous system

$$\dot{\gamma}^i(t) = X^i(\gamma^1(t), \dots, \gamma^n(t)), \qquad i = 1, \dots, n$$

whose existence, uniqueness, and smooth dependence on initial data are the content of the fundamental theorem of ordinary differential equations ([5, The Fundamental Theorem for Autonomous Systems]); the name *integral curve* recalls that solving such a system is "integrating" it. The manifold consequences follow at once.

Proposition 5.6.3: Local Existence

Let X be a smooth vector field on M. Each $p \in M$ has an $\varepsilon > 0$ and a smooth integral curve $\gamma\colon (-\varepsilon, \varepsilon) \to M$ of X starting at p.

***Proof*:**
In a chart at p the integral-curve condition is the autonomous system above, whose right-hand side X^i is smooth; the existence theorem for ODEs ([5, Existence and Uniqueness for Autonomous Systems]) provides a smooth solution on a small interval about 0 with $\gamma(0) = p$, completing the proof.

Affine reparametrizations rescale and translate integral curves in the expected way.

Lemma 5.6.4: Rescaling

If $\gamma\colon I \to M$ is an integral curve of X, then for $a \in \mathbb{R}$ the curve $\tilde{\gamma}(t) = \gamma(at)$ on $\tilde{I} = \{t \mid at \in I\}$ is an integral curve of aX.

***Proof*:**
For f smooth near $\tilde{\gamma}(t_0)$, the chain rule and $\gamma' = X_\gamma$ give

$$\tilde{\gamma}'(t_0)f = \frac{d}{dt}\bigg|_{t_0} f(\gamma(at)) = a\,(f \circ \gamma)'(at_0) = a\,\gamma'(at_0)f = a\,X_{\tilde{\gamma}(t_0)}f$$

so $\tilde{\gamma}'(t_0) = (aX)_{\tilde{\gamma}(t_0)}$, completing the proof.

Lemma 5.6.5: Translation

If $\gamma\colon I \to M$ is an integral curve of X, then for $b \in \mathbb{R}$ the curve $\hat{\gamma}(t) = \gamma(t+b)$ on $\hat{I} = \{t \mid t + b \in I\}$ is also an integral curve of X.

***Proof*:**
By the chain rule $\hat{\gamma}'(t) = \gamma'(t+b) = X_{\gamma(t+b)} = X_{\hat{\gamma}(t)}$, completing the proof.

Integral curves transport along f-related fields, which is the pointwise root of the naturality of flows.

Proposition 5.6.6: Naturality of Integral Curves

For a smooth $f\colon M \to N$, $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ are f-related if and only if f carries integral curves of X to integral curves of Y: for every integral curve γ of X, $f \circ \gamma$ is an integral curve of Y.

***Proof*:**
If X, Y are f-related and γ an integral curve of X, then $\sigma = f \circ \gamma$ satisfies

$$\sigma'(t) = T_{\gamma(t)}f(\gamma'(t)) = T_{\gamma(t)}f(X_{\gamma(t)}) = Y_{f(\gamma(t))} = Y_{\sigma(t)}$$

so σ is an integral curve of Y. Conversely, given $p \in M$, an integral curve γ of X starting at p has $f \circ \gamma$ an integral curve of Y starting at $f(p)$, so $Y_{f(p)} = (f \circ \gamma)'(0) = T_p f(X_p)$; as p was arbitrary, X and Y are f-related, completing the proof.

5.7 Flow

A vector field moves the whole manifold: run every point along its integral curve for time t, and the resulting maps assemble into a one-parameter family of diffeomorphisms, the *flow*. When all integral curves are defined for all time, this family is an $\mathbb{R}$-action; in general it is defined only on part of $\mathbb{R} \times M$.

Suppose first that the integral curve $\theta^{(p)}\colon \mathbb{R} \to M$ starting at each p is defined for all time. Define $\theta_t\colon M \to M$ by $\theta_t(p) = \theta^{(p)}(t)$, sliding each point t units along its curve. By the translation lemma (lemma 5.6.5), $t \mapsto \theta^{(p)}(t+s)$ is the integral curve starting at $\theta^{(p)}(s)$, so uniqueness gives $\theta^{(q)}(t) = \theta^{(p)}(t+s)$ with $q = \theta^{(p)}(s)$, that is

$$\theta_t \circ \theta_s = \theta_{t+s}, \qquad \theta_0 = \mathrm{id}$$

making $\theta\colon \mathbb{R} \times M \to M$ an action of $\mathbb{R}$.

Definition 5.7.1: Global Flow

A *global flow* (or one-parameter group action) on M is a smooth left $\mathbb{R}$-action $\theta\colon \mathbb{R} \times M \to M$: for all $s, t \in \mathbb{R}$ and $p \in M$,

$$\theta(t, \theta(s,p)) = \theta(s+t, p), \qquad \theta(0,p) = p$$

Writing $\theta_t(p) = \theta(t,p)$, each θ_t is a diffeomorphism with inverse θ_{-t}. Every smooth global flow comes from a vector field.

Definition 5.7.2: Infinitesimal Generator

The *infinitesimal generator* of a global flow θ is the rough vector field $X_p = (\theta^{(p)})'(0)$.

Proposition 5.7.3: Generators of Global Flows

The infinitesimal generator X of a smooth global flow θ is a smooth vector field, and each $\theta^{(p)}$ is an integral curve of X.

***Proof*:**
For smoothness, by proposition 5.3.1 it suffices that Xf is smooth for each $f \in C^\infty(M)$; and

$$Xf(p) = X_p f = (\theta^{(p)})'(0) f = \left.\frac{\partial}{\partial t}\right|_{(0,p)} f(\theta(t,p))$$

is a partial derivative of the smooth function $f \circ \theta$, hence smooth in p. For the integral-curve property, fix t_0 and set $q = \theta_{t_0}(p)$. The group law gives $\theta^{(q)}(t) = \theta^{(p)}(t+t_0)$, so for f smooth near q,

$$X_q f = (\theta^{(q)})'(0) f = \left.\frac{d}{dt}\right|_0 f(\theta^{(p)}(t+t_0)) = (\theta^{(p)})'(t_0) f$$

i.e. $(\theta^{(p)})'(t_0) = X_{\theta^{(p)}(t_0)}$, completing the proof.

Example 5.7.4: Global Flows

The fields of example 5.6.2 generate global flows:

$$V = \frac{\partial}{\partial x}\colon\ \tau_t(x,y) = (x+t, y), \qquad W = x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\colon\ \theta_t(x,y) = (x\cos t - y\sin t,\ x\sin t + y\cos t)$$

Dropping the all-time hypothesis, every global flow still has a generator, but a vector field need not generate a *global* one: an integral curve can run off the manifold or to infinity in finite time.

Example 5.7.5: Fields Without Global Flows

1. On $M = \mathbb{R}^2 \setminus \{0\}$ with $V = \partial/\partial x$, the integral curve from $(-1, 0)$ is $\gamma(t) = (t-1, 0)$, which cannot extend past $t = 1$: as $t \nearrow 1$ it would have to reach $(0, 0) \notin M$, so no integral curve is defined at $t = 1$.
2. On $M = \mathbb{R}^2$ with $W = x^2\, \partial/\partial x$, the integral curve from $(1, 0)$ solves $x' = x^2$, $x(0) = 1$, giving $x(t) = 1/(1-t)$, which blows up as $t \nearrow 1$; the curve escapes to infinity in finite time.

The right domain is a flow domain: open, and an interval about 0 in each time slice.

Definition 5.7.6: Flow Domain and Local Flow

A *flow domain* is an open $\mathcal{D} \subseteq \mathbb{R} \times M$ such that $\mathcal{D}^{(p)} = \{t \mid (t, p) \in \mathcal{D}\}$ is an open interval containing 0 for each p. A *local flow* is a smooth $\theta \colon \mathcal{D} \to M$ on a flow domain with $\theta(0, p) = p$ and, whenever $s \in \mathcal{D}^{(p)}$, $t \in \mathcal{D}^{(\theta(s,p))}$, and $s + t \in \mathcal{D}^{(p)}$,

$$\theta(t, \theta(s, p)) = \theta(s + t, p)$$

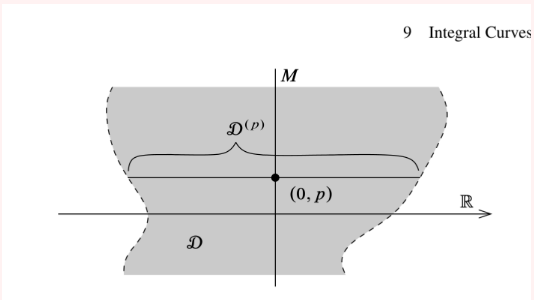


Figure 5.1: A flow domain.

The partial group law makes this a groupoid action rather than a group action. Writing $\theta_t(p) = \theta(t, p)$ and $M_t = \{p \mid (t, p) \in \mathcal{D}\}$, the generator is again $X_p = (\theta^{(p)})'(0)$ when θ is smooth.

Proposition 5.7.7: Generators of Local Flows

The infinitesimal generator of a smooth local flow is a smooth vector field, and each $\theta^{(p)}$ is an integral curve of it.

***Proof*:**
Both claims are local in t and p, and the argument of proposition 5.7.3 uses the group law only for s, t small enough to lie in the flow domain, so it applies verbatim, completing the proof.

Maximal integral curves and flows are those admitting no extension.

Definition 5.7.8: Maximal Integral Curve and Flow

A *maximal integral curve* extends to no integral curve on a larger interval; a *maximal flow* extends to no flow on a larger flow domain.

Assembling the local integral curves gives the central theorem: each vector field has a unique maximal flow.

Theorem 5.7.9: Fundamental Theorem of Flows

Every smooth vector field X on M has a unique smooth maximal flow $\theta\colon \mathcal{D} \to M$ with infinitesimal generator X. Moreover:

1. each $\theta^{(p)}\colon \mathcal{D}^{(p)} \to M$ is the unique maximal integral curve of X from p;
2. if $s \in \mathcal{D}^{(p)}$, then $\mathcal{D}^{(\theta(s,p))} = \mathcal{D}^{(p)} - s$;
3. each M_t is open, and $\theta_t\colon M_t \to M_{-t}$ is a diffeomorphism with inverse θ_{-t}.

***Proof*:**
In a chart the ODE $\dot{\gamma} = X(\gamma)$ has, by the existence-uniqueness theorem ([5, Existence and Uniqueness for Autonomous Systems]), a unique solution through each point; uniqueness is chart independent, so any two integral curves of X agreeing at one time agree on their common domain. Hence the integral curves from p glue to a unique maximal one $\theta^{(p)}\colon \mathcal{D}^{(p)} \to M$ on the union $\mathcal{D}^{(p)}$ of their domains, an open interval about 0; this gives (1). Set $\mathcal{D} = \left\{(t,p) \;\middle|\; t \in \mathcal{D}^{(p)}\right\}$ and $\theta(t,p) = \theta^{(p)}(t)$. The translation lemma and uniqueness give, for $s \in \mathcal{D}^{(p)}$, that $t \mapsto \theta^{(p)}(t+s)$ is the maximal integral curve from $\theta(s,p)$, whence $\mathcal{D}^{(\theta(s,p))} = \mathcal{D}^{(p)} - s$ and the group law $\theta_t \circ \theta_s = \theta_{t+s}$ where defined, proving (2). Smooth dependence on initial conditions ([5, The Fundamental Theorem for Autonomous Systems]) makes $\mathcal{D}$ open and θ smooth, so θ is a smooth flow with generator X; the group law then makes each θ_t a diffeomorphism $M_t \to M_{-t}$ with inverse θ_{-t}, proving (3). Maximality and uniqueness of θ follow from those of the $\theta^{(p)}$, completing the proof.

Definition 5.7.10: Flow of a Vector Field

The maximal flow of the previous theorem is the *flow of X*.

Flows are natural under f-related fields, the flow-level form of proposition 5.6.6.

Proposition 5.7.11: Naturality of Flows

Let $f\colon M \to N$ be smooth with $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ f-related, and let θ, η be their flows. Then for each t, $f(M_t) \subseteq N_t$ and $\eta_t \circ f = f \circ \theta_t$ on M_t:

$$\begin{tikzcd} M_t \arrow[r, "f"] \arrow[d, "\theta_t"'] & N_t \arrow[d, "\eta_t"] \\ M_{-t} \arrow[r, "f"'] & N_{-t} \end{tikzcd}$$

***Proof*:**
For $p \in M_t$, the curve $f \circ \theta^{(p)}$ is an integral curve of Y (proposition 5.6.6) starting at $f(p)$ and defined on $\mathcal{D}^{(p)} \ni t$, so by uniqueness and maximality it is a restriction of $\eta^{(f(p))}$; thus $t \in N^{(f(p))}$, i.e. $f(p) \in N_t$, and $\eta_t(f(p)) = f(\theta^{(p)}(t)) = f(\theta_t(p))$, completing the proof.

Corollary 5.7.12: Diffeomorphism Invariance of Flows

If $f\colon M \to N$ is a diffeomorphism and θ is the flow of X, then the flow of f_*X is $\eta_t = f \circ \theta_t \circ f^{-1}$ on $N_t = f(M_t)$.

***Proof*:**
As f_*X is f-related to X (definition 5.4.5), proposition 5.7.11 gives $\eta_t \circ f = f \circ \theta_t$, i.e. $\eta_t = f \circ \theta_t \circ f^{-1}$, and applying it to f^{-1} shows the domains match, $N_t = f(M_t)$, completing the proof.

5.8 Complete Vector Fields

A vector field is complete when its flow is global: no integral curve escapes in finite time. Completeness cannot always be read off a field, but two large classes are complete for structural reasons.

Definition 5.8.1: Complete Vector Field

A smooth vector field is *complete* if it generates a global flow, equivalently if every maximal integral curve is defined on all of $\mathbb{R}$.

The examples of example 5.7.5 are incomplete because a curve reaches an edge or blows up. A uniform lower bound on the existence times rules this out.

Lemma 5.8.2: Uniform Time Lemma

Let X have flow θ. If there is $\varepsilon > 0$ with $(-\varepsilon, \varepsilon) \subseteq \mathcal{D}^{(p)}$ for every p, then X is complete.

***Proof*:**
Suppose some $\mathcal{D}^{(p)}$ has a finite upper bound b. Choose $t_0 \in (b-\varepsilon, b)$ and set $q = \theta_{t_0}(p)$; the integral curve from q is defined on $(-\varepsilon, \varepsilon)$, so by the translation lemma (lemma 5.6.5) $\theta^{(p)}$ extends to $t_0 + \varepsilon > b$, contradicting maximality. Thus each $\mathcal{D}^{(p)}$ is unbounded above, and likewise below, so X is complete, completing the proof.

Theorem 5.8.3: Compact Support Implies Complete

Every compactly supported smooth vector field on a manifold is complete.

***Proof*:**
Let $K = \operatorname{supp} X$. Each $p \in K$ lies in a flow-box neighbourhood on which the flow is defined for $|t| < \varepsilon_p$ (proposition 5.6.3 with smooth dependence); finitely many cover K, giving a uniform $\varepsilon > 0$ with $(-\varepsilon, \varepsilon) \subseteq \mathcal{D}^{(q)}$ for all $q \in K$. For $q \notin K$ one has $X_q = 0$, so $\theta^{(q)} \equiv q$ is defined for all time. Hence $(-\varepsilon, \varepsilon) \subseteq \mathcal{D}^{(q)}$ for every q, and X is complete by lemma 5.8.2, completing the proof.

Corollary 5.8.4: Fields on Compact Manifolds are Complete

Every smooth vector field on a compact manifold is complete.

***Proof*:**
On a compact M every field has support contained in the compact M, hence is compactly supported and complete by theorem 5.8.3, completing the proof.

The obstruction to completeness is always escape from compact sets, a fact worth isolating.

Lemma 5.8.5: Escape Lemma

If $\gamma\colon I \to M$ is a maximal integral curve of X whose domain has a finite upper bound b, then for each $t_0 \in I$ the image $\gamma([t_0, b))$ lies in no compact subset of M.

***Proof*:**
If $\gamma([t_0, b)) \subseteq K$ for a compact K, the uniform-time argument of theorem 5.8.3 gives $\varepsilon > 0$ with $(-\varepsilon, \varepsilon) \subseteq \mathcal{D}^{(q)}$ for all $q \in K$. Taking $t_1 \in (\max(t_0, b-\varepsilon), b)$ and $q = \gamma(t_1) \in K$, the translation lemma extends γ past b, contradicting maximality; so no such K exists, completing the proof.

Incompleteness is thus a feature of the manifold, not repairable in general: $x^2\,\partial/\partial x$ is incomplete on $\mathbb{R}$ but complete on the one-point compactification S^1, yet a compactification adding singular points is rarely available or well-behaved. On a Lie group the left-invariant vector fields are complete, their flows the one-parameter subgroups; this belongs to the structure theory of [16].

Flows now discharge a debt left in chapter 1. The collar neighbourhood theorem was stated there and its proof deferred to this chapter, since the construction is nothing but the flow of a field pointing into the manifold.

***Proof of theorem 1.2.20, deferred from chapter 1*:**

An inward-pointing field. Call $v \in T_pM$ at $p \in \partial M$ *inward-pointing* if it is not tangent to ∂M and its last coordinate is positive in some (hence any) boundary chart $\varphi\colon U \to \mathbb{H}^n$. In such a chart the field $\partial/\partial x_n$ is inward-pointing along $U \cap \partial M$. Cover ∂M by boundary charts U_α, take a smooth partition of unity $\{\rho_\alpha\}$ subordinate to $\{U_\alpha\}$ (theorem 1.5.5), and set $X = \sum_\alpha \rho_\alpha\, \varphi_\alpha^*(\partial/\partial x_n)$, a smooth field defined near ∂M. At each $p \in \partial M$ the last coordinate of X_p in any boundary chart is a convex combination of positive numbers, hence positive, so X is inward-pointing along all of ∂M. Extend X to a compactly supported field on M by multiplying with a bump function equal to 1 near ∂M; it is then complete by theorem 5.8.3, with flow θ.

The flow is a diffeomorphism near the boundary. Define

$$c\colon \partial M \times [0,\infty) \to M, \qquad c(x,t) = \theta_t(x),$$

smooth by theorem 5.7.9. Its differential at $(x,0)$ is injective: it restricts to the identity on $T_x(\partial M)$ and sends $\partial/\partial t$ to X_x, which is inward-pointing and so not tangent to ∂M; the two contributions span T_xM by dimension count. Since $\partial M \times \{0\}$ is a properly embedded submanifold on which c is injective, the inverse function theorem applied along it gives an $\varepsilon\colon \partial M \to (0,\infty)$, smooth after averaging, such that c restricted to $W = \{(x,t) : 0 \le t < \varepsilon(x)\}$ is a diffeomorphism onto an open neighbourhood of ∂M in M. Injectivity on all of W (rather than locally) follows because two flow lines through distinct boundary points are disjoint, the flow being a bijection at each time.

Normalizing the interval. The map $(x,t) \mapsto (x, t/\varepsilon(x))$ is a diffeomorphism from W to $\partial M \times [0,1)$ carrying $\partial M \times \{0\}$ to itself. Composing its inverse with c gives the required diffeomorphism $\partial M \times [0,1) \to M$ onto a neighbourhood of ∂M, restricting to the identity on ∂M.

5.9 Flowout

The flow straightens a vector field. Starting from a hypersurface transverse to X, the flow sweeps out a neighbourhood in which X is a single coordinate field; this is the canonical local form of a nonvanishing vector field.

Theorem 5.9.1: Flowout Theorem

Let X be a smooth vector field on M, $S \subseteq M$ an embedded hypersurface, and suppose X is nowhere tangent to S. Then there is a flow domain $O \subseteq \mathbb{R} \times S$ containing $\{0\} \times S$ and a diffeomorphism $\Phi\colon O \to \Phi(O)$ onto an open subset of M, $\Phi(t,s) = \theta_t(s)$, carrying $\partial/\partial t$ to X.

***Proof*:**
Let $\Phi(t,s) = \theta(t,s)$ be the flow restricted to $\mathbb{R} \times S$, defined on the flow domain $O = (\mathbb{R} \times S) \cap \mathcal{D}$. At $(0,s)$ its differential sends $\partial/\partial t$ to X_s and T_sS identically into T_sM; since $X_s \notin T_sS$ by transversality, these span T_sM, so $T_{(0,s)}\Phi$ is an isomorphism. By the inverse function theorem Φ is a local diffeomorphism near $\{0\} \times S$, and shrinking O makes it a diffeomorphism onto an open set. As $t \mapsto \Phi(t,s)$ is an integral curve of X, Φ carries $\partial/\partial t$ to X, completing the proof.

Corollary 5.9.2: Canonical Form of a Nonvanishing Field

If $X_p \neq 0$, there is a chart $(s^1, \ldots, s^n)$ centred at p in which $X = \partial/\partial s^1$.

***Proof*:**
Choose a chart at p in which $X_p = \partial/\partial x^1|_p$ and let S be the slice $\{x^1 = 0\}$, a hypersurface with $X_p \notin T_pS$; shrinking S to a neighbourhood of p, X stays transverse to it by continuity. The flowout $\Phi(t, s)$ of theorem 5.9.1 is a diffeomorphism carrying $\partial/\partial t$ to X; its inverse is a chart in which $X = \partial/\partial s^1$ (with $s^1 = t$), completing the proof.

5.10 The Lie Derivative

The flow of X lets one differentiate another vector field along X: transport Y back by the flow and take the t-derivative at 0. The result is a vector field, and it is exactly the Lie bracket, so the algebraic bracket and the dynamical flow are two faces of one object.

Definition 5.10.1: Lie Derivative

Let $X \in \mathfrak{X}(M)$ have flow θ. The *Lie derivative* of $Y \in \mathfrak{X}(M)$ along X is the vector field

$$(L_X Y)_p = \left.\frac{d}{dt}\right|_{t=0} \big((\theta_{-t})_* Y\big)_p = \left.\frac{d}{dt}\right|_{t=0} T\theta_{-t}\big(Y_{\theta_t(p)}\big)$$

the derivative at 0 of the curve $t \mapsto T\theta_{-t}(Y_{\theta_t(p)})$ in T_pM.

Theorem 5.10.2: The Lie Derivative is the Bracket

For all $X, Y \in \mathfrak{X}(M)$, $L_X Y = [X, Y]$.

***Proof*:**
Work in a chart, writing θ_t and X, Y by their coordinate expressions. Since $\theta_{-t} \circ \theta_t = \mathrm{id}$, the chain rule gives $D\theta_{-t}(\theta_t(p)) = [D\theta_t(p)]^{-1}$, so

$$\big((\theta_{-t})_* Y\big)_p = [D\theta_t(p)]^{-1}\, Y(\theta_t(p))$$

Differentiate at $t = 0$, where $\theta_0 = \mathrm{id}$ and $D\theta_0 = I$. From $\partial_t \theta_t|_0 = X$ come $\frac{d}{dt}\big|_0 Y(\theta_t(p)) = DY(p)\, X(p)$ and $\frac{d}{dt}\big|_0 D\theta_t(p) = DX(p)$, hence $\frac{d}{dt}\big|_0 [D\theta_t(p)]^{-1} = -DX(p)$. By the product rule,

$$(L_X Y)_p = -DX(p)\, Y(p) + DY(p)\, X(p)$$

whose i-th component is $\sum_j \big(X^j\, \partial Y^i/\partial x^j - Y^j\, \partial X^i/\partial x^j\big) = [X, Y]^i$ by proposition 5.5.2, so $L_X Y = [X, Y]$, completing the proof.

The identification makes the bracket dynamical: $[X, Y]$ measures the failure of Y to be carried to itself by the flow of X, and its vanishing is the commuting of flows.

Theorem 5.10.3: Commuting Fields and Commuting Flows

For $X, Y \in \mathfrak{X}(M)$ with flows θ, η, the following are equivalent:

1. $[X, Y] = 0$;
2. Y is invariant under the flow of X: $(\theta_t)_* Y = Y$ wherever defined;
3. the flows commute: $\theta_t \circ \eta_s = \eta_s \circ \theta_t$ where both sides are defined.

***Proof*:**
The family $t \mapsto (\theta_{-t})_* Y$ satisfies $\frac{d}{dt}(\theta_{-t})_* Y = (\theta_{-t})_*(L_X Y)$: differentiating at $t = t_0$ and using the group law $\theta_{-t} = \theta_{-t_0} \circ \theta_{-(t-t_0)}$ reduces it to the $t_0 = 0$ case, which is definition 5.10.1. If $[X, Y] = 0$ then $L_X Y = 0$ (theorem 5.10.2), so this derivative vanishes and $(\theta_{-t})_* Y$ is constant, equal to its value Y at $t = 0$; thus (1) $\Rightarrow$ (2). Conversely (2) gives $L_X Y = \frac{d}{dt}\big|_0 (\theta_{-t})_* Y = 0$, so (2) $\Rightarrow$ (1). Finally $(\theta_t)_* Y = Y$ says θ_t carries Y to itself, hence carries integral curves of Y to integral curves of Y; by naturality of flows (proposition 5.7.11) applied to the diffeomorphism θ_t, this is $\theta_t \circ \eta_s = \eta_s \circ \theta_t$, giving (2) $\Leftrightarrow$ (3), completing the proof.

A commuting frame is therefore a coordinate frame, the multi-field companion of the straightening corollary 5.9.2.

Theorem 5.10.4: Canonical Form of a Commuting Frame

If $X_1, \ldots, X_k \in \mathfrak{X}(M)$ are independent at p and pairwise commute, $[X_i, X_j] = 0$, then there is a chart $(s^1, \ldots, s^n)$ centred at p in which $X_i = \partial/\partial s^i$ for $i = 1, \ldots, k$.

***Proof*:**
Let θ^i be the flow of X_i; by theorem 5.10.3 the θ^i commute. Complete $X_1|_p, \ldots, X_k|_p$ to a basis of $T_p M$ with coordinate vectors, and let S be the $(n-k)$-slice they span through p. Define

$$\Phi(t^1, \ldots, t^k, s) = \theta^1_{t^1} \circ \cdots \circ \theta^k_{t^k}(s), \qquad s \in S$$

near 0. Its differential at 0 sends $\partial/\partial t^i$ to $X_i|_p$ and $\partial/\partial s^j$ to the completing vectors, a basis, so Φ is a local diffeomorphism. Because the flows commute, θ^i_t can be moved to the front, showing $\Phi(\cdots + \varepsilon \text{ in slot } i \cdots)$ is the flow of X_i, so $\Phi_*(\partial/\partial t^i) = X_i$. The inverse of Φ is the required chart with $s^i = t^i$, completing the proof.

When the fields do not commute the frame cannot be straightened, and the obstruction is measured by the brackets $[X_i, X_j]$. Whether a k-plane field spanned by such fields is nonetheless tangent to a family of submanifolds is the integrability question answered by the Frobenius theorem (chapter 6).

5.11 Application: The Classification of 1-Manifolds

The flow of a nowhere-vanishing field parametrizes a manifold by time, and in dimension one that parametrization exhausts the manifold: every compact connected 1-manifold is an interval or a circle.

This is the base case of the classification of manifolds, and its corollary (a compact 1-manifold has an even number of boundary points) is the counting principle behind the intersection theory of chapter 8.

Theorem 5.11.1: Classification of 1-Manifolds

A smooth compact connected 1-manifold with boundary is diffeomorphic to the circle S^1 if its boundary is empty, and to the interval $[0,1]$ otherwise.

***Proof*:**
Equip M with a Riemannian metric, patched from the Euclidean metrics of finitely many charts by a partition of unity (theorem 1.5.5). Call a smooth curve $\gamma\colon J \to M$ *unit-speed* if $|\gamma'(t)| = 1$ throughout. Such a curve has nonvanishing velocity, so in dimension one it is an immersion, hence a local diffeomorphism onto an open subset of M, carrying interior points of J to $M \setminus \partial M$ and an endpoint of J to ∂M.

A unit tangent vector at a point determines a unit-speed curve through it, unique wherever two are both defined, as both solve the same autonomous first-order equation with the same initial data (theorem 5.7.9). Let $\gamma\colon J \to M$ be a maximal unit-speed curve, J an interval. Its image is open; it is also closed, for if $\gamma(t_k) \to q$ with $t_k \to \sup J$ (a limit exists by compactness), a unit-speed curve through q agrees with γ near $\sup J$, so either γ extends past $\sup J$ (impossible by maximality unless $q \in \partial M$ and $\sup J \in J$) or γ returns to an earlier value. As M is connected and $\gamma(J)$ is nonempty, open, and closed, γ is onto.

If γ is injective it is a bijective local diffeomorphism, hence a diffeomorphism $J \xrightarrow{\sim} M$; compactness forces $J = [0, L]$ closed, its endpoints the two boundary points, so $M \cong [0,1]$. Otherwise let $t_2 = \inf\{t \mid \gamma(t) \in \gamma(J_{<t})\}$ be the first return, with $\gamma(t_1) = \gamma(t_2) =: q$ for some $t_1 < t_2$. The velocities $\gamma'(t_1), \gamma'(t_2)$ are unit vectors of the line T_qM; were they opposite, the images of small arcs about t_1 and t_2 would coincide, producing a return before t_2 and contradicting its minimality, so $\gamma'(t_1) = \gamma'(t_2)$. Uniqueness then gives $\gamma(t + t_2 - t_1) = \gamma(t)$ for all t: the curve is periodic of least period $\ell = t_2 - t_1$ and descends to a diffeomorphism $\mathbb{R}/\ell\mathbb{Z} \xrightarrow{\sim} M$, so $M \cong S^1$ with $\partial M = \emptyset$, completing the classification.

Corollary 5.11.2: Boundary Parity of Compact 1-Manifolds

A compact 1-manifold with boundary has an even number of boundary points.

***Proof*:**
A compact 1-manifold has finitely many components, each a compact connected 1-manifold; by theorem 5.11.1 each is a circle, contributing no boundary points, or an interval, contributing exactly two, so the total is even.

This parity is the entire secret of intersection theory: the parameter space of a homotopy is one dimension up, and the two ends of the homotopy sit on its boundary, where they must balance.

6

Distributions and the Frobenius Theorem

A vector field prescribes a line at each point, and its integral curves knit those lines into curves filling the manifold (section 5.7). Raising the dimension, a *distribution* prescribes a k-plane at each point, and one asks the same question: is there a family of k-dimensional submanifolds tangent to it everywhere? For a single field the answer is always yes; for a plane field it is not, and the Frobenius theorem identifies exactly when. The criterion is bracket-closure: the plane field must be closed under the Lie bracket, and this *involutivity* is both necessary and sufficient for tangent submanifolds to exist. The theorem packages the commuting-frame straightening of section 5.10 into a coordinate-free integrability condition; it is the manifold input on which the integration of Lie subalgebras to subgroups rests ([16]).

Section 6.1 sets up distributions, involutivity, and integral manifolds and proves the easy half; section 6.2 proves the Frobenius theorem via a flat chart; and section 6.3 assembles the integral manifolds into a foliation.

6.1 Distributions

A distribution is a field of tangent planes, the plane-field analogue of a vector field.

Definition 6.1.1: Distribution

A *distribution* of rank k on M assigns to each p a k-dimensional subspace $D_p \subseteq T_pM$, varying smoothly: each point has a neighbourhood on which D is spanned by k smooth vector fields $X_1, \ldots, X_k$, independent at every point. Such a tuple is a *local frame* for D. A vector field X is a *section* of D, written $X \in \Gamma(D)$, if $X_p \in D_p$ for all p.

The bracket of two sections need not be a section; when it always is, the distribution is involutive.

Definition 6.1.2: Involutive Distribution

A distribution D is *involutive* if $[X, Y] \in \Gamma(D)$ whenever $X, Y \in \Gamma(D)$.

Involutivity need only be checked on a local frame, because the bracket is tensorial in its failure to be linear over functions.

Proposition 6.1.3: Involutivity on a Frame

A distribution D is involutive if and only if every local frame $(X_1, \ldots, X_k)$ satisfies $[X_i, X_j] \in \Gamma(D)$ for all i, j.

***Proof*:**
Necessity is immediate. For sufficiency, any two sections are locally $X = \sum_i f^i X_i$ and $Y = \sum_j g^j X_j$, and by the bracket expansion (proposition 5.5.3)

$$[X, Y] = \sum_{i,j} f^i g^j \, [X_i, X_j] + \sum_j (X g^j) \, X_j - \sum_i (Y f^i) \, X_i$$

The last two sums are sections of D (combinations of the X_i), and the first is too if each $[X_i, X_j] \in \Gamma(D)$; hence $[X, Y] \in \Gamma(D)$, completing the proof.

The submanifolds a distribution might be tangent to are its integral manifolds.

Definition 6.1.4: Integral Manifold

An *integral manifold* of a distribution D is a connected immersed k-dimensional submanifold $N \subseteq M$ with $T_pN = D_p$ for every $p \in N$. The distribution D is *integrable* if every point of M lies in an integral manifold.

Integrability forces involutivity: on an integral manifold the sections of D are tangent fields, and tangent fields bracket-close.

Proposition 6.1.5: Integrable Implies Involutive

Every integrable distribution is involutive.

***Proof*:**
Let $X, Y \in \Gamma(D)$ and $p \in M$, lying in an integral manifold $\iota\colon N \hookrightarrow M$. Since $D_q = T_qN$ for $q \in N$, the restrictions $X|_N, Y|_N$ are sections of TN, i.e. smooth vector fields $\tilde{X}, \tilde{Y}$ on N to which X, Y are ι-related. By naturality of the bracket (proposition 5.5.4), $[X, Y]$ is ι-related to $[\tilde{X}, \tilde{Y}]$, so $[X, Y]_p = T_p\iota\big([\tilde{X}, \tilde{Y}]_p\big) \in T_pN = D_p$. As p was arbitrary, $[X, Y] \in \Gamma(D)$, completing the proof.

Two examples frame the theorem to come: one integrable, one not.

Example 6.1.6: Integrable and Non-Integrable Plane Fields

1. *(Slices.)* On $\mathbb{R}^n$, $D = \operatorname{span}(\partial/\partial x^1, \ldots, \partial/\partial x^k)$ is integrable: through each point the k-slice $\{x^{k+1} = c^{k+1}, \ldots, x^n = c^n\}$ is an integral manifold. It is involutive, the coordinate fields commuting (exercise 5.5.1 (2)).
2. *(A contact plane field.)* On $\mathbb{R}^3$, $D = \operatorname{span}(\partial/\partial x,\ \partial/\partial y + x\,\partial/\partial z)$ is a rank-2 distribution with
$$[\partial/\partial x,\ \partial/\partial y + x\,\partial/\partial z] = \partial/\partial z$$
which at each point is not in $D_p = \operatorname{span}((1, 0, 0), (0, 1, x))$. So D is not involutive, hence not integrable (proposition 6.1.5): no surface is tangent to D everywhere.

The Frobenius theorem is the converse of proposition 6.1.5: the bracket obstruction of the second example is the only one.

6.2 The Frobenius Theorem

Involutivity is not only necessary for integrability but sufficient. The proof turns an involutive frame into a commuting one, which the commuting-frame canonical form of section 5.10 straightens to coordinate fields; the coordinate slices are then the integral manifolds.

Theorem 6.2.1: Frobenius Theorem

A smooth distribution is integrable if and only if it is involutive. Moreover, an involutive rank-k distribution D is *flat*: each point has a chart $(s^1, \ldots, s^n)$ in which $D = \operatorname{span}(\partial/\partial s^1, \ldots, \partial/\partial s^k)$, so the slices $\{s^{k+1} = c^{k+1}, \ldots, s^n = c^n\}$ are integral manifolds.

***Proof*:**
Integrability implies involutivity by proposition 6.1.5, and the flat form exhibits integral manifolds, so it remains to derive the flat chart from involutivity.

A commuting frame. Fix p and a chart $(x^1, \ldots, x^n)$ at p chosen so that D_p is complementary to $\operatorname{span}(\partial/\partial x^{k+1}|_p, \ldots, \partial/\partial x^n|_p)$. Then the projection $\pi\colon D_q \to \operatorname{span}(\partial/\partial x^1, \ldots, \partial/\partial x^k)$ onto the first k coordinates is an isomorphism for q near p, and there are unique sections $Y_i \in \Gamma(D)$ with

$\pi(Y_i) = \partial/\partial x^i$, namely

$$Y_i = \frac{\partial}{\partial x^i} + \sum_{a>k} b_i^a \frac{\partial}{\partial x^a}$$

for smooth b_i^a. The Y_i span D near p. Their brackets have no $\partial/\partial x^l$ component for $l \le k$: since $Y_i^l = \delta_i^l$ and $Y_j^l = \delta_j^l$ are constant, $[Y_i, Y_j]^l = Y_i(Y_j^l) - Y_j(Y_i^l) = 0$. But D is involutive, so $[Y_i, Y_j] \in \Gamma(D)$, and a section of D is determined by its first k components (where π is an isomorphism); one with all of them zero is zero. Hence $[Y_i, Y_j] = 0$: the frame commutes.

The flat chart. The $Y_1|_p, \ldots, Y_k|_p$ are independent and pairwise commute, so by the canonical form for a commuting frame (theorem 5.10.4) there is a chart $(s^1, \ldots, s^n)$ centred at p with $Y_i = \partial/\partial s^i$ for $i \le k$. Then $D = \operatorname{span}(Y_1, \ldots, Y_k) = \operatorname{span}(\partial/\partial s^1, \ldots, \partial/\partial s^k)$, and each slice $\{s^{k+1} = c^{k+1}, \ldots, s^n = c^n\}$ has tangent space $\operatorname{span}(\partial/\partial s^1, \ldots, \partial/\partial s^k) = D$ at each of its points, so it is an integral manifold through the corresponding point. Thus D is integrable and flat, completing the proof.

The theorem reduces a nonlinear question, the existence of tangent submanifolds, to the linear-algebraic test of bracket-closure on a frame. The contact plane field of example 6.1.6 fails the test, and so no surface is tangent to it: the planes twist too fast to be integrated, the prototype of a genuinely non-integrable field.

6.3 Foliations

The flat charts of an involutive distribution overlap consistently, and the slices through a point chain together into a single maximal integral manifold. The manifold is thereby partitioned into these *leaves*, a foliation.

Definition 6.3.1: Foliation

A *foliation* of dimension k on M is a partition $\mathcal{F}$ of M into connected immersed k-dimensional submanifolds, the *leaves*, such that each point has a flat chart $(s^1, \ldots, s^n)$ in which each leaf meets the chart domain in a union of slices $\{s^{k+1} = c^{k+1}, \ldots, s^n = c^n\}$.

Theorem 6.3.2: Global Frobenius Theorem

An involutive rank-k distribution D on M determines a k-dimensional foliation whose leaves are the maximal connected integral manifolds of D; through each point passes exactly one leaf.

***Proof*:**
By the Frobenius theorem each point has a flat chart, in which the slices are integral manifolds; call these the *plaques*. On an overlap of two flat charts, a plaque of one is an integral manifold tangent to D, hence meets each plaque of the other in a subset open-and-closed in their overlap (the last $n - k$ coordinates of the second chart are constant along the first plaque), so plaques match up along overlaps. Declare $q \sim q'$ if a finite chain of plaques joins them; this is an equivalence relation, and each class L, given the topology and smooth structure in which the

plaques are open submanifolds, is a connected immersed k-dimensional integral manifold. It is maximal: any connected integral manifold through a point of L is a union of plaques (an integral manifold is locally a plaque in each flat chart), hence contained in L. Distinct classes are disjoint, so the leaves partition M with one through each point, and the flat charts are exactly the foliation charts, completing the proof.

A leaf carries a topology finer than the subspace one, yet a smooth map that lands in a leaf is smooth into the leaf. This corestriction property, promised when immersed submanifolds were introduced (chapter 3), is what the immersed integral manifolds of the Frobenius theorem satisfy, and it is why a subgroup cut out as an integral manifold receives the maps into it smoothly.

Definition 6.3.3: Weakly Embedded Submanifold

An immersed submanifold $S \subseteq M$ (definition 3.2.13) is *weakly embedded* if every smooth map $f\colon N \to M$ with $f(N) \subseteq S$ has smooth corestriction $f\colon N \to S$ for the manifold structure of S.

Theorem 6.3.4: Leaves Are Weakly Embedded

Every leaf L of a foliation $\mathcal{F}$ of M is weakly embedded: if $f\colon N \to M$ is smooth with $f(N) \subseteq L$, then $f\colon N \to L$ is smooth.

***Proof*:**
Fix $q \in N$ and set $p = f(q) \in L$. Take a flat chart (U, s) at p (definition 6.3.1), so that $L \cap U$ is a union of the slices (plaques) $\{s^{k+1} = c^{k+1}, \dots, s^n = c^n\}$. Being a manifold (definition 3.2.13), L is second countable, and the plaques comprising the open subset $L \cap U$ of L are pairwise disjoint and open in L, hence countably many. Write $c^{(0)}, c^{(1)}, \dots$ for their distinct values of $(s^{k+1}, \dots, s^n)$, with p on the plaque P_0 of value $c^{(0)}$.

Choose a connected neighbourhood V of q with $f(V) \subseteq U$. The map $g = (s^{k+1}, \dots, s^n) \circ f\colon V \to \mathbb{R}^{n-k}$ is continuous and takes values in the countable set $\{c^{(0)}, c^{(1)}, \dots\}$, because $f(V) \subseteq L \cap U$. A continuous map from the connected space V into a countable subset of $\mathbb{R}^{n-k}$ is constant, since a nonconstant one would have a connected image with more than one point, hence uncountable. So $g \equiv c^{(0)}$ and $f(V) \subseteq P_0$.

The plaque P_0 is a slice, an embedded submanifold of M, and it is an open submanifold of L. The corestriction of the smooth map $f|_V$ to the embedded P_0 is smooth (proposition 3.2.18), and composing with the inclusion of P_0 as an open subset of L makes $f|_V\colon V \to L$ smooth. As q was arbitrary, $f\colon N \to L$ is smooth, completing the proof.

The dense leaf of the Kronecker foliation (example 6.3.5) is thus weakly embedded, though its topology is strictly finer than the subspace one: a smooth map into the torus that happens to land in the leaf still descends to a smooth map into it.

Example 6.3.5: Foliations

1. *(Fibres of a submersion.)* For a submersion $f\colon M \to N$, the kernels $D_p = \ker T_p f$ form an involutive distribution whose leaves are the connected components of the fibres $f^{-1}(q)$ (corollary 3.2.10); the product $\mathbb{R}^k \times \mathbb{R}^{n-k}$ foliated by $\mathbb{R}^k \times \{c\}$ is the model.

2. *(Kronecker foliation.)* On the torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$, the field $X = \partial/\partial x + \alpha\,\partial/\partial y$ spans a rank-1 distribution; for irrational α each leaf is the image of a line of slope α, which is dense in T^2 (exercise 4.3.1 (5)). The leaves are immersed but not embedded, and the leaf space $T^2/\mathcal{F}$ is not Hausdorff: a foliation need not fibre the manifold.

Exercise 6.3.1

1. Show that every rank-1 distribution is involutive, and hence integrable, recovering the local existence of integral curves.
2. Show that for a submersion $f\colon M \to N$ the kernel distribution $D_p = \ker T_p f$ is involutive.
3. For $D = \operatorname{span}(\partial/\partial x,\, \partial/\partial y + x\,\partial/\partial z)$ on $\mathbb{R}^3$, verify D is non-involutive and conclude no embedded surface is tangent to D at every point.

Part II

Differential Topology

Part I built the local theory of smooth manifolds: charts, the rank theorem, vector fields and their flows. Part II turns to the global question it makes possible, what smooth maps reveal about the *shape* of a manifold. The tools are two genericity principles. Sard's theorem says that the critical values of a smooth map are negligible, so regular values, where preimages are submanifolds, are the rule; transversality says that two submanifolds can always be perturbed to meet cleanly. From these come the invariants of the subject: the *degree* of a map, counting preimages with sign; the *index* of a vector field and the Poincaré-Hopf theorem, equating it with the Euler characteristic; Morse theory, reading the topology of a manifold off the critical points of a function; and cobordism, the equivalence relation under which manifolds bound. Each rests on the measure-zero and transversality arguments of the first two chapters.

7

Genericity and the Whitney Theorems

The failure of a smooth map to be a submersion is confined to a small set: Sard's theorem says the critical values, the images of the points where the rank drops, form a set of measure zero. Regular values are therefore dense, and their preimages are submanifolds (corollary 3.2.10), so the level-set constructions apply to almost every value. The measure-zero language also underlies integration on manifolds (chapter 13) and Whitney's approximation theorems.

The chapter's second theme, *transversality*, is the companion genericity principle: a map is transversal to a submanifold when, wherever the two meet, the image of the derivative and the tangent space of the submanifold together span the ambient tangent space, and transversal preimages are again submanifolds. Sard's theorem makes transversality *generic*: any map can be perturbed to meet a fixed submanifold transversally, and this genericity is the engine behind the intersection and degree invariants of the chapters that follow.

7.1 Measure Zero

Measure zero is the right notion of negligible for this chapter: it is preserved by smooth maps between equidimensional manifolds and needs no measure to define, only covers by cubes of small total volume.

Definition 7.1.1: Measure Zero

A subset $A \subseteq \mathbb{R}^n$ has *measure zero* if for every $\varepsilon > 0$ there is a countable cover $A \subseteq \bigcup_i W_i$ by cubes with $\sum_i \operatorname{vol}(W_i) < \varepsilon$.

A C^1 map cannot enlarge a measure-zero set, because it is locally Lipschitz.

Lemma 7.1.2: Smooth Image of a Measure-Zero Set

If $U \subseteq \mathbb{R}^n$ is open, $f\colon U \to \mathbb{R}^n$ is C^1, and $A \subseteq U$ has measure zero, then $f(A)$ has measure zero.

***Proof*:**
By second countability U is a countable union of open cubes with compact closure in U, so it suffices to show $f(A \cap Q)$ has measure zero for one such cube Q. On the compact $\overline{Q}$ the derivative is bounded, so the mean value inequality makes f Lipschitz there with some constant c: a set of diameter d maps to one of diameter $\leq cd$. Given $\varepsilon > 0$, cover $A \cap Q$ by cubes W_i of side s_i with $\sum_i \operatorname{vol}(W_i) < \varepsilon$. Each W_i has diameter $s_i\sqrt{n}$, so $f(A \cap Q \cap W_i)$ (which lies in $\overline{Q}$, where f is c-Lipschitz) sits in a cube of side $cs_i\sqrt{n}$, of volume $(c\sqrt{n})^n \operatorname{vol}(W_i)$. Hence $f(A \cap Q)$ is covered with total volume $\leq (c\sqrt{n})^n \sum_i \operatorname{vol}(W_i) < (c\sqrt{n})^n\varepsilon$; as ε was arbitrary, $f(A \cap Q)$ has measure zero, completing the proof.

Measure zero passes to manifolds through charts, consistently because transitions are C^1.

Definition 7.1.3: Measure Zero on a Manifold

A subset A of a smooth n-manifold M has *measure zero* if $\varphi(A \cap U)$ has measure zero in $\mathbb{R}^n$ for every chart (U, φ).

Proposition 7.1.4: One Atlas Suffices

If $A \subseteq M$ has $\varphi_\alpha(A \cap U_\alpha)$ of measure zero for every chart of some atlas $\{(U_\alpha, \varphi_\alpha)\}$, then A has measure zero.

***Proof*:**
Let (V, ψ) be any chart. For each α, $\psi(A \cap U_\alpha \cap V) = (\psi \circ \varphi_\alpha^{-1})\big(\varphi_\alpha(A \cap U_\alpha \cap V)\big)$ is the C^1 image of a measure-zero set, hence measure zero (lemma 7.1.2). By second countability the cover $\{U_\alpha\}$ of $A \cap V$ has a countable subcover, so $\psi(A \cap V)$ is a countable union of measure-zero sets, hence measure zero; since (V, ψ) was arbitrary, A has measure zero, completing the proof.

Proposition 7.1.5: Measure Zero under Smooth Maps

If $f\colon M \to N$ is smooth with $\dim M \leq \dim N$ and $A \subseteq M$ has measure zero, then $f(A)$ has measure zero.

***Proof*:**
Cover M by countably many charts $(U_\alpha, \varphi_\alpha)$ and N by charts (V_β, ψ_β). Fix charts and consider the coordinate representation $F = \psi_\beta \circ f \circ \varphi_\alpha^{-1}\colon \mathbb{R}^{\dim M} \to \mathbb{R}^{\dim N}$. View the measure-zero set $\varphi_\alpha(A \cap U_\alpha \cap f^{-1}(V_\beta))$ inside $\mathbb{R}^{\dim N}$ through the inclusion into the first $\dim M$ coordinates – still measure zero, as $\dim M \leq \dim N$ – and extend F to a C^1 endomorphism of $\mathbb{R}^{\dim N}$ by precomposing with the projection onto those coordinates. This endomorphism carries the (measure-zero) embedded set to ψ_β of the corresponding piece of $f(A)$, which is therefore measure zero (lemma 7.1.2). Countably many such pieces cover $f(A)$, so it has measure zero by

proposition 7.1.4, completing the proof.

7.2 Sard's Theorem

Sard's theorem is proved by reducing to a map of Euclidean domains and inducting on the dimension of the source, stratifying the critical set by the order to which the derivatives vanish. Two analytic inputs are cited: Taylor's inequality ([4]) and the measure-zero form of Fubini's theorem ([14, Fubini-Tonelli's Theorem for Complete Measures]) – a σ-compact subset of $\mathbb{R}^m$ every slice $\{t\} \times \mathbb{R}^{m-1}$ of which has measure zero is itself measure zero. The critical-value sets below are continuous images of σ-compact critical sets, hence σ-compact, so this form applies.

Theorem 7.2.1: Sard's Theorem

Let $f\colon M \to N$ be a smooth map of manifolds. Then the set of critical values of f has measure zero in N; in particular its complement, the regular values, is dense.

***Proof*:**
Covering M and N by countably many charts reduces the claim, by proposition 7.1.4, to the Euclidean statement: for smooth $f\colon U \to \mathbb{R}^m$ on open $U \subseteq \mathbb{R}^n$ with critical set $C = \{x \mid \mathrm{rank} Df(x) < m\}$, the image $f(C)$ has measure zero. Proceed by induction on n; the case $n = 0$ is trivial, $f(C)$ being at most a point. Stratify C by

$$C_i = \{x \in U \mid \text{all partial derivatives of } f \text{ of order} \leq i \text{ vanish at } x\}$$

so $C \supseteq C_1 \supseteq C_2 \supseteq \cdots$, and split $f(C)$ into $f(C \setminus C_1)$, the $f(C_i \setminus C_{i+1})$, and $f(C_k)$ for a large k.

Step 1: $f(C \setminus C_1)$ has measure zero. At $p \in C \setminus C_1$ some first partial is nonzero, say $\partial f^1/\partial x^1(p) \neq 0$. Then $h(x) = (f^1(x), x^2, \ldots, x^n)$ has nonsingular Jacobian at p, hence is a diffeomorphism near p, and $g = f \circ h^{-1}$ preserves the first coordinate: $g(t, y) = (t, g^t(y))$ with $g^t\colon \mathbb{R}^{n-1} \to \mathbb{R}^{m-1}$ smooth. A point (t, y) is critical for g iff y is critical for g^t, so the critical values of g over the slice $\{t\} \times \mathbb{R}^{m-1}$ are the critical values of g^t, which have measure zero by the inductive hypothesis in dimension $n-1$. By the measure-zero Fubini theorem the critical values of g, and so $f(C \setminus C_1)$ near p, have measure zero; countably many such neighbourhoods cover $C \setminus C_1$.

Step 2: $f(C_i \setminus C_{i+1})$ has measure zero, $i \geq 1$. At $p \in C_i \setminus C_{i+1}$ some $(i+1)$-st partial is nonzero, so some i-th partial w has $\partial w/\partial x^1(p) \neq 0$. Then $h(x) = (w(x), x^2, \ldots, x^n)$ is a diffeomorphism near p carrying C_i into the hyperplane $\{x^1 = 0\}$, since w vanishes on C_i. The restriction of $f \circ h^{-1}$ to that hyperplane is a smooth map of an $(n-1)$-dimensional domain whose critical set contains $h(C_i)$, so by induction its critical values, hence $f(C_i \setminus C_{i+1})$ near p, have measure zero.

Step 3: $f(C_k)$ has measure zero for $k \geq n/m$. On C_k all partials up to order k vanish, so Taylor's inequality ([4]) gives, on a compact cube $Q \subseteq U$, a constant c with $|f(x+h) - f(x)| \leq c|h|^{k+1}$ for $x \in C_k \cap Q$ and small h. Subdivide Q (side δ) into l^n subcubes of side δ/l. A subcube meeting C_k has its image in a cube of side $c'(1/l)^{k+1}$, of volume $\leq c''\, l^{-m(k+1)}$; there are at most l^n such subcubes, so $f(C_k \cap Q)$ is covered with total volume $\leq c''\, l^{n-m(k+1)}$. For $k+1 > n/m$ this exponent is negative, so the bound tends to 0 as $l \to \infty$; hence $f(C_k \cap Q)$, and so $f(C_k)$, has measure zero.

Taking $k \geq n/m$, $C = (C \setminus C_1) \cup \bigcup_{i=1}^{k-1}(C_i \setminus C_{i+1}) \cup C_k$, and each image has measure zero by Steps 1–3, so $f(C)$ has measure zero. This completes the induction and the proof; density of the regular values follows since the complement of a measure-zero set is dense.

Corollary 7.2.2: Regular Values are Dense

The regular values of a smooth $f : M \to N$ are dense in N; if f is proper, they are also open, hence a dense open set.

***Proof*:**
Density is the last line of theorem 7.2.1. If f is proper, the critical value set is closed (the image of the closed critical set under a proper, hence closed, map), so the regular values are open, completing the proof.

Sard also bounds the size of an image whenever the source is too small to fill the target.

Corollary 7.2.3: Small-Dimensional Images are Negligible

If $f : M \to N$ is smooth with $\dim M < \dim N$, then $f(M)$ has measure zero; in particular f is not surjective.

***Proof*:**
Every $T_p f : T_p M \to T_{f(p)} N$ has rank at most $\dim M < \dim N$, so is not surjective; thus every point of M is critical and $f(M)$ is the set of critical values, of measure zero by theorem 7.2.1. A measure-zero set is not all of N, so f is not surjective, completing the proof.

A C^1 curve $\mathbb{R} \to \mathbb{R}^2$ therefore cannot fill a square: a space-filling curve is necessarily not continuously differentiable.

Exercise 7.2.1

1. Show that the set of critical *points* need not have measure zero, even though the critical *values* do: give a smooth $f : \mathbb{R} \to \mathbb{R}$ whose critical set is all of $\mathbb{R}$.
2. Deduce from corollary 7.2.3 that no smooth map $S^1 \to S^2$ is surjective, and that every smooth $f : S^1 \to S^2$ misses an open set.
3. Let $f : M \to \mathbb{R}$ be smooth. Show that for almost every $c \in \mathbb{R}$ the level set $f^{-1}(c)$ is either empty or an embedded hypersurface of M.

7.3 Transversality

Sard's theorem says a smooth map meets a chosen point cleanly for almost every point. Transversality is the same principle for a submanifold in place of a point: two submanifolds, or a map and a submanifold, meet *cleanly* when their tangent data together fill the ambient tangent space, and theorem 7.2.1 will show that clean meeting is the generic case. Clean intersections are exactly the ones that are again submanifolds, with codimensions adding.

The linear model comes first. Subspaces $U, W \subseteq V$ are *transverse* in V if $U + W = V$. The sum need not be direct: in $\mathbb{R}^3$ the planes $\langle e_1, e_2\rangle$ and $\langle e_2, e_3\rangle$ are transverse yet meet in the line $\langle e_2\rangle$. For transverse U, W, dimension counting gives $\dim(U \cap W) = \dim U + \dim W - \dim V$ (in general $\dim(U + W)$ replaces $\dim V$), which in codimensions ($\mathrm{codim}U = \dim V - \dim U$) reads

$$\mathrm{codim}(U \cap W) = \mathrm{codim}U + \mathrm{codim}W.$$

Codimension counts constraints, and a transverse intersection imposes the constraints of each factor independently, so the counts add. Two planes in $\mathbb{R}^3$ (codim1 each) meet transversally in a line (codim2); two lines in $\mathbb{R}^3$ (codim2 each) would need codim4, which is impossible in a 3-space, so lines in $\mathbb{R}^3$ can meet only degenerately, never transversally except when disjoint.

Imposing this on every tangent space is the definition.

Definition 7.3.1: Transverse Submanifolds

Regular submanifolds $K, L \subseteq M$ are *transverse*, written $K \pitchfork L$, if

$$T_pK + T_pL = T_pM \qquad \text{for every } p \in K \cap L.$$

The condition is vacuous where K and L are disjoint, so disjoint submanifolds count as transverse. Where they meet, tangential transversality forces the intersection to be a submanifold of the predicted codimension.

Proposition 7.3.2: A Transverse Intersection Is a Submanifold

If $K, L \subseteq M$ are transverse regular submanifolds, then $K \cap L$ is a regular submanifold with $\mathrm{codim}(K \cap L) = \mathrm{codim}K + \mathrm{codim}L$, and $T_p(K \cap L) = T_pK \cap T_pL$ at each $p \in K \cap L$.

***Proof*:**
Being a submanifold is local and detected by defining maps (theorem 3.2.11): near $p \in K \cap L$ choose neighbourhoods U, V and submersions $f: U \to \mathbb{R}^{\mathrm{codim}K}$, $g: V \to \mathbb{R}^{\mathrm{codim}L}$ with $K \cap U = f^{-1}(0)$ and $L \cap V = g^{-1}(0)$. Set $(f, g): U \cap V \to \mathbb{R}^{\mathrm{codim}K + \mathrm{codim}L}$, $(f, g)(x) = (f(x), g(x))$. Then

$$(f, g)^{-1}(0) = f^{-1}(0) \cap g^{-1}(0) = K \cap L \cap U \cap V,$$

so it suffices to show 0 is a regular value of (f, g). At $q \in (f, g)^{-1}(0)$ the derivative $D_q(f, g) = \begin{pmatrix} D_qf \\ D_qg \end{pmatrix}$ has kernel $\ker D_qf \cap \ker D_qg = T_qK \cap T_qL$ (proposition 3.2.16). Transversality gives $T_qK + T_qL = T_qM$, so

$$\dim(T_qK \cap T_qL) = \dim T_qK + \dim T_qL - \dim T_qM = \dim M - \mathrm{codim}K - \mathrm{codim}L.$$

The kernel therefore has dimension $\dim M - (\mathrm{codim}K + \mathrm{codim}L)$, so $D_q(f, g)$ has full rank $\mathrm{codim}K + \mathrm{codim}L$ onto $\mathbb{R}^{\mathrm{codim}K + \mathrm{codim}L}$; thus 0 is a regular value. Hence $K \cap L$ is a regular submanifold of codimension $\mathrm{codim}K + \mathrm{codim}L$, with tangent space the kernel $T_qK \cap T_qL$, completing the proof.

Transverse intersections manufacture manifolds, sometimes exotic ones.

Example 7.3.3: Brieskorn's Exotic Spheres

In $\mathbb{C}^5 \setminus \{0\}$, intersect a Brieskorn variety with the unit sphere:

$$\Sigma_k = \left\{ z \in \mathbb{C}^5 \;\middle|\; z_1^2 + z_2^2 + z_3^2 + z_4^3 + z_5^{6k-1} = 0 \right\} \cap \left\{ z \in \mathbb{C}^5 \;\middle|\; \textstyle\sum_{i=1}^5 |z_i|^2 = 1 \right\}, \qquad k \in \{1, \dots, 28\}.$$

The two real hypersurfaces meet transversally: the field $\frac{1}{2} z_1 \partial_{z_1} + \frac{1}{2} z_2 \partial_{z_2} + \frac{1}{2} z_3 \partial_{z_3} + \frac{1}{3} z_4 \partial_{z_4} + \frac{1}{6k-1} z_5 \partial_{z_5}$ is tangent to the first variety but transverse to the sphere, so their tangent spaces span. Each Σ_k is thus a smooth 7-manifold, and each is homeomorphic to S^7; but the 28 of them realize the 28 distinct oriented diffeomorphism classes of the 7-sphere, no two diffeomorphic. These are the *exotic 7-spheres* of Brieskorn. (In real coordinates $z = x + iy$, $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$.)

7.4 Transversal Maps and Preimages

Replacing the two submanifolds by smooth maps loses nothing and gains the generality the rest of the chapter needs. The inclusion of a submanifold is a special case, and the inverse-image constructions of chapter 3 become the special case where one map is a constant.

Definition 7.4.1: Transverse Maps

Smooth maps $f\colon K \to M$ and $g\colon L \to M$ are *transverse*, written $f \pitchfork g$, if

$$\operatorname{im}(D_a f) + \operatorname{im}(D_b g) = T_p M \qquad \text{whenever } f(a) = g(b) = p.$$

The two derivatives $D_a f\colon T_a K \to T_p M$ and $D_b g\colon T_b L \to T_p M$ share the codomain $T_p M$, so the sum makes sense. The most frequent case is a single map $f\colon N \to M$ against a submanifold $S \subseteq M$, meaning f transverse to the inclusion $\iota\colon S \hookrightarrow M$; there $\operatorname{im} D_b \iota = T_b S$, and the condition reads $\operatorname{im}(D_a f) + T_{f(a)} S = T_{f(a)} M$ at each $a \in f^{-1}(S)$. When $S = \{y\}$ is a point, $T_y S = 0$ and transversality becomes $\operatorname{im} D_a f = T_y M$ for all $a \in f^{-1}(y)$, i.e. y is a regular value: *regularity is transversality to a point.*

The reduction that proves the level-set theorem proves the general statement. Near $p \in S$ pick a submersion $\varphi\colon U \to \mathbb{R}^{\operatorname{codim} S}$ with $S \cap U = \varphi^{-1}(0)$; then near a point $a \in f^{-1}(S)$,

$$f^{-1}(S) = (\varphi \circ f)^{-1}(0),$$

so $f^{-1}(S)$ is cut out by $\varphi \circ f$, and it is a submanifold as soon as 0 is a regular value of $\varphi \circ f$. Since $D_a(\varphi \circ f) = D_{f(a)} \varphi \circ D_a f$ and $\ker D_{f(a)} \varphi = T_{f(a)} S$, the composite $\varphi \circ f$ is a submersion at a exactly when $\operatorname{im} D_a f$ surjects onto $T_{f(a)} M / T_{f(a)} S$, which is the transversality condition

7.4.2 Transversality Criterion

$$\operatorname{im}(D_a f) + T_{f(a)} S = T_{f(a)} M.$$

Proposition 7.4.3: The Preimage of a Transverse Submanifold

Let $S \subseteq M$ be an embedded submanifold and $f\colon N \to M$ a smooth map transverse to S. Then $f^{-1}(S)$ is an embedded submanifold of N with $\operatorname{codim} f^{-1}(S) = \operatorname{codim} S$.

***Proof*:**
Let $k = \operatorname{codim} S$. Fix $a \in f^{-1}(S)$ and, using that S is embedded, a neighbourhood U of $f(a)$ and a submersion $\varphi\colon U \to \mathbb{R}^k$ with $S \cap U = \varphi^{-1}(0)$. On $f^{-1}(U)$,

$$(\varphi \circ f)^{-1}(0) = f^{-1}(\varphi^{-1}(0)) = f^{-1}(S \cap U) = f^{-1}(S) \cap f^{-1}(U),$$

so it suffices that 0 be a regular value of $\varphi \circ f$. Let $p \in (\varphi \circ f)^{-1}(0)$ and $z \in \mathbb{R}^k = T_0\mathbb{R}^k$. As φ is a submersion there is $y \in T_{f(p)}M$ with $D_{f(p)}\varphi(y) = z$; transversality writes $y = y_0 + D_pf(v)$ with $y_0 \in T_{f(p)}S$ and $v \in T_pN$. Since $\varphi \equiv 0$ on $S \cap U$ and $T_{f(p)}S = \ker D_{f(p)}\varphi$, we get $D_{f(p)}\varphi(y_0) = 0$, so

$$D_p(\varphi \circ f)(v) = D_{f(p)}\varphi\big(D_pf(v)\big) = D_{f(p)}\varphi(y) = z.$$

Thus $D_p(\varphi \circ f)$ is surjective, 0 is a regular value, and theorem 3.2.11 makes $f^{-1}(S)$ an embedded submanifold of codimension k, completing the proof.

A submersion is transverse to everything, so the corollary needs no separate argument.

Corollary 7.4.4: Submersions Pull Back Submanifolds

If $f\colon N \to M$ is a submersion and $S \subseteq M$ an embedded submanifold, then f is transverse to S and $f^{-1}(S)$ is an embedded submanifold with $\operatorname{codim} f^{-1}(S) = \operatorname{codim} S$.

***Proof*:**
A submersion has $\operatorname{im} D_pf = T_{f(p)}M$, so $\operatorname{im} D_pf + T_{f(p)}S = T_{f(p)}M$ trivially; apply proposition 7.4.3.

The tangent space of the preimage is read off the same criterion.

Corollary 7.4.5: Tangent Space of a Transverse Preimage

With $f \pitchfork S$ and $W = f^{-1}(S)$ as above, $T_pW = (D_pf)^{-1}(T_{f(p)}S)$ at each $p \in W$.

***Proof*:**
Locally $W = (\varphi \circ f)^{-1}(0)$ with φ a submersion cutting out S, so $T_pW = \ker D_p(\varphi \circ f) = \ker\big(D_{f(p)}\varphi \circ D_pf\big) = (D_pf)^{-1}(\ker D_{f(p)}\varphi) = (D_pf)^{-1}(T_{f(p)}S)$.

For two general maps there is no ambient submanifold to invert. The remedy is to build the space the intersection ought to live on. If $f\colon K \to M$ and $g\colon L \to M$, the set of matched pairs

$$K\,{}_f\!\times_g L = \{(a, b) \in K \times L \mid f(a) = g(b)\}$$

is the *fibered product* of f and g; when the maps are understood it is written $K \times_M L$. It reduces to a preimage in the previous sense by cutting the product $K \times L$ with the diagonal.

Proposition 7.4.6: The Fibered Product of Transverse Maps

If $f\colon K \to M$ and $g\colon L \to M$ are transverse, then $K \times_M L$ is a regular submanifold of $K \times L$ of codimension $\dim M$, and it carries smooth projections to K and L making

$$\begin{tikzcd} K \times_M L \arrow[r, "\pi_L"] \arrow[d, "\pi_K"'] & L \arrow[d, "g"] \\ K \arrow[r, "f"'] & M \end{tikzcd}$$

commute.

***Proof*:**
Consider $f \times g\colon K \times L \to M \times M$ and the diagonal $\Delta_M \subseteq M \times M$. A pair (a, b) lies in $(f \times g)^{-1}(\Delta_M)$ iff $f(a) = g(b)$, so $(f \times g)^{-1}(\Delta_M) = K \times_M L$. It suffices, by proposition 7.4.3, to show $f \times g$ is transverse to Δ_M; since $\operatorname{codim}\Delta_M = \dim M$, the codimension statement follows.

Fix (a, b) with $f(a) = g(b) = p$. The tangent space of the diagonal is $T_{(p,p)}\Delta_M = \{(u, u) \mid u \in T_pM\}$, and $\operatorname{im} D_{(a,b)}(f \times g) = \{(D_af\, v,\ D_bg\, w) \mid v \in T_aK,\ w \in T_bL\}$. Given an arbitrary $(A, B) \in T_pM \times T_pM$, transversality of f, g provides v, w with $D_af\, v - D_bg\, w = A - B$. Set $u = A - D_af\, v = B - D_bg\, w$. Then

$$(A, B) = (D_af\, v,\ D_bg\, w) + (u, u) \in \operatorname{im} D_{(a,b)}(f \times g) + T_{(p,p)}\Delta_M,$$

so $f \times g \pitchfork \Delta_M$. Hence $K \times_M L$ is a regular submanifold of codimension $\dim M$. The ambient projections $K \times L \to K$ and $K \times L \to L$ restrict to the smooth maps π_K, π_L, and $f \circ \pi_K = g \circ \pi_L$ by the defining equation, completing the proof.

Example 7.4.7: Fibered Products

1. The name is explained by the trivial case. For $\pi_1\colon M \times Z_1 \to M$ and $\pi_2\colon M \times Z_2 \to M$ the projections (submersions, hence transverse), the fibered product is

$$(M \times Z_1) \times_M (M \times Z_2) \cong M \times Z_1 \times Z_2,$$

the family of product fibers $Z_1 \times Z_2$ over M.

2. Let $p\colon S^3 \to S^2$ be the Hopf map. It is a submersion, so transverse to itself, and the fibered product $S^3 \times_{S^2} S^3$ is a smooth 4-manifold whose map to S^2 has fiber $S^1 \times S^1$ over each point, a torus bundle pulled back from the Hopf fibration $S^1 \to S^3 \to S^2$, the archetypal nontrivial *fiber bundle* of chapter 11.

3. Let S^1 lie in the xy-plane and $(S^1)'$ in the yz-plane of $\mathbb{R}^3$. Wherever they meet, $\dim(TS^1 + T(S^1)') \le 2 < 3$, so they can *never* meet transversally in $\mathbb{R}^3$, even when $S^1 \cap (S^1)'$ happens to be a manifold. Transversality is a property of the pair *together with the ambient space*, not of the intersection alone.

Transversality to the fibers of a product recovers, and sharpens, the description of a graph.

Theorem 7.4.8: Global Characterization of Graphs

Let $S \subseteq M \times N$ be an embedded submanifold, with π_M, π_N the projections. The following are equivalent:

1. S is the graph of a smooth map $f\colon M \to N$;
2. $\pi_M|_S\colon S \to M$ is a diffeomorphism;
3. for every $p \in M$, S meets $\{p\} \times N$ transversally in exactly one point.

When they hold, $f = \pi_N \circ (\pi_M|_S)^{-1}$.

***Proof*:**
(1) $\Rightarrow$ (2). If $S = \{(x, f(x)) \mid x \in M\}$, then $\pi_M|_S(x, f(x)) = x$ is a bijection with smooth inverse $x \mapsto (x, f(x))$, hence a diffeomorphism.

(2) $\Rightarrow$ (3). Since $\pi_M|_S$ is a bijection, exactly one point (p, q) of S lies over each p, so $S \cap (\{p\} \times N) = \{(p, q)\}$. At (p, q), the isomorphism $D(\pi_M|_S)\colon T_{(p,q)}S \to T_pM$ shows $D\pi_M$ is injective on $T_{(p,q)}S$, so $T_{(p,q)}S \cap (\{0\} \times T_qN) = T_{(p,q)}S \cap \ker D\pi_M = 0$. As $\dim S = \dim M$, dimensions give $T_{(p,q)}S \oplus (\{0\} \times T_qN) = T_pM \times T_qN = T_{(p,q)}(M \times N)$, which is transversality to $\{p\} \times N$.

(3) $\Rightarrow$ (1). Exactly one point over each p makes $\pi_M|_S$ a bijection, defining a set map $f\colon M \to N$ by $S = \{(p, f(p)) \mid p \in M\}$. At the point (p, q) over p, the transverse intersection $S \cap (\{p\} \times N)$ is a submanifold of dimension $\dim S - \dim M$; being the single point (p, q), it is 0-dimensional, forcing $\dim S = \dim M$ and $T_{(p,q)}S \cap (\{0\} \times T_qN) = 0$. Hence $D(\pi_M|_S)$ is an isomorphism at every point, and $\pi_M|_S$ is a smooth bijective local diffeomorphism, so a diffeomorphism. Then $f = \pi_N \circ (\pi_M|_S)^{-1}$ is smooth and S is its graph, completing the proof.

Dropping to one point gives the local statement, the transversality form of the implicit function theorem.

Corollary 7.4.9: Local Characterization of Graphs

Let $S \subseteq M \times N$ be an immersed submanifold with $\dim S = \dim M$, and $(p, q) \in S$. If S meets $\{p\} \times N$ transversally at (p, q), then a neighbourhood of (p, q) in S is the graph of a smooth map $f\colon U \to N$ defined on a neighbourhood U of p.

***Proof*:**
Since $\dim S = \dim M$, transversality $T_{(p,q)}S + (\{0\} \times T_qN) = T_pM \times T_qN$ forces $\dim\big(T_{(p,q)}S \cap (\{0\} \times T_qN)\big) = \dim S + \dim N - (\dim M + \dim N) = 0$, so $D(\pi_M|_S)\colon T_{(p,q)}S \to T_pM$ is an isomorphism of equidimensional spaces. By the inverse function theorem $\pi_M|_S$ restricts to a diffeomorphism from a neighbourhood V of (p, q) in S onto a neighbourhood U of p; then $f = \pi_N \circ (\pi_M|_S)^{-1}\colon U \to N$ is smooth and V is its graph, completing the proof.

7.5 Stability and Genericity

Two features single transversality out as the right notion of clean intersection. It is *stable*, unchanged by small perturbations of the maps, and it is *generic*, achievable by an arbitrarily small perturbation of any map. Stability follows from compactness and genericity from Sard's theorem; together they say that among all smooth maps the transverse ones form a large, robust class. The perturbations are packaged as homotopies.

Definition 7.5.1: Smooth Homotopy

A *smooth homotopy* between smooth maps $f_0, f_1 : M \to N$ is a smooth map $F : M \times [0,1] \to N$ with $F(\cdot, 0) = f_0$ and $F(\cdot, 1) = f_1$; write $f_t = F(\cdot, t)$.

Definition 7.5.2: Stability

A property of a smooth map $f : M \to N$ is *stable* if for every smooth homotopy f_t with $f_0 = f$ there is an $\varepsilon > 0$ such that f_t has the property for all $t < \varepsilon$.

On a compact source, being an immersion is stable, because injectivity of the derivative is the non-vanishing of a minor.

Proposition 7.5.3: Stability of Immersions and Submersions

On a compact manifold M, the properties of $f : M \to N$ being an immersion, and of being a submersion, are each stable.

***Proof*:**
Take the immersion case; the submersion case is identical with rows and columns exchanged. Let f_t be a homotopy with $f_0 = f$ an immersion. At each $p \in M$, $D_p f$ is injective, so in local coordinates its Jacobian has a nonvanishing $m \times m$ minor ($m = \dim M$). That minor is a continuous function of (p, t) and is nonzero at $(p, 0)$, so it is nonzero on a neighbourhood $U_p \times [0, \varepsilon_p)$, where $D_q f_t$ is injective. The sets $U_p \times \{0\}$ cover the compact $M \times \{0\}$; a finite subcover has a smallest ε, and f_t is an immersion for all $t < \varepsilon$, completing the proof.

Theorem 7.5.4: Stability of Transversality on Compact Sources

Let K be compact and $L \subseteq M$ a closed embedded submanifold. If $f : K \to M$ is transverse to L, then transversality to L is stable under homotopies of f.

***Proof*:**
Let f_t be a homotopy with $f_0 = f$, and $F : K \times [0,1] \to M$ the homotopy map. Cover K by neighbourhoods on which f_t stays transverse for a short time, then use compactness.

Where f misses L: for $p \notin f^{-1}(L)$, $f(p)$ lies in the open set $M \setminus L$ (here L closed is used), so $F^{-1}(M \setminus L)$ is open and contains $(p, 0)$; choose $U_p \times [0, \varepsilon_p)$ inside it, on which f_t avoids L and is vacuously transverse.

Where f meets L: for $p \in f^{-1}(L)$, pick near $f(p)$ a submersion φ with $L = \varphi^{-1}(0)$. By the transversality criterion (box 7.4.2), transversality of f_t at a point of $(\varphi \circ f_t)^{-1}(0)$ is exactly $\varphi \circ f_t$ being a submersion there, i.e. a nonvanishing maximal minor of $D(\varphi \circ f_t)$. That minor is continuous in (p, t) and nonzero at $(p, 0)$, so nonzero on some $U_p \times [0, \varepsilon_p)$. The U_p cover the compact K; a finite subcover with smallest ε makes f_t transverse to L for all $t < \varepsilon$, completing the proof.

Genericity is the deeper half, and it is where Sard's theorem enters. The tool is a whole family of perturbations at once.

Definition 7.5.5: Smooth Family of Maps

A *smooth family* of maps $N \to M$ parametrized by a manifold S is a smooth map $F \colon N \times S \to M$; write $F_s = F(\cdot, s)$.

The next theorem is what makes genericity work: if the *total* family is transverse to X, then almost every member is. It is often called the parametric transversality, or Thom transversality, theorem.

Theorem 7.5.6: Parametric Transversality

Let $X \subseteq M$ be an embedded submanifold and $F \colon N \times S \to M$ a smooth family transverse to X. Then F_s is transverse to X for almost every $s \in S$.

***Proof*:**
Since $F \pitchfork X$, the preimage $W = F^{-1}(X)$ is an embedded submanifold of $N \times S$ (proposition 7.4.3). Let $\pi \colon N \times S \to S$ be the projection and restrict it to $\pi|_W \colon W \to S$. By Sard's theorem (theorem 7.2.1) almost every $s \in S$ is a regular value of $\pi|_W$; it suffices to show that each such s makes F_s transverse to X.

Fix a regular value s, a point $p \in F_s^{-1}(X)$, and $q = F_s(p) \in X$. The goal is

$$T_q M = T_q X + \operatorname{im} D_p F_s.$$

Three facts are in hand. First, $F \pitchfork X$ gives $T_q M = T_q X + \operatorname{im} D_{(p,s)} F$. Second, s a regular value of $\pi|_W$ gives $D_{(p,s)}(\pi|_W)(T_{(p,s)} W) = T_s S$. Third, $T_{(p,s)} W = (D_{(p,s)} F)^{-1}(T_q X)$ (corollary 7.4.5). Now take any $w \in T_q M$. By the first fact,

$$w = a + D_{(p,s)} F(v_1, e_1), \qquad a \in T_q X,\ (v_1, e_1) \in T_p N \times T_s S.$$

By the second fact there is $(v_2, e_1) \in T_{(p,s)} W$ with the same S-component e_1 (the projection π reads off that component). By the third fact $D_{(p,s)} F(v_2, e_1) \in T_q X$. Splitting off the S-direction, and writing $\iota_s \colon N \to N \times S$, $\iota_s(x) = (x, s)$, so that $F \circ \iota_s = F_s$,

$$D_{(p,s)} F(v_1, e_1) = \underbrace{D_{(p,s)} F(v_2, e_1)}_{\in T_q X} + D_{(p,s)} F(v_1 - v_2, 0), \qquad D_{(p,s)} F(v_1 - v_2, 0) = D_p F_s(v_1 - v_2).$$

Hence $w = a + D_{(p,s)} F(v_2, e_1) + D_p F_s(v_1 - v_2) \in T_q X + \operatorname{im} D_p F_s$, which is the required transversality, completing the proof.

To perturb a *given* map one needs a family through it that is itself transverse to X. When the target is $\mathbb{R}^n$ this is immediate: $F(x,s) = f(x) + s$ on $N \times \mathbb{R}^n$ is a submersion, hence transverse to everything, and $F_0 = f$. For a general target the family is built inside a tubular neighbourhood, which is where the embedding theory of section 7.6 is borrowed.

Theorem 7.5.7: Transversality Homotopy Theorem

Let $X \subseteq M$ be an embedded submanifold. Every smooth map $f\colon N \to M$ is homotopic to a smooth map transverse to X.

***Proof*:**
Embed M in some $\mathbb{R}^K$ (theorem 7.6.1) and let $\pi\colon T \to M$ be the retraction of a tubular neighbourhood $T \subseteq \mathbb{R}^K$ of M (theorem 7.8.1). For each x, $f(x) + b \in T$ once $b \in \mathbb{R}^K$ is small, so on the open unit ball $B \subseteq \mathbb{R}^K$ set

$$F\colon N \times B \to M, \qquad F(x,b) = \pi\big(f(x) + \varepsilon\, b\big),$$

with $\varepsilon > 0$ chosen (using compact exhaustion, or a smooth $\varepsilon(x)$) so that $f(x) + \varepsilon b \in T$ throughout. For fixed x the map $b \mapsto f(x) + \varepsilon b$ is a submersion onto a neighbourhood of $f(x)$ in $\mathbb{R}^K$, and π is a submersion on T, so F is a submersion; in particular $F \pitchfork X$. By parametric transversality (theorem 7.5.6) there is b, arbitrarily close to 0, with F_b transverse to X. Finally $F_0(x) = \pi(f(x)) = f(x)$ since $f(x) \in M$ and π fixes M, so $t \mapsto F_{tb}$ is a homotopy from f to the transverse map F_b, completing the proof.

Transversality of two maps $f\colon N \to M$ and $g\colon N' \to M$ is transversality of $f \times g$ to the diagonal $\Delta_M \subseteq M \times M$, so the compact-source stability and the genericity theorems apply to pairs of maps as well, perturbing either factor.

Exercise 7.5.1

1. Let $C \subseteq \mathbb{R}^n$ be closed and $f(x) = d(x, C)$ the distance to C, a continuous function with $f^{-1}(0) = C$. Let $M = \Gamma(f) \subseteq \mathbb{R}^{n+1}$ be its graph and $N = \mathbb{R}^n \times \{0\}$. Show M and N are topological manifolds with $M \cap N = C \times \{0\} \cong C$, so that the transverse-intersection theorem fails outright for topological manifolds: C can be any closed set.

7.6 Whitney's Immersion and Embedding Theorems

The definition of a manifold names no ambient space. Whitney's theorem shows none is lost by demanding one: every n-manifold is a submanifold of a Euclidean space whose dimension depends only on n. The proof is the first real dividend of Sard's theorem. Build some embedding into a large $\mathbb{R}^N$ by hand, then project it to lower and lower dimensions, discarding at each step only a measure-zero set of bad directions.

Theorem 7.6.1: Whitney Immersion and Embedding

Every smooth n-manifold M admits a smooth immersion into $\mathbb{R}^{2n}$ and a proper smooth embedding into $\mathbb{R}^{2n+1}$. In particular M is diffeomorphic to a closed submanifold of $\mathbb{R}^{2n+1}$.

***Proof*:**
Take M compact first.

Step 1: an embedding into some $\mathbb{R}^N$. Choose a finite atlas (U_i, φ_i), $i = 1, \ldots, k$, and a partition of unity $\{\rho_i\}$ with $\operatorname{supp}\rho_i \subseteq U_i$ (theorem 1.5.5). Each $\rho_i\varphi_i$, extended by 0 off U_i, is a smooth map $M \to \mathbb{R}^n$, and

$$F = (\rho_1\varphi_1, \ldots, \rho_k\varphi_k, \rho_1, \ldots, \rho_k)\colon M \to \mathbb{R}^{k(n+1)}$$

is an embedding. It is injective: if $F(p) = F(q)$, pick i with $\rho_i(p) > 0$; then $\rho_i(q) = \rho_i(p) > 0$, so $p, q \in U_i$, and $\rho_i(p)\varphi_i(p) = \rho_i(q)\varphi_i(q)$ forces $\varphi_i(p) = \varphi_i(q)$, hence $p = q$. It is an immersion: where $\rho_i > 0$ the coordinates recover $\varphi_i = (\rho_i\varphi_i)/\rho_i$, so dF_p is injective. An injective immersion of a compact manifold is an embedding (proposition 3.1.17).

Step 2: lowering the dimension. Let $M \subseteq \mathbb{R}^N$ with $N > 2n+1$, and for a unit vector v let π_v be orthogonal projection onto $v^\perp \cong \mathbb{R}^{N-1}$. As $\ker \pi_v = \mathbb{R}v$, the restriction $\pi_v|_M$ is an immersion unless $v \in T_pM$ for some p, and is injective unless $v = \pm(p-q)/|p-q|$ for distinct $p, q \in M$. The tangent directions are the image of the unit tangent bundle $UM = \{(p, w) \mid w \in T_pM,\ |w| = 1\}$, a compact $(2n-1)$-manifold, under $(p, w) \mapsto w$; the secant directions are the image of the $2n$-manifold $(M \times M) \setminus \Delta$ under $(p, q) \mapsto (p-q)/|p-q|$. Both are smooth images of manifolds of dimension $< N-1 = \dim S^{N-1}$, hence have measure zero in S^{N-1} (corollary 7.2.3). A unit vector outside their union makes $\pi_v|_M$ an injective immersion, so an embedding, into $\mathbb{R}^{N-1}$; iterating drops the target to $\mathbb{R}^{2n+1}$. Ignoring injectivity, a v avoiding only the $(2n-1)$-dimensional tangent directions exists whenever $2n-1 < N-1$, so the projection continues to $\mathbb{R}^{2n}$ and gives an immersion there. On a compact manifold every embedding is proper.

The noncompact case. For general second-countable M a smooth proper function exists: for a partition of unity $\{\rho_i\}$ subordinate to a locally finite cover by precompact charts, $f = \sum_i i\,\rho_i$ satisfies $f(x) > m$ whenever $x \notin \bigcup_{i \le m} U_i$, so each sublevel set $\{f \le m\}$ is compact. Because an n-manifold has covering dimension n, its open covers refine into $n+1$ locally finite families with pairwise-disjoint members ([7]); running Step 1 on each family yields a smooth injective immersion $G\colon M \to \mathbb{R}^L$, so (G, f) is a proper injective immersion, hence a closed embedding $M \hookrightarrow \mathbb{R}^{L+1}$. Replacing f by a generic distance-squared function $x \mapsto |x-a|^2$ on the image (proper, with compact sublevel sets, and Morse) keeps (G, f) a proper closed embedding, so we may take the last coordinate f to be Morse, with $\operatorname{Crit} f$ discrete. Projecting $\mathbb{R}^{L+1}$ along directions orthogonal to the f-axis retains f, hence properness, and can lose injectivity only through a secant with $f(p) = f(q)$. Away from the discrete set $\operatorname{Crit} f \times \operatorname{Crit} f$ the locus $\{(p, q) \mid f(p) = f(q),\ p \neq q\}$ is a $(2n-1)$-manifold, since $(p, q) \mapsto f(p) - f(q)$ is a submersion there; its secant image, the countably many critical-to-critical secants, and the $(2n-1)$-dimensional tangent directions together have measure zero in each sphere. The projection therefore reduces to a proper embedding in $\mathbb{R}^{2n+1}$, and, in one further unrestricted projection, to an immersion in $\mathbb{R}^{2n}$, completing the proof.

Corollary 7.6.2: Manifolds Are Euclidean Submanifolds

Every smooth n-manifold is diffeomorphic to a closed embedded submanifold of $\mathbb{R}^{2n+1}$.

***Proof*:**
The embedding of theorem 7.6.1 is proper, so its image is closed (a proper map into a locally compact Hausdorff space is closed) and it is a diffeomorphism onto that image.

Whitney later sharpened the bounds by one, to an immersion in $\mathbb{R}^{2n-1}$ and an embedding in $\mathbb{R}^{2n}$, through the far subtler *Whitney trick* for cancelling double points; the elementary projection above gives the $2n$ and $2n+1$ bounds, which suffice throughout this book.

7.7 The Normal Bundle

To record how a submanifold sits inside the ambient manifold, one keeps at each point the directions transverse to it. Assembled over the submanifold these form the normal bundle, the object the tubular neighbourhood theorem identifies with an actual neighbourhood.

Definition 7.7.1: Normal Bundle

Let $S \subseteq M$ be an embedded submanifold. The *normal space* to S at p is the quotient $N_pS = T_pM/T_pS$, and the *normal bundle* is $NS = \bigsqcup_{p\in S} N_pS$ with the projection $\nu\colon NS \to S$ sending N_pS to p.

The quotient needs no metric; a Riemannian metric, when present, identifies N_pS with the orthogonal complement $(T_pS)^\perp \subseteq T_pM$.

Proposition 7.7.2: The Normal Bundle Is a Vector Bundle

NS is a smooth manifold of dimension $\dim M$, and $\nu\colon NS \to S$ is a smooth vector bundle of rank $\mathrm{codim} S$: every $p \in S$ has a neighbourhood V in S with $\nu^{-1}(V) \cong V \times \mathbb{R}^{\mathrm{codim} S}$ linearly on fibers. The zero section $p \mapsto 0 \in N_pS$ is an embedding of S.

***Proof*:**
Around p take a slice chart (W, ψ), $\psi(W \cap S) = \psi(W) \cap (\mathbb{R}^s \times 0)$ with $s = \dim S$. For each $q \in W \cap S$ the coordinate vectors $\partial_{s+1}, \dots, \partial_n$ at q project to a basis of $N_qS = T_qM/T_qS$, giving a bijection

$$\Phi\colon \nu^{-1}(W \cap S) \to (W \cap S) \times \mathbb{R}^{\mathrm{codim} S}, \qquad (q, [w]) \mapsto (q, w^{s+1}, \dots, w^n),$$

linear on each fiber. On the overlap of two slice charts the transition is smooth, and the change in the last-coordinate identification is the induced isomorphism of quotients T_qM/T_qS, whose matrix is a smooth invertible function of q; so the Φ form a smooth vector-bundle atlas. The total dimension is $s + \mathrm{codim} S = \dim M$, and the zero section is S. This is the first instance of the general vector bundles of chapter 11.

7.8 Tubular Neighbourhoods

An embedded submanifold sits inside the ambient manifold no more tightly than its normal directions allow: a neighbourhood of it looks like a neighbourhood of the zero section of its normal bundle, and collapses smoothly back onto it. This retraction is the tool the transversality homotopy theorem (theorem 7.5.7) uses to perturb a map into general position, and it underlies the Thom class and Poincaré duality later.

Theorem 7.8.1: Tubular Neighbourhood Theorem

Let $S \subseteq M$ be an embedded submanifold. Then S admits a *tubular neighbourhood*: an open set $U \subseteq M$ containing S, diffeomorphic to the total space of the normal bundle of S by a diffeomorphism fixing S pointwise, together with a smooth retraction $\pi\colon U \to S$ (a submersion, with $\pi|_S = \mathrm{id}_S$; under the diffeomorphism it is the normal-bundle projection). In particular an embedded submanifold $M \subseteq \mathbb{R}^K$ has an open neighbourhood $T \subseteq \mathbb{R}^K$ carrying a smooth submersive retraction $\pi\colon T \to M$.

***Proof*:**

The Euclidean case $S \subseteq \mathbb{R}^K$. Identify N_pS with the orthogonal complement $(T_pS)^\perp \subseteq \mathbb{R}^K$ and form

$$E\colon NS \to \mathbb{R}^K, \qquad E(p, v) = p + v.$$

At a zero-section point $(p, 0)$ the derivative is the identity on $T_pS \oplus N_pS = \mathbb{R}^K$ (tangent directions include, normal directions map by $v \mapsto v$), hence an isomorphism; by the inverse function theorem E is a local diffeomorphism near the zero section. For S compact it is injective on $\{(p, v) \mid |v| < \varepsilon\}$ for small ε: otherwise there would be $(p_i, v_i) \neq (p'_i, v'_i)$ with $E(p_i, v_i) = E(p'_i, v'_i)$ and $|v_i|, |v'_i| \to 0$, and a convergent subsequence would produce a point of the zero section at which E is not locally injective, against the previous line. Then $T = E(\{(p, v) \mid |v| < \varepsilon\})$ is open, E maps the ε-disc bundle diffeomorphically onto T, and $\pi = \nu \circ E^{-1}\colon T \to S$ is the nearest-point retraction, a submersion because ν is. For noncompact S a positive smooth function $\varepsilon(p)$ replaces the constant, obtained from a partition of unity.

The general case $S \subseteq M$. Embed M in some $\mathbb{R}^K$ (theorem 7.6.1) and let $r_M\colon T_M \to M$ be the Euclidean tubular retraction of M from the case just proved. Realise the normal bundle of S in M inside $\mathbb{R}^K$ as $N_pS = (T_pS)^\perp \cap T_pM$, and set

$$\Psi\colon NS \to M, \qquad \Psi(p, v) = r_M(p + v),$$

defined for $|v|$ small enough that $p + v \in T_M$. Then $\Psi(p, 0) = r_M(p) = p$, and for $w \in T_pS$ and $u \in N_pS$,

$$D\Psi_{(p,0)}(w, u) = D(r_M)_p(w + u) = w + u,$$

since $w + u \in T_pM$ and $D(r_M)_p$ is the orthogonal projection onto T_pM; thus $D\Psi_{(p,0)} = \mathrm{id}_{T_pM}$. As in the Euclidean case the inverse function theorem and a compactness (or smooth $\varepsilon(p)$) argument make Ψ a diffeomorphism from an ε-disc bundle in NS onto an open $U \subseteq M$ containing S, fixing S; the retraction is $\pi = \nu \circ \Psi^{-1}$, a submersion, completing the proof.

7.9 Whitney's Approximation Theorem

Sard and the tubular neighbourhood together smooth continuous maps. The mechanism is first visible for maps into $\mathbb{R}^k$, where a partition of unity averages local constant approximations; a general target is reduced to this by embedding it and retracting.

Theorem 7.9.1: Whitney Approximation

Let M, N be smooth manifolds and $f\colon M \to N$ continuous. Then f is homotopic to a smooth map, and the homotopy may be taken relative to any closed set $A \subseteq M$ on a neighbourhood of which f is already smooth. When $N = \mathbb{R}^k$, the smooth map can be taken within any prescribed positive continuous function of f in norm.

***Proof*:**

Target $\mathbb{R}^k$. Let $\varepsilon\colon M \to (0,\infty)$ be continuous. Each p has a neighbourhood U_p with $|f(x)-f(p)| < \varepsilon(x)$ for $x \in U_p$ (continuity of f and ε). Take a partition of unity $\{\rho_i\}$ subordinate to a locally finite refinement $\{U_{p_i}\}$ (theorem 1.5.5) and set $g = \sum_i \rho_i\, f(p_i)$, a smooth map. Then

$$|g(x) - f(x)| = \Big|\sum_i \rho_i(x)\big(f(p_i) - f(x)\big)\Big| \le \sum_i \rho_i(x)\,|f(p_i) - f(x)| < \varepsilon(x),$$

the last step because $\rho_i(x) \neq 0$ only for $x \in U_{p_i}$. The straight-line homotopy $(1-t)f + tg$ stays within ε of f, so $f \simeq g$. If f is smooth on a neighbourhood W of the closed set A, choose a smooth χ that is 1 near A and supported in W, and replace g by $\chi f + (1-\chi)g$: still smooth and ε-close, and equal to f near A, so the homotopy is relative to A.

General target. Embed N in some $\mathbb{R}^K$ (theorem 7.6.1) with tubular neighbourhood T and retraction $r\colon T \to N$ (theorem 7.8.1). Pick a continuous $\delta\colon N \to (0,\infty)$ so that the δ-ball around each $y \in N$ lies in T. Viewing f as a map into $\mathbb{R}^K$, the first part gives a smooth $g_0\colon M \to \mathbb{R}^K$ with $|g_0(x) - f(x)| < \delta(f(x))$, so $g_0(x) \in T$; set $g = r \circ g_0$, a smooth map into N. The segment from $f(x)$ to $g_0(x)$ lies in T, so $t \mapsto r\big((1-t)f(x) + tg_0(x)\big)$ is a homotopy in N from $r(f(x)) = f(x)$ to $g(x)$. Taking $g_0 = f$ near A as above makes it relative to A, completing the proof.

Exercise 7.9.1

1. No compact n-manifold embeds in $\mathbb{R}^n$.
2. Compute the normal bundle of the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$, and show it is trivial.
3. Deduce from theorem 7.9.1 that two smooth maps $f_0, f_1\colon M \to N$ that are continuously homotopic are *smoothly* homotopic.

8

Intersection and Degree Theory

Transversality (section 7.4) turns intersection into a numerical invariant. When a map $f\colon M \to N$ is transverse to a closed submanifold $Z \subseteq N$ of complementary dimension, $f^{-1}(Z)$ is a finite set of points; counting them, with signs once everything is oriented, gives an integer unchanged under homotopy. This *intersection number* is the common source of the *degree* of a map, the Euler characteristic, and the Lefschetz fixed-point count. Its invariance is the parity of section 5.11: the parameter space of a homotopy is one dimension up, and its boundary carries the two ends of the homotopy, where the count must balance. From the signed count the chapter builds the degree, then the Lefschetz number and the index of a vector field, closing with the Poincaré-Hopf theorem, which equates the total index of a vector field with the Euler characteristic $\chi(M) = I(\Delta, \Delta)$.

8.1 Mod-2 Intersection Theory

The transversality theory of chapter 7 makes the preimage $f^{-1}(Z)$ of a submanifold a manifold of predictable dimension whenever $f \pitchfork Z$. This section extracts the first topological invariant from that fact. When M is compact and Z is closed of complementary dimension, $f^{-1}(Z)$ is a finite set, and the parity of its size is unchanged as f varies through a homotopy. The mechanism is the Boundary Theorem: a homotopy fills in a compact one-dimensional cobordism between the two preimages, and a compact 1-manifold has an even number of boundary points (corollary 5.11.2). Counting preimages modulo 2 therefore descends to homotopy classes, and the resulting invariants are the mod-2 intersection number and, when Z is a single point, the mod-2 degree.

Throughout, manifolds are boundaryless unless a boundary is named, and a map $f\colon M \to N$ and a closed submanifold $Z \subseteq N$ are *complementary* when $\dim M + \dim Z = \dim N$; the source M is compact. For such data a transverse f has $f^{-1}(Z)$ a compact 0-manifold, a finite set.

The Boundary Theorem rests on one geometric fact: the preimage of a transverse submanifold under a map on a manifold with boundary is again a manifold with boundary, and its boundary is exactly the part lying over the boundary of the source. The transverse-preimage theorem of chapter 7 handles interior points; boundary points need one added transversality hypothesis, after which they contribute a genuine boundary.

Lemma 8.1.1: Preimage Under a Map on a Manifold with Boundary

Let W be a manifold with boundary, N a boundaryless manifold, and $Z \subseteq N$ a closed embedded submanifold. Suppose $F\colon W \to N$ is transverse to Z and its restriction $\partial F = F|_{\partial W}\colon \partial W \to N$ is also transverse to Z. Then $F^{-1}(Z)$ is an embedded submanifold with boundary of W, with

$$\partial\big(F^{-1}(Z)\big) = F^{-1}(Z) \cap \partial W, \qquad \operatorname{codim} F^{-1}(Z) = \operatorname{codim} Z$$

the codimensions being taken in W and N respectively.

***Proof*:**
Write $k = \operatorname{codim} Z$ and $d = \dim W$. The statement is local, so fix $a \in F^{-1}(Z)$ and, using that Z is embedded, a neighbourhood U of $F(a)$ together with a submersion $\varphi\colon U \to \mathbb{R}^k$ with $Z \cap U = \varphi^{-1}(0)$. On $F^{-1}(U)$ the preimage is the zero set of $g = \varphi \circ F$, and transversality of F to Z at a point p is exactly surjectivity of $D_p g$, that is 0 being a regular value of g (the computation is that of proposition 7.4.3).

Step 1: Interior points. If $a \in \operatorname{int} W$, then near a the map F is defined on a boundaryless manifold and $F \pitchfork Z$, so proposition 7.4.3 makes $F^{-1}(Z)$ an embedded submanifold of codimension k near a, with no boundary contributed.

Step 2: Boundary points. Let $a \in F^{-1}(Z) \cap \partial W$. Choose a boundary chart carrying a neighbourhood of a to the half-space $H^d = \big\{x \in \mathbb{R}^d \;\big|\; x_d \geq 0\big\}$ with $a \mapsto 0$ and $\partial W \mapsto \{x \mid x_d = 0\}$. In this chart F is, by the definition of smoothness on a manifold with boundary, the restriction of a smooth map defined on an open neighbourhood of 0 in $\mathbb{R}^d$; hence $g = \varphi \circ F$ extends to a smooth $\widetilde{g}\colon V \to \mathbb{R}^k$ on an open $V \ni 0$ in $\mathbb{R}^d$. Transversality of F to Z at a makes $D_0\widetilde{g}$ surjective, so 0 is a regular value of $\widetilde{g}$ and $C = \widetilde{g}^{-1}(0)$ is, near 0, an embedded submanifold of dimension $d - k$ (proposition 7.4.3).

Transversality of ∂F to Z at a says that the restriction of $D_0\widetilde{g}$ to the hyperplane $H_0 = \{x \mid x_d = 0\} = T_0\partial W$ is surjective onto $\mathbb{R}^k$. Since $T_0C = \ker D_0\widetilde{g}$, surjectivity of $D_0\widetilde{g}|_{H_0}$ forces $T_0C \cap H_0$ to have dimension $\dim T_0C - 1$, that is $T_0C \not\subseteq H_0$; concretely the coordinate x_d has nonzero differential along C at 0. Thus $x_d|_C$ is a submersion near 0, and

$$F^{-1}(Z) \cap H^d = C \cap \{x \mid x_d \geq 0\} = (x_d|_C)^{-1}\big([0,\infty)\big)$$

near a is the preimage of a half-line under a submersion from a $(d-k)$-manifold, hence an embedded $(d-k)$-manifold with boundary whose boundary is $(x_d|_C)^{-1}(0) = C \cap \partial W = F^{-1}(Z) \cap \partial W$.

Steps 1 and 2 exhibit $F^{-1}(Z)$ locally as a submanifold with boundary of the stated codimension, with boundary points precisely those of $F^{-1}(Z) \cap \partial W$, completing the proof.

The codimension-complementary case specializes this to what the next result needs: when Z is complementary to ∂W, the preimage is one dimensional, and a compact 1-manifold has a controlled boundary.

Theorem 8.1.2: The Boundary Theorem

Let W be a compact manifold with boundary and N a boundaryless manifold, and let $Z \subseteq N$ be a closed submanifold with $\dim \partial W + \dim Z = \dim N$. If $F\colon W \to N$ is transverse to Z and its boundary restriction ∂F is transverse to Z, then $F^{-1}(Z) \cap \partial W$ is a finite set of even cardinality.

***Proof*:**
By lemma 8.1.1, $F^{-1}(Z)$ is a submanifold with boundary of W of dimension

$$\dim F^{-1}(Z) = \dim W - \operatorname{codim} Z = \dim W - (\dim N - \dim Z) = 1$$

where the last equality uses $\dim \partial W = \dim W - 1$ and $\dim \partial W + \dim Z = \dim N$. Its boundary is $F^{-1}(Z) \cap \partial W$. As Z is closed and F continuous, $F^{-1}(Z)$ is closed in the compact W, hence compact. A compact 1-manifold with boundary has an even number of boundary points (corollary 5.11.2), and these are exactly the points of $F^{-1}(Z) \cap \partial W$, so that set is finite of even cardinality, as we sought to show.

The Boundary Theorem is the source of the homotopy invariance below. Read it as a conservation law: the points of $F^{-1}(Z)$ lying over the boundary of W are the endpoints of a compact system of arcs and circles inside W, and endpoints of arcs come in pairs. Whatever a homotopy does in the interior, the count it leaves on the two ends must balance modulo 2. Both transversality hypotheses are used: $F \pitchfork Z$ makes the interior a 1-manifold, and $\partial F \pitchfork Z$ makes the ends genuine boundary points rather than tangencies.

Fix now the setting of a compact source. For $f \pitchfork Z$ with complementary dimensions the preimage $f^{-1}(Z)$ is a compact 0-manifold (proposition 7.4.3), a finite set, and its parity is the invariant to isolate.

Definition 8.1.3: Mod-2 Intersection Number of a Transverse Map

Let M be compact, N boundaryless, and $Z \subseteq N$ a closed submanifold with $\dim M + \dim Z = \dim N$. For a smooth map $f\colon M \to N$ transverse to Z, the *mod-2 intersection number* is

$$I_2(f, Z) = \# f^{-1}(Z) \bmod 2 \ \in \ \mathbb{Z}/2$$

the number of preimage points counted modulo 2.

The definition asks only for a count, so it is completely explicit once a transverse map is in hand. The content, deferred to the next theorem, is that the count does not depend on which transverse map in a homotopy class is used.

Example 8.1.4: A Curve Meeting a Curve on the Torus

Let $N = T^2 = S^1 \times S^1$, let $M = S^1$, and let $f\colon S^1 \to T^2$, $f(z) = (z, 1)$, parametrize the horizontal circle $S^1 \times \{1\}$. Take $Z = \{1\} \times S^1$, a closed 1-submanifold, so $\dim M + \dim Z = 1+1 = 2 = \dim N$. A point $f(z) = (z,1)$ lies in Z exactly when $z = 1$, so $f^{-1}(Z) = \{1\}$ is a single point.

At $z = 1$ the image $Df(T_1 S^1)$ is the tangent line to $S^1 \times \{1\}$, while $T_{(1,1)}Z$ is the tangent line to $\{1\} \times S^1$; these are the two coordinate directions of $T_{(1,1)}T^2$ and together span it, so $f \pitchfork Z$. Hence $I_2(f, Z) = 1$. The two circles cross once, and modulo 2 they cannot be separated: no matter how f is deformed, an odd number of crossings with Z persists.

Homotopy invariance is where the count becomes a topological invariant. A homotopy of maps $M \to N$ is a map on the cylinder $W = M \times [0,1]$, whose boundary is the two ends; if the homotopy is transverse to Z, the Boundary Theorem forces the two end-counts to agree modulo 2. The only technical point is that a homotopy may be perturbed to be transverse to Z without disturbing its ends, which are already transverse.

Lemma 8.1.5: Boundary-Fixing Transversality

Let W be a compact manifold with boundary, N boundaryless, and $Z \subseteq N$ a closed submanifold, and let $F\colon W \to N$ be smooth with $\partial F = F|_{\partial W}$ transverse to Z. Then F is homotopic, through maps agreeing with F on ∂W, to a smooth map F' transverse to Z; the boundary restriction $\partial F' = \partial F$ is then still transverse to Z.

***Proof*:**

Embed N in some $\mathbb{R}^K$ (theorem 7.6.1) and let $\pi\colon T \to N$ be the retraction of a tubular neighbourhood $T \subseteq \mathbb{R}^K$ (theorem 7.8.1). Choose a smooth function $\varepsilon\colon W \to [0,\infty)$ vanishing exactly on ∂W and positive on the interior: patch the boundary-chart functions $x_d \geq 0$ (the last half-space coordinate, which vanishes on ∂W) and the constant 1 on interior charts with a subordinate partition of unity (chapter 1); the result is nonnegative, vanishes on ∂W, and is positive at every interior point, where some chart contributes a positive value. Shrinking ε by a small constant (using compactness of W), arrange $F(x) + \varepsilon(x)\, b \in T$ for all $x \in W$ and all b in the open unit ball $B \subseteq \mathbb{R}^K$, and set

$$F_b(x) = \pi\big(F(x) + \varepsilon(x)\, b\big)$$

a smooth family with $F_0 = F$.

Step 1: The family is a submersion over the interior. For $x \in \operatorname{int} W$ one has $\varepsilon(x) > 0$, so $b \mapsto F(x) + \varepsilon(x) b$ is a submersion onto a neighbourhood of $F(x)$, and π is a submersion; hence the map $(x, b) \mapsto F_b(x)$ on $\operatorname{int} W \times B$ is a submersion, in particular transverse to Z.

Step 2: A generic parameter works. By parametric transversality (theorem 7.5.6) there is b, arbitrarily close to 0, with $F_b|_{\operatorname{int} W}$ transverse to Z. On ∂W, where $\varepsilon = 0$, one has $F_b(x) = \pi(F(x)) = F(x)$ since $F(x) \in N$ and π fixes N; thus F_b agrees with F on ∂W. At a boundary point a, transversality of F_b to Z is automatic once $\partial F_b = \partial F$ is transverse there, because $\operatorname{im} D_a(\partial F_b) \subseteq \operatorname{im} D_a F_b$ enlarges the image only further.

Set $F' = F_b$. Then $F' \pitchfork Z$ on all of W: on the interior by Step 2, and at each boundary point because $\partial F' = \partial F$ is transverse to Z by hypothesis, which as noted forces transversality of F' there. The homotopy $s \mapsto F_{sb}$ is stationary on ∂W, so F' is homotopic to F rel ∂W with $\partial F' = \partial F$, completing the proof.

With a homotopy available transverse to Z and fixed on its already-transverse ends, the invariance of the count is now a direct reading of the Boundary Theorem.

Theorem 8.1.6: Homotopy Invariance of the Mod-2 Intersection Number

Let M be compact, N boundaryless, and $Z \subseteq N$ a closed submanifold with $\dim M + \dim Z = \dim N$. If $f_0, f_1 \colon M \to N$ are homotopic and each transverse to Z, then $I_2(f_0, Z) = I_2(f_1, Z)$.

***Proof*:**
Choose a homotopy $F \colon W \to N$ on the cylinder $W = M \times [0,1]$, with $F(x,0) = f_0(x)$ and $F(x,1) = f_1(x)$. Then W is compact with $\partial W = M \times \{0\} \sqcup M \times \{1\}$, and its boundary restriction is $\partial F = f_0 \sqcup f_1$, transverse to Z by hypothesis. Also $\dim \partial W + \dim Z = \dim M + \dim Z = \dim N$, so Z is complementary to ∂W.

By lemma 8.1.5 the homotopy may be taken transverse to Z without altering its already-transverse boundary values: replace F by a map, homotopic rel ∂W, with $F \pitchfork Z$ and $\partial F = f_0 \sqcup f_1$ still transverse to Z. The Boundary Theorem (theorem 8.1.2) applies, so $F^{-1}(Z) \cap \partial W$ has even cardinality. This set is

$$\big(f_0^{-1}(Z) \times \{0\}\big) \sqcup \big(f_1^{-1}(Z) \times \{1\}\big)$$

of size $\#f_0^{-1}(Z) + \#f_1^{-1}(Z)$. Evenness of the sum forces $\#f_0^{-1}(Z) \equiv \#f_1^{-1}(Z) \pmod 2$, that is $I_2(f_0, Z) = I_2(f_1, Z)$, as we sought to show.

The count is a homotopy invariant, so it measures the map's homotopy class, not the map. This is the promised conservation law made precise: crossings with Z can be created or destroyed only in pairs, because each pair is born or dies at the two ends of an arc in the cylinder. The invariant is coarse, valued in $\mathbb{Z}/2$, and the price of that coarseness is that no orientation is needed anywhere; the finer $\mathbb{Z}$-valued theory, which does need orientations, is taken up in section 8.3.

Homotopy invariance also removes the transversality restriction from the definition. An arbitrary continuous map is not transverse to anything in particular, but it is homotopic to one that is, and the invariant of the transverse representative can be transferred back.

Every continuous map $f \colon M \to N$ is homotopic to a smooth map (smooth approximation, chapter 7), and every smooth map is homotopic to one transverse to Z (theorem 7.5.7). So f is homotopic to a smooth map transverse to Z, and the following definition and proposition make the transfer legitimate.

Definition 8.1.7: Mod-2 Intersection Number of a Continuous Map

Let M be compact, N boundaryless, and $Z \subseteq N$ closed with $\dim M + \dim Z = \dim N$. For any continuous $f \colon M \to N$, choose a smooth map g homotopic to f and transverse to Z, and set

$$I_2(f, Z) = I_2(g, Z) \ \in\ \mathbb{Z}/2$$

Proposition 8.1.8: Well-Definedness of the Extended Intersection Number

The value $I_2(f, Z)$ of definition 8.1.7 is independent of the chosen transverse representative g.

***Proof*:**
Let g and g' both be smooth, homotopic to f, and transverse to Z. Then g is homotopic to g' (through f), and both are transverse to Z, so theorem 8.1.6 gives $I_2(g, Z) = I_2(g', Z)$. The common value is thus independent of the representative, completing the proof.

By construction $I_2(f, Z)$ depends only on the homotopy class of f, and for a smooth transverse f it recovers $\#f^{-1}(Z) \bmod 2$; homotopy invariance now holds for all continuous maps at once. A continuous but nowhere-differentiable loop $f\colon S^1 \to T^2$, for instance, still has a well-defined $I_2(f, Z)$ with the meridian Z of example 8.1.4, computed from any smooth approximation.

When both objects are submanifolds of N, the intersection number becomes a symmetric pairing. For a compact submanifold $X \subseteq N$ the inclusion $\iota_X\colon X \hookrightarrow N$ is a map from a compact manifold, and applying the intersection number to it counts how X meets Z.

Definition 8.1.9: Mod-2 Intersection Number of Submanifolds

Let $X, Z \subseteq N$ be submanifolds of a boundaryless manifold N with X compact, Z closed, and $\dim X + \dim Z = \dim N$. The *mod-2 intersection number* of X and Z is

$$I_2(X, Z) = I_2(\iota_X, Z) \in \mathbb{Z}/2$$

where $\iota_X\colon X \hookrightarrow N$ is the inclusion. When $X \pitchfork Z$ (transverse as submanifolds), $I_2(X, Z) = \#(X \cap Z) \bmod 2$.

Proposition 8.1.10: Symmetry of the Mod-2 Intersection Number

Let $X, Z \subseteq N$ be compact submanifolds of complementary dimension. Then $I_2(X, Z) = I_2(Z, X)$.

***Proof*:**
Two submanifolds are transverse as submanifolds precisely when the inclusion of either is transverse to the other, and this condition is symmetric in X and Z. When $X \pitchfork Z$ the intersection $X \cap Z$ is a finite set, and

$$I_2(X, Z) = \#(X \cap Z) \bmod 2 = \#(Z \cap X) \bmod 2 = I_2(Z, X)$$

the count being literally the same set of points read from either side.

In general neither inclusion need be transverse. The mod-2 intersection number is unchanged under an isotopy of *either* factor: for the map being intersected this is theorem 8.1.6, and for the target submanifold the same Boundary-Theorem argument applies to the trace of the isotopy in $N \times [0, 1]$. Now perturb ι_X to an embedding onto a copy X' of X transverse to Z (theorem 7.5.7); moving X to X' in whichever slot it occupies,

$$I_2(X, Z) = I_2(X', Z) = \#(X' \cap Z) \bmod 2 = \#(Z \cap X') \bmod 2 = I_2(Z, X') = I_2(Z, X),$$

the inner equalities because $X' \pitchfork Z$ is a transverse pair counted the same from either side, completing the proof.

Symmetry says the pairing does not remember which factor was perturbed. It is the mod-2 form of the fact that an intersection point belongs to both submanifolds at once, and it has immediate geometric bite.

Example 8.1.11: Homology of Curves on the Torus

On $T^2 = S^1 \times S^1$ take the two coordinate circles $A = S^1 \times \{1\}$ and $B = \{1\} \times S^1$, each a compact 1-submanifold complementary to the other.

1. A and B meet transversally in the single point $(1, 1)$, exactly as in example 8.1.4, so $I_2(A, B) = 1$. Symmetry gives $I_2(B, A) = 1$ as well.
2. Consequently neither circle bounds. If $A = \partial W$ for a compact surface $W \subseteq T^2$ with boundary, the inclusion of A would extend over W, forcing $I_2(A, B) = 0$ (the extension argument is theorem 8.1.2); but it is 1. For the same reason A and B cannot be pulled apart: any homotopy leaves an odd number of crossings.
3. The self-pairing $I_2(A, A)$ is 0. The circle $A = S^1 \times \{1\}$ is disjoint from its pushoff $A' = S^1 \times \{-1\}$, and ι_A is homotopic to the inclusion of A', which is transverse to A with empty intersection; thus $I_2(A, A) = I_2(A', A) = 0$.

The two circles are the generators of the first homology of the torus, and the pairing I_2 reproduces the mod-2 intersection form: A and B pair to 1, each pairs with itself to 0.

The last invariant of the section is the mod-2 degree, the intersection number against a single point. Its definition presupposes that the count $I_2(f, \{y\})$ does not depend on the point y. When N is connected this is true, and the reason is that any point of N can be slid to any other by a diffeomorphism isotopic to the identity, along which the count cannot change.

Lemma 8.1.12: Homogeneity of a Connected Manifold

Let N be a connected boundaryless manifold. For any two points $y_0, y_1 \in N$ there is a diffeomorphism $h\colon N \to N$, smoothly isotopic to the identity, with $h(y_0) = y_1$.

***Proof*:**

Step 1: A local push in Euclidean space. Let $a, c \in \mathbb{R}^n$ with $|c| < 1$, and let $b = a + c$. Choose a bump function $\rho\colon \mathbb{R}^n \to [0, 1]$ (chapter 1) equal to 1 on the closed ball $\overline{B}(a, |c|)$ and supported in $B(a, 1)$, and set $X = \rho\, c$, a vector field with the constant direction c. Since X vanishes outside the compact ball $\overline{B}(a, 1)$ it is complete (chapter 5), so its flow $\{\varphi_t\}_{t\in\mathbb{R}}$ consists of diffeomorphisms of $\mathbb{R}^n$, each equal to the identity outside $B(a, 1)$, with $\varphi_0 = \mathrm{id}$. For $t \in [0, 1]$ the point $a + tc$ lies in $\overline{B}(a, |c|)$, where $\rho \equiv 1$, so the trajectory through a solves $\dot{x} = c$ there: $\varphi_t(a) = a + tc$, and $\varphi_1(a) = a + c = b$. Thus $\{\varphi_t\}_{t\in[0,1]}$ is an isotopy from id to a diffeomorphism carrying a to b.

Step 2: An equivalence relation. Declare $y \sim y'$ when some diffeomorphism of N isotopic to the identity carries y to y'. This is reflexive (the identity), symmetric (an inverse isotopy), and transitive (concatenate isotopies), so it is an equivalence relation on N.

Step 3: The classes are open. Fix $y \in N$ and a chart carrying a neighbourhood of y diffeomorphically onto $\mathbb{R}^n$. Given y' near y, the chart sends y, y' to points a, b with $|b - a| < 1$, and Step 1 produces an isotopy of $\mathbb{R}^n$ supported in a ball inside the chart image, carrying a to b. Because it is the identity outside a compact subset of the chart image, it extends by the identity to an isotopy of N carrying y to y'. Hence every point near y is equivalent to y, and each class is open.

Step 4: Connectedness. The classes partition N into open sets, and N is connected, so there is a single class. In particular $y_0 \sim y_1$, which is the assertion, completing the proof.

Proposition 8.1.13: The Point-Count Is Independent of the Point

Let M be compact, N connected, both boundaryless, with $\dim M = \dim N$, and let $f\colon M \to N$ be continuous. Then $I_2(f, \{y\})$ is independent of $y \in N$. For a smooth f and a regular value y, it equals $\# f^{-1}(y) \bmod 2$.

***Proof*:**
A point $\{y\} \subseteq N$ is a closed submanifold of dimension 0, complementary to M since $\dim M + 0 = \dim N$, so $I_2(f, \{y\})$ is defined. It suffices to treat smooth f, replacing a continuous map by a smooth homotopic one without changing any value (proposition 8.1.8). For smooth f a point y is a regular value exactly when $f \pitchfork \{y\}$, and then $I_2(f, \{y\}) = \# f^{-1}(y) \bmod 2$.

Fix a regular value y_0 of f, which exists by Sard's theorem (theorem 7.2.1), and let $y_1 \in N$ be arbitrary. By lemma 8.1.12 there is a diffeomorphism $h\colon N \to N$, isotopic to the identity, with $h(y_0) = y_1$; write h_t for the isotopy, $h_0 = \mathrm{id}$, $h_1 = h$. Then $h_t \circ f$ is a homotopy from f to $h \circ f$, so homotopy invariance gives

$$I_2(f, \{y_1\}) = I_2(h \circ f, \{y_1\})$$

Since h is a diffeomorphism with $h(y_0) = y_1$, the map $h \circ f$ is transverse to $\{y_1\}$ exactly where f is transverse to $\{y_0\}$, and $(h \circ f)^{-1}(y_1) = f^{-1}(y_0)$. As y_0 is a regular value of f, both sides are transverse counts and

$$I_2(h \circ f, \{y_1\}) = \#(h \circ f)^{-1}(y_1) \bmod 2 = \# f^{-1}(y_0) \bmod 2$$

Combining, $I_2(f, \{y_1\}) = \# f^{-1}(y_0) \bmod 2$ for every $y_1 \in N$. The right side does not involve y_1, so $I_2(f, \{y\})$ is the same for all y, and it equals $\# f^{-1}(y) \bmod 2$ at any regular value, completing the proof.

Definition 8.1.14: Mod-2 Degree

Let M be compact, N connected, both boundaryless, with $\dim M = \dim N$. The *mod-2 degree* of a continuous map $f\colon M \to N$ is

$$\deg_2 f = I_2(f, \{y\}) \in \mathbb{Z}/2$$

for any $y \in N$, well defined by proposition 8.1.13. For smooth f and a regular value y, $\deg_2 f = \# f^{-1}(y) \bmod 2$.

The mod-2 degree records a single bit: whether f hits a generic point an even or odd number of times. It is a homotopy invariant, since $I_2(f, \{y\})$ is, so homotopic maps have the same degree, and a map of odd degree is never homotopic to a constant, whose degree is 0.

Example 8.1.15: The Degree of $z \mapsto z^k$ on the Circle

View $S^1 \subseteq \mathbb{C}$ and let $f\colon S^1 \to S^1$, $f(z) = z^k$ for an integer k. For $k \neq 0$ the derivative in the angular coordinate $z = e^{i\theta}$, $f(z) = e^{ik\theta}$, multiplies velocities by $k \neq 0$, so every point of S^1 is a regular value. The fibre over $w = e^{i\alpha}$ is the set of k-th roots $\{e^{i(\alpha+2\pi j)/k} \mid 0 \le j < |k|\}$, of size $|k|$. Hence

$$\deg_2(z \mapsto z^k) = |k| \bmod 2 = k \bmod 2$$

For $k = 0$ the map is constant; a point y other than its value is a regular value with empty fibre, so $\deg_2 = 0$, again $k \bmod 2$. Thus $z \mapsto z^k$ has mod-2 degree 1 for k odd and 0 for k even. In particular the identity ($k = 1$) has degree 1 and the squaring map ($k = 2$) has degree 0, so squaring is homotopic to a constant through maps $S^1 \to S^1$ only if its degree vanishes, which it does; the identity, of degree 1, is not.

Exercise 8.1.1

1. Compute the mod-2 degree of the antipodal map $a\colon S^n \to S^n$, $a(x) = -x$, and of a constant map $S^n \to S^n$ (for $n \ge 1$). Which pairs of these maps must fail to be homotopic?

2. Let M be compact and N connected, both boundaryless with $\dim M = \dim N$, and let $f\colon M \to N$ be continuous. Show that if f is not surjective then $\deg_2 f = 0$. Deduce that any map homotopic to a constant has mod-2 degree 0.

3. Let W be a compact manifold with boundary $\partial W = X$, let N be boundaryless, and let $Z \subseteq N$ be closed with $\dim X + \dim Z = \dim N$. Suppose $G\colon W \to N$ is smooth and its restriction $g = G|_X$ is transverse to Z. Show that $I_2(g, Z) = 0$.

4. On $T^2 = S^1 \times S^1$ let $A = S^1 \times \{1\}$ and $B = \{1\} \times S^1$. Using $I_2(A, B) = 1$, show that neither A nor B is the boundary of a compact surface-with-boundary in T^2, and that A and B cannot be made disjoint by an isotopy. Contrast the situation of two disjoint embedded circles on S^2.

5. Let $f\colon M \to N$ and $g\colon N \to P$ be continuous maps between compact connected boundaryless manifolds of one common dimension. Prove the multiplicativity $\deg_2(g \circ f) = \deg_2(g) \cdot \deg_2(f)$ in $\mathbb{Z}/2$.

8.2 Winding Numbers and the Jordan-Brouwer Theorem

Mod-2 intersection theory (section 8.1) turns a geometric count into a homotopy invariant. Its first payoff is a number older than topology itself: whether a closed hypersurface encloses a point off it. The mod-2 winding number decides this by counting, modulo two, how a direction map wraps around the point, and the answer proves the Jordan-Brouwer separation theorem in every dimension. The same parity bookkeeping, applied to maps that commute with the antipodal involution, yields the Borsuk-Ulam theorem and, as a corollary, the Ham Sandwich theorem. Every result in this section is the mod-2 degree of a single sphere-valued map, read in a different geometric disguise.

The one tool imported from section 8.1 is the mod-2 degree $\deg_2 g \in \mathbb{Z}/2$ of a map $g\colon M \to S^k$ from a compact k-manifold: it is the number of preimages of a regular value counted modulo two, is independent of that value, and is a homotopy invariant. The one tool imported from earlier chapters is the parity of the boundary of a compact 1-manifold, $\#\partial N \equiv 0 \pmod 2$ (corollary 5.11.2).

Definition 8.2.1: Mod-2 Winding Number

Let M be a compact $(n-1)$-manifold with $n \geq 2$, and let $f\colon M \to \mathbb{R}^n$ be a smooth map whose image avoids a point $z \in \mathbb{R}^n$. The *direction map* of f about z is

$$u_{f,z}\colon M \to S^{n-1}, \qquad u_{f,z}(x) = \frac{f(x)-z}{|f(x)-z|}$$

The *mod-2 winding number* of f about z is the mod-2 degree of the direction map,

$$W_2(f,z) := \deg_2\left(u_{f,z}\right) \in \mathbb{Z}/2$$

When $X \subseteq \mathbb{R}^n$ is a compact hypersurface and $z \notin X$, its winding number about z is $W_2(X,z) := W_2(\iota_X, z)$, the winding number of the inclusion $\iota_X\colon X \hookrightarrow \mathbb{R}^n$; the direction map is written $u_z(x) = (x-z)/|x-z|$.

The direction map records where each point of M sits as seen from z, and its mod-2 degree records how many times, modulo two, this view sweeps around the whole sphere of directions. For M a hypersurface enclosing z, the view covers every direction an odd number of times; for z far outside, the surface subtends only part of the sky and no direction is covered an odd number of times. The circle makes both cases explicit.

Example 8.2.2: Winding of the Circle

Take $n=2$, $M = S^1$, and $X = S^1 \subseteq \mathbb{R}^2$ the unit circle, so $u_z\colon S^1 \to S^1$.

For $z = 0$ the direction map is $u_0(x) = x$, the identity of S^1, whose only regular value v has the single preimage v; hence $\deg_2 u_0 = 1$ and $W_2(X,0) = 1$. The origin is enclosed.

For z with $|z| > 1$, the circle lies to one side of z: every point of X is within angular distance $\arcsin(1/|z|) < \pi/2$ of the direction $-z/|z|$ pointing from z toward the origin, so u_z misses the open half of S^1 centred on $z/|z|$. A missed value is a regular value with empty preimage, so $\deg_2 u_z = 0$ and $W_2(X,z) = 0$. Points outside the circle are not enclosed, and the winding number sees the difference.

The dividing line between the two computations is that the outside point is joined to infinity by a path missing X, while the inside point is not. The next lemma turns this into the mechanism behind every winding-number vanishing: a map that extends across a filling of its domain winds trivially.

Lemma 8.2.3: Winding Numbers and Bounding

Let W be a compact manifold with boundary $\partial W = M$, and let $F\colon W \to \mathbb{R}^n$ be smooth with $z \notin F(W)$. Then $W_2(F|_M, z) = 0$.

***Proof*:**
Because $z \notin F(W)$, the direction map extends over W: the smooth map

$$U\colon W \to S^{n-1}, \qquad U(y) = \frac{F(y) - z}{|F(y) - z|}$$

restricts on $\partial W = M$ to $u := u_{F|_M, z}$. By theorem 7.2.1 choose a point $p \in S^{n-1}$ that is a regular value of both U and $u = U|_{\partial W}$. Then $U^{-1}(p)$ is a compact 1-manifold with boundary

$$\partial\big(U^{-1}(p)\big) = U^{-1}(p) \cap \partial W = u^{-1}(p)$$

so $\#u^{-1}(p)$ is even (corollary 5.11.2). Hence $\deg_2 u = 0$, that is $W_2(F|_M, z) = 0$, as we sought to show.

This is the winding-number face of the Boundary Theorem of section 8.1: a count taken on the boundary of a compact region vanishes because it balances across the region. The circle example is the case $W = \overline{B^2}$, F the inclusion, and z outside the disc. Read the other way, the lemma is a criterion: if the winding number is nonzero, then f does *not* extend to a z-avoiding map on any filling, and in particular z cannot be pushed to infinity without crossing the image. This is what makes the winding number decide the inside-outside question, once its dependence on z is understood. The dependence is the mildest possible.

Lemma 8.2.4: Winding by Ray Counting

Let $X \subseteq \mathbb{R}^n$ be a compact hypersurface, $z \notin X$, and $v \in S^{n-1}$. If the open ray $R_{z,v} = \{z + tv \mid t > 0\}$ is transverse to X, then v is a regular value of u_z and

$$W_2(X, z) \equiv \#\big(X \cap R_{z,v}\big) \pmod 2$$

***Proof*:**
A point $x \in X$ satisfies $u_z(x) = v$ if and only if $x - z = |x - z|\, v$, that is $x = z + tv$ with $t = |x - z| > 0$, so $u_z^{-1}(v) = X \cap R_{z,v}$. The map u_z is the radial projection $y \mapsto (y - z)/|y - z|$ restricted to X; its differential at such an x has kernel the radial line $\mathbb{R}\,(x - z)$ and is an isomorphism on any complement onto $T_v S^{n-1}$. Thus $d(u_z)_x$ is injective on $T_x X$ precisely when $T_x X$ does not contain the radial direction v, that is when $T_x X + \mathbb{R}v = \mathbb{R}^n$, which is transversality of the ray to X at x. So v is a regular value, and $W_2(X, z) = \deg_2 u_z = \#u_z^{-1}(v) \bmod 2 = \#(X \cap R_{z,v}) \bmod 2$ (section 8.1), completing the proof.

The ray count makes the winding number computable and exposes both how it stays constant and how it jumps. Sliding z without crossing X deforms the direction map continuously and cannot change its degree; sliding z across X adds or removes exactly one intersection on a suitable ray.

Proposition 8.2.5: The Winding Number Is Locally Constant

Let $X \subseteq \mathbb{R}^n$ be a compact hypersurface. The function $z \mapsto W_2(X, z)$ is locally constant on $\mathbb{R}^n \setminus X$, hence constant on each connected component. It equals 0 for every z outside a ball containing X.

***Proof*:**
If z_0 and z_1 lie in a ball disjoint from X, the segment $z_s = (1-s)z_0 + sz_1$ stays in $\mathbb{R}^n \setminus X$, and $(x, s) \mapsto (x - z_s)/|x - z_s|$ is a homotopy from u_{z_0} to u_{z_1}. Homotopic maps have equal mod-2 degree (section 8.1), so $W_2(X, z_0) = W_2(X, z_1)$; thus $W_2(X, -)$ is locally constant, and constant on components. If $X \subseteq B_r(0)$ and $|z| > r$, choose $v = z/|z|$: the ray $R_{z,v}$ points away from the ball and misses X, so $\#(X \cap R_{z,v}) = 0$ and $W_2(X, z) = 0$ by lemma 8.2.4, completing the proof.

Lemma 8.2.6: Crossing a Hypersurface Flips the Winding Number

Let $X \subseteq \mathbb{R}^n$ be a compact hypersurface and ℓ a line transverse to X. If $z_0, z_1 \in \ell \setminus X$ are chosen so that the segment between them meets X in exactly one point, then

$$W_2(X, z_0) + W_2(X, z_1) = 1 \quad \text{in } \mathbb{Z}/2$$

***Proof*:**
Let v be the unit direction of ℓ pointing from z_1 toward z_0, and set $t_0 = |z_0 - z_1| > 0$, so $z_0 = z_1 + t_0 v$. Because ℓ is transverse to X, so is each ray along it, and v is a regular value of both u_{z_0} and u_{z_1} (lemma 8.2.4). The two rays are collinear:

$$R_{z_1,v} = \{z_1 + tv \mid t > 0\} = \big(z_1, z_0\big] \sqcup R_{z_0,v}, \qquad R_{z_0,v} = \{z_1 + tv \mid t > t_0\}$$

so $X \cap R_{z_1,v} = \big(X \cap (z_1, z_0]\big) \sqcup \big(X \cap R_{z_0,v}\big)$. The half-open segment $(z_1, z_0]$ is the part of $[z_0, z_1]$ not equal to z_1, and since $z_0 \notin X$ while the segment meets X in its single interior crossing, $X \cap (z_1, z_0]$ is that one point. Hence $\#(X \cap R_{z_1,v}) = \#(X \cap R_{z_0,v}) + 1$, and reducing modulo two, $W_2(X, z_1) = W_2(X, z_0) + 1$ by lemma 8.2.4, completing the proof.

Local constancy and the crossing rule are the two halves of a separation theorem. The winding number is constant on each region cut out by X and changes by one on stepping across X, so it can take at most two values, and it takes both. That X cuts $\mathbb{R}^n$ into exactly two regions, and not more, is the content of Jordan-Brouwer; the crossing rule gives the lower bound of two, and a tubular neighbourhood gives the upper bound.

Theorem 8.2.7: Jordan-Brouwer Separation

Let $X \subseteq \mathbb{R}^n$ $(n \geq 2)$ be a compact connected smooth hypersurface. Then $\mathbb{R}^n \setminus X$ has exactly two connected components,

$$D_0 = \{z \mid W_2(X, z) = 0\}, \qquad D_1 = \{z \mid W_2(X, z) = 1\}$$

of which D_0 is unbounded and D_1 is bounded. Each D_i is open, and X is the topological boundary of each: $\partial D_0 = \partial D_1 = X$.[a]

[a] Camille Jordan stated the case $n = 2$ (the closed curve) in 1887, and L. E. J. Brouwer proved the general-dimensional separation around 1911; the winding-number proof given here is the smooth-category streamlining of Brouwer's.

Proof:
By proposition 8.2.5 the function $W_2(X, -)$ is constant on each component of $\mathbb{R}^n \setminus X$, so D_0 and D_1 are unions of components. Both are nonempty: the region outside a ball containing X has $W_2 = 0$, and applying lemma 8.2.6 to a line transverse to X (a generic line, by theorem 7.2.1) produces a point with $W_2 = 1$. Thus $\mathbb{R}^n \setminus X$ has at least two components.

Step 1: X is two-sided. By the tubular neighbourhood theorem (theorem 7.8.1) there is an open neighbourhood N of X and a diffeomorphism of N with the total space of the normal bundle $\nu \to X$, carrying X to the zero section; ν is a real line bundle, since $\dim \mathbb{R}^n - \dim X = 1$. Then $N \setminus X$ is diffeomorphic to ν minus its zero section, which fibers over X with fiber the two rays $\mathbb{R} \setminus \{0\}$. If $N \setminus X$ were disconnected, fiberwise negation, a homeomorphism exchanging the two rays in each fiber, would exchange its components, so each component would meet every fiber in exactly one ray; choosing that ray would be a nowhere-vanishing section of ν, forcing ν to be trivial. Suppose then, for contradiction, that ν is nontrivial; the above shows $N \setminus X$ is connected, so $W_2(X, -)$ is constant on it. But for any $x \in X$ a short segment through x transverse to X has its two endpoints in $N \setminus X$ on opposite sides, and by lemma 8.2.6 they carry different winding numbers, a contradiction. Hence ν is trivial: $N \setminus X \cong X \times (\mathbb{R} \setminus \{0\})$ splits into two connected collars N_+ and N_-, each diffeomorphic to $X \times \mathbb{R}$.

Step 2: at most two components. Let C be any component of $\mathbb{R}^n \setminus X$; it is open. Its closure meets X: otherwise C would be open and closed in the connected space $\mathbb{R}^n$, forcing $C = \mathbb{R}^n$, impossible as $X \neq \emptyset$. Pick $x \in \overline{C} \cap X$. Since N is a neighbourhood of x and C is disjoint from X, the set C meets $N \setminus X = N_+ \sqcup N_-$; as $N_\pm$ are connected and disjoint from X, whichever one C meets is contained in C. So every component contains N_+ or N_-, and with only two collars there are at most two components. Combined with the lower bound, $\mathbb{R}^n \setminus X$ has exactly two components, distinguished by $W_2 \in \{0, 1\}$, so they are D_0 and D_1.

Step 3: bounded, unbounded, and boundary. Choose r with $X \subseteq B_r(0)$. The set $\mathbb{R}^n \setminus \overline{B_r(0)}$ is connected (as $n \geq 2$) and has $W_2 = 0$, so it lies in D_0; hence D_0 is unbounded, and $D_1 \subseteq B_r(0)$ is bounded. Finally, the two collars N_+, N_- carry different winding numbers (their points near a common $x \in X$ are on opposite sides of X, lemma 8.2.6), so one lies in D_0 and the other in D_1. Thus every $x \in X$ is a limit of points of D_0 and of points of D_1, giving $X \subseteq \partial D_0 \cap \partial D_1$. Conversely $\partial D_i \subseteq X$, because D_i is open and its complement $X \sqcup D_{1-i}$ has the open set D_{1-i}

contributing no boundary points. Hence $\partial D_0 = \partial D_1 = X$, completing the proof.

A compact connected smooth hypersurface therefore behaves exactly like the sphere: it has a well-defined inside and outside, and it is the full common boundary of both. The proof extracts more than separation. Step 1 shows such a hypersurface is *two-sided*, its normal bundle trivial, and hence (with the standard orientation of $\mathbb{R}^n$) orientable, a fact that fails for hypersurfaces in other ambient manifolds, where the normal bundle may be nontrivial, as for the one-sided $\mathbb{RP}^{n-1} \subseteq \mathbb{RP}^n$. The smoothness is doing real work: the topological Jordan-Brouwer theorem for continuous embeddings holds as well, but its proof runs through the homology of the complement rather than a tubular neighbourhood, and belongs to algebraic topology ([11, Generalized Jordan Curve Theorem]). The winding number gives the separation and, at the same time, names the two pieces.

Definition 8.2.8: Inside and Outside

For a compact connected smooth hypersurface $X \subseteq \mathbb{R}^n$, the bounded component $D_1 = \{z \mid W_2(X,z) = 1\}$ of $\mathbb{R}^n \setminus X$ is the *inside* of X, and the unbounded component $D_0 = \{z \mid W_2(X,z) = 0\}$ is the *outside*. A point $z \notin X$ is inside X if and only if $W_2(X,z) = 1$.

This inside-outside criterion is a decision procedure: to test whether z lies inside X, shoot a generic ray from z and count its transverse crossings with X; the point is inside precisely when the count is odd (lemma 8.2.4). It is the higher-dimensional even-odd rule familiar for polygons in the plane, and it is independent of the ray because the winding number is.

The same parity idea, transferred from a point in $\mathbb{R}^n$ to the antipodal involution of a sphere, proves the Borsuk-Ulam theorem. The bridge is a single fact about maps that anticommute with the antipode.

Definition 8.2.9: Antipodal Map

A map $g\colon S^n \to S^k$ is *antipodal* (or *odd*) if $g(-x) = -g(x)$ for all x.

The mod-2 degree of an antipodal self-map is forced to be odd.

Lemma 8.2.10: Antipodal Self-Maps Have Odd Degree

Every smooth antipodal map $A\colon S^m \to S^m$ has $\deg_2 A = 1$, for all $m \geq 0$.

***Proof*:**
Induction on m.

Base $m = 0$. An antipodal map $A\colon S^0 \to S^0$ sends the two points $\{1, -1\}$ to $A(1) = a$ and $A(-1) = -a$, a bijection; each point of S^0 is a regular value with a single preimage, so $\deg_2 A = 1$.

Reduction to the equator ($m \geq 1$). Let $\Sigma = S^{m-1} = S^m \cap \{x_{m+1} = 0\}$ be the equator, with closed hemispheres $D_\pm = S^m \cap \{\pm x_{m+1} \geq 0\}$, so $\partial D_+ = \Sigma$ and $D_- = -D_+$. The restriction $A|_\Sigma\colon \Sigma \to S^m$ has image of measure zero (its domain has dimension $m - 1 < m$, theorem 7.2.1), so it misses a point q, and by antipodal symmetry of its image it misses $-q$ too. The complement $S^m \setminus \{q, -q\}$ deformation-retracts, antipodally and equivariantly, onto the great sphere $\{x \perp q\}$ along meridians. Composing $A|_\Sigma$ with this retraction and then with an antipodal-preserving rotation carrying $\{x \perp q\}$ to Σ, then extending the resulting homotopy of $A|_\Sigma$ over a symmetric collar of Σ (theorem 7.8.1), produces a smooth antipodal map homotopic to A that carries Σ into Σ. Since homotopic maps have equal mod-2 degree, assume from now on $A(\Sigma) \subseteq \Sigma$, and write $\widehat{A} := A|_\Sigma \colon \Sigma \to \Sigma$, itself antipodal, with $\deg_2 \widehat{A} = 1$ by the inductive hypothesis.

A parity count. Set $B := A|_{D_+} \colon D_+ \to S^m$. Fix a point y in the open upper hemisphere and an arc $\gamma \subseteq S^m$ from y to $-y$ meeting the equator Σ in exactly one point w, chosen generically so that B is transverse to γ, that $y, -y$ are regular values of B, and that w is a regular value of $\widehat{A}$ (theorem 7.2.1). Then $N := B^{-1}(\gamma)$ is a compact 1-manifold with boundary

$$\partial N = B^{-1}(y) \sqcup B^{-1}(-y) \sqcup \big(B^{-1}(\gamma) \cap \Sigma\big)$$

and $B^{-1}(\gamma) \cap \Sigma = \widehat{A}^{-1}(\gamma \cap \Sigma) = \widehat{A}^{-1}(w)$, since $\widehat{A}$ maps Σ into Σ and $\gamma \cap \Sigma = \{w\}$. By corollary 5.11.2 the total count $\#\partial N$ is even, so

$$\#B^{-1}(y) + \#B^{-1}(-y) \equiv \#\widehat{A}^{-1}(w) \pmod 2$$

Conclusion. Because $A(\Sigma) \subseteq \Sigma$ avoids y and $-y$, the preimages $A^{-1}(y)$ lie in the open hemispheres. Those in D_+ are exactly $B^{-1}(y)$; each x in the open lower hemisphere with $A(x) = y$ corresponds under $x \mapsto -x$ to a point $-x \in D_+$ with $A(-x) = -y$, that is to a point of $B^{-1}(-y)$. Hence, at the regular value y,

$$\deg_2 A = \#A^{-1}(y) \equiv \#B^{-1}(y) + \#B^{-1}(-y) \equiv \#\widehat{A}^{-1}(w) = \deg_2 \widehat{A} = 1 \pmod 2$$

completing the induction.

The lemma says an odd symmetry cannot be unwound: no antipodal self-map of a sphere is mod-2 nullhomotopic. Its immediate corollary is that the antipodal symmetry cannot be pushed down a dimension.

Corollary 8.2.11: No Antipodal Map to a Lower Sphere

For $n \geq 1$ there is no continuous antipodal map $g \colon S^n \to S^{n-1}$.

***Proof*:**
Suppose $g \colon S^n \to S^{n-1}$ is continuous and antipodal. Smooth approximation followed by symmetrization replaces it by a smooth antipodal map: if g' is a smooth approximation with $|g' - g|$ small, then $g'(x) - g'(-x)$ is close to $2g(x) \neq 0$, so $x \mapsto (g'(x) - g'(-x))/|g'(x) - g'(-x)|$ is smooth, antipodal, and defined everywhere. Assume then g smooth. Its restriction to the equator $\Sigma = S^{n-1} \subseteq S^n$ is an antipodal self-map $h = g|_\Sigma \colon S^{n-1} \to S^{n-1}$, so $\deg_2 h = 1$ by lemma 8.2.10. But h extends over the upper hemisphere disc D^n (via $g|_{D^n} \colon D^n \to S^{n-1}$, with $\partial D^n = \Sigma$): for a common regular value p of $g|_{D^n}$ and h, the preimage $(g|_{D^n})^{-1}(p)$ is a compact 1-manifold with boundary $h^{-1}(p)$, so $\#h^{-1}(p)$ is even (corollary 5.11.2) and $\deg_2 h = 0$. This contradicts $\deg_2 h = 1$, so no such g exists, completing the proof.

Borsuk-Ulam is the geometric restatement of this impossibility. If a map to $\mathbb{R}^n$ separated every antipodal pair, the normalized difference would be exactly the forbidden antipodal map to S^{n-1}.

Theorem 8.2.12: Borsuk-Ulam

Every continuous map $f\colon S^n \to \mathbb{R}^n$ ($n \geq 1$) identifies some antipodal pair: there exists $x \in S^n$ with $f(x) = f(-x)$.

***Proof*:**
Suppose $f(x) \neq f(-x)$ for every $x \in S^n$. Then

$$u\colon S^n \to S^{n-1}, \qquad u(x) = \frac{f(x) - f(-x)}{|f(x) - f(-x)|}$$

is a well-defined continuous map, and it is antipodal, since $u(-x) = (f(-x) - f(x))/|f(-x) - f(x)| = -u(x)$. This contradicts corollary 8.2.11. Hence some x has $f(x) = f(-x)$, as we sought to show.

Borsuk-Ulam says a continuous field of n real quantities on the n-sphere cannot separate every antipodal pair: somewhere the two ends of a diameter read identically. In the proof, the failure of the conclusion manufactures exactly the antipodal map to a lower sphere that odd degree forbids. Two consequences follow: an existence theorem it hands to measure theory, and a concrete reading on the 2-sphere.

The Ham Sandwich theorem is the statement that a single cut bisects several masses at once. It is Borsuk-Ulam with the sphere of directions reinterpreted as the sphere of affine hyperplanes.

Corollary 8.2.13: Ham Sandwich Theorem

Let $A_1, \ldots, A_n$ be Lebesgue-measurable subsets of $\mathbb{R}^n$, each of finite measure. Then there is an affine hyperplane $H \subseteq \mathbb{R}^n$ that simultaneously bisects every A_i: each of the two closed half-spaces bounded by H contains exactly half the measure of each A_i.[a]

[a]Named for the problem of halving a sandwich of bread, ham, and bread with one straight cut; the general statement and this Borsuk-Ulam proof are due to Stone and Tukey (1942).

***Proof*:**
Parametrize oriented affine hyperplanes by the sphere $S^n \subseteq \mathbb{R}^{n+1} = \mathbb{R}^n \times \mathbb{R}$. To a unit vector (a, b) with $a \in \mathbb{R}^n$, $b \in \mathbb{R}$, associate the closed half-space $H^+_{(a,b)} = \{x \in \mathbb{R}^n \mid a \cdot x \geq b\}$, and define

$$f\colon S^n \to \mathbb{R}^n, \qquad f(a, b) = \bigl(\mu_1(a, b), \ldots, \mu_n(a, b)\bigr), \quad \mu_i(a, b) = \mathrm{vol}\bigl(A_i \cap H^+_{(a,b)}\bigr)$$

Each μ_i is continuous: as (a, b) varies, the half-space varies, and dominated convergence applies because a hyperplane has Lebesgue measure zero, so no mass sits on the boundary in the limit. By theorem 8.2.12 there is (a, b) with $f(a, b) = f(-(a, b))$. Negating (a, b) replaces the half-space by its closure-complementary one (the other side of the same hyperplane), so $\mu_i(-(a, b)) = |A_i| - \mu_i(a, b)$, again because the bounding hyperplane carries no measure. The equality $\mu_i(a, b) = \mu_i(-(a, b))$ therefore reads $\mu_i(a, b) = |A_i|/2$ for every i. Finally $a \neq 0$: if $a = 0$ then $(a, b) = (0, \pm 1)$ and each half-space is all of $\mathbb{R}^n$ or empty, forcing $\mu_i \in \{0, |A_i|\}$, incompatible with $\mu_i = |A_i|/2$ unless every A_i is null (in which case any hyperplane bisects). Thus $H = \{x \mid a \cdot x = b\}$ is a genuine hyperplane bisecting all the A_i, completing the proof.

The measurable sets need not be connected or convex; the one hyperplane splits each in half regardless. The proof is a template: any n continuous mass distributions on $\mathbb{R}^n$ are simultaneously bisected, because the bisection condition is exactly the antipodal coincidence that Borsuk-Ulam guarantees. The planar case is the sandwich itself, two regions and one line.

Example 8.2.14: Equal Temperature and Pressure at Antipodes

Model the surface of the Earth as S^2 and suppose temperature $T\colon S^2 \to \mathbb{R}$ and barometric pressure $P\colon S^2 \to \mathbb{R}$ vary continuously. The pair $f = (T, P)\colon S^2 \to \mathbb{R}^2$ satisfies the hypotheses of theorem 8.2.12 with $n = 2$, so there is a point $x \in S^2$ with $f(x) = f(-x)$: at x and its antipode $-x$ the temperature and the pressure agree simultaneously. No smoothness or genericity of T and P is needed, only continuity; the conclusion is forced by the topology of the sphere, not by any feature of the weather. The same statement with a single scalar (temperature alone) is the $n = 1$ case applied to a great circle: some pair of antipodal points on any circle has equal temperature.

Exercise 8.2.1

1. Let $S^{n-1} \subseteq \mathbb{R}^n$ be the unit sphere. Show directly that $W_2(S^{n-1}, z) = 1$ when $|z| < 1$ and $W_2(S^{n-1}, z) = 0$ when $|z| > 1$, using the bounding lemma for the outside and the ray count for the inside.
2. Let M be a compact $(n-1)$-manifold and $f\colon M \to \mathbb{R}^n$ smooth. Show that if z_0 and z_1 lie in the same connected component of $\mathbb{R}^n \setminus f(M)$, then $W_2(f, z_0) = W_2(f, z_1)$. (This is why the winding number depends on z only through its component.)
3. Deduce from the proof of theorem 8.2.7 that a compact connected smooth hypersurface $X \subseteq \mathbb{R}^n$ is orientable, by exhibiting a consistent orientation of T_xX from a global unit normal.
4. Use the Borsuk-Ulam theorem to prove that no continuous map $g\colon S^n \to \mathbb{R}^n$ is injective, and conclude that S^n does not embed in $\mathbb{R}^n$.
5. State and prove the planar Ham Sandwich theorem: any two bounded measurable regions $A_1, A_2 \subseteq \mathbb{R}^2$ are simultaneously bisected by a line. Then show separately that a single bounded measurable region can be bisected by a line in every prescribed direction.

8.3 Oriented Intersection Theory

The mod-2 theory of section 8.1 counts transverse preimages, but only their parity survives. Orienting M, N, and Z promotes that count to a signed integer, the oriented intersection number $I(f, Z) \in \mathbb{Z}$, which refines $I_2(f, Z)$ and carries strictly more information: from it come a degree, a Lefschetz number, and an Euler characteristic. This section fixes the orientation conventions that attach a sign to each transverse intersection, defines $I(f, Z)$, and proves it a homotopy invariant by the same cobordism argument as before, now tracking signs through the oriented Boundary Theorem. The number then transfers to submanifolds, where it acquires a symmetry $I(X, Z) = \pm I(Z, X)$ and, on the diagonal, computes the self-intersection that section 8.6 will name the Euler characteristic.

Everything below rests on a single bookkeeping device: an orientation of a vector space is a choice of positive ordered basis up to positive determinant, and for a direct sum $U = V \oplus W$ the notation $[V] \oplus [W] = [U]$ means that a positive basis of V followed by a positive basis of W is a positive basis of U. Reordering the summands, or reversing one of them, changes $[U]$ by the sign of the corresponding permutation or by -1. The three conventions the theory needs are all instances of the same rule, and they are chosen once so that they never fight one another.

8.3.1 Convention: Orientation Sign Rules Fix oriented manifolds throughout, with orientations recorded fibrewise as above.

1. *Product orientation.* $M \times N$ is oriented so that a positive frame of T_pM followed by a positive frame of T_qN is a positive frame of $T_{(p,q)}(M \times N)$; that is, $[T_pM] \oplus [T_qN] = [T_{(p,q)}(M \times N)]$.
2. *Boundary orientation.* A frame of $T_x(\partial M)$ is positive exactly when prefixing it with an outward-pointing vector gives a positive frame of T_xM (outward normal first).
3. *Normal orientation.* For an oriented submanifold $Z \subseteq N$ of the oriented N, the normal space $\nu_y(Z) = T_yN/T_yZ$ is oriented so that $[\nu_y(Z)] \oplus [T_yZ] = [T_yN]$ (normal direction first).

The normal orientation is the pattern to remember: the complementary directions come first, the submanifold second. The boundary orientation is the case $Z = \partial M$ with the outward normal as the one complementary direction, and the sign rule below is the same choice applied to $df_x(T_xM)$ as the complementary piece. With the conventions fixed, a transverse map orients its preimage.

Definition 8.3.2: Orientation of a Preimage

Let M and N be oriented, let $Z \subseteq N$ be a closed oriented submanifold without boundary, and let $f\colon M \to N$ be transverse to Z. By proposition 7.4.3, $S = f^{-1}(Z)$ is a submanifold of M with $\operatorname{codim}_M S = \operatorname{codim}_N Z$, and $T_xS = (df_x)^{-1}(T_{f(x)}Z)$ at each $x \in S$ (corollary 7.4.5). Transversality makes the induced map

$$df_x\colon \nu_x(S) = T_xM/T_xS \longrightarrow \nu_{f(x)}(Z) = T_{f(x)}N/T_{f(x)}Z$$

an isomorphism. The *preimage orientation* of S is defined by giving $\nu_x(S)$ the orientation pulled back from the normal orientation of $\nu_{f(x)}(Z)$ through this isomorphism, and then orienting T_xS by the direct-sum rule $[\nu_x(S)] \oplus [T_xS] = [T_xM]$.

The construction reads off in one line for two oriented curves crossing in an oriented surface. If $\dim M = \dim Z = 1$ and $\dim N = 2$, then at a transverse crossing x the space T_xS is zero, $\nu_x(S) = T_xM$, and the orientation of the point is a sign: it records whether the isomorphism $df_x\colon T_xM \to \nu_{f(x)}(Z)$ matches the two chosen normal orientations. That sign is the object the whole theory counts, so it is worth a name of its own.

Definition 8.3.3: The Sign of a Transverse Intersection

Let M, N, Z be oriented with $Z \subseteq N$ closed and without boundary, let $f\colon M \to N$ be transverse to Z, and suppose the dimensions are *complementary*, $\dim M + \dim Z = \dim N$. For $x \in f^{-1}(Z)$ transversality gives the direct sum

$$df_x(T_xM) \oplus T_{f(x)}Z = T_{f(x)}N$$

and df_x is injective on T_xM. The *intersection sign* of f at x is

$$\operatorname{sign}(x) = \begin{cases} +1 & \text{if } [df_x(T_xM)] \oplus [T_{f(x)}Z] = [T_{f(x)}N] \\ -1 & \text{otherwise} \end{cases}$$

where $df_x(T_xM)$ carries the orientation pushed forward from T_xM. This is the preimage orientation of definition 8.3.2 in the zero-dimensional case, read as an element of $\{\pm 1\}$.

The sign compares two ways of building the orientation of $T_{f(x)}N$: the one carried into N along f and the one already on Z. Where the map's image reinforces the ambient orientation the crossing counts positively, where it opposes it the crossing counts negatively. Summing these signs over a compact source gives a finite integer.

Definition 8.3.4: Oriented Intersection Number

Let M be compact and oriented, N oriented, and $Z \subseteq N$ a closed oriented submanifold without boundary, with $\dim M + \dim Z = \dim N$. For $f\colon M \to N$ transverse to Z, the set $f^{-1}(Z)$ is a compact 0-manifold, hence finite, and the *oriented intersection number* is

$$I(f, Z) = \sum_{x \in f^{-1}(Z)} \operatorname{sign}(x) \in \mathbb{Z}$$

Reducing each sign modulo 2 collapses ± 1 to 1 and recovers the mod-2 count of section 8.1, so $I(f, Z) \equiv I_2(f, Z) \pmod 2$. The integer sees more: it distinguishes a map that crosses Z twice with cancelling signs from one that misses Z altogether, whereas I_2 cannot. The next example carries out both the computation and the refinement.

Example 8.3.5: Winding on the Torus

Write the torus as $T^2 = S^1 \times S^1$ with $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ and coordinates (x, y), oriented by $[\partial_x] \oplus [\partial_y]$. Fix the meridian $Z = \{0\} \times S^1$, a closed oriented 1-submanifold with positive tangent ∂_y, complementary to any map from a 1-manifold. For an integer $k \neq 0$ let

$$f_k\colon S^1 \to T^2, \qquad f_k(\theta) = (k\theta, 0)$$

with S^1 oriented by ∂_θ. Then $f_k(\theta) \in Z$ iff $k\theta \equiv 0 \pmod{2\pi}$, giving the $|k|$ points $\theta_j = 2\pi j/k$, $j = 0, \ldots, |k| - 1$. At each, $df_k(\partial_\theta) = k\,\partial_x$, transverse to $T_{f_k(\theta_j)}Z = \mathbb{R}\,\partial_y$ because $k \neq 0$. The intersection sign compares $[k\,\partial_x] \oplus [\partial_y]$ with the torus orientation $[\partial_x] \oplus [\partial_y]$: for $k > 0$ this is orientation-preserving and the sign is $+1$, for $k < 0$ it is -1. All $|k|$ points share the sign of k, so

$$I(f_k, Z) = k$$

The map wraps the first circle k times, and the oriented count records the winding with its sense.

Modulo 2 this is k mod 2, exactly $I_2(f_k, Z)$: the mod-2 number cannot tell f_2 (two cancelling-in-parity crossings) from a constant map, but $I(f_2, Z) = 2 \neq 0$ does. This integer is the degree of the circle map $\theta \mapsto k\theta$, which section 8.4 recovers as intersection with a point.

Homotopy invariance is what makes $I(f, Z)$ a topological rather than an incidental quantity, and its proof is the oriented refinement of the cobordism argument behind section 8.1. The engine is the signed count on a compact 1-manifold, where the classification of section 5.11 forces the signs to cancel. For a compact oriented 1-manifold C and a boundary point $x \in \partial C$, the boundary orientation of box 8.3.1 assigns x the sign $\sigma_\partial(x) = +1$ if the positive unit tangent of C points out of C at x, and $\sigma_\partial(x) = -1$ if it points inward.

Theorem 8.3.6: Oriented Boundary Theorem

Let C be a compact oriented 1-manifold with boundary. Then the boundary signs sum to zero,

$$\sum_{x \in \partial C} \sigma_\partial(x) = 0$$

***Proof*:**
By the classification of compact 1-manifolds (section 5.11), C is a finite disjoint union of components, each diffeomorphic to the circle S^1 or to the closed interval $[0, 1]$. A circle component has empty boundary and contributes nothing. On an interval component, the orientation is a single coherent choice of positive tangent direction along the arc; that direction points outward at one endpoint and inward at the other, since the two outward directions at the ends of an arc are opposite. So each interval contributes $(+1) + (-1) = 0$. Summing over components gives 0, as we sought to show.

This refines the parity statement of corollary 5.11.2, which records only that $\#\partial C$ is even: here the signed sum vanishes on the nose. To run the homotopy argument the boundary signs of a preimage must be matched to intersection signs, and the two differ by a single universal sign, computed once in the following lemma.

Lemma 8.3.7: Boundary of an Oriented Preimage

Let W be an oriented manifold with boundary, N oriented, and $Z \subseteq N$ a closed oriented submanifold without boundary. Let $F\colon W \to N$ be transverse to Z with $\partial F = F|_{\partial W}$ also transverse to Z, and suppose $\dim F^{-1}(Z) = 1$. Then $S = F^{-1}(Z)$ is a 1-manifold with $\partial S = (\partial F)^{-1}(Z)$, and at each $x \in \partial S$ the boundary sign of S (with its preimage orientation) and the intersection sign of ∂F are related by

$$\sigma_\partial(x) = (-1)^{\operatorname{codim} Z} \operatorname{sign}_{\partial F}(x)$$

***Proof*:**
Write $c = \operatorname{codim} Z$, so $\dim W = c + 1$ and $\dim \partial W = c$. By the preimage theorem for manifolds with boundary (lemma 8.1.1), the two transversality hypotheses make S a 1-manifold meeting ∂W transversally in $\partial S = (\partial F)^{-1}(Z)$. Fix $x \in \partial S$, set $y = F(x)$, and let $q = (dF_x \bmod T_y Z)\colon T_x W \to \nu_y(Z)$; transversality makes q surjective with kernel $T_x S$.

Step 1: Adapted bases. Choose a positive basis $u_1, \dots, u_c$ of $\nu_y(Z)$ and lift it to $b_1, \dots, b_c \in T_xW$ with $q(b_i) = u_i$; let t be the positive unit tangent of S at x, so $q(t) = 0$. The preimage orientation is defined by $[\nu_x(S)] \oplus [T_xS] = [T_xW]$, which says exactly that $(b_1, \dots, b_c, t)$ is a positive basis of T_xW. Let $n \in T_xW$ be outward at x and $(w_1, \dots, w_c)$ a positive basis of $T_x(\partial W)$; the boundary convention makes $(n, w_1, \dots, w_c)$ positive as well, so the change of basis P from $(b_1, \dots, b_c, t)$ to $(n, w_1, \dots, w_c)$ has $\det P > 0$.
Step 2: The two signs. Writing $t = t_0 + \alpha n$ with $t_0 \in T_x(\partial W)$, the positive tangent points outward exactly when $\alpha > 0$, so $\sigma_\partial(x) = \operatorname{sign}(\alpha)$. The map ∂F is transverse to Z with complementary dimensions, and its intersection sign is $+1$ iff $q|_{T_x(\partial W)}$ carries the boundary orientation of ∂W to the normal orientation of $\nu_y(Z)$; hence $\operatorname{sign}_{\partial F}(x) = \operatorname{sign}(\det A)$, where A is the matrix of $q|_{T_x(\partial W)}$ from (w_j) to (u_i). Because $\partial F \pitchfork Z$, the map $q|_{T_x(\partial W)}$ is onto, so A is invertible.
Step 3: One determinant. Expand the new basis in the old one: $n = \sum_i B_i b_i + Dt$ and $w_j = \sum_i A_{ij} b_i + c_j t$ (the coefficient of b_i in w_j is A_{ij} because $q(w_j) = \sum_i A_{ij} u_i$). Reading these as the columns of P and moving the first column past the remaining c,

$$\det P = (-1)^c \det \begin{pmatrix} A & B \\ c^\mathsf{T} & D \end{pmatrix} = (-1)^c \det(A)\,(D - c^\mathsf{T} A^{-1} B)$$

by the Schur complement. Solving $t = \sum_j \beta_j w_j + \alpha n$ for its b_i- and t-coefficients gives $A\beta + \alpha B = 0$ and $c^\mathsf{T}\beta + \alpha D = 1$, whence $\alpha^{-1} = D - c^\mathsf{T} A^{-1} B$. Therefore $\det P = (-1)^c \det(A)/\alpha$, and $\det P > 0$ forces $\operatorname{sign}(\alpha)\operatorname{sign}(\det A) = (-1)^c$, that is $\sigma_\partial(x) = (-1)^{\operatorname{codim} Z} \operatorname{sign}_{\partial F}(x)$, completing the proof.

The universal sign $(-1)^{\operatorname{codim} Z}$ is the same at every boundary point, so it factors out of any sum and cannot obstruct a cancellation. With it in hand the invariance is immediate.

Theorem 8.3.8: Homotopy Invariance of the Oriented Intersection Number

Let M be compact and oriented, N oriented, and $Z \subseteq N$ a closed oriented submanifold without boundary, with $\dim M + \dim Z = \dim N$. If $f_0, f_1 \colon M \to N$ are homotopic and each transverse to Z, then $I(f_0, Z) = I(f_1, Z)$.

***Proof*:**
Let $m = \dim M$.
Step 1: A transverse cobordism. Extend a homotopy from f_0 to f_1 to a smooth map $F \colon M \times [0,1] \to N$ with $F(\cdot, 0) = f_0$ and $F(\cdot, 1) = f_1$. The boundary values are already transverse to Z, so by boundary-fixing transversality (lemma 8.1.5) we may homotope F rel $M \times \{0,1\}$ to make $F \pitchfork Z$ while fixing f_0 and f_1. Then $S = F^{-1}(Z)$ is compact ($M \times [0,1]$ is compact and Z closed) of dimension $(m+1) - \operatorname{codim} Z = 1$, a compact oriented 1-manifold with

$$\partial S = f_1^{-1}(Z) \times \{1\} \;\sqcup\; f_0^{-1}(Z) \times \{0\}$$

Step 2: Summing the boundary. Give $M \times [0,1]$ the product orientation and S its preimage orientation. The oriented Boundary Theorem (theorem 8.3.6) gives $\sum_{x \in \partial S} \sigma_\partial(x) = 0$, and lemma 8.3.7 with $\operatorname{codim} Z = m$ turns this into

$$\sum_{x \in \partial S} \operatorname{sign}_{\partial F}(x) = 0$$

Step 3: The two ends. The outward normals of $M \times \{1\}$ and $M \times \{0\}$ are $+\partial_t$ and $-\partial_t$, so with the product orientation the boundary orientation of $M \times \{1\}$ is $(-1)^m$ times that of M and the boundary orientation of $M \times \{0\}$ is $-(-1)^m$ times it. Reversing the orientation of the source negates every intersection sign, so $\operatorname{sign}_{\partial F} = (-1)^m \operatorname{sign}_{f_1}$ on the top face and $\operatorname{sign}_{\partial F} = (-1)^{m+1} \operatorname{sign}_{f_0}$ on the bottom. Substituting,

$$0 = (-1)^m \sum_{f_1^{-1}(Z)} \operatorname{sign}_{f_1} + (-1)^{m+1} \sum_{f_0^{-1}(Z)} \operatorname{sign}_{f_0} = (-1)^m \big(I(f_1, Z) - I(f_0, Z) \big)$$

Since $(-1)^m \neq 0$, $I(f_0, Z) = I(f_1, Z)$, as we sought to show.

Invariance does two things at once. It says $I(f, Z)$ depends only on the homotopy class of f, and it lets the number be defined for maps that are not transverse to Z at all: perturb them until they are. The same argument, run with Z moved instead of f, shows $I(f, Z)$ is unchanged when Z is carried across an ambient isotopy, since sliding Z by an orientation-preserving diffeomorphism h_1 isotopic to the identity is the same as replacing f by $h_1^{-1} \circ f \simeq f$. Both freedoms are used below.

Definition 8.3.9: Oriented Intersection Number of an Arbitrary Map

Let M be compact and oriented, N oriented, and $Z \subseteq N$ a closed oriented submanifold without boundary of complementary dimension. For *any* smooth $f \colon M \to N$, choose $g \simeq f$ with $g \pitchfork Z$ (possible by theorem 7.5.7) and set

$$I(f, Z) = I(g, Z)$$

The choice does not matter: two transverse maps homotopic to f are homotopic to each other, so theorem 8.3.8 makes their intersection numbers equal. A continuous f is first replaced by a smooth homotopic map, exactly as in the mod-2 theory of section 8.1, so $I(f, Z)$ extends to continuous maps and remains a homotopy invariant. In particular the number attaches to submanifolds through their inclusions.

Definition 8.3.10: Oriented Intersection Number of Submanifolds

Let X be a compact oriented submanifold and Z a closed oriented submanifold without boundary of the oriented manifold N, with $\dim X + \dim Z = \dim N$. The *oriented intersection number* of X and Z is

$$I(X, Z) = I(i_X, Z), \qquad i_X \colon X \hookrightarrow N$$

When $X \pitchfork Z$ the inclusion is transverse and

$$I(X, Z) = \sum_{x \in X \cap Z} \operatorname{sign}(x), \qquad \operatorname{sign}(x) = +1 \iff [T_x X] \oplus [T_x Z] = [T_x N]$$

When X and Z are both compact, each may be counted against the other, and the two counts differ only by the price of reordering a frame of X past a frame of Z.

Theorem 8.3.11: Symmetry of the Intersection Number

Let X and Z be compact oriented submanifolds of the oriented manifold N with $\dim X + \dim Z = \dim N$. Then

$$I(X, Z) = (-1)^{(\dim X)(\dim Z)} I(Z, X)$$

***Proof*:**
Transversality is symmetric ($X \pitchfork Z$ iff $Z \pitchfork X$) and generic, so after a small ambient isotopy of X we may assume $X \pitchfork Z$; by the isotopy invariance noted after theorem 8.3.8 this changes neither $I(X, Z)$ nor $I(Z, X)$. Now X and Z meet in the same finite set $X \cap Z$ from both points of view. At a point x of it, transversality gives $T_xX \oplus T_xZ = T_xN$, and the two signs are

$$\operatorname{sign}_{I(X,Z)}(x) = +1 \iff [T_xX] \oplus [T_xZ] = [T_xN], \qquad \operatorname{sign}_{I(Z,X)}(x) = +1 \iff [T_xZ] \oplus [T_xX] = [T_xN]$$

Passing a positive frame of T_xX (of length $\dim X$) across a positive frame of T_xZ (of length $\dim Z$) is a permutation of sign $(-1)^{(\dim X)(\dim Z)}$, so $\operatorname{sign}_{I(X,Z)}(x) = (-1)^{(\dim X)(\dim Z)} \operatorname{sign}_{I(Z,X)}(x)$ at every x. Summing over $X \cap Z$ gives $I(X, Z) = (-1)^{(\dim X)(\dim Z)} I(Z, X)$, completing the proof.

The symmetry has teeth even when $X = Z$. Taking $Z = X$ forces $I(X, X) = (-1)^{(\dim X)^2} I(X, X)$, so a compact oriented submanifold of odd dimension whose double fills the ambient dimension has intersection number zero with itself. To make sense of $I(X, X)$ at all, note that X is never transverse to itself (it meets itself in all of X); the number is defined only through the perturbation of definition 8.3.9, and computing it is the first place the tubular neighbourhood theorem does real work.

Definition 8.3.12: Self-Intersection Number

Let X be a compact oriented submanifold of the oriented manifold N with $2 \dim X = \dim N$. The *self-intersection number* of X is $I(X, X) = I(i_X, X)$, the intersection number of X with itself computed by perturbing the inclusion to a transverse map.

The perturbation has a canonical shape. A tubular neighbourhood identifies a neighbourhood of X in N with the normal bundle $\nu(X)$, and a section of $\nu(X)$ pushes X off itself; the pushed-off copy meets X exactly at the zeros of the section, and the signs agree. So the self-intersection number is a count of zeros, the invariant a bundle carries in its own right.

Proposition 8.3.13: Self-Intersection via the Normal Bundle

Let X be a compact oriented submanifold of the oriented N with $2 \dim X = \dim N$, and let $\nu(X)$ be its normal bundle, an oriented bundle of rank $\dim X$ over X. Then $I(X, X)$ equals the number of zeros, counted with signs, of any section $s\colon X \to \nu(X)$ transverse to the zero section. In particular $I(X, X)$ depends only on $\nu(X)$; it is the *Euler number* of $\nu(X)$.

***Proof*:**
Step 1: Push X off itself. By the tubular neighbourhood theorem (theorem 7.8.1) there is an open set $U \supseteq X$ in N and a diffeomorphism $\Phi\colon \nu(X) \to U$ carrying the zero section to X by the identity. Let s be a section transverse to the zero section; rescaling by a small constant

leaves its zeros and their transversality untouched and places its image in the domain of Φ, so $g = \Phi \circ s \colon X \to U \subseteq N$ is defined. Sliding s to the zero section, $\Phi \circ (\tau s)$ for $\tau \in [0,1]$, is a homotopy from g to i_X, so $I(X,X) = I(g,X)$ by definition 8.3.9.

Step 2: The count. Since Φ is a diffeomorphism carrying the zero section to X, $g(x) \in X$ iff $s(x) = 0$, and $g \pitchfork X$ there iff s is transverse to the zero section. Hence $g^{-1}(X)$ is the zero set of s, and $I(g,X) = \sum_{s(x)=0} \operatorname{sign}_g(x)$.

Step 3: Signs match. At a zero x of s the vertical derivative $ds_x \colon T_xX \to \nu_x(X)$ is a well-defined isomorphism (transversality to the zero section), and dg_x sends T_xX to the graph of ds_x inside $T_xN = T_xX \oplus \nu_x(X)$. Deforming that graph to its vertical part through subspaces still complementary to T_xX leaves the intersection sign unchanged, so $\operatorname{sign}_g(x)$ equals the sign of $[\nu_x(X)] \oplus [T_xX] = [T_xN]$ read through ds_x, which is $+1$ iff ds_x preserves orientation. This is exactly the sign of the zero x. Therefore $I(X,X) = \sum_{s(x)=0} \operatorname{sign}(\det ds_x)$. Any two transverse sections are homotopic through sections, and small rescalings keep them inside U, so theorem 8.3.8 makes the signed count independent of s: it is an invariant of $\nu(X)$, the Euler number, completing the proof.

The zero section is the model case, and the diagonal is the case that matters for the rest of the chapter.

Example 8.3.14: The Zero Section and the Diagonal

1. *Zero section of a bundle.* For an oriented rank-k bundle $E \to X$ over a compact oriented X of dimension k, the zero section $X_0 \subseteq E$ is a compact oriented submanifold with $2 \dim X_0 = \dim E$ and normal bundle $\nu(X_0) \cong E$. So $I(X_0, X_0)$ is the Euler number of E: the signed count of zeros of a generic section of E. Taking $E = TS^2$ and $s = \operatorname{grad}(\text{height})$, the only zeros are the north and south poles, each nondegenerate with two eigenvalues of one sign (a source and a sink), so each contributes $\operatorname{sign}(\det ds) = +1$ and the Euler number is 2.

2. *The diagonal.* For a compact oriented M of dimension m, the diagonal $\Delta = \{(p,p) \mid p \in M\} \subseteq M \times M$ is a compact oriented submanifold with $\dim \Delta = m$ and $2 \dim \Delta = \dim(M \times M)$. Its tangent space at (p,p) is $\{(v,v) \mid v \in T_pM\}$, and the map $(v,w) \mapsto w - v$ identifies $\nu_{(p,p)}(\Delta) = T_{(p,p)}(M \times M)/T_{(p,p)}\Delta$ with T_pM; hence $\nu(\Delta) \cong TM$. So $I(\Delta,\Delta)$ is the Euler number of TM, the signed count of zeros of a vector field on M. For $M = S^2$ this is 2 by part (1), so $I(\Delta,\Delta) = 2$ in $S^2 \times S^2$. This integer, the self-intersection of the diagonal, is the quantity section 8.6 will name the Euler characteristic $\chi(M)$ and equate with the total index of any vector field.

The two computations are the same computation: the tubular neighbourhood turns the diagonal into the zero section of TM, and the Euler number of TM is what a vector field measures. Poincaré-Hopf is then the statement that this self-intersection equals the sum of indices, which section 8.6 proves. One sign convention deserves a closing remark, because it explains why so many geometric intersection numbers turn out to be positive.

8.3.15 Note: Complex Intersections Are Positive A complex vector space has a preferred orientation: any complex basis $e_1, \dots, e_n$, read as the real basis $e_1, ie_1, \dots, e_n, ie_n$, is positive, and any two complex bases give the same real orientation. For complex subspaces $V \oplus W = U$ this preferred orientation satisfies $[V] \oplus [W] = [U]$. Consequently, if X and Z are complex submanifolds of a complex manifold, of complementary real dimension and meeting transversally, then every intersection sign is $+1$ and $I(X, Z) = \#(X \cap Z) \geq 0$. Two distinct lines in the complex projective plane, for instance, meet transversally in a single point, so their intersection number is $+1$ rather than -1.

Exercise 8.3.1

1. On the torus $T^2 = S^1 \times S^1$, oriented by $[\partial_x] \oplus [\partial_y]$, let $a = S^1 \times \{q\}$ (oriented by ∂_x) and $b = \{p\} \times S^1$ (oriented by ∂_y). Show that a and b meet transversally in one point, compute $I(a, b)$ and $I(b, a)$, and verify theorem 8.3.11 directly.
2. Show that for $f \colon M \to N$ with M compact oriented and Z closed oriented of complementary dimension, the oriented intersection number reduces to the mod-2 number, $I(f, Z) \equiv I_2(f, Z) \pmod 2$. Deduce that a nonzero $I_2(f, Z)$ forces $I(f, Z) \neq 0$, but not conversely.
3. Let X be a compact oriented submanifold of an oriented N with $2 \dim X = \dim N$. Using theorem 8.3.11, show that $I(X, X) = 0$ whenever $\dim X$ is odd. Illustrate with a circle embedded in a surface.
4. Determine how $I(f, Z)$ changes when the orientation of M is reversed, when that of Z is reversed, and when that of N is reversed. (Hold the other two fixed each time and track the intersection sign of a single transverse point.)
5. Compute the self-intersection of the diagonal $\Delta \subseteq T^2 \times T^2$. Identify $\nu(\Delta) \cong T(T^2)$, exhibit a nowhere-zero section, and conclude $I(\Delta, \Delta) = 0$. Reconcile this with exercise 8.3.1 (3) (does it apply?) and with the value $\chi(T^2) = 0$.

8.4 The Degree of a Map

The oriented intersection number of section 8.3 measures a map $f \colon M \to N$ against a fixed target Z. Taking Z to be a single point strips the target of all geometry: what survives is an integer attached to f alone, recording how many times f covers a typical value of N, counted with sign. That integer is the *degree*. This section extracts it from the intersection theory already in hand, records its formal properties, and spends them: the fundamental theorem of algebra and the Brouwer fixed-point theorem fall out immediately, the Hopf theorem shows the degree is the *only* homotopy invariant of a self-map of a sphere, and the same construction applied to a pair of submanifolds produces their linking number.

Fix compact oriented n-manifolds M and N with N connected, and a smooth map $f \colon M \to N$. A point $y \in N$ is a regular value exactly when f is transverse to the 0-dimensional submanifold $\{y\}$, so the machinery of section 8.3 applies verbatim with $Z = \{y\}$. The point y carries its positive orientation as a 0-manifold, and $f^{-1}(y)$ is a compact 0-manifold, hence a finite set of points, each of which receives a sign.

Definition 8.4.1: Brouwer Degree

Let M and N be compact oriented n-manifolds with N connected, and let $f\colon M \to N$ be smooth. For a regular value $y \in N$ the preimage $f^{-1}(y)$ is finite (by corollary 3.2.10 together with compactness of M), and the *degree of f at y* is

$$\deg(f, y) \;=\; \sum_{x \in f^{-1}(y)} \operatorname{sign}\det df_x$$

where $\operatorname{sign}\det df_x = +1$ if $df_x\colon T_xM \to T_yN$ is orientation-preserving and -1 if it is orientation-reversing. This is the oriented intersection number $I(f, \{y\})$ of section 8.3. Once theorem 8.4.4 shows the value is independent of y, it is written $\deg f$ and called the *degree* of f.

The sign of a point $x \in f^{-1}(y)$ is exactly the intersection sign of the standing orientation convention (definition 8.3.3): the isomorphism df_x is $+1$ when it carries the orientation of T_xM to that of T_yN. When M and N are connected of the same dimension and f is a diffeomorphism, every value is regular and the single preimage of each point contributes ± 1 uniformly, so the degree is $+1$ for an orientation-preserving diffeomorphism and -1 for an orientation-reversing one. The first concrete computation is this diffeomorphism case.

Example 8.4.2: Degree of the Identity and a Reflection

The identity $\mathrm{id}_N\colon N \to N$ has every point as a regular value, with a single preimage on which $d(\mathrm{id}_N)_y$ is the identity of T_yN; hence $\deg(\mathrm{id}_N, y) = +1$ for every y, so $\deg \mathrm{id}_N = 1$.

Let $r\colon S^n \to S^n$ be the reflection $r(x_0, x_1, \ldots, x_n) = (-x_0, x_1, \ldots, x_n)$, the restriction of the linear reflection R of $\mathbb{R}^{n+1}$ in the hyperplane $x_0 = 0$. It is a diffeomorphism, so every $y \in S^n$ is regular with one preimage. Orient S^n as the boundary of the unit disc with the outward-normal-first convention: a basis $(v_1, \ldots, v_n)$ of $T_xS^n = x^\perp$ is positive precisely when $(x, v_1, \ldots, v_n)$ is positive in $\mathbb{R}^{n+1}$. Since $\det R = -1$, applying R to the positive frame $(x, v_1, \ldots, v_n)$ yields a *negative* frame $(r(x), Rv_1, \ldots, Rv_n)$ of $\mathbb{R}^{n+1}$, so $(dr_xv_1, \ldots, dr_xv_n)$ is a negative basis of $T_{r(x)}S^n$. Thus dr_x reverses orientation, every point contributes -1, and $\deg r = -1$.

That $\deg(f, y)$ does not depend on the regular value y is the first substantive claim, and it is what promotes $\deg(f, y)$ to an invariant $\deg f$ of the map. The argument moves a chosen regular value to any other by a diffeomorphism of N that can be deformed to the identity, so the following lemma is the engine.

Lemma 8.4.3: Homogeneity Lemma

Let N be a connected smooth manifold (without boundary) and let $y, y' \in N$. Then there is a diffeomorphism $h\colon N \to N$, smoothly isotopic to id_N, with $h(y) = y'$. In particular h is orientation-preserving.

***Proof*:**
By lemma 8.1.12 there is a diffeomorphism $h\colon N \to N$, smoothly isotopic to id_N, with $h(y) = y'$; write $\{h_t\}$ for the isotopy, $h_0 = \mathrm{id}_N$ and $h_1 = h$. It remains to check that h preserves orientation. For each $x \in N$ the determinant $\det d(h_t)_x$ is continuous in t, never zero because each h_t is a diffeomorphism, and equal to $+1$ at $t = 0$; it therefore stays positive, so $h = h_1$ preserves orientation, completing the proof.

Theorem 8.4.4: The Degree Is Well Defined and Homotopy Invariant

Let M, N be compact oriented n-manifolds with N connected, and $f\colon M \to N$ smooth.

1. The value $\deg(f, y)$ is the same for every regular value $y \in N$; call it $\deg f$.
2. If $f \simeq g$ are smoothly homotopic maps $M \to N$, then $\deg f = \deg g$.

***Proof*:**
Part (2) is the homotopy invariance of the oriented intersection number $I(f, \{y\})$ established in section 8.3, applied to the fixed target $Z = \{y\}$; it rests, as there, on the signed boundary count of a compact oriented 1-manifold vanishing (theorem 8.3.6), the signed refinement of the mod-2 parity used for $\deg_2$.

For part (1), let y and y' be two regular values. By the Homogeneity Lemma (lemma 8.4.3) there is an orientation-preserving diffeomorphism $h\colon N \to N$, isotopic to id_N, with $h(y') = y$. Two computations combine.

Naturality. The point x lies in $f^{-1}(y) = f^{-1}(h(y'))$ if and only if $h^{-1}(f(x)) = y'$, i.e. $x \in (h^{-1} \circ f)^{-1}(y')$, and since h^{-1} is a diffeomorphism, y' is a regular value of $h^{-1} \circ f$ with the same preimage set. At each such x,

$$\operatorname{sign}\det d(h^{-1} \circ f)_x = \operatorname{sign}\bigl(\det dh^{-1}_{f(x)} \cdot \det df_x\bigr) = \operatorname{sign}\det df_x$$

because h^{-1} preserves orientation. Summing gives $\deg(h^{-1} \circ f,\, y') = \deg(f, y)$.

Homotopy. Because h is isotopic to id_N, the maps $h^{-1} \circ f$ and f are homotopic (compose f with the isotopy), so part (2) gives $\deg(h^{-1} \circ f, y') = \deg(f, y')$.

Combining, $\deg(f, y) = \deg(f, y')$, as we sought to show.

The degree turns a homotopy problem into arithmetic. Two maps with different degrees cannot be homotopic, and a single integer, computed at one convenient regular value, decides the question for all of them. The Homogeneity Lemma also isolates the one geometric fact that makes this possible: a connected manifold looks the same near any point, so no regular value is distinguished, and the count cannot depend on which one is used.

The algebraic behaviour of the degree under composition and under reversal of orientation is recorded next; each part is a direct reading of the signed count.

Proposition 8.4.5: Basic Properties of the Degree

Let M, N, P be compact oriented n-manifolds with N, P connected, and let $f\colon M \to N$, $g\colon N \to P$ be smooth.

1. $\deg(g \circ f) = \deg(g)\deg(f)$.
2. Reversing the orientation of M (or of N) negates the degree: with $-M$ denoting M reoriented, $\deg(f\colon -M \to N) = -\deg f$.
3. If f is not surjective, then $\deg f = 0$.
4. $\deg \mathrm{id}_N = 1$.

***Proof*:**
(1). By theorem 7.2.1 choose $z \in P$ that is a regular value of both g and $g \circ f$. Write $g^{-1}(z) = \{y_1, \ldots, y_m\}$. If $x \in (g \circ f)^{-1}(z)$ then $d(g \circ f)_x = dg_{f(x)} \circ df_x$ is invertible, forcing df_x invertible; thus every y_j is a regular value of f, and $(g \circ f)^{-1}(z) = \bigsqcup_j f^{-1}(y_j)$. Now

$$\deg(g \circ f) = \sum_j \sum_{x \in f^{-1}(y_j)} \operatorname{sign}\big(\det dg_{y_j} \cdot \det df_x\big) = \sum_j \operatorname{sign}(\det dg_{y_j}) \sum_{x \in f^{-1}(y_j)} \operatorname{sign}\det df_x$$

The inner sum is $\deg(f, y_j) = \deg f$ by theorem 8.4.4, so this equals $\big(\sum_j \operatorname{sign}\det dg_{y_j}\big)\deg f = \deg(g)\deg(f)$.

(2). Reversing the orientation of M reverses the ambient frame at every x, hence flips each $\operatorname{sign}\det df_x$; reversing that of N has the same effect. Either negates the whole sum.

(3). If $y \notin \operatorname{im} f$, then y is vacuously a regular value with $f^{-1}(y) = \emptyset$, so $\deg(f, y) = 0$; by theorem 8.4.4, $\deg f = 0$.

(4). Immediate from example 8.4.2, completing the proof.

Property (3) is the one exploited most often: a map of nonzero degree is forced to hit every value, so degree computations are existence proofs in disguise. Property (1) makes the degree a homomorphism from composition to multiplication, and it computes the antipodal map at a stroke.

Example 8.4.6: Degree of the Antipodal Map

The antipodal map $a\colon S^n \to S^n$, $a(x) = -x$, is the restriction of $-I$ on $\mathbb{R}^{n+1}$, which factors as the product of the $n+1$ coordinate reflections $r_0, r_1, \ldots, r_n$, where r_i negates the i-th coordinate. Each r_i has degree -1 by the computation in example 8.4.2, so by proposition 8.4.5(1),

$$\deg a = (\deg r_0)(\deg r_1)\cdots(\deg r_n) = (-1)^{n+1}$$

Thus the antipodal map is orientation-preserving on odd-dimensional spheres and orientation-reversing on even-dimensional ones. This parity is the seed of the hairy ball phenomenon: on S^{2n} the antipodal map is not homotopic to the identity, since their degrees -1 and $+1$ differ.

The first payoff is algebraic. A complex polynomial extends to a self-map of the sphere, and its degree there is visibly its algebraic degree, so a map of positive degree cannot avoid the value 0.

Theorem 8.4.7: Fundamental Theorem of Algebra

Every complex polynomial $p(z) = a_k z^k + a_{k-1} z^{k-1} + \cdots + a_0$ with $k \geq 1$ and $a_k \neq 0$ has a root in $\mathbb{C}$.

***Proof*:**
Identify S^2 with the one-point compactification $\mathbb{C} \cup \{\infty\} = \mathcal{P}^1$, using the two stereographic charts z and $w = 1/z$. The polynomial p defines a map $P: S^2 \to S^2$ with $P(\infty) = \infty$: in the chart w near ∞, writing the target coordinate $\zeta = 1/P$,

$$\zeta = \frac{w^k}{a_k + a_{k-1} w + \cdots + a_0 w^k}$$

which is smooth at $w = 0$ because the denominator equals $a_k \neq 0$ there. So P is smooth, and its degree is defined.

Step 1: Homotope to the leading term. For $t \in [0,1]$ set $p_t(z) = a_k z^k + t\,(a_{k-1} z^{k-1} + \cdots + a_0)$. Each p_t is a polynomial of degree exactly k with leading coefficient a_k, so it extends as above to a smooth map $P_t: S^2 \to S^2$ fixing ∞, and the same denominator computation shows $(z,t) \mapsto P_t(z)$ is a smooth homotopy. Since $P_1 = P$ and $P_0(z) = a_k z^k$, homotopy invariance (theorem 8.4.4) gives $\deg P = \deg P_0$.

Step 2: Compute the degree of $z \mapsto a_k z^k$. Take the regular value $w_0 = a_k$. The equation $a_k z^k = a_k$ reads $z^k = 1$, whose solutions are the k distinct k-th roots of unity (elementary in polar form, no appeal to the theorem being proved); the point ∞ is not a preimage since $P_0(\infty) = \infty$. At each root z, the map P_0 is holomorphic with derivative $k a_k z^{k-1} \neq 0$, and a nonzero complex-linear map has real determinant $|k a_k z^{k-1}|^2 > 0$, so dP_0 preserves orientation and each of the k preimages contributes $+1$. Hence $\deg P_0 = k$.

Therefore $\deg P = k \geq 1$, so P is surjective by proposition 8.4.5(3). In particular $0 \in \operatorname{im} P$, and since $P(\infty) = \infty \neq 0$ the point mapping to 0 lies in $\mathbb{C}$, giving $z_0 \in \mathbb{C}$ with $p(z_0) = 0$, completing the proof.

The proof says something sharper than existence: counted with multiplicity, p takes each generic value exactly k times, because the degree is k and every preimage sign is $+1$. Holomorphicity is doing the orientation bookkeeping for free, which is why the complex degree never cancels. The same surjectivity mechanism, run on a manifold rather than a polynomial, yields the Brouwer fixed-point theorem.

Lemma 8.4.8: No Smooth Retraction of the Disc onto its Boundary

There is no smooth map $\rho: D^n \to S^{n-1}$ with $\rho(x) = x$ for all $x \in S^{n-1} = \partial D^n$.

***Proof*:**
Suppose such a retraction ρ exists. Define $H\colon S^{n-1}\times[0,1]\to S^{n-1}$ by $H(x,t)=\rho(tx)$. This is smooth, and $H(x,1)=\rho(x)=x$ while $H(x,0)=\rho(0)$ is a constant. Thus $\mathrm{id}_{S^{n-1}}$ is homotopic to a constant map. But $\deg\mathrm{id}_{S^{n-1}}=1$ (example 8.4.2) while a constant map is not surjective and so has degree 0 (proposition 8.4.5(3)); homotopy invariance (theorem 8.4.4) forces $1=0$, a contradiction. For $n=1$ the target S^0 is disconnected: a retraction $\rho\colon D^1\to S^0$ would fix the two endpoints ±1, so its image contains both, yet $\rho(D^1)$ is connected as the continuous image of the interval and $\{\pm1\}$ is not connected, again a contradiction. Hence no retraction exists, completing the proof.

Theorem 8.4.9: Brouwer Fixed-Point Theorem

Every smooth map $g\colon D^n\to D^n$ has a fixed point.

***Proof*:**
Suppose $g(x)\neq x$ for every $x\in D^n$. For each x, the ray starting at $g(x)$ and passing through x meets S^{n-1} in a unique point beyond x; let $\rho(x)$ be that point,

$$\rho(x)=x+s(x)\,\frac{x-g(x)}{|x-g(x)|}$$

where $s(x)\geq0$ is the nonnegative root of $|\rho(x)|^2=1$. The discriminant of this quadratic in s is strictly positive (since $|x|\leq1$ and the direction is a unit vector), so s is a smooth function of x, and ρ is smooth. If $x\in S^{n-1}$ then $|x|=1$ gives $s(x)=0$, so $\rho(x)=x$. Thus ρ is a smooth retraction $D^n\to S^{n-1}$, contradicting lemma 8.4.8. Therefore g has a fixed point, as we sought to show.

The disc has no interesting self-maps to speak of, yet it cannot be continuously retracted onto its edge, and that single obstruction pins every self-map to a fixed point. The smoothness hypothesis is removable: a continuous self-map is uniformly approximated by a smooth one (theorem 7.5.7 and the approximation theorems of chapter 7), and a limit of near-fixed-points is a fixed point. The degree has now proved two classical theorems by the same move, non-vanishing forces surjectivity. The Hopf theorem asserts the converse organizing principle for spheres: the degree is a complete invariant.

Theorem 8.4.10: Hopf Degree Theorem

Two smooth maps $f,g\colon S^n\to S^n$ are smoothly homotopic if and only if $\deg f=\deg g$. Consequently the degree is a bijection from the set of homotopy classes of self-maps of S^n to $\mathbb{Z}$.

***Proof*:**
The forward implication is homotopy invariance (theorem 8.4.4). For the converse, the case $n=1$ admits a self-contained argument, and the general case is deferred to the framed cobordism construction of chapter 10, whose machinery is not yet available.

The case $n=1$. Write $S^1=\mathbb{R}/\mathbb{Z}$ with covering map $\pi\colon\mathbb{R}\to S^1$, $\pi(s)=e^{2\pi is}$. For a smooth $f\colon S^1\to S^1$, the composite $f\circ\pi\colon\mathbb{R}\to S^1$ has simply connected domain, so it lifts to a smooth $F\colon \mathbb{R} \to \mathbb{R}$ with $\pi \circ F = f \circ \pi$. Since $\pi(F(s+1)) = f(\pi(s)) = \pi(F(s))$, the difference $F(s+1) - F(s)$ is a continuous integer, hence a constant $d \in \mathbb{Z}$. Counting the signed solutions of $F(s) \equiv a \pmod 1$ for a regular value $\pi(a)$ shows $\deg f = F(1) - F(0) = d$: as s runs over $[0,1]$, F climbs from $F(0)$ to $F(0)+d$, crossing each level $a + \mathbb{Z}$ a net d times with sign $\operatorname{sign} F'$.

Now suppose $\deg f = \deg g = d$, with lifts F, G satisfying $F(s+1) = F(s) + d$ and $G(s+1) = G(s) + d$. The straight-line homotopy $F_u = (1-u)F + uG$ satisfies $F_u(s+1) = F_u(s) + d$, so each F_u descends through π to a smooth map $f_u\colon S^1 \to S^1$; this is a homotopy from f to g.

The general case. The complete argument is the Pontryagin framed cobordism correspondence developed in chapter 10: a regular value of f has a finite preimage which, framed by the derivative of f, represents a framed cobordism class, and $\deg f$ is exactly that class in the framed cobordism group of a point, $\mathbb{Z}$. Two maps of equal degree have framed-cobordant preimages, and a framed cobordism between them is precisely a homotopy. The verification of this dictionary requires the tubular-neighbourhood surgery of that construction, so it is deferred there rather than sketched here.

For S^1 the argument is concrete: the degree is the number of times the circle wraps around itself, read off from the lift, and two maps that wrap the same number of times are slid into one another by interpolating their lifts. The general statement identifies $\pi_n(S^n)$ with $\mathbb{Z}$ by the degree, the differential-topological counterpart of the Hurewicz isomorphism $\pi_n(S^n) \cong \mathbb{Z}$ of algebraic topology ([11, Connectivity and Bottom Homotopy of Spheres]); the two agree because both count preimages of a point. That the degree classifies self-maps of spheres is what makes it the natural target for obstruction and index arguments, the vector-field index of section 8.6 among them.

The final construction of the section is a degree attached not to a single map but to a pair of disjoint submanifolds. When their dimensions are one short of complementary in the ambient space, the unit vector pointing from one to the other sweeps out a self-map of a sphere, and its degree measures how the two are intertwined.

Definition 8.4.11: Linking Number

Let X and Y be disjoint compact oriented boundaryless submanifolds of $\mathbb{R}^n$ with $\dim X + \dim Y = n - 1$. The *Gauss map* (or direction map)

$$u\colon X \times Y \to S^{n-1}, \qquad u(x,y) = \frac{y-x}{|y-x|}$$

is well defined because $X \cap Y = \emptyset$ forces $y \neq x$. Its domain $X \times Y$ is a compact oriented manifold (product orientation) of dimension $\dim X + \dim Y = n-1 = \dim S^{n-1}$, so the *linking number* is

$$\operatorname{lk}(X,Y) = \deg u \ \in \ \mathbb{Z}$$

Because the degree is a homotopy invariant, $\operatorname{lk}(X,Y)$ is unchanged when X and Y are moved by any isotopy of $\mathbb{R}^n$ that keeps them disjoint: deforming X or Y deforms u through Gauss maps, and the degree cannot jump. The linking number is therefore an invariant of the isotopy class of the disjoint pair, the simplest numerical measure of entanglement. The Hopf link and its unlinked counterpart show it is nonzero.

Example 8.4.12: The Hopf Link

Work in $\mathbb{R}^3$, so $n = 3$ and $\dim X = \dim Y = 1$. Take the unit circle in the xy-plane and a unit circle in the xz-plane centred at $(1, 0, 0)$,

$$X = \{(\cos s, \sin s, 0) \mid s \in \mathbb{R}/2\pi\mathbb{Z}\}, \qquad Y = \{(1 + \cos t, 0, \sin t) \mid t \in \mathbb{R}/2\pi\mathbb{Z}\}$$

which are disjoint. To find the preimage under u of the direction $v = (0, 0, 1)$, solve $Y(t)-X(s) = \lambda v$ with $\lambda > 0$. The y-coordinate gives $\sin s = 0$, so $s \in \{0, \pi\}$; the x-coordinate gives $1 + \cos t = \cos s$. For $s = \pi$ this forces $\cos t = -2$, impossible; for $s = 0$ it forces $\cos t = 0$, and then $\lambda = \sin t > 0$ selects $t = \pi/2$. So $u^{-1}(v) = \{(0, \pi/2)\}$ is a single point, at which a direct check shows du is an isomorphism, so v is a regular value and $|\operatorname{lk}(X, Y)| = 1$.

By contrast, place two small circles in disjoint balls about $(-2, 0, 0)$ and $(2, 0, 0)$. Every difference $y - x$ then has positive first coordinate, so u misses the open hemisphere $\{v_0 < 0\}$ of S^2; being non-surjective, u has degree 0 (proposition 8.4.5(3)), so this unlinked pair has $\operatorname{lk} = 0$. Since the linking number is an isotopy invariant, the two configurations cannot be deformed one into the other through disjoint pairs: the Hopf link does not come apart.

The direction u hides a more geometric description: $\operatorname{lk}(X, Y)$ equals the signed number of times Y crosses any compact oriented surface bounded by X, as the Hopf example illustrates with Y piercing the disc spanning X exactly once. The two viewpoints are reconciled by the extension principle behind homotopy invariance, that a Gauss map extending over a bounding manifold has degree zero. The linking number is also symmetric up to a sign fixed by the ambient dimension.

Proposition 8.4.13: Symmetry of the Linking Number

For disjoint compact oriented submanifolds $X, Y \subseteq \mathbb{R}^n$ with $\dim X + \dim Y = n - 1$,

$$\operatorname{lk}(Y, X) = (-1)^{n+(\dim X)(\dim Y)} \operatorname{lk}(X, Y)$$

In particular, for two disjoint circles in $\mathbb{R}^3$ the linking number is symmetric, $\operatorname{lk}(Y, X) = \operatorname{lk}(X, Y)$.

***Proof*:**

The Gauss map for (Y, X) is $u_{Y,X}(y, x) = (x-y)/|x-y| = -\,u_{X,Y}(x, y)$. Thus $u_{Y,X} = \alpha \circ u_{X,Y} \circ \sigma$, where $\sigma\colon Y \times X \to X \times Y$ is the swap and $\alpha\colon S^{n-1} \to S^{n-1}$ is the antipodal map. The swap has degree $(-1)^{(\dim X)(\dim Y)}$ (reordering the two blocks of an oriented product basis), and the antipodal map of S^{n-1} has degree $(-1)^n$ by example 8.4.6. By proposition 8.4.5(1), $\operatorname{lk}(Y, X) = \deg \alpha \cdot \deg u_{X,Y} \cdot \deg \sigma = (-1)^{n+(\dim X)(\dim Y)} \operatorname{lk}(X, Y)$. For $n = 3$ and $\dim X = \dim Y = 1$ the exponent is $3 + 1 = 4$, even, so the linking number is symmetric, completing the proof.

The linking number completes this section: the degree, born as an intersection count against a point, reappears as an invariant of two submanifolds against each other, computed by the direction map into a sphere. The same integer will return in section 8.6 as the index of a vector field, where the sphere is swept out not by the direction between two manifolds but by the direction of the field near one of its zeros.

Exercise 8.4.1

1. Compute the degree of the power map $f: S^1 \to S^1$, $f(z) = z^k$ (for $k \in \mathbb{Z}$), directly from the definition by choosing a regular value and summing the signs. Confirm the answer against the lift $F(s) = ks$ under $\pi(s) = e^{2\pi i s}$.
2. Let $f: S^n \to S^n$ be smooth with no fixed point. Show that f is homotopic to the antipodal map, and deduce that any smooth self-map of S^n whose degree differs from $(-1)^{n+1}$ has a fixed point.
3. Show that a smooth map $f: S^n \to S^n$ that factors through the equator, i.e. $f = \iota \circ g$ for some $g: S^n \to S^{n-1}$ and the inclusion $\iota: S^{n-1} \hookrightarrow S^n$, has degree 0.
4. Prove that if $\deg f \neq 0$ then f is surjective, and use this to give a one-line proof that a polynomial map $p: \mathcal{P}^1 \to \mathcal{P}^1$ of positive degree is onto.
5. Using the Hopf degree theorem for $n = 1$, show that there are infinitely many homotopy classes of smooth maps $S^1 \to S^1$, and that a map of odd degree is not homotopic to any map of even degree.
6. Let $X, Y \subseteq \mathbb{R}^n$ be disjoint compact oriented submanifolds with $\dim X + \dim Y = n - 1$. Show that if $X = \partial W$ for a compact oriented submanifold $W \subseteq \mathbb{R}^n$ (with boundary) disjoint from Y, then $\mathrm{lk}(X, Y) = 0$.

8.5 The Lefschetz Fixed-Point Theorem

The oriented intersection number of the previous sections counts how two submanifolds, or a map and a submanifold, meet. Applied to a single self-map $f: M \to M$ it answers a question with no intersections in sight: does f return some point to itself? The device is to replace f by a pair of submanifolds of $M \times M$ whose meetings are exactly the fixed points of f, the diagonal and the graph, and to read the fixed-point question off their oriented intersection number. That number is a homotopy invariant, it obstructs fixed-point-free maps, and for a map whose fixed points are nondegenerate it is a signed count computed one point at a time.

Throughout, M is a compact oriented n-manifold without boundary and $f: M \to M$ is smooth. The product $M \times M$ is a $2n$-manifold, oriented by the product orientation. Two compact n-dimensional submanifolds sit inside it: the *diagonal* $\Delta = \{(x, x) \mid x \in M\}$ and the *graph* $\Gamma_f = \{(x, f(x)) \mid x \in M\}$. Each is the image of an embedding of M, namely $x \mapsto (x, x)$ and $x \mapsto (x, f(x))$, and each is oriented by transporting the orientation of M along that embedding. Their dimensions are complementary, $n + n = 2n$, so the oriented intersection number $I(\Delta, \Gamma_f)$ of section 8.3 is defined.

Definition 8.5.1: Lefschetz Number

The *Lefschetz number* of a smooth map $f: M \to M$ on a compact oriented manifold is

$$L(f) = I(\Delta, \Gamma_f)$$

the oriented intersection number in $M \times M$ of the diagonal Δ with the graph Γ_f.

Because oriented intersection numbers are homotopy invariant (section 8.3) and a homotopy f_t drags the graph Γ_{f_t} along with it, $L(f)$ depends only on the homotopy class of f. It is an integer attached to f that cannot change as f is deformed.

The whole construction is engineered around one incidence.

Proposition 8.5.2: Fixed Points as an Intersection

The assignment $x \mapsto (x, x)$ is a bijection from the fixed-point set of f onto $\Delta \cap \Gamma_f$. In particular Δ and Γ_f meet if and only if f has a fixed point.

***Proof*:**
A point of Δ has the form (x, x) and a point of Γ_f has the form $(x', f(x'))$. They coincide precisely when $x = x'$ and $x = f(x') = f(x)$, that is, when $f(x) = x$. The assignment $x \mapsto (x, x)$ therefore sends fixed points bijectively to $\Delta \cap \Gamma_f$, completing the proof.

The oriented intersection number vanishes when the two submanifolds are disjoint, so a nonzero Lefschetz number forces Δ and Γ_f to meet.

Theorem 8.5.3: Lefschetz Fixed-Point Theorem

Let M be compact and oriented and $f\colon M \to M$ smooth. If $L(f) \neq 0$, then f has a fixed point.

***Proof*:**
Suppose f has no fixed point. By proposition 8.5.2 the diagonal and the graph are then disjoint, $\Delta \cap \Gamma_f = \emptyset$. Disjoint submanifolds are vacuously transverse, and the oriented intersection number is the signed count of intersection points (section 8.3), an empty sum. Hence $L(f) = I(\Delta, \Gamma_f) = 0$. The contrapositive is the assertion, as we sought to show.

The theorem converts a fixed-point problem into the computation of an integer insensitive to deformation. Where the Brouwer fixed-point theorem (section 8.4) forced a fixed point on the disc from the impossibility of retracting it to its boundary, the Lefschetz number performs the same service for a closed manifold: a single nonzero invariant rules out every fixed-point-free representative of the homotopy class. The converse fails, a map with a fixed point can still have $L(f) = 0$, but the obstruction is often computable when the map itself is not. The antipodal map $a\colon S^2 \to S^2$, $a(x) = -x$, has no fixed point, since $-x = x$ has no solution on the sphere; proposition 8.5.2 makes $\Delta \cap \Gamma_a$ empty, so $L(a) = 0$.

The map that meets itself everywhere is the identity, and its Lefschetz number is forced to be an intersection number of the diagonal with itself.

Corollary 8.5.4: The Identity Map

For the identity map, $\Gamma_{\mathrm{id}_M} = \Delta$, so

$$L(\mathrm{id}_M) = I(\Delta, \Delta)$$

the self-intersection number of the diagonal.

***Proof*:**
The graph of the identity is $\{(x,x) \mid x \in M\} = \Delta$, and the orientation transported along $x \mapsto (x, \mathrm{id}_M(x)) = (x,x)$ is the orientation of Δ. Thus $L(\mathrm{id}_M) = I(\Delta, \Gamma_{\mathrm{id}_M}) = I(\Delta, \Delta)$, as we sought to show.

The self-intersection number $I(\Delta, \Delta)$ is the integer that section 8.6 adopts as the definition of the *Euler characteristic* $\chi(M)$; the identity map is the bridge, $L(\mathrm{id}_M) = \chi(M)$. Homotopy invariance then reads: every map homotopic to the identity shares the Lefschetz number $\chi(M)$, so if $\chi(M) \neq 0$, every such map has a fixed point. The computation in example 8.5.10 below gives $\chi(S^2) = 2$, and the antipodal map, with $L(a) = 0 \neq 2$, cannot be homotopic to the identity on S^2.

To turn $L(f)$ from an obstruction into a number one can compute, the fixed points must be countable and each must carry a well-defined sign. Both hold when the graph meets the diagonal transversally, and transversality has a linear-algebra criterion at a fixed point.

Definition 8.5.5: Lefschetz Fixed Point

A fixed point p of $f\colon M \to M$ is a *Lefschetz fixed point* (or *nondegenerate*) if the differential $df_p\colon T_pM \to T_pM$ has no eigenvalue equal to 1, equivalently if $df_p - \mathrm{id}$ is invertible. The map f is a *Lefschetz map* if all its fixed points are Lefschetz.

At a fixed point $f(p) = p$, so df_p is an endomorphism of T_pM and the eigenvalue condition makes sense. It is exactly transversality in $M \times M$.

Proposition 8.5.6: Transversality of the Graph and the Diagonal

Let p be a fixed point of f. Under the canonical identification $T_{(p,p)}(M \times M) = T_pM \oplus T_pM$,

$$T_{(p,p)}\Delta = \{(v,v) \mid v \in T_pM\}, \qquad T_{(p,p)}\Gamma_f = \{(v, df_pv) \mid v \in T_pM\}$$

and Γ_f is transverse to Δ at (p,p) if and only if p is a Lefschetz fixed point.

***Proof*:**
The diagonal is the image of $x \mapsto (x,x)$, whose differential at p is $v \mapsto (v,v)$; the graph is the image of $x \mapsto (x, f(x))$, whose differential at p is $v \mapsto (v, df_pv)$. This gives the two tangent spaces displayed, each of dimension n.
Transversality at (p,p) means $T_{(p,p)}\Delta + T_{(p,p)}\Gamma_f = T_pM \oplus T_pM$. The two summands have dimensions adding to $n + n = 2n = \dim(T_pM \oplus T_pM)$, so the sum is everything if and only if it is direct, that is, if and only if $T_{(p,p)}\Delta \cap T_{(p,p)}\Gamma_f = 0$. A nonzero vector in the intersection is a pair $(v,v) = (v, df_pv)$ with $v \neq 0$, which forces $df_pv = v$: a nonzero 1-eigenvector of df_p. Thus the intersection is nonzero exactly when 1 is an eigenvalue of df_p, and transversality holds exactly when it is not, which is the definition of a Lefschetz fixed point (definition 8.5.5), completing the proof.

Transversality of complementary-dimensional submanifolds makes their intersection a 0-manifold, so the fixed points of a Lefschetz map are isolated; compactness of M, hence of Δ, then makes them finite in number. Each contributes to $I(\Delta, \Gamma_f)$ its local intersection sign, and that sign is computed by the same matrix that detected transversality.

Definition 8.5.7: Local Lefschetz Number

The *local Lefschetz number* of f at a Lefschetz fixed point p is

$$L_p(f) = \operatorname{sign}\det(df_p - \mathrm{id}) \in \{+1, -1\}$$

The determinant is that of an endomorphism of T_pM, so $L_p(f)$ needs no choice of basis or orientation of the tangent space; it is intrinsic to f at p. The next lemma identifies it with the oriented intersection sign that $I(\Delta, \Gamma_f)$ assigns to (p, p), which is where the orientation of M enters.

Lemma 8.5.8: The Local Intersection Sign

Let p be a Lefschetz fixed point of f. The oriented intersection sign of Δ and Γ_f at (p, p), taken in the order (Δ, Γ_f), equals $L_p(f) = \operatorname{sign}\det(df_p - \mathrm{id})$.

***Proof*:**
Fix a positively oriented basis $e_1, \dots, e_n$ of T_pM, and let A be the matrix of df_p in this basis, with I the $n \times n$ identity matrix. The product orientation of $T_pM \oplus T_pM$ has the positively oriented basis $(e_1, 0), \dots, (e_n, 0), (0, e_1), \dots, (0, e_n)$. By proposition 8.5.6, a positively oriented basis of $T_{(p,p)}\Delta$ is $(e_1, e_1), \dots, (e_n, e_n)$ and a positively oriented basis of $T_{(p,p)}\Gamma_f$ is $(e_1, df_pe_1), \dots, (e_n, df_pe_n)$, these being the images of $e_1, \dots, e_n$ under the orienting embeddings. The intersection sign in the order (Δ, Γ_f) is the sign of the determinant of the frame obtained by concatenating these two bases, expressed in the product-orientation basis (section 8.3). Written in $n \times n$ blocks, that frame is

$$\begin{pmatrix} I & I \\ I & A \end{pmatrix}$$

whose first block column lists the Δ vectors (e_i, e_i) and whose second lists the Γ_f vectors (e_i, Ae_i). Subtracting the first block row from the second leaves $\begin{pmatrix} I & I \\ 0 & A - I \end{pmatrix}$, whose determinant is $\det(A - I) = \det(df_p - \mathrm{id})$. Its sign is the intersection sign, as we sought to show.

Summing the local signs over the finitely many fixed points recovers the global invariant.

Theorem 8.5.9: The Lefschetz-Hopf Theorem

Let M be compact and oriented and let $f\colon M \to M$ be a Lefschetz map. Then f has finitely many fixed points, and

$$L(f) = \sum_{p\,:\,f(p)=p} \operatorname{sign}\det(df_p - \mathrm{id}) = \sum_{p\,:\,f(p)=p} L_p(f)$$

***Proof*:**
By proposition 8.5.6, f being a Lefschetz map means $\Gamma_f \pitchfork \Delta$ at every point of $\Delta \cap \Gamma_f$. Two complementary-dimensional submanifolds meeting transversally do so in a 0-manifold; as a closed subset of the compact manifold Δ, the intersection $\Delta \cap \Gamma_f$ is compact, hence finite. Proposition 8.5.2 identifies it with the fixed-point set of f, which is therefore finite.

For a transverse intersection of complementary dimension, the oriented intersection number is the sum of the local intersection signs over the intersection points (section 8.3):

$$L(f) = I(\Delta, \Gamma_f) = \sum_{(p,p)\in\Delta\cap\Gamma_f} (\text{sign at } (p,p))$$

By lemma 8.5.8 the sign at (p,p) is $\operatorname{sign}\det(df_p - \mathrm{id}) = L_p(f)$. Summing over the fixed points gives the stated formula, completing the proof.

The theorem is the fixed-point analogue of counting the preimages of a regular value: a global homotopy invariant is pinned down by purely local data, the linearization of f at each fixed point. The identity map is excluded, since every point is fixed and none is nondegenerate, which is exactly why computing $\chi(M) = L(\mathrm{id}_M)$ requires perturbing the identity to a Lefschetz map first, the route through vector fields that section 8.6 takes.

Example 8.5.10: Rotations of the Sphere

Let $R_\theta\colon S^2 \to S^2$ be the rotation by a fixed angle $\theta \notin 2\pi\mathbb{Z}$ about the vertical axis. Its fixed points are the two poles N and S, where the axis meets the sphere. At each pole the differential is the rotation by θ of the tangent plane, so in an orthonormal basis

$$df_N = df_S = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Hence

$$\det(df_p - \mathrm{id}) = (\cos\theta - 1)^2 + \sin^2\theta = 2(1-\cos\theta) > 0$$

for $\theta \notin 2\pi\mathbb{Z}$, so both poles are Lefschetz fixed points with local number $L_N(R_\theta) = L_S(R_\theta) = +1$. By theorem 8.5.9,

$$L(R_\theta) = 1 + 1 = 2$$

Each R_θ is homotopic to id_{S^2} through the rotations $R_{s\theta}$, $s \in [0,1]$, so homotopy invariance gives $L(\mathrm{id}_{S^2}) = 2$. By corollary 8.5.4 this is the self-intersection number of the diagonal, so $\chi(S^2) = I(\Delta,\Delta) = 2$.

The computation recovers a fact about the sphere from data at two points, and it exposes the role of degeneracy: as $\theta \to 0$ the two fixed points remain the poles, but $\det(df_p - \mathrm{id}) = 2(1-\cos\theta) \to 0$, so nondegeneracy is lost in the limit, where the identity is degenerate at every point. The invariant $L = 2$ is unchanged by the degeneration.

The identity is not the only degenerate map, but degeneracy is never forced: an arbitrarily small perturbation makes every fixed point nondegenerate.

Proposition 8.5.11: Genericity of Lefschetz Maps

Every smooth map $f\colon M \to M$ on a compact manifold is homotopic to a Lefschetz map. Consequently $L(f)$ is always the finite signed count of theorem 8.5.9 for a suitable representative of its homotopy class.

***Proof*:**
Embed M as a closed submanifold of some $\mathbb{R}^N$ (Whitney, chapter 7), and let $\pi\colon U \to M$ be the retraction of a tubular neighbourhood $U \subseteq \mathbb{R}^N$ of M (theorem 7.8.1); it is a submersion with $\pi|_M = \mathrm{id}_M$. Since M is compact, there is an open ball $B \subseteq \mathbb{R}^N$ about 0 with $f(x) + a \in U$ for all $x \in M$ and $a \in B$. Define

$$F\colon M \times B \to M \times M, \qquad F(x,a) = \big(x,\ \pi(f(x)+a)\big)$$

and write $f_a = \pi(f(\cdot)+a)\colon M \to M$, so that $F(\cdot,a) = \gamma_{f_a}$ is the graph map of f_a and $f_0 = f$.
Step 1: F is transverse to Δ. Fix (x,a) with $F(x,a) \in \Delta$; then $f_a(x) = x$. Holding x fixed and varying a, the differential of $a \mapsto \pi(f(x)+a)$ is $d\pi_{f(x)+a}$ restricted to the $\mathbb{R}^N$ directions, which is surjective onto T_xM because π is a submersion; these directions contribute all of $\{0\} \oplus T_xM$ to the image of $dF_{(x,a)}$. Holding a fixed and varying x contributes vectors of the form $(w,*)$ with w ranging over T_xM. Subtracting a vertical vector from such a $(w,*)$ yields $(w,0)$, so the image of $dF_{(x,a)}$ contains both $\{0\} \oplus T_xM$ and $T_xM \oplus \{0\}$, hence is all of $T_xM \oplus T_xM$. Thus F is a submersion at (x,a), and in particular $F \pitchfork \Delta$.
Step 2: perturb. By the parametric transversality theorem (theorem 7.5.6), for almost every $a \in B$ the map $F(\cdot,a) = \gamma_{f_a}$ is transverse to Δ. Choose such an a; then $\Gamma_{f_a} \pitchfork \Delta$, so by proposition 8.5.6 every fixed point of f_a is Lefschetz, that is, f_a is a Lefschetz map. The straight-line homotopy $t \mapsto \pi(f(\cdot)+ta)$, $t \in [0,1]$, connects $f_0 = f$ to f_a within the smooth maps $M \to M$, so $f \simeq f_a$. Homotopy invariance of L together with theorem 8.5.9 gives the final clause, completing the proof.

Every homotopy class of self-map thus has a representative whose Lefschetz number is a genuine count, and the count is the same across the class. The invariant carries no more information than the homotopy class, but it extracts from that class a computable integer.

8.5.12 Remark: The Homological Reading A second Lefschetz number is built from the action of f on homology rather than from intersection theory, as the alternating sum $\sum_i (-1)^i \operatorname{tr}\big(f_*\colon H_i(M) \to H_i(M)\big)$ of the traces of the induced maps. The homological Lefschetz fixed-point theorem of [11] evaluates it, for a Lefschetz map, as a signed count over the fixed points of f, so it vanishes exactly when $I(\Delta,\Gamma_f)$ does and carries the same fixed-point obstruction. The two are not identical, however. The homological count sums the fixed-point index $\operatorname{sign}\det(\mathrm{id} - df_p)$, whereas lemma 8.5.8 computes the intersection sign in the order (Δ,Γ_f) as $\operatorname{sign}\det(df_p - \mathrm{id})$; since $\det(\mathrm{id} - df_p) = (-1)^n \det(df_p - \mathrm{id})$, these differ by a global sign set by the dimension:

$$I(\Delta,\Gamma_f) = (-1)^n \sum_i (-1)^i \operatorname{tr}(f_*)$$

The factor $(-1)^n$ is the price of ordering the diagonal before the graph and is invisible for even n. For $f = \mathrm{id}_M$ the maps f_* are the identity, their traces are the Betti numbers b_i, and the alternating sum is $\sum_i (-1)^i b_i$; a closed odd-dimensional manifold has vanishing Euler characteristic, so the sign drops out and this is the homological form of the identity $\chi(M) = I(\Delta,\Delta)$ recorded in section 8.6.

Exercise 8.5.1

1. Let M be compact and oriented and suppose $f\colon M \to M$ is homotopic to a map with no fixed

points. Show that $L(f) = 0$. Deduce that the antipodal map $a\colon S^2 \to S^2$, $a(x) = -x$, is not homotopic to the identity.

2. Let M be compact and oriented of dimension n, and let $f \equiv q$ be a constant map. Show that q is the only fixed point, that it is a Lefschetz fixed point, and that $L(f) = (-1)^n$. In particular $L(f) = 1$ when M is even-dimensional.

3. A matrix $A \in M_2(\mathbb{Z})$ with $\det(A - I) \neq 0$ induces a smooth map $f_A\colon T^2 \to T^2$ on the torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$, $f_A(x) = Ax \bmod \mathbb{Z}^2$. Show that every fixed point of f_A is Lefschetz, that f_A has exactly $|\det(A - I)|$ fixed points, and that $L(f_A) = \det(A - I)$. Evaluate this for the "cat map" $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.

4. Let p be a Lefschetz fixed point of f, and let k be the number of real eigenvalues of df_p (counted with algebraic multiplicity) that are less than 1. Show that $L_p(f) = (-1)^k$.

5. Let M be compact and oriented and $f \simeq \mathrm{id}_M$. Show that $L(f) = I(\Delta, \Delta) = \chi(M)$. Exhibit a fixed-point-free self-map of the torus T^2 that is homotopic to the identity, and explain why this is consistent with the Lefschetz fixed-point theorem.

8.6 Vector Fields and the Poincaré-Hopf Theorem

Oriented intersection theory attached a signed integer to a transverse pair of complementary submanifolds, and the sections before this one turned that integer into the degree of a map and the Lefschetz number of a self-map. One transverse pair is left to count. A vector field is a section of the tangent bundle, and it meets the zero section exactly at its zeros; counting those zeros with signs produces a local invariant, the *index*, while summing the indices over the manifold produces a single global integer. The Poincaré-Hopf theorem identifies that sum with the self-intersection number of the diagonal, the Euler characteristic, and so ties the local behaviour of a field near its equilibria to the global topology of the space it lives on. The hairy ball theorem and a topological Gauss-Bonnet formula fall out at once.

Throughout, M is a compact boundaryless manifold and, where signs are counted, oriented (per the standing conventions of this chapter).

A vector field v vanishes at its *equilibria.* Away from a zero the normalized field $v/|v|$ is a well-defined unit vector, and as one circles a small sphere around an isolated zero this unit vector sweeps out a map of spheres. The number of times it wraps around is the local invariant this section is built on.

Definition 8.6.1: Index of an Isolated Zero

Let v be a smooth vector field on an n-manifold M with an *isolated zero* at p: that is, $v(p) = 0$ and v has no other zero on some neighbourhood of p. Choose a chart (U, φ) with $\varphi(p) = 0$; the coordinate frame $\partial/\partial x^1, \dots, \partial/\partial x^n$ trivializes TM over U and expresses v as a map $\tilde{v}\colon \varphi(U) \to \mathbb{R}^n$ with $\tilde{v}(0) = 0$ and no other zero on a closed ball $\overline{B}_\varepsilon$. The *local Gauss map*

$$g_\varepsilon\colon S^{n-1}_\varepsilon \to S^{n-1}, \qquad g_\varepsilon(x) = \frac{\tilde{v}(x)}{|\tilde{v}(x)|}$$

is a smooth map between compact oriented manifolds of equal dimension with connected target, so it has a degree (section 8.4). The *index* of v at p is

$$\operatorname{ind}_p(v) := \deg(g_\varepsilon) \in \mathbb{Z}$$

The index measures the winding of the direction field around the zero, and the sign records whether the field turns the same way as, or opposite to, the standard sphere. The definition names two choices, the chart and the radius; the next proposition shows the index depends on neither, so it is an invariant of the field near the point.

Before proving that, here are the indices that occur for planar fields, the pictures worth carrying.

Example 8.6.2: Indices of Planar Vector Fields

Identify $\mathbb{R}^2$ with $\mathbb{C}$, so a planar field is a map $\mathbb{C} \to \mathbb{C}$ with an isolated zero at the origin, and its index is the degree of the induced self-map of S^1.

1. *Source* $v(x, y) = (x, y)$, that is $v(z) = z$. The Gauss map is the identity, of degree $+1$, so $\operatorname{ind}_0(v) = +1$.
2. *Sink* $v(x, y) = (-x, -y)$, that is $v(z) = -z$. The Gauss map is the antipodal map of S^1, of degree $(-1)^2 = +1$, so $\operatorname{ind}_0(v) = +1$.
3. *Saddle* $v(x, y) = (x, -y)$. The linear part has determinant -1; the Gauss map has degree -1, so $\operatorname{ind}_0(v) = -1$.
4. *Rotation* $v(x, y) = (-y, x)$, that is $v(z) = iz$. The determinant is $+1$ and $\operatorname{ind}_0(v) = +1$.
5. *Higher order* $v(z) = z^k$ for $k \geq 1$. On S^1 this is $\theta \mapsto k\theta$, of degree k, so $\operatorname{ind}_0(v) = k$; the conjugate field $v(z) = \bar{z}^k$ has index $-k$.

The first four have determinant ± 1 and their index is the sign of that determinant; the last shows that a degenerate zero can carry any integer index.

The pattern in the first four items is general and is the computational heart of the whole theory. Say the zero p is *nondegenerate* if the *linearization* $Dv_p\colon T_pM \to T_pM$ is invertible. This linearization is well-defined at a zero: since $v(p) = 0$, a change of coordinates multiplies the first-order term of v by conjugation, leaving $\det(Dv_p)$ and its sign unambiguous.

Proposition 8.6.3: The Index Is Well-Defined

The index $\mathrm{ind}_p(v)$ of an isolated zero does not depend on the radius ε or on the chart used to compute it. If the zero is nondegenerate, then

$$\mathrm{ind}_p(v) = \operatorname{sign}\det(Dv_p)$$

***Proof*:**
Work in the chart, so $\tilde{v}\colon \overline{B}_\varepsilon \to \mathbb{R}^n$ has 0 as its only zero.

Step 1: Independence of the radius. For $0 < r \leq \varepsilon$ the field $\tilde{v}$ is nonzero on the closed annulus $r \leq |x| \leq \varepsilon$. Pulling each sphere back to the unit sphere by $x \mapsto \rho x$, the maps $x \mapsto \tilde{v}(\rho x)/|\tilde{v}(\rho x)|$ for $\rho \in [r, \varepsilon]$ give a homotopy $S^{n-1} \to S^{n-1}$ between the normalized fields on the radius-r and radius-ε spheres. Since $x \mapsto \rho x$ preserves orientation, these represent g_r and g_ε; homotopic maps have equal degree, so $\deg(g_r) = \deg(g_\varepsilon)$.

Step 2: Independence of the chart. A change of chart carries $\tilde{v}$ to its pushforward $w = Dg \circ \tilde{v} \circ g^{-1}$ under a diffeomorphism g of neighbourhoods of 0 in $\mathbb{R}^n$ fixing 0, and it suffices to show $\mathrm{ind}_0(w) = \mathrm{ind}_0(\tilde{v})$. Set $A = Dg_0$ and $g_s(x) = g(sx)/s$ for $s \in (0, 1]$, with $g_0 := A$; this is a smooth family of diffeomorphisms fixing 0, interpolating from $g_1 = g$ to the linear map A. The pushforwards $w_s = (g_s)_*\tilde{v}$ form a continuous family, each with 0 as its only zero on a small ball; by compactness they are bounded away from 0 on a small enough sphere, so their Gauss maps are all homotopic and $\mathrm{ind}_0(w) = \mathrm{ind}_0(A_*\tilde{v})$. Finally A and, when g preserves orientation, the identity lie in the same path component of $\mathrm{GL}(n, \mathbb{R})$; a path through $\mathrm{GL}(n, \mathbb{R})$ from A to id homotopes $A_*\tilde{v}$ to $\tilde{v}$ without introducing zeros on a small sphere, giving $\mathrm{ind}_0(A_*\tilde{v}) = \mathrm{ind}_0(\tilde{v})$. If instead g reverses orientation, it reverses the orientations of both the domain sphere and the target sphere of the Gauss map at once, and the degree is unchanged; the index is therefore independent of the chart with no orientation hypothesis at all.

Step 3: The nondegenerate formula. If $A = Dv_p$ is invertible, then $\tilde{v}(x) = Ax + o(|x|)$, so on a small sphere g_ε is homotopic to $x \mapsto Ax/|Ax|$. The linear map $x \mapsto Ax$ carries S^{n-1} to an ellipsoid, and radial projection back to S^{n-1} has degree $+1$ if $\det A > 0$ and -1 if $\det A < 0$, since $\mathrm{GL}^+(n, \mathbb{R})$ is path-connected and every element of $\mathrm{GL}^-(n, \mathbb{R})$ is a reflection composed with an element of $\mathrm{GL}^+(n, \mathbb{R})$. Hence $\mathrm{ind}_p(v) = \operatorname{sign}\det(Dv_p)$, completing the proof.

Step 2 does more than tidy the definition: it shows the index makes sense on any manifold, orientable or not, because an orientation-reversing coordinate change flips the two spheres together. The index is thus a genuine local invariant of the field, and the nondegenerate formula reduces its computation to a determinant. The general index of a possibly degenerate zero is recovered from nondegenerate ones by perturbation, which is the mechanism the Poincaré-Hopf proof will exploit.

To sum indices into a statement, the target of that sum must be named. The right target is a single integer attached to M itself, and this chapter has already built the object that carries it: the self-intersection number of the diagonal.

Definition 8.6.4: Euler Characteristic

Let M be a compact oriented manifold and $\Delta \subseteq M \times M$ its diagonal. The *Euler characteristic* of M is the self-intersection number of the diagonal,

$$\chi(M) := I(\Delta, \Delta)$$

Equivalently, since the graph of the identity is the diagonal, $\Gamma_{\mathrm{id}_M} = \Delta$, it is the Lefschetz number of the identity map, $\chi(M) = L(\mathrm{id}_M)$ (section 8.3, section 8.5).

Defining $\chi(M)$ this way makes it visibly a homotopy-invariant integer, and one intrinsic to M: it needs no triangulation, no cell structure, and no metric, only the diagonal and the intersection pairing. The Lefschetz formula $\chi(M) = L(\mathrm{id}_M)$ is the special case of the fixed-point theory of the previous section where every point is a fixed point, so the local fixed-point indices of the identity, once Poincaré-Hopf is proved, will turn out to be exactly the indices of a vector field.

8.6.5 Remark: The Homological Euler Characteristic The reader who knows singular homology will recognize $\chi(M)$ as the alternating sum of Betti numbers,

$$\chi(M) = \sum_{i=0}^{\dim M} (-1)^i b_i, \qquad b_i = \operatorname{rank} H_i(M)$$

The identification of $I(\Delta, \Delta)$ with this sum runs through the Künneth theorem and Poincaré duality of [11, Künneth Formula for Homology, Poincaré Duality], though the assembly itself is a recorded gap in that book. It is not needed here, where the intersection-theoretic definition is self-contained. Because the index sum below is defined from local charts alone, both the definition of χ and the Poincaré-Hopf theorem extend verbatim to compact manifolds that are not orientable, where the homological sum (with field coefficients) is the invariant they compute.

The circle is the first check, and it can be run directly from the definition.

Example 8.6.6: The Euler Characteristic of the Circle

For $M = S^1$ the diagonal $\Delta \subseteq S^1 \times S^1 = T^2$ is the $(1,1)$-curve $\{(\theta, \theta)\}$. Push it off itself to $\Delta' = \{(\theta, \theta + \pi)\}$; the pushed-off curve is homotopic to Δ and disjoint from it, since $\theta = \theta + \pi$ has no solution on S^1. Hence $\chi(S^1) = I(\Delta, \Delta) = I(\Delta', \Delta) = 0$. The circle has a nowhere-vanishing vector field, the unit tangent $\partial/\partial\theta$, and its vanishing Euler characteristic is exactly what will let that field exist.

Everything is now in place. The Poincaré-Hopf theorem says the local index sum of any vector field with isolated zeros equals this global invariant, so in particular the sum does not depend on the field.

Theorem 8.6.7: Poincaré-Hopf

Let M be a compact oriented manifold and v a smooth vector field on M with only isolated zeros. Then

$$\sum_{p:\, v(p)=0} \operatorname{ind}_p(v) = \chi(M)$$

In particular the sum is independent of the field v.

***Proof*:**
The compactness of M makes the zero set of v finite, so the sum is finite. The argument computes $\chi(M) = I(\Delta, \Delta)$ by pushing the diagonal off itself along v.

Step 1: The normal bundle of the diagonal is the tangent bundle. At a point $(p, p) \in \Delta$,

$$T_{(p,p)}(M \times M) = T_pM \oplus T_pM, \qquad T_{(p,p)}\Delta = \{(u, u) \mid u \in T_pM\}$$

so the normal space $N_{(p,p)}\Delta = T_{(p,p)}(M \times M)/T_{(p,p)}\Delta$ is carried isomorphically to T_pM by $[(u, w)] \mapsto w - u$. These isomorphisms assemble to a vector bundle isomorphism $N\Delta \cong TM$ over the identification $\Delta \cong M$, $(p, p) \mapsto p$. By the tubular neighbourhood theorem (theorem 7.8.1) there is a diffeomorphism Φ from a neighbourhood of the zero section in TM onto a neighbourhood of Δ in $M \times M$, restricting to $\Delta \cong M$ on the zero section. Replacing v by εv for small $\varepsilon > 0$ (which changes neither the zeros nor the indices, as $v/|v|$ is unchanged) puts the image of $v \colon M \to TM$ inside this neighbourhood, so

$$\Delta_v := \Phi\big(\{(p, v(p)) : p \in M\}\big) \subseteq M \times M$$

is a submanifold, a pushoff of Δ, with $\Delta_v \cap \Delta = \Phi(\{v = 0\})$: the pushed-off diagonal meets the diagonal exactly over the zeros of v. As v is homotopic to the zero field through tv, the submanifold Δ_v is homotopic to Δ, and homotopy invariance of the oriented intersection number (section 8.3) gives $I(\Delta, \Delta) = I(\Delta, \Delta_v)$.

Step 2: A nondegenerate zero contributes its index. Suppose first that every zero of v is nondegenerate. Then v, as a section of TM, is transverse to the zero section, so Δ is transverse to Δ_v and $I(\Delta, \Delta_v)$ is the signed count of the finitely many intersection points. Fix a zero p and an oriented chart, trivializing TM near p as $B \times \mathbb{R}^n$ with the zero section $B \times \{0\}$; to first order Δ_v is the graph $\{(x, Dv_p\, x) \mid x\}$. The tangent spaces $\mathbb{R}^n \times \{0\}$ and $\{(u, Dv_p u) \mid u\}$ span the ambient $\mathbb{R}^n \times \mathbb{R}^n$ precisely because Dv_p is invertible, and with the orientation conventions fixed in section 8.3 the sign of the intersection at p is $\operatorname{sign}\det(Dv_p)$. By proposition 8.6.3 this sign is $\operatorname{ind}_p(v)$. Summing,

$$\chi(M) = I(\Delta, \Delta_v) = \sum_{p:\, v(p)=0} \operatorname{sign}\det(Dv_p) = \sum_{p:\, v(p)=0} \operatorname{ind}_p(v)$$

Step 3: The general case by perturbation. Let v have isolated but possibly degenerate zeros $p_1, \ldots, p_k$, and choose disjoint closed balls B_j around them, in coordinates, on which v has no other zero. Inside a ball, v reads as $\tilde{v} \colon B_j \to \mathbb{R}^n$; take a bump function β equal to 1 near the centre and 0 near ∂B_j, and (by Sard's theorem, theorem 7.2.1) a regular value ξ of $\tilde{v}$ with

$0 < |\xi| < \min\{|\tilde{v}(x)| \mid \beta(x) < 1\}$, the minimum positive because $\tilde{v}$ vanishes only at the centre, where $\beta = 1$. The field v' agreeing with v outside the balls and equal to $\tilde{v} - \beta\xi$ inside is smooth, equals v on each ∂B_j, is nonzero wherever $\beta < 1$ (there $|\tilde{v}| > |\xi| \geq |\beta\xi|$), and has only nondegenerate zeros in the cores (there $\tilde{v} = \xi$ is a regular value, so $D\tilde{v}$ is invertible). The degree of $v/|v| = v'/|v'|$ on ∂B_j equals $\mathrm{ind}_{p_j}(v)$ and, by additivity of the degree over the interior zeros, also equals $\sum_{q \in B_j : v'(q)=0} \mathrm{ind}_q(v')$. Summing over j,

$$\sum_{p:\, v(p)=0} \mathrm{ind}_p(v) = \sum_{q:\, v'(q)=0} \mathrm{ind}_q(v') = \chi(M)$$

by Step 2 applied to the nondegenerate field v', as we sought to show.

The theorem is a bridge between two registers. On the left is differential data, the linearizations of a field at its equilibria, computed determinant by determinant; on the right is a topological invariant of M that does not mention any field. That the left side is forced to equal the right regardless of the field is the content, and it is what makes the index sum useful: to compute $\chi(M)$ one may exhibit any convenient field, and to constrain the fields on M one may compute $\chi(M)$ by any means. The running example of the sphere shows both directions at work.

Example 8.6.8: The Euler Characteristic of Spheres

On $S^n \subseteq \mathbb{R}^{n+1}$ take the field $v(p) = e_{n+1} - \langle e_{n+1}, p\rangle\, p$, the tangential part of the constant vector e_{n+1} (the gradient of the height function). Its zeros are the points where e_{n+1} is normal to the sphere, namely the poles $N = e_{n+1}$ and $S = -e_{n+1}$. Near S the field points radially outward, a source, so $\mathrm{ind}_S(v) = +1$; near N it points radially inward, a sink, whose Gauss map is the antipodal map of S^{n-1}, so $\mathrm{ind}_N(v) = (-1)^n$. By theorem 8.6.7,

$$\chi(S^n) = \mathrm{ind}_S(v) + \mathrm{ind}_N(v) = 1 + (-1)^n = \begin{cases} 2, & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$$

So even spheres have Euler characteristic 2 and odd spheres have 0, matching $\chi(S^1) = 0$ from example 8.6.6.

Example 8.6.9: The Torus and the Genus-g Surface

The n-torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ carries the nowhere-vanishing field $\partial/\partial\theta^1$, which has no zeros, so theorem 8.6.7 gives $\chi(T^n) = 0$ for every $n \geq 1$.

For a closed orientable surface Σ_g of genus g, stand it upright in $\mathbb{R}^3$ and take the downhill field of the height function. On a surface a source (a minimum) and a sink (a maximum) each have index $+1$, and a saddle has index -1, following example 8.6.2. The height function on Σ_g has one minimum, one maximum, and $2g$ saddles (one at the top and one at the bottom of each of the g holes), so

$$\chi(\Sigma_g) = 1 + 1 - 2g = 2 - 2g$$

The case $g = 1$ recovers $\chi(T^2) = 0$, and $g = 0$ recovers $\chi(S^2) = 2$.

The even-sphere computation has an immediate consequence that predates the general theorem and gave the subject one of its names. If $\chi(M) \neq 0$, then no vector field on M can avoid vanishing, for a nowhere-vanishing field has no zeros and its (empty) index sum is 0.

Theorem 8.6.10: Hairy Ball Theorem

The even-dimensional sphere S^{2n} admits no nowhere-vanishing smooth vector field. Equivalently, every smooth vector field tangent to S^{2n} vanishes somewhere.

***Proof*:**
A nowhere-vanishing field has isolated zeros vacuously, so theorem 8.6.7 applies and gives $0 = \sum \mathrm{ind}_p(v) = \chi(S^{2n})$. But $\chi(S^{2n}) = 2$ by example 8.6.8, a contradiction. Hence no such field exists, as we sought to show.

The name is the two-dimensional picture: one cannot comb the hair flat on a sphere without leaving a cowlick. The obstruction is entirely the nonvanishing of χ, and the odd spheres show it is sharp. On $S^{2n-1} \subseteq \mathbb{R}^{2n} = \mathbb{C}^n$ the field $v(z) = iz$ is tangent, since $\langle iz, z\rangle = 0$, and nowhere zero, since $|iz| = |z| = 1$; in real coordinates it is $(x_1, \dots, x_{2n}) \mapsto (-x_2, x_1, -x_4, x_3, \dots)$. Thus odd spheres are combable, in agreement with $\chi(S^{2n-1}) = 0$. The general principle behind both cases is the converse direction, due to Hopf: a compact manifold admits a nowhere-vanishing field if and only if $\chi(M) = 0$. Poincaré-Hopf is the forward half, and the converse is developed in [11].

For a hypersurface in Euclidean space the index sum can be read off a single map, the field of unit normals, and this yields the topological core of the Gauss-Bonnet theorem. Let $M^n \subseteq \mathbb{R}^{n+1}$ be a compact boundaryless hypersurface, oriented by a choice of unit normal.

Definition 8.6.11: Gauss Map

The *Gauss map* of an oriented hypersurface $M^n \subseteq \mathbb{R}^{n+1}$ is

$$G\colon M \to S^n, \qquad G(p) = \nu(p)$$

sending each point to its outward unit normal.

Since M and S^n are compact oriented n-manifolds and S^n is connected, G has a degree (section 8.4). For the round sphere $S^n \subseteq \mathbb{R}^{n+1}$ itself the outward normal at p is p, so G is the identity and $\deg G = 1$. The theorem below reads this against the Euler characteristic.

Theorem 8.6.12: Topological Gauss-Bonnet

Let $M^n \subseteq \mathbb{R}^{n+1}$ be a compact boundaryless hypersurface with Gauss map G. Then

$$\chi(M) = \big(1 + (-1)^n\big) \deg G$$

In particular, for n even, $\deg G = \frac{1}{2}\chi(M)$; for n odd both sides vanish.

***Proof*:**
Choose a unit vector $a \in S^n$ that is a regular value of G and for which $-a$ is also a regular value; almost every a qualifies by Sard's theorem (theorem 7.2.1). On M consider the height function

$h_a(x) = \langle a, x\rangle$ and its gradient field

$$w(p) = a - \langle a, \nu(p)\rangle\, \nu(p)$$

the tangential projection of a. Then $w(p) = 0$ exactly when a is normal to M at p, that is when $\nu(p) = \pm a$, so the zeros of w are the finite set $G^{-1}(a) \cup G^{-1}(-a)$.

Differentiating and projecting to T_pM at a zero p (the normal component does not affect the index), the linearization is

$$Dw_p = -\langle a, \nu(p)\rangle\, dG_p \colon T_pM \to T_pM$$

where $dG_p \colon T_pM \to T_{G(p)}S^n$ is identified with an endomorphism of T_pM through $T_{G(p)}S^n = \nu(p)^\perp = T_pM$. Because a is a regular value of G, dG_p is invertible at each such p, so every zero of w is nondegenerate and, by proposition 8.6.3,

$$\operatorname{ind}_p(w) = \operatorname{sign}\det(Dw_p) = \big(-\langle a, \nu(p)\rangle\big)^n \operatorname{sign}\det(dG_p)$$

At a zero with $\nu(p) = -a$ the scalar $-\langle a, \nu(p)\rangle$ is $+1$, so $\operatorname{ind}_p(w) = \operatorname{sign}\det(dG_p)$; at a zero with $\nu(p) = a$ it is -1, so $\operatorname{ind}_p(w) = (-1)^n \operatorname{sign}\det(dG_p)$. The degree of G, computed at the regular value a and again at the regular value $-a$, is

$$\deg G = \sum_{p\in G^{-1}(a)} \operatorname{sign}\det(dG_p) = \sum_{p\in G^{-1}(-a)} \operatorname{sign}\det(dG_p)$$

since the degree is independent of the regular value (section 8.4). Summing the indices and applying theorem 8.6.7,

$$\chi(M) = \sum_{p\in G^{-1}(a)} (-1)^n \operatorname{sign}\det(dG_p) + \sum_{p\in G^{-1}(-a)} \operatorname{sign}\det(dG_p) = (-1)^n \deg G + \deg G$$

which is $(1 + (-1)^n)\deg G$, completing the proof.

For n even the formula reads $\chi(M) = 2\deg G$; the sphere checks it, with $\chi(S^{2m}) = 2$ and $\deg G = 1$. For n odd it says only $0 = 0$, consistent with the vanishing of the Euler characteristic of an odd-dimensional closed manifold. The content is that the degree of the normal map, an integer computed by counting how the tangent planes turn, is half a homotopy invariant of M.

8.6.13 Foreshadowing: The Curvature Form of Gauss-Bonnet The degree of the Gauss map is also an integral: for a surface, $\deg G = \frac{1}{4\pi}\int_M K\, dA$ with K the Gaussian curvature, so $\int_M K\, dA = 2\pi\,\chi(M)$, the classical Gauss-Bonnet theorem. The passage from the topological count above to the curvature integral requires a metric and its second fundamental form, which this book does not develop; the metric form is proved in [6, Global Gauss–Bonnet]. What is established here is the topological skeleton on which that theorem hangs: the total curvature is a homotopy invariant because it computes the degree of a map of spheres.

Exercise 8.6.1

1. Compute the index at the origin of the planar vector fields $v(z) = z^k$ and $v(z) = \bar{z}^k$ for $k \geq 1$, and of $v(x, y) = (x^2 - y^2, 2xy)$. Deduce that a single isolated zero can carry any integer index.
2. Let M and N be compact oriented manifolds. Show that if v on M and w on N have isolated zeros, then the field (v, w) on $M \times N$ has isolated zeros with $\mathrm{ind}_{(p,q)}(v, w) = \mathrm{ind}_p(v) \cdot \mathrm{ind}_q(w)$, and conclude the product formula $\chi(M \times N) = \chi(M)\,\chi(N)$.
3. Using that $\mathrm{ind}_p(-v) = (-1)^n \,\mathrm{ind}_p(v)$ for a zero of a field on an n-manifold, prove that every compact odd-dimensional manifold has $\chi(M) = 0$.
4. Let f be a smooth function on M with a nondegenerate critical point at p, and let λ be its Morse index (the number of negative eigenvalues of the Hessian). Show that the gradient field ∇f (for any Riemannian metric) has a nondegenerate zero at p with $\mathrm{ind}_p(\nabla f) = (-1)^\lambda$.
5. Use theorem 8.6.7 and the product formula of exercise 8.6.1 (2) to decide, for each pair, whether $S^{2n} \times S^{2m}$ and $S^{2n} \times S^{2m+1}$ admit a nowhere-vanishing vector field, exhibiting one where it exists.
6. The torus T^2 embeds in $\mathbb{R}^3$ as the surface of revolution of a circle. Compute $\deg G$ for its Gauss map two ways: directly from theorem 8.6.12 using $\chi(T^2)$, and by identifying the points where the outward normal equals a fixed generic direction.

9

Morse Theory

A single well-chosen function reads the topology of a manifold off its critical points. Morse theory studies the functions whose critical points are as simple as possible (*nondegenerate*) and recovers from them the shape of the manifold: a handle for each critical point, a cell of dimension its *index*, and inequalities bounding the Betti numbers below the counts of critical points. The engine is the gradient flow of chapter 5, which between critical values retracts one sublevel set onto another and at a critical value of index λ attaches a λ-cell. Existence rests on Sard's theorem (theorem 7.2.1): almost every height function on an embedded manifold is Morse. The alternating count $\sum_{\lambda}(-1)^{\lambda}c_{\lambda} = \chi(M)$ is the Poincaré-Hopf theorem (section 8.6) applied to the gradient, and the Morse inequalities refine it against the Betti numbers of [11, Betti Number And Torsion Coefficients]. The chapter closes with *Morse homology*, the complex generated by the critical points whose boundary counts gradient trajectories and whose homology is that of the manifold, the finite-dimensional model whose infinite-dimensional analogue is Floer theory.

9.1 Morse Functions and the Morse Lemma

A smooth function on a manifold records topological information at the points where its differential vanishes. This section isolates the local data that make that information usable: the nondegeneracy of a critical point and its index. The central result is the *Morse Lemma*, a normal form showing that near a nondegenerate critical point f is, in suitable coordinates, a sum of signed squares, with the number of negative squares equal to the index.

The earlier chapters counted the preimages of a *regular* value, where the level set is a submanifold (corollary 3.2.10). Morse theory looks at the opposite locus, the critical points, and reads the topology of M off their number and type.

Definition 9.1.1: Critical Point and Critical Value

Let $f\colon M \to \mathbb{R}$ be smooth. A point $p \in M$ is a *critical point* of f if the differential $df_p\colon T_pM \to \mathbb{R}$ is the zero map; otherwise p is a *regular point*. The image $f(p)$ of a critical point is a *critical value*, and a real number that is the image of no critical point is a *regular value*. Write

$$\operatorname{Crit}(f) := \{p \in M \mid df_p = 0\}$$

for the set of critical points.

In a chart (U, φ) the condition $df_p = 0$ says that all first partial derivatives of $\widehat{f} := f \circ \varphi^{-1}$ vanish at $\varphi(p)$, so $\operatorname{Crit}(f)$ is the common zero set of the coordinate derivatives and, being the zero set of the continuous section df of T^*M, is closed. A critical point carries no first-order information, so to classify it one turns to the second-order term. On a general manifold there is no coordinate-free second derivative of a function, but at a critical point the ambiguity in the naive second derivative cancels, and a well-defined symmetric bilinear form survives.

Definition 9.1.2: Hessian at a Critical Point

Let p be a critical point of $f\colon M \to \mathbb{R}$. The *Hessian* of f at p is

$$\operatorname{Hess}_p f\colon T_pM \times T_pM \to \mathbb{R}, \qquad \operatorname{Hess}_p f(v, w) = v(\widetilde{W} f)$$

where $\widetilde{W}$ is any smooth vector field near p with $\widetilde{W}_p = w$, and the tangent vector v acts as a derivation on the smooth function $\widetilde{W} f$.

Proposition 9.1.3: The Hessian Is a Well-Defined Symmetric Form

At a critical point p, the value $\operatorname{Hess}_p f(v, w)$ is independent of the chosen extension $\widetilde{W}$, and $\operatorname{Hess}_p f$ is a symmetric bilinear form. In a chart (U, φ) with coordinates $\varphi = (x^1, \dots, x^n)$ and $\widehat{f} = f \circ \varphi^{-1}$,

$$\operatorname{Hess}_p f\Big(\tfrac{\partial}{\partial x^i}\big|_p, \tfrac{\partial}{\partial x^j}\big|_p\Big) = \frac{\partial^2 \widehat{f}}{\partial x^i\, \partial x^j}(\varphi(p))$$

the ordinary Hessian matrix of $\widehat{f}$ at $\varphi(p)$.

***Proof*:**

Fix a chart (U, φ) around p and write $v = \sum_i v^i\, \partial/\partial x^i|_p$ and $\widetilde{W} = \sum_j W^j\, \partial/\partial x^j$ near p, so that $W^j(p) = w^j$ are the components of w. In the chart $\widetilde{W} f = \sum_j W^j\, \partial_j \widehat{f}$, and applying v as a derivation,

$$v(\widetilde{W} f) = \sum_{i,j} v^i\, \partial_i\big(W^j\, \partial_j \widehat{f}\big)(\varphi(p)) = \sum_{i,j} v^i \Big(\partial_i W^j \cdot \partial_j \widehat{f} + W^j\, \partial_i \partial_j \widehat{f}\Big)(\varphi(p))$$

Since p is critical, $\partial_j \widehat{f}(\varphi(p)) = 0$ for every j, so the first term drops and

$$\operatorname{Hess}_p f(v, w) = \sum_{i,j} v^i w^j\, \partial_i \partial_j \widehat{f}(\varphi(p))$$

The right side involves only the components of v and w, not the extension $\widetilde{W}$, and it is symmetric because $\partial_i\partial_j\widehat{f} = \partial_j\partial_i\widehat{f}$. Setting $v = \partial/\partial x^i|_p$ and $w = \partial/\partial x^j|_p$ reads off the matrix entry, as we sought to show.

The invariant description hides the same cancellation. Extending v to a field $\widetilde{V}$ as well,

$$\mathrm{Hess}_p f(v,w) - \mathrm{Hess}_p f(w,v) = \widetilde{V}_p(\widetilde{W}f) - \widetilde{W}_p(\widetilde{V}f) = df_p\big([\widetilde{V},\widetilde{W}]_p\big) = 0$$

by definition of the Lie bracket (chapter 5) and $df_p = 0$, so symmetry is exactly the vanishing of the differential on the bracket. The coordinate formula also shows how the Hessian responds to a change of chart: at a critical point the first derivatives that would otherwise contribute vanish, so if $y = (y^1, \ldots, y^n)$ is a second chart with transition Jacobian $J = (\partial x^i/\partial y^k)$ at p, the Hessian matrices satisfy $H_y = J^\top H_x J$. The Hessian is therefore a genuine bilinear form on T_pM, transforming as a symmetric $(0,2)$-tensor at the single point p.

Because the two matrices are congruent, their rank and signature are chart-independent, and these are the two invariants that classify a critical point.

Definition 9.1.4: Nondegeneracy, Morse Index, Morse Function

A critical point p of f is *nondegenerate* if $\mathrm{Hess}_p f$ is a nondegenerate bilinear form, that is, its matrix in one (hence every) chart is invertible. The *Morse index* $\lambda(p)$ is the dimension of a maximal subspace of T_pM on which $\mathrm{Hess}_p f$ is negative definite. The function f is a *Morse function* if every critical point is nondegenerate. For a Morse function, c_λ denotes the number of critical points of index λ.

By Sylvester's law of inertia a symmetric matrix has a well-defined number of negative eigenvalues, invariant under the congruence $H \mapsto J^\top HJ$; since a change of chart acts on the Hessian by exactly such a congruence, the Morse index equals the number of negative eigenvalues of the Hessian matrix in any chart, counted with multiplicity, and does not depend on the chart. Nondegeneracy is the vanishing of the nullity, so a nondegenerate critical point has $\lambda(p) + \mu(p) = n$ with $\mu(p)$ the number of positive eigenvalues. The extreme indices are the familiar cases: $\lambda(p) = 0$ means $\mathrm{Hess}_p f$ is positive definite and p is a local minimum, $\lambda(p) = n$ means it is negative definite and p is a local maximum, and intermediate indices are saddles.

The running example shows all of this.

Example 9.1.5: The Height Function on S^n

Let $h\colon S^n \to \mathbb{R}$, $h(x) = x_{n+1} = \langle x, e_{n+1}\rangle$, the height in the last coordinate, with $S^n \subseteq \mathbb{R}^{n+1}$ the unit sphere. At $p \in S^n$ the tangent space is $T_pS^n = p^\perp$, and $dh_p(v) = \langle e_{n+1}, v\rangle$ for $v \in p^\perp$. This vanishes for all $v \in p^\perp$ precisely when $e_{n+1} \perp p^\perp$, that is $e_{n+1} \in \mathbb{R}p$, so the critical points are the two poles $N = e_{n+1}$ and $S = -e_{n+1}$, with critical values $+1$ and -1.

Near S, project to the first n coordinates: the map $u \mapsto (u, -\sqrt{1-|u|^2})$ for $|u| < 1$ is a chart carrying $u = 0$ to S, in which h becomes

$$\widehat{h}(u) = -\sqrt{1-|u|^2} = -1 + \tfrac{1}{2}|u|^2 + O(|u|^4)$$

so the Hessian matrix at $u = 0$ is $\partial_i\partial_j\big(\frac{1}{2}\sum_k u_k^2\big) = \delta_{ij}$, the identity. It is positive definite, so S is a nondegenerate critical point of index 0, a minimum. Near N the chart $u \mapsto (u, +\sqrt{1-|u|^2})$ gives $\widehat{h}(u) = 1 - \frac{1}{2}|u|^2 + O(|u|^4)$, Hessian $-\delta_{ij}$, negative definite, so N is nondegenerate of index

n, a maximum. Thus h is a Morse function with exactly two critical points, and

$$c_0 = 1, \qquad c_n = 1, \qquad c_\lambda = 0 \ \text{ otherwise}$$

The alternating sum $\sum_\lambda (-1)^\lambda c_\lambda = 1 + (-1)^n$ matches $\chi(S^n)$ from example 8.6.8, a coincidence the later sections will explain.

The height function on S^n was, near each pole, exactly a positive or negative sum of squares once the higher-order terms were discarded. The Morse Lemma says the higher-order terms can always be absorbed by a change of coordinates: at any nondegenerate critical point f agrees with its second-order model on the nose, in a suitable chart.

Theorem 9.1.6: Morse Lemma

Let p be a nondegenerate critical point of $f\colon M \to \mathbb{R}$, of Morse index $\lambda = \lambda(p)$. Then there is a chart (U, φ) centered at p, with coordinates $\varphi = (x^1, \dots, x^n)$ and $\varphi(p) = 0$, in which

$$f = f(p) - (x^1)^2 - \cdots - (x^\lambda)^2 + (x^{\lambda+1})^2 + \cdots + (x^n)^2$$

throughout U.

***Proof*:**
Step 1: Reduction to a Euclidean model. Choose any chart centered at p; then f corresponds to a smooth function, still written f, on a convex open ball V around 0 in $\mathbb{R}^n$, with $f(0) = f(p)$ and $Df(0) = 0$ since p is critical. Subtract the constant to assume $f(0) = 0$; the constant is restored at the end. The Hessian matrix $\big(\partial_i \partial_j f(0)\big)$ is symmetric and, by nondegeneracy, invertible.

Step 2: A smooth quadratic expansion. Because V is star-shaped about 0 and $f(0) = 0$, the fundamental theorem of calculus along the segment $t \mapsto tx$ gives, for $x \in V$,

$$f(x) = \int_0^1 \frac{d}{dt} f(tx)\, dt = \sum_i x_i \int_0^1 \partial_i f(tx)\, dt = \sum_i x_i\, g_i(x)$$

with $g_i(x) := \int_0^1 \partial_i f(tx)\, dt$ smooth and $g_i(0) = \partial_i f(0) = 0$. Applying the same identity to each g_i,

$$g_i(x) = \sum_j x_j\, h_{ij}(x), \qquad h_{ij}(x) := \int_0^1 \partial_j g_i(tx)\, dt$$

smooth, so $f(x) = \sum_{i,j} x_i x_j\, h_{ij}(x)$. Replacing h_{ij} by $\frac{1}{2}(h_{ij} + h_{ji})$ leaves the sum unchanged, so assume $h_{ij} = h_{ji}$. Differentiating $f = \sum x_i x_j h_{ij}$ twice at 0 gives $\partial_i \partial_j f(0) = 2h_{ij}(0)$, so the symmetric matrix $\big(h_{ij}(0)\big) = \frac{1}{2}\big(\partial_i \partial_j f(0)\big)$ is invertible.

Step 3: Inductive diagonalization. The claim is that for each $r = 1, \dots, n+1$ there is a coordinate system $y = (y_1, \dots, y_n)$ near 0, fixing 0, in which

$$f = \varepsilon_1 y_1^2 + \cdots + \varepsilon_{r-1} y_{r-1}^2 + \sum_{i,j \ge r} y_i y_j\, H_{ij}(y)$$

with each $\varepsilon_k \in \{\pm 1\}$, the H_{ij} smooth and symmetric, and the trailing block $\big(H_{ij}(0)\big)_{i,j\geq r}$ invertible. For $r = 1$ this is Step 2 with $y = x$, $H_{ij} = h_{ij}$.

Assume the form holds for some $r \leq n$. The block $\big(H_{ij}(0)\big)_{i,j\geq r}$ is an invertible symmetric matrix, hence nonzero, so a linear change of the coordinates $y_r, \ldots, y_n$ (fixing $y_1, \ldots, y_{r-1}$) makes $H_{rr}(0) \neq 0$: if some diagonal entry $H_{kk}(0) \neq 0$ permute it into position r, and otherwise pick $H_{k\ell}(0) \neq 0$ with $k < \ell$ and substitute $y_\ell \mapsto y_k + y_\ell$, which turns the (k, k) entry into $2H_{k\ell}(0) \neq 0$. Then H_{rr} is nonzero on a neighbourhood of 0, and $g(y) := \sqrt{|H_{rr}(y)|}$ is smooth and nonvanishing there. Put $\varepsilon_r := \operatorname{sign} H_{rr}(0)$ and introduce

$$z_r := g(y)\Big(y_r + \sum_{i>r} y_i \,\frac{H_{ir}(y)}{H_{rr}(y)}\Big), \qquad z_i := y_i \quad (i \neq r)$$

The Jacobian $\partial z/\partial y$ at 0 fixes every coordinate but the r-th and has $\partial z_r/\partial y_r(0) = g(0) \neq 0$, so it is invertible and, by the inverse function theorem, $z = (z_1, \ldots, z_n)$ is a coordinate system near 0 fixing 0. Since $g^2 = |H_{rr}|$ and $\varepsilon_r|H_{rr}| = H_{rr}$,

$$\varepsilon_r z_r^2 = H_{rr}\, y_r^2 + 2y_r \sum_{i>r} y_i H_{ir} + \frac{\big(\sum_{i>r} y_i H_{ir}\big)^2}{H_{rr}}$$

The trailing block of the inductive form expands as $\sum_{i,j\geq r} y_i y_j H_{ij} = H_{rr} y_r^2 + 2y_r \sum_{i>r} y_i H_{ir} + \sum_{i,j>r} y_i y_j H_{ij}$, so subtracting the previous display,

$$f = \varepsilon_1 z_1^2 + \cdots + \varepsilon_r z_r^2 + \sum_{i,j>r} z_i z_j\, H'_{ij}(z), \qquad H'_{ij} := H_{ij} - \frac{H_{ir}H_{jr}}{H_{rr}}$$

with $z_i = y_i$ for $i > r$, and H'_{ij} smooth and symmetric. The Hessian of f at 0 in the z-coordinates is block diagonal, $\operatorname{diag}(2\varepsilon_1, \ldots, 2\varepsilon_r) \oplus \big(2H'_{ij}(0)\big)_{i,j>r}$, and it is congruent to the original invertible Hessian because a diffeomorphism fixing p acts on the Hessian by congruence. An invertible block-diagonal matrix has invertible blocks, so $\big(H'_{ij}(0)\big)_{i,j>r}$ is invertible, completing the induction. At $r = n + 1$,

$$f = \varepsilon_1 z_1^2 + \cdots + \varepsilon_n z_n^2, \qquad \varepsilon_k \in \{\pm 1\}$$

Step 4: The number of negative squares is the index. Reorder the coordinates so that the indices with $\varepsilon_k = -1$ come first, say there are m of them, and rename the reordered coordinates $x^1, \ldots, x^n$. Restoring the constant,

$$f = f(p) - (x^1)^2 - \cdots - (x^m)^2 + (x^{m+1})^2 + \cdots + (x^n)^2$$

The Hessian of the right side at 0 is $\operatorname{diag}(-2, \ldots, -2, +2, \ldots, +2)$ with m negative entries. This is congruent to $\operatorname{Hess}_p f$, so by Sylvester's law of inertia the two have the same number of negative eigenvalues; hence $m = \lambda(p)$. Renaming m as λ gives the stated normal form, completing the proof.

The Morse Lemma says the second-order Taylor data of f at a nondegenerate critical point is a complete local invariant: the germ of f there is determined, up to a change of coordinates, by the single integer λ. This is why analytic data, the critical points and their indices, will control topology in the sections that follow. It also settles the local geometry at once.

Corollary 9.1.7: Nondegenerate Critical Points Are Isolated

A nondegenerate critical point p of f has a neighbourhood containing no other critical point. In particular, a Morse function on a compact manifold has only finitely many critical points, and every c_λ is finite.

***Proof*:**
In the Morse chart of theorem 9.1.6 the differential is

$$df = -2x^1\,dx^1 - \cdots - 2x^\lambda\,dx^\lambda + 2x^{\lambda+1}\,dx^{\lambda+1} + \cdots + 2x^n\,dx^n$$

which vanishes only at $x = 0$, that is at p. So p is the only critical point in that chart, and $\mathrm{Crit}(f)$ is discrete. It is also closed, being the zero set of the section df. When M is compact, a closed discrete subset of a compact space is finite, so $\mathrm{Crit}(f)$ is finite and each c_λ is finite, completing the proof.

The second running picture is the torus, where a single height function already exhibits all the indices between 0 and n.

Example 9.1.8: The Standing Torus

Place the torus upright in $\mathbb{R}^3$ as the surface of revolution about the y-axis, with $0 < a < b$,

$$\Phi(\theta,\psi) = \big((b + a\cos\psi)\cos\theta,\ a\sin\psi,\ (b + a\cos\psi)\sin\theta\big)$$

so that $\Phi\colon (\mathbb{R}/2\pi\mathbb{Z})^2 \to T^2$ is a diffeomorphism and (θ,ψ) are local angle coordinates. Take the height function $f = z$, the third coordinate, so $f\circ\Phi = (b + a\cos\psi)\sin\theta$. Its partial derivatives are

$$\partial_\theta(f\circ\Phi) = (b + a\cos\psi)\cos\theta, \qquad \partial_\psi(f\circ\Phi) = -a\sin\psi\,\sin\theta$$

Since $b + a\cos\psi > 0$, the first vanishes exactly when $\cos\theta = 0$, i.e. $\theta \in \{\frac{\pi}{2}, \frac{3\pi}{2}\}$, and then $\sin\theta = \pm 1 \neq 0$ forces $\sin\psi = 0$, i.e. $\psi \in \{0,\pi\}$. The four critical points are

$$(\theta,\psi) \in \left\{\tfrac{\pi}{2}, \tfrac{3\pi}{2}\right\} \times \{0,\pi\}$$

At a critical point $\sin\psi = 0$, so the mixed partial $\partial_\theta\partial_\psi(f\circ\Phi) = -a\sin\psi\cos\theta$ is zero and the Hessian in the (θ,ψ) chart is diagonal, with entries

$$\partial_\theta^2(f\circ\Phi) = -(b + a\cos\psi)\sin\theta, \qquad \partial_\psi^2(f\circ\Phi) = -a\cos\psi\,\sin\theta$$

Evaluating gives the four signatures:

1. $(\frac{3\pi}{2}, 0)$, height $-(b+a)$: entries $(b+a,\ a)$, both positive, index 0, the bottom of the outer rim.
2. $(\frac{3\pi}{2}, \pi)$, height $-(b-a)$: entries $(b-a,\ -a)$, signs $(+,-)$, index 1, the bottom of the hole.
3. $(\frac{\pi}{2}, \pi)$, height $b-a$: entries $(-(b-a),\ a)$, signs $(-,+)$, index 1, the top of the hole.

4. $(\frac{\pi}{2}, 0)$, height $b+a$: entries $(-(b+a),\ -a)$, both negative, index 2, the top of the outer rim.

Every Hessian is nonsingular, so f is a Morse function, with indices $0, 1, 1, 2$ and

$$c_0 = 1, \qquad c_1 = 2, \qquad c_2 = 1$$

The alternating sum is $\sum_\lambda (-1)^\lambda c_\lambda = 1-2+1 = 0 = \chi(T^2)$, again matching the Euler characteristic; section 8.6 will show why.

Exercise 9.1.1

1. *A degenerate critical point.* Let $f\colon \mathbb{R}^2 \to \mathbb{R}$, $f(x,y) = x^3 - 3xy^2$ (the monkey saddle). Show that the origin is a critical point, compute $\operatorname{Hess}_0 f$, and conclude that f is not a Morse function.
2. *Extreme indices.* Using the Morse Lemma, show that a nondegenerate critical point of index 0 is a strict local minimum and one of index n is a strict local maximum. Conversely, show that a nondegenerate critical point at which f has a local minimum has index 0.
3. *Tilted height.* Fix $a \in \mathbb{R}^{n+1}$ with $a \neq 0$ and let $f_a\colon S^n \to \mathbb{R}$, $f_a(x) = \langle x, a\rangle$. Show that the critical points of f_a are $\pm a/|a|$, that both are nondegenerate, of indices 0 and n, and hence that f_a is Morse.
4. *Chart independence.* Let p be a critical point of f and let (U,φ), (U',ψ) be two charts around p. Show that the two Hessian matrices are congruent through the Jacobian of the transition map at p, and deduce that both nondegeneracy and the Morse index are independent of the chart.
5. *Max and min on a compact manifold.* Let f be a Morse function on a compact manifold M^n. Show that f attains its maximum at a critical point of index n and its minimum at a critical point of index 0, and deduce $c_0 \geq 1$ and $c_n \geq 1$.

9.2 Existence of Morse Functions

Section 9.1 produced Morse functions on the sphere and the standing torus by hand. Neither construction generalizes: a smooth function chosen at random need not be Morse, and on an arbitrary manifold there is no evident formula to write one down. The way past this is to embed the manifold in a Euclidean space (theorem 7.6.1) and measure squared distance to a point. These distance functions form an N-parameter family, and Sard's theorem (theorem 7.2.1) forces almost every member to be Morse. Every compact manifold therefore carries a Morse function, and the same mechanism shows the Morse functions are dense among all smooth functions.

Fix an embedded submanifold $M \subseteq \mathbb{R}^N$ of dimension n; by Whitney's theorem (theorem 7.6.1) every smooth manifold occurs this way. All inner products and lengths below are the standard ones on $\mathbb{R}^N$.

Definition 9.2.1: Distance-Square Function

For $p \in \mathbb{R}^N$ the *distance-square function* $f_p\colon M \to \mathbb{R}$ is the restriction to M of

$$f_p(x) = |x-p|^2 = \sum_{k=1}^{N}(x_k - p_k)^2$$

Each f_p is smooth, and on a compact M it attains a minimum and a maximum, so it always has at least two critical points. The question is whether those critical points, and any others, are nondegenerate. The first step is to locate them.

The critical points of f_p have a purely geometric description: they are the points of M seen from p along a normal direction.

Lemma 9.2.2: Critical Points of the Distance-Square Function

A point $q \in M$ is a critical point of f_p if and only if $p - q$ is orthogonal to T_qM. Equivalently, the critical points of f_p are exactly the points q for which $(q,\, p-q)$ lies in the normal bundle

$$\nu = \left\{(x, v) \in M \times \mathbb{R}^N \;\middle|\; v \perp T_xM\right\}$$

the realization of NM (proposition 7.7.2) by orthogonal complements.

***Proof*:**
Near q choose a local parametrization $\psi\colon \Omega \to \mathbb{R}^N$, a diffeomorphism from an open $\Omega \subseteq \mathbb{R}^n$ onto an open subset of M (the inverse of a chart), with $\psi(u) = q$. Its partial derivatives $\partial_i\psi := \partial\psi/\partial u^i$ span T_qM. Writing $g = f_p \circ \psi$,

$$\partial_i g(u) = 2\,\langle \psi(u) - p,\ \partial_i\psi(u)\rangle$$

which vanishes for all i precisely when $\psi(u) - p$ is orthogonal to $\operatorname{span}\{\partial_i\psi\} = T_qM$. Since q was arbitrary, q is a critical point of f_p if and only if $p - q \perp T_qM$. In that case $v := p - q$ lies in $(T_qM)^\perp$, so $(q, v) \in \nu$; conversely any $(q, v) \in \nu$ with $p = q + v$ makes $p - q = v$ normal, hence q critical, as we sought to show.

The orthogonal complement $(T_xM)^\perp \subseteq \mathbb{R}^N$ canonically represents the quotient $T_x\mathbb{R}^N/T_xM$, so ν is a smooth manifold of dimension $n + (N - n) = N$ and carries the normal-bundle structure of proposition 7.7.2. Its points package a point of M together with a normal direction, which is exactly the data of a critical point of some f_p.

Example 9.2.3: Distance-Square Functions on the Sphere
Take $M = S^n \subseteq \mathbb{R}^{n+1}$ and $p \neq 0$. On the sphere $|x| = 1$, so

$$f_p(x) = |x|^2 - 2\langle x, p\rangle + |p|^2 = 1 + |p|^2 - 2\langle x, p\rangle$$

which is an affine function of the height $\langle x, p\rangle$ in the direction p. The tangent space is $T_xS^n = x^\perp$, so lemma 9.2.2 makes q critical exactly when $p - q \in (q^\perp)^\perp = \mathbb{R}q$, that is $q = \pm p/|p|$. At the near point $q_+ = p/|p|$ the value is $(|p| - 1)^2$, a minimum, and at the far point $q_- = -p/|p|$ it is

$(|p|+1)^2$, a maximum. These are the two critical points of the height function of section 9.1, of indices 0 and n; both are nondegenerate, so f_p is Morse for every $p \neq 0$. The single excluded value $p = 0$ gives the constant $f_0 \equiv 1$, whose every point is a degenerate critical point.

The example shows the pattern in general. The distance-square function is Morse except when p is special, and the special p form a set of measure zero (for the sphere, the single centre). To make this precise the degeneracy of a critical point must be tied to a single map, whose critical values will be the bad p.

Definition 9.2.4: Endpoint Map

The *endpoint map* of $M \subseteq \mathbb{R}^N$ is

$$E\colon \nu \to \mathbb{R}^N, \qquad E(x,v) = x + v$$

from the normal bundle ν to the ambient space.

E is smooth, being the restriction to ν of vector addition $\mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}^N$. Its domain ν and its target $\mathbb{R}^N$ both have dimension N, so at each point E is either a local diffeomorphism or singular. By lemma 9.2.2 the critical points of f_p are exactly the points of $E^{-1}(p)$: a fibre of E over p records every point of M from which p lies in a normal direction. The next result identifies degeneracy with the singularities of E.

Lemma 9.2.5: Degeneracy and the Endpoint Map

Let $(q,v) \in \nu$ and $p = E(q,v) = q + v$, so that q is a critical point of f_p. Then q is a nondegenerate critical point of f_p if and only if (q,v) is a regular point of the endpoint map E. Moreover the nullity of $\operatorname{Hess}_q f_p$ equals the dimension of $\ker D_{(q,v)}E$.

***Proof*:**
Work in a local parametrization $\psi\colon \Omega \to \mathbb{R}^N$ with $\psi(u) = q$, and choose a smooth orthonormal frame $e_1(u), \dots, e_r(u)$ (with $r = N - n$) of the normal spaces $(T_{\psi(u)}M)^\perp$, which exists near u because ν is a smooth vector bundle (proposition 7.7.2). The maps $(u,t) \mapsto (\psi(u), \sum_a t_a e_a(u))$ are coordinates on ν near (q,v), in which

$$E(u,t) = \psi(u) + \sum_{a=1}^{r} t_a\, e_a(u)$$

with columns $\partial_i E = \partial_i \psi + \sum_a t_a\, \partial_i e_a$ and $\partial E/\partial t_a = e_a$.

Step 1: the Hessian in these terms. In the basis $\partial_1\psi, \dots, \partial_n\psi$ of T_qM the Hessian $\operatorname{Hess}_q f_p$ is represented by the matrix $\big(\partial_i\partial_j g\big)$, $g = f_p \circ \psi$, which is well defined because q is critical (section 9.1). Differentiating $\partial_i g = 2\langle \psi - p, \partial_i \psi\rangle$ gives

$$\tfrac{1}{2}\,\partial_i\partial_j g = \langle \partial_i\psi, \partial_j\psi\rangle + \langle \psi - p, \partial_i\partial_j\psi\rangle$$

At the critical point $\psi - p = -v = -\sum_a t_a e_a$, so the second term is $-\sum_a t_a \langle e_a, \partial_i\partial_j\psi\rangle$. Because

$\langle e_a, \partial_j\psi\rangle = 0$ identically, differentiating in u^i yields $\langle e_a, \partial_i\partial_j\psi\rangle = -\langle \partial_i e_a, \partial_j\psi\rangle$, and therefore

$$H_{ij} := \tfrac{1}{2}\,\partial_i\partial_j g = \Big\langle \partial_i\psi + \textstyle\sum_a t_a\,\partial_i e_a,\ \partial_j\psi \Big\rangle = \langle \partial_i E,\ \partial_j\psi\rangle$$

Thus $\mathrm{Hess}_q\, f_p = 2H$ with $H_{ij} = \langle \partial_i E, \partial_j\psi\rangle$.

Step 2: H singular if and only if DE singular. Suppose first $\xi \in \ker H$, $\xi \neq 0$, and set $w = \sum_i \xi_i\,\partial_i E$. Then $\langle w, \partial_j\psi\rangle = \sum_i \xi_i H_{ij} = 0$ for all j, so $w \perp T_qM$; hence $w = \sum_a \tau_a e_a$ for some τ, and $\sum_i \xi_i\,\partial_i E - \sum_a \tau_a e_a = 0$ is a nontrivial linear relation among the columns of DE, which is therefore singular. Conversely, suppose $\sum_i \xi_i\,\partial_i E + \sum_a \tau_a e_a = 0$ with $(\xi,\tau) \neq 0$. Pairing with $\partial_j\psi$ and using $\langle e_a, \partial_j\psi\rangle = 0$ gives $\sum_i \xi_i H_{ij} = 0$ for all j. If $\xi = 0$ then $\sum_a \tau_a e_a = 0$ with $\tau \neq 0$, impossible since the e_a are independent; so $\xi \neq 0$ lies in $\ker H$, and H is singular. The assignment $\xi \mapsto (\xi,\tau)$, with τ determined by $\sum_a \tau_a e_a = -\sum_i \xi_i \partial_i E$, is a linear isomorphism $\ker H \to \ker DE$, so the two nullities agree.

Since $\mathrm{Hess}_q\, f_p = 2H$, its nondegeneracy is equivalent to $\det H \neq 0$, hence to DE being nonsingular at (q,v), that is to (q,v) being a regular point of E; and the nullities coincide, completing the proof.

The two lemmas reduce the Morse condition to a single statement about the endpoint map: f_p is Morse exactly when every point of $E^{-1}(p)$ is a regular point of E, which is to say p is a regular value of E. Sard's theorem gives almost every such p.

Theorem 9.2.6: Existence of Morse Functions

Let $M \subseteq \mathbb{R}^N$ be an embedded submanifold. For almost every $p \in \mathbb{R}^N$ (every p outside a set of Lebesgue measure zero) the distance-square function $f_p\colon M \to \mathbb{R}$ is a Morse function. Consequently every compact smooth manifold admits a Morse function.

***Proof*:**
By lemma 9.2.2 the critical points of f_p are the points of $E^{-1}(p)$, and by lemma 9.2.5 such a critical point is degenerate if and only if E is singular there. Hence f_p is Morse if and only if no point of $E^{-1}(p)$ is a critical point of E, that is if and only if p is a regular value of $E\colon \nu \to \mathbb{R}^N$. The domain ν and target $\mathbb{R}^N$ both have dimension N, so Sard's theorem (theorem 7.2.1) makes the set of critical values of E measure zero; every p in its complement is a regular value, and for those p the function f_p is Morse.

For the second statement, let M be a compact smooth n-manifold. By Whitney's theorem (theorem 7.6.1) M is diffeomorphic to a submanifold of $\mathbb{R}^{2n+1}$; transport the smooth structure and apply the first part to obtain a regular value p of the corresponding endpoint map. Then f_p is Morse on M, so M carries a Morse function, completing the proof.

The critical points of f_p are the feet of the perpendiculars dropped from p to M, and a critical point degenerates exactly when p sits at a point where nearby normals focus. The critical values of E are, in classical terminology, the *focal points* of M: for the sphere they are the single centre, in agreement with example 9.2.3. Sard's theorem says the focal points are negligible, so a generic observation point

p sees M through nondegenerate critical points alone. The argument needs no metric on M and no analysis beyond Sard; the Riemannian structure it uses lives entirely in the ambient $\mathbb{R}^N$, imported by the embedding.

The same distance-square functions are proper on a closed embedded M, since a sublevel set $\{f_p \leq c\}$ is closed and bounded; for a noncompact M the proper embedding of theorem 7.6.1 therefore yields a *proper* Morse function, whose sublevel sets are compact. This is the property the fundamental theorems of the next section will use to run the gradient flow.

Existence is only the beginning: the construction already shows Morse functions are not rare but typical. Any smooth function becomes Morse after an arbitrarily small linear perturbation.

Proposition 9.2.7: Genericity of Morse Functions

Let $M \subseteq \mathbb{R}^N$ be an embedded submanifold and $f\colon M \to \mathbb{R}$ smooth. For almost every $a \in \mathbb{R}^N$ the function

$$f_a\colon M \to \mathbb{R}, \qquad f_a(x) = f(x) + \langle a, x\rangle$$

is Morse. If M is compact, the Morse functions form a dense subset of $C^\infty(M)$.

***Proof*:**
A function on $\mathbb{R}^n$ is Morse exactly when its gradient is transverse to $\{0\}$: the zeros of the gradient are the critical points, and the derivative of the gradient at a zero is the Hessian, which is onto $\mathbb{R}^n$ precisely when it is nonsingular. This local criterion drives the argument.

Step 1: a transverse family in one chart. Cover M by countably many local parametrizations $\psi\colon \Omega \to \mathbb{R}^N$, $\Omega \subseteq \mathbb{R}^n$ open, with images covering M; such a countable cover exists because M is second countable. Fix one and write $\tilde{f} = f \circ \psi$. In the coordinates u the perturbed function is $\tilde{f}(u) + \langle a, \psi(u)\rangle$, with gradient

$$\Phi(u, a) := \nabla \tilde{f}(u) + D\psi(u)^\top a \in \mathbb{R}^n$$

where $D\psi(u)$ is the $N \times n$ Jacobian. Regard $\Phi\colon \Omega \times \mathbb{R}^N \to \mathbb{R}^n$ as a smooth family (definition 7.5.5) with parameter a. Because ψ is an immersion, $D\psi(u)$ has rank n, so its transpose $D\psi(u)^\top\colon \mathbb{R}^N \to \mathbb{R}^n$ is surjective; hence $\partial\Phi/\partial a = D\psi(u)^\top$ is onto at every point, Φ is a submersion, and in particular $\Phi \pitchfork \{0\}$.

Step 2: a generic parameter works. By parametric transversality (theorem 7.5.6) applied to Φ and the point $\{0\} \subseteq \mathbb{R}^n$, for almost every a the map $\Phi(\cdot, a) \pitchfork \{0\}$. For such an a and any critical point u of f_a in this chart, $\Phi(u, a) = 0$ and $D_u\Phi(u, a)$ is onto, and $D_u\Phi(u, a) = \mathrm{Hess}(\tilde{f} + \langle a, \psi\rangle)(u)$ is exactly the coordinate Hessian of f_a at u; so every critical point of f_a in the chart is nondegenerate.

Step 3: assembling the charts. For each of the countably many charts the set of bad a has measure zero, so their union has measure zero. Any a outside it makes f_a Morse in every chart, hence at every critical point of M, so f_a is Morse. When M is compact it is bounded, so $x \mapsto \langle a, x\rangle$ is C^∞-small on M when $|a|$ is small; since the good set of a has full measure it meets every ball about 0, and choosing such an a makes f_a both Morse and arbitrarily close to f. Thus every smooth function is a limit of Morse functions, as we sought to show.

Taking $f \equiv 0$ recovers a fact worth isolating: on a compact $M \subseteq \mathbb{R}^N$ the linear height functions $x \mapsto \langle a, x \rangle$ are Morse for almost every direction a, which is why the ad hoc height functions of section 9.1 were Morse for the generic placement of the manifold. Genericity is the reason Morse theory applies to every manifold at once: the hypothesis "f is Morse" excludes nothing, since it can always be arranged by an unnoticeable perturbation.

9.2.8 Remark: Morse Functions Are Open On a compact manifold the Morse functions are not only dense but open in the C^2 topology, so "Morse" is a genuinely generic condition. A Morse function has finitely many critical points: they are isolated (section 9.1) and form a closed, hence compact, subset of M. Near each, the Hessian is nonsingular, an open condition that a C^2-small perturbation preserves while moving the critical point only slightly; away from the critical points df is bounded away from 0 by compactness, so a small perturbation creates no new ones. A nearby function is therefore again Morse.

Exercise 9.2.1

1. For $S^n \subseteq \mathbb{R}^{n+1}$ and a point $p \neq 0$, find all critical points of $f_p(x) = |x - p|^2$, compute their Morse indices, and confirm that f_p is Morse. What goes wrong when $p = 0$?
2. For $M = S^n$, show that the origin is a critical value of the endpoint map E by exhibiting, for each $x \in S^n$, a normal vector v with $E(x, v) = 0$. Explain how this is consistent with theorem 9.2.6.
3. Show that the normal bundle of $S^n \subseteq \mathbb{R}^{n+1}$ is $\nu = \{(x, tx) \mid x \in S^n,\ t \in \mathbb{R}\}$ and that $E(x, t) = (1 + t)x$. Deduce that the set of critical values of E is exactly $\{0\}$, and reconcile this with example 9.2.3.
4. Let $M \subseteq \mathbb{R}^N$ be compact. Using proposition 9.2.7, show that the height function $h_a(x) = \langle a, x \rangle$ is Morse on M for almost every $a \in \mathbb{R}^N$, and show that the critical points of h_a are exactly the points of M at which a is a normal vector.
5. Show that a Morse function on a compact manifold has only finitely many critical points, and conclude that the counts c_λ are all finite.

9.3 The Two Fundamental Theorems

The chapter opening promised that a Morse function's downhill flow builds the manifold one handle at a time. This section proves the two statements that make the promise precise. The first says that where f has no critical value the sublevel sets do not change topological type: M^a is a deformation retract of M^b. The second says that crossing a single critical value of index λ changes the homotopy type in exactly one way, by attaching a λ-cell. Both are driven by the flow of one vector field, a *pseudo-gradient* for f, which decreases f away from the critical points and near each of them looks like the downhill field of the Morse normal form. Constructing that field without a Riemannian metric comes first.

A Riemannian gradient would do the job: its flow moves in the direction of steepest descent and vanishes exactly at the critical points. This book develops no metric, so the gradient is replaced by any field with the two properties the arguments actually use, negativity of $X(f)$ off the critical set and a standard model at each critical point.

Definition 9.3.1: Pseudo-Gradient

Let $f\colon M \to \mathbb{R}$ be a Morse function. A *pseudo-gradient* for f is a smooth vector field X on M such that

1. $X(f) < 0$ at every noncritical point of f; and
2. around each critical point p there is a Morse chart $(u_1, \dots, u_n)$, with $u(p) = 0$ and

$$f = f(p) - \xi^2 + \eta^2, \qquad \xi^2 = u_1^2 + \cdots + u_\lambda^2, \quad \eta^2 = u_{\lambda+1}^2 + \cdots + u_n^2$$

($\lambda = \lambda(p)$ the Morse index), in which X is the model field

$$X = \sum_{i=1}^{\lambda} u_i \frac{\partial}{\partial u_i} - \sum_{i=\lambda+1}^{n} u_i \frac{\partial}{\partial u_i}$$

The two conditions carve out the flow's behaviour in the two regimes the theorems treat separately. Off the critical set condition (1) forces f strictly downhill along the flow, so a flow line crosses each regular level exactly once and can be used to slide one sublevel set onto a lower one. At a critical point condition (2) pins the field to the negative Euclidean gradient of the normal form: its linearization is the diagonal matrix with λ entries $+1$ and $n-\lambda$ entries -1, a nondegenerate rest point whose unstable directions are the λ "negative" coordinates ξ and whose stable directions are the $n-\lambda$ "positive" coordinates η. The index λ is thus visible in the flow as the dimension of the set of points that run away from p. Direct computation confirms the sign in (1) for the model: since $\partial f/\partial u_i = -2u_i$ for $i \le \lambda$ and $+2u_i$ for $i > \lambda$,

$$X(f) = \sum_{i=1}^{\lambda}(-2u_i)(u_i) + \sum_{i=\lambda+1}^{n}(2u_i)(-u_i) = -2\xi^2 - 2\eta^2 = -2|u|^2$$

which is negative except at $u = 0$, so the two conditions are compatible on the overlap of a Morse chart with the noncritical set.

The single critical point of index 1 shows the flow lines a pseudo-gradient produces.

Example 9.3.2: The Saddle Model

Take $f = c - u_1^2 + u_2^2$ on $\mathbb{R}^2$, a nondegenerate critical point of index 1 at the origin, and the model pseudo-gradient $X = u_1\,\partial_{u_1} - u_2\,\partial_{u_2}$. Its flow is

$$\psi_t(u_1, u_2) = (e^t u_1,\ e^{-t} u_2)$$

so along a flow line $f(\psi_t(u_1, u_2)) = c - e^{2t}u_1^2 + e^{-2t}u_2^2$, a strictly decreasing function of t off the origin. The u_1-axis is the unstable manifold, on which points flow out to infinity and $f \to -\infty$; the u_2-axis is the stable manifold, on which points flow into the origin and $f \downarrow c$; every other flow line is a hyperbola $u_1 u_2 = \text{const}$ that descends past the saddle. The one negative direction u_1 carries the index, matching $\lambda = 1$.

Conditions (1) and (2) can always be met at once. The construction glues local models with a partition of unity, arranged so that the model at each critical point is unchanged by the gluing.

Proposition 9.3.3: Existence of Pseudo-Gradients

Every Morse function $f\colon M \to \mathbb{R}$ on a compact manifold admits a pseudo-gradient.

***Proof*:**
Because M is compact and the critical points of a Morse function are isolated (section 9.1), the critical set $\{p_1, \dots, p_k\}$ is finite.

Step 1: Local fields. Around each p_j fix a Morse chart (U_j, u) and let X_j be the model field of definition 9.3.1(2), pulled back to U_j; there $X_j(f) = -2|u|^2$, negative off p_j. Shrinking the U_j, take them pairwise disjoint, and choose smaller open sets $U_j' \subseteq U_j$ with $p_j \in U_j'$ and $\overline{U_j'} \subseteq U_j$. The complement $K = M \setminus \bigcup_j U_j'$ is compact and contains no critical point. For each $q \in K$ choose a chart with coordinates $(y_1, \dots, y_n)$; since $df_q \neq 0$, some $\partial f/\partial y_i(q) \neq 0$, and the constant field

$$X_q = -\operatorname{sign}\left(\tfrac{\partial f}{\partial y_i}(q)\right) \frac{\partial}{\partial y_i}$$

has $X_q(f)(q) = -|\partial f/\partial y_i(q)| < 0$, hence $X_q(f) < 0$ on an open neighbourhood V_q of q. Shrinking V_q, arrange $V_q \cap \overline{U_j'} = \emptyset$ for every j, which is possible because $q \notin \overline{U_j'}$. Finitely many of the V_q cover the compact set K; call them $V_1, \dots, V_m$ with fields $X_1', \dots, X_m'$, each satisfying $X_\alpha'(f) < 0$ throughout V_α and $V_\alpha \cap \bigcup_j \overline{U_j'} = \emptyset$.

Step 2: Patching. The sets $\{U_j\}_j \cup \{V_\alpha\}_\alpha$ form a finite open cover of M. Let $\{\rho_j\}_j \cup \{\sigma_\alpha\}_\alpha$ be a smooth partition of unity subordinate to it (theorem 1.5.5), with $\operatorname{supp} \rho_j \subseteq U_j$ and $\operatorname{supp} \sigma_\alpha \subseteq V_\alpha$. Set

$$X = \sum_j \rho_j \, X_j + \sum_\alpha \sigma_\alpha \, X_\alpha'$$

a smooth vector field on M.

Step 3: Verification. On U_j' every σ_α vanishes (its support misses $\overline{U_j'}$) and every $\rho_{j'}$ with $j' \neq j$ vanishes (disjoint supports), so $\rho_j \equiv 1$ there and $X = X_j$ on U_j'. Thus X meets condition (2) in the chart U_j' at each p_j. For condition (1), take any noncritical q. Each summand contributes $\rho_j X_j(f) \leq 0$ (as $X_j(f) = -2|u|^2 \leq 0$) and $\sigma_\alpha X_\alpha'(f) \leq 0$, so $X(f)(q) \leq 0$; and at least one coefficient is positive at q with its field's f-derivative strictly negative there. Indeed, if $\sigma_\alpha(q) > 0$ then $q \in V_\alpha$ and $X_\alpha'(f)(q) < 0$; otherwise only some $\rho_j(q) > 0$, whence $q \in U_j$ with $q \neq p_j$ (as q is noncritical), and $X_j(f)(q) = -2|u(q)|^2 < 0$. Either way one summand is strictly negative and the rest are nonpositive, giving $X(f)(q) < 0$. Hence X is a pseudo-gradient, completing the proof.

The metric-free construction changes nothing downstream: the flow of X decreases f, its rest points are the critical points of f with the expected local dynamics, and that is all the fundamental theorems use. On $S^n \subseteq \mathbb{R}^{n+1}$ with $f(p) = p_{n+1}$ the tangential field $X(p) = p_{n+1}\, p - e_{n+1}$ already satisfies condition (1), with $X(f) = p_{n+1}^2 - 1 \leq 0$ vanishing only at the two poles; the existence proof adjusts it near the poles to the normal-form model, but the flow one pictures on the round sphere, sliding every non-pole point down toward the south pole, is exactly the pseudo-gradient flow.

The first fundamental theorem is now within reach. It isolates the principle behind the whole subject:

topology can only change where f has a critical value, so on any interval free of critical values the sublevel sets are interchangeable.

Theorem 9.3.4: Deformation Retraction Below a Regular Interval

Let $f\colon M \to \mathbb{R}$ be smooth on a compact manifold M, and let $a < b$ be such that $f^{-1}([a,b])$ contains no critical point of f. Then M^a is a strong deformation retract of M^b. In particular the inclusion $M^a \hookrightarrow M^b$ is a homotopy equivalence.

***Proof*:**
The statement asks only for a downhill flow that moves at unit speed across the slab $A = f^{-1}([a,b])$; a pseudo-gradient, renormalized, gives it.

Step 1: A unit-speed field. Let X be a pseudo-gradient for f (proposition 9.3.3 applies, as M is compact; when f is only assumed smooth, any field with $X(f) < 0$ on the critical-point-free set A serves, and one is built exactly as in Step 1 of that proof). The slab A is compact and disjoint from the closed critical set, so there is a smooth $\rho\colon M \to [0,1]$ with $\rho \equiv 1$ on A and $\operatorname{supp}\rho$ contained in the open set $M \setminus \operatorname{Crit}(f)$, on which $X(f) < 0$. Define

$$Y = -\frac{\rho}{X(f)}\,X$$

on $\operatorname{supp}\rho$ (where $X(f) < 0$ makes the coefficient smooth) and $Y = 0$ elsewhere; the two agree on the overlap, so Y is a smooth vector field on M. By construction $Y(f) = -\rho$, which equals -1 on A and lies in $[-1,0]$ everywhere. Since M is compact, Y is complete (corollary 5.8.4), with flow $\psi\colon \mathbb{R} \times M \to M$ (theorem 5.7.9).

Step 2: The retraction. Along a flow line, $\frac{d}{dt} f(\psi_t(q)) = Y(f)(\psi_t(q)) = -\rho(\psi_t(q)) \le 0$, so f is nonincreasing under the forward flow. If $f(q) = s \in [a,b]$, then as long as $\psi_t(q)$ stays in A the derivative is exactly -1, so $f(\psi_t(q)) = s - t$ for $t \in [0, s-a]$, and at $t = s - a$ the point reaches level a. Define $R\colon [0,1] \times M^b \to M$ by

$$R(\tau, q) = \begin{cases} q, & f(q) \le a \\ \psi\big(\tau(f(q) - a),\, q\big), & a \le f(q) \le b \end{cases}$$

The two formulas agree where $f(q) = a$ (there $\tau(f(q) - a) = 0$ and $\psi_0 = \mathrm{id}$), so R is continuous. For $a \le f(q) \le b$ one has $f(R(\tau,q)) = f(q) - \tau(f(q) - a) \le f(q) \le b$, so R maps into M^b; for $f(q) \le a$ it fixes $q \in M^a$. Thus $R(0,\cdot) = \mathrm{id}_{M^b}$, each $R(\tau,\cdot)$ restricts to the identity on M^a, and $R(1,q) = \psi_{f(q)-a}(q) \in f^{-1}(a) \subseteq M^a$ when $f(q) \ge a$ while $R(1,q) = q$ when $f(q) \le a$. Hence $R(1,\cdot)$ retracts M^b onto M^a and R is a strong deformation retraction, as we sought to show.

The theorem is the reason a single number, the value of f, organizes the topology. As a increases through an interval of regular values the space M^a grows but its homotopy type is frozen; it can jump only when a passes a critical value. Below the minimum $M^a = \emptyset$, and once $a \ge \max f$ one has $M^a = M$, so the entire manifold is assembled by watching what happens at the finitely many critical levels in between. The same flow gives slightly more than homotopy equivalence.

9.3.5 Remark: The Product Structure of a Regular Slab Under the hypotheses of theorem 9.3.4, if in addition a and b are regular values, then $f^{-1}([a,b])$ is diffeomorphic to the product $f^{-1}(a) \times [a,b]$. The unit-speed field Y from Step 2 has $Y(f) = -1$ on the slab, so $f(\psi_{-t}(x)) = a + t$ for $x \in f^{-1}(a)$ and $t \in [0, b-a]$; the map $(x,t) \mapsto \psi_{-t}(x)$ is then a diffeomorphism $f^{-1}(a) \times [0, b-a] \to f^{-1}([a,b])$, with smooth inverse $y \mapsto (\psi_{f(y)-a}(y),\ f(y)-a)$, and $f^{-1}(a)$ is a manifold by corollary 3.2.10. This presents M^b as M^a with an external collar glued along its boundary $f^{-1}(a)$; absorbing the collar (theorem 1.2.20) makes M^b diffeomorphic to M^a. Crossing no critical value thus changes the sublevel set neither up to homotopy nor up to diffeomorphism.

The height function on the sphere shows the theorem and its limit in one picture.

Example 9.3.6: Sublevel Sets of the Height Function on S^n

Let $f(p) = p_{n+1}$ on S^n, with critical points the south pole S (index 0, value -1) and north pole N (index n, value $+1$). For $-1 < a < b < 1$ the slab $f^{-1}([a,b])$ contains no critical point, so theorem 9.3.4 makes M^a a deformation retract of M^b: every sublevel set strictly between the poles is a closed n-disk (the spherical cap $\{p_{n+1} \le a\}$), and they are all diffeomorphic by the remark above. The homotopy type jumps only at the two poles: $M^a = \emptyset$ for $a < -1$, a point appears at $a = -1$, the cap persists up to $a = 1$, and at the last instant the top cell caps it off to the whole sphere.

At a critical value the sublevel set does change, and the second fundamental theorem says precisely how: it grows by a cell whose dimension is the Morse index. This is the mechanism that reads a cell structure off the critical points.

Theorem 9.3.7: Cell Attachment at a Critical Point

Let $f\colon M \to \mathbb{R}$ be a Morse function on a compact manifold, and let p be a critical point of index λ with $f(p) = c$, the only critical point on the level c. Then for all sufficiently small $\varepsilon > 0$ there is a map $\varphi\colon \partial D^\lambda \to M^{c-\varepsilon}$ such that $M^{c+\varepsilon}$ has the homotopy type of $M^{c-\varepsilon}$ with a λ-cell attached along φ:

$$M^{c+\varepsilon} \simeq M^{c-\varepsilon} \cup_\varphi e^\lambda$$

***Proof*:**

The idea is to push a dimple into f over the critical point so that the critical value drops below $c - \varepsilon$; theorem 9.3.4 then retracts $M^{c+\varepsilon}$ onto the modified sublevel set, which is $M^{c-\varepsilon}$ together with the desired cell.

Step 1: The Morse chart and the cell. By the Morse Lemma (theorem 9.1.6) fix a chart $(u_1, \ldots, u_n)$ on a neighbourhood U of p, with $u(p) = 0$ and

$$f = c - \xi^2 + \eta^2, \qquad \xi^2 = u_1^2 + \cdots + u_\lambda^2, \quad \eta^2 = u_{\lambda+1}^2 + \cdots + u_n^2$$

Choose $\varepsilon > 0$ so small that (i) $f^{-1}([c-\varepsilon, c+\varepsilon])$ is compact and p is its only critical point, and (ii) the closed ellipsoid $E = \{\xi^2 + 2\eta^2 \le 2\varepsilon\}$ lies in $u(U)$. Condition (i) is available because the critical points are finite (section 9.1), hence the critical values other than c stay a positive distance from c. Set

$$e^\lambda = \{u \in U \mid \eta = 0,\ \xi^2 \le \varepsilon\}$$

a smooth λ-disk on which $f = c - \xi^2 \in [c-\varepsilon, c]$. Its boundary $\partial e^\lambda = \{\eta = 0,\ \xi^2 = \varepsilon\}$ lies in $f^{-1}(c-\varepsilon) \subseteq M^{c-\varepsilon}$, while its interior has $f > c - \varepsilon$; so e^λ is a cell attached to $M^{c-\varepsilon}$ along the sphere ∂e^λ, and φ will be that inclusion.

Step 2: The modified function. Pick a smooth $\mu\colon [0,\infty) \to \mathbb{R}$ with $\mu(0) > \varepsilon$, with $\mu(t) = 0$ for $t \ge 2\varepsilon$, and with $-1 < \mu'(t) \le 0$ throughout. Define $F\colon M \to \mathbb{R}$ by $F = f$ off E and

$$F = f - \mu(\xi^2 + 2\eta^2)$$

on U; the two formulas agree where $\xi^2 + 2\eta^2 \ge 2\varepsilon$ (there $\mu = 0$), and E is compact in U, so F is smooth on M and equals f outside the compact set E.

F has the same sublevel set at $c{+}\varepsilon$. Since $\mu \ge 0$, $F \le f$, so $\{f \le c{+}\varepsilon\} \subseteq \{F \le c{+}\varepsilon\}$. Conversely, off E equality $F = f$ gives the reverse inclusion, while inside E one has $\eta^2 \le \frac{1}{2}(\xi^2{+}2\eta^2) \le \varepsilon$, hence $f = c{-}\xi^2{+}\eta^2 \le c{+}\eta^2 \le c{+}\varepsilon$ automatically. Therefore $F^{-1}(-\infty, c{+}\varepsilon] = f^{-1}(-\infty, c{+}\varepsilon] = M^{c+\varepsilon}$.

F has the same critical points. Off E, $F = f$. Inside U, using $\partial(\xi^2 + 2\eta^2)/\partial u_i = 2u_i$ for $i \le \lambda$ and $4u_i$ for $i > \lambda$,

$$\frac{\partial F}{\partial u_i} = -2(1+\mu')\,u_i\ (i \le \lambda), \qquad \frac{\partial F}{\partial u_i} = 2(1-2\mu')\,u_i\ (i > \lambda)$$

and $1 + \mu' > 0$, $1 - 2\mu' \ge 1 > 0$, so dF vanishes only at $u = 0$. Thus p is the sole critical point of F in U, and $F(p) = c - \mu(0) < c - \varepsilon$.

Step 3: Retract onto the lower modified sublevel set. The set $F^{-1}([c-\varepsilon, c+\varepsilon])$ is compact (closed in $M^{c+\varepsilon}$) and contains no critical point of F: the only candidate is p, and $F(p) < c - \varepsilon$. Applying theorem 9.3.4 to F with $a = c - \varepsilon$, $b = c + \varepsilon$ shows that $F^{-1}(-\infty, c-\varepsilon]$ is a deformation retract of $F^{-1}(-\infty, c+\varepsilon] = M^{c+\varepsilon}$. Write

$$F^{-1}(-\infty, c-\varepsilon] = M^{c-\varepsilon} \cup H, \qquad H = \{x \mid F(x) \le c - \varepsilon < f(x)\}$$

Since $F < f$ only where $\mu > 0$, that is inside the open ellipsoid $\{\xi^2 + 2\eta^2 < 2\varepsilon\}$, the handle H is contained in that ellipsoid. The cell lies in the lower set: on e^λ one has $\eta = 0$ and $F = c - (\xi^2 + \mu(\xi^2))$, and $g(t) = t + \mu(t)$ has $g(0) = \mu(0) > \varepsilon$ and $g'(t) = 1 + \mu'(t) > 0$, so $g \ge g(0) > \varepsilon$ and $F < c - \varepsilon$ on all of e^λ. Thus $M^{c-\varepsilon} \cup e^\lambda \subseteq F^{-1}(-\infty, c-\varepsilon]$.

Step 4: Retract the handle onto the cell. It remains to deformation retract $M^{c-\varepsilon} \cup H$ onto $M^{c-\varepsilon} \cup e^\lambda$; composing with Step 3 then finishes the proof. Define $G\colon [0,1] \times (M^{c-\varepsilon} \cup H) \to M^{c-\varepsilon} \cup H$ as the identity outside the chart U, and inside U in three regions of (ξ, η), in each case scaling only the η-coordinate:

1. $\xi^2 \le \varepsilon$: put $G(\tau, u) = (\xi,\ (1-\tau)\eta)$.
2. $\varepsilon \le \xi^2$ and $\eta^2 \ge \xi^2 - \varepsilon$: put $G(\tau, u) = (\xi,\ \kappa_\tau\,\eta)$ with $\kappa_\tau = (1-\tau) + \tau\sqrt{(\xi^2-\varepsilon)/\eta^2}$.
3. $\eta^2 \le \xi^2 - \varepsilon$: put $G(\tau, u) = u$.

The formulas patch continuously. On $\xi^2 = \varepsilon$ the square root vanishes, so region (2) gives $(\xi, (1-\tau)\eta)$, matching region (1). On $\eta^2 = \xi^2 - \varepsilon$ the square root is 1, so $\kappa_\tau = 1$ and region (2) is the identity, matching region (3). Where regions (1) and (3) overlap one has $\eta = 0$ and $\xi^2 = \varepsilon$, and both give $(\xi, 0)$. Every point actually moved lies inside the open ellipsoid: a moved point of H has $\mu > 0$, hence $\xi^2 + 2\eta^2 < 2\varepsilon$, and the only moved points of $M^{c-\varepsilon}$ inside U would need $f \leq c - \varepsilon$ with $\eta \neq 0$ and $\eta^2 > \xi^2 - \varepsilon$, which is impossible. So G is the identity near ∂U and glues with the identity on the rest of M.

The homotopy stays in the set. Scaling η down decreases η^2, hence decreases $f = c - \xi^2 + \eta^2$, and increases $\mu(\xi^2 + 2\eta^2)$ (as μ is nonincreasing), so it decreases $F = f - \mu$; starting from $F \leq c - \varepsilon$, each track stays in $F^{-1}(-\infty, c-\varepsilon]$. At $\tau = 1$: region (1) sends u to $(\xi, 0)$ with $\xi^2 \leq \varepsilon$, a point of e^λ; region (2) sends u to a point with $\eta^2 = \xi^2 - \varepsilon$, hence $f = c - \varepsilon$, a point of $M^{c-\varepsilon}$; region (3) fixes points of $M^{c-\varepsilon}$. Thus $G(1, \cdot)$ maps $M^{c-\varepsilon} \cup H$ into $M^{c-\varepsilon} \cup e^\lambda$. It fixes that subset pointwise: e^λ meets U in region (1) with $\eta = 0$, where G is the identity, and the points of $M^{c-\varepsilon}$ inside U lie in region (3) or on its boundary, again fixed. Hence G is a strong deformation retraction of $M^{c-\varepsilon} \cup H$ onto $M^{c-\varepsilon} \cup e^\lambda$.

Composing the retraction of Step 3 with G deformation retracts $M^{c+\varepsilon}$ onto $M^{c-\varepsilon} \cup e^\lambda$, so $M^{c+\varepsilon} \simeq M^{c-\varepsilon} \cup_\varphi e^\lambda$ with $\varphi\colon \partial e^\lambda \hookrightarrow M^{c-\varepsilon}$, completing the proof.

The two extremes of the index show the statement. When $\lambda = 0$ the boundary ∂D^0 is empty, so attaching a 0-cell means adding a disjoint point: crossing an index-0 critical value gives birth to a new component, appearing as an n-disk that then grows. When $\lambda = n$ the cell is top-dimensional and its attachment fills a spherical hole; crossing an index-n value caps off a boundary sphere. Between these, an index-λ critical point threads a λ-cell through the manifold along the λ negative directions of its Hessian, and the attaching map records how the boundary sphere of that cell sits in the sublevel set already built.

Two critical points may share a level. The proof adapts verbatim.

9.3.8 Remark: Several Critical Points on One Level If the level c carries several critical points $p_1, \ldots, p_r$ of indices $\lambda_1, \ldots, \lambda_r$, choose disjoint Morse charts and a single ε small enough for all of them. The modified function F dimples each chart independently, its handles $H_1, \ldots, H_r$ are disjoint, and the region-by-region retraction of theorem 9.3.7 runs simultaneously in each chart. The conclusion becomes

$$M^{c+\varepsilon} \simeq M^{c-\varepsilon} \cup_{\varphi_1} e^{\lambda_1} \cup \cdots \cup_{\varphi_r} e^{\lambda_r}$$

so all the cells at one level attach at once. One may also separate the critical values: replacing f by $f + \sum_j \varepsilon_j \chi_j$, with χ_j a bump equal to 1 near p_j and supported in a small ball there, shifts $f(p_j)$ to $f(p_j) + \varepsilon_j$ without moving the critical point or changing its Hessian, and for small ε_j introduces no new critical points; a generic choice of the ε_j makes all critical values distinct. Nothing is lost, then, by assuming one critical point per level when convenient.

Running the two theorems from the bottom up recovers a cell structure on the sphere and on the torus, the base cases behind the general decomposition.

Example 9.3.9: Building S^n and the Torus from Critical Points

For the height function on S^n (example 9.3.6) there are two critical values. Passing the minimum at -1 (index 0) attaches a 0-cell to the empty set, giving a point; the point thickens to a disk $M^a \simeq \mathrm{pt}$ as a climbs the regular interval $(-1, 1)$, by theorem 9.3.4. Passing the maximum at $+1$ (index n) attaches an n-cell along $\varphi\colon \partial D^n = S^{n-1} \to \mathrm{pt}$, the constant map. Hence

$$S^n \simeq \mathrm{pt} \cup_\varphi e^n = D^n/\partial D^n$$

the standard two-cell structure, one 0-cell and one n-cell.

Stand the torus T^2 on end in $\mathbb{R}^3$ and take the height function, with critical values at a minimum (index 0), two saddles (index 1), and a maximum (index 2). The sublevel sets pass, in order, through: the empty set; a point, growing to a disk; a cylinder after the first saddle attaches a 1-cell to the disk; a space homotopy equivalent to a wedge $S^1 \vee S^1$ after the second saddle attaches a second 1-cell; and finally T^2 itself after the maximum attaches the 2-cell. The counts $c_0 = 1$, $c_1 = 2$, $c_2 = 1$ are the numbers of cells in each dimension.

Exercise 9.3.1

1. Stand the torus T^2 upright in $\mathbb{R}^3$ and let f be the height function, with critical values $a_0 < a_1 < a_2 < a_3$ of indices $0, 1, 1, 2$. For a value t in each of the five ranges (below a_0; between a_0 and a_1; between a_1 and a_2; between a_2 and a_3; above a_3), give the homotopy type of the sublevel set M^t, citing theorem 9.3.4 and theorem 9.3.7 at each step.

2. On $S^n \subseteq \mathbb{R}^{n+1}$ with $f(p) = p_{n+1}$, consider the tangential field $X(p) = p_{n+1}\, p - e_{n+1}$. Verify that $X(p) \in T_pS^n$, that $X(f) = p_{n+1}^2 - 1 \leq 0$ with equality only at the poles, and by linearizing at each pole identify the Morse indices 0 and n. Why does X fail, as written, to be a pseudo-gradient in the strict sense of definition 9.3.1, and what does the existence proof do about it?

3. Using the models near a critical point, prove directly that crossing an index-0 critical value of a Morse function on a compact n-manifold adds a disjoint closed n-disk to the sublevel set (a new component is born), and that crossing an index-n value attaches an n-cell along a map $S^{n-1} \to M^{c-\varepsilon}$ that caps off a boundary sphere. Deduce that just above its minimum value, $M^{c+\varepsilon}$ is a disjoint union of n-disks, one per index-0 critical point on the lowest level.

4. Let $a < b$ be regular values of a Morse function f on a compact manifold with no critical value in $[a, b]$. Using the unit-speed field of theorem 9.3.4, show that $f^{-1}([a, b]) \cong f^{-1}(a) \times [a, b]$, and deduce by absorbing the collar (theorem 1.2.20) that M^b is diffeomorphic, not just homotopy equivalent, to M^a. Show also that $M^t = M$ for every $t \geq \max f$.

5. Let f be a Morse function on a compact connected manifold M^n. Show that f attains a minimum and a maximum, that these are nondegenerate critical points of indices 0 and n respectively, and hence that $c_0 \geq 1$ and $c_n \geq 1$.

6. The height function on the circle $S^1 \subseteq \mathbb{R}^2$ has one minimum (index 0) and one maximum (index 1). Trace the two fundamental theorems to obtain $S^1 \simeq e^0 \cup_\varphi e^1$, and describe the attaching map $\varphi\colon \partial D^1 = S^0 \to \{\mathrm{pt}\}$. Compare with the standard CW structure of the circle.

9.4 Handle and CW Decompositions

Section 9.3 isolated two local moves. Across an interval of regular values the sublevel set does not change up to diffeomorphism (theorem 9.3.4), and across a single critical value of index λ it acquires a λ-cell (theorem 9.3.7). Sweeping f from its minimum to its maximum and applying these moves in turn builds M one piece at a time, a single piece for each critical point. This section makes that assembly precise in both of its registers: the smooth picture, in which a *handle* $D^\lambda \times D^{n-\lambda}$ is glued on at each critical point, and its homotopy counterpart, in which a cell of dimension λ is attached. Two payoffs follow at once. A manifold carrying a Morse function with only two critical points is a sphere, and complex projective space inherits its standard cell structure from a single explicit height function.

Throughout, M is a compact boundaryless manifold of dimension n and $f\colon M \to \mathbb{R}$ is a Morse function, with c_λ critical points of index λ.

The smooth object attached at a critical point is a thickened cell. The λ-cell of theorem 9.3.7 is the descending disk of the critical point, the set of points that flow down away from it; thickening it in the complementary directions produces a piece of the manifold with a standard product shape.

Definition 9.4.1: Handle

A *λ-handle* of dimension n is a copy of the product $D^\lambda \times D^{n-\lambda}$ of closed disks, where $0 \le \lambda \le n$. Its constituents are

- the *core* $D^\lambda \times \{0\}$ and the *cocore* $\{0\} \times D^{n-\lambda}$;
- the *attaching region* $\partial D^\lambda \times D^{n-\lambda} = S^{\lambda-1} \times D^{n-\lambda}$ and, inside it, the *attaching sphere* $S^{\lambda-1} \times \{0\}$;
- the *belt sphere* $\{0\} \times S^{n-\lambda-1}$.

To *attach a λ-handle* to a compact n-manifold-with-boundary W along a smooth embedding $\varphi\colon S^{\lambda-1} \times D^{n-\lambda} \to \partial W$ is to form the quotient

$$W \cup_\varphi (D^\lambda \times D^{n-\lambda})$$

in which each point of the attaching region is identified with its image under φ, and then to smooth the resulting corner along $\varphi(S^{\lambda-1} \times \{0\})$. The number λ is the *index* of the handle.

The index of a handle is the dimension of its core, and it will match the Morse index of the critical point that produces it. A 0-handle is a disk D^n attached along the empty set $S^{-1} \times D^n = \emptyset$, so it simply starts a new component; an n-handle is a disk D^n attached along its entire boundary sphere $S^{n-1} \times \{0\}$, so it caps off a boundary component. The intermediate handles interpolate between these extremes.

Example 9.4.2: Handles on Surfaces

For $n = 2$ there are three handle types.

1. A 0-*handle* $D^0 \times D^2 = D^2$ is a disk, attached along the empty set; it begins a surface.
2. A 1-*handle* $D^1 \times D^1$ is a band, attached along $S^0 \times D^1$, a pair of arcs on the existing

boundary; gluing it connects two boundary arcs, adding a strip.

3. A 2-*handle* $D^2 \times D^0 = D^2$ is a disk, attached along $S^1 \times \{0\}$, an entire boundary circle; it caps a hole.

Stand the torus T^2 upright in $\mathbb{R}^3$ and take the height function, whose critical points have indices $0, 1, 1, 2$ (a bottom minimum, two saddles, a top maximum). Building T^2 by increasing height attaches, in order, one 0-handle (a disk at the bottom), two 1-handles (a band at each saddle), and one 2-handle (a disk capping the top). The two bands, glued to the boundary circle of the first disk, produce a surface with one boundary circle onto which the final disk fits, closing up the torus.

The claim implicit in the surface example is that passing a critical value of index λ attaches a λ-handle, not a λ-cell alone. This is the smooth strengthening of theorem 9.3.7.

Theorem 9.4.3: Handle Attachment

Let c be a critical value of f whose only critical point is p, of index λ, and let $\varepsilon > 0$ be small enough that $[c-\varepsilon, c+\varepsilon]$ contains no other critical value. Then $M^{c+\varepsilon}$ is diffeomorphic to $M^{c-\varepsilon}$ with a λ-handle attached,

$$M^{c+\varepsilon} \cong M^{c-\varepsilon} \cup_\varphi (D^\lambda \times D^{n-\lambda})$$

along an embedding $\varphi\colon S^{\lambda-1} \times D^{n-\lambda} \to \partial M^{c-\varepsilon}$ whose core $D^\lambda \times \{0\}$ is the descending disk of p.

***Proof*:**
Since p is the only critical point on $f^{-1}(c)$ and critical points are isolated, shrink ε so that $f^{-1}[c-\varepsilon, c+\varepsilon]$ is compact with p its only critical point. By the Morse Lemma (theorem 9.1.6) there are coordinates $(u, v) = (x_1, \ldots, x_\lambda, x_{\lambda+1}, \ldots, x_n)$ centered at p with

$$f = c - |u|^2 + |v|^2$$

on a ball $B = \{|u|^2 + |v|^2 \le 2\varepsilon\}$; shrink ε once more so that B lies inside the chart.

Step 1: The descending disk and its handle. The *descending disk* $D_- = \{v = 0,\ |u| \le \sqrt{\varepsilon}\}$ is a smoothly embedded λ-disk on which $f = c - |u|^2$, so $f \le c$ on D_- with $f = c - \varepsilon$ exactly on $\partial D_- = \{v = 0,\ |u| = \sqrt{\varepsilon}\}$. Hence $\partial D_- \subseteq f^{-1}(c-\varepsilon) = \partial M^{c-\varepsilon}$, while the open disk $D_- \setminus \partial D_-$ lies above $M^{c-\varepsilon}$ inside $M^{c+\varepsilon}$. The disk D_- is contractible, so its normal bundle in $M^{c+\varepsilon}$ is trivial; by the tubular neighbourhood theorem (theorem 7.8.1) it has a closed neighbourhood

$$H \cong D^\lambda \times D^{n-\lambda}, \qquad D_- \cong D^\lambda \times \{0\}$$

meeting $\partial M^{c-\varepsilon}$ in a tubular neighbourhood of ∂D_-, namely the attaching region $\varphi(S^{\lambda-1} \times D^{n-\lambda})$ with φ the induced embedding into $\partial M^{c-\varepsilon}$.

Step 2: Filling in the complement. On $f^{-1}[c-\varepsilon, c+\varepsilon]$ minus a neighbourhood of p there are no critical points, so a pseudo-gradient for f (adapted near p to the coordinates above) is nonvanishing there, and its downward flow carries $M^{c+\varepsilon}$ minus the open handle H diffeomorphically onto $M^{c-\varepsilon}$, fixing $\partial M^{c-\varepsilon}$ away from the attaching region; this is the deformation of theorem 9.3.4 run in the complement of the handle. Reassembling the two pieces exhibits $M^{c+\varepsilon}$ as $M^{c-\varepsilon}$ with H glued along φ. Rounding the corner along $\varphi(S^{\lambda-1} \times \{0\})$, the standard smoothing of a manifold-with-corners, restores a smooth boundary $f^{-1}(c+\varepsilon)$ and gives the stated diffeomorphism, as we sought to show.

The core of the attached handle is exactly the λ-cell of theorem 9.3.7: collapsing the disk factor $D^{n-\lambda}$ to a point retracts H onto its core, which recovers the homotopy statement from the smooth one. The two theorems are the same geometric event told at two resolutions, and the finer smooth version is what a handle decomposition records.

Corollary 9.4.4: Handle Decomposition

Every compact manifold admits a handle decomposition: it is built from the empty set by attaching finitely many handles, one λ-handle for each critical point of index λ of any Morse function on it.

***Proof*:**
A Morse function exists (section 9.2) and, being Morse on a compact manifold, has finitely many critical points. Order the critical values $c_1 < \cdots < c_k$ and choose regular values $a_0 < c_1 < a_1 < \cdots < c_k < a_k$, so that $M^{a_0} = \emptyset$ and $M^{a_k} = M$. Passing from $M^{a_{i-1}}$ to M^{a_i} crosses the single value c_i; if the critical points at that level are $p_1, \dots, p_m$ with indices $\lambda_1, \dots, \lambda_m$, then theorem 9.4.3, applied to disjoint charts about the p_j, attaches an m-tuple of handles of those indices simultaneously. After k stages every critical point has contributed exactly one handle of its index, and $M^{a_k} = M$, completing the proof.

The homotopy consequence is cleaner still, because at the level of homotopy type a handle is indistinguishable from its core cell, and the attaching data may be simplified freely. The result is that a Morse function equips M with the homotopy type of a CW complex whose cells are indexed by the critical points.

Theorem 9.4.5: Morse CW Structure

Let f be a Morse function on the compact manifold M, with c_λ critical points of index λ. Then M has the homotopy type of a CW complex with exactly c_λ cells of dimension λ, one for each critical point of index λ.

***Proof*:**
Step 1: Separate the critical values. A small modification makes the critical values distinct without changing the critical points or their indices: for each critical point p_i choose a bump function β_i equal to 1 near p_i and supported in a chart on which p_i is the only critical point, and replace f by $f + \sum_i t_i \beta_i$ with the t_i generic and small. Near p_i this adds a constant, preserving the critical point and its Hessian; on the compact region where $d\beta_i \neq 0$ the gradient of f is bounded away from 0, so for $|t_i|$ small no new critical points appear. Choose the t_i so that the values $f(p_i) + t_i$ are pairwise distinct.

Step 2: Induction over critical values. Order the critical values $c_1 < \cdots < c_k$, each carrying one critical point p_i of index λ_i, and pick regular values $a_0 < c_1 < a_1 < \cdots < c_k < a_k$ with $M^{a_0} = \emptyset$,

$M^{a_k} = M$. The claim is that for each i there is a CW complex K_i with one λ_j-cell for each $j \le i$ and a homotopy equivalence $M^{a_i} \simeq K_i$. The base case $K_0 = \emptyset$ is immediate. Assume $M^{a_{i-1}} \simeq K_{i-1}$ via a homotopy equivalence h. By theorem 9.3.7, crossing c_i attaches a λ_i-cell,

$$M^{a_i} \simeq M^{a_{i-1}} \cup_\varphi e^{\lambda_i}, \qquad \varphi\colon S^{\lambda_i - 1} \to M^{a_{i-1}}$$

Two facts from homotopy theory ([11]) finish the step. First, attaching a cell is invariant under homotopy equivalence of the base and under homotopy of the attaching map: the equivalence h carries $M^{a_{i-1}} \cup_\varphi e^{\lambda_i}$ to $K_{i-1} \cup_{h\varphi} e^{\lambda_i}$ up to homotopy equivalence. Second, cellular approximation homotopes $h\varphi$ to a cellular map $\psi\colon S^{\lambda_i - 1} \to K_{i-1}$, and homotopic attaching maps yield homotopy-equivalent adjunction spaces, so

$$M^{a_i} \simeq K_{i-1} \cup_\psi e^{\lambda_i} =: K_i$$

a genuine CW complex with one new λ_i-cell.

Step 3: Conclusion. Compactness bounds the number of critical points, so the induction terminates at $i = k$ with $M = M^{a_k} \simeq K_k$, a finite CW complex carrying exactly c_λ cells of dimension λ, one per index-λ critical point, completing the proof.

The theorem converts the analytic census of critical points into a topological model of M. Every homotopy invariant of M can now be computed from a CW complex with c_λ cells in dimension λ; the Euler characteristic and the Betti numbers in particular are constrained by the counts c_λ, and section 9.5 extracts those constraints. The immediate use is negative: a Morse function with few critical points forces M to be simple. The extreme case, two critical points, forces a sphere.

Theorem 9.4.6: Reeb's Theorem

If a compact manifold M of dimension n admits a Morse function with exactly two critical points, then M is homeomorphic to S^n.

***Proof*:**
Let f have exactly two critical points. Since M is compact, f attains a minimum at some p_0 (value a) and a maximum at some p_1 (value b), both critical; as there are only two critical points, these are they, and $a < b$. A nondegenerate minimum has index 0 and a nondegenerate maximum has index n, so p_0 has index 0 and p_1 has index n.

Step 1: The sublevel and superlevel sets are disks. By the Morse Lemma (theorem 9.1.6) at p_0 there are coordinates with $f = a + |x|^2$, so for small $\varepsilon > 0$ the set $\{f \le a + \varepsilon\}$ is the coordinate ball $\{|x|^2 \le \varepsilon\}$. Because a is the global minimum, attained only at the isolated p_0, for small ε the whole sublevel set $M^{a+\varepsilon}$ is this ball, so $M^{a+\varepsilon} \cong D^n$. Applying the same argument to $-f$ at p_1 gives $\{f \ge b - \varepsilon\} \cong D^n$.

Step 2: Glue the two disks. The values $a + \varepsilon$ and $b - \varepsilon$ are both regular and the interval $[a + \varepsilon, b - \varepsilon]$ contains no critical value, so the regular-slab product structure (box 9.3.5) gives a diffeomorphism $M^{b-\varepsilon} \cong M^{a+\varepsilon} \cong D^n$ carrying boundary to boundary. Thus M is the union of

the two closed n-disks
$$D_- = M^{b-\varepsilon}, \qquad D_+ = \{f \geq b - \varepsilon\}$$
glued along their common boundary $f^{-1}(b-\varepsilon) \cong S^{n-1}$. Under the identifications $D_\pm \cong D^n$ this gluing is a homeomorphism $g\colon S^{n-1} \to S^{n-1}$ of boundary spheres.

Step 3: Two disks glued give a sphere. Extend g radially to $\bar{g}\colon D^n \to D^n$ by $\bar{g}(x) = |x|\, g(x/|x|)$ for $x \neq 0$ and $\bar{g}(0) = 0$; this is a homeomorphism (the Alexander trick). Writing the standard sphere as the union $S^n = D^n_- \cup_{\mathrm{id}} D^n_+$ of two hemispheres glued by the identity on the equator, define $\Phi\colon M \to S^n$ to be the identity on D_+ and $\bar{g}$ on D_-. On the shared boundary the two prescriptions agree, since $\bar{g}$ restricts to g there, so Φ is a well-defined continuous bijection of compact Hausdorff spaces, hence a homeomorphism. Therefore $M \cong S^n$, as we sought to show.

The conclusion is a homeomorphism, not a diffeomorphism, and the gap is real. The radial extension $\bar{g}$ need not be smooth at the origin, so the gluing may produce a smooth structure on S^n that is not the standard one.

9.4.7 Remark: Exotic Spheres A manifold built from two disks glued along their boundary is a *twisted sphere*, always homeomorphic to S^n by theorem 9.4.6 but not always diffeomorphic to it. John Milnor's 1956 construction of smooth 7-manifolds that carry Morse functions with exactly two critical points, yet are not diffeomorphic to the standard S^7, exhibits exactly this phenomenon and opened the study of exotic smooth structures. Reeb's theorem is therefore sharp: the critical-point count controls the homeomorphism type of M but not its diffeomorphism type.

Compactness entered the CW structure theorem only to keep the number of critical points finite and to run the gradient flow. A proper function bounded below has compact sublevel sets, so both hold locally, one critical level at a time, and the same construction then reads a cell structure off a possibly infinite set of critical points.

Theorem 9.4.8: Morse CW Structure for a Proper Function

Let f be a Morse function on a manifold M that is proper and bounded below, with c_λ critical points of index λ. Then M has the homotopy type of a CW complex with exactly c_λ cells of dimension λ, one for each critical point of index λ.

***Proof*:**
The two fundamental theorems used to build the CW structure need compactness only of the slabs they act on, and properness makes those slabs compact. For regular values $a < b$ the slab $f^{-1}([a,b])$ is compact (proposition 3.2.20), so when it contains no critical point the downhill flow of theorem 9.3.4 stays inside it and exists for the finite time the retraction needs even where M is not compact, giving the deformation retraction $M^a \hookrightarrow M^b$. The local cell attachment of theorem 9.3.7 likewise takes place in a compact slab $f^{-1}([c-\varepsilon, c+\varepsilon])$ about a critical value, compact by properness as well. Properness also controls the critical set: each compact sublevel set $f^{-1}([\inf f, b])$ meets it in finitely many isolated nondegenerate points, so the critical values form a discrete set, bounded below, with finitely many below any level. Two cases arise.

Finitely many critical values $c_1 < \cdots < c_k$. Pick a regular value a exceeding every critical value. No critical value lies above a, so the downhill flow retracts M onto M^a: a point with $f \geq a$ reaches level a in finite time inside the compact slab between the two levels, so M^a is a deformation retract of M (equal to M when M is compact, where the maximum is the top critical value). The sublevel set $M^a = f^{-1}([\inf f, a])$ is compact, so the induction of theorem 9.4.5 applies to it verbatim and produces a finite CW complex K with one λ-cell for each index-λ critical point and a homotopy equivalence $M^a \simeq K$. Hence $M \simeq M^a \simeq K$.

Infinitely many critical values. Now $c_1 < c_2 < \cdots$ with $c_i \to \infty$, since they cannot accumulate, and regular values may be chosen with $a_0 < c_1 < a_1 < c_2 < \cdots$ and $a_i \to \infty$. Then $M = \bigcup_i M^{a_i}$ with $M^{a_i} \subseteq \operatorname{int} M^{a_{i+1}}$ (as $a_i < a_{i+1}$), so M carries the colimit topology of this exhaustion, and $M \times I$ that of the $M^{a_i} \times I$. The induction of theorem 9.4.5 gives finite complexes $K_0 \subseteq K_1 \subseteq \cdots$, with K_i obtained from K_{i-1} by attaching one λ_j-cell for each index-λ_j critical point at level $c_j \leq c_i$, together with homotopy equivalences $h_i\colon M^{a_i} \to K_i$. By that construction h_{i+1} restricts on M^{a_i} to a map homotopic to h_i, the two differing only over the cell attached in crossing c_{i+1}.

Each level $f^{-1}(a_i)$ is regular, so a critical-point-free slab $f^{-1}([a_i, a_i+\delta])$ gives a collar $f^{-1}(a_i) \times [a_i, a_i+\delta]$ of M^{a_i} in $M^{a_{i+1}}$ (box 9.3.5). The collar makes $(M^{a_{i+1}} \times \{0\}) \cup (M^{a_i} \times I)$ a retract of $M^{a_{i+1}} \times I$, so the pair $(M^{a_{i+1}}, M^{a_i})$ has the homotopy extension property ([11, Condition For Homotopy Extension]), and the CW pair (K_{i+1}, K_i) has it as well ([11, Homotopy Extension For CW Pair]). Homotopy extension along the first inclusion replaces h_{i+1}, within its homotopy class, by one with $h_{i+1}|_{M^{a_i}} = h_i$ exactly, so the h_i assemble into a single map $h\colon M \to K = \bigcup_i K_i$.

To see that h is a homotopy equivalence, factor it as the inclusion of M into its mapping cylinder $W = (M \times I \sqcup K)/\big((x,1) \sim h(x)\big)$ followed by the collapse $\rho\colon W \to K$ onto the far end, a deformation retraction, so that $W \simeq K$. Since $h|_{M^{a_i}} = h_i$, the cylinder is the increasing union $W = \bigcup_i W_i$ of the mapping cylinders $W_i = (M^{a_i} \times I \sqcup K_i)/\sim$, and W carries the colimit topology of this exhaustion, being a quotient of $M \times I \sqcup K$. Each $W_i \hookrightarrow W_{i+1}$ is a cofibration, glued from the two cofibrations of the previous paragraph, and the near-end inclusion $M^{a_i} \hookrightarrow W_i$ is a cofibration whose composite with $\rho|_{W_i}\colon W_i \to K_i$ is h_i. Being a homotopy equivalence, it exhibits M^{a_i} as a deformation retract of W_i ([11, homotopy extension and deformation Retraction]).

These retractions can be chosen to nest, which is what pins the retracting homotopies together across stages. Suppose $r_i\colon W_i \times I \to W_i$ retracts W_i onto M^{a_i} rel M^{a_i}, reaching M^{a_i} at time $1 - 2^{-i-1}$ and resting there afterwards. The inclusion of $W_i \cup M^{a_{i+1}}$ into W_{i+1} is again a cofibration, so the homotopy equal to r_i on W_i and constant on $M^{a_{i+1}}$ (consistent on the overlap M^{a_i}, where r_i rests) extends over W_{i+1} from the identity by homotopy extension. Following it on $[1-2^{-i-1}, 1-2^{-i-2}]$ by a deformation retraction of W_{i+1} onto $M^{a_{i+1}}$ rel $M^{a_{i+1}}$ ([11, homotopy extension and deformation Retraction] again) yields r_{i+1}, which fixes $M^{a_{i+1}}$, rests there after time $1 - 2^{-i-2}$, and restricts to r_i on $W_i \times I$ exactly, the added stage fixing $M^{a_i} \subseteq M^{a_{i+1}}$. By the colimit topology of $W \times I$ the r_i assemble into one deformation retraction of W onto $M = \bigcup_i M^{a_i}$: a point of W_i has been carried into M^{a_i} by time $1 - 2^{-i-1}$ and is fixed by every later stage. Hence $M \simeq W \simeq K$, a CW complex with exactly c_λ cells of dimension λ. In both cases M has the homotopy type of such a complex, completing the proof.

The positive use of the CW structure is to read a manifold's cells off an explicit function. The model computation is complex projective space, where a single height function is Morse with one critical point in each even index and none in odd degrees.

Example 9.4.9: A Morse Function on $\mathcal{P}^n$

Fix distinct weights $0, 1, \ldots, n$ and define $f\colon \mathcal{P}^n \to \mathbb{R}$ by

$$f([z_0 : \cdots : z_n]) = \frac{\sum_{j=0}^{n} j\,|z_j|^2}{\sum_{j=0}^{n} |z_j|^2}$$

which is well-defined because numerator and denominator scale the same way under $z \mapsto \lambda z$.

Critical points. Work in the affine chart $U_k = \{z_k \neq 0\}$ with coordinates $w_j = z_j/z_k$ for $j \neq k$. Setting $S = \sum_{j\neq k} |w_j|^2$,

$$f = \frac{k + \sum_{j\neq k} j\,|w_j|^2}{1+S}$$

Differentiating with the Wirtinger derivative $\partial/\partial\bar{w}_j$ and using $f(1+S) = k + \sum_{j\neq k} j|w_j|^2$,

$$\frac{\partial f}{\partial \bar{w}_j} = \frac{(j-f)\,w_j}{1+S}$$

This vanishes for every $j \neq k$ only if each $w_j = 0$ or $f = j$. Distinct j cannot both satisfy $f = j$, so at most one w_{j_0} is nonzero; but with only slots k and j_0 active, f lies strictly between k and j_0 and never equals j_0. Hence all $w_j = 0$: the only critical point in U_k is $P_k = [0 : \cdots : 1 : \cdots : 0]$. As the charts U_k cover $\mathcal{P}^n$, the critical points are exactly $P_0, \ldots, P_n$.

Indices. Near P_k expand f in the coordinates w_j. With $s = \sum_{j\neq k} |w_j|^2$,

$$f = \Big(k + \sum_{j\neq k} j|w_j|^2\Big)(1 - s + O(s^2)) = k + \sum_{j\neq k}(j-k)\,|w_j|^2 + O(|w|^4)$$

Writing $w_j = x_j + iy_j$, the quadratic part is $\sum_{j\neq k}(j-k)(x_j^2 + y_j^2)$, a nondegenerate form whose negative directions are the two real coordinates x_j, y_j for each $j < k$. There are k such indices j, so the form has $2k$ negative directions and P_k is a nondegenerate critical point of index $2k$.

Thus f is a Morse function with one critical point of index 2λ for each $\lambda = 0, 1, \ldots, n$ and none of odd index. By theorem 9.4.5, $\mathcal{P}^n$ has the homotopy type of a CW complex with exactly one cell in each even dimension $0, 2, 4, \ldots, 2n$ and no odd-dimensional cells, which is its standard cell structure.

Because $\mathcal{P}^n$ has no two critical points of adjacent index, every cell is attached to a lower skeleton by a map that cannot cancel it, and the CW structure is as economical as the topology permits. Section 9.5 formalizes this as the statement that such a Morse function is *perfect*.

Handles of adjacent index can sometimes be slid apart and cancelled, which is how a handle decomposition is simplified. The main theorem of that theory deserves a statement, though its proof is beyond this book.

9.4.10 Remark: The h-Cobordism Theorem Two closed manifolds M_0, M_1 are *h-cobordant* if there is a compact manifold W with $\partial W = M_0 \sqcup M_1$ for which both inclusions $M_i \hookrightarrow W$ are homotopy equivalences. The *h-cobordism theorem* of Smale states that if such a W is simply connected of dimension $n \geq 6$, then W is diffeomorphic to the cylinder $M_0 \times [0,1]$; in particular $M_0 \cong M_1$. The proof is a handle calculus: one uses a Morse function on W without boundary-critical points, then cancels its handles in pairs by the Whitney trick, which requires the simple connectivity and the dimension bound. A consequence is the generalized Poincaré conjecture in high dimensions, that a smooth manifold homotopy equivalent to S^n $(n \geq 5)$ is homeomorphic to S^n. The Whitney-trick argument is research-level and lies outside this series (no book develops handle-trading or the h-cobordism theorem; recorded gap). It is stated here only to place the handle decomposition in its larger setting.

Exercise 9.4.1

1. The height function $f(x) = x_{n+1}$ on $S^n \subseteq \mathbb{R}^{n+1}$ has exactly two critical points. Identify them and their indices, describe the resulting handle decomposition of S^n, and read off its CW structure. Reconcile the answer with theorem 9.4.6.
2. Specialize example 9.4.9 to $n = 1$. Show that the Morse function on $\mathcal{P}^1$ has exactly two critical points, of indices 0 and 2, and conclude from theorem 9.4.6 that $\mathcal{P}^1$ is homeomorphic to S^2.
3. On $\mathbb{RP}^n$ define $g([x_0 : \cdots : x_n]) = \big(\sum_j j\, x_j^2\big)/\big(\sum_j x_j^2\big)$ with real homogeneous coordinates. Show that g is Morse with critical points the coordinate points P_k, that P_k has index k, and deduce that $\mathbb{RP}^n$ has a CW structure with exactly one cell in each dimension $0, 1, \ldots, n$.
4. Using theorem 9.4.5 and the fact that the Euler characteristic of a finite CW complex is the alternating sum of its cell counts ([11, Euler's Characteristic]), compute $\chi(\mathcal{P}^n)$ and $\chi(\mathbb{RP}^n)$.
5. Give the handle decomposition and the CW structure of T^2 produced by the height function of example 9.4.2 (indices $0, 1, 1, 2$), and check that the alternating cell count gives $\chi(T^2) = 0$.
6. Let M be a compact connected n-manifold and f a Morse function with exactly two critical points. Prove directly, without invoking theorem 9.4.6, that the two indices must be 0 and n.

9.5 The Morse Inequalities

The handle and CW decompositions of section 9.4 record a Morse function's critical points as cells, one cell of dimension λ for each critical point of index λ. This section reads off the numerical consequences. The alternating sum $\sum_\lambda (-1)^\lambda c_\lambda$ equals the Euler characteristic, whatever Morse function produced the counts c_λ; and each individual count is bounded below by a Betti number, with a sharper family of alternating bounds tying the two sequences together. The first statement is an equality, the Morse-Poincaré identity; the second is the pair of Morse inequalities.

Throughout, M is a compact manifold without boundary of dimension n, f a Morse function on it, and c_λ the number of critical points of index λ. Since f is Morse its critical points are isolated (section 9.1) and M is compact, so each c_λ is finite and $c_\lambda = 0$ for $\lambda > n$.

The Euler characteristic already appeared in section 8.6 as the sum of the indices of a vector field. A Morse function comes with a distinguished vector field, the pseudo-gradient of section 9.3, whose zeros are exactly the critical points of f. Feeding it into Poincaré-Hopf turns the index sum into a count of critical points, weighted by a sign that the Morse Lemma pins down.

Theorem 9.5.1: The Morse-Poincaré Identity

Let f be a Morse function on a compact n-manifold M without boundary. Then

$$\sum_{\lambda=0}^{n}(-1)^{\lambda}c_{\lambda}=\chi(M)$$

where $\chi(M)$ is the Euler characteristic of M.

***Proof*:**
Let X be a pseudo-gradient for f (section 9.3), so $X(f)<0$ away from the critical points and $X=0$ exactly at them. Set $V=-X$, the ascending field, which has the same zeros as X, namely the critical points of f, and these are finite in number.

Step 1: the local model. Fix a critical point p of index λ. By the Morse Lemma (theorem 9.1.6) there are coordinates $(x_1,\dots,x_n)$ centred at p in which

$$f=c-x_1^2-\cdots-x_\lambda^2+x_{\lambda+1}^2+\cdots+x_n^2$$

In these coordinates the pseudo-gradient is modelled on the linear field

$$X_0=\sum_{i\le\lambda}x_i\,\partial_{x_i}-\sum_{i>\lambda}x_i\,\partial_{x_i}$$

which indeed satisfies $X_0(f)=-2(x_1^2+\cdots+x_n^2)<0$ off the origin, matching the pseudo-gradient sign convention. Thus near p the ascending field $V=-X$ is modelled on $-X_0$, whose linearization at the origin is the diagonal matrix with λ entries equal to -1 and $n-\lambda$ entries equal to $+1$.

Step 2: the local index. The origin is a nondegenerate zero of $-X_0$, since $\det(D(-X_0)_0)=(-1)^{\lambda}\cdot 1^{n-\lambda}\neq 0$. The index of a vector field at a nondegenerate zero is the sign of the determinant of its linearization (section 8.6), so

$$\operatorname{ind}_p V=\operatorname{sign}\det(D(-X_0)_0)=(-1)^{\lambda}$$

Each critical point of index λ therefore contributes $(-1)^{\lambda}$ to the index sum of V.

Step 3: summation. The zeros of V are the critical points of f, all nondegenerate, and M is compact, so Poincaré-Hopf (theorem 8.6.7) applies:

$$\chi(M)=\sum_{p}\operatorname{ind}_p V=\sum_{\lambda=0}^{n}(-1)^{\lambda}c_{\lambda}$$

the middle sum ranging over the zeros p of V, grouped by index in the last equality. This is the claimed identity, completing the proof.

The identity says the Euler characteristic is computable from any Morse function by a single signed count, with no reference to homology. It reproves, in Morse language, that χ is a topological invariant: the left side is manifestly integer-valued and additive in the critical data, the right side manifestly independent of f, and the theorem forces the two to agree. The same content reappears as the top equality of the strong inequalities below, where $\lambda = n$ collapses the alternating bounds to this equation. Two elementary Morse functions already exercise it.

Example 9.5.2: The Euler Characteristic of Spheres and the Torus

The height function on $S^n \subseteq \mathbb{R}^{n+1}$ has exactly two critical points, a minimum of index 0 (the south pole) and a maximum of index n (the north pole), so $c_0 = c_n = 1$ and all other counts vanish. The identity gives

$$\chi(S^n) = (-1)^0 + (-1)^n = 1 + (-1)^n$$

equal to 2 when n is even and 0 when n is odd, matching the count from section 8.6.

The standing torus $T^2 \subseteq \mathbb{R}^3$, resting on a plane with the height function, has one minimum (the bottom, index 0), two saddles (the top of the hole and the point where the inner wall meets the plane, both index 1), and one maximum (the top, index 2): $c_0 = 1$, $c_1 = 2$, $c_2 = 1$. Then

$$\chi(T^2) = 1 - 2 + 1 = 0$$

as expected for a closed orientable surface of genus one.

The identity constrains only the alternating sum; it says nothing about the individual counts. The height function on S^2 can be perturbed to grow extra critical points, always in pairs, keeping $1 - 2 + 1 = 0$ unchanged. What bounds each c_λ from below is the homology of M: a class in $H_\lambda(M)$ must be built from a λ-cell, hence from a critical point of index λ, so c_λ cannot fall below $b_\lambda = \operatorname{rank} H_\lambda(M;\mathbb{Z})$, the λ-th Betti number ([11, Betti Number And Torsion Coefficients]). Making this precise, and sharpening it, is an exercise in linear algebra on the cellular chain complex.

The mechanism is entirely formal, so it is worth isolating it from the manifold theory. The CW structure of section 9.4 produces a chain complex of vector spaces, one basis vector per critical point; the following lemma extracts the inequalities from any such complex.

Lemma 9.5.3: Morse Inequalities for a Chain Complex

Let $\mathbb{F}$ be a field and let

$$0 \to C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_1} C_0 \to 0$$

be a chain complex of finite-dimensional $\mathbb{F}$-vector spaces. Write $c_\lambda = \dim_{\mathbb{F}} C_\lambda$ and $b_\lambda = \dim_{\mathbb{F}} H_\lambda(C_\bullet)$. Then for every $\lambda \geq 0$,

$$\sum_{i=0}^{\lambda} (-1)^{\lambda - i} c_i \geq \sum_{i=0}^{\lambda} (-1)^{\lambda - i} b_i$$

with equality when $\lambda = n$, and in particular $c_\lambda \geq b_\lambda$ for every λ.

***Proof*:**
Let $r_\lambda = \text{rank}\partial_\lambda = \dim_{\mathbb{F}} \text{im}\partial_\lambda$, with the conventions $r_0 = 0$ and $r_{n+1} = 0$ (there is no ∂_0 and no ∂_{n+1}). Rank-nullity applied to $\partial_\lambda \colon C_\lambda \to C_{\lambda-1}$ gives $\dim\ker\partial_\lambda = c_\lambda - r_\lambda$, and since $H_\lambda = \ker\partial_\lambda/\text{im}\partial_{\lambda+1}$,

$$b_\lambda = (c_\lambda - r_\lambda) - r_{\lambda+1}$$

Rearranged, $c_\lambda - b_\lambda = r_\lambda + r_{\lambda+1} \geq 0$, which is the weak inequality $c_\lambda \geq b_\lambda$.

For the alternating sums, substitute this expression and telescope:

$$\sum_{i=0}^{\lambda}(-1)^{\lambda-i}(c_i - b_i) = \sum_{i=0}^{\lambda}(-1)^{\lambda-i}(r_i + r_{i+1})$$

In the right-hand sum the coefficient of r_i for $1 \leq i \leq \lambda$ is $(-1)^{\lambda-i} + (-1)^{\lambda-(i-1)} = 0$, the term r_0 vanishes, and the only surviving term is $r_{\lambda+1}$ from $i = \lambda$. Hence

$$\sum_{i=0}^{\lambda}(-1)^{\lambda-i}c_i - \sum_{i=0}^{\lambda}(-1)^{\lambda-i}b_i = r_{\lambda+1} \geq 0$$

which is the stated inequality. When $\lambda = n$ the surplus r_{n+1} is 0, so the two alternating sums are equal, as we sought to show.

Applying the lemma to the cellular chain complex of the CW structure turns these into statements about M.

Theorem 9.5.4: The Morse Inequalities

Let f be a Morse function on a compact n-manifold M without boundary, with c_λ critical points of index λ, and let $b_\lambda = \text{rank}H_\lambda(M;\mathbb{Z})$ be the Betti numbers. Then:

1. (*Weak inequalities*) $c_\lambda \geq b_\lambda$ for every $\lambda \geq 0$.
2. (*Strong inequalities*) for every $\lambda \geq 0$,

$$c_\lambda - c_{\lambda-1} + \cdots + (-1)^\lambda c_0 \geq b_\lambda - b_{\lambda-1} + \cdots + (-1)^\lambda b_0$$

with equality at $\lambda = n$, which reproduces $\sum_\lambda(-1)^\lambda c_\lambda = \chi(M)$.

***Proof*:**
By theorem 9.4.5 the manifold M has the homotopy type of a CW complex K with exactly c_λ cells of dimension λ. Its cellular chain complex with rational coefficients,

$$C_\lambda = H_\lambda(K^{(\lambda)}, K^{(\lambda-1)};\mathbb{Q}) \cong \mathbb{Q}^{c_\lambda}$$

is a chain complex of finite-dimensional $\mathbb{Q}$-vector spaces with $\dim_{\mathbb{Q}} C_\lambda = c_\lambda$, and its homology is the singular homology of K, hence of M (cellular homology computes singular homology, and singular homology is a homotopy invariant, [11, Cellular Homology, Homotopy Equivalence, Then Isomorphism]). By the universal coefficient theorem $\dim_{\mathbb{Q}} H_\lambda(M;\mathbb{Q}) = \text{rank}H_\lambda(M;\mathbb{Z}) = b_\lambda$ ([11, Universal Coefficient Theorem For Homology]). Applying lemma 9.5.3 with $\mathbb{F} = \mathbb{Q}$ to this complex yields both families of inequalities, and the case $\lambda = n$ gives $\sum_\lambda(-1)^\lambda c_\lambda =$

$\sum_\lambda (-1)^\lambda b_\lambda = \chi(M)$, the last equality being the homological formula for the Euler characteristic ([11, Euler's Characteristic]). This completes the proof.

Summing the weak inequalities, the total number of critical points of any Morse function is at least $\sum_\lambda b_\lambda$, so a manifold with large homology forces many critical points regardless of the function chosen. The excess is governed by the boundary maps: from the proof, $c_\lambda - b_\lambda = r_\lambda + r_{\lambda+1}$, and summing over λ gives $\sum_\lambda c_\lambda - \sum_\lambda b_\lambda = 2\sum_\lambda r_\lambda$, an even number. Geometrically, the surplus critical points come in cancelling pairs of consecutive index, one creating a cell that the next kills homologically, exactly the two extra critical points a dimple adds to the height function on S^2.

The argument runs over any field, not only $\mathbb{Q}$. With $\mathbb{F} = \mathbb{F}_p$ the Betti numbers b_λ are replaced by $\dim_{\mathbb{F}_p} H_\lambda(M;\mathbb{F}_p)$, which can be strictly larger when $H_*(M;\mathbb{Z})$ has p-torsion, giving inequalities the rational ones miss. The real projective plane is the standard instance: $H_1(\mathbb{RP}^2;\mathbb{Z}) = \mathbb{Z}/2$ contributes nothing to the rational Betti numbers but makes $\dim_{\mathbb{F}_2} H_\lambda(\mathbb{RP}^2;\mathbb{F}_2) = 1$ in each degree $\lambda = 0, 1, 2$, so over $\mathbb{F}_2$ every Morse function on $\mathbb{RP}^2$ has at least one critical point of each index.

When the weak inequalities are equalities the function is as economical as its topology allows, and it computes the homology outright.

Definition 9.5.5: Perfect Morse Function

A Morse function f on M is *perfect* over a field $\mathbb{F}$ if $c_\lambda = \dim_{\mathbb{F}} H_\lambda(M;\mathbb{F})$ for every λ, that is, if the weak Morse inequalities over $\mathbb{F}$ are all equalities. When $\mathbb{F}$ is understood, or when f is perfect over every field, f is called *perfect*.

By the identity $c_\lambda - b_\lambda = r_\lambda + r_{\lambda+1}$, perfection over $\mathbb{F}$ is equivalent to $r_\lambda = \operatorname{rank}\partial_\lambda = 0$ for all λ, that is, to the vanishing of every cellular boundary map: the cellular complex has zero differential, so $H_\lambda(M;\mathbb{F}) = C_\lambda$ and the Betti numbers are read straight off the critical points. Not every manifold admits a perfect Morse function over $\mathbb{Z}$ (torsion obstructs it, as $\mathbb{RP}^2$ shows), but a simple index condition guarantees perfection when it holds.

Proposition 9.5.6: The Lacunary Principle

If a Morse function f has no two critical points of consecutive index, that is, if for every λ at least one of c_λ, $c_{\lambda+1}$ is zero, then f is perfect over every field, and $b_\lambda = c_\lambda$ for all λ. In particular this holds whenever all critical points have even index.

***Proof*:**
Work over an arbitrary field $\mathbb{F}$ and consider the cellular boundary $\partial_\lambda \colon C_\lambda \to C_{\lambda-1}$, where $\dim C_\lambda = c_\lambda$. If $c_\lambda = 0$ then $C_\lambda = 0$ and $\partial_\lambda = 0$; if $c_{\lambda-1} = 0$ then $C_{\lambda-1} = 0$ and again $\partial_\lambda = 0$. The hypothesis makes one of these hold for every λ, so every boundary map vanishes, $H_\lambda(M;\mathbb{F}) = C_\lambda$, and $b_\lambda = \dim_{\mathbb{F}} C_\lambda = c_\lambda$. Since $\mathbb{F}$ was arbitrary, f is perfect over every field. When all critical points have even index, no two indices are consecutive, so the hypothesis holds, completing the proof.

The name is Morse's: a sequence with gaps is *lacunary*, and a lacunary set of indices leaves the boundary maps no room to act. The principle converts a purely geometric fact, that the critical points sit in non-adjacent dimensions, into a complete homology computation. Complex projective space is the model case.

Example 9.5.7: The Betti Numbers of Complex Projective Space

The Morse function on $\mathcal{P}^n$ constructed in section 9.4 has exactly one critical point of index $2k$ for each $k = 0, 1, \ldots, n$, and none of odd index: $c_{2k} = 1$ for $0 \le k \le n$ and $c_\lambda = 0$ for λ odd. All indices are even, so the Lacunary Principle (proposition 9.5.6) applies and the function is perfect. Reading off,

$$b_\lambda = \dim H_\lambda(\mathcal{P}^n; \mathbb{Z}) = \begin{cases} 1 & \lambda = 0, 2, 4, \ldots, 2n \\ 0 & \text{otherwise} \end{cases}$$

so the Poincaré polynomial is $\sum_\lambda b_\lambda t^\lambda = 1 + t^2 + t^4 + \cdots + t^{2n}$. The Euler characteristic is $\chi(\mathcal{P}^n) = \sum_\lambda (-1)^\lambda b_\lambda = n + 1$, the number of critical points, each of even index and so counted with sign $+1$. The homology of $\mathcal{P}^n$ is thereby computed from a single height-type function, without a chain-level boundary calculation.

The same reasoning gives the homology of any manifold carrying a Morse function whose indices skip, among them the even-dimensional complex Grassmannians and, over $\mathbb{F}_2$, the real projective spaces. Where the indices are forced to be consecutive, as for a surface of genus $g \ge 1$, the inequalities remain strict lower bounds and the boundary maps carry genuine information.

Exercise 9.5.1

1. Let Σ_g be the closed orientable surface of genus g, with $H_0 = H_2 = \mathbb{Z}$ and $H_1 = \mathbb{Z}^{2g}$. Suppose a Morse function on Σ_g has a single minimum and a single maximum. Using theorem 9.5.4 and theorem 9.5.1, show that it has exactly $2g$ critical points of index 1, and that in general any Morse function on Σ_g has at least $2g$ index-1 critical points.
2. Show directly from the strong Morse inequalities that the weak inequalities follow, by adding the strong inequality at index λ to the one at index $\lambda - 1$. Deduce also that the strong inequalities contain the Morse-Poincaré identity.
3. For $n \ge 2$, use the height function on S^n and proposition 9.5.6 to compute the Betti numbers of S^n. Explain why the argument fails for $n = 1$, and check by hand that the height function on S^1 is nonetheless perfect.
4. The mod-2 homology of $\mathbb{R}\mathrm{P}^2$ is $H_\lambda(\mathbb{R}\mathrm{P}^2; \mathbb{F}_2) = \mathbb{F}_2$ for $\lambda = 0, 1, 2$. Show that every Morse function on $\mathbb{R}\mathrm{P}^2$ has at least three critical points, and that the rational Morse inequalities alone give only the bound "at least one". Reconcile the two counts using $\chi(\mathbb{R}\mathrm{P}^2) = 1$.
5. Prove that for any Morse function the total surplus $\sum_\lambda c_\lambda - \sum_\lambda b_\lambda$ is a non-negative even integer, and interpret the value 0.
6. Define the Morse number $\gamma(M)$ to be the minimum, over all Morse functions on M, of the total number of critical points. Show $\gamma(M) \ge \sum_\lambda \dim_{\mathbb{F}} H_\lambda(M; \mathbb{F})$ for every field $\mathbb{F}$, that $\gamma(S^n) = 2$, and that $\gamma(\mathcal{P}^n) = n + 1$.

9.6 Morse Homology

The Morse inequalities of section 9.5 bound the Betti numbers below the critical-point counts but stop short of computing the homology. This section produces the homology itself from the same data. The index-λ critical points of a Morse function freely generate an abelian group C_λ; a boundary operator $\partial\colon C_\lambda \to C_{\lambda-1}$ counts, with signs, the gradient trajectories running from a critical point to one of index one lower; and the homology of the resulting complex is the homology of M. Read against this, the weak Morse inequality $c_\lambda \geq b_\lambda$ becomes the statement that a free complex has homology of rank at most its number of generators.

Making the count well defined is the work of the section. The trajectories between two critical points form a manifold whose dimension is the difference of their indices, provided the descending and ascending sets meet transversally; this is the Morse-Smale condition, and it holds after an arbitrarily small perturbation of the pseudo-gradient. When the index difference is one the trajectory count is finite; when it is two the trajectories form a compact 1-manifold whose ends are the broken trajectories, and the resulting even boundary count is exactly $\partial^2 = 0$. We close by identifying the complex with the cellular complex of the handle decomposition, which forces its homology to be that of M, and by pointing to the infinite-dimensional analogue that motivated the whole construction.

Throughout, M is compact without boundary, f is a Morse function, and X is the pseudo-gradient of section 9.3: a vector field with $X(f) < 0$ off the critical points which, in a Morse chart around each critical point p of index λ (theorem 9.1.6), coincides with the linear field

$$\xi(x) = (x_1, \ldots, x_\lambda, -x_{\lambda+1}, \ldots, -x_n) \tag{9.1}$$

in the coordinates $(x_1, \ldots, x_n)$ that put f in normal form $f = f(p) - x_1^2 - \cdots - x_\lambda^2 + x_{\lambda+1}^2 + \cdots + x_n^2$. A direct check gives $\xi(f) = -2(x_1^2 + \cdots + x_n^2)$, negative off the origin, so ξ is a legitimate local pseudo-gradient. Since M is compact, X is complete and its flow φ_t (theorem 5.7.9) is defined for all time.

The starting observation is that every trajectory of the flow limits onto critical points. Along a non-constant trajectory $f(\varphi_t(q))$ is strictly decreasing and bounded, so its ω-limit set (the set of subsequential limits of $\varphi_t(q)$ as $t \to +\infty$) is a nonempty, flow-invariant subset of a single level set on which f is constant; a flow-invariant set on which f is constant consists of critical points, and as these are isolated (theorem 9.1.6) the limit is a single critical point. The same holds as $t \to -\infty$. Each trajectory therefore emanates from one critical point and terminates at another of strictly smaller f-value.

Definition 9.6.1: Stable and Unstable Manifolds

Let p be a critical point of f. Its *unstable manifold* and *stable manifold* for the pseudo-gradient X are

$$W^u(p) = \{q \in M \mid \varphi_t(q) \to p \text{ as } t \to -\infty\}, \qquad W^s(p) = \{q \in M \mid \varphi_t(q) \to p \text{ as } t \to +\infty\}$$

the points swept out from p under backward and forward flow.

The unstable manifold collects the points that descend from p, the stable manifold those that ascend to it. Their dimensions are read off the local model: in the coordinates of (9.1) the flow is $\varphi_t(x) = (e^t x^-, e^{-t} x^+)$ with $x^- = (x_1, \ldots, x_\lambda)$ and $x^+ = (x_{\lambda+1}, \ldots, x_n)$, so a nearby point tends to the origin backward in time exactly when $x^+ = 0$ and forward in time exactly when $x^- = 0$. The negative-definite directions of the Hessian are the unstable ones.

Proposition 9.6.2: Descending and Ascending Disks

For a critical point p of index λ, the unstable manifold $W^u(p)$ is the image of an injective immersion of $\mathbb{R}^\lambda$, and the stable manifold $W^s(p)$ is the image of an injective immersion of $\mathbb{R}^{n-\lambda}$. In particular $\dim W^u(p) = \lambda$ and $\dim W^s(p) = n - \lambda$, and $W^u(p) \cap W^s(p) = \{p\}$.

***Proof*:**
The two statements are identical after reversing time (replacing X by $-X$ exchanges W^u and W^s and sends an index-λ point to an index-$(n-\lambda)$ point), so it suffices to treat $W^u(p)$.

Step 1: The local unstable set is an embedded λ-disk. In the Morse chart, the flow $\varphi_t(x) = (e^t x^-, e^{-t} x^+)$ shows that a point of the chart tends to the origin as $t \to -\infty$ if and only if $x^+ = 0$. Thus, on a chart ball B, the local unstable set is $W^u_{\mathrm{loc}}(p) = \{x \in B \mid x^+ = 0\}$, the coordinate λ-disk $D^- = \{x^+ = 0\}$, which is an embedded submanifold of dimension λ.

Step 2: The global unstable manifold is the forward flowout of the local disk. If $q \in W^u(p)$ then $\varphi_t(q) \to p$ as $t \to -\infty$, so $\varphi_{-T}(q) \in W^u_{\mathrm{loc}}(p)$ for some large T, whence $q = \varphi_T(\varphi_{-T}(q)) \in \varphi_T(D^-)$. Conversely D^- is forward-flow-invariant into $W^u(p)$: a point of D^- stays in D^- under backward flow and hence tends to p, and so does its image under any φ_T. Therefore

$$W^u(p) = \bigcup_{T \geq 0} \varphi_T(D^-)$$

Each nonconstant unstable trajectory crosses the boundary sphere $S = \partial D^-_\varepsilon = \{x^+ = 0 \mid |x^-| = \varepsilon\}$ exactly once, since $|x^-(t)| = e^t |x^-(0)|$ is strictly increasing along the flow. Hence the map $\Psi \colon S \times \mathbb{R} \to M$, $\Psi(s,t) = \varphi_t(s)$, is injective, and it is an immersion because φ_t is a diffeomorphism and X is transverse to S (as $X(f) < 0$ while f is constant on the level containing S up to first order, the flow leaves S). Its image is $W^u(p) \setminus \{p\}$. Adjoining p, and identifying $S \times \mathbb{R}$ together with the cone point with the open disk $\mathbb{R}^\lambda$, exhibits $W^u(p)$ as the image of an injective immersion of $\mathbb{R}^\lambda$.

Step 3: The two disks meet only at p. A point in $W^u(p) \cap W^s(p)$ has φ_t converging to p in both time directions; along its trajectory f is both increasing (backward from p) and decreasing (forward to p), forcing f constant, so the trajectory is the rest point p. Thus $W^u(p) \cap W^s(p) = \{p\}$, completing the proof.

The unstable manifold of a maximum is (an immersed copy of) the whole manifold minus lower strata, and the unstable manifold of a minimum is the single point. Between these extremes the index measures how many independent directions of descent leave the critical point. The disks are only *immersed*, not embedded: an unstable disk may wind through the manifold and accumulate on itself and on lower critical points, and controlling that accumulation is the content of the compactness theorem below.

Example 9.6.3: Height on the Sphere

Let $f\colon S^2 \to \mathbb{R}$ be the height function $f(x, y, z) = z$, with critical points the north pole N (index 2, a maximum) and the south pole S (index 0, a minimum). The pseudo-gradient flow slides points down meridians. The unstable manifold of N is everything that flows down from the top,

$$W^u(N) = S^2 \setminus \{S\}, \qquad W^s(N) = \{N\}$$

an immersed (here embedded) 2-disk and a point, matching $\lambda(N) = 2$ and $2 - \lambda(N) = 0$. Symmetrically $W^u(S) = \{S\}$ and $W^s(S) = S^2 \setminus \{N\}$. The two nontrivial disks $W^u(N) = S^2 \setminus \{S\}$ and $W^s(S) = S^2 \setminus \{N\}$ are distinct open sets whose intersection $W^u(N) \cap W^s(S) = S^2 \setminus \{N, S\}$ is a proper subset of each, an intersection which is automatically transverse since each is open.

For the trajectory count to define a manifold, the descending disk from one critical point and the ascending disk to another must meet transversally. This is a genericity condition, not an automatic one, and it is the hypothesis under which the whole construction runs.

Definition 9.6.4: Morse-Smale Condition

The pair (f, X) is *Morse-Smale* if for every pair of critical points a, b the unstable manifold of a is transverse to the stable manifold of b,

$$W^u(a) \pitchfork W^s(b)$$

Transversality of an immersed submanifold means transversality at every intersection point, using its immersed tangent space. On S^2 with the height function (example 9.6.3) the condition holds trivially, since every nonempty intersection of an unstable with a stable manifold involves an open set. The condition begins to bite when three or more critical points allow trajectories to run in cascade, and its force is dimensional, as the next result shows.

Theorem 9.6.5: Genericity of the Morse-Smale Condition

For a fixed Morse function f, the pseudo-gradients X for which (f, X) is Morse-Smale are dense: given any pseudo-gradient X_0 and any neighbourhood of it in the C^∞ topology, some X in that neighbourhood is Morse-Smale.

***Proof*:**

Since M is compact the critical points are finite in number (theorem 9.1.6), so there are finitely many ordered pairs (a, b), and it suffices to arrange $W^u(a) \pitchfork W^s(b)$ for each pair by an arbitrarily small perturbation of X. For a single pair the slices S^u_a and S^s_b constructed below are compact, so this condition is open as well as dense, and the finitely many open dense conditions can be met in turn, each perturbation small enough to preserve those already arranged.

Step 1: Reduce transversality to a level set. Fix $a \neq b$; the case $a = b$ holds by proposition 9.6.2. Choose a regular value α of f with $f(b) < \alpha < f(a)$ and $f^{-1}(\alpha)$ containing no critical point; the level $V = f^{-1}(\alpha)$ is a compact hypersurface (corollary 3.2.10). Because $X(f) \neq 0$ on V, the flow crosses V transversally, and the flow-invariant manifolds $W^u(a)$ and $W^s(b)$ each meet V

transversally in submanifolds

$$S_a^u = W^u(a) \cap V, \qquad S_b^s = W^s(b) \cap V$$

of dimensions $\lambda(a)-1$ and $n-\lambda(b)-1$. As $W^u(a)$ is the flowout of S_a^u and $W^s(b)$ the flowout of S_b^s, and both tangent spaces contain the common flow direction X, transversality $W^u(a) \pitchfork W^s(b)$ in M is equivalent to transversality $S_a^u \pitchfork S_b^s$ in V.

Step 2: A finite-dimensional family of perturbations. Embed V in some $\mathbb{R}^K$ (theorem 7.6.1) and let $Y_1, \dots, Y_k$ be the tangential components of the constant coordinate vector fields of $\mathbb{R}^K$; at each point of V they span the tangent space TV. For $s = (s_1, \dots, s_k)$ in a small ball $\Omega \subseteq \mathbb{R}^k$ let ψ_s be the time-one flow on V of $\sum_i s_i Y_i$, a smooth family of diffeomorphisms with $\psi_0 = \mathrm{id}$ and

$$\left.\frac{\partial}{\partial s}\right|_{s=0} \psi_s(x) = (Y_1(x), \dots, Y_k(x))$$

spanning T_xV for every x. Each ψ_s is realized by a perturbation of the pseudo-gradient inside a collar of V disjoint from the critical points, adjusting the holonomy across the collar; the perturbation is C^∞-small when s is, and does not disturb the local models near the critical points, so X remains a pseudo-gradient. Under this perturbation the trace S_a^u is replaced by $\psi_s(S_a^u)$.

Step 3: Parametric transversality. Consider $F\colon S_a^u \times \Omega \to V$, $F(x,s) = \psi_s(x)$. For fixed x the derivative in s at $s=0$ is surjective onto T_xV, so F is a submersion, hence transverse to S_b^s. By the parametric transversality theorem (theorem 7.5.6) applied to F and the submanifold $S_b^s \subseteq V$, the map $F_s\colon S_a^u \to V$, $x \mapsto \psi_s(x)$, is transverse to S_b^s for almost every $s \in \Omega$, that is $\psi_s(S_a^u) \pitchfork S_b^s$. Any such s arbitrarily close to 0 gives a Morse-Smale-improving perturbation of X for the pair (a,b) that is as C^∞-small as desired. Arranging the finitely many pairs in turn, each perturbation small enough to preserve the transversalities already achieved (open conditions, by compactness of the slices), completes the proof.

Morse-Smale is thus a generic hypothesis, purchased by a small perturbation and thereafter assumed. Its payoff is that the sets of trajectories between critical points are manifolds of predictable dimension.

Definition 9.6.6: Moduli of Trajectories

For critical points $a \neq b$ the *trajectory space* is the intersection

$$\mathcal{M}(a,b) = W^u(a) \cap W^s(b)$$

the points lying on a flow line from a to b. The flow φ_t acts on $\mathcal{M}(a,b)$ by translation along trajectories; the quotient

$$\mathcal{L}(a,b) = \mathcal{M}(a,b)/\mathbb{R}$$

is the *moduli space of unparametrized trajectories* from a to b.

A point of $\mathcal{M}(a,b)$ is a point of M lying on a trajectory running out of a and into b; a point of $\mathcal{L}(a,b)$ is the whole trajectory. The $\mathbb{R}$-action is free because no point of $\mathcal{M}(a,b)$ is critical, so the quotient loses exactly the one dimension of the flow direction.

Proposition 9.6.7: Dimension of the Trajectory Moduli

Let (f, X) be Morse-Smale and $a \neq b$. Then $\mathcal{M}(a, b)$ is a submanifold of M of dimension $\lambda(a) - \lambda(b)$, and $\mathcal{L}(a, b)$ is a manifold of dimension

$$\dim \mathcal{L}(a, b) = \lambda(a) - \lambda(b) - 1$$

In particular $\mathcal{L}(a, b) = \varnothing$ unless $\lambda(a) > \lambda(b)$, and when $\lambda(a) - \lambda(b) = 1$ it is a 0-manifold.

***Proof*:**
Represent an unparametrized trajectory by its unique crossing of a separating level. Fix a regular value α with $f(b) < \alpha < f(a)$ as in the previous proof, and let $V = f^{-1}(\alpha)$, $S_a^u = W^u(a) \cap V$, $S_b^s = W^s(b) \cap V$. Each trajectory from a to b crosses V exactly once, since f is strictly decreasing along it from $f(a)$ to $f(b)$, so the crossing map identifies $\mathcal{L}(a, b)$ with $S_a^u \cap S_b^s$. The Morse-Smale condition makes this intersection transverse in V, hence a submanifold of dimension

$$\dim S_a^u + \dim S_b^s - \dim V = (\lambda(a) - 1) + (n - \lambda(b) - 1) - (n - 1) = \lambda(a) - \lambda(b) - 1$$

The full trajectory space $\mathcal{M}(a, b)$ is the flowout $\mathcal{L}(a, b) \times \mathbb{R}$ of this crossing set, of dimension one greater. Negativity of the dimension forces $\mathcal{L}(a, b)$ empty unless $\lambda(a) > \lambda(b)$, and the value $\lambda(a) - \lambda(b) = 1$ gives a 0-manifold, as claimed.

The dimension formula counts directions: the number of independent descending directions at a, minus the number the trajectory must still lose to arrive at b, minus one for the freedom to slide along the flow. When the indices differ by one, the moduli space is a discrete set of trajectories, and its cardinality is what the boundary operator will count. The formula also records that the flow decreases the index, never raising it, which is why ∂ will drop degree by exactly one.

The discrete count is only well defined once the discrete set is finite, and the vanishing $\partial^2 = 0$ needs the one-dimensional moduli spaces to close up into compact 1-manifolds. Both follow from a single compactness principle: a sequence of trajectories degenerates only by breaking at intermediate critical points.

Theorem 9.6.8: Compactness and Broken Trajectories

Let (f, X) be Morse-Smale on the compact manifold M.

1. If $\lambda(a) - \lambda(b) = 1$, then $\mathcal{L}(a, b)$ is a finite set.
2. If $\lambda(a) - \lambda(c) = 2$, then $\mathcal{L}(a, c)$ is a 1-manifold that admits a compactification to a compact 1-manifold with boundary $\overline{\mathcal{L}(a, c)}$, with

$$\partial \overline{\mathcal{L}(a, c)} = \bigsqcup_{\lambda(b) = \lambda(a) - 1} \mathcal{L}(a, b) \times \mathcal{L}(b, c)$$

   the pairs of trajectories forming a broken trajectory from a through b to c.

***Proof*:**
Step 1: Limits of trajectories are broken trajectories. Let (γ_j) be a sequence in $\mathcal{L}(a,c)$. Off the critical points, reparametrize each by f-value: write $\gamma_j(u)$ for the point of the trajectory at height $u \in (f(c), f(a))$, well defined since f is strictly monotone along a trajectory. On any closed interval $[u_1, u_2] \subseteq (f(c), f(a))$ containing no critical value, $X(f)$ is bounded away from zero, so the reparametrized velocity $\gamma_j'(u) = X(\gamma_j(u))/X(f)(\gamma_j(u))$ is uniformly bounded; as M is compact, Arzelà-Ascoli extracts a subsequence converging uniformly on $[u_1, u_2]$. Exhausting $(f(c), f(a))$ minus the finitely many intermediate critical values by such intervals and diagonalizing produces a subsequence, still written (γ_j), converging on that punctured interval to a limit γ_∞; because X is smooth and the convergence is uniform, γ_∞ solves the flow equation and is a trajectory on each maximal interval between successive critical values it meets.

Step 2: The limit passes through critical points. As u decreases to an intermediate critical value β, the points $\gamma_\infty(u)$ lie in the compact set M and have $f = u \to \beta$; any subsequential limit is a point of $f^{-1}(\beta)$ toward which the flow tends, hence (by the ω-limit argument preceding definition 9.6.1) a critical point at level β. Thus γ_∞ is a concatenation of trajectory pieces joining a descending chain of critical points $a = p_0, p_1, \ldots, p_m = c$ with $f(p_0) > f(p_1) > \cdots > f(p_m)$. By proposition 9.6.7 each nonconstant piece $p_i \to p_{i+1}$ satisfies $\lambda(p_i) > \lambda(p_{i+1})$, so the indices strictly decrease along the chain, and the total drop is $\lambda(a) - \lambda(c)$.

Step 3: Finiteness when the index difference is one. If $\lambda(a) - \lambda(b) = 1$ the chain can drop the index by only 1, which forbids any intermediate critical point; hence every limit of a sequence in $\mathcal{L}(a,b)$ is again an element of $\mathcal{L}(a,b)$. So $\mathcal{L}(a,b)$ is compact, and being a 0-manifold (proposition 9.6.7) it is finite.

Step 4: The boundary when the index difference is two. If $\lambda(a) - \lambda(c) = 2$ the only chains are the unbroken one ($m = 1$) and the once-broken chains $a \to b \to c$ with an intermediate b of index $\lambda(a) - 1$ ($m = 2$); a longer chain would drop the index by more than 2. So the possible limits of $\mathcal{L}(a,c)$ that are not themselves in $\mathcal{L}(a,c)$ are exactly the pairs $(\gamma_1, \gamma_2) \in \mathcal{L}(a,b) \times \mathcal{L}(b,c)$, a finite set by part (1). Adjoining these limit points compactifies the 1-manifold $\mathcal{L}(a,c)$; the converse direction, that each broken pair is the limit of a unique end of $\mathcal{L}(a,c)$ and that the compactification is a smooth manifold with boundary there, is the *gluing theorem*. Its proof is an implicit-function-theorem argument on the space of approximate trajectories, of a piece with the analysis of Floer's equation; it lies outside the scope of this book, and the standard references are Schwarz, *Morse Homology*, and Audin and Damian, *Morse Theory and Floer Homology*. Granting it, $\overline{\mathcal{L}(a,c)}$ is a compact 1-manifold whose boundary is the stated disjoint union of broken trajectories, as claimed.

The picture is the same one that organized intersection theory in chapter 8: a 1-parameter family closes up into a compact 1-manifold, and its two-ended arcs pair the degenerations at its boundary. Here the degenerations are broken trajectories, and the pairing is about to force the identity $\partial^2 = 0$.

Definition 9.6.9: The Morse Complex

Let (f, X) be Morse-Smale. The *Morse complex* has chain groups the free abelian groups on the critical points,

$$C_\lambda = C_\lambda(f, X) = \mathbb{Z}\big\langle \text{critical points of index } \lambda \big\rangle$$

and boundary operator $\partial \colon C_\lambda \to C_{\lambda-1}$ defined on a generator a by

$$\partial a = \sum_{\lambda(b)=\lambda-1} n(a, b)\, b$$

where $n(a, b) \in \mathbb{Z}$ is the signed count of the finite moduli space $\mathcal{L}(a, b)$. With $\mathbb{Z}/2$ coefficients $n(a, b)$ is the number of points of $\mathcal{L}(a, b)$ modulo two.

The signs deserve a word. Each trajectory in $\mathcal{L}(a, b)$ carries a sign comparing a chosen orientation of the unstable disk $W^u(a)$ with the transported orientation along the flow; making these signs consistent across all critical points is the theory of *coherent orientations*, whose construction is sketched in the remark below. Over $\mathbb{Z}/2$ no choice is needed and the complex is unconditional; the signs may then be ignored throughout, which loses torsion information.

Theorem 9.6.10: The Boundary Squares to Zero

For a Morse-Smale pair (f, X), $\partial \circ \partial = 0$.

***Proof*:**
Fix generators a of index λ and c of index $\lambda - 2$. The coefficient of c in $\partial^2 a$ is

$$\sum_{\lambda(b)=\lambda-1} n(a, b)\, n(b, c)$$

which counts, with signs, the pairs $(\gamma_1, \gamma_2) \in \mathcal{L}(a, b) \times \mathcal{L}(b, c)$, that is the broken trajectories from a through some index-$(\lambda - 1)$ point b to c. By theorem 9.6.8 these pairs are exactly the boundary points of the compact 1-manifold $\overline{\mathcal{L}(a, c)}$.

Work first over $\mathbb{Z}/2$. A compact 1-manifold with boundary has an even number of boundary points (corollary 5.11.2), so the total number of broken trajectories from a to c is even, giving

$$\sum_{\lambda(b)=\lambda-1} \#\mathcal{L}(a, b) \cdot \#\mathcal{L}(b, c) \equiv 0 \pmod 2$$

Hence $\partial^2 = 0$ over $\mathbb{Z}/2$.

Over $\mathbb{Z}$ the coherent orientations are chosen so that the two endpoints of each arc of $\overline{\mathcal{L}(a, c)}$ carry opposite signs (a coherent orientation is built precisely to guarantee this; see box 9.6.12). Summing the signs of all boundary points of a compact oriented 1-manifold cancels arc by arc, so the signed count of broken trajectories is zero, and $\partial^2 = 0$ over $\mathbb{Z}$, completing the proof.

The identity $\partial^2 = 0$ is the boundary parity of corollary 5.11.2 read through the moduli spaces: the ways a two-step descent from a to c can degenerate come in cancelling pairs. The complex (C_*, ∂) is therefore a genuine chain complex, and its homology is the object the section set out to build.

Definition 9.6.11: Morse Homology

The *Morse homology* of the Morse-Smale pair (f, X) is the homology of its complex,

$$HM_\lambda(f, X) = \frac{\ker\big(\partial\colon C_\lambda \to C_{\lambda-1}\big)}{\operatorname{im}\big(\partial\colon C_{\lambda+1} \to C_\lambda\big)}$$

9.6.12 Remark: Coherent Orientations An orientation of each unstable manifold $W^u(a)$ induces, at a trajectory $\gamma \in \mathcal{L}(a,b)$ with $\lambda(a) - \lambda(b) = 1$, a sign ± 1: the flow identifies a neighbourhood of γ in $W^u(a)$ with $\gamma \times W^u(b)$ up to the stable directions of b, and the sign records whether the chosen orientations agree. The count $n(a,b)$ is the sum of these signs. The choice of orientations of the $W^u(a)$ is a coherent orientation, and the flow-out compatibility along the ends of a two-dimensional moduli space is exactly what makes the two boundary points of each arc cancel, as used in theorem 9.6.10. The construction is elementary but bookkeeping-heavy; the references above carry it out in full, and it is the finite-dimensional case of the orientation of determinant lines in Floer theory.

Example 9.6.13: The Standing Torus

Let T^2 stand upright in $\mathbb{R}^3$ and let f be the height, with critical points the bottom minimum p (index 0), the lower saddle q_1 and upper saddle q_2 (each index 1), and the top maximum r (index 2). A slight tilt of the torus makes the pair Morse-Smale (in the perfectly upright position a gradient trajectory joins the two saddles, an index-preserving connection that the tilt removes); the critical points and their indices are unchanged. The complex is

$$C_0 = \mathbb{Z}\langle p\rangle, \qquad C_1 = \mathbb{Z}\langle q_1, q_2\rangle, \qquad C_2 = \mathbb{Z}\langle r\rangle$$

From each saddle two descending trajectories run to the minimum, one down each side, so $\#\mathcal{L}(q_i, p) = 2$; with coherent orientations the two sides carry opposite signs and $n(q_i, p) = 0$, and over $\mathbb{Z}/2$ likewise $2 \equiv 0$. Thus $\partial|_{C_1} = 0$. Symmetrically two trajectories run from the maximum to each saddle and cancel, so $\partial r = 0$. The complex has zero boundary, and

$$HM_0 = \mathbb{Z}, \qquad HM_1 = \mathbb{Z}^2, \qquad HM_2 = \mathbb{Z}$$

which is $H_*(T^2)$. The count $c_\lambda = b_\lambda$ for every λ makes the height function a perfect Morse function (section 9.5); the vanishing of ∂ is the chain-level face of that perfection. The four broken trajectories $r \to q_i \to p$ through each saddle are the boundary points of the 1-manifold $\overline{\mathcal{L}(r,p)}$, and their even number is the mechanism of $\partial^2 = 0$ made concrete.

The torus recovers the homology it should, but with a boundary operator that happens to vanish. To see the complex compute homology that its generators alone cannot, the boundary must be nonzero, and the identification with singular homology is what guarantees it always does the right thing.

Theorem 9.6.14: Morse Homology is Singular Homology

For any Morse-Smale pair (f, X) on a compact manifold M, there is an isomorphism

$$HM_\lambda(f, X) \cong H_\lambda(M; \mathbb{Z})$$

for every λ. In particular the Morse homology is independent of the Morse function and the pseudo-gradient.

***Proof*:**
By the handle decomposition (theorem 9.4.5) M has the homotopy type of a CW complex $\mathcal{X}$ with exactly one λ-cell e_a for each index-λ critical point a. For a Morse-Smale pair each cell is realized geometrically: e_a is the unstable manifold $W^u(a)$, with characteristic map the closed disk $\overline{W^u(a)}$ attached along the lower-index cells. That the closures $\overline{W^u(a)}$ assemble into a genuine CW structure is itself a theorem, and a nontrivial one, since an unstable disk is only immersed and its closure can accumulate on the lower critical points; its proof rests on the same compactness and gluing of broken trajectories as theorem 9.6.8, and is carried out in the references of box 9.6.12. Granting it, the cellular chain group $C_\lambda^{\text{cell}}(\mathcal{X})$ is the free abelian group on the index-λ critical points, canonically the Morse chain group $C_\lambda(f, X)$.

Step 1: The cellular boundary counts trajectories. The cellular boundary map has matrix entry $[e_a : e_b]$ equal to the degree of the composite

$$S^{\lambda-1} = \partial\overline{W^u(a)} \xrightarrow{\text{attach}} \mathcal{X}^{(\lambda-1)} \longrightarrow \mathcal{X}^{(\lambda-1)}/\mathcal{X}^{(\lambda-2)} \longrightarrow S_b^{\lambda-1}$$

where the last map collapses every $(\lambda-1)$-cell but e_b. The attaching sphere $\partial\overline{W^u(a)}$ is carried by the flow onto the descending trajectories from a, and a point of $S_b^{\lambda-1}$ near the centre of e_b has preimages exactly the trajectories from a to b, that is the points of $\mathcal{L}(a, b)$, each counted with the sign comparing the flowed orientation of $W^u(a)$ to that of $W^u(b)$. This is the local degree at each preimage, so the total degree is the signed count $n(a, b)$. Hence the cellular boundary and the Morse boundary agree, $[e_a : e_b] = n(a, b)$, and the two complexes are identical.

Step 2: Cellular homology is singular homology. The cellular chain complex of a CW complex computes its singular homology ([11, Cellular Homology]), and $\mathcal{X}$ is homotopy equivalent to M, so

$$HM_\lambda(f, X) = H_\lambda^{\text{cell}}(\mathcal{X}) \cong H_\lambda(\mathcal{X}) \cong H_\lambda(M; \mathbb{Z})$$

The right-hand side depends on neither f nor X, so the Morse homology is an invariant of M, completing the proof.

The Morse homology recovers the singular homology of M, so the number of index-λ generators bounds the rank of $H_\lambda(M)$: the weak Morse inequality $c_\lambda \geq b_\lambda$ reappears as $\dim_{\mathbb{Q}}(C_\lambda \otimes \mathbb{Q}) \geq \dim_{\mathbb{Q}}(HM_\lambda \otimes \mathbb{Q})$, and the alternating sum $\sum(-1)^\lambda c_\lambda = \sum(-1)^\lambda b_\lambda = \chi(M)$ is the Euler characteristic of the complex (theorem 9.5.1). The homological content of the critical points, glimpsed as bounds in section 9.5, is now an equality of groups. The complex also sees torsion that the Betti numbers miss, as the next example shows.

Example 9.6.15: The Projective Plane

The real projective plane $\mathbb{R}P^2$ carries a Morse function with one critical point in each index $0, 1, 2$, mirroring its CW structure with one cell per dimension. Writing x_0, x_1, x_2 for the generators, the boundary operator is read off the attaching maps of that CW structure via theorem 9.6.14: the 1-cell attaches to the 0-cell by a constant map, so $n(x_1, x_0) = 0$, and the 2-cell attaches to the 1-cell by the degree-two map $S^1 \to \mathbb{R}P^1$, so $n(x_2, x_1) = 2$. The complex is

$$\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$$

in degrees $2, 1, 0$. Its homology is

$$HM_0 = \mathbb{Z}, \qquad HM_1 = \mathbb{Z}/2\mathbb{Z}, \qquad HM_2 = 0$$

reproducing $H_*(\mathbb{R}P^2; \mathbb{Z})$, torsion and all. The two index-1 and index-2 generators would allow $b_1 = b_2 = 1$; the boundary $n(x_2, x_1) = 2$ collapses them to a single 2-torsion class, an outcome invisible to the critical-point counts alone but forced by the trajectory count.

The Morse complex is the finite-dimensional model of a construction that reaches much further. When the manifold is replaced by an infinite-dimensional space of paths or loops and the Morse function by an energy or action functional, the critical points become geodesics or periodic orbits, the gradient trajectories become solutions of an elliptic partial differential equation, and the same count of connecting trajectories defines Floer homology. The finiteness, compactness, and $\partial^2 = 0$ arguments carry over almost verbatim, with the gluing and orientation analysis this section deferred taking centre stage. Morse homology is where those ideas can be seen whole, on a manifold small enough to hold in view.

Exercise 9.6.1

1. Let f be a Morse function on S^n with exactly two critical points (a minimum and a maximum, as furnished by a height function). Write down its Morse complex, compute ∂, and identify the Morse homology. Confirm it agrees with $H_*(S^n; \mathbb{Z})$.

2. A Morse-Smale pair on a 5-manifold has an index-4 critical point a and an index-1 critical point b. What is the dimension of $\mathcal{M}(a, b)$, and of $\mathcal{L}(a, b)$? For which index pairs $(\lambda(a), \lambda(b))$ does a generic trajectory count $n(a, b)$ enter the boundary operator?

3. Call a Morse function *perfect* (over $\mathbb{Q}$) if $c_\lambda = b_\lambda$ for all λ. Using theorem 9.6.14, show that for a perfect Morse function the rational boundary operator vanishes, and conversely that $\partial \otimes \mathbb{Q} = 0$ forces perfection.

4. Verify by hand that the complex $\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$ of example 9.6.15 satisfies $\partial^2 = 0$, and compute its homology in each degree. Then explain, in one sentence, why the same manifold $\mathbb{R}P^2$ has a Morse complex over $\mathbb{Z}/2$ with $\partial = 0$.

5. Show that $\sum_\lambda (-1)^\lambda \operatorname{rank} C_\lambda = \sum_\lambda (-1)^\lambda \operatorname{rank} HM_\lambda$ for any finite complex of free abelian groups, and deduce $\sum_\lambda (-1)^\lambda c_\lambda = \chi(M)$ directly from theorem 9.6.14, without invoking Poincaré-Hopf.

6. Suppose $\lambda(a) - \lambda(c) = 2$ and the compact 1-manifold $\overline{\mathcal{L}(a, c)}$ has three arcs and no circles. How many broken trajectories $a \to b \to c$ are there in total, and what does this force about

the parity of $\sum_b \#\mathcal{L}(a,b) \cdot \#\mathcal{L}(b,c)$?

10

Cobordism and the Pontryagin-Thom Construction

Two manifolds are cobordant when together they bound a third. This is the crudest equivalence a manifold admits, coarser than diffeomorphism and coarser than homotopy equivalence, and precisely for that reason it is computable: the resulting invariant is an abelian group, and Thom identified it completely. The relation is the natural endpoint of the machinery of this part. Transversality (chapter 7) made preimages of regular values into submanifolds; the degree (chapter 8) counted the preimage of a point; a one-parameter family sweeps out a cobordism between the preimages at its ends. Cobordism is what the boundary operator of chapter 7 measures once one stops tracking anything but the bounding relation itself.

The chapter's engine is the refinement that remembers a normal framing. A framed submanifold of a sphere collapses to a map into a lower sphere, and a regular value of a map into a sphere returns a framed submanifold; Pontryagin's theorem is that these two passages are mutually inverse. The homotopy groups of spheres, the most classical and most stubborn invariants in topology, are therefore framed cobordism groups of manifolds. The construction is entirely geometric, built from the transversality and tubular-neighbourhood theorems already in hand, and it settles a debt: the Hopf degree theorem for maps $S^n \to S^n$, stated in chapter 8 and proved there only for $n = 1$, falls out as the zero-dimensional case.

Where the answers become genuinely homotopy-theoretic, the account hands off to the companion volume. Thom's computation of the cobordism ring, the Freudenthal stabilization that turns framed cobordism into the stable stems, and the spectrum-level reformulation all live in [11, Thom's Theorem, Freudenthal Suspension Theorem, The Pontryagin-Thom Isomorphism]; they are stated here with their geometric representatives and cited. What this chapter proves is the bridge: the dictionary between framed manifolds and maps of spheres, and the geometry that makes it an isomorphism.

10.1 Cobordism and the Unoriented Cobordism Ring

Two closed manifolds are declared equivalent when, placed side by side, they form the boundary of one compact manifold a dimension higher. This relation converts the qualitative property "is a boundary" into an algebraic invariant: disjoint union makes the equivalence classes into a group, Cartesian product makes the graded collection into a ring, and Thom computed that ring completely. This section builds the unoriented cobordism ring, computes it in low degrees, and states Thom's classification. The framed refinement of section 10.2 and the Pontryagin-Thom construction of section 10.3 will read the homotopy of spheres off the same geometric idea.

Definition 10.1.1: Cobordant Manifolds

Let M_0 and M_1 be closed n-manifolds. A *cobordism* from M_0 to M_1 is a compact $(n+1)$-manifold with boundary W together with a diffeomorphism $\partial W \cong M_0 \sqcup M_1$. When such a W exists, M_0 and M_1 are *cobordant*, written $M_0 \sim M_1$. A closed manifold M is *null-cobordant* (or *bounds*) if $M = \partial W$ for a compact W, equivalently if $M \sim \emptyset$.

The prototype is already in hand. If $f \colon P \to \mathbb{R}$ is proper with regular values $a < b$, the preimage $f^{-1}([a, b])$ is a compact manifold whose boundary is $f^{-1}(a) \sqcup f^{-1}(b)$ (proposition 3.2.20), so the two level sets are cobordant, cobounding the slab between them. The sphere is the cleanest instance: $S^n = \partial D^{n+1}$ is the boundary of the closed disc, so S^n is null-cobordant for every n. A cobordism is therefore a smooth interpolation, one dimension up, between two manifolds; the relation asks not whether M_0 and M_1 are diffeomorphic but whether they can be joined through such an interpolation.

Before the classes can be added, the relation must be shown to partition closed manifolds into equivalence classes at all.

Proposition 10.1.2: Cobordism Is an Equivalence Relation

Cobordism is an equivalence relation on the set of closed n-manifolds.

***Proof*:**
Reflexivity. For a closed M the cylinder $M \times [0, 1]$ is a compact $(n+1)$-manifold with $\partial(M \times [0, 1]) = M \times \{0\} \sqcup M \times \{1\} \cong M \sqcup M$, since M is boundaryless. Thus $M \sim M$.

Symmetry. The relation is stated symmetrically: $M_0 \sqcup M_1 \cong M_1 \sqcup M_0$, so a single W witnesses both $M_0 \sim M_1$ and $M_1 \sim M_0$.

Transitivity. Suppose $\partial W = M_0 \sqcup M_1$ and $\partial W' = M_1 \sqcup M_2$. By the collar neighbourhood theorem (theorem 1.2.20) the boundary component M_1 has a collar $M_1 \times [0, 1)$ in each of W and W'. Glue W and W' along M_1 by identifying these collars, matching $M_1 \times [0, 1) \subseteq W$ with $M_1 \times (-1, 0] \subseteq W'$ across $M_1 \times \{0\}$; the collars give the smooth structure across the seam, producing a compact $(n+1)$-manifold $W'' = W \cup_{M_1} W'$. The two copies of M_1 cancel, leaving $\partial W'' = M_0 \sqcup M_2$, so $M_0 \sim M_2$. Cobordism is therefore an equivalence relation, as we sought to show.

The collar is exactly what a naive gluing lacks: without it the union of W and W' along M_1 is only a topological manifold, with a possible corner along the seam. Integrating an inward-pointing field to produce the collar is what smooths the join, and this is the single technical device that makes the algebra below work.

Definition 10.1.3: Unoriented Cobordism Group

The *n-th unoriented cobordism group* $\mathfrak{N}_n$ is the set of cobordism classes $[M]$ of closed n-manifolds, with addition

$$[M_0] + [M_1] = [M_0 \sqcup M_1]$$

The identity is the class $[\emptyset]$ of null-cobordant manifolds.

Addition is well defined: if $M_0 \sim M_0'$ via W_0 and $M_1 \sim M_1'$ via W_1, then $W_0 \sqcup W_1$ is a cobordism from $M_0 \sqcup M_1$ to $M_0' \sqcup M_1'$, so the sum depends only on the classes. It is associative and commutative because $\sqcup$ is, and $[M] + [\emptyset] = [M]$ because $M \sqcup \emptyset = M$. Only the inverse remains, and it is the first surprise of the theory: each element is its own inverse.

Example 10.1.4: The Cobordism Group of Points

A closed 0-manifold is a finite set of points. Two such sets M_0, M_1 are cobordant when $M_0 \sqcup M_1$ is the boundary of a compact 1-manifold W, and by the boundary parity of compact 1-manifolds (corollary 5.11.2) that boundary has an even number of points. Hence $|M_0| + |M_1|$ is even, so $|M_0| \equiv |M_1| \pmod 2$. Conversely, if $|M_0| \equiv |M_1| \pmod 2$, pair the points of $M_0 \sqcup M_1$ and join each pair by an arc: a disjoint union of closed intervals is a compact 1-manifold with the required boundary. Cardinality modulo 2 is therefore a complete invariant, and

$$\mathfrak{N}_0 \cong \mathbb{F}_2$$

generated by the class of a single point [pt]. In particular a single point does not bound: a compact 1-manifold has an even number of endpoints, never one.

Proposition 10.1.5: Every Cobordism Class Is 2-Torsion

For every closed manifold M, $2[M] = 0$ in $\mathfrak{N}_n$; that is, $M \sqcup M$ is null-cobordant.

***Proof*:**
The cylinder $M \times [0,1]$ is a compact $(n+1)$-manifold with boundary $M \times \{0\} \sqcup M \times \{1\} \cong M \sqcup M$. Thus $M \sqcup M = \partial(M \times [0,1])$ is null-cobordant, so $[M] + [M] = [M \sqcup M] = 0$, completing the proof.

Each $\mathfrak{N}_n$ is therefore an abelian group in which every element has order dividing 2, that is, a vector space over $\mathbb{F}_2$. The cobordism relation on n-manifolds is precisely equality in this group: $[M] = [N]$ holds if and only if $[M] + [N] = 0$, which by 2-torsion is $[M \sqcup N] = 0$, that is, $M \sqcup N$ bounds. The single reflexive cobordism $M \times [0,1]$ has collapsed the distinction between an element and its negative, and it is this feature that pins the coefficient ring of the whole theory to $\mathbb{F}_2$.

Disjoint union is only half of the natural algebra on manifolds; Cartesian product gives a multiplication, and it respects cobordism, so the groups assemble into a graded ring.

Definition 10.1.6: Unoriented Cobordism Ring

The *unoriented cobordism ring* is the graded group

$$\mathfrak{N}_\bullet = \bigoplus_{n\geq 0} \mathfrak{N}_n$$

with multiplication induced by Cartesian product, $[M]\cdot[N] = [M\times N]$. The unit is the class $[\mathrm{pt}] \in \mathfrak{N}_0$ of a point.

Multiplication is well defined: if $M \sim M'$ via a cobordism W, then $W \times N$ is a cobordism from $M \times N$ to $M' \times N$, since $\partial(W\times N) = \partial W \times N = (M \sqcup M')\times N \cong (M\times N)\sqcup(M'\times N)$ because N is boundaryless. Product is associative and commutative up to the diffeomorphism $M\times N \cong N \times M$, and $M \times \mathrm{pt} = M$, so $\mathfrak{N}_\bullet$ is a commutative graded $\mathbb{F}_2$-algebra. It records, in one object, which manifolds bound and how products of manifolds behave under the bounding relation.

To find a single nonzero class beyond the point one must produce a manifold that does not bound, and Euler characteristic detects the obstruction. The key fact is a parity constraint on boundaries.

Lemma 10.1.7: The Euler Characteristic of a Boundary Is Even

If a closed manifold M is null-cobordant, then $\chi(M)$ is even.

For a manifold that need not be orientable, $\chi(M)$ is taken to be the index sum $\sum_p \mathrm{ind}_p(v)$ of any vector field with nondegenerate zeros; the index is defined from local charts alone and is independent of orientation, so Poincaré-Hopf and this definition extend verbatim to the non-orientable case (section 8.6, box 8.6.5).

***Proof*:**
Write $n = \dim M$ and let W be a compact $(n+1)$-manifold with $\partial W = M$.

Step 1: The odd case. If n is odd then $\chi(M) = 0$ with no hypothesis on bounding. Indeed, for a field v with nondegenerate zeros, $-v$ has the same zeros and $\mathrm{ind}_p(-v) = (-1)^n \mathrm{ind}_p(v)$, so summing gives $\chi(M) = (-1)^n\chi(M) = -\chi(M)$, whence $\chi(M) = 0$ (section 8.6). Zero is even, so assume henceforth that n is even.

Step 2: The double. Glue two copies W_+ and W_- of W along their common boundary M by the collar neighbourhood theorem (theorem 1.2.20), forming the closed $(n+1)$-manifold $DW = W_+ \cup_M W_-$. Since n is even, DW has odd dimension $n+1$, so $\chi(DW) = 0$ by Step 1 applied to DW. Let $\rho\colon DW \to DW$ be the reflection interchanging W_+ and W_- and fixing M pointwise; on a collar $C \cong M \times (-1,1)$ of the seam $M = M\times\{0\}$, with $W_\pm$ containing $M\times[0,1)$ and $M \times (-1,0]$, it is $\rho(x,t) = (x,-t)$.

Step 3: A symmetric field. Choose a vector field w on M with nondegenerate zeros, so that $\sum_p \mathrm{ind}_p(w) = \chi(M)$ (theorem 8.6.7). On C define

$$V(x,t) = w(x) + t\,\frac{\partial}{\partial t}$$

Its only zeros in C are the points $(p,0)$ with $w(p) = 0$, and at each the derivative is the block matrix $Dw_p \oplus (1)$, so $\mathrm{ind}_{(p,0)}(V) = \operatorname{sign}\det(Dw_p \oplus 1) = \mathrm{ind}_p(w)$. Hence the seam contributes $\sum_C \mathrm{ind}(V) = \chi(M)$. The field V on C is ρ-invariant, since $\rho_*\big(t\,\partial/\partial t\big)(x,-t) = -t\,\partial/\partial t$ evaluated at $(x, -t)$ returns $t\,\partial/\partial t$, and w is unchanged. Extend V over $W_+ \setminus C$ to a field with nondegenerate zeros and set $V := \rho_* V$ on $W_- \setminus C$; this yields a ρ-invariant field on DW with nondegenerate zeros, unchanged on C.

Step 4: Counting. The reflection ρ interchanges the zeros of V in $W_+ \setminus C$ with those in $W_- \setminus C$, and because the index is diffeomorphism invariant and orientation independent (section 8.6), corresponding zeros have equal index. Their total contribution is therefore $2J$ for an integer J. Poincaré-Hopf on the closed manifold DW now gives

$$0 = \chi(DW) = \sum \operatorname{ind}(V) = J + \chi(M) + J = \chi(M) + 2J$$

so $\chi(M) = -2J$ is even, completing the proof.

The lemma reads the local geometry of a filling against the global topology of its boundary: a manifold that bounds cannot have odd Euler characteristic, because doubling the filling produces an odd-dimensional closed manifold, whose vanishing Euler characteristic forces the boundary's to be even. One nonorientable surface violates the constraint, and that single example generates the first interesting cobordism group.

Example 10.1.8: Spheres Bound, but the Projective Plane Does Not

Spheres. Every sphere bounds a disc, $S^n = \partial D^{n+1}$, so $[S^n] = 0$ in $\mathfrak{N}_n$ for all n. More generally $S^m \times S^n = \partial(D^{m+1} \times S^n)$ is null-cobordant, since $\partial(D^{m+1} \times S^n) = S^m \times S^n$ with S^n boundaryless.

The projective plane. The Euler characteristic of $\mathbb{RP}^2$ is odd. The double cover $\pi\colon S^2 \to \mathbb{RP}^2$ is a local diffeomorphism; a vector field v on $\mathbb{RP}^2$ with nondegenerate zeros pulls back to the π-related field $\tilde{v}$ on S^2, whose zeros are the two preimages of each zero of v, each with the same index as the zero below it because π is a local diffeomorphism there. Summing indices,

$$\chi(S^2) = \sum_q \operatorname{ind}_q(\tilde{v}) = 2 \sum_p \operatorname{ind}_p(v) = 2\,\chi(\mathbb{RP}^2)$$

and $\chi(S^2) = 2$ (example 8.6.8) gives $\chi(\mathbb{RP}^2) = 1$. Since 1 is odd, lemma 10.1.7 shows $\mathbb{RP}^2$ is not null-cobordant. Thus $[\mathbb{RP}^2] \neq 0$ in $\mathfrak{N}_2$, and it is the first closed manifold met in this book that provably does not bound.

The projective plane also shows that Euler characteristic is not the whole story, because it is a cobordism invariant only modulo 2: cobordant manifolds have Euler characteristics of the same parity (a boundary has even χ, and χ is additive under disjoint union), but not necessarily equal. The full system of $\mathbb{F}_2$-valued cobordism invariants, the Stiefel-Whitney numbers, is developed in [11, Stiefel-Whitney Number, Stiefel-Whitney Numbers Are a Complete Unoriented Cobordism Invariant], and Thom's theorem is that they determine the cobordism class completely.

Theorem 10.1.9: Thom's Theorem

The unoriented cobordism ring is a polynomial algebra over $\mathbb{F}_2$,

$$\mathfrak{N}_\bullet \cong \mathbb{F}_2[x_i : i \geq 2,\ i \neq 2^j - 1]$$

with one generator x_i in each degree $i \geq 2$ that is not of the form $2^j - 1$. The degree-2 generator may be taken to be $x_2 = [\mathbb{RP}^2]$.

***Proof Key ideas*:**
The computation is homotopy-theoretic and lies outside this book's scope. The Pontryagin-Thom construction of section 10.3 identifies $\mathfrak{N}_n$ with a homotopy group of the Thom spectrum MO; the ring $\pi_\bullet(MO)$ is then evaluated using the Steenrod algebra and Wu's formula, and the answer is the stated polynomial ring. Both the spectrum-level Pontryagin-Thom isomorphism $\mathfrak{N}_n \cong \pi_n(MO)$ and its evaluation are developed in [11, The Pontryagin-Thom Isomorphism, Thom's Theorem]. The Steenrod-algebra evaluation of $\pi_*(MO)$ is there deferred to the external literature.

The theorem turns the classification of manifolds up to cobordism into a problem in polynomial algebra over $\mathbb{F}_2$. In low degrees it reads off at once. There are no generators in degrees 1 and 3 (both of the form $2^j - 1$), and none of degree 0, so

$$\mathfrak{N}_0 \cong \mathbb{F}_2, \qquad \mathfrak{N}_1 = 0, \qquad \mathfrak{N}_2 \cong \mathbb{F}_2, \qquad \mathfrak{N}_3 = 0$$

with $\mathfrak{N}_0$ generated by [pt] and $\mathfrak{N}_2$ by $[\mathbb{R}\mathrm{P}^2]$. The vanishing of $\mathfrak{N}_1$ recovers the fact that every closed curve bounds, and the value of $\mathfrak{N}_2$ says that a closed surface bounds if and only if its Euler characteristic is even, so among the closed surfaces exactly the odd-χ ones fail to bound. The degree-4 part $\mathfrak{N}_4 \cong \mathbb{F}_2^2$ is spanned by $x_2^2 = [\mathbb{R}\mathrm{P}^2 \times \mathbb{R}\mathrm{P}^2]$ and a new generator x_4, showing that products of low-dimensional generators do not exhaust the ring. The geometric content of the isomorphism, that a cobordism class is nothing but a homotopy class of collapse maps, is the subject of the rest of this chapter.

Exercise 10.1.1

1. Prove that $\mathfrak{N}_1 = 0$: every closed 1-manifold is null-cobordant. Use the classification of compact 1-manifolds (theorem 5.11.1).
2. Show that if closed n-manifolds M and N are cobordant, then $\chi(M) \equiv \chi(N) \pmod 2$. Deduce that for n even the map $[M] \mapsto \chi(M) \bmod 2$ is a well-defined homomorphism $\mathfrak{N}_n \to \mathbb{F}_2$, and that it is nonzero on $\mathfrak{N}_2$.
3. Show that $[S^m \times S^n] = 0$ in $\mathfrak{N}_{m+n}$ in two ways: from the ring structure using $[S^m] = 0$, and by exhibiting an explicit compact manifold that $S^m \times S^n$ bounds.
4. Prove that for closed manifolds $\chi(M \times N) = \chi(M)\,\chi(N)$ by counting the zeros of a product vector field. Conclude that $\mathbb{R}\mathrm{P}^2 \times \mathbb{R}\mathrm{P}^2$ is not null-cobordant, so $[\mathbb{R}\mathrm{P}^2]^2 \neq 0$ in $\mathfrak{N}_\bullet$.
5. Prove that for closed n-manifolds M and N, the cobordism relation $M \sim N$ holds if and only if $[M] = [N]$ in $\mathfrak{N}_n$. Explain where two-torsion (proposition 10.1.5) is used.

10.2 Framed Submanifolds and Framed Cobordism

Section 10.1 recorded of a manifold only whether it bounds, discarding every trace of how it might sit inside an ambient space. The Pontryagin-Thom construction recovers the homotopy of spheres from cobordism, and to do so it needs one piece of embedding data that the bare relation throws away: a trivialization of the normal bundle. This section introduces that data, a *framing*, and builds the framed analogue of section 10.1: framed submanifolds, the product neighbourhood a framing determines, the framed cobordism relation, and the group Ω_n^{fr} it defines. These are the objects that the collapse construction of section 10.3 will match, one for one, with maps of spheres.

Throughout, submanifolds are compact and embedded, and the normal bundle $\nu(N)$ of an n-submanifold $N \subseteq M^{n+k}$ is the rank-k vector bundle whose fiber $\nu_p(N) = T_pM/T_pN$ is the normal space at p (proposition 7.7.2).

Definition 10.2.1: Framing and Framed Submanifold

Let $N^n \subseteq M^{n+k}$ be a compact submanifold of codimension k. A *framing* of N is a smooth trivialization of its normal bundle $\nu(N)$: an ordered k-tuple $\varphi = (v_1, \dots, v_k)$ of smooth normal sections $v_i \colon N \to \nu(N)$ whose values $v_1(p), \dots, v_k(p)$ form a basis of $\nu_p(N)$ at every $p \in N$. The pair (N, φ) is a *framed submanifold* of M; its codimension is k. Two framings of N are regarded as the same when they are homotopic through framings, that is joined by a smooth path φ_s $(s \in [0,1])$ of framings.

A framing is a smoothly varying choice of ordered basis for the directions normal to N. Fixing a Euclidean metric on the ambient space, the sections v_i can be represented by vectors in T_pM orthogonal to T_pN, and Gram-Schmidt orthonormalization is a deformation through framings, so a framing may always be assumed orthonormal without changing its homotopy class. The reason framings are counted only up to homotopy is that the invariant to be built, framed cobordism, cannot see a deformation of the framing: a homotopy of framings is itself realized by a cobordism (example 10.2.9 and Exercise). Not every submanifold admits a framing; a framing exists precisely when $\nu(N)$ is trivial, which is why the theory selects the submanifolds whose normal bundle has been trivialized, not just oriented.

The most important source of framed submanifolds is the one that drives the whole construction: regular level sets come framed for free. If $f \colon M^{n+k} \to \mathbb{R}^k$ is smooth and 0 is a regular value, then $N = f^{-1}(0)$ is an n-submanifold (corollary 3.2.10), and the differential $df_p \colon T_pM \to \mathbb{R}^k$ descends to an isomorphism $\nu_p(N) = T_pM/T_pN \cong \mathbb{R}^k$; pulling the standard basis $e_1, \dots, e_k$ of $\mathbb{R}^k$ back through these isomorphisms gives a framing $\varphi_f = (df^{-1}e_1, \dots, df^{-1}e_k)$ of N. This *pullback framing* is the geometric data determined by f, and reconstructing f from it is exactly the Pontryagin-Thom correspondence of section 10.3.

Example 10.2.2: The Outward Framing of a Sphere

The sphere $S^n \subseteq \mathbb{R}^{n+1}$ has codimension $k = 1$, so a framing is a single nowhere-zero normal section. The normal space at $p \in S^n$ is the line $\nu_p(S^n) = \mathbb{R}p$, and the outward unit normal

$$v(p) = p$$

spans it, giving the *outward framing* $\varphi_{\mathrm{out}} = (v)$. It is the pullback framing of $f(x) = |x|^2 - 1$, for which $df_p = 2\langle p, - \rangle$ carries p to 2; up to the positive scale, $\varphi_f = \varphi_{\mathrm{out}}$. Reversing the sign gives the *inward framing* $\varphi_{\mathrm{in}} = (-v)$. Because $\nu(S^n)$ is a trivial line bundle over the connected base S^n, its nowhere-zero sections fall into exactly two homotopy classes, distinguished by sign, so φ_{out} and φ_{in} are the only two framings of S^n up to homotopy, and they are exchanged by the reflection $v \mapsto -v$.

A framing does more than trivialize the normal bundle abstractly: through the tubular neighbourhood theorem it thickens N to a product $N \times \mathbb{R}^k$ sitting inside M. This product neighbourhood is the technical hinge of the construction, and the collapse map of section 10.3 is built directly from it.

Proposition 10.2.3: Framings and Product Neighbourhoods

Let (N, φ) be a framed submanifold of M^{n+k} with N compact. Then there is an open neighbourhood U of N in M and a diffeomorphism

$$\Phi\colon N \times \mathbb{R}^k \to U, \qquad \Phi(p, 0) = p$$

whose derivative along the zero section carries the standard basis of $\mathbb{R}^k$ to the framing: $D_{(p,0)}\Phi(0, e_i) = v_i(p)$ for each i.

***Proof*:**
By the tubular neighbourhood theorem (theorem 7.8.1) there is an open neighbourhood U of N in M and a diffeomorphism ψ from the total space of $\nu(N)$ onto U, fixing N pointwise (the zero section maps to N) and equal along the zero section to the inclusion of normal directions. The framing φ is a bundle isomorphism

$$\tau\colon N \times \mathbb{R}^k \to \nu(N), \qquad \tau(p, a) = \sum_{i=1}^{k} a_i\, v_i(p)$$

which is linear on fibers and carries (p, e_i) to $v_i(p)$. The composite $\Phi = \psi \circ \tau$ is a diffeomorphism onto U with $\Phi(p, 0) = \psi(p, 0) = p$, and along the zero section its derivative sends $(0, e_i)$ to $v_i(p)$, since τ sends it to $v_i(p) \in \nu_p(N)$ and ψ is the normal inclusion there. This is the required product neighbourhood, as we sought to show.

Read backward, Φ presents U as the framed submanifold thickened in its normal directions, with the i-th coordinate line $\{p\} \times \mathbb{R}e_i$ following the framing vector v_i. The projection $N \times \mathbb{R}^k \to \mathbb{R}^k$ then measures a point of U by its normal displacement, and composing it with the diffeomorphism gives a map $U \to \mathbb{R}^k$ vanishing exactly on N. Extending this map by a constant off U is the collapse construction of section 10.3; the framing is what makes the target $\mathbb{R}^k$, and hence the sphere S^k, well defined.

With framings in hand, the cobordism relation of section 10.1 refines to one that remembers them. A cobordism between framed submanifolds is a cobordism of the underlying manifolds carrying a framing of its own that restricts to the given framings at the two ends.

Definition 10.2.4: Framed Cobordism

Fix a codimension k. Two framed n-submanifolds (N_0, φ_0) and (N_1, φ_1) of $\mathbb{R}^{n+k}$ are *framed cobordant* if there is a compact $(n+1)$-submanifold with boundary $W \subseteq \mathbb{R}^{n+k} \times [0, 1]$ such that:

1. W meets the boundary slabs $\mathbb{R}^{n+k} \times \{0\}$ and $\mathbb{R}^{n+k} \times \{1\}$ transversally, in $N_0 \times \{0\}$ and $N_1 \times \{1\}$ respectively, and in no other points of the two slabs; and
2. W carries a framing $\Phi = (V_1, \ldots, V_k)$ of its normal bundle in $\mathbb{R}^{n+k} \times [0, 1]$ that restricts to φ_0 on $N_0 \times \{0\}$ and to φ_1 on $N_1 \times \{1\}$.

Write $(N_0, \varphi_0) \sim (N_1, \varphi_1)$, and call W a *framed cobordism* between them.

The dimension bookkeeping is the same as for the underlying manifolds: W has dimension $n+1$ in the ambient $\mathbb{R}^{n+k+1}$, so its normal bundle again has rank k, and a framing of it is again a k-tuple of independent normal sections. The transversality clause is what lets the framing be read off at the ends. Because W meets each slab transversally, its normal directions there lie in the slab $\mathbb{R}^{n+k}$ itself, so the vectors V_i restrict on $N_0 \times \{0\}$ and $N_1 \times \{1\}$ to framings of N_0 and N_1 inside $\mathbb{R}^{n+k}$; requiring these to be φ_0 and φ_1 is the content of the second clause. A framed submanifold that is framed cobordant to the empty set, meaning it forms the entire boundary of a framed W contained in $\mathbb{R}^{n+k} \times [0,1]$ with all of ∂W in the slab $\mathbb{R}^{n+k} \times \{0\}$, is called *framed null-cobordant*.

Proposition 10.2.5: Framed Cobordism Is an Equivalence Relation

Framed cobordism is an equivalence relation on framed n-submanifolds of $\mathbb{R}^{n+k}$.

***Proof*:**
Reflexive. For a framed (N,φ), the product cylinder $W = N \times [0,1] \subseteq \mathbb{R}^{n+k} \times [0,1]$ has boundary $N \times \{0,1\}$, meets both slabs transversally, and its normal bundle is the pullback of $\nu(N)$; framing it by $\Phi(p,t) = \varphi(p)$, constant in t, restricts to φ at both ends. Hence $(N,\varphi) \sim (N,\varphi)$.

Symmetric. The reflection $\rho\colon \mathbb{R}^{n+k} \times [0,1] \to \mathbb{R}^{n+k} \times [0,1]$, $\rho(x,t) = (x, 1-t)$, is a diffeomorphism carrying a framed cobordism W from (N_0,φ_0) to (N_1,φ_1) onto a framed cobordism $\rho(W)$ from (N_1,φ_1) to (N_0,φ_0); the framing is transported by ρ and its restrictions at the two ends are interchanged. Hence $\sim$ is symmetric.

Transitive. Let W be a framed cobordism from (N_0,φ_0) to (N_1,φ_1) and W' one from (N_1,φ_1) to (N_2,φ_2). Stacking the two intervals and rescaling, gluing W and W' along their common boundary component N_1 produces a compact $(n+1)$-manifold $W \cup_{N_1} W'$, smoothed across the seam using a collar of N_1 exactly as for unframed cobordism (section 10.1). The two framings agree on N_1, both restricting to φ_1, so they concatenate to a single framing of the glued manifold, constant across the collar. The result is a framed cobordism from (N_0,φ_0) to (N_2,φ_2), so $\sim$ is transitive, completing the proof.

Framed cobordism therefore partitions framed n-submanifolds into classes, and disjoint union descends to an addition on them. Placing two framed submanifolds in disjoint balls of $\mathbb{R}^{n+k}$ makes their union framed, and the union of two framed cobordisms is a framed cobordism, so the operation is well defined on classes.

Definition 10.2.6: Framed Cobordism Group

Fix a codimension k. The *framed cobordism group* Ω_n^{fr} is the set of framed cobordism classes of framed n-submanifolds of $\mathbb{R}^{n+k}$, with addition

$$[N_0,\varphi_0] + [N_1,\varphi_1] = [N_0 \sqcup N_1,\ \varphi_0 \sqcup \varphi_1]$$

given by disjoint union (the two summands placed in disjoint balls of $\mathbb{R}^{n+k}$) and identity element $[\emptyset]$, the class of the empty submanifold.

The additive notation anticipates a group, and Ω_n^{fr} is one. Only the existence of inverses requires an argument, and it is the argument that reveals why reversing a framing matters.

Proposition 10.2.7: Inverses and the Group Structure

Ω_n^{fr} is an abelian group. The inverse of $[N, \varphi]$ with $\varphi = (v_1, v_2, \ldots, v_k)$ is

$$-[N, \varphi] = [N, \varphi^-], \qquad \varphi^- = (-v_1, v_2, \ldots, v_k)$$

the class of the same submanifold with one framing vector reversed.

***Proof*:**
The monoid axioms. Disjoint union is associative and commutative on the level of framed submanifolds, and $\emptyset$ is a two-sided identity, so addition on classes is associative with identity $[\emptyset]$. Commutativity of the classes needs one word more: the two disjoint balls holding the summands can be isotoped past each other inside $\mathbb{R}^{n+k}$ (which is connected and of dimension at least 1), and the trace of a framed isotopy is a framed cobordism, so $[N_0] + [N_1] = [N_1] + [N_0]$.

Inverses. Fix the Euclidean metric and assume φ orthonormal, with v_1 a unit normal field along N. It suffices to show $(N, \varphi) \sqcup (N, \varphi^-)$ is framed null-cobordant. For small $\varepsilon > 0$ define

$$\Psi\colon N \times [0, \pi] \to \mathbb{R}^{n+k} \times [0, 1], \qquad \Psi(p, \theta) = \big(p + \varepsilon(1 - \cos\theta)\, v_1(p),\ \varepsilon \sin\theta\big)$$

For ε small this is an embedding: its image lies in the tubular neighbourhood of N (theorem 7.8.1), and a point of the image determines p by the nearest-point retraction, then $1 - \cos\theta$ from the v_1-displacement and $\sin\theta$ from the last coordinate, hence $\theta \in [0, \pi]$. Let $W = \Psi(N \times [0, \pi])$, a compact $(n+1)$-manifold with boundary. Its last coordinate $\varepsilon \sin\theta$ vanishes only at $\theta = 0$ and $\theta = \pi$, where $\partial_\theta(\varepsilon \sin\theta) = \pm\varepsilon \neq 0$, so W lies in $\mathbb{R}^{n+k} \times [0, \varepsilon] \subseteq \mathbb{R}^{n+k} \times [0, 1]$ and meets the slab $\mathbb{R}^{n+k} \times \{0\}$ transversally in its two boundary components

$$\Psi(N \times \{0\}) = N \times \{0\}, \qquad \Psi(N \times \{\pi\}) = (N + 2\varepsilon v_1) \times \{0\}$$

The second is the copy of N pushed off along v_1, disjoint from the first, and translation within $\mathbb{R}^{n+k}$ is a framed cobordism, so this boundary is $(N, \varphi) \sqcup (N, \varphi^-)$ once the framings are computed.

To frame W, note that its tangent space at $\Psi(p, \theta)$ is spanned by $\partial_\theta \Psi = (\varepsilon \sin\theta\, v_1, \varepsilon \cos\theta)$ together with the near-horizontal images of T_pN. The vectors $v_2(p), \ldots, v_k(p)$, having no last coordinate and being orthogonal to v_1 and to T_pN, stay normal to W for small ε. In the plane spanned by $(v_1, 0)$ and the last coordinate direction $(0, 1)$, the tangent $\partial_\theta \Psi$ points along $(\sin\theta, \cos\theta)$, whose unit normal is $(\cos\theta, -\sin\theta)$; thus

$$V_1(p, \theta) = \big(\cos\theta\, v_1(p),\ -\sin\theta\big)$$

is a unit normal to W, orthogonal to $v_2, \ldots, v_k$. Then $\Phi = (V_1, v_2, \ldots, v_k)$ is a framing of W. At $\theta = 0$ it restricts to $(v_1, v_2, \ldots, v_k) = \varphi$, and at $\theta = \pi$ to $(-v_1, v_2, \ldots, v_k) = \varphi^-$; both restrictions have vanishing last coordinate, so they are framings of N inside $\mathbb{R}^{n+k}$. Hence W is a framed null-cobordism of $(N, \varphi) \sqcup (N, \varphi^-)$, giving $[N, \varphi] + [N, \varphi^-] = 0$. So every class has an inverse, and Ω_n^{fr} is an abelian group, as we sought to show.

The fold Ψ is the mechanism behind the group's relations: a strand of N is lifted into the collar

$\mathbb{R}^{n+k} \times (0, \varepsilon]$ and brought back down, and as it turns through the angle π the framing vector v_1 rotates to $-v_1$. Reversing a framing vector is therefore invisible to the group only after the strand has been folded, which is precisely why the inverse of a framed manifold is its reflection rather than the manifold with no framing at all. The construction also settles the smallest case completely.

Example 10.2.8: Framed Points and Ω_0^{fr}

Take $n = 0$. A framed 0-submanifold of $\mathbb{R}^k$ is a finite set of points, each carrying a framing of its normal space, and the normal space of a point $p \in \mathbb{R}^k$ is all of $\mathbb{R}^k$. A framing of the point is thus an ordered basis $(v_1, \ldots, v_k)$ of $\mathbb{R}^k$, an element of $\mathrm{GL}_k(\mathbb{R})$. Since $\mathrm{GL}_k(\mathbb{R})$ has exactly two path components, distinguished by the sign of the determinant, a framed point carries a well-defined *sign* $\varepsilon(p) = \operatorname{sign}\det(v_1, \ldots, v_k) \in \{+1, -1\}$, and up to homotopy of framings this sign is the only invariant of the framed point.

A positively framed point and a negatively framed point cobound: they are $(N, \varphi) \sqcup (N, \varphi^-)$ for N a single point, so proposition 10.2.7 makes them framed null-cobordant, and a $+1$ point cancels a -1 point. The class of a framed 0-manifold is therefore its signed count $\sum_p \varepsilon(p) \in \mathbb{Z}$, and this count is a complete invariant, giving

$$\Omega_0^{\mathrm{fr}} \cong \mathbb{Z}$$

The isomorphism sends the class of a single positively framed point to 1. This signed count is the germ of the degree, and section 10.4 builds the Hopf degree theorem on it.

The rank-one case, framed circles, already shows that a framing carries information the bare manifold cannot, and it is the first place the homotopy of spheres appears.

Example 10.2.9: The Two Framings of the Circle

Consider first the circle in the plane, $S^1 \subseteq \mathbb{R}^2$, of codimension $k = 1$. As in example 10.2.2 it has exactly two framings up to homotopy, the outward φ_{out} and the inward $\varphi_{\mathrm{in}} = \varphi_{\mathrm{out}}^-$. Both are framed null-cobordant: the upper hemisphere of the unit sphere in $\mathbb{R}^2 \times [0, 1]$, with boundary the equator $S^1 \times \{0\}$ and framed by its outer (respectively inner) unit normal, is a framed null-cobordism of $(S^1, \varphi_{\mathrm{out}})$ (respectively φ_{in}). At codimension 1, then, every framed circle in the plane bounds, and Ω_1^{fr} in codimension 1 collapses.

Room to distinguish framings appears one codimension up. Regard $S^1 \subseteq \mathbb{R}^3$, codimension $k = 2$, with normal bundle of rank 2. A framing turns as one traverses the circle, and framings up to homotopy differ by an integer, the number of full rotations of the normal 2-frame; the untwisted framing, obtained by spanning a disc $D^2 \subseteq \mathbb{R}^3$ and restricting a framing of the disc, extends over that disc and so is framed null-cobordant. The framing that makes exactly one rotation (the left-invariant framing of the circle as a Lie group) does not bound. Under the correspondence of section 10.3 the framed circles of codimension 2 realize Ω_1^{fr} at codimension 2, which is $\pi_{1+2}(S^2) = \pi_3(S^2) \cong \mathbb{Z}$: the winding number of the framing is the Hopf invariant, so the once-rotation framing maps to a *generator* (the Hopf map η, of Hopf invariant 1) and has *infinite* order, while the w-fold twist is $w \in \mathbb{Z}$, no nonzero twist bounding. The torsion relation $2\eta = 0$ is not present at this codimension; it emerges only after *stabilization*. Pushing the circle into higher codimension collapses $\pi_3(S^2) = \mathbb{Z}$ onto the stable group $\pi_1^s \cong \mathbb{Z}/2$ (section 10.5; the stable value is cited to [11, The First Stable Stem is $\mathbb{Z}/2$]), where two copies of the twisted circle cobound.

A framing is the surplus of embedding data over bare cobordism, and Ω_n^{fr} is where it accumulates. The framed circle shows the surplus is nonzero: the once-twisted circle in $\mathbb{R}^3$ bounds no framed surface and survives in the group, though the unframed circle bounds a disc. This is the first sign that the homotopy of spheres has been encoded in framed manifolds.

Exercise 10.2.1

1. Show that $S^n \subseteq \mathbb{R}^{n+1}$ has exactly two framings up to homotopy, the outward φ_{out} and the inward φ_{in}, and that they are exchanged by the reflection $v \mapsto -v$. Conclude from proposition 10.2.7 that $[S^n, \varphi_{\text{out}}]$ and $[S^n, \varphi_{\text{in}}]$ are inverse classes in Ω_n^{fr}.

2. Let $f\colon \mathbb{R}^{n+k} \to \mathbb{R}^k$ be smooth with 0 a regular value, and $N = f^{-1}(0)$. Show that df induces a framing φ_f of N (the pullback framing) by trivializing $\nu_p(N) = T_pN^\perp \cong \mathbb{R}^k$ through df_p. Compute φ_f for $f(x) = |x|^2 - 1$ and identify it with a framing of example 10.2.2.

3. Prove directly that Ω_n^{fr} is abelian by exhibiting a framed cobordism from $(N_0, \varphi_0) \sqcup (N_1, \varphi_1)$ to $(N_1, \varphi_1) \sqcup (N_0, \varphi_0)$. (Slide the two disjoint balls of $\mathbb{R}^{n+k}$ past one another and take the trace of the isotopy.)

4. Let φ_s ($s \in [0, 1]$) be a homotopy of framings of a fixed compact $N \subseteq \mathbb{R}^{n+k}$, from φ_0 to φ_1. Construct a framed cobordism from (N, φ_0) to (N, φ_1), and conclude that the framed cobordism class of (N, φ) depends only on the homotopy class of φ.

5. In $\mathbb{R}^k$, place a positively framed point at a and a negatively framed point at $b \neq a$. Write down explicitly a framed cobordism between $\{a, b\}$ (with these framings) and $\emptyset$, and explain how it specializes the fold Ψ of proposition 10.2.7 to the case $N = \text{pt}$. Deduce that the signed count $\Omega_0^{\text{fr}} \to \mathbb{Z}$ of example 10.2.8 is additive.

10.3 The Pontryagin-Thom Construction

The dictionary promised in the chapter opening now becomes precise. Two constructions run in opposite directions. A framed submanifold of a sphere collapses onto a map of spheres, by crushing everything outside a framed tubular neighbourhood to a point; and a regular value of a map of spheres has a preimage that is a submanifold, framed by the derivative of the map. The task of this section is to show that both passages descend to equivalence classes, framed cobordism on one side and homotopy on the other, and that once so descended they are mutually inverse. The homotopy groups of spheres are thereby identified with framed cobordism groups of manifolds, an identification proved entirely with the transversality and tubular-neighbourhood machinery already in hand.

Throughout, the sphere S^{n+k} is the one-point compactification $\mathbb{R}^{n+k} \cup \{\infty\}$, and ∞ is its basepoint; likewise $S^k = \mathbb{R}^k \cup \{\infty\}$ with basepoint ∞. A framed submanifold $N \subseteq \mathbb{R}^{n+k}$ of codimension k (definition 10.2.1) then sits inside S^{n+k} away from ∞. Because a compact submanifold, and any framed cobordism between two of them, occupies a bounded region of $\mathbb{R}^{n+k}$, framed cobordism computed in $\mathbb{R}^{n+k}$ agrees with framed cobordism computed in S^{n+k}; the two versions of Ω_n^{fr} coincide, and either may be used. Write Ω_n^{fr} for the codimension-k framed cobordism group (definition 10.2.6), and

$$\pi_{n+k}(S^k) = \big[\,(S^{n+k}, \infty),\ (S^k, \infty)\,\big]$$

for the set of based homotopy classes of based maps. For $k \geq 2$ the target is simply connected and based homotopy agrees with free homotopy, so basepoints may be dropped without changing the set ([11]); the constructions below respect the basepoint in any case.

One reduction is used repeatedly and is recorded once. Every continuous map $S^{n+k} \to S^k$ is homotopic to a smooth one, and two smooth maps that are continuously homotopic are smoothly homotopic, both by Whitney approximation (theorem 7.9.1, applied to a homotopy relative to the two ends where the map is already smooth). Consequently $\pi_{n+k}(S^k)$ is represented by smooth maps up to smooth homotopy, and only smooth maps and smooth homotopies appear from here on.

Definition 10.3.1: The Collapse Map of a Framed Submanifold

Let $N^n \subseteq S^{n+k}$ be a compact framed submanifold with framing $\mathbf{v} = (v_1, \dots, v_k)$. By the tubular neighbourhood theorem (theorem 7.8.1) together with the framing trivialization $\nu(N) \cong N \times \mathbb{R}^k$ (definition 10.2.1, proposition 7.7.2), fix, for some $\varepsilon > 0$, a diffeomorphism

$$\varphi\colon N \times B_\varepsilon \to U, \qquad B_\varepsilon = \left\{ t \in \mathbb{R}^k \;\middle|\; |t| < \varepsilon \right\},$$

onto an open neighbourhood U of N in S^{n+k}, with $\varphi(x, 0) = x$ and with fiber derivative $\partial_{t^i}\varphi(x, 0) = v_i(x)$. Fix a radial diffeomorphism $\alpha\colon B_\varepsilon \to \mathbb{R}^k$, $\alpha(t) = \lambda(|t|)\, t$, equal to the identity on the closed half-ball $\overline{B_{\varepsilon/2}}$ and with $1/\lambda(r)$ vanishing to infinite order as $r \to \varepsilon$. The *collapse map* of $(N, \mathbf{v})$ is

$$p(N)\colon S^{n+k} \to S^k, \qquad p(N)(y) = \begin{cases} \alpha\big(\pi_2(\varphi^{-1}(y))\big), & y \in U, \\ \infty, & y \notin U, \end{cases}$$

where $\pi_2\colon N \times B_\varepsilon \to B_\varepsilon$ is the projection.

The map is smooth. On U it is a composite of smooth maps into $\mathbb{R}^k \subseteq S^k$, and on the open complement of $\overline{U}$ it is the constant ∞; only the points of ∂U require checking. In the chart $w\colon S^k \setminus \{0\} \to \mathbb{R}^k$, $w(u) = u/|u|^2$, $w(\infty) = 0$, a point $y \in U$ with $s = \pi_2(\varphi^{-1}(y))$ has

$$w\big(p(N)(y)\big) = \frac{\alpha(s)}{|\alpha(s)|^2} = \frac{s}{|s|^2} \cdot \frac{1}{\lambda(|s|)}$$

and $|s| \to \varepsilon$ as $y \to \partial U$. Near the rim $\lambda(r) = \exp\big(1/(\varepsilon^2 - r^2)\big)$, so $1/\lambda(|s|) = \exp\big(-1/(\varepsilon^2 - |s|^2)\big)$ together with all of its derivatives tends to 0, while $s/|s|^2$ stays bounded; the displayed expression therefore extends by the constant $0 = w(\infty)$ across ∂U to a smooth function, so $p(N)$ is smooth. The basepoint ∞ of S^{n+k} lies outside the bounded set U, and $p(N)(\infty) = \infty$, so $p(N)$ is based.

Two properties are read directly off the construction, and they are what make the collapse map the inverse of the preimage construction below. First, $\alpha^{-1}(0) = 0$, so $p(N)^{-1}(0) = \varphi(N \times \{0\}) = N$; and since α is a diffeomorphism fixing 0 while $\pi_2 \circ \varphi^{-1}$ is a submersion on U restricting to an isomorphism $\nu_x(N) \to \mathbb{R}^k$ on each normal space, 0 is a regular value of $p(N)$. Second, because $\alpha = \mathrm{id}$ near 0 its derivative there is the identity, so the framing of N that $d\,p(N)$ induces at 0, namely the preimage under $d\,p(N)_x|_{\nu_x}$ of the standard basis of $T_0\mathbb{R}^k$, is exactly $\mathbf{v}$. The collapse map thus encodes both the submanifold and its framing.

The homotopy class $[p(N)] \in \pi_{n+k}(S^k)$ turns out to depend only on the framed cobordism class of N, not on the choices of ε, of φ, or of α; this is lemma 10.3.6 below, of which choice-independence is the special case of the constant cobordism $N \times [0, 1]$.

Example 10.3.2: Collapsing Framed Points

Take $n = 0$, so $N \subseteq S^k$ is a finite framed 0-manifold: a finite set of points $a_1, \dots, a_r$, each equipped with a framing of its k-dimensional normal space $\nu_{a_j}(N) = T_{a_j} S^k$, that is, an ordered basis, whose sign $\epsilon_j \in \{\pm 1\}$ records whether it agrees with a fixed orientation of S^k (example 10.2.8). Disjoint balls $\varphi(\{a_j\} \times B_\varepsilon)$ around the points give the tubular neighbourhood, and the collapse map $p(N)\colon S^k \to S^k$ sends each ball onto S^k by the framing chart and everything else to ∞.

The value $0 \in \mathbb{R}^k$ is regular with $p(N)^{-1}(0) = \{a_1, \dots, a_r\}$, and the sign of $\det d\,p(N)_{a_j}$ is ϵ_j. So $\deg p(N) = \sum_{j=1}^r \epsilon_j$. A single positively framed point gives a degree-one map, homotopic to the identity; the empty framed manifold gives the constant map ∞, of degree 0. The signed count of the framed points is precisely the degree of the collapse map, the case $n = 0$ that section 10.4 turns into a proof of the Hopf degree theorem.

The reverse passage starts from a map and produces a framed manifold. A regular value has a preimage that is a submanifold of the right dimension, and the derivative of the map trivializes its normal bundle.

Definition 10.3.3: The Framed Preimage of a Regular Value

Let $g\colon S^{n+k} \to S^k$ be smooth and let $y \in \mathbb{R}^k \subseteq S^k$ be a regular value of g (theorem 7.2.1 provides one, and $y \neq \infty$ may be assumed). The *framed preimage* is the submanifold $N = g^{-1}(y)$ together with the framing described below. By the regular level set theorem (corollary 3.2.10), N is a compact boundaryless submanifold of dimension $(n+k) - k = n$. At each $x \in N$ the derivative $dg_x\colon T_x S^{n+k} \to T_y S^k$ has kernel $T_x N$ and induces an isomorphism

$$\overline{dg_x}\colon \nu_x(N) = T_x S^{n+k}/T_x N \xrightarrow{\ \sim\ } T_y S^k \cong \mathbb{R}^k.$$

Fixing the standard basis $e_1, \dots, e_k$ of $T_y \mathbb{R}^k = \mathbb{R}^k$, the *induced framing* of N is $\mathbf{v} = (v_1, \dots, v_k)$ with $v_i(x) = \overline{dg_x}^{-1}(e_i)$, a smoothly varying trivialization of $\nu(N)$.

The framing is the differential-topological content of "regular value": surjectivity of dg_x is exactly the statement that the normal space maps isomorphically to $\mathbb{R}^k$, and inverting that isomorphism turns the standard frame of the target into a frame of the normal bundle of the preimage. The construction of definition 10.3.1 was rigged so that its collapse map returns this framing: for a framed N, the framed preimage of 0 under $p(N)$ is N with its original framing.

Example 10.3.4: The Framed Preimage of a Power Map

Identify $S^1 = \mathbb{R}/\mathbb{Z}$ and take $g\colon S^1 \to S^1$, $g(z) = z^d$ for an integer $d \neq 0$ (here $n = 0$, $k = 1$). Every value is regular, since g is a submersion. For a regular value y, the preimage $g^{-1}(y)$ is the set of $|d|$ solutions of $z^d = y$, evenly spaced on the circle. At each solution x the derivative $g'(x) = d\,x^{d-1}$ multiplies the positively oriented unit tangent by d, so the induced framing of the normal line (here $T_x S^1$ itself, since $\nu_x = T_x S^1$ for a 0-manifold in a 1-manifold) has sign $\operatorname{sign}(d)$ at every point. The framed preimage is thus $|d|$ points each of sign $\operatorname{sign}(d)$, a framed 0-manifold whose signed count is d. This matches example 10.3.2 in reverse: the map $z \mapsto z^d$ is the collapse map of d signed points, and its framed preimage recovers them.

Both constructions must be shown to respect equivalence. The first lemma sends homotopy to framed cobordism and thereby makes the preimage well defined on homotopy classes; the second sends framed cobordism to homotopy and makes the collapse well defined on cobordism classes.

The idea for the first is that a homotopy between two maps is itself a map on the cylinder, and the preimage of a regular value of that map is a manifold whose boundary is exactly the two preimages at the ends.

Lemma 10.3.5: Homotopic Maps Have Framed-Cobordant Preimages

Let $g_0, g_1 : S^{n+k} \to S^k$ be smoothly homotopic, and let y_i be a regular value of g_i for $i = 0, 1$. Then the framed preimages $g_0^{-1}(y_0)$ and $g_1^{-1}(y_1)$ are framed cobordant. In particular the framed cobordism class of $g^{-1}(y)$ depends neither on the regular value y nor on the map g within its homotopy class.

***Proof*:**
Step 1: reduce to a common regular value. By the homogeneity of the sphere (lemma 8.4.3) there is a diffeomorphism $h\colon S^k \to S^k$, isotopic to the identity, with $h(y_1) = y_0$. Then $h \circ g_1$ is homotopic to g_1 (compose g_1 with the isotopy), has y_0 as a regular value, and $(h \circ g_1)^{-1}(y_0) = g_1^{-1}(y_1)$; the induced framings differ only by the isomorphism dh_{y_1} of $\mathbb{R}^k$, which, being isotopic to the identity through linear isomorphisms, gives a homotopic and hence framed-cobordant framing. Replacing g_1 by $h \circ g_1$, assume $y_0 = y_1 =: y$.

Step 2: the cylinder preimage is a framed cobordism. Let $G\colon S^{n+k} \times [0,1] \to S^k$ be a smooth homotopy with $G(\cdot, 0) = g_0$ and $G(\cdot, 1) = g_1$. Its boundary restriction $G|_\partial = g_0 \sqcup g_1$ has y as a regular value, hence is transverse to the point $\{y\}$; by boundary-fixing transversality (lemma 8.1.5), G is homotopic, through maps fixed on $S^{n+k} \times \{0, 1\}$, to a smooth map for which y is a regular value of G itself. Replacing G by this map leaves g_0, g_1, and their framed preimages unchanged, and now y is a regular value of both G and $G|_\partial$. By the regular preimage theorem for manifolds with boundary (proposition 3.2.19),

$$W = G^{-1}(y) \subseteq S^{n+k} \times [0,1]$$

is a compact $(n+1)$-manifold with boundary

$$\partial W = W \cap \big(S^{n+k} \times \{0,1\}\big) = g_0^{-1}(y) \times \{0\} \ \sqcup\ g_1^{-1}(y) \times \{1\}$$

Step 3: the framing and its restriction. At each $x \in W$ the derivative dG_x has kernel T_xW and induces an isomorphism $\nu_x(W) \to T_yS^k \cong \mathbb{R}^k$; inverting it on the standard basis $e_1, \dots, e_k$ frames $\nu(W)$, exactly as in definition 10.3.3. On the end $S^{n+k} \times \{i\}$ the restriction $G|_{S^{n+k}\times\{i\}} = g_i$ has the same derivative in the sphere directions, so this framing restricts on $g_i^{-1}(y)$ to the framing induced by dg_i. Finally, $y \neq \infty$ implies W avoids $G^{-1}(\infty) \supseteq \{\infty\} \times [0,1]$, so $W \subseteq (S^{n+k} \setminus \{\infty\}) \times [0,1] = \mathbb{R}^{n+k} \times [0,1]$ is a framed cobordism in the sense of definition 10.2.4. Hence $g_0^{-1}(y)$ and $g_1^{-1}(y)$ are framed cobordant, as we sought to show.

The lemma is what makes the correspondence well posed on one side: the preimage construction, which a priori depends on a choice of smooth representative and of regular value, lands in a single framed cobordism class. Taking $g_0 = g_1$ shows different regular values of the same map give framed-cobordant preimages; taking distinct homotopic maps shows the class is a homotopy invariant.

The converse lemma runs the same construction one dimension up. A framed cobordism between two submanifolds is a framed submanifold of the cylinder, and its collapse map is a homotopy between the collapse maps of the ends.

Lemma 10.3.6: Framed-Cobordant Submanifolds Have Homotopic Collapse Maps

Let $N_0, N_1 \subseteq S^{n+k}$ be framed cobordant compact submanifolds. Then their collapse maps $p(N_0)$ and $p(N_1)$ are homotopic. In particular $[p(N)]$ is independent of the choices of tubular neighbourhood and radial reparametrization made in definition 10.3.1, so $[p(N)]$ depends only on the framed cobordism class of N.

***Proof*:**
Let $W^{n+1} \subseteq \mathbb{R}^{n+k} \times [0,1]$ be a framed cobordism with $\partial W = N_0 \times \{0\} \sqcup N_1 \times \{1\}$, its framing restricting to those of N_0 and N_1. By the collar neighbourhood theorem (theorem 1.2.20) the cobordism may be taken to be a product near the ends: $W \cap \big(\mathbb{R}^{n+k} \times [0,\delta)\big) = N_0 \times [0,\delta)$ and $W \cap \big(\mathbb{R}^{n+k} \times (1-\delta,1]\big) = N_1 \times (1-\delta,1]$, with the framing constant in the collar coordinate.

Step 1: a product-collared tube of W. Apply the tubular neighbourhood theorem (theorem 7.8.1) to W in the interior $\mathbb{R}^{n+k} \times (0,1)$ and, in the two collars, use the product of the given tube of N_i with the interval; a partition of unity in the collar coordinate splices these into a single diffeomorphism

$$\Phi\colon W \times B_\varepsilon \to \mathcal{U}, \qquad \mathcal{U} \subseteq S^{n+k} \times [0,1] \text{ open},$$

onto a neighbourhood of W, framed and equal to the product $\varphi_i \times \mathrm{id}$ in the two collars. Composing with the radial α of definition 10.3.1 defines the collapse map of W,

$$P\colon S^{n+k} \times [0,1] \to S^k, \qquad P(y,u) = \begin{cases} \alpha\big(\pi_2(\Phi^{-1}(y,u))\big), & (y,u) \in \mathcal{U}, \\ \infty, & (y,u) \notin \mathcal{U}, \end{cases}$$

smooth by the same rim estimate as in definition 10.3.1.

Step 2: the ends. In the collar $u \in [0,\delta)$ the tube is the product $\varphi_0 \times \mathrm{id}$, so $\Phi^{-1}(y,u) = (\varphi_0^{-1}(y), u; t)$ and $P(y,u) = \alpha(\pi_2(\varphi_0^{-1}(y))) = p(N_0)(y)$, independent of u; likewise $P(\cdot,u) = p(N_1)$ for u near 1. Thus $u \mapsto P(\cdot,u)$ is a homotopy from $p(N_0)$ to $p(N_1)$, and it fixes the basepoint since $\{\infty\} \times [0,1]$ lies outside the bounded set $\mathcal{U}$. Hence $p(N_0) \simeq p(N_1)$.

For the independence claim, take $W = N \times [0,1]$ with the constant framing but with the two given collapse data (tube and α) installed at the two ends; the construction above interpolates them, so the two collapse maps of N are homotopic, completing the proof.

With both passages descending to equivalence classes, they may be compared. Define

$$P\colon \Omega_n^{\mathrm{fr}} \to \pi_{n+k}(S^k), \quad P[N,\mathbf{v}] = [p(N)], \qquad Q\colon \pi_{n+k}(S^k) \to \Omega_n^{\mathrm{fr}}, \quad Q[g] = [g^{-1}(y), \mathbf{v}_{dg}],$$

well defined by lemma 10.3.6 and lemma 10.3.5 respectively. The theorem is that they are inverse to one another.

Theorem 10.3.7: The Pontryagin-Thom Bijection

For $n \geq 0$ and $k \geq 1$, the maps P and Q are mutually inverse. Hence the collapse construction is a bijection

$$\Omega_n^{\mathrm{fr}} \xrightarrow{\cong} \pi_{n+k}(S^k) = \left[\, S^{n+k}, S^k \,\right]$$

between the group of framed cobordism classes of codimension-k framed n-manifolds and the $(n+k)$-th homotopy group of S^k.

***Proof*:**

Step 1: $Q \circ P = \mathrm{id}$. Let $(N, \mathbf{v})$ be framed. By definition 10.3.1 and the two properties recorded after it, 0 is a regular value of $p(N)$ with $p(N)^{-1}(0) = N$, and the framing that $d\,p(N)$ induces at 0 is $\mathbf{v}$. Thus $Q(P[N, \mathbf{v}]) = Q[p(N)] = [N, \mathbf{v}]$.

Step 2: $P \circ Q = \mathrm{id}$. Let $g\colon S^{n+k} \to S^k$ be smooth with regular value 0, and $N = g^{-1}(0)$ framed by dg. It suffices to prove $g \simeq p(N)$ for a suitable choice of the collapse data, since $[p(N)]$ is independent of that choice (lemma 10.3.6).

A tube adapted to g. Choose any tubular retraction $\pi_N\colon U_0 \to N$ (theorem 7.8.1) and form $\Xi = (\pi_N, g)\colon U_0 \to N \times \mathbb{R}^k$. At $x \in N$ the derivative $d\Xi_x$ is the identity on T_xN (since $\pi_N|_N = \mathrm{id}$ and $dg_x|_{T_xN} = 0$) and carries $\nu_x(N)$ isomorphically to $\mathbb{R}^k$ (since $d\pi_N$ kills ν_x and dg_x is onto), hence is an isomorphism. By the inverse function theorem and compactness of N, Ξ is a diffeomorphism from a neighbourhood U of N onto an open set containing $N \times \{0\}$; shrink so its image contains $N \times B_\varepsilon$ and set $\varphi = \Xi^{-1}|_{N \times B_\varepsilon}$. Then $g(\varphi(x,t)) = t$ and $\varphi(x, 0) = x$, and the framing φ induces is $v_i(x) = \overline{dg_x}^{-1}(e_i)$, exactly the dg-framing of N. Build $p(N)$ from this φ and an α that is the identity on $\overline{B_{\varepsilon/2}}$.

The two maps agree near N. On the smaller tube $U' = \varphi(N \times B_{\varepsilon/2})$ one has $g = \pi_2 \circ \varphi^{-1}$ with values in $B_{\varepsilon/2}$, where $\alpha = \mathrm{id}$, so $p(N) = \alpha \circ g = g$ on the closure $\overline{U'}$.

Away from N the target is contractible. Set $C = S^{n+k} \setminus U'$. Since $g^{-1}(0) = N \subseteq U'$, the map g sends C into $S^k \setminus \{0\}$; and $p(N)$ sends C into $S^k \setminus \{0\}$ as well, being $\alpha(t)$ with $|t| \geq \varepsilon/2$ on $C \cap U$ and ∞ off U. The stereographic diffeomorphism $\Theta\colon S^k \setminus \{0\} \to \mathbb{R}^k$ identifies this common target with a convex set. On C the maps g and $p(N)$ agree along $\partial C = \partial U' \subseteq \overline{U'}$, so the straight-line homotopy

$$H_\sigma(y) = \Theta^{-1}\big((1-\sigma)\,\Theta g(y) + \sigma\,\Theta p(N)(y)\big), \qquad y \in C,\ \ \sigma \in [0,1],$$

stays in $S^k \setminus \{0\}$ (convexity of $\mathbb{R}^k$) and is constant on ∂C (where $g = p(N)$). Extending H_σ by the common value $g = p(N)$ on $\overline{U'}$ gives a smooth homotopy of S^{n+k}, from g to $p(N)$, fixing the basepoint. Hence $g \simeq p(N)$ and $P(Q[g]) = [g]$.

Steps 1 and 2 show $Q \circ P = \mathrm{id}$ and $P \circ Q = \mathrm{id}$, so P is a bijection with inverse Q, completing the proof.

A homotopy class of maps $S^{n+k} \to S^k$, an object of algebraic topology with no evident geometric content, is the same datum as a framed n-manifold up to framed cobordism, an object built by hand out of transversality and tubular neighbourhoods. Nothing in the proof used any homotopy theory beyond the definition of $\pi_{n+k}(S^k)$: Sard's theorem produced the preimages, the boundary preimage theorem made a homotopy into a cobordism, and the tubular neighbourhood theorem ran the collapse. The identification is an isomorphism of groups once both sides are given their natural additions, disjoint union of framed manifolds on the left and the track sum on the right; disjoint framed submanifolds placed in disjoint balls collapse to the sum of the two maps, so P carries $[N_0] + [N_1]$ to $[p(N_0)] + [p(N_1)]$, the additivity worked out in the exercises.

The construction sits inside the spectrum-level picture of the companion volume. Letting the codimension k grow (section 10.5) stabilizes both sides, and the stable value of $\pi_{n+k}(S^k)$ is the n-th stable stem π_n^s; the unoriented analogue, in which framings are dropped, computes the cobordism ring through the Thom spectrum MO. Both are instances of the general principle that a Thom spectrum's homotopy groups are cobordism groups, developed in [11, Thom's Theorem, The Pontryagin-Thom Isomorphism, Oriented Bordism and the Spectrum MSO]. What this section proves is the differential-topological bijection at a fixed codimension, from which those statements follow by stabilization.

Exercise 10.3.1

1. Show that the empty framed manifold $\emptyset \subseteq S^{n+k}$ has collapse map the constant map at the basepoint ∞, and that under the bijection of theorem 10.3.7 the identity element $[\emptyset] \in \Omega_n^{\mathrm{fr}}$ corresponds to the class of the constant map. Conclude that a framed submanifold is null-cobordant (bounds a framed manifold) if and only if its collapse map is null-homotopic.
2. Let $g\colon S^{n+k} \to S^k$ be smooth and let y, y' be two regular values in $\mathbb{R}^k$. Using only lemma 8.4.3 and definition 10.3.3 (not the full bijection), show directly that $g^{-1}(y)$ and $g^{-1}(y')$ are framed cobordant.
3. On $N = S^1 \subseteq \mathbb{R}^2 \subseteq S^3$ (codimension $k = 2$), the two framings of example 10.2.9 give collapse maps $S^3 \to S^2$. Explain why one is null-homotopic and the other is not, and identify the non-trivial class with the generator of $\pi_3(S^2) \cong \mathbb{Z}$ (the Hopf map η); cite [11, The Homotopy of S^2 via the Hopf Fibration] for the value of the group.
4. Define the sum of two based maps $f_0, f_1\colon S^{n+k} \to S^k$ by pinching S^{n+k} along an equator into a wedge $S^{n+k} \vee S^{n+k}$ and applying f_0 and f_1 to the two summands. Show that if $N_0, N_1 \subseteq S^{n+k}$ are framed submanifolds lying in the two open hemispheres, then $p(N_0 \sqcup N_1) \simeq p(N_0) + p(N_1)$, so that P is a homomorphism from $(\Omega_n^{\mathrm{fr}}, \sqcup)$ to $(\pi_{n+k}(S^k), +)$.
5. Where in the proof of theorem 10.3.7 is it used that $\pi_{n+k}(S^k)$ may be computed with smooth maps and smooth homotopies rather than continuous ones? Name the two places and the theorem that permits the reduction.

10.4 The Hopf Degree Theorem via Framed 0-Manifolds

The Pontryagin-Thom bijection of section 10.3 matches framed cobordism classes of n-manifolds with homotopy classes of maps between spheres. Its lowest case, $n = 0$, is the one that can be computed by hand: a framed 0-manifold is a finite set of points each carrying a sign, so Ω_0^{fr} is a signed point count, the group $\mathbb{Z}$. Read through the bijection, that integer is the degree of a self-map of a sphere.

This section makes the identification precise and spends it. The degree turns out to be a complete homotopy invariant of self-maps of S^m in every dimension, which is the Hopf degree theorem: theorem 8.4.10 proved it for $m = 1$ and deferred the general case to the framed cobordism machinery now in hand.

A framed submanifold of dimension 0 in S^m is a finite set of points, and definition 10.2.1 attaches to each a framing, an ordered basis of its normal space. A point p has trivial tangent space, so its normal space in S^m is all of T_pS^m, and a framing at p is an ordered basis of T_pS^m. Comparing that basis with a fixed orientation of S^m collapses the framing to a single bit of data, a sign. Example 10.2.8 already read Ω_0^{fr} as the resulting signed count; the proposition below proves the identification and fixes the sign, which is the quantity the degree computation requires.

Proposition 10.4.1: Framed 0-Manifolds Are Signed Points

Fix the standard orientation of S^m.

1. A framing of a point $p \in S^m$ is an ordered basis of T_pS^m. Two framings of p are homotopic through framings if and only if they induce the same orientation of T_pS^m; write $\varepsilon_p = +1$ when the framing is positively oriented and $\varepsilon_p = -1$ otherwise.
2. The *signed count* $c(N, F) = \sum_{p \in N} \varepsilon_p$ depends only on the framed cobordism class of the framed 0-manifold (N, F), and

$$c\colon \Omega_0^{\mathrm{fr}} \to \mathbb{Z}$$

is a group isomorphism.

***Proof*:**
Step 1: A single framed point. The framings of a fixed point p are the ordered bases of T_pS^m, that is, the elements of $\mathrm{GL}_m(\mathbb{R})$ once a reference basis is chosen. A homotopy through framings of p is a path in $\mathrm{GL}_m(\mathbb{R})$, and $\mathrm{GL}_m(\mathbb{R})$ has exactly two path components, distinguished by the sign of the determinant, hence by the orientation the basis induces. So two framings of p are homotopic through framings precisely when they induce the same orientation, and ε_p is well defined. Such a homotopy is itself a framed cobordism: the cylinder $\{p\} \times [0, 1] \subseteq S^m \times [0, 1]$, framed by the path of bases, cobounds the two framed points.

Step 2: The signed count is a framed cobordism invariant. Let (N_0, F_0) and (N_1, F_1) cobound a framed 1-manifold (W, F) in $S^m \times [0, 1]$, with W meeting the ends transversally and $\partial W = (N_0 \times \{0\}) \sqcup (N_1 \times \{1\})$. Using the collar of the boundary (theorem 7.8.1), arrange the framing to be a product near each end: the framing vectors $F_1, \ldots, F_m$ are tangent to the slices $S^m \times \{s\}$ and constant in s there. Orient $S^m \times [0, 1]$ by the standard orientation of S^m followed by $\partial/\partial s$. The framing orients W: at each $w \in W$ choose the unit tangent $\tau(w)$ making $(F_1(w), \ldots, F_m(w), \tau(w))$ a positive basis of $T_w(S^m \times [0, 1])$. This choice is nowhere zero, so it orients the compact 1-manifold W.

A compact oriented 1-manifold is a disjoint union of arcs and circles (theorem 5.11.1), and the boundary signs of an oriented compact 1-manifold sum to zero (theorem 8.3.6), so

$$\sum_{q \in \partial W} \sigma_q = 0$$

where $\sigma_q = +1$ if τ points out of W at q and $\sigma_q = -1$ if it points in. Near a top endpoint

$q \in N_1 \times \{1\}$ the framing is a product, so positivity of $(F_1, \ldots, F_m, \tau)$ forces the $\partial/\partial s$-component of $\tau(q)$ to have sign ε_q; since the outward direction there is $+\partial/\partial s$, this gives $\sigma_q = \varepsilon_q$. At a bottom endpoint $q \in N_0 \times \{0\}$ the outward direction is $-\partial/\partial s$, so $\sigma_q = -\varepsilon_q$. Substituting,

$$0 = \sum_{q \in \partial W} \sigma_q = \sum_{q \in N_1} \varepsilon_q - \sum_{q \in N_0} \varepsilon_q = c(N_1, F_1) - c(N_0, F_0)$$

so c is constant on framed cobordism classes. It is additive under disjoint union by inspection, hence a homomorphism $\Omega_0^{\mathrm{fr}} \to \mathbb{Z}$.

Step 3: c is an isomorphism. For surjectivity, k positively framed points give $c = k$ and a single negatively framed point gives $c = -1$, so every integer is attained. For injectivity, suppose $c(N, F) = 0$. Then N has equally many positive and negative points; pair each positive point with a negative one and join the pair by an embedded arc that dips into the interior $S^m \times (0, 1)$ and returns to $S^m \times \{0\}$, the arcs chosen disjoint. The normal bundle of an arc is trivial, and traversing the U-turn reverses the s-direction, hence the orientation used to sign a framing; the two endpoint framings, of opposite sign $\varepsilon_{p_+} = -\varepsilon_{p_-}$, therefore correspond under the trivialization to bases in the same component of $\mathrm{GL}_m(\mathbb{R})$, so a path of bases connects them and frames the arc. The union of these framed arcs is a framed 1-manifold with boundary N, exhibiting (N, F) as null-cobordant. Thus $\ker c = 0$, and c is an isomorphism, as we sought to show.

The proposition strips framed 0-dimensional cobordism to arithmetic: a framed point is a plus or a minus, disjoint union adds, and the only relation is that a plus and a minus in the same sphere annihilate. The isomorphism c is the invariant every later identification passes through, and the sign ε_p is the local datum the degree will give.

Example 10.4.2: Canceling and Persistent Point Pairs

Two points $p, q \in S^m$ with opposite framings, $\varepsilon_p = +1$ and $\varepsilon_q = -1$, form a framed 0-manifold of signed count 0. By proposition 10.4.1 it is null-cobordant, and the null-cobordism is explicit: the U-turn arc that rises from p into $S^m \times (0, 1)$ and descends to q, carrying a path of frames whose orientation reverses across the turn. Two points with the *same* sign have count $\pm 2 \neq 0$ and bound nothing; no framed arc can join them, since the two ends of a single framed arc based at $S^m \times \{0\}$ always carry opposite signs. The signed count is exactly the obstruction to canceling points against one another.

The identification of Ω_0^{fr} with $\mathbb{Z}$ needs nothing on the framed side. The content is that the general Pontryagin-Thom bijection, specialized to $n = 0$, turns this signed count into the degree of a map. A regular value of a self-map of S^m has a finite preimage, and the derivative frames it; the sign of that frame is the local degree sign.

Theorem 10.4.3: The Hopf Degree Theorem in All Dimensions

For every $m \geq 1$, two smooth maps $f, g\colon S^m \to S^m$ are smoothly homotopic if and only if $\deg f = \deg g$. Equivalently, the degree

$$\deg\colon [\, S^m, S^m \,] = \pi_m(S^m) \longrightarrow \mathbb{Z}$$

is a bijection.

***Proof*:**
Let $y \in S^m$ be a regular value of g (theorem 7.2.1), and give T_yS^m a positively oriented basis $(w_1, \ldots, w_m)$, the standard framing of the point y. By definition 10.3.3 the preimage $g^{-1}(y) = \{x_1, \ldots, x_r\}$ is a framed 0-manifold, the framing at each x being the pullback $\big((dg_x)^{-1}w_1, \ldots, (dg_x)^{-1}w_m\big)$ of the framing of y through the isomorphism $dg_x\colon T_xS^m \to T_yS^m$. Its sign is

$$\varepsilon_x = \operatorname{sign}\det\big[(dg_x)^{-1}w_1, \ldots, (dg_x)^{-1}w_m\big] = \operatorname{sign}\det(dg_x)^{-1} = \operatorname{sign}\det dg_x$$

since $(w_1, \ldots, w_m)$ is positive and a matrix and its inverse have determinants of the same sign. Summing over the preimage and comparing with definition 8.4.1,

$$c\big(g^{-1}(y)\big) = \sum_{x \in g^{-1}(y)} \operatorname{sign}\det dg_x = \deg g$$

so the signed count of the framed preimage is the degree.

The Pontryagin-Thom bijection (theorem 10.3.7), specialized to $n = 0$ and $k = m$, is the map

$$Q\colon \pi_m(S^m) \to \Omega_0^{\mathrm{fr}}, \qquad Q[g] = \big[g^{-1}(y)\big]$$

and it is a bijection. Composing with the isomorphism $c\colon \Omega_0^{\mathrm{fr}} \to \mathbb{Z}$ of proposition 10.4.1 gives a bijection $c \circ Q\colon \pi_m(S^m) \to \mathbb{Z}$, which by the computation above sends $[g]$ to $c(g^{-1}(y)) = \deg g$. A composite of bijections is a bijection, so deg is a bijection $\pi_m(S^m) \to \mathbb{Z}$. In particular deg is injective: $\deg f = \deg g$ forces $[f] = [g]$, that is, $f \simeq g$. The converse is the homotopy invariance of the degree (theorem 8.4.4), completing the proof.

This closes theorem 8.4.10. The general-dimensional case, deferred in section 8.4 to the framed cobordism machinery, is exactly the composite $c \circ Q$ computed here, and the lift argument given there for $m = 1$ is its lowest instance. The two isomorphisms carry all the weight: Q is transversality (a regular value of a homotopy frames a cobordism, theorem 10.3.7), and c is the arithmetic of signed points. The degree is neither an accident of intersection counting nor of the algebraic topology of $\pi_m(S^m)$; it is the same integer seen through a framed manifold. That deg is surjective recovers the fact that every integer occurs as a degree, and that it is injective is the classification: the homotopy classes of self-maps of S^m are indexed by $\mathbb{Z}$, one class per degree, in every dimension.

Reading the identification $c(g^{-1}(y)) = \deg g$ on its own gives the degree a geometric meaning independent of the choice of regular value.

Corollary 10.4.4: Degree as a Framed Cobordism Invariant

For a smooth map $f\colon S^m \to S^m$ and any regular value y, the framed preimage $f^{-1}(y)$ has framed cobordism class

$$\left[f^{-1}(y)\right] = \deg f \in \Omega_0^{\mathrm{fr}} \cong \mathbb{Z}$$

In particular $\deg f$ is independent of the regular value y and invariant under smooth homotopy of f.

***Proof*:**
The identity $c(f^{-1}(y)) = \deg f$ is theorem 10.4.3, and c is the isomorphism of proposition 10.4.1, so $[f^{-1}(y)] = \deg f$ under it. The class $[f^{-1}(y)]$ does not depend on y and is a homotopy invariant because $Q[f] = [f^{-1}(y)]$ is well defined on homotopy classes (theorem 10.3.7); hence so is $\deg f$, as we sought to show.

The corollary re-proves the well-definedness of the degree (theorem 8.4.4) from the framed side: independence of the regular value and homotopy invariance are one fact, that the framed preimage has a well-defined cobordism class, rather than two applications of the boundary theorem. The degree, born in section 8.4 as an oriented intersection count against a point, is now a framed cobordism invariant.

Three consistency checks tie the framed picture to the invariants of chapter 8. First, reducing the signed count modulo 2 discards each sign, since every $\varepsilon_x \equiv 1 \pmod 2$, so

$$\deg f \equiv \# f^{-1}(y) \equiv \deg_2 f \pmod 2$$

the mod-2 degree of definition 8.1.14: it is the image of $\deg f$ under the reduction $\Omega_0^{\mathrm{fr}} \to \mathfrak{N}_0 \cong \mathbb{F}_2$ that forgets the framing (definition 10.1.3). Second, a non-surjective map f has an empty framed preimage over any omitted value, hence $\deg f = c(\emptyset) = 0$; this is the framed reading of the Brouwer principle that a map of nonzero degree is onto. Third, the index of an isolated zero of a vector field (definition 8.6.1) is the degree of a local Gauss map $S^{m-1} \to S^{m-1}$, so theorem 10.4.3 makes it a complete invariant of that Gauss map up to homotopy; the homotopy invariance of the degree feeding the Poincaré-Hopf index sum (section 8.6) is the surjectivity-plus-injectivity established here.

Example 10.4.5: Degrees as Signed Point Counts
Each degree computation of section 8.4 becomes a signed count of framed points.

1. The identity id_{S^m} has every y as a regular value with single preimage y and $d(\mathrm{id})_y = \mathrm{id}$, so $\varepsilon_y = +1$ and $c(\{y\}) = +1 = \deg \mathrm{id}_{S^m}$.
2. A reflection $r\colon S^m \to S^m$ has single preimage $r^{-1}(y)$, at which dr reverses orientation, so $\varepsilon = -1$ and $c = -1 = \deg r$ (example 8.4.2).
3. The map $z \mapsto z^k$ on $S^2 = \mathcal{P}^1$ has, over a regular value, k preimages, each a point where the map is holomorphic and hence orientation-preserving, so every $\varepsilon = +1$ and $c = k = \deg$. The framed 0-manifold is k positively framed points.

In each case the signed count of the framed preimage returns the degree already computed by intersection theory, and the framed viewpoint reads it as a class in $\Omega_0^{\mathrm{fr}} \cong \mathbb{Z}$.

Exercise 10.4.1

1. Compute the degree of the antipodal map $a\colon S^m \to S^m$, $a(x) = -x$, as a signed count of framed points: identify the framed preimage of a regular value and its sign, and confirm the answer $\deg a = (-1)^{m+1}$ of example 8.4.6.
2. Show that a smooth map $f\colon S^m \to S^m$ that is not surjective is null-homotopic. (Compute $\deg f$ from the framed preimage of an omitted value, then apply theorem 10.4.3.)
3. Let $p, q \in S^m$ be framed points. Show that $\{p, q\}$ cobounds a framed arc in $S^m \times [0, 1]$ if and only if $\varepsilon_p = -\varepsilon_q$, and deduce that two same-signed framed points are never framed-cobordant to the empty manifold.
4. Show that the mod-2 degree $\deg_2 f$ of a self-map of S^m is the reduction of $\deg f$ modulo 2, and exhibit two self-maps of S^2 with the same mod-2 degree that are not homotopic. Conclude that $\deg_2$ is a strictly coarser invariant than deg.
5. Using framed preimages, re-derive the multiplicativity $\deg(g \circ f) = \deg g \cdot \deg f$ for smooth $f, g\colon S^m \to S^m$. (Take a common regular value of g and $g \circ f$, and track the sign at each nested preimage point.)

10.5 Stabilization and the Stable Homotopy of Spheres

The Pontryagin-Thom bijection of theorem 10.3.7 identifies, for each codimension k, the homotopy group $\pi_{n+k}(S^k)$ with the framed cobordism group of framed n-submanifolds of $\mathbb{R}^{n+k}$; section 10.4 read the case $n = 0$ as the Brouwer degree. Both sides of that bijection still carry the codimension k as a parameter. This section removes it. Raising k by one is a geometric operation on framed submanifolds (adjoin a coordinate to the ambient space and a constant vector to the framing), and under the bijection it matches the suspension of maps on the homotopy side. Freudenthal's theorem says that both families of groups stabilize once k is large, and the common value is a single codimension-free invariant, the n-th stable stem π_n^s. The upshot is that framed cobordism computes the entire stable homotopy of spheres.

Throughout, write $\Omega_n^{\mathrm{fr}}(\mathrm{codim} k)$ for the group of framed cobordism classes of framed n-submanifolds of $\mathbb{R}^{n+k}$ (definition 10.2.6), displaying the codimension k that definition 10.2.6 held fixed. A framing of such a submanifold N is an ordered k-tuple $(v_1, \ldots, v_k)$ of sections spanning the normal bundle $\nu(N)$ fibrewise.

Definition 10.5.1: Suspension of a Framed Submanifold

Let $(N, (v_1, \ldots, v_k))$ be a framed n-submanifold of $\mathbb{R}^{n+k}$. Let $\iota\colon \mathbb{R}^{n+k} \hookrightarrow \mathbb{R}^{n+k+1}$ be the inclusion $\iota(x) = (x, 0)$ of the first $n+k$ coordinates, and let $e = (0, \ldots, 0, 1)$ be the last standard basis vector. The *suspension* (or *stabilization*) of $(N, (v_i))$ is the framed n-submanifold

$$\sigma(N, (v_i)) := \big(\iota(N),\ (\iota_* v_1, \ldots, \iota_* v_k, e)\big)$$

of $\mathbb{R}^{n+k+1}$, of codimension $k+1$. Here $\iota_* v_i$ is v_i regarded as a vector in $\mathbb{R}^{n+k+1}$ via ι; since $\iota(N) \subseteq \mathbb{R}^{n+k} \times \{0\}$, the constant field e is everywhere normal to $\iota(N)$ and independent of $\iota_* v_1, \ldots, \iota_* v_k$, so the $k+1$ fields span $\nu(\iota(N))$.

The construction is compatible with framed cobordism and with disjoint union, so it descends to a group homomorphism

$$\sigma\colon \Omega_n^{\mathrm{fr}}(\mathrm{codim} k) \longrightarrow \Omega_n^{\mathrm{fr}}(\mathrm{codim} k+1)$$

the *stabilization map*. Indeed, if (N_0, F_0) and (N_1, F_1) cobound a framed cobordism $W \subseteq \mathbb{R}^{n+k} \times [0,1]$ with framing G, then $\iota(W) \subseteq \mathbb{R}^{n+k+1} \times [0,1]$ with the augmented framing $(\iota_* G, e)$ is a framed cobordism between $\sigma(N_0, F_0)$ and $\sigma(N_1, F_1)$, because e is constant, hence normal to $\iota(W)$ and to its boundary, and tangent to the $[0,1]$-slices. Compatibility with disjoint union is immediate from the injectivity of ι, since σ acts separately on each component.

The stabilization map is often already an isomorphism at small k. For $n = 0$, a framed 0-manifold in $\mathbb{R}^k$ is a finite set of points, each carrying a sign $\varepsilon(x) = \operatorname{sign}\det[v_1(x) \cdots v_k(x)]$, and $\Omega_0^{\mathrm{fr}}(\mathrm{codim} k) \cong \mathbb{Z}$ by the signed count (proposition 10.4.1). Suspension sends a signed point $x \in \mathbb{R}^k$ with framing $(v_1, \ldots, v_k)$ to $(x, 0) \in \mathbb{R}^{k+1}$ with framing $(v_1, \ldots, v_k, e)$, and appending e as a final column changes the framing determinant from $\det[v_1 \cdots v_k]$ to

$$\det\begin{pmatrix} [v_1 \cdots v_k] & 0 \\ 0 & 1 \end{pmatrix} = \det[v_1 \cdots v_k]$$

so the sign, and hence the signed count, is unchanged. Thus $\sigma\colon \Omega_0^{\mathrm{fr}}(\mathrm{codim} k) \to \Omega_0^{\mathrm{fr}}(\mathrm{codim} k+1)$ is the identity map $\mathbb{Z} \to \mathbb{Z}$ for every $k \geq 1$: the 0-th group is stable from the outset.

To see why σ deserves the name suspension, recall the homotopy suspension homomorphism $\Sigma\colon \pi_m(S^k) \to \pi_{m+1}(S^{k+1})$. Under the identification $S^m = (\mathbb{R}^m)_+$ (one-point compactification, with ∞ the basepoint), the reduced suspension of a based map $g\colon S^m \to S^k$ is modelled by $g \times \mathrm{id}_{\mathbb{R}}$ followed by one-point compactification,

$$\Sigma g\colon (\mathbb{R}^m \times \mathbb{R})_+ \to (\mathbb{R}^k \times \mathbb{R})_+, \qquad (x, s) \mapsto (g(x), s)$$

with basepoints sent to the basepoint ([11, The Suspension Homomorphism, Reduced Suspension as a Union of Cones]). Freudenthal's theorem controls when Σ is an isomorphism.

The Freudenthal suspension theorem ([11, Freudenthal Suspension Theorem, Well-Definedness and Stabilization of the Stable Stems]) states that $\Sigma\colon \pi_{n+k}(S^k) \to \pi_{n+k+1}(S^{k+1})$ is surjective for $n + k = 2k - 1$ and an isomorphism for $n + k \leq 2k - 2$, that is, for $k \geq n + 2$. The groups $\pi_{n+k}(S^k)$ therefore stabilize: for $k \geq n+2$ they are all canonically isomorphic, to a single group

$$\pi_n^s := \operatorname*{colim}_k\ \pi_{n+k}(S^k)$$

the n-th *stable stem*, or *stable homotopy group of spheres*. The colimit is attained at finite stage $k = n+2$. On the geometric side, define the *stable framed cobordism group* $\Omega_n^{\mathrm{fr}} := \operatorname{colim}_k \Omega_n^{\mathrm{fr}}(\mathrm{codim} k)$

along the stabilization maps. The next theorem shows these two constructions agree, and that on the framed side too the colimit is reached at finite stage.

Theorem 10.5.2: Framed Cobordism Is the Stable Stem

For every $n \geq 0$ the Pontryagin-Thom bijection is compatible with stabilization: the square

$$\begin{tikzcd} \Omega_n^{\mathrm{fr}}(\mathrm{codim} k) \arrow[r, "\cong"] \arrow[d, "\sigma"] & \pi_{n+k}(S^k) \arrow[d, "\Sigma"] \\ \Omega_n^{\mathrm{fr}}(\mathrm{codim} k+1) \arrow[r, "\cong"] & \pi_{n+k+1}(S^{k+1}) \end{tikzcd}$$

commutes, its horizontal arrows being the bijection of theorem 10.3.7. Consequently the stabilization map σ is an isomorphism for $k \geq n+2$, and

$$\Omega_n^{\mathrm{fr}} \cong \pi_n^s$$

In particular $\Omega_n^{\mathrm{fr}}(\mathrm{codim} k) \cong \pi_n^s$ already for every $k \geq n+2$.

***Proof*:**

Step 1: The bijection is natural in the codimension. Write $P\colon \Omega_n^{\mathrm{fr}}(\mathrm{codim} k) \to \pi_{n+k}(S^k)$ for the Pontryagin-Thom map of theorem 10.3.7, sending a framed $N \subseteq \mathbb{R}^{n+k}$ to the class of its collapse map $p(N)$ (definition 10.3.1), where $S^{n+k} = (\mathbb{R}^{n+k})_+$. The framing gives, via the tubular neighbourhood theorem (theorem 7.8.1), a diffeomorphism $\varphi\colon N \times \mathbb{R}^k \xrightarrow{\cong} U$ onto an open neighbourhood U of N in $\mathbb{R}^{n+k}$, with $\varphi(x,0) = x$; setting $q := \pi_{\mathbb{R}^k} \circ \varphi^{-1}\colon U \to \mathbb{R}^k$, the collapse map is

$$p(N)(z) = \begin{cases} q(z), & z \in U \\ \infty, & z \notin U \end{cases}$$

and $p(N)(\infty) = \infty$.

Now compute the collapse map of σN. Its framing is $(\iota_* v_1, \ldots, \iota_* v_k, e)$, and the associated framing chart is

$$\varphi \times \mathrm{id}_{\mathbb{R}}\colon N \times \mathbb{R}^{k+1} \longrightarrow U \times \mathbb{R}, \qquad (x,t,s) \mapsto (\varphi(x,t), s)$$

a diffeomorphism onto the open neighbourhood $U \times \mathbb{R}$ of $\iota(N)$ in $\mathbb{R}^{n+k+1}$, because the last framing vector e contributes exactly the coordinate s. Its projection to the framing coordinates is $(z,s) \mapsto (q(z), s)$, so with $S^{n+k+1} = (\mathbb{R}^{n+k+1})_+$ and $S^{k+1} = (\mathbb{R}^{k+1})_+$,

$$p(\sigma N)(z,s) = \begin{cases} (q(z), s), & (z,s) \in U \times \mathbb{R} \\ \infty, & \text{otherwise} \end{cases}$$

When $z \notin U$ one has $p(N)(z) = \infty$, so in all cases $p(\sigma N)(z,s) = (p(N)(z), s)$, read in $(\mathbb{R}^k \times \mathbb{R})_+ = S^{k+1}$ with the convention that any point over ∞ is the basepoint. This is precisely the model $(p(N) \times \mathrm{id}_{\mathbb{R}})_+$ of the reduced suspension. Hence $p(\sigma N) = \Sigma\, p(N)$, and therefore

$$P(\sigma[N]) = [p(\sigma N)] = [\Sigma\, p(N)] = \Sigma[p(N)] = \Sigma\, P([N])$$

which is commutativity of the square.

Step 2: Freudenthal stabilizes both columns. By the Freudenthal suspension theorem ([11, Freudenthal Suspension Theorem]), the right-hand vertical map $\Sigma\colon \pi_{n+k}(S^k) \to \pi_{n+k+1}(S^{k+1})$ is an isomorphism for $k \geq n+2$. Since the horizontal maps P are bijections (theorem 10.3.7) and the square commutes by Step 1, the left-hand vertical map σ is an isomorphism for the same range $k \geq n+2$.

Step 3: Pass to the colimit. The two towers $\{\Omega_n^{\mathrm{fr}}(\mathrm{codim} k)\}$ and $\{\pi_{n+k}(S^k)\}$, with connecting maps σ and Σ, are isomorphic by the commuting squares of Step 1. Their colimits are therefore isomorphic:

$$\Omega_n^{\mathrm{fr}} = \operatorname*{colim}_k \Omega_n^{\mathrm{fr}}(\mathrm{codim} k) \cong \operatorname*{colim}_k \pi_{n+k}(S^k) = \pi_n^s$$

By Step 2 every connecting map is an isomorphism once $k \geq n+2$, so each colimit equals its stage-$(n+2)$ term, giving $\Omega_n^{\mathrm{fr}}(\mathrm{codim} k) \cong \pi_n^s$ for all $k \geq n+2$, as we sought to show.

The geometry of framed manifolds thus computes the stable homotopy groups of spheres. These groups are hard to compute, and the isomorphism transports that difficulty rather than removing it. The stable stems π_n^s are finite for $n \geq 1$ and their values grow erratically, so Ω_n^{fr} inherits that intricacy. What the isomorphism gives is a dictionary: a framed manifold with a specified framing names a stable homotopy class, and framed cobordisms are homotopies. Cartesian product of framed submanifolds descends to a graded-ring structure on $\Omega_\bullet^{\mathrm{fr}}$, the framed analogue of the unoriented cobordism ring of definition 10.1.6; under theorem 10.5.2 this product corresponds to the composition product on the stable stems $\pi_\bullet^s$, the ring structure of the sphere spectrum (the composition product; [11] constructs the spectrum but not yet the product, a recorded gap), so a product of framed representatives represents the product class.

Example 10.5.3: The Low Stable Stems

The first few stable stems, with a framed representative for each and the group value cited to [11].

1. $\pi_0^s \cong \mathbb{Z}$. A framed 0-manifold is a finite signed point set, and the isomorphism is the signed count (proposition 10.4.1); a single positively-framed point generates. Under theorem 10.4.3 this signed count is the degree, so $\pi_0^s \cong \mathbb{Z}$ recovers the Brouwer degree.

2. $\pi_1^s \cong \mathbb{Z}/2$. The generator is η, the stable class of the Hopf map $S^3 \to S^2$. Its framed representative is the circle S^1 with its non-bounding (left-invariant, or Lie-group) framing, the framing of example 10.2.9 that does not extend over a bounding surface. The relation $2\eta = 0$ is visible framed: two copies of the non-bounding circle cobound a framed surface.

3. $\pi_2^s \cong \mathbb{Z}/2$. The generator is the composite η^2, represented by the torus $T^2 = S^1 \times S^1$ carrying the product (Lie-group) framing.

4. $\pi_3^s \cong \mathbb{Z}/24$. The generator is ν, the stable class of the quaternionic Hopf map $S^7 \to S^4$; a regular value has framed preimage S^3, and this is the Lie-group framing of $S^3 = SU(2)$, which represents a generator of $\mathbb{Z}/24$. The earlier generator reappears here as $\eta^3 = 12\nu$, an element of order 2.

The examples show both faces of the invariant. The stable stem is a homotopy-theoretic quantity, yet each class is pinned to a specific manifold and framing, so questions about π_n^s become questions about which framed manifolds cobound. The passage from $\pi_0^s = \mathbb{Z}$ to the torsion groups $\pi_1^s, \pi_2^s, \pi_3^s$ is the passage from degree theory to higher homotopy.

The final step situates the geometric construction inside the spectrum formalism developed in the sibling volume. A Thom space collects the compactified normal data of all framed (or all unframed) submanifolds at once, and its stable homotopy is the corresponding cobordism group. Two instances organize the chapter.

10.5.4 The Thom Spectrum and the Sphere Spectrum The geometric Pontryagin-Thom construction has a homotopy-theoretic refinement, developed in [11, Thom Spectrum, Thom's Theorem]. The unoriented normal data of n-submanifolds assembles into the Thom spectrum MO, and Thom's theorem is the isomorphism

$$\mathfrak{N}_n \cong \pi_n(MO)$$

identifying the unoriented cobordism group of theorem 10.1.9 with a stable homotopy group of MO. The framed analogue trivializes the normal data, replacing MO by the sphere spectrum $\mathbb{S}$, and gives

$$\Omega_n^{\mathrm{fr}} \cong \pi_n(\mathbb{S}) = \pi_n^s$$

the spectrum-level form of theorem 10.5.2. The construction of this chapter is the geometric form of these isomorphisms: transversality builds preimages, framings trivialize normal bundles, and cobordisms are homotopies. The spectra MO and $\mathbb{S}$, the Thom isomorphism, and the computations behind the stable stems are the province of [11]; the oriented ring $\Omega_\bullet^{SO}$, complex cobordism, and the chromatic machinery that computes deep in $\pi_\bullet^s$ belong to the planned volume on stable homotopy theory.

Exercise 10.5.1

1. Using the signed-count description of $\Omega_0^{\mathrm{fr}}(\mathrm{codim}k) \cong \mathbb{Z}$ (proposition 10.4.1), verify directly that the stabilization map σ is the identity on $\mathbb{Z}$ for every $k \geq 1$, and explain why this is consistent with $\pi_0^s \cong \mathbb{Z}$ being reached at the earliest possible stage.
2. For fixed n, determine the smallest codimension k at which the Freudenthal theorem guarantees $\pi_{n+k}(S^k) \cong \pi_n^s$. Evaluate it for $n = 1, 2, 3$, and name the stable value in each case (values cited to [11]).
3. The framed group $\Omega_1^{\mathrm{fr}} \cong \pi_1^s \cong \mathbb{Z}/2$ is nontrivial, yet the unoriented cobordism group $\mathfrak{N}_1$ is zero. Reconcile the two facts. Which manifold represents the nontrivial framed class, and what data does the unoriented class forget?
4. Given $\pi_3^s \cong \mathbb{Z}/24$ generated by ν, and the relation $\eta^3 = 12\nu$, compute the order of η^3 in π_3^s. Is η^3 zero? Show that your answer is consistent with η having order 2.
5. Using the isomorphisms of box 10.5.4, compute $\pi_0(\mathbb{S})$, $\pi_1(\mathbb{S})$, $\pi_1(MO)$, and $\pi_2(MO)$, identifying a generator wherever the group is nonzero. (Use theorem 10.1.9 for the unoriented values.)

Part III

Bundles, Forms, and Integration

In this part, we will start exploring another that can be put on manifolds that will give information at each point which shall be glued together in a local manner similarly to how we consructed the tangnet bundle. In fact, it is a generalization of the tangent bundle, in . So far, we have structure that are locally $\mathbb{R}^n$ with regularity C^k for $k \in \{0, 1, ..., n, ..., \infty\}$ that are also Hausdorff and paracompact. Equivalently, we studied structures that can be constructed by taking open subsets, and gluing parts of them via C^k maps ($k \in \{0, 1, ..., n, ..., \infty\}$) in such a way that the spaces remain Hausdorff. In this section, we will now take a look at spaces that are locally the Cartesian product of an open subset of our manifold M and a *fiber*. We will get to a formal definition in the next chapter, we will for now simply give the high-level intuition on this new type of structure. Take for example $E = S^1 \times (-1, 1)$. This space comes equipped with a natural projection $\pi : E \to S^1$. Then for any open $U \subseteq S^1$ and consider $\pi^{-1}(U)$, we want it to be isomorphic to $U \times (-1, 1)$ in such a way that fibers of U remain above U, that is we want this diagram to commute:

$$\begin{tikzcd}
\pi^{-1}(U) \arrow[rr] & & U \times (-1,1) \arrow[dl] \\
& U \arrow[ul] &
\end{tikzcd}$$

The isomorphism in question will be part of the regularity information. The space E is called the *fiber bundle*. The type of fiber will determine the type of fiber bundles we are working with: for example they may be vector-spaces (giving us vector-bundles), or groups (giving us group bundles), or circle (giving us Hopf bundles), and so on. Fiber bundles will turn out give us a lot of information about manifolds and allow us to not only define many new manifolds, but define many concepts on manifolds like length and distance, measure, curvature, angle, and so forth.

11

Fiber and Vector Bundles

A vector bundle is a family of vector spaces, one over each point of a manifold, varying smoothly and locally trivially yet globally, perhaps, twisted. The tangent bundle is the first example, and the rest of this book runs on the bundles built from it: the cotangent bundle, the tensor bundles, and the exterior bundles that carry differential forms. This chapter develops the machinery that manufactures them. It builds the general notion of a fiber bundle, isolates the transition data that records the twisting, and then develops the functorial operations, dual, direct sum, tensor product, exterior power, and pullback, that produce new bundles from old.

The scope is deliberately that of differential topology. What is developed here is the construction apparatus: how a bundle is assembled from local pieces and how the standard operations act on it. The finer questions, the classification of bundles by homotopy classes of maps into a Grassmannian, their characteristic classes, and the K-theory that organizes them, are the province of the companion volumes and are cited, not developed. The cotangent and tensor bundles that these constructions produce are taken up in the next chapter, where the differential forms they carry begin the road to de Rham cohomology.

11.1 Fiber Bundles

A product $B \times F$ projects onto B with every fiber a copy of F. A fiber bundle is the same picture made local: a space that looks like $B \times F$ over small pieces of the base but may be glued together with a twist, so that no global product structure need exist. The Möbius band is the standard warning. Cut it along its core circle into two arcs; over each arc it is a rectangle, a product of the arc with an interval, but the two rectangles are reattached with a flip, and the result is not the cylinder $S^1 \times (-1, 1)$. The flip is recorded by a single transition map, and the whole of bundle theory is the bookkeeping of such transitions.

Definition 11.1.1: Fiber Bundle

Let B and F be smooth manifolds and let G be a Lie group acting smoothly and effectively on F (on the left). A *smooth fiber bundle* over B with *fiber* F and *structure group* G is a smooth manifold E (the *total space*) with a smooth surjection $\pi\colon E \to B$ (the *projection*) together with an open cover $\{U_\alpha\}_{\alpha\in A}$ of B and diffeomorphisms (*local trivializations*)

$$\Phi_\alpha\colon \pi^{-1}(U_\alpha) \xrightarrow{\cong} U_\alpha \times F, \qquad \pi_1 \circ \Phi_\alpha = \pi,$$

such that on each overlap $U_\alpha \cap U_\beta$ the composite $\Phi_\alpha \circ \Phi_\beta^{-1}$ has the form

$$\Phi_\alpha \circ \Phi_\beta^{-1}(x, f) = \bigl(x,\ g_{\alpha\beta}(x) \cdot f\bigr)$$

for a smooth map $g_{\alpha\beta}\colon U_\alpha \cap U_\beta \to G$, the *transition function*. The fiber over p is $E_p := \pi^{-1}(p)$, diffeomorphic to F; the cover with its trivializations is a *bundle atlas*.

The condition $\pi_1 \circ \Phi_\alpha = \pi$ is what makes a trivialization fiber preserving: it carries each fiber E_p into $\{p\} \times F$ rather than smearing it across the base. Without it a diffeomorphism $\pi^{-1}(U) \to U \times F$ could permute fibers and retain no bundle meaning. The structure group G is the group in which the ambiguity of a fiber lives: two trivializations over the same point differ by the action of an element of G, and requiring $g_{\alpha\beta}$ to land in a chosen G is a restriction on how much twisting is allowed. For a vector bundle G will be $\mathrm{GL}_k(\mathbb{R})$; for an oriented one, $\mathrm{GL}_k^+(\mathbb{R})$; for a Riemannian one, $O(k)$.

Differentiating the overlap identities recorded above yields the constraints that any family of transition functions must satisfy. On a triple overlap the two ways of passing from the γ-trivialization to the α-trivialization agree, and on a single chart the transition to itself is trivial.

Proposition 11.1.2: The Cocycle Conditions

The transition functions of a fiber bundle satisfy, on the indicated overlaps,

$$g_{\alpha\alpha} = e, \qquad g_{\alpha\beta}\, g_{\beta\alpha} = e, \qquad g_{\alpha\beta}(x)\, g_{\beta\gamma}(x) = g_{\alpha\gamma}(x) \quad (x \in U_\alpha \cap U_\beta \cap U_\gamma),$$

where products are taken in G. The last identity is the *cocycle condition*, and it implies the first two.

***Proof*:**
On $U_\alpha \cap U_\beta \cap U_\gamma$ the equality $\Phi_\alpha \circ \Phi_\gamma^{-1} = (\Phi_\alpha \circ \Phi_\beta^{-1}) \circ (\Phi_\beta \circ \Phi_\gamma^{-1})$ reads, on the second coordinate, $g_{\alpha\gamma}(x) \cdot f = g_{\alpha\beta}(x) \cdot \bigl(g_{\beta\gamma}(x) \cdot f\bigr)$ for every $f \in F$. Because G acts effectively, an element of G is determined by its action on F, so $g_{\alpha\gamma}(x) = g_{\alpha\beta}(x)\, g_{\beta\gamma}(x)$. Setting $\gamma = \alpha$ and using $\Phi_\alpha \circ \Phi_\alpha^{-1} = \mathrm{id}$ gives $g_{\alpha\alpha} = e$; then $\beta = \gamma = \alpha$ collapses to nothing, while α, β, α gives $g_{\alpha\beta}\, g_{\beta\alpha} = g_{\alpha\alpha} = e$.

Example 11.1.3: Fiber Bundles

1. The *product bundle* $E = B \times F$ with $\pi = \pi_1$ and the single trivialization id. All transition functions are e; the structure group can be taken trivial.
2. The *Möbius band* over S^1 with fiber $(-1, 1)$ and structure group $\{\pm 1\} \subseteq \mathrm{GL}_1(\mathbb{R})$. Cover

S^1 by two arcs U_1, U_2 meeting in two components; set $g_{12} = +1$ on one component and $g_{12} = -1$ on the other. The sign flip on one overlap is exactly the twist, and it cannot be undone by rechoosing trivializations.

3. The *tangent bundle* TM (definition 2.4.1) is a fiber bundle with fiber $\mathbb{R}^n$ and structure group $\mathrm{GL}_n(\mathbb{R})$; its transition functions are the Jacobians of the chart changes.

4. A smooth *covering map* $\pi\colon E \to B$ is a fiber bundle with discrete fiber and structure group a permutation group; here G acts on a discrete F.

5. The *frame bundle* of a rank-k vector bundle has fiber $\mathrm{GL}_k(\mathbb{R})$ and structure group $\mathrm{GL}_k(\mathbb{R})$ acting on itself by left translation; its fiber over p is the set of ordered bases of E_p. This is the model *principal* bundle, in which fiber and structure group coincide.

The cocycle conditions are not only necessary; they are sufficient. A fiber bundle can be assembled from base, fiber, and a compatible family of transition functions alone, with no total space given in advance. This is the construction that lets one build the bundles of this book, tensor and exterior bundles among them, by prescribing how local pieces glue.

Theorem 11.1.4: Fiber Bundle Construction

Let B be a smooth manifold, F a smooth manifold, and G a Lie group acting smoothly and effectively on F. Let $\{U_\alpha\}_{\alpha\in A}$ be an open cover of B and let $g_{\alpha\beta}\colon U_\alpha \cap U_\beta \to G$ be smooth maps satisfying the cocycle condition

$$g_{\alpha\beta}(x)\, g_{\beta\gamma}(x) = g_{\alpha\gamma}(x) \qquad (x \in U_\alpha \cap U_\beta \cap U_\gamma).$$

Then there is a smooth fiber bundle $\pi\colon E \to B$ with fiber F, structure group G, and these transition functions, and it is unique up to isomorphism of bundles over B.

***Proof*:**
On the disjoint union $\coprod_{\alpha\in A} U_\alpha \times F$ declare

$$(\alpha, x, f) \sim (\beta, y, f') \iff x = y \in U_\alpha \cap U_\beta \text{ and } f' = g_{\beta\alpha}(x)\cdot f.$$

This is an equivalence relation. Reflexivity uses $g_{\alpha\alpha} = e$ (proposition 11.1.2); symmetry uses $g_{\alpha\beta} = g_{\beta\alpha}^{-1}$; and if $(\alpha, x, f) \sim (\beta, x, f')$ and $(\beta, x, f') \sim (\gamma, x, f'')$ then $f'' = g_{\gamma\beta}(x) f' = g_{\gamma\beta}(x) g_{\beta\alpha}(x) f = g_{\gamma\alpha}(x) f$ by the cocycle condition, so $(\alpha, x, f) \sim (\gamma, x, f'')$.

Let $E = \big(\coprod_\alpha U_\alpha \times F\big)/{\sim}$ with the quotient topology, q the quotient map, and $\pi[\alpha, x, f] = x$, well defined since equivalent triples share their base point. For each α the composite $U_\alpha \times F \hookrightarrow \coprod U_\alpha \times F \xrightarrow{q} E$ is injective (distinct points of $U_\alpha \times F$ are never $\sim$-related to each other, as $g_{\alpha\alpha} = e$) with image $\pi^{-1}(U_\alpha)$, and open, since its saturation $\{(\beta, x, g_{\beta\alpha}(x) f) : (\alpha, x, f) \in U_\alpha \times F\}$ is open in each slice by continuity of $g_{\beta\alpha}$. Define

$$\Phi_\alpha\colon \pi^{-1}(U_\alpha) \to U_\alpha \times F, \qquad \Phi_\alpha[\alpha, x, f] = (x, f),$$

a homeomorphism inverse to $q|_{U_\alpha\times F}$, with $\pi_1 \circ \Phi_\alpha = \pi$.

The transition functions are the prescribed ones. For $x \in U_\alpha \cap U_\beta$, $\Phi_\alpha \Phi_\beta^{-1}(x, f) = \Phi_\alpha[\beta, x, f] =$

$\Phi_\alpha[\alpha, x, g_{\alpha\beta}(x)f] = (x, g_{\alpha\beta}(x)f)$, smooth in (x, f) since $g_{\alpha\beta}$ is smooth and the action is smooth.

Smooth structure. Give E the smooth structure for which each Φ_α is a diffeomorphism: charts are $(\varphi \times \mathrm{id}_F) \circ \Phi_\alpha$ for charts φ of U_α, and two such agree smoothly on overlaps because the $\Phi_\alpha\Phi_\beta^{-1}$ are diffeomorphisms. The topology is Hausdorff: two points with distinct base images are separated by preimages of separating opens in B, while two points over the same x lie in a common $\pi^{-1}(U_\alpha) \cong U_\alpha \times F$, which is Hausdorff. Paracompactness (hence the manifold axioms) follows from that of B and F over a locally finite refinement. Then π is a smooth surjection and each Φ_α a trivialization, so E is a smooth fiber bundle with the stated data.

Uniqueness. If $\pi' \colon E' \to B$ is another bundle with the same cover and transition functions, with trivializations Φ'_α, define $H \colon E \to E'$ over each U_α by $H = (\Phi'_\alpha)^{-1} \circ \Phi_\alpha$. On an overlap the two definitions agree, since $(\Phi'_\alpha)^{-1}\Phi_\alpha = (\Phi'_\beta)^{-1}(\Phi'_\beta\Phi'^{-1}_\alpha)(\Phi_\alpha\Phi_\beta^{-1})\Phi_\beta = (\Phi'_\beta)^{-1}\Phi_\beta$, the two inner factors twisting by the mutually inverse $g_{\beta\alpha}$ and $g_{\alpha\beta}$. Thus H is a well-defined diffeomorphism with $\pi' \circ H = \pi$ and H fiber-linear where relevant, an isomorphism of bundles.

The theorem changes how a bundle is specified. Rather than exhibit a total space, it suffices to give a cover and a cocycle of transition functions; the space is then reconstructed. Every construction in this chapter takes this form, applying a functorial operation to the transition functions of known bundles and invoking the reconstruction to name the result.

That is one direction of traffic: data in, bundle out. The other direction is the one a geometer meets more often. A map appears, and the question is whether it is a bundle at all. The answer is unreasonably cheap. Two conditions suffice, neither of them mentioning local triviality, and together they say that a family of compact manifolds cannot change its diffeomorphism type as the parameter varies.

Definition 11.1.5: Proper Map

A continuous map $f \colon M \to N$ is *proper* if $f^{-1}(K)$ is compact for every compact $K \subseteq N$.

Theorem 11.1.6: Ehresmann's Theorem

Let $f \colon M \to N$ be a proper submersion of smooth manifolds. Then f is a smooth fiber bundle: every $y \in N$ has a neighbourhood V and a diffeomorphism

$$\Phi \colon f^{-1}(V) \to V \times f^{-1}(y), \qquad \pi_1 \circ \Phi = f.$$

In particular, if N is connected, then all fibers of f are diffeomorphic.

The proof is a single idea carried out carefully. A submersion lets one lift the coordinate directions of the base to vector fields upstairs; flowing along the lifts drags one fiber onto its neighbours, and properness is exactly what stops the flow from running off the end of the manifold before it arrives.

***Proof*:**

Reduction to a cube. Fix $y \in N$ and a chart $\psi \colon V \to Q := (-1, 1)^n$ with $\psi(y) = 0$, where $n = \dim N$. Write $f_i := \psi_i \circ f$ for the components of f read in this chart. It suffices to produce the trivialization over V, since the statement is local on the base.

Lifting the coordinate directions. Each $p \in f^{-1}(V)$ has, by the theorem 3.1.11 applied to the submersion f, a chart in which f is a projection; in such a chart the coordinate field $\partial/\partial y_i$ of the base lifts to a field f-related to it in the sense of definition 5.4.1. Cover $f^{-1}(V)$ by such chart domains W_α with local lifts X_i^α, take a smooth partition of unity $\{\rho_\alpha\}$ subordinate to the cover (theorem 1.5.5), and set $X_i := \sum_\alpha \rho_\alpha X_i^\alpha$. Being f-related to $\partial/\partial y_i$ is a linear condition on the field at each point, and the ρ_α sum to 1, so

$$df_p(X_i(p)) = \frac{\partial}{\partial y_i}\bigg|_{f(p)} \qquad \text{for every } p \in f^{-1}(V).$$

Properness makes the lifts flow long enough. Let γ be a maximal integral curve of X_i with $\gamma(0) = p$, defined on $\mathcal{D} \subseteq \mathbb{R}$. Since X_i is f-related to $\partial/\partial y_i$, the composite $\psi \circ f \circ \gamma$ solves $\dot{u} = e_i$, so

$$\psi(f(\gamma(t))) = \psi(f(p)) + te_i,$$

a straight line in Q traversed at unit speed. Suppose $\mathcal{D}$ has finite upper endpoint b while this line stays in a compact $K \subseteq Q$ for $t \in [0, b)$. Then $\gamma([0, b))$ lies in $f^{-1}(\psi^{-1}(K))$, which is compact because f is proper, contradicting lemma 5.8.5. Hence the flow θ^i of X_i is defined for exactly as long as the straight line above remains in Q. This is the only step at which properness is used, and it is the whole of its contribution.

The trivialization. Write $F := f^{-1}(y)$, compact by properness. For $u = (u_1, \ldots, u_n) \in Q$ and $q \in F$ set

$$\Psi(u, q) := \theta^n_{u_n} \circ \cdots \circ \theta^1_{u_1}(q).$$

Each flow in turn is defined by the previous step: applying $\theta^1_{u_1}$ moves the base point from 0 along e_1 to $u_1 e_1$, then $\theta^2_{u_2}$ along e_2, and so on, the base point tracing the staircase from 0 to u inside the cube Q, which is convex in each coordinate direction separately. Composition of the flow identities gives $\psi(f(\Psi(u, q))) = u$, so Ψ maps $Q \times F$ into $f^{-1}(V)$ over ψ^{-1}. It is smooth, being a composite of flows, which are smooth in theorem 5.7.9. The map

$$\Psi^{-1}(p) = \big(\psi(f(p)),\ \theta^1_{-u_1} \circ \cdots \circ \theta^n_{-u_n}(p)\big), \qquad u := \psi(f(p)),$$

is smooth for the same reason and inverts Ψ, since each $\theta^i_{-u_i}$ undoes $\theta^i_{u_i}$ and the second component lands in F: its base point is $u - u_n e_n - \cdots - u_1 e_1 = 0$. Then $\Phi := (\psi^{-1} \times \mathrm{id}_F) \circ \Psi^{-1}$ is the required diffeomorphism $f^{-1}(V) \to V \times F$, and $\pi_1 \circ \Phi = f$ by construction. Together with definition 11.1.1 this exhibits f as a fiber bundle with fiber F.

Connected base. The relation "$f^{-1}(y) \cong f^{-1}(y')$" has open equivalence classes in N, by the local triviality just proved, so on a connected N there is one class.

Families whose members have boundary are common — a family of discs, or the piece of a degenerating family cut out by a sphere — and the theorem covers them, provided the boundary submerses too. Without that proviso the fibers can change: the height function on a solid cylinder capped at one end is a proper submersion on the interior, but its fibers jump from a disc to a point.

Corollary 11.1.7: Ehresmann for Manifolds with Boundary

Let M be a smooth manifold with boundary and $f\colon M \to N$ a proper map such that both f and its restriction $f|_{\partial M}$ are submersions. Then f is a locally trivial fiber bundle whose fiber is a compact manifold with boundary, and $f|_{\partial M}$ is a fiber bundle with fiber the boundary of that fiber.

***Proof*:**
Run the proof of theorem 11.1.6, choosing the lifts X_i tangent to ∂M along ∂M. This is possible precisely because $f|_{\partial M}$ is a submersion: near a boundary point, lift $\partial/\partial y_i$ first along $f|_{\partial M}$ to a field on ∂M, then extend it into a boundary chart, and patch with the partition of unity as before, which preserves tangency along ∂M since that is a linear condition. A field tangent to ∂M has a flow carrying ∂M to itself and M to itself, so the flows θ^i stay in M and restrict to ∂M. The rest of the argument is unchanged, and applying it to $f|_{\partial M}$ gives the second claim.

The two hypotheses are both needed, and each fails in an instructive way. The inclusion of an open disc into $\mathbb{R}^2$ is a submersion but not proper, and is not a bundle: its fibers over interior points are single points and its fibers outside are empty. The map $\mathbb{R} \to \mathbb{R}$, $x \mapsto x^3$, is proper but not a submersion at the origin, and is not a bundle either, though for a duller reason. Ehresmann's theorem is the engine behind every statement that a smooth family of compact manifolds is topologically locally constant, and it is what permits the monodromy of such a family to be defined at all.

11.2 Application: Jet Bundles

The tangent space packages the first-order behavior of a map in a coordinate-free way: df_p is the linear part of f at p, independent of the charts used to compute it. Nothing distinguishes first order in principle, and the same packaging carries every order. The resulting object, the *jet*, records the Taylor polynomial of a map to a fixed order intrinsically, and jets assemble into a fiber bundle whose sections are exactly the systems of partial derivatives that a differential equation constrains. This section constructs that bundle; it is the natural home of the higher-order transversality that drives singularity theory.

Fix smooth manifolds M^m and N^n and a point $p \in M$. Two smooth maps f, g defined near p with $f(p) = g(p) = y$ *agree to order* k at p if, in some (equivalently any) charts around p and y, their partial derivatives at p coincide through order k. The chart-independence is the content of the multivariable chain rule: a change of chart acts on the array of derivatives through order k by an invertible polynomial substitution, so vanishing of the difference of derivative arrays is preserved.

Definition 11.2.1: Jet and Jet Bundle

The equivalence class of f under agreement to order k at p is the *k-jet* $j^k_p f$ of f at p; its *source* is p and its *target* is $f(p)$. The *jet bundle* is the set

$$J^k(M,N) = \left\{ j^k_p f \;\middle|\; p \in M,\ f \text{ smooth near } p \right\}$$

with the projection $\pi^k\colon J^k(M,N) \to M \times N,\ \ j^k_p f \mapsto (p, f(p))$.

In charts a k-jet is recorded by its target together with the derivative arrays $\big(\partial^{|\alpha|} f^i/\partial x^\alpha\big)(p)$ for $1 \le |\alpha| \le k$, so a fiber of π^k is a finite-dimensional vector space of coordinate polynomial data. The transition functions, the polynomial substitutions induced by chart changes, are smooth, so the construction of theorem 11.1.4 applies.

Proposition 11.2.2: The Jet Bundle Is a Fiber Bundle

$J^k(M,N)$ is a smooth fiber bundle over $M \times N$ whose fiber is the vector space $P^k_{m,n} = \bigoplus_{j=1}^k \operatorname{Sym}^j(\mathbb{R}^m)^* \otimes \mathbb{R}^n$ of vectors of homogeneous polynomial maps $\mathbb{R}^m \to \mathbb{R}^n$ of degrees 1 through k. In particular $J^1(M,N)$ is the external hom bundle $\operatorname{Hom}(\operatorname{pr}_M^* TM, \operatorname{pr}_N^* TN)$ over $M \times N$, with fiber the linear maps $\mathbb{R}^m \to \mathbb{R}^n$.

***Proof*:**
Charts (U,x) of M and (V,y) of N give a bijection $\Phi_{U,V} \colon (\pi^k)^{-1}(U \times V) \to (U \times V) \times P^k_{m,n}$ sending $j^k_p f$ to $(p, f(p))$ together with the derivative arrays of $y \circ f \circ x^{-1}$ at $x(p)$, degree by degree. A second pair of charts changes the arrays by the chain rule: the degree-j component transforms by a polynomial in the derivatives of the two chart changes through order j, invertibly and smoothly in the base point. These are the transition functions of a fiber bundle, so theorem 11.1.4 produces the smooth structure and $\Phi_{U,V}$ are its trivializations. For $k=1$ the fiber is $(\mathbb{R}^m)^* \otimes \mathbb{R}^n = \operatorname{Hom}(\mathbb{R}^m, \mathbb{R}^n)$ and the degree-one array is the Jacobian, so $J^1(M,N)$ is that external hom bundle as claimed.

A smooth map $f \colon M \to N$ prolongs to its *k-jet section* $j^k f \colon M \to J^k(M,N)$, $p \mapsto j^k_p f$, a smooth section of $J^k(M,N) \to M$ along the graph. Differential conditions on f, that it be an immersion, that it have a singularity of a prescribed type, become the condition that $j^k f$ meet a fixed submanifold $Z \subseteq J^k(M,N)$. Thom's jet transversality theorem asserts that for generic f the section $j^k f$ is transverse to Z, so that the singular set has the codimension predicted by $\operatorname{codim} Z$; the proof is the parametric transversality of chapter 7 applied to the jet prolongation, and the statement is what makes genericity arguments in singularity theory quantitative. The intersection-theoretic count of such singularities is the degree and Euler-class bookkeeping of chapter 8.

11.3 Vector Bundles

A vector bundle is a fiber bundle whose fibers are vector spaces and whose transitions are linear. The linearity is the whole point: it lets fiberwise linear algebra, sums, duals, tensor products, be performed over the base and glued into new bundles, and it is what the next sections exploit.

Definition 11.3.1: Vector Bundle

A *smooth real vector bundle of rank* k over a smooth manifold M is a smooth manifold E with a smooth surjection $\pi\colon E \to M$ such that

1. each fiber $E_p = \pi^{-1}(p)$ carries the structure of a k-dimensional real vector space; and
2. each $p \in M$ has a neighborhood U and a diffeomorphism $\Phi\colon \pi^{-1}(U) \to U \times \mathbb{R}^k$ with $\pi_1 \circ \Phi = \pi$ whose restriction $\Phi|_{E_q}\colon E_q \to \{q\} \times \mathbb{R}^k$ is a linear isomorphism for each $q \in U$.

Such a Φ is a *local trivialization*; E is *trivial* if one exists over all of M. Replacing $\mathbb{R}^k$ by $\mathbb{C}^k$ and linear by complex-linear gives a *complex vector bundle*.

A vector bundle is a fiber bundle with fiber $\mathbb{R}^k$ and structure group $\mathrm{GL}_k(\mathbb{R})$: the fiberwise linearity forces the transition maps into the general linear group.

Proposition 11.3.2: Transition Functions of a Vector Bundle

Let $\Phi\colon \pi^{-1}(U) \to U \times \mathbb{R}^k$ and $\Psi\colon \pi^{-1}(V) \to V \times \mathbb{R}^k$ be local trivializations of a smooth rank-k bundle with $U \cap V \neq \emptyset$. Then there is a smooth map $\tau_{\Phi\Psi}\colon U \cap V \to \mathrm{GL}_k(\mathbb{R})$ with

$$\Phi \circ \Psi^{-1}(p, v) = \big(p,\ \tau_{\Phi\Psi}(p)\, v\big).$$

***Proof*:**
Since $\pi_1 \circ \Phi = \pi = \pi_1 \circ \Psi$, the composite fixes the base: $\Phi\Psi^{-1}(p, v) = (p, \sigma(p, v))$ for a smooth $\sigma\colon (U \cap V) \times \mathbb{R}^k \to \mathbb{R}^k$. For fixed p the map $v \mapsto \sigma(p, v)$ is the composite of two linear fiber isomorphisms $\Phi|_{E_p} \circ (\Psi|_{E_p})^{-1}$, hence linear and invertible, so $\sigma(p, v) = \tau_{\Phi\Psi}(p)v$ for a unique $\tau_{\Phi\Psi}(p) \in \mathrm{GL}_k(\mathbb{R})$. Its entries are $\tau_{\Phi\Psi}(p)^i_j = \sigma(p, e_j)^i$, smooth in p because σ is smooth; and $\tau_{\Phi\Psi}$ lands in $\mathrm{GL}_k(\mathbb{R})$, an open subset of $\mathbb{R}^{k^2}$, so it is smooth as a map into the group.

Example 11.3.3: Vector Bundles

1. The *product bundle* $M \times \mathbb{R}^k$, always trivial.
2. The *tangent bundle* TM (definition 2.4.1), rank n, with the Jacobians of chart changes as transition functions.
3. The *tautological line bundle* $\gamma^1 \to \mathbb{R}\mathrm{P}^n$: over a line $\ell \in \mathbb{R}\mathrm{P}^n$ the fiber is the line itself,

   $$\gamma^1 = \left\{(\ell, v) \in \mathbb{R}\mathrm{P}^n \times \mathbb{R}^{n+1} \;\middle|\; v \in \ell\right\},$$

   with $\pi(\ell, v) = \ell$. Over the standard chart $U_i = \{[x] : x_i \neq 0\}$ the map $(\ell, v) \mapsto (\ell, v_i)$ trivializes it, and for $n = 1$ this is the Möbius bundle; γ^1 is nontrivial for every $n \geq 1$: a nowhere-zero section would assign each line ℓ a nonzero $s(\ell) \in \ell$, so writing $s([x]) = f(x)\, x$ for $x \in S^n$ makes $f\colon S^n \to \mathbb{R}$ odd (as $[x] = [-x]$ forces $f(-x) = -f(x)$) and nowhere zero, impossible on the connected S^n; thus γ^1 has no global frame (corollary 11.4.4).

To recognize a vector bundle it is enough to produce fiberwise trivializations with smooth linear transitions; a total-space topology need not be given in advance. This is the linear refinement of the fiber bundle construction, and it is what builds bundles from fiberwise data.

Lemma 11.3.4: Vector Bundle Chart Lemma

Let M be a smooth manifold, and for each $p \in M$ let E_p be a real vector space of fixed dimension k. Set $E = \coprod_p E_p$ and let $\pi\colon E \to M$ send E_p to p. Suppose given an open cover $\{U_\alpha\}$ of M and, for each α, a bijection

$$\Phi_\alpha\colon \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^k$$

restricting to a linear isomorphism $E_p \xrightarrow{\cong} \{p\} \times \mathbb{R}^k$ on each fiber, such that for all α, β the map $\Phi_\alpha \Phi_\beta^{-1}$ has the form $(p, v) \mapsto (p, \tau_{\alpha\beta}(p)v)$ for a smooth $\tau_{\alpha\beta}\colon U_\alpha \cap U_\beta \to \mathrm{GL}_k(\mathbb{R})$. Then E carries a unique smooth manifold structure making it a rank-k vector bundle over M with the Φ_α as smooth local trivializations.

***Proof*:**
The maps $\tau_{\alpha\beta}$ satisfy the cocycle condition automatically: on a triple overlap the Φ_α are honest bijections from the common set $\pi^{-1}(U_\alpha \cap U_\beta \cap U_\gamma)$, so $\Phi_\alpha\Phi_\gamma^{-1} = (\Phi_\alpha\Phi_\beta^{-1})(\Phi_\beta\Phi_\gamma^{-1})$, which reads $\tau_{\alpha\gamma} = \tau_{\alpha\beta}\tau_{\beta\gamma}$. Theorem 11.1.4, applied with fiber $\mathbb{R}^k$ and structure group $\mathrm{GL}_k(\mathbb{R})$ acting by matrix multiplication, produces a smooth fiber bundle E' over M with transition functions $\tau_{\alpha\beta}$ and trivializations Φ'_α. The bijection $E \to E'$ that is $(\Phi'_\alpha)^{-1} \circ \Phi_\alpha$ over each U_α is well defined (the two definitions differ by the common twist $\tau_{\beta\alpha}$) and carries the transported smooth structure to E; each fiber isomorphism $\Phi_\alpha|_{E_p}$ being linear, the induced fiber structure on E' is the given vector-space structure on E_p, and $\mathrm{GL}_k(\mathbb{R})$-valued transitions make it a vector bundle. Uniqueness is that of theorem 11.1.4.

Example 11.3.5: Bundles Built from Fiberwise Data

1. *The tautological bundle over a Grassmannian.* Over the Grassmannian $\mathrm{Gr}_k(\mathbb{R}^n)$ of k-planes in $\mathbb{R}^n$, set the fiber over a plane P to be P itself. Over the chart where a k-plane is a graph of a linear map $\mathbb{R}^k \to \mathbb{R}^{n-k}$ the plane varies smoothly, giving trivializing bijections with smooth $\mathrm{GL}_k(\mathbb{R})$-transitions; lemma 11.3.4 makes $\gamma^k \to \mathrm{Gr}_k(\mathbb{R}^n)$ a rank-k bundle. For $k = 1$ it is the tautological line bundle over $\mathbb{R}\mathrm{P}^{n-1}$.

2. *The determinant bundle.* From a rank-k bundle E with transitions $\tau_{\alpha\beta}$, the maps $\det \tau_{\alpha\beta}\colon U_\alpha \cap U_\beta \to \mathrm{GL}_1(\mathbb{R})$ form a cocycle and, by the lemma, a line bundle $\det E$. Its triviality is exactly the orientability of E (section 11.7).

11.4 Local and Global Sections of Vector Bundles

A section chooses one vector from each fiber, smoothly. Sections are to a bundle what functions are to a manifold, and the failure of a bundle to admit a nowhere-zero section is the first measure of its twisting.

Definition 11.4.1: Section

A *section* of $\pi\colon E \to M$ over $U \subseteq M$ is a smooth map $\sigma\colon U \to E$ with $\pi \circ \sigma = \mathrm{id}_U$; it is *global* if $U = M$. The sections over U form a module $\Gamma(U, E)$ over $C^\infty(U)$ under fiberwise operations, with the *zero section* $\zeta(p) = 0_{E_p}$ as its identity. A section is *nowhere zero* if $\sigma(p) \neq 0_{E_p}$ for all p.

Definition 11.4.2: Local and Global Frame

An ordered k-tuple $(\sigma_1, \dots, \sigma_k)$ of sections over U is a *local frame* if $(\sigma_1(p), \dots, \sigma_k(p))$ is a basis of E_p for every $p \in U$; it is a *global frame* if $U = M$. A smooth frame is one whose sections are smooth. A bundle admitting a global smooth frame is exactly a trivial bundle, as the next result shows; a manifold whose tangent bundle is trivial is *parallelizable*.

Local frames and local trivializations are two views of the same datum. A trivialization names a basis of each fiber by pulling back the standard basis of $\mathbb{R}^k$; conversely a frame lists a basis, and reading coordinates against it trivializes the bundle.

Proposition 11.4.3: Frames and Trivializations Correspond

Let $\pi\colon E \to M$ be a smooth rank-k bundle and $U \subseteq M$ open. A smooth local trivialization $\Phi\colon \pi^{-1}(U) \to U \times \mathbb{R}^k$ and a smooth local frame $(\sigma_1, \dots, \sigma_k)$ over U determine each other by

$$\sigma_i(p) = \Phi^{-1}(p, e_i), \qquad \Phi\Big(\textstyle\sum_i v^i \sigma_i(p)\Big) = (p, v),$$

and this is a bijection between smooth local trivializations over U and smooth local frames over U.

***Proof*:**
Given Φ, each $\sigma_i = \Phi^{-1}(\,\cdot\,, e_i)$ is smooth with $\pi\sigma_i = \mathrm{id}$, and $\Phi|_{E_p}$ being a linear isomorphism carries $(e_1, \dots, e_k)$ to a basis, so (σ_i) is a smooth frame. Conversely, given a smooth frame, define Φ by the displayed formula; it is fiberwise linear and bijective. To see Φ is a diffeomorphism, compare it with any smooth trivialization Ψ over a neighborhood of a point: writing $\sigma_i^j = \pi_2^j(\Psi\sigma_i)$ for the components of the frame against Ψ, one has $\Psi\Phi^{-1}(p, v) = (p, S(p)v)$ with $S(p) = (\sigma_i^j(p))_{j,i}$. Each σ_i^j is smooth and $S(p) \in \mathrm{GL}_k(\mathbb{R})$ since $(\sigma_i(p))$ is a basis; matrix inversion being smooth, $\Psi\Phi^{-1}$ and its inverse are smooth, so Φ is a diffeomorphism. The two passages are mutually inverse by construction.

Corollary 11.4.4: Trivial Bundles Are Framed Bundles

A smooth vector bundle is trivial if and only if it admits a global smooth frame.

***Proof*:**
Apply proposition 11.4.3 with $U = M$: a global frame yields a global trivialization and conversely.

Local frames are never obstructed; only global ones are. Any independent partial frame near a point extends to a full local frame, so trivializations exist around every point, as the definition of a bundle already promised, and the extension is by the same linear-algebra move that completes a basis.

Proposition 11.4.5: Completion of Local Frames

Let $\pi\colon E \to M$ be a smooth rank-k bundle and $(\sigma_1, \dots, \sigma_j)$ smooth sections over a neighborhood of p whose values at p are linearly independent. Then there are smooth sections $\sigma_{j+1}, \dots, \sigma_k$ near p completing them to a smooth local frame on some neighborhood of p. In particular any nonzero vector in E_p is the value at p of a section belonging to a smooth local frame.

***Proof*:**
Trivialize over a neighborhood U of p by Φ, and read the σ_i as smooth maps $s_i = \pi_2\Phi\sigma_i\colon U \to \mathbb{R}^k$; the $s_i(p)$ are independent. Extend $(s_1(p), \dots, s_j(p))$ to a basis of $\mathbb{R}^k$ by constant vectors $w_{j+1}, \dots, w_k$ and set $\sigma_i = \Phi^{-1}(\,\cdot\,, w_i)$ for $i > j$. The matrix with columns $s_1(p), \dots, s_j(p), w_{j+1}, \dots, w_k$ is invertible; invertibility is an open condition, so the sections stay independent on a neighborhood of p, where they form a smooth local frame. Given a nonzero $v \in E_p$, take $j = 1$ and σ_1 any section with $\sigma_1(p) = v$ (extend $\Phi^{-1}(\,\cdot\,, \pi_2\Phi(v))$).

11.5 Principal and Associated Bundles

A bundle's twisting is recorded entirely by its structure group, and there is a bundle whose fiber *is* that group: the principal bundle. Every bundle with a given structure group, vector bundles among them, is manufactured from a single principal bundle by attaching a fiber on which the group acts. The structure group, not the fiber, is then the primary datum, and geometric structures on a bundle become reductions of that group.

Definition 11.5.1: Principal Bundle

Let G be a Lie group. A *principal G-bundle* is a fiber bundle $\pi\colon P \to B$ with fiber G together with a smooth right action $P \times G \to P$, $(p, g) \mapsto p \cdot g$, that preserves fibers ($\pi(p \cdot g) = \pi(p)$) and is simply transitive on each fiber, such that the local trivializations $\Phi_\alpha\colon \pi^{-1}(U_\alpha) \to U_\alpha \times G$ are G-equivariant: $\Phi_\alpha(p \cdot g) = \Phi_\alpha(p) \cdot g$, where G acts on $U_\alpha \times G$ by right translation in the second factor. Its transition functions take values in G acting on the fiber G by left translation.

Each fiber is a *G-torsor*: a copy of G that has forgotten its identity element, on which G acts simply transitively. Choosing an identity in each fiber is exactly a trivialization, so a principal bundle is trivial precisely when it has a global section. This is the sharp contrast with vector bundles, which always carry the zero section: for a principal bundle, the existence of a section is the whole obstruction.

Proposition 11.5.2: Principal Triviality and Sections

A principal G-bundle $\pi\colon P \to B$ is trivial if and only if it admits a global smooth section.

***Proof*:**
A trivialization $\Phi\colon P \to B \times G$ gives the section $s(b) = \Phi^{-1}(b, e)$. Conversely, a global section s gives a map $\Psi\colon B \times G \to P$, $\Psi(b, g) = s(b) \cdot g$, which is a diffeomorphism (simple transitivity makes it a fiberwise bijection, and it is smooth with smooth inverse $p \mapsto (\pi(p), \delta(p))$ where $\delta(p)$ is the unique g with $s(\pi(p)) \cdot g = p$); it is G-equivariant, so $\Phi = \Psi^{-1}$ is an equivariant trivialization.

The motivating example turns a vector bundle into a principal bundle by remembering its frames.

Example 11.5.3: The Frame Bundle
The *frame bundle* $\mathrm{Fr}(E)$ of a rank-k vector bundle $E \to M$ has fiber over p the set of ordered bases (frames) of E_p (section 11.4). The group $\mathrm{GL}_k(\mathbb{R})$ acts on the right by change of basis: a frame $u = (v_1, \ldots, v_k)$ and $A \in \mathrm{GL}_k(\mathbb{R})$ give $u \cdot A = \left(\sum_i v_i A^i_1, \ldots, \sum_i v_i A^i_k\right)$. The action is free and simply transitive on each fiber, since one frame is carried to another by a unique change-of- basis matrix. A smooth local frame of E is a smooth local section of $\mathrm{Fr}(E)$ (proposition 11.4.3) and yields an equivariant trivialization, so $\mathrm{Fr}(E)$ is a principal $\mathrm{GL}_k(\mathbb{R})$-bundle. By proposition 11.5.2 it is trivial if and only if E has a global frame, that is if and only if E is trivial (corollary 11.4.4).

Attaching a fiber to a principal bundle reverses the construction and recovers, from one principal bundle, every bundle with that structure group.

Definition 11.5.4: Associated Bundle

Let $\pi\colon P \to B$ be a principal G-bundle and let G act smoothly on the left of a manifold F. The *associated bundle* is

$$P \times_G F = (P \times F)/\!\sim, \qquad (p \cdot g,\ f) \sim (p,\ g \cdot f),$$

with projection $[p, f] \mapsto \pi(p)$.

Proposition 11.5.5: Associated Bundles Are Fiber Bundles

$P \times_G F$ is a smooth fiber bundle over B with fiber F and the transition functions of P acting on F. If F is a vector space and G acts linearly, it is a vector bundle; for $F = \mathbb{R}^k$ and $G = \mathrm{GL}_k(\mathbb{R})$ acting standardly, applied to $P = \mathrm{Fr}(E)$, the map

$$\mathrm{Fr}(E) \times_{\mathrm{GL}_k(\mathbb{R})} \mathbb{R}^k \xrightarrow{\cong} E, \qquad [\,(v_1, \ldots, v_k),\ a\,] \mapsto \sum_i a^i v_i$$

is an isomorphism of vector bundles.

***Proof*:**
Over U_α a trivialization $\Phi_\alpha(p) = (\pi(p), \theta_\alpha(p))$ of P gives a bijection $[p, f] \mapsto (\pi(p),\ \theta_\alpha(p) \cdot f)$ onto $U_\alpha \times F$, well defined on $\sim$-classes because $\theta_\alpha(p \cdot g) = \theta_\alpha(p)g$ cancels the g^{-1} from $g \cdot f$. Two such compose to $(b, f) \mapsto (b, \tau_{\alpha\beta}(b) \cdot f)$ with $\tau_{\alpha\beta}$ the transition functions of P, now acting on F; these are smooth and satisfy the cocycle condition, so theorem 11.1.4 (with structure group the image of G in the diffeomorphisms of F, which acts effectively) makes $P \times_G F$ a fiber bundle, a vector bundle when the action is linear. For the frame-bundle map, well-definedness is the identity $[uA, A^{-1}a] \mapsto (uA)(A^{-1}a) = u(a)$ (a frame u read as the isomorphism $\mathbb{R}^k \to E_p$, $a \mapsto \sum a^i v_i$), which is a fiberwise linear isomorphism, smooth in local frames; so it is a bundle isomorphism.

Thus a vector bundle is exactly a linear representation $\mathbb{R}^k$ of $\mathrm{GL}_k(\mathbb{R})$ attached to its frame bundle, and the tensor, dual, and exterior bundles of section 11.8 are the other representations of $\mathrm{GL}_k(\mathbb{R})$ attached to the same principal bundle. The structure group is the organizing datum; restricting it encodes geometry.

Definition 11.5.6: Reduction of Structure Group

Let $H \leq G$ be a closed subgroup. A *reduction* of a principal G-bundle P to H is a principal H-sub-bundle $Q \subseteq P$, equivalently a bundle atlas of P whose transition functions take values in H.

Example 11.5.7: Geometric Structures as Reductions

Reductions of the frame bundle $\mathrm{Fr}(E)$ of a rank-k bundle encode geometric structure on E.

1. An *orientation* is a reduction of $\mathrm{GL}_k(\mathbb{R})$ to $\mathrm{GL}_k^+(\mathbb{R})$: the oriented frames (section 11.7).
2. A *fiber metric* is a reduction to the orthogonal group $\mathrm{O}(k)$, the bundle of orthonormal frames. Because $\mathrm{O}(k) \hookrightarrow \mathrm{GL}_k(\mathbb{R})$ is a deformation retract (Gram-Schmidt), this reduction always exists, made precise and proved in proposition 11.5.8.
3. An *almost-complex structure* on a rank-$2m$ bundle is a reduction to $\mathrm{GL}_m(\mathbb{C}) \subseteq \mathrm{GL}_{2m}(\mathbb{R})$.

The orthogonal reduction is the one every bundle carries, and it deserves a proof rather than an appeal to a deformation retract: a fiber metric is exactly a smoothly varying bundle of orthonormal frames sitting inside the frame bundle.

Proposition 11.5.8: The Orthonormal Frame Bundle

Let $E \to M$ be a smooth rank-k vector bundle with a fiber metric g (lemma 11.6.5). The set $\mathrm{O}_g(E) \subseteq \mathrm{Fr}(E)$ of g-orthonormal frames is a smooth principal $\mathrm{O}(k)$-subbundle of the frame bundle, an $\mathrm{O}(k)$-reduction of $\mathrm{Fr}(E)$ (definition 11.5.6). Since a fiber metric always exists, every smooth vector bundle admits such a reduction.

***Proof*:**
Near any point take a smooth local frame $(\sigma_1, \ldots, \sigma_k)$ of E (proposition 11.4.5) and apply the Gram-Schmidt process with respect to g: the resulting frame $(e_1, \ldots, e_k)$ is smooth, each e_i

being a smooth expression in the σ_j and the inner products $g(\sigma_a, \sigma_b)$, the normalization dividing by the positive norms produced at each step, on which the square root is smooth. This is a smooth local section of $\mathrm{O}_g(E)$, and a local section of a principal bundle yields an equivariant local trivialization by the construction of proposition 11.5.2, here $(u, A) \mapsto (e_1, \ldots, e_k)(u) \cdot A$, for the right $\mathrm{O}(k)$-action restricted from the $\mathrm{GL}_k(\mathbb{R})$-action on $\mathrm{Fr}(E)$. Two g-orthonormal frames over an overlap differ by the change-of-basis matrix carrying one orthonormal basis to another, an element of $\mathrm{O}(k)$, so the transition functions take values in $\mathrm{O}(k)$. Thus $\mathrm{O}_g(E)$ is a principal $\mathrm{O}(k)$-bundle whose bundle atlas has $\mathrm{O}(k)$-valued transitions inside those of $\mathrm{Fr}(E)$, an $\mathrm{O}(k)$-reduction of the frame bundle (definition 11.5.6). Existence of g (lemma 11.6.5) gives the last claim, completing the proof.

The classification of principal G-bundles over B by homotopy classes of maps into a classifying space BG, and the resulting characteristic classes, are the natural next step; they are homotopy-theoretic and are developed in [11, Classification of Principal G-Bundles, Characteristic Classes are the Cohomology of BG] (with the K-theory of the vector-bundle case in the companion volume). This chapter gives their geometric input: the bundles, their frames, and the operations that build new bundles from old.

11.6 Bundle Homomorphisms

A map of bundles should carry fibers to fibers linearly. Such maps form the morphisms of the category of vector bundles, and, when their rank is locally constant, they cut out the sub-bundles and quotient bundles from which every later construction is assembled.

Definition 11.6.1: Bundle Homomorphism

Let $\pi \colon E \to M$ and $\pi' \colon E' \to M'$ be vector bundles. A smooth map $F \colon E \to E'$ is a *bundle homomorphism covering* $f \colon M \to M'$ if $\pi' \circ F = f \circ \pi$ and each restriction $F|_{E_p} \colon E_p \to E'_{f(p)}$ is linear. When $M = M'$ and $f = \mathrm{id}_M$ it is a *homomorphism over* M. It is a *bundle isomorphism* if it has a bundle-homomorphism inverse; over M this holds exactly when every $F|_{E_p}$ is an isomorphism.

Example 11.6.2: Bundle Homomorphisms

1. A smooth map $T \colon M \to \mathrm{Mat}_{\ell \times k}(\mathbb{R})$ gives a homomorphism $M \times \mathbb{R}^k \to M \times \mathbb{R}^\ell$ over M, $(p, v) \mapsto (p, T(p)v)$, an isomorphism precisely where $T(p)$ is invertible.

2. The differential $df \colon TM \to TN$ of a smooth $f \colon M \to N$ is a bundle homomorphism covering f.

3. For a Lie group G, left translations trivialize TG, so $TG \cong G \times \mathfrak{g}$: every Lie group is parallelizable, and TG is isomorphic over G to a product bundle.

A homomorphism over M need not have kernels and images that are bundles, since fiber ranks can jump. Constancy of rank is exactly what prevents the jump.

Definition 11.6.3: Sub-bundle

A *sub-bundle* of $\pi\colon E \to M$ is a subset $D \subseteq E$ that is a vector bundle with the restricted projection, such that each $D_p = D \cap E_p$ is a linear subspace and D is locally spanned by part of a frame of E: near each point there is a frame $(\sigma_1, \ldots, \sigma_k)$ of E with $(\sigma_1, \ldots, \sigma_r)$ a frame of D.

Proposition 11.6.4: Kernel and Image of a Constant-Rank Map

Let $F\colon E \to E'$ be a bundle homomorphism over M of locally constant rank r. Then $\ker F = \coprod_p \ker(F|_{E_p})$ and $\operatorname{im} F = \coprod_p \operatorname{im}(F|_{E_p})$ are sub-bundles of E and E', of ranks $\operatorname{rank} E - r$ and r.

***Proof*:**
Fix p; trivialize E, E' near p and read F as a smooth matrix-valued map $A(x)$ of constant rank r. Choosing bases so that $A(p)$ has an invertible $r \times r$ block, that block stays invertible nearby, and Gaussian elimination smooth in x produces frames adapting A: smooth frames $(\varepsilon_1, \ldots, \varepsilon_k)$ of E and $(\varepsilon'_1, \ldots, \varepsilon'_\ell)$ of E' near p with $F\varepsilon_i = \varepsilon'_i$ for $i \leq r$ and $F\varepsilon_i = 0$ for $i > r$. Then $(\varepsilon_{r+1}, \ldots, \varepsilon_k)$ frames $\ker F$ and $(\varepsilon'_1, \ldots, \varepsilon'_r)$ frames $\operatorname{im} F$ near p, exhibiting both as sub-bundles.

A sub-bundle has a complement, and every fiberwise-exact sequence of bundles splits. The tool is a fiber metric, a smoothly varying inner product, which exists on any bundle by the same partition-of-unity patching that builds Riemannian metrics; here it is used only to name orthogonal complements, the geometry proper being the subject of the Riemannian volume.

Lemma 11.6.5: Existence of a Fiber Metric

Every smooth vector bundle $\pi\colon E \to M$ admits a *fiber metric*: a smooth section g of positive-definite symmetric bilinear forms, assigning each E_p an inner product g_p varying smoothly in p.

***Proof*:**
Take a trivializing cover $\{U_\alpha\}$ and a subordinate smooth partition of unity $\{\rho_\alpha\}$ (theorem 1.5.5). Each trivialization Φ_α pulls back the Euclidean inner product on $\mathbb{R}^k$ to an inner product g^α on $E|_{U_\alpha}$. Set $g = \sum_\alpha \rho_\alpha g^\alpha$. The sum is locally finite and smooth; each summand is symmetric and positive semidefinite, and at each p some $\rho_\alpha(p) > 0$ with g^α_p positive definite, so g_p is positive definite.

Proposition 11.6.6: Splitting of Bundle Sequences

Let $0 \to E' \xrightarrow{\iota} E \xrightarrow{\rho} E'' \to 0$ be a sequence of bundle homomorphisms over M that is exact on each fiber. Then $\iota(E')$ is a sub-bundle with complement, and $E \cong E' \oplus E''$. In particular a sub-bundle $D \subseteq E$ has a complementary sub-bundle $D^\perp$ with $E \cong D \oplus D^\perp$.

***Proof*:**
Fiberwise exactness makes ι injective and ρ surjective of constant ranks, so by proposition 11.6.4 $\iota(E') = \ker\rho$ is a sub-bundle. Fix a fiber metric g (lemma 11.6.5) and let $D^\perp = (\ker\rho)^\perp$ be the fiberwise orthogonal complement; it is the image of the constant-rank bundle map $E \to E$ given by orthogonal projection, hence a sub-bundle, and $\rho|_{D^\perp}: D^\perp \to E''$ is a fiberwise isomorphism, so a bundle isomorphism. Then $v \mapsto (\iota^{-1}\,\mathrm{pr}_{\ker\rho}\, v,\ \rho v)$ is an isomorphism $E \xrightarrow{\cong} E' \oplus E''$.

The sub-bundle and quotient machinery reproduces a bundle already in hand, and doing so sharpens what the earlier construction meant.

Example 11.6.7: The Normal Bundle as a Quotient
Let $N \subseteq M$ be an embedded submanifold. Each T_pN is a subspace of T_pM, so TN is a sub-bundle of the restriction $TM|_N$, and proposition 11.6.6 gives the short exact sequence

$$0 \to TN \to TM|_N \to \nu(N) \to 0$$

whose quotient $\nu(N) = TM|_N/TN$ is the *normal bundle* of N (definition 7.7.1); its dual $\nu(N)^*$, the annihilator of TN inside $T^*M|_N$, is the *conormal bundle*. A fiber metric on $TM|_N$ (lemma 11.6.5) splits the sequence and identifies $\nu(N)$ with the orthogonal complement of TN, the form in which the tubular neighbourhood theorem (theorem 7.8.1) uses it; the deeper geometry of connections and curvature on the normal bundle belongs to [6]. That a sub-bundle, its quotient, and a metric splitting manufacture an object of independent geometric weight is the construction machinery at work.

11.7 Oriented Bundles

An orientation of a vector space is a choice, consistent under change of basis, between its two classes of ordered bases: those reached from a fixed one by a positive-determinant change. Bundles inherit the notion fiber by fiber, and the content is whether the choice can be made to vary continuously over the base. When it can, the bundle is oriented; the tangent bundle's case is the orientability of the manifold.

Recall that the ordered bases of a k-dimensional real vector space V $(k \geq 1)$ fall into two classes under positive-determinant change of basis; an *orientation* of V is a choice of one class, and a linear isomorphism is *orientation preserving* when its matrix in oriented bases has positive determinant. The standard orientation of $\mathbb{R}^k$ is the class of $(e_1, \dots, e_k)$.

Definition 11.7.1: Oriented Vector Bundle

An *orientation* of a rank-k bundle $\pi: E \to M$ $(k \geq 1)$ is a choice of orientation μ_p of each fiber E_p that is *locally constant*: each point has a local trivialization $\Phi: \pi^{-1}(U) \to U \times \mathbb{R}^k$ carrying μ_p to the standard orientation of $\mathbb{R}^k$ for every $p \in U$. A bundle is *orientable* if an orientation exists, and *oriented* once one is chosen; an orientable connected bundle has exactly two orientations, μ and $-\mu$.

Orientability is a condition on transition functions, and equivalently the triviality of a single line bundle.

Proposition 11.7.2: Characterizations of Orientability

For a rank-k bundle $E \to M$ the following are equivalent:

1. E is orientable;
2. E admits a bundle atlas whose transition functions take values in $\mathrm{GL}_k^+(\mathbb{R})$;
3. the determinant line bundle $\det E = \Lambda^k E$ (section 11.8) is trivial.

***Proof*:**
(1)⇒(2): the trivializations realizing an orientation have transitions preserving the standard orientation of $\mathbb{R}^k$, hence of positive determinant. (2)⇒(1): declaring μ_p to be the pullback of the standard orientation under any atlas trivialization at p is well defined, as the transitions have positive determinant, and locally constant by construction. (2)⇔(3): the transition functions of $\det E$ are $\det \tau_{\alpha\beta}$, so an atlas with $\tau_{\alpha\beta} \in \mathrm{GL}_k^+$ is exactly one making them positive. Given such an atlas, the local generators $\det \Phi_\alpha$ of $\det E$ differ on overlaps by the positive factors $\det \tau_{\alpha\beta}$, so patching them with a subordinate partition of unity, $s = \sum_\alpha \rho_\alpha \det \Phi_\alpha$, gives a nowhere-zero section: at each point the contributing generators are positive multiples of one another, so the sum cannot vanish. Hence $\det E$ is trivial. Conversely a nowhere-zero section of the line bundle $\det E$ (equivalently its triviality, corollary 11.4.4) rescales to a consistent positive generator, giving an atlas with positive transitions.

Definition 11.7.3: Oriented Manifold

A smooth manifold is *orientable* if its tangent bundle TM is orientable, and *oriented* once an orientation of TM is fixed. Concretely M is oriented by an atlas whose chart transition Jacobians have positive determinant everywhere.

Example 11.7.4: Orientability

1. $\mathbb{R}^k$-bundles that are trivial are orientable; S^n is an orientable manifold, oriented by the outward normal (its tangent bundle sits in a trivial $\mathbb{R}^{n+1}$ with an oriented complement).
2. The Möbius bundle $\gamma^1 \to \mathbb{R}\mathrm{P}^1$ is not orientable: its single nontrivial transition has determinant -1, and $\det \gamma^1 = \gamma^1$ is nontrivial. Hence $\mathbb{R}\mathrm{P}^2$, which contains a Möbius band with $T\mathbb{R}\mathrm{P}^2$ restricting nontrivially, is a non-orientable manifold; more generally $\mathbb{R}\mathrm{P}^n$ is orientable iff n is odd.
3. Every complex vector bundle is canonically oriented as a real bundle: complex bases give real bases whose changes, lying in $\mathrm{GL}_k(\mathbb{C}) \subseteq \mathrm{GL}_{2k}^+(\mathbb{R})$, have positive real determinant $|\det_{\mathbb{C}}|^2 > 0$.

The obstruction to orientability is a single cohomology class, the first Stiefel-Whitney class $w_1(E) \in H^1(M; \mathbb{Z}/2)$, which vanishes exactly when E is orientable; its construction and that of the higher characteristic classes belong to [11, Stiefel-Whitney Classes and the Total Class, First Stiefel-Whitney Class is the Obstruction to Orientability]. What is needed here is only the geometric criterion above, through which orientability feeds the integration theory of the next chapter.

11.8 Constructions on Vector Bundles

Linear algebra manufactures new vector spaces from old, dual, direct sum, tensor product, exterior power, and each construction is natural: an isomorphism of the inputs induces one of the outputs, compatibly with composition. Naturality is exactly what is needed to run the construction fiber by fiber over a base. The transition functions of a bundle are isomorphisms of $\mathbb{R}^k$; applying a natural construction to them produces a new cocycle, and the chart lemma turns that cocycle into a bundle. This section makes the principle precise and then reads off the bundles that carry the rest of the book.

Call a rule T assigning to each finite-dimensional real vector space V a vector space $T(V)$, and to each isomorphism $A\colon V \to W$ an isomorphism $T(A)\colon T(V) \to T(W)$, a *smooth functor* if $T(\mathrm{id}) = \mathrm{id}$, $T(AB) = T(A)T(B)$, and the induced map $\mathrm{GL}(V) \to \mathrm{GL}(T(V))$, $A \mapsto T(A)$, is smooth. Contravariant and multivariable versions are defined the same way, with $T(AB) = T(B)T(A)$ in the contravariant case.

Theorem 11.8.1: Bundle Functors

Let T be a smooth functor and $E \to M$ a smooth vector bundle. Then there is a smooth vector bundle $T(E) \to M$ with fibers $T(E)_p = T(E_p)$, whose local trivializations are $T(\Phi_\alpha)$ (applied fiberwise) and whose transition functions are $T(\tau_{\alpha\beta})$ for a covariant T, and $T(\tau_{\alpha\beta}^{-1})$ for a contravariant one. A bundle homomorphism $E \to E'$ covering id_M that is a fiberwise isomorphism induces one $T(E) \to T(E')$; multivariable smooth functors produce bundles from tuples of bundles the same way.

***Proof*:**
Set $T(E)_p = T(E_p)$ and, over each U_α, use Φ_α to identify $E_p \cong \mathbb{R}^k$ and hence $T(E_p) \cong T(\mathbb{R}^k)$; call the resulting bijection $\Psi_\alpha\colon T(E)|_{U_\alpha} \to U_\alpha \times T(\mathbb{R}^k)$. On an overlap,

$$\Psi_\alpha \Psi_\beta^{-1}(p, w) = \big(p,\ T(\tau_{\alpha\beta}(p))\, w\big),$$

since the fiber identifications differ by $\Phi_\alpha\Phi_\beta^{-1} = \tau_{\alpha\beta}$ and T is applied fiberwise. The map $p \mapsto T(\tau_{\alpha\beta}(p))$ is smooth, being the composite of the smooth $\tau_{\alpha\beta}$ with the smooth $A \mapsto T(A)$, and lands in $\mathrm{GL}(T(\mathbb{R}^k))$; functoriality gives the cocycle condition $T(\tau_{\alpha\beta})T(\tau_{\beta\gamma}) = T(\tau_{\alpha\gamma})$. Lemma 11.3.4 then makes $T(E)$ a smooth vector bundle with the Ψ_α as trivializations. For a contravariant T the identity $T(\tau_{\alpha\beta}) = T(\tau_{\beta\alpha}^{-1})$ turns the cocycle around, and $T(\tau_{\alpha\beta}^{-1})$ is the transition. A fiberwise isomorphism $F\colon E \to E'$ acts as $T(F_p)$ on fibers; in local frames this is T of a smooth GL-valued matrix, hence a smooth bundle isomorphism. The multivariable case is identical, applying the functor to the tuple $(\tau_{\alpha\beta}, \sigma_{\alpha\beta}, \dots)$.

Every standard operation is a smooth functor, so each yields a bundle. Their fibers are computed pointwise, which is what makes them usable: a section of $\mathrm{Hom}(E, F)$ is a smoothly varying linear map $E_p \to F_p$, a section of $\Lambda^k E^*$ is a smoothly varying alternating form, and so on.

Proposition 11.8.2: Standard Bundle Constructions

For smooth vector bundles E, F over M of ranks k, ℓ, the following are smooth vector bundles, with the stated fibers, transition functions, and ranks:

1. the *dual bundle* E^*, fiber $(E_p)^*$, transitions $(\tau_{\alpha\beta}^{-1})^\top$ (contravariant);
2. the *Whitney sum* $E \oplus F$, fiber $E_p \oplus F_p$, rank $k + \ell$, transitions $\tau_{\alpha\beta} \oplus \sigma_{\alpha\beta}$;
3. the *tensor product* $E \otimes F$, fiber $E_p \otimes F_p$, rank $k\ell$, transitions $\tau_{\alpha\beta} \otimes \sigma_{\alpha\beta}$;
4. the *homomorphism bundle* $\operatorname{Hom}(E, F) \cong E^* \otimes F$, fiber $\operatorname{Hom}(E_p, F_p)$;
5. the *exterior powers* $\Lambda^j E$, fiber $\Lambda^j E_p$, rank $\binom{k}{j}$, transitions $\Lambda^j \tau_{\alpha\beta}$, and the *symmetric powers* $S^j E$;
6. the *determinant bundle* $\det E = \Lambda^k E$, a line bundle with transitions $\det \tau_{\alpha\beta}$.

***Proof*:**
Each is the image under theorem 11.8.1 of the corresponding smooth functor of vector spaces: the dual $V \mapsto V^*$ (contravariant), the direct sum, the tensor product, Hom, and the exterior and symmetric powers. The fibers are the functor applied to the fibers; the transition functions are the functor applied to $\tau_{\alpha\beta}$ (its inverse for the contravariant dual, whence $(\tau_{\alpha\beta}^{-1})^\top$); and the ranks are the dimensions of the functor's value on $\mathbb{R}^k$, namely $k\ell$ for the tensor product and $\binom{k}{j}$ for Λ^j. The determinant is Λ^k of a rank-k bundle, a line bundle whose single transition entry is $\det \tau_{\alpha\beta}$.

The dual illustrates the contravariant twist. A basis of E_p has a dual basis of $(E_p)^*$, and if fibers are reglued by τ then dual bases are reglued by $(\tau^{-1})^\top$, so that the pairing $\langle \xi, v\rangle$ is preserved across trivializations. This is why the cotangent bundle, the dual of the tangent bundle, transforms by inverse-transpose Jacobians, the covariant transformation law of classical tensor calculus.

Base change is the remaining operation, and it goes the other way: a map of bases pulls a bundle back rather than pushing it forward.

Proposition 11.8.3: Pullback Bundle

Let $\pi\colon E \to M$ be a smooth rank-k bundle and $f\colon N \to M$ smooth. Then

$$f^*E = \{(q, e) \in N \times E \mid f(q) = \pi(e)\}$$

is a smooth rank-k bundle over N with fiber $(f^*E)_q = E_{f(q)}$ and projection $(q, e) \mapsto q$. If E has transition functions $\tau_{\alpha\beta}$ over $\{U_\alpha\}$, then f^*E has $\tau_{\alpha\beta} \circ f$ over $\{f^{-1}(U_\alpha)\}$. The map $(q, e) \mapsto e$ is a bundle homomorphism $f^*E \to E$ covering f, fiberwise an isomorphism.

***Proof*:**
Over $f^{-1}(U_\alpha)$ define $\tilde{\Phi}_\alpha(q, e) = (q, \pi_2\Phi_\alpha(e))$, a fiberwise-linear bijection onto $f^{-1}(U_\alpha) \times \mathbb{R}^k$; the overlaps compose to $(q, v) \mapsto (q, \tau_{\alpha\beta}(f(q))v)$, smooth and $\mathrm{GL}_k(\mathbb{R})$-valued and cocyclic since f is smooth. Lemma 11.3.4 gives the bundle structure, and $(q, e) \mapsto e$ is fiberwise the identity

$E_{f(q)} \to E_{f(q)}$, covering f.

Pullback is functorial, $(g \circ f)^* E \cong f^* g^* E$ and $\mathrm{id}^* E = E$, and commutes with the fiber constructions: $f^*(E \otimes F) \cong f^* E \otimes f^* F$ and likewise for $\oplus$, $(\,\cdot\,)^*$, and Λ^j. A homotopy-invariance theorem, that homotopic maps pull a bundle back to isomorphic bundles, holds over a paracompact base and is the entry point to the classification of bundles; it is proved in [11, Homotopy Invariance of Pullback Bundles, Classification of Vector Bundles over a Paracompact Base], and only its statement is needed downstream.

These constructions produce exactly the bundles the next chapter runs on. From the tangent bundle TM they build the *cotangent bundle* $T^*M = (TM)^*$, whose sections are the differential 1-forms; the *tensor bundles* $T^r_s M = (TM)^{\otimes r} \otimes (T^*M)^{\otimes s}$, whose sections are the tensor fields; and the *exterior bundles* $\Lambda^k T^*M$, whose sections are the differential k-forms that carry integration and de Rham cohomology. The constructions are now complete; chapter 12 applies them to the tangent bundle.

12

The Cotangent Bundle and Differential Forms

The constructions of chapter 11, applied to the tangent bundle, produce the objects of this chapter. Dualizing gives the cotangent bundle, whose sections are the differential 1-forms; tensoring gives the tensor bundles, whose sections are the tensor fields of classical geometry; and the exterior powers give the bundles whose sections are the differential forms that the rest of the book integrates and differentiates. The pointwise linear algebra of alternating tensors is settled once and for all in [2] and is not repeated here; what this chapter builds is the calculus of these fields on a manifold.

That calculus has one central operator. The exterior derivative d raises the degree of a form by one, squares to zero, and specializes in low dimensions to the gradient, the curl, and the divergence of vector calculus, unifying three operators and their integral theorems into a single identity. Its square being zero forces every exact form to be closed; whether the converse holds is a question about the shape of the manifold, and the gap between closed and exact is what de Rham cohomology (chapter 14) will measure. The chapter closes by singling out the top-degree forms and the orientations they encode, which is precisely the data that integration (chapter 13) uses.

The order of business is dictated by that arc: the cotangent bundle and the differential, then line integrals as the first taste of integrating a form, then tensor fields, the exterior algebra of forms, the exterior derivative with its invariant formula and Cartan's identity, and finally orientation and volume forms. Throughout, the algebra is cited and the calculus is developed here.

12.1 The Cotangent Bundle and 1-Forms

The tangent bundle records velocities; its dual records the linear measurements one can make of a velocity. This section dualizes the tangent bundle of chapter 11 to build the cotangent bundle, whose sections are the covector fields, or 1-forms. The differential of a function is the first of these: the coordinate-free residue of the gradient, defined with no metric in sight. The section closes with the pullback of covectors, an operation available for every smooth map and running opposite to it, in contrast with the pushforward of vectors, which needs a diffeomorphism.

A tangent vector at p eats a function and returns a number; a covector at p eats a tangent vector and returns a number. The covectors at p are therefore the elements of the dual space $(T_pM)^*$, and assembling these duals over all p is the dual-bundle construction of section 11.8 applied to TM.

Definition 12.1.1: Cotangent Bundle

Let M be a smooth n-manifold. The *cotangent bundle* of M is the dual bundle

$$T^*M = (TM)^*$$

of the tangent bundle (definition 2.4.1), formed by the dual construction of proposition 11.8.2. It is a smooth vector bundle of rank n over M; its fiber over p is the *cotangent space* $T_p^*M = (T_pM)^*$, the space of *covectors* at p. If the tangent bundle has transition functions $\tau_{\alpha\beta}$, the Jacobians of the chart transitions, then T^*M has transition functions $(\tau_{\alpha\beta}^{-1})^\mathsf{T}$.

The inverse-transpose transition law is the classical statement that covectors transform covariantly, oppositely to the vectors they pair against, so that the pairing $\langle \xi, v\rangle$ of a covector with a vector is chart-independent (section 11.8). A covector is a coordinate-free linear functional on velocities, not itself a velocity, and this is the distinction the two transition laws enforce.

A tangent vector lives at a single point; a vector field chooses one smoothly at every point. The covector analogue is a field of covectors, and it is the central object of the chapter.

Definition 12.1.2: Covector Field (1-Form)

A *covector field*, or *differential* 1*-form*, on M is a smooth section (definition 11.4.1) of the cotangent bundle: a smooth map $\omega\colon M \to T^*M$ with $\pi \circ \omega = \mathrm{id}_M$, assigning to each p a covector $\omega_p \in T_p^*M$. The 1-forms form a module over $C^\infty(M)$ under the pointwise operations $(f\omega + g\eta)_p = f(p)\omega_p + g(p)\eta_p$, written

$$\Omega^1(M) = \Gamma(T^*M)$$

Sections of T^*M are to 1-forms what functions are to 0-forms: a smooth function is a section of the trivial line bundle, and $\Omega^0(M) = C^\infty(M)$ under this reading. The rest of the section produces the 1-forms one computes with, beginning with the one attached to a function.

In $\mathbb{R}^n$ the gradient ∇f packages the partial derivatives of f into a vector field, but it is a vector field only because the dot product is used to identify vectors with covectors. Without a metric the partials assemble instead into a covector field, and this metric-free object is the differential.

Definition 12.1.3: Differential of a Function

Let $f \in C^\infty(M)$. The *differential* of f is the 1-form df whose value at p is the covector $df_p \in T_p^*M$ given by

$$df_p(v) = v(f), \qquad v \in T_pM$$

Thus df_p sends a tangent vector to the directional derivative of f along it.

That df is a smooth section, and not just a rough one, is not yet visible from the definition; it becomes clear once df is written in a coordinate frame. Computing with 1-forms needs a local frame for T^*M, and the coordinate functions give it: each coordinate x^i on a chart domain is a smooth function, so it has a differential $dx^i = d(x^i)$, a 1-form on that domain.

Proposition 12.1.4: Coordinate Covector Frame

Let (U, φ) be a chart of M with coordinates $\varphi = (x^1, \dots, x^n)$. The differentials $dx^1, \dots, dx^n$ form a smooth local frame for T^*M over U, dual to the coordinate vector fields:

$$dx^i\left(\frac{\partial}{\partial x^j}\right) = \delta^i_j$$

Every 1-form ω is written over U as $\omega = \sum_{i=1}^n \omega_i\, dx^i$ with component functions $\omega_i = \omega\big(\partial/\partial x^i\big)$, and ω is smooth on U if and only if each component ω_i is smooth. In particular, for $f \in C^\infty(M)$,

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x^i}\, dx^i$$

so df is a smooth 1-form.

***Proof*:**
Step 1: Duality. By the definition of the differential and of the coordinate vector fields, for each $p \in U$,

$$dx^i_p\left(\frac{\partial}{\partial x^j}\bigg|_p\right) = \frac{\partial}{\partial x^j}\bigg|_p (x^i) = \frac{\partial x^i}{\partial x^j}(p) = \delta^i_j$$

Since $\big(\partial/\partial x^j|_p\big)_j$ is a basis of T_pM, the covectors $\big(dx^i_p\big)_i$ are its dual basis in T_p^*M, and so form a basis of the fiber at every $p \in U$.

Step 2: Smoothness of the frame. The coordinate vector fields trivialize T^*M over U by $\xi \mapsto \big(p, (\xi(\partial/\partial x^j|_p))_j\big)$ for $\xi \in T_p^*M$ (section 11.4). In this trivialization dx^i is the constant section with component vector e_i, by Step 1, hence smooth. So $(dx^1, \dots, dx^n)$ is a smooth local frame (definition 11.4.2) dual to $(\partial/\partial x^i)$.

Step 3: Components. Any covector $\xi \in T_p^*M$ and the covector $\sum_i \xi(\partial/\partial x^i|_p)\, dx^i_p$ agree on each basis vector $\partial/\partial x^j|_p$ by Step 1, hence are equal. Applying this to $\xi = \omega_p$ gives $\omega = \sum_i \omega_i\, dx^i$ over U with $\omega_i = \omega(\partial/\partial x^i)$. The components ω_i are the coordinates of ω in the smooth frame (dx^i), and a section is smooth if and only if its coordinates in a smooth frame are smooth (section 11.4), giving the smoothness criterion.

Step 4: The differential. Apply Step 3 to $\omega = df$. Its components are $df_i = df(\partial/\partial x^i) =$

$(\partial/\partial x^i)(f) = \partial f/\partial x^i$, each smooth, so $df = \sum_i (\partial f/\partial x^i)\, dx^i$ is a smooth 1-form, as claimed.

Here and throughout the chapter a repeated index, one upper and one lower, is summed over its range (the summation convention), though the summation sign is often kept explicit. The formula $df = \sum_i (\partial f/\partial x^i)\, dx^i$ is the coordinate-free content of the total-derivative notation of calculus: dx^i is now a genuine covector field, not an infinitesimal, and $\partial f/\partial x^i$ is its i-th component. Taking $f = x^j$ recovers $d(x^j) = dx^j$, so the two uses of the symbol dx^j agree.

Example 12.1.5: Differentials on $\mathbb{R}^2$ and on the Sphere

1. On $\mathbb{R}^2$ with coordinates (x, y), let $f(x, y) = x^2 + y^2$. Then

$$df = 2x\, dx + 2y\, dy$$

   At the point $(1, 0)$ this is $df_{(1,0)} = 2\, dx$, so on a tangent vector $v = a\, \partial/\partial x + b\, \partial/\partial y$ it returns $df_{(1,0)}(v) = 2a$: the differential of f measures the component of v pointing radially outward, and ignores the tangential component b.

2. Let $h = z|_{S^2}$ be the height function on the unit sphere $S^2 \subseteq \mathbb{R}^3$, the restriction of the third coordinate z. For $p \in S^2$ and $v \in T_pS^2 \subseteq T_p\mathbb{R}^3$,

$$dh_p(v) = v(z) = dz_p(v)$$

   so dh_p is the restriction of dz_p to T_pS^2. It vanishes exactly when T_pS^2 is horizontal, that is at the two poles $(0, 0, \pm 1)$; everywhere else $dh_p \neq 0$. The points where the differential of a function vanishes are its critical points, and h has precisely two.

A smooth map $f\colon M \to N$ does not in general push a vector field on M forward to a vector field on N (chapter 5): a value at a point of N not in the image is undefined, and a point with two preimages may receive two competing vectors. Covectors run the other way and meet no such obstruction. A covector on N pulls back to a covector on M by precomposition with the tangent map, with no hypothesis on f at all.

Definition 12.1.6: Pullback of a Covector Field

Let $f\colon M \to N$ be smooth and $\omega \in \Omega^1(N)$. The *pullback* $f^*\omega$ is the covector field on M whose value at p is

$$(f^*\omega)_p = \omega_{f(p)} \circ T_pf \in T_p^*M$$

where $T_pf\colon T_pM \to T_{f(p)}N$ is the tangent map (definition 2.2.4); explicitly $(f^*\omega)_p(v) = \omega_{f(p)}(T_pf\, v)$ for $v \in T_pM$.

The pullback is computed through a single identity: it commutes with the differential. This one rule, together with linearity over functions, reduces every pullback to a coordinate calculation and shows along the way that $f^*\omega$ is smooth.

Proposition 12.1.7: Pullback Commutes with the Differential

Let $f\colon M \to N$ be smooth. For every $g \in C^\infty(N)$,

$$f^*(dg) = d(g \circ f) \tag{12.1}$$

Consequently $f^*\colon \Omega^1(N) \to \Omega^1(M)$ is $\mathbb{R}$-linear and satisfies $f^*(h\,\omega) = (h \circ f)\, f^*\omega$ for $h \in C^\infty(N)$; and in coordinates $\psi = (y^1, \dots, y^m)$ on N, with $\omega = \sum_j w_j\, dy^j$ and $f^j = y^j \circ f$ the components of f,

$$f^*\omega = \sum_j (w_j \circ f)\, d(f^j)$$

which is a smooth 1-form. In particular $f^*\omega \in \Omega^1(M)$.

***Proof*:**
Step 1: Naturality of the differential. Fix $p \in M$ and $v \in T_pM$. Using the definition of the pullback, then the differential, then the defining property of the tangent map $T_pf(v)(g) = v(g \circ f)$ (definition 2.2.4),

$$(f^*(dg))_p(v) = dg_{f(p)}(T_pf\, v) = (T_pf\, v)(g) = v(g \circ f) = d(g \circ f)_p(v)$$

Since p and v are arbitrary, $f^*(dg) = d(g \circ f)$, which is (12.1).

Step 2: Linearity over functions. For each p the map $\xi \mapsto \xi \circ T_pf$ is linear on $T^*_{f(p)}N$, so f^* is $\mathbb{R}$-linear. For $h \in C^\infty(N)$,

$$(f^*(h\omega))_p = (h\omega)_{f(p)} \circ T_pf = h(f(p))\, (\omega_{f(p)} \circ T_pf) = (h \circ f)(p)\, (f^*\omega)_p$$

so $f^*(h\omega) = (h \circ f)\, f^*\omega$.

Step 3: Coordinate formula and smoothness. Near $f(p)$ write $\omega = \sum_j w_j\, dy^j$. By Steps 1 and 2,

$$f^*\omega = \sum_j (w_j \circ f)\, f^*(dy^j) = \sum_j (w_j \circ f)\, d(y^j \circ f) = \sum_j (w_j \circ f)\, d(f^j)$$

Each $w_j \circ f$ is smooth and each $d(f^j)$ is a smooth 1-form (proposition 12.1.4), so $f^*\omega$ is a smooth 1-form, completing the proof.

Formula (12.1) makes $\Omega^1(-)$ a contravariant functor: pullback along a composite factors as $(g \circ f)^* = f^* \circ g^*$ and the identity pulls back to the identity, as the exercises verify. This contravariance, and the fact that the pullback needs no hypothesis on the map, are the structural reasons that differential forms rather than vector fields are the objects integrated over a manifold and assembled into de Rham cohomology (chapter 14). The next section takes the first step in that direction by integrating a 1-form along a curve.

Example 12.1.8: Pullback under Polar Coordinates
Let $\Phi\colon (0, \infty) \times \mathbb{R} \to \mathbb{R}^2$ be the polar-coordinate map $\Phi(r, \theta) = (r\cos\theta, r\sin\theta)$, so its component functions are $x \circ \Phi = r\cos\theta$ and $y \circ \Phi = r\sin\theta$. By (12.1),

$$\Phi^*(dx) = d(r\cos\theta) = \cos\theta\, dr - r\sin\theta\, d\theta, \qquad \Phi^*(dy) = d(r\sin\theta) = \sin\theta\, dr + r\cos\theta\, d\theta$$

Pulling back two combinations then gives

$$\Phi^*(x\,dx + y\,dy) = r\,dr, \qquad \Phi^*(x\,dy - y\,dx) = r^2\,d\theta$$

the second computation using $x \circ \Phi = r\cos\theta$ and $y \circ \Phi = r\sin\theta$ to clear the coefficients. The form $(x\,dy - y\,dx)/r^2$ therefore pulls back to $d\theta$, which is why it is called the angle form; it reappears in section 12.2.

Exercise 12.1.1

1. On $\mathbb{R}^3$ with coordinates (x, y, z), compute the differential of $f(x, y, z) = xy + z^2$. Then evaluate $df_{(1,2,3)}$ on the tangent vector $v = 2\,\partial/\partial x - \partial/\partial y + \partial/\partial z$.
2. Show that the differential obeys the Leibniz rule $d(fg) = f\,dg + g\,df$ for all $f, g \in C^\infty(M)$, and deduce that $d(f^2) = 2f\,df$.
3. Let $\gamma\colon \mathbb{R} \to \mathbb{R}^2$ be $\gamma(t) = (\cos t, \sin t)$ and let $\omega = x\,dy - y\,dx$. Compute the pullback $\gamma^*\omega$.
4. Let (U, φ) and (V, ψ) be overlapping charts with coordinates x^i and y^j. Using proposition 12.1.4, prove the covector transformation law
$$dx^i = \sum_j \frac{\partial x^i}{\partial y^j}\,dy^j$$
on $U \cap V$, and explain how the change-of-coframe matrix relates to the transition functions of T^*M recorded in definition 12.1.1.
5. For smooth maps $M \xrightarrow{f} N \xrightarrow{g} P$ and a 1-form $\omega \in \Omega^1(P)$, prove that $(g \circ f)^*\omega = f^*(g^*\omega)$ and that $\mathrm{id}_M^*\omega = \omega$. Conclude that $\Omega^1(-)$ is a contravariant functor.

12.2 Line Integrals and Conservative Fields

A 1-form assigns to each tangent vector a number, so it is exactly the object that can be integrated against the velocity of a curve. Pairing the form with $\gamma'(t)$ at each time and integrating over the parameter interval produces the line integral, the first and simplest instance of integrating a differential form. This section builds that integral, checks that it sees only the oriented image of the curve and not its parametrization, and proves the fundamental theorem of line integrals: the integral of a differential df reads off the change in f between the endpoints. That theorem forces the line integral of an exact form to depend on nothing but the endpoints, and the converse turns out to hold as well, giving a clean characterization of the 1-forms that are gradients. The residue of the discussion is a necessary condition, closedness, whose failure to be sufficient is the first appearance of the topology that chapter 14 will measure.

Throughout, M is a smooth n-manifold and a *curve* in M is a smooth map $\gamma\colon [a, b] \to M$; its *velocity* at t is the tangent vector $\gamma'(t) = T_t\gamma\big(\frac{d}{dt}\big) \in T_{\gamma(t)}M$, the image of the standard basis vector of $T_t[a, b]$ under the tangent map. A *piecewise-smooth curve* is a continuous $\gamma\colon [a, b] \to M$ for which there is a partition $a = t_0 < t_1 < \cdots < t_k = b$ with each restriction $\gamma|_{[t_{i-1}, t_i]}$ smooth; the one-sided velocities at the breakpoints need not agree.

The pairing $\omega_{\gamma(t)}(\gamma'(t))$ of a 1-form ω (definition 12.1.2) with the velocity is a real number depending smoothly on t: in a chart (U, x) where $\omega = \sum_i a_i\, dx^i$ and $\gamma^i = x^i \circ \gamma$, it equals $\sum_i a_i(\gamma(t))\, \dot\gamma^i(t)$, a smooth function of t. It is therefore a legitimate integrand.

Definition 12.2.1: Line Integral of a 1-Form

Let ω be a 1-form on M and $\gamma\colon [a,b] \to M$ a smooth curve. The *line integral* of ω along γ is

$$\int_\gamma \omega = \int_a^b \omega_{\gamma(t)}\big(\gamma'(t)\big)\, dt$$

For a piecewise-smooth γ with partition $a = t_0 < \cdots < t_k = b$, set $\int_\gamma \omega = \sum_{i=1}^k \int_{\gamma|_{[t_{i-1},t_i]}} \omega$.

Equivalently, the integrand is the pullback: $\gamma^*\omega$ (definition 12.1.6) is the 1-form $\omega_{\gamma(t)}(\gamma'(t))\, dt$ on $[a,b]$, and $\int_\gamma \omega = \int_{[a,b]} \gamma^*\omega$ is the ordinary integral of that pullback. In the chart coordinates above,

$$\int_\gamma \omega = \int_a^b \sum_i a_i(\gamma(t))\, \dot\gamma^i(t)\, dt \tag{12.2}$$

which is the expression used in every computation. When ω is read as a force field on M, the pairing $\omega(\gamma')$ is the rate at which the force does work along the trajectory, and $\int_\gamma \omega$ is the total work done in moving from $\gamma(a)$ to $\gamma(b)$; this is the mechanical origin of the construction and the reason the endpoint dependence studied below is called conservation of energy.

The first example shows the mechanics and already exhibits the phenomenon that organizes the rest of the section: the value can depend on the route, not just its ends.

Example 12.2.2: A Path-Dependent Line Integral

On $\mathbb{R}^2$ take $\omega = y\, dx$ and integrate from $(0,0)$ to $(1,1)$ along two curves.

Along the diagonal $\gamma(t) = (t,t)$, $t \in [0,1]$, one has $\dot\gamma^1 = 1$ and $y(\gamma(t)) = t$, so by (12.2)

$$\int_\gamma \omega = \int_0^1 t \cdot 1\, dt = \tfrac{1}{2}$$

Along the staircase σ that runs first from $(0,0)$ to $(1,0)$, then from $(1,0)$ to $(1,1)$, the first leg has $y = 0$ so contributes 0, and the second leg has $x \equiv 1$ so $\dot\sigma^1 = 0$ and again contributes 0; hence $\int_\sigma \omega = 0$.

The two curves share their endpoints, yet $\int_\gamma \omega = \frac{1}{2} \neq 0 = \int_\sigma \omega$. The line integral of $y\, dx$ is not determined by the endpoints alone, so this 1-form is not a differential of any function on $\mathbb{R}^2$, a point the fundamental theorem below makes precise.

The line integral is attached to the parametrized curve γ, but its value should be a property of the oriented arc swept out, unchanged if the same arc is traced at a different speed. Reparametrizing by an orientation-preserving diffeomorphism of parameter intervals leaves the value fixed; reversing the orientation flips the sign.

Proposition 12.2.3: Reparametrization Invariance

Let ω be a 1-form on M, let $\gamma\colon [a,b] \to M$ be a piecewise-smooth curve, and let $\varphi\colon [c,d] \to [a,b]$ be a diffeomorphism. Then

$$\int_{\gamma\circ\varphi} \omega = \begin{cases} \int_\gamma \omega & \text{if } \varphi \text{ is orientation-preserving } (\varphi(c) = a,\ \varphi(d) = b) \\ -\int_\gamma \omega & \text{if } \varphi \text{ is orientation-reversing } (\varphi(c) = b,\ \varphi(d) = a) \end{cases}$$

***Proof*:**
It suffices to treat a smooth γ; the piecewise case follows by summing over the smooth pieces. Write $\eta = \gamma\circ\varphi$. By the chain rule the velocity of the reparametrized curve is $\eta'(s) = \varphi'(s)\,\gamma'(\varphi(s))$, so, using linearity of the covector $\omega_{\eta(s)} = \omega_{\gamma(\varphi(s))}$,

$$\omega_{\eta(s)}\big(\eta'(s)\big) = \varphi'(s)\,\omega_{\gamma(\varphi(s))}\big(\gamma'(\varphi(s))\big)$$

Let $g(t) = \omega_{\gamma(t)}(\gamma'(t))$, so the right-hand side is $\varphi'(s)\,g(\varphi(s))$. Then

$$\int_{\gamma\circ\varphi} \omega = \int_c^d g(\varphi(s))\,\varphi'(s)\,ds$$

and the substitution $t = \varphi(s)$, $dt = \varphi'(s)\,ds$ converts this to $\int_{\varphi(c)}^{\varphi(d)} g(t)\,dt$. If φ is orientation-preserving the limits are a and b, giving $\int_a^b g\,dt = \int_\gamma \omega$; if orientation-reversing the limits are b and a, giving $\int_b^a g\,dt = -\int_\gamma \omega$, as we sought to show.

Reparametrization invariance is what lets the line integral be written $\int_\gamma \omega$ with γ standing for the oriented curve rather than a chosen parametrization, and the sign rule records that reversing a curve negates its integral. This sign will drive the loop criterion for conservative forms below, where a path and its reverse are concatenated into a loop.

The central computational fact is that integrating the differential of a function recovers the function's net change. This is the manifold form of the fundamental theorem of calculus, and it is proved by pulling the integrand back to the parameter interval, where the ordinary theorem applies.

Theorem 12.2.4: Fundamental Theorem for Line Integrals

Let $f \in C^\infty(M)$ and let $\gamma\colon [a,b] \to M$ be a piecewise-smooth curve. Then

$$\int_\gamma df = f(\gamma(b)) - f(\gamma(a))$$

In particular the integral of the exact form df depends only on the endpoints of γ, and vanishes when γ is closed.

***Proof*:**
Assume first that γ is smooth. By the definition of the differential (definition 12.1.3), $(df)_{\gamma(t)}$ acts on a tangent vector by differentiation, so

$$(df)_{\gamma(t)}\big(\gamma'(t)\big) = \gamma'(t)(f) = \frac{d}{dt}\big(f \circ \gamma\big)(t)$$

the last equality being the definition of the velocity as the tangent map applied to $\frac{d}{dt}$. Hence the integrand of $\int_\gamma df$ is the derivative of the smooth real-valued function $f \circ \gamma$ on $[a, b]$, and the ordinary fundamental theorem of calculus gives

$$\int_\gamma df = \int_a^b \frac{d}{dt}\big(f \circ \gamma\big)(t)\, dt = (f \circ \gamma)(b) - (f \circ \gamma)(a)$$

which is the claim for smooth γ. For a piecewise-smooth γ with partition $a = t_0 < \cdots < t_k = b$, summing the smooth case over the pieces telescopes:

$$\int_\gamma df = \sum_{i=1}^{k} \big(f(\gamma(t_i)) - f(\gamma(t_{i-1}))\big) = f(\gamma(t_k)) - f(\gamma(t_0)) = f(\gamma(b)) - f(\gamma(a))$$

since γ is continuous across the breakpoints. When γ is closed, $\gamma(b) = \gamma(a)$ and the difference is zero, completing the proof.

The theorem says that an exact form carries no information beyond a potential function, and reading it against a curve simply differences that potential. Two immediate consequences deserve names, because their converse is the substance of the section: the integral of df is the same for any two curves with common endpoints, and it is zero around any loop. A 1-form for which the line integral has this endpoint-only dependence is the object that classical mechanics calls conservative, the potential being the potential energy.

Definition 12.2.5: Conservative 1-Form

A 1-form ω on M is *conservative* if the line integral $\int_\gamma \omega$ depends only on the endpoints $\gamma(a)$ and $\gamma(b)$ of the piecewise-smooth curve γ: whenever two such curves have the same initial and terminal points, their integrals agree.

Example 12.2.6: The Coordinate Form dx^i Is Conservative
On $\mathbb{R}^n$ the coordinate 1-form dx^i is the differential of the i-th coordinate function, so theorem 12.2.4 gives $\int_\gamma dx^i = x^i(\gamma(b)) - x^i(\gamma(a))$ for every curve, an endpoint-only quantity. Thus dx^i is conservative. By contrast $y\,dx$ is not conservative, as example 12.2.2 exhibits two curves with equal endpoints and unequal integrals.

By theorem 12.2.4 every exact form is conservative. The converse holds, and the proof is constructive: given a conservative form, integrating it out from a fixed basepoint manufactures a potential. Path-independence is precisely what makes the resulting function well-defined, and its differential recovers the form.

Theorem 12.2.7: Conservative Equals Exact

Let ω be a 1-form on the smooth manifold M. The following are equivalent.

1. ω is exact: $\omega = df$ for some $f \in C^\infty(M)$.
2. ω is conservative (definition 12.2.5).
3. $\int_\gamma \omega = 0$ for every piecewise-smooth closed curve γ in M.

***Proof*:**
Step 1: (1) ⇒ (2). If $\omega = df$, then by theorem 12.2.4 $\int_\gamma \omega = f(\gamma(b)) - f(\gamma(a))$ depends only on the endpoints, so ω is conservative.

Step 2: (2) ⇔ (3). Suppose ω is conservative and $\gamma\colon [a,b] \to M$ is closed, so $\gamma(a) = \gamma(b) = p$. The constant curve c_p at p has the same endpoints and integral 0, so path-independence forces $\int_\gamma \omega = \int_{c_p} \omega = 0$; this is (3). Conversely assume (3) and let $\gamma_1, \gamma_2\colon [0,1] \to M$ be piecewise-smooth with $\gamma_1(0) = \gamma_2(0)$ and $\gamma_1(1) = \gamma_2(1)$. Let $\overline{\gamma_2}(t) = \gamma_2(1-t)$ be the reverse of γ_2; by proposition 12.2.3 (applied to $\varphi(t) = 1 - t$) one has $\int_{\overline{\gamma_2}} \omega = -\int_{\gamma_2} \omega$. The concatenation γ that traverses γ_1 and then $\overline{\gamma_2}$ is a piecewise-smooth closed curve, and by additivity over the pieces (definition 12.2.1)

$$0 = \int_\gamma \omega = \int_{\gamma_1} \omega + \int_{\overline{\gamma_2}} \omega = \int_{\gamma_1} \omega - \int_{\gamma_2} \omega$$

so $\int_{\gamma_1} \omega = \int_{\gamma_2} \omega$, which is (2).

Step 3: (2) ⇒ (1). Assume ω conservative. The construction is carried out on each connected component of M separately; since components of a manifold are open, a function defined smoothly on each is smooth on M, so it suffices to treat M connected. A connected manifold is path-connected, and any two of its points are joined by a piecewise-smooth curve (chain together the straight coordinate segments of a finite chart cover of a connecting continuous path). Fix a basepoint p_0 and define

$$f(p) = \int_{\gamma_p} \omega$$

where γ_p is any piecewise-smooth curve from p_0 to p; by hypothesis (2) this is independent of the choice of γ_p, so $f\colon M \to \mathbb{R}$ is well-defined.

It remains to show f is smooth with $df = \omega$. Fix p and a chart (U, x) centered at p with $\omega = \sum_i a_i\, dx^i$ on U. For small s let $\sigma_s\colon [0,s] \to U$ be the i-th coordinate segment through p, the curve whose x^i-coordinate is t and whose other coordinates are those of p. Concatenating γ_p with σ_s and using path-independence,

$$f(\sigma_s(s)) - f(p) = \int_{\sigma_s} \omega = \int_0^s a_i(\sigma_s(t))\, dt$$

by (12.2), since along σ_s only the i-th coordinate varies. Dividing by s and letting $s \to 0$, the right-hand side tends to $a_i(p)$ by continuity of a_i (the same computation with the oppositely oriented segment gives the limit from the other side), so $\partial f/\partial x^i(p) = a_i(p)$. As $p \in U$ and i were arbitrary, $\partial f/\partial x^i = a_i$ on U for each i; the partials are smooth, so f is smooth there, and $df = \sum_i a_i\, dx^i = \omega$ on U (proposition 12.1.4). Since the charts cover M, f is smooth with $df = \omega$, so ω is exact. This closes the cycle (1) $\Rightarrow$ (2) $\Leftrightarrow$ (3) $\Rightarrow$ (1), establishing the equivalence, as we sought to show.

The theorem reduces the analytic question "is this force conservative" to the algebraic one "is this form a differential", and hands back a construction of the potential. The potential is unique up to a locally constant function: if $df = dg = \omega$ then $d(f - g) = 0$, so $f - g$ has vanishing differential and is constant on each component of M.

A necessary condition for exactness is visible in coordinates and is far easier to check than producing a potential. If $\omega = df$ then $a_i = \partial f/\partial x^i$ in every chart, and the equality of mixed second partials of the smooth function f forces the components to satisfy a symmetry.

Proposition 12.2.8: Exact Forms Are Closed

If a 1-form ω on M is exact, then in every chart (U, x) its components $\omega = \sum_i a_i\, dx^i$ satisfy

$$\frac{\partial a_i}{\partial x^j} = \frac{\partial a_j}{\partial x^i} \qquad \text{for all } i, j \tag{12.3}$$

A 1-form satisfying (12.3) in every chart is called *closed*.

***Proof*:**
Write $\omega = df$. On a chart, $a_i = \partial f/\partial x^i$ by proposition 12.1.4, so

$$\frac{\partial a_i}{\partial x^j} = \frac{\partial^2 f}{\partial x^j\, \partial x^i} = \frac{\partial^2 f}{\partial x^i\, \partial x^j} = \frac{\partial a_j}{\partial x^i}$$

where the middle equality is the symmetry of second partial derivatives of the smooth function f. This is (12.3), completing the proof.

The condition (12.3) is the coordinate form of a single invariant equation, $d\omega = 0$, once the exterior derivative d of a form is defined in section 12.5; there closedness is recast without reference to a chart, and shown to be automatic for exact forms in every degree. Whether the converse holds, whether every closed form is exact, is a different matter entirely. It is false in general, and the failure is not a defect of the analysis but a feature of the manifold. The standard example is the angle form on the punctured plane.

Example 12.2.9: The Angle Form on $\mathbb{R}^2 \setminus \{0\}$
On $M = \mathbb{R}^2 \setminus \{0\}$ define the 1-form

$$\omega = \frac{-y\, dx + x\, dy}{x^2 + y^2}$$

with components $a_1 = -y/(x^2 + y^2)$ and $a_2 = x/(x^2 + y^2)$, both smooth on M.

The form is closed. A direct computation gives

$$\frac{\partial a_2}{\partial x} = \frac{(x^2+y^2) - x(2x)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}, \qquad \frac{\partial a_1}{\partial y} = \frac{-(x^2+y^2) + y(2y)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

so (12.3) holds and ω is closed.

The form is not exact. Integrate around the unit circle $\gamma(t) = (\cos t, \sin t)$, $t \in [0, 2\pi]$. Here $\dot\gamma^1 = -\sin t$, $\dot\gamma^2 = \cos t$, and $x^2 + y^2 = 1$, so the integrand of (12.2) is

$$a_1\dot\gamma^1 + a_2\dot\gamma^2 = (-\sin t)(-\sin t) + (\cos t)(\cos t) = 1$$

whence $\int_\gamma \omega = \int_0^{2\pi} 1\,dt = 2\pi$. The line integral around a closed curve is nonzero, so by theorem 12.2.7 the form ω is not conservative and therefore not exact.

The obstruction is the hole at the origin. On any region of M that omits a ray from the origin, for instance the right half-plane $\{x > 0\}$, the smooth function $\theta = \arctan(y/x)$ satisfies $d\theta = \omega$ (a computation identical to the closedness check), so ω is locally exact; it measures the polar angle, which increases by 2π once around the origin and admits no single-valued smooth branch on all of M. The number $\frac{1}{2\pi}\int_\gamma \omega$ is the winding number of γ about the origin, and its nonvanishing for the unit circle detects the puncture. The precise statement, that on $\mathbb{R}^2 \setminus \{0\}$ the closed forms modulo exact forms are one-dimensional and generated by ω, is the first computation of de Rham cohomology in chapter 14.

These conditions form a hierarchy. Exact is the strongest condition, closed is the necessary condition, and the two coincide precisely when the manifold has no holes for a loop to wind around. The gap between them is a topological invariant of M; naming it, computing it, and proving that it is a topological invariant is the programme of chapter 14, for which the closed-versus-exact distinction introduced here is the point of departure.

Exercise 12.2.1

1. On $\mathbb{R}^2$ compute $\int_\gamma (x\,dy - y\,dx)$ along the upper unit semicircle $\gamma(t) = (\cos t, \sin t)$, $t \in [0, \pi]$, from $(1, 0)$ to $(-1, 0)$. Interpret the answer geometrically.
2. Show that the 1-form $\omega = (2xy + z)\,dx + x^2\,dy + x\,dz$ on $\mathbb{R}^3$ is exact by producing a potential f with $df = \omega$, and use theorem 12.2.4 to evaluate $\int_\gamma \omega$ for any piecewise-smooth curve γ from $(0, 0, 0)$ to $(1, 1, 1)$.
3. Let $\gamma\colon [a, b] \to M$ be a piecewise-smooth curve and let $\overline{\gamma}(t) = \gamma(a + b - t)$ be its reverse. Prove directly from proposition 12.2.3 that $\int_{\overline{\gamma}} \omega = -\int_\gamma \omega$ for every 1-form ω.
4. Let $\omega = (-y\,dx + x\,dy)/(x^2 + y^2)$ be the angle form of example 12.2.9. Compute $\oint_{C_R} \omega$ where C_R is the circle of radius $R > 0$ about the origin traversed once counterclockwise, and observe that the value is independent of R. Then explain, without a further computation, why $\oint_\beta \omega = 0$ when β is the boundary of a square that does not contain the origin.
5. Decide whether $\omega = e^x \sin y\,dx + e^x \cos y\,dy$ on $\mathbb{R}^2$ is exact, and if so exhibit a potential. Verify first that ω is closed.
6. Prove that the conservative 1-forms on M form an $\mathbb{R}$-linear subspace of $\Omega^1(M)$, and that on a connected manifold the potential of a conservative form is unique up to an additive constant.

12.3 Tensor Fields

The differential 1-forms of section 12.1 are the sections of a single bundle, T^*M. Tensoring copies of TM and T^*M together produces bundles whose sections carry several vector or covector arguments at once: the metrics, endomorphism fields, and higher tensors of classical geometry. This section builds these tensor fields from the constructions of section 11.8, equips them with the two operations that survive from pointwise multilinear algebra, contraction and pullback, and proves the criterion that recognizes a tensor field by a linearity test rather than a transformation law. The Riemannian metric, the first tensor field beyond a form or a vector field, closes the section as the motivating example.

The pointwise algebra of tensors on a single vector space, the tensor product, its multilinear interpretation, contraction, and the basis for $V^{\otimes r} \otimes (V^*)^{\otimes s}$, is developed once in [2] and is applied here fiberwise. What this section develops is the bundle version: the smoothly varying family of tensors and the differential-topological operations on it.

Definition 12.3.1: Tensor Field

Let M be a smooth n-manifold with tangent bundle TM (definition 2.4.1) and cotangent bundle T^*M (section 12.1). For integers $r, s \geq 0$ the *tensor bundle of type* (r,s) is

$$T^r_s M = (TM)^{\otimes r} \otimes (T^*M)^{\otimes s},$$

the smooth vector bundle produced from TM and T^*M by the tensor construction of section 11.8 (proposition 11.8.2). Its fiber at p is $(T_pM)^{\otimes r} \otimes (T_p^*M)^{\otimes s}$. A *tensor field of type* (r,s), or an (r,s)*-tensor field*, is a smooth section $A \in \Gamma(T^r_s M)$; write $\mathcal{T}^r_s(M) = \Gamma(T^r_s M)$ for the space of such fields. A field of type $(0,s)$ is called *covariant*, one of type $(r,0)$ *contravariant*.

The multilinear algebra of [2, Tensors as Multilinear Maps] identifies the fiber $(T^r_s M)_p = (T_pM)^{\otimes r} \otimes (T_p^*M)^{\otimes s}$ with the space of real multilinear maps

$$A_p \colon \underbrace{T_p^*M \times \cdots \times T_p^*M}_{r} \times \underbrace{T_pM \times \cdots \times T_pM}_{s} \longrightarrow \mathbb{R},$$

covector arguments feeding the contravariant slots and vector arguments the covariant ones. This is the picture to keep in mind: an (r,s)-tensor field assigns to each point a multilinear gadget eating r covectors and s vectors. The low types recover familiar objects. A $(0,0)$-field is a smooth function; a $(1,0)$-field is a vector field, $\mathcal{T}^1_0(M) = \mathfrak{X}(M)$; and a $(0,1)$-field is a 1-form (definition 12.1.2), $\mathcal{T}^0_1(M) = \Omega^1(M)$. Covariant fields of higher order act on tuples of vector fields to give functions, which is the source of the characterization proved below.

In a chart (U,x), abbreviate $\partial_i = \partial/\partial x^i$ for the coordinate vector fields and dx^i for the coordinate covector fields dual to them (proposition 12.1.4). The tensor products $\partial_{i_1} \otimes \cdots \otimes \partial_{i_r} \otimes dx^{j_1} \otimes \cdots \otimes dx^{j_s}$ frame $T^r_s M$ over U, so every (r,s)-tensor field is

$$A = \sum A^{i_1 \cdots i_r}_{j_1 \cdots j_s}\, \partial_{i_1} \otimes \cdots \otimes \partial_{i_r} \otimes dx^{j_1} \otimes \cdots \otimes dx^{j_s}$$

on U, the sum running over all index tuples, with component functions $A^{i_1 \cdots i_r}_{j_1 \cdots j_s} = A(dx^{i_1}, \ldots, dx^{i_r}, \partial_{j_1}, \ldots, \partial_{j_s})$. Because the tensor bundle is built by the chart lemma from the transition functions of TM (theorem 11.8.1), a rough section is smooth if and only if its components are smooth functions in every chart. The components of a $(1,1)$-field, for instance, carry one Jacobian and one inverse Jacobian between charts, the mixed transformation law of classical tensor calculus.

Example 12.3.2: Endomorphism Fields

The type $(1,1)$ is the first genuinely new case. Pointwise, $T_pM \otimes T_p^*M$ is the space $\mathrm{End}(T_pM)$ of linear endomorphisms of T_pM, the decomposable tensor $v \otimes \xi$ acting as $w \mapsto \xi(w)\,v$ ([2, $V^* \otimes W$ and Linear Maps]). A $(1,1)$-tensor field is therefore a smoothly varying endomorphism of the tangent spaces, a section of $\mathrm{End}(TM) = T^1_1 M$.

Two examples on $M = \mathbb{R}^2$ with standard coordinates (x, y). The *identity field* $\delta = \partial_x \otimes dx + \partial_y \otimes dy$ acts as $\delta_p = \mathrm{id}_{T_p\mathbb{R}^2}$ at every point. The *rotation field*

$$J = \partial_y \otimes dx - \partial_x \otimes dy$$

sends $\partial_x \mapsto \partial_y$ and $\partial_y \mapsto -\partial_x$, a quarter turn of each tangent plane, and satisfies $J^2 = -\delta$. Both have smooth (constant) components, so both are smooth tensor fields. The rotation field is the tangential form of multiplication by i on $\mathbb{C} = \mathbb{R}^2$, the model for the almost-complex structures of Hodge theory.

Contraction is the one operation internal to a single tensor bundle. It trades a contravariant slot against a covariant one by evaluating the covector part on the vector part, lowering both r and s by one.

Definition 12.3.3: Contraction

Fix $1 \le a \le r$ and $1 \le b \le s$. The (a,b)-*contraction* is the bundle homomorphism $\mathcal{C}^a_b \colon T^r_s M \to T^{r-1}_{s-1} M$ acting on decomposable tensors by

$$\mathcal{C}^a_b\big(v_1 \otimes \cdots \otimes v_r \otimes \xi^1 \otimes \cdots \otimes \xi^s\big) = \xi^b(v_a)\; v_1 \otimes \cdots \widehat{v_a} \cdots \otimes v_r \otimes \xi^1 \otimes \cdots \widehat{\xi^b} \cdots \otimes \xi^s,$$

the hats marking the omitted factors, and extended by linearity. Applied to an (r,s)-tensor field A it gives the *contraction* $\mathcal{C}^a_b A \in \mathcal{T}^{r-1}_{s-1}(M)$, whose components are obtained from those of A by summing over one repeated index,

$$(\mathcal{C}^1_1 A)^{i_2 \cdots i_r}_{j_2 \cdots j_s} = \sum_k A^{k\, i_2 \cdots i_r}_{k\, j_2 \cdots j_s},$$

and similarly for other slots.

The evaluation pairing $V \otimes V^* \to \mathbb{R}$, $v \otimes \xi \mapsto \xi(v)$, is defined without reference to a basis, so $\mathcal{C}^a_b$ is independent of the chart used to write it; the summed index in coordinates is a computation of this intrinsic pairing ([2, Universal Property Of Tensor Product (Over Commutative Rings)]). Smoothness is immediate: the contracted components are sums of the smooth components of A, hence smooth. The simplest instance is the $(1,1) \to (0,0)$ contraction, which sends an endomorphism field $A = \sum A^i_j\, \partial_i \otimes dx^j$ to the function $\mathcal{C}^1_1 A = \sum_i A^i_i$, the trace of the endomorphism A_p at each point. For the identity field of example 12.3.2 this is the constant $n = 2$, and for the rotation field J it is 0, matching the trace of a quarter turn. That the answer does not depend on the coordinates is the coordinate-free content of the trace, verified directly in exercise 12.3.1 (2).

Covariant tensor fields, unlike vector fields, transfer along an arbitrary smooth map. A covector pulls back by precomposition with the differential (definition 12.1.6); feeding each argument of a covariant tensor through the differential does the same for tensors of every covariant order.

Definition 12.3.4: Pullback of a Covariant Tensor Field

Let $f\colon M \to N$ be smooth and $A \in \mathcal{T}^0_s(N)$ a covariant s-tensor field on N. Its *pullback* $f^*A \in \mathcal{T}^0_s(M)$ is the covariant s-tensor field on M defined pointwise by

$$(f^*A)_p(v_1,\dots,v_s) = A_{f(p)}\big(df_p(v_1),\dots,df_p(v_s)\big), \qquad v_1,\dots,v_s \in T_pM,$$

where $df_p\colon T_pM \to T_{f(p)}N$ is the differential of f at p.

For $s = 1$ this is the covector pullback $(f^*\omega)_p = \omega_{f(p)} \circ df_p$ of definition 12.1.6, and in particular $f^*(dg) = d(g \circ f)$ for $g \in C^\infty(N)$ (proposition 12.1.7). Contravariant tensors have no such pullback: the differential df_p pushes vectors forward, not back, so vector fields transfer only along diffeomorphisms. The asymmetry is exactly the contravariance of the dual construction (section 11.8), and it is why the tensors one integrates over an oriented manifold, the differential forms of the next section, are covariant. Smoothness of f^*A and its formal properties are the content of the next result.

Proposition 12.3.5: Functoriality of Pullback

Let $f\colon M \to N$ and $h\colon N \to P$ be smooth. Then f^*A is a smooth covariant tensor field for every $A \in \mathcal{T}^0_s(N)$, and pullback satisfies

$$\mathrm{id}_M^* = \mathrm{id}, \qquad (h \circ f)^* = f^* \circ h^*, \qquad f^*(A \otimes B) = f^*A \otimes f^*B,$$

the last for $A \in \mathcal{T}^0_s(N)$ and $B \in \mathcal{T}^0_t(N)$.

***Proof*:**
Smoothness. Fix charts (U, x) near $p \in M$ and (V, y) near $f(p)$ with $f(U) \subseteq V$, and write $A = \sum A_{j_1\cdots j_s}\, dy^{j_1} \otimes \cdots \otimes dy^{j_s}$ on V with smooth components $A_{j_1\cdots j_s}$. Since $f^*(dy^j) = d(y^j \circ f)$ (proposition 12.1.7) and $d(y^j \circ f) = \sum_i \frac{\partial(y^j\circ f)}{\partial x^i}\, dx^i$ has smooth coefficients, applying the definition to the coordinate fields gives

$$(f^*A)_{i_1\cdots i_s} = \sum_{j_1,\dots,j_s} (A_{j_1\cdots j_s} \circ f)\, \frac{\partial(y^{j_1} \circ f)}{\partial x^{i_1}} \cdots \frac{\partial(y^{j_s} \circ f)}{\partial x^{i_s}},$$

a sum of products of smooth functions. Hence f^*A has smooth components in every chart and is a smooth tensor field.

The three identities. That $\mathrm{id}_M^* = \mathrm{id}$ is immediate, since $d(\mathrm{id})_p = \mathrm{id}$. For composition, the chain rule $d(h \circ f)_p = dh_{f(p)} \circ df_p$ gives, for $C \in \mathcal{T}^0_s(P)$ and $v_1,\dots,v_s \in T_pM$,

$$\big((h \circ f)^*C\big)_p(v_1,\dots,v_s) = C_{h(f(p))}\big(dh\, df(v_1),\dots,dh\, df(v_s)\big) = (h^*C)_{f(p)}\big(df(v_1),\dots,df(v_s)\big),$$

which is $\big(f^*(h^*C)\big)_p(v_1,\dots,v_s)$; thus $(h \circ f)^* = f^* \circ h^*$. Finally, on decomposable arguments the tensor product of covariant tensors evaluates factor by factor, so

$$\big(f^*(A\otimes B)\big)_p(v_1,\dots,v_{s+t}) = (A\otimes B)_{f(p)}(df\, v_1,\dots,df\, v_{s+t}) = A_{f(p)}(df\, v_1,\dots)\cdot B_{f(p)}(df\, v_{s+1},\dots),$$

which equals $(f^*A \otimes f^*B)_p(v_1, \ldots, v_{s+t})$. This proves all three identities, as we sought to show.

The identity $(h \circ f)^* = f^* \circ h^*$ reverses the order of composition, marking pullback as a contravariant operation on covariant tensor fields, and the compatibility with $\otimes$ shows it respects the whole tensor algebra. Both facts are used constantly in the next two sections, where forms are pulled back and their wedge products and exterior derivatives are checked to commute with pullback.

The definition of a tensor field asks for a smooth section of a bundle, an object living over M. In practice a tensor field is nearly always presented instead by how it acts on vector fields and 1-forms, and one wants to know when such an action comes from a section. The answer is clean and is the working test for tensoriality: the action must be linear not over $\mathbb{R}$ alone but over the ring $C^\infty(M)$ of smooth functions.

Proposition 12.3.6: The Tensor Characterization

Let M be a smooth manifold. A map

$$\mathcal{A}\colon \underbrace{\Omega^1(M) \times \cdots \times \Omega^1(M)}_{r} \times \underbrace{\mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M)}_{s} \longrightarrow C^\infty(M)$$

is induced by an (r, s)-tensor field A, in the sense that

$$\mathcal{A}(\omega^1, \ldots, \omega^r, X_1, \ldots, X_s)(p) = A_p\big(\omega^1_p, \ldots, \omega^r_p, (X_1)_p, \ldots, (X_s)_p\big)$$

for all p, if and only if $\mathcal{A}$ is $C^\infty(M)$*-multilinear*, that is, linear over $C^\infty(M)$ in each argument. The tensor field A is then unique.

***Proof*:**
Necessity. If A is a tensor field, define $\mathcal{A}$ by the displayed formula. Each A_p is $\mathbb{R}$-multilinear, and for $h \in C^\infty(M)$ one has $(hX_k)_p = h(p)(X_k)_p$, so

$$\mathcal{A}(\ldots, hX_k, \ldots)(p) = A_p(\ldots, h(p)(X_k)_p, \ldots) = h(p)\,\mathcal{A}(\ldots, X_k, \ldots)(p),$$

giving $\mathcal{A}(\ldots, hX_k, \ldots) = h\,\mathcal{A}(\ldots, X_k, \ldots)$; the covector slots are identical. Thus $\mathcal{A}$ is $C^\infty(M)$-multilinear. Its output is smooth: in a chart it is a sum of products of the smooth components of A with those of the arguments.

For the converse, the crux is that $C^\infty(M)$-linearity forces the value at a point to depend only on the arguments at that point. This is proved in two steps and then assembled.

Step 1: Locality. If some argument vanishes on an open set U, then $\mathcal{A}(\ldots)$ vanishes on U. Suppose the k-th argument is a vector field X_k with $X_k \equiv 0$ on U, and fix $p \in U$. Choose $\varphi \in C^\infty(M)$ with $\varphi(p) = 1$ and $\operatorname{supp}\varphi \subseteq U$. Then $\varphi X_k \equiv 0$ on all of M: it vanishes on U because X_k does, and off U because φ does. By $C^\infty(M)$-linearity in the k-th slot,

$$0 = \mathcal{A}(\ldots, \varphi X_k, \ldots) = \varphi\,\mathcal{A}(\ldots, X_k, \ldots),$$

and evaluating at p, where $\varphi(p) = 1$, gives $\mathcal{A}(\ldots, X_k, \ldots)(p) = 0$. As $p \in U$ was arbitrary the function vanishes on U; the argument for a covector slot is the same with X_k replaced by a

1-form. By multilinearity, if two argument tuples agree on U slot by slot then the outputs agree on U.

Step 2: Pointwise dependence. If some argument vanishes at the single point p, then $\mathcal{A}(\dots)(p) = 0$. Suppose $X_k(p) = 0$. Take a chart (V, x) about p and write $X_k = \sum_i a^i \partial_i$ on V, so $a^i(p) = 0$. Choose $\varphi \in C^\infty(M)$ with $\operatorname{supp}\varphi \subseteq V$ and $\varphi \equiv 1$ on a neighbourhood W of p. The globally defined fields $\widetilde{\partial}_i = \varphi\, \partial_i$ and functions $\widetilde{a}^i = \varphi a^i$ (each extended by zero outside V) satisfy

$$\varphi^2 X_k = \sum_i \widetilde{a}^i\, \widetilde{\partial}_i$$

everywhere: both sides vanish off V, and on V the right side is $\varphi^2 \sum_i a^i \partial_i$. Since $\varphi^2 \equiv 1$ on W, the fields $\varphi^2 X_k$ and X_k agree on W, so Step 1 gives $\mathcal{A}(\dots, X_k, \dots)(p) = \mathcal{A}(\dots, \varphi^2 X_k, \dots)(p)$. Expanding by $C^\infty(M)$-linearity,

$$\mathcal{A}(\dots, \varphi^2 X_k, \dots)(p) = \sum_i \widetilde{a}^i(p)\, \mathcal{A}(\dots, \widetilde{\partial}_i, \dots)(p) = 0,$$

because $\widetilde{a}^i(p) = \varphi(p) a^i(p) = a^i(p) = 0$. Hence the value at p is zero. Again the covector slots are handled identically, writing a 1-form as $\sum_i b_i\, dx^i$ near p.

Step 3: Assembling the tensor field. Fix p and let $\xi^1, \dots, \xi^r \in T_p^*M$ and $v_1, \dots, v_s \in T_pM$. Extend each to a global field: for $v_l = \sum_i c^i\, \partial_i|_p$ set $X_l = \sum_i (\varphi c^i)(\varphi \partial_i)$ with φ a bump equal to 1 near p, so $(X_l)_p = v_l$, and likewise produce $\omega^m \in \Omega^1(M)$ with $\omega^m_p = \xi^m$. Define

$$A_p(\xi^1, \dots, \xi^r, v_1, \dots, v_s) = \mathcal{A}(\omega^1, \dots, \omega^r, X_1, \dots, X_s)(p)$$

By Step 2 the right side is unchanged if any extension is replaced by another with the same value at p (their difference vanishes at p), so A_p is well defined, and it is $\mathbb{R}$-multilinear because $\mathcal{A}$ is, in particular, $\mathbb{R}$-multilinear. Thus A_p is an element of the fiber $(T^r_s M)_p$ under the identification of multilinear functionals with tensors ([2, Tensors as Multilinear Maps]). Smoothness: over a chart the components $A(dx^{i_1}, \dots, \partial_{j_s})$ equal, on the neighbourhood W where $\varphi \equiv 1$, the smooth functions $\mathcal{A}$ of the bump-extended coordinate frames, so every component is smooth and A is a smooth tensor field. It induces $\mathcal{A}$ by construction, and uniqueness holds because the displayed formula forces the value of A_p on every tuple of arguments. This completes the proof.

The characterization turns the question "is this expression a tensor field?" into the mechanical check of $C^\infty(M)$-linearity in each slot, performed without ever choosing coordinates. The contrast with the Lie bracket is instructive. The bracket $(X, Y) \mapsto [X, Y]$ is $\mathbb{R}$-bilinear but not $C^\infty(M)$-bilinear: by proposition 5.5.3, $[X, fY] = f[X, Y] + (Xf)\, Y$, and the extra term $(Xf)\, Y$ obstructs C^∞-linearity, so the bracket defines no tensor field. A Riemannian metric $(X, Y) \mapsto g(X, Y)$ and the curvature $(X, Y, Z) \mapsto R(X, Y)Z$ of the Riemannian geometry volume, by contrast, are C^∞-multilinear and hence are tensor fields; this is the sense in which curvature is a pointwise invariant while the bracket is not.

The same criterion works with the scalar target $C^\infty(M)$ replaced by the sections of a vector bundle and the arguments by sections of vector bundles. This is the form tensoriality takes for a bundle-valued expression, such as the second fundamental form of a submanifold or the difference of two connections, each a $C^\infty(M)$-multilinear map of vector fields with values in a bundle.

Proposition 12.3.7: The Bundle-Valued Tensor Characterization

Let $E_1, \dots, E_s$ and F be smooth vector bundles over M, with section spaces $\Gamma(E_1), \dots, \Gamma(E_s), \Gamma(F)$. A map

$$\mathcal{A} \colon \Gamma(E_1) \times \cdots \times \Gamma(E_s) \longrightarrow \Gamma(F)$$

is induced by a smooth bundle homomorphism, in the sense that there is a smooth section A of $\mathrm{Hom}(E_1 \otimes \cdots \otimes E_s,\ F)$ (proposition 11.8.2) with

$$\mathcal{A}(\sigma_1, \dots, \sigma_s)(p) = A_p\big((\sigma_1)_p, \dots, (\sigma_s)_p\big)$$

for all p, if and only if $\mathcal{A}$ is $C^\infty(M)$-multilinear. The bundle map A is then unique.

***Proof*:**
The argument is that of proposition 12.3.6, with vector fields and 1-forms replaced throughout by sections of the E_j and the scalar value replaced by a value in the fiber F_p.

Necessity. A section A of $\mathrm{Hom}(E_1 \otimes \cdots \otimes E_s, F)$ is a smoothly varying multilinear map A_p, so $\mathcal{A}(\sigma_1, \dots, \sigma_s)(p) = A_p((\sigma_1)_p, \dots, (\sigma_s)_p)$ is $C^\infty(M)$-multilinear, since $(h\sigma_j)_p = h(p)(\sigma_j)_p$ pulls $h(p)$ out of the j-th slot, and it is a smooth section of F, being in local frames a sum of products of the smooth components of A with those of the arguments.

Sufficiency. Suppose $\mathcal{A}$ is $C^\infty(M)$-multilinear. Locality and pointwise dependence hold verbatim as in proposition 12.3.6: multiplying an argument by a bump function φ with $\varphi(p) = 1$, supported where that argument vanishes, shows the value at p depends only on the arguments near p. Writing an argument in a smooth local frame of E_j (proposition 11.4.5) with coefficients vanishing at p shows it depends only on the values at p. Fixing p and $v_j \in (E_j)_p$, extend each v_j to a section σ_j with $(\sigma_j)_p = v_j$ (bump-multiply a frame element, proposition 11.4.5) and set

$$A_p(v_1, \dots, v_s) = \mathcal{A}(\sigma_1, \dots, \sigma_s)(p) \in F_p$$

Pointwise dependence makes this independent of the extensions, and multilinearity of $\mathcal{A}$ makes A_p multilinear, hence an element of the fiber $\mathrm{Hom}(E_1 \otimes \cdots \otimes E_s, F)_p$. Reading arguments and value in local frames, the components of A are the smooth functions $\mathcal{A}$ of the frame sections, so A is a smooth section. It induces $\mathcal{A}$, and the displayed formula forces it, giving uniqueness. This completes the proof.

The metric named in that contrast is the first tensor field a manifold needs beyond its vector fields and forms, and it is where the covariant, symmetric, positive-definite case earns a name.

Definition 12.3.8: Riemannian Metric

A *Riemannian metric* on a smooth manifold M is a covariant 2-tensor field $g \in \mathcal{T}^0_2(M)$ that is *symmetric*, $g_p(u, v) = g_p(v, u)$, and *positive definite*, $g_p(v, v) > 0$ for every $v \neq 0$, at each point p. Equivalently, g_p is an inner product on T_pM varying smoothly in p. In a chart, $g = \sum_{i,j} g_{ij}\, dx^i \otimes dx^j$ with $g_{ij} = g_{ji}$ and $(g_{ij}(p))$ a positive-definite symmetric matrix.

A Riemannian metric is precisely a fiber metric on the tangent bundle TM in the sense of lemma 11.6.5. That lemma, applied to the bundle $E = TM$, therefore settles existence at once.

Corollary 12.3.9: Existence of Riemannian Metrics

Every smooth manifold admits a Riemannian metric.

***Proof*:**
Apply lemma 11.6.5 to $E = TM$. Concretely, take a cover of M by charts, pull back the Euclidean inner product through each chart to get a metric g^α on the chart domain, and patch by a subordinate smooth partition of unity $\{\rho_\alpha\}$ (theorem 1.5.5), setting $g = \sum_\alpha \rho_\alpha g^\alpha$. Each summand is symmetric and positive semidefinite, and at every p some $\rho_\alpha(p) > 0$ with g^α_p positive definite, so g_p is positive definite, completing the proof.

Positive-definiteness is preserved by the sum because a convex combination of inner products is again an inner product (exercise 12.3.1 (5)), the pointwise fact behind the patching. Existence is a purely differential-topological statement; the geometry a metric carries, lengths of curves, geodesics, and curvature, is developed in the Riemannian geometry volume and is not needed here. The metric gives this book examples of tensor fields and, through lemma 11.6.5, the orthogonal complements used to split bundles.

Example 12.3.10: The Round Metric on the Sphere

On $\mathbb{R}^n$ the *Euclidean metric* is $g_{\text{eucl}} = \sum_i dx^i \otimes dx^i$, with constant components $g_{ij} = \delta_{ij}$; it is symmetric and positive definite, so it is a Riemannian metric. Its restriction to a submanifold pulls back through the inclusion, giving the *round metric* on the sphere.

Compute it on $S^2 \subseteq \mathbb{R}^3$ using the parametrization $\sigma(\theta, \varphi) = (\sin\theta\cos\varphi,\ \sin\theta\sin\varphi,\ \cos\theta)$ of the complement of the poles. Pulling the coordinate differentials back through σ (proposition 12.1.7),

$$\begin{aligned}
\sigma^* dx &= \cos\theta\cos\varphi\, d\theta - \sin\theta\sin\varphi\, d\varphi,\\
\sigma^* dy &= \cos\theta\sin\varphi\, d\theta + \sin\theta\cos\varphi\, d\varphi,\\
\sigma^* dz &= -\sin\theta\, d\theta
\end{aligned}$$

and therefore, by proposition 12.3.5,

$$\sigma^* g_{\text{eucl}} = (\sigma^* dx)^{\otimes 2} + (\sigma^* dy)^{\otimes 2} + (\sigma^* dz)^{\otimes 2}$$

The coefficient of $d\theta \otimes d\theta$ is $\cos^2\theta(\cos^2\varphi + \sin^2\varphi) + \sin^2\theta = 1$; the coefficient of $d\varphi \otimes d\varphi$ is $\sin^2\theta(\sin^2\varphi + \cos^2\varphi) = \sin^2\theta$; and the two cross terms cancel. Hence

$$\sigma^* g_{\text{eucl}} = d\theta \otimes d\theta + \sin^2\theta\, d\varphi \otimes d\varphi,$$

a symmetric 2-tensor field, positive definite wherever $\sin\theta \neq 0$, that is, away from the poles, where the parametrization is a diffeomorphism onto its image. This is the round metric of S^2, the running example's first metric.

Exercise 12.3.1

1. Let $A = \sum A^i_j\, \partial_i \otimes dx^j$ be a $(1,1)$-tensor field written in a chart (U, x), and let $(U, \widetilde{x})$ be a second chart on the same domain. Using $\partial_i = \sum_a \frac{\partial \widetilde{x}^a}{\partial x^i}\, \widetilde{\partial}_a$ and $dx^j = \sum_b \frac{\partial x^j}{\partial \widetilde{x}^b}\, d\widetilde{x}^b$, derive the transformation law for the components $\widetilde{A}^a_b$ of A in the second chart.

2. Show that the trace $\sum_i A^i_i$ of a $(1,1)$-tensor field is independent of the chart, so that the contraction $\mathcal{C}^1_1 A$ of definition 12.3.3 is a well-defined function on M. Do this in two ways: directly from the transformation law of exercise 12.3.1 (1), and by identifying the answer with the value of the tensor characterization proposition 12.3.6.

3. The Lie bracket defines a map $B\colon \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$, $B(X, Y) = [X, Y]$. Decide whether B is $C^\infty(M)$-bilinear, hence whether it is a $(1,2)$-tensor field. On $M = \mathbb{R}$ with $X = Y = \partial_x$ and $f = x$, compute $[X, fY]$ and $f[X, Y]$ explicitly and use the result of proposition 12.3.6 to justify your conclusion.

4. Let $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ be the polar map $f(r, \theta) = (r\cos\theta,\ r\sin\theta)$. Compute the pullback $f^* g_{\mathrm{eucl}}$ of the Euclidean metric $g_{\mathrm{eucl}} = dx \otimes dx + dy \otimes dy$, and identify the region where it is a Riemannian metric.

5. Let g and h be Riemannian metrics on M and let $t \in [0,1]$. Show that $tg + (1-t)h$ is a Riemannian metric. Explain how this fact underlies the partition-of- unity construction of corollary 12.3.9.

12.4 Differential Forms and the Wedge Product

The cotangent bundle of section 12.1 carries the 1-forms; its exterior powers carry forms of every degree. This section assembles the differential k-forms as sections of $\Lambda^k T^*M$, imports the pointwise exterior algebra of theorem 0.5.9 fiber by fiber, and equips the resulting spaces with the two algebraic operations the exterior calculus runs on: the wedge product, which multiplies forms and raises degree, and the interior product, which contracts a form against a vector field and lowers it. Pullback along a smooth map completes the picture, making $\Omega^\bullet$ contravariant and giving the change-of-variables rule that integration will use.

Definition 12.4.1: Differential k-Form

Let M be a smooth n-manifold and $0 \le k \le n$. A *differential k-form* on M is a smooth section of the exterior bundle $\Lambda^k T^*M$, the exterior-power construction of section 11.8 applied to the cotangent bundle $T^*M = (TM)^*$ of definition 2.4.1. The space of k-forms is

$$\Omega^k(M) = \Gamma\big(\Lambda^k T^*M\big)$$

a module over $C^\infty(M)$. By convention $\Omega^0(M) = C^\infty(M)$, and $\Omega^k(M) = 0$ for $k > n$; the full space of forms is $\Omega^\bullet(M) = \bigoplus_{k=0}^n \Omega^k(M)$.

A k-form assigns to each point p an alternating k-linear form ω_p on the tangent space T_pM: it eats k tangent vectors and returns a number, changes sign under a transposition of its arguments, and vanishes whenever two arguments coincide. For $k = 1$ this recovers the covector fields of definition 12.1.2, and for $k = 0$ the smooth functions. The alternating condition forces $\Omega^k(M) = 0$ once k exceeds n, since an alternating form in more arguments than the dimension of the space must vanish; the top degree $k = n$ is one-dimensional fiberwise, and is the degree that orientation and integration (chapter 13) use.

A form of degree ≥ 2 is built from covectors by a product. The exterior algebra of a single vector space gives that product, and applying it fiber by fiber multiplies forms.

Definition 12.4.2: Wedge Product of Forms

For $\omega \in \Omega^k(M)$ and $\eta \in \Omega^l(M)$ the *wedge product* $\omega \wedge \eta \in \Omega^{k+l}(M)$ is the form

$$(\omega \wedge \eta)_p = \omega_p \wedge \eta_p$$

the right-hand product being the exterior multiplication of the algebra $\Lambda^\bullet T_p^*M$ (theorem 0.5.9), taken in the *determinant normalization*: a product of 1-forms evaluates on tangent vectors by

$$(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \ldots, v_k) = \det\big[\alpha^i(v_j)\big]_{i,j=1}^k \tag{12.4}$$

which fixes the normalization of the product in every degree.

For a multi-index $I = (i_1, \ldots, i_k)$ with $1 \leq i_1 < \cdots < i_k \leq n$, write $dx^I = dx^{i_1} \wedge \cdots \wedge dx^{i_k}$, and let $\sum_I$ denote a sum over strictly increasing multi-indices. In a chart the wedge is then computed by the coordinate rule

$$\Big(\sum_I a_I\, dx^I\Big) \wedge \Big(\sum_J b_J\, dx^J\Big) = \sum_{I,J} a_I b_J\, dx^I \wedge dx^J$$

where $dx^I \wedge dx^J$ is 0 when I and J share an index and otherwise $\pm dx^K$ for the increasing rearrangement K of their union, the sign being that of the sorting permutation. These products of coordinate covector fields are the frame in which every form is written.

Proposition 12.4.3: Coordinate Frame for k-Forms

Let (U, x) be a chart on M, with coordinate vector fields $\partial_i = \partial/\partial x^i$ and coordinate covector fields dx^i. The forms $\{dx^I \,|\, I \text{ increasing}\}$ constitute a smooth local frame for $\Lambda^k T^*M$ over U, so $\Lambda^k T^*M$ has rank $\binom{n}{k}$, and every $\omega \in \Omega^k(M)$ is written over U as

$$\omega = \sum_I a_I\, dx^I, \qquad a_I = \omega(\partial_{i_1}, \ldots, \partial_{i_k})$$

with each $a_I \in C^\infty(U)$. A rough k-form is smooth if and only if its components a_I are smooth in every chart.

***Proof*:**
At each $p \in U$ the covectors $dx_p^1, \ldots, dx_p^n$ are the basis of T_p^*M dual to $\partial_1|_p, \ldots, \partial_n|_p$ (proposition 12.1.4). By the basis theorem for the exterior algebra of a vector space (theorem 0.5.9), the exterior products dx_p^I over increasing I form a basis of $\Lambda^k T_p^*M$, whose dimension is therefore $\binom{n}{k}$; hence $\omega_p = \sum_I a_I(p)\, dx_p^I$ for unique scalars $a_I(p)$. Evaluating dx^J on $\partial_{i_1}, \ldots, \partial_{i_k}$ by (12.4) gives $\det[dx^{j_a}(\partial_{i_b})] = \det[\delta_{i_b}^{j_a}] = \delta_I^J$, so pairing ω_p against $\partial_{i_1}|_p, \ldots, \partial_{i_k}|_p$ isolates the coefficient, $a_I(p) = \omega_p(\partial_{i_1}|_p, \ldots, \partial_{i_k}|_p)$.
The frame $\{dx^I\}$ is smooth: it is the image of the smooth local frame $\{dx^i\}$ for T^*M under the

fiberwise exterior-power construction (theorem 11.8.1), which carries a smooth frame to a smooth frame. Smoothness of ω as a section is therefore equivalent to smoothness of its components in this frame, by the frame criterion for sections (section 11.4), completing the proof.

Example 12.4.4: Forms on $\mathbb{R}^3$

On $\mathbb{R}^3$ with coordinates (x, y, z) the bundles $\Lambda^k T^*\mathbb{R}^3$ have ranks $\binom{3}{k} = 1, 3, 3, 1$ for $k = 0, 1, 2, 3$:

- $\Omega^0 = C^\infty(\mathbb{R}^3)$, the smooth functions;
- Ω^1 has frame dx, dy, dz; a 1-form is $P\,dx + Q\,dy + R\,dz$;
- Ω^2 has frame $dy \wedge dz,\ dz \wedge dx,\ dx \wedge dy$; a 2-form is $A\,dy \wedge dz + B\,dz \wedge dx + C\,dx \wedge dy$;
- Ω^3 has frame $dx \wedge dy \wedge dz$; a 3-form is $h\,dx \wedge dy \wedge dz$.

The three coefficients (A, B, C) of a 2-form and the single coefficient h of a 3-form are the device by which vector calculus encodes a vector field as a 2-form and a scalar as a 3-form; the wedge and the exterior derivative of section 12.5 make this correspondence precise. Evaluating $dx \wedge dy$ on $u = \partial_x + \partial_z$ and $v = \partial_y$ gives, by (12.4), $dx(u)\,dy(v) - dx(v)\,dy(u) = (1)(1) - (0)(0) = 1$.

The structural identities of the product are inherited from the pointwise algebra rather than proved afresh.

Proposition 12.4.5: The Algebra of Forms

For all $\omega \in \Omega^k(M)$, $\eta \in \Omega^l(M)$, $\zeta \in \Omega^m(M)$, and $f \in C^\infty(M)$:

1. the wedge is $\mathbb{R}$-bilinear and $C^\infty(M)$-bilinear: $(f\omega) \wedge \eta = f(\omega \wedge \eta) = \omega \wedge (f\eta)$;
2. it is associative: $(\omega \wedge \eta) \wedge \zeta = \omega \wedge (\eta \wedge \zeta)$;
3. it is graded-anticommutative: $\omega \wedge \eta = (-1)^{kl}\,\eta \wedge \omega$.

Thus $\Omega^\bullet(M)$ is an associative, graded-anticommutative $C^\infty(M)$-algebra, and a 1-form satisfies $\alpha \wedge \alpha = 0$.

***Proof*:**

The wedge is defined fiberwise, so every identity that holds in the exterior algebra of the single space T_p^*M holds for forms once it holds at each p; theorem 0.5.9 gives these identities pointwise.

1. $\mathbb{R}$-bilinearity is pointwise linearity of the product in each argument, and $((f\omega) \wedge \eta)_p = f(p)\,\omega_p \wedge \eta_p = f(p)(\omega \wedge \eta)_p$, with the same computation on the other side, gives $C^\infty(M)$-bilinearity.
2. Associativity holds in $\Lambda^\bullet T_p^*M$ at every p, hence for the sections.
3. Graded-anticommutativity $\omega_p \wedge \eta_p = (-1)^{kl}\,\eta_p \wedge \omega_p$ holds at every p; taking $\eta = \omega$ of odd degree gives $\omega_p \wedge \omega_p = -\omega_p \wedge \omega_p$, hence 0, and in particular $\alpha \wedge \alpha = 0$ for a 1-form.

Smoothness is the only bundle-level point. In a chart the coordinate rule expresses the components of $\omega \wedge \eta$ as sums of products $a_I b_J$ of the smooth components of ω and η, hence $\omega \wedge \eta \in \Omega^{k+l}(M)$, completing the proof.

Associativity lets a wedge of several factors be written without parentheses, and graded-anticommutativity is the source of every sign in the calculus: swapping two 1-forms introduces a minus sign, while a 2-form commutes past everything.

Example 12.4.6: A Wedge of 1-Forms

For $\alpha = a_1\,dx + a_2\,dy + a_3\,dz$ and $\beta = b_1\,dx + b_2\,dy + b_3\,dz$ on $\mathbb{R}^3$, the coordinate rule with $dx \wedge dx = 0$ and $dy \wedge dx = -dx \wedge dy$ gives

$$\alpha \wedge \beta = (a_2b_3 - a_3b_2)\,dy \wedge dz + (a_3b_1 - a_1b_3)\,dz \wedge dx + (a_1b_2 - a_2b_1)\,dx \wedge dy$$

so the three coefficients of $\alpha\wedge\beta$ are the 2×2 minors of the array with rows (a_1, a_2, a_3) and (b_1, b_2, b_3), which are the components of the cross product $a \times b$ under the identification of example 12.4.4. In particular $\alpha \wedge \alpha = 0$, the vanishing of a determinant with a repeated row.

Wedging raises degree by inserting a covector; the adjoint operation removes one, by feeding a tangent vector into a form.

Definition 12.4.7: Interior Product

Let $X \in \mathfrak{X}(M)$ be a vector field. The *interior product* (or *contraction*) with X is the map $\iota_X : \Omega^k(M) \to \Omega^{k-1}(M)$ given for $k \geq 1$ by

$$(\iota_X\omega)(v_1, \ldots, v_{k-1}) = \omega(X, v_1, \ldots, v_{k-1})$$

and by $\iota_X\omega = 0$ for $\omega \in \Omega^0(M)$. Thus $\iota_X\omega$ is ω with its first argument filled by X.

On a product of 1-forms the contraction is computed by expanding (12.4) along the column indexed by X: for $\alpha^1, \ldots, \alpha^k \in \Omega^1(M)$,

$$\iota_X(\alpha^1 \wedge \cdots \wedge \alpha^k) = \sum_{i=1}^{k} (-1)^{i-1}\,\alpha^i(X)\,\alpha^1 \wedge \cdots \wedge \widehat{\alpha^i} \wedge \cdots \wedge \alpha^k \tag{12.5}$$

where the hat marks an omitted factor. Evaluating the left side on $v_1, \ldots, v_{k-1}$ produces the determinant of the matrix whose first column is $(\alpha^i(X))_i$ and whose remaining columns are $(\alpha^i(v_j))$; cofactor expansion along that first column is exactly the right side.

Proposition 12.4.8: The Interior Product is an Antiderivation

Let $X, Y \in \mathfrak{X}(M)$, $f \in C^\infty(M)$, $\omega \in \Omega^k(M)$, and $\eta \in \Omega^l(M)$.

1. ι_X is $\mathbb{R}$-linear in the form and $C^\infty(M)$-linear in the field: $\iota_{fX+Y} = f\,\iota_X + \iota_Y$.
2. $\iota_X \circ \iota_X = 0$; consequently $\iota_X\iota_Y = -\iota_Y\iota_X$.
3. ι_X is an *antiderivation* of degree -1:

$$\iota_X(\omega \wedge \eta) = (\iota_X\omega) \wedge \eta + (-1)^k\,\omega \wedge (\iota_X\eta)$$

***Proof*:**

1. The defining formula is linear in ω, since ω_p enters linearly, and $\mathbb{R}$-linear in the vector slot; as $(fX+Y)_p = f(p)X_p + Y_p$ and ω_p is multilinear, $(\iota_{fX+Y}\omega)(v_1,\dots) = \omega(fX+Y, v_1,\dots) = f\,\omega(X,v_1,\dots) + \omega(Y,v_1,\dots)$, which is $(f\iota_X + \iota_Y)\omega$.

2. For $\omega \in \Omega^k(M)$,
$$(\iota_X\iota_X\omega)(v_1,\dots,v_{k-2}) = (\iota_X\omega)(X,v_1,\dots,v_{k-2}) = \omega(X,X,v_1,\dots,v_{k-2}) = 0$$
since ω is alternating and its first two arguments agree. Applying $\iota_Z^2 = 0$ to $Z = X+Y$ and expanding by (1) gives $0 = \iota_X^2 + \iota_X\iota_Y + \iota_Y\iota_X + \iota_Y^2 = \iota_X\iota_Y + \iota_Y\iota_X$.

3. Both sides are $\mathbb{R}$-bilinear in (ω,η) and computed pointwise, so it suffices to verify the identity on forms that span Ω^k and Ω^l in a chart: the decomposables $\omega = \alpha^1\wedge\cdots\wedge\alpha^k$ and $\eta = \beta^1\wedge\cdots\wedge\beta^l$ with the α^i,β^j among the dx^i (proposition 12.4.3). For these, $\omega\wedge\eta = \alpha^1\wedge\cdots\wedge\alpha^k\wedge\beta^1\wedge\cdots\wedge\beta^l$, and (12.5) over all $k+l$ factors splits into two sums. The first k terms sum to $(\iota_X\omega)\wedge\eta$. In the remaining l terms the sign attached to the j-th β is $(-1)^{k+j-1} = (-1)^k(-1)^{j-1}$, so those terms sum to $(-1)^k\,\omega\wedge(\iota_X\eta)$. Adding the two recovers the antiderivation identity. Since both sides are additive in ω and η and every form is locally a sum of such decomposables, the identity holds for all forms, as we sought to show.

The interior product is the degree-lowering counterpart of the wedge, and it pairs with the exterior derivative of section 12.5 in Cartan's formula. Its antiderivation rule is the graded Leibniz rule the wedge already carries, with the sign now tracking the degree of the factor the vector field passes.

Example 12.4.9: Contraction on $\mathbb{R}^3$
Let $X = f\,\partial_x + g\,\partial_y + h\,\partial_z$ on $\mathbb{R}^3$ and let $\mu = dx\wedge dy\wedge dz$ be the standard 3-form. By (12.5),
$$\iota_X\mu = f\,dy\wedge dz - g\,dx\wedge dz + h\,dx\wedge dy = f\,dy\wedge dz + g\,dz\wedge dx + h\,dx\wedge dy$$
the 2-form whose coefficients are the components of X under the identification of example 12.4.4. This is the flux form of X: contracting the volume form with a field turns the field into the 2-form later integrated over surfaces. Contracting once more gives $\iota_X\iota_X\mu = 0$, as proposition 12.4.8 requires.

A smooth map has no well-defined pushforward of forms, but it has a pullback: covectors compose with the differential (definition 12.1.6), and the exterior power carries this to every degree.

Definition 12.4.10: Pullback of a Differential Form

Let $f\colon N\to M$ be smooth and $\omega\in\Omega^k(M)$. The *pullback* $f^*\omega\in\Omega^k(N)$ is defined by
$$(f^*\omega)_p(v_1,\dots,v_k) = \omega_{f(p)}\big(df_p(v_1),\dots,df_p(v_k)\big)$$
for $v_1,\dots,v_k\in T_pN$, with $f^*g = g\circ f$ for $g\in\Omega^0(M)$. For $k=1$ this is the covector pullback of definition 12.1.6.

Proposition 12.4.11: Naturality of the Pullback

Let $f\colon N \to M$ and $h\colon P \to N$ be smooth. The pullback $f^*\colon \Omega^\bullet(M) \to \Omega^\bullet(N)$ is a degree-preserving $\mathbb{R}$-algebra homomorphism:

1. f^* is $\mathbb{R}$-linear and $f^*(\omega \wedge \eta) = f^*\omega \wedge f^*\eta$;
2. $f^*(g\,\omega) = (g \circ f)\, f^*\omega$ for $g \in C^\infty(M)$;
3. in coordinates $f^*\big(\sum_I a_I\, dx^I\big) = \sum_I (a_I \circ f)\, d(x^{i_1} \circ f) \wedge \cdots \wedge d(x^{i_k} \circ f)$, and in particular $f^*\omega$ is smooth;
4. pullback is functorial: $(f \circ h)^* = h^* \circ f^*$ and $\mathrm{id}_M^* = \mathrm{id}$.

***Proof*:**
Pointwise, $(f^*\omega)_p = \big(\Lambda^k(df_p)^*\big)(\omega_{f(p)})$, where $(df_p)^*\colon T^*_{f(p)}M \to T^*_pN$ is the transpose of df_p and $\Lambda^k(df_p)^*$ its k-th exterior power. The assignment $A \mapsto \Lambda^\bullet A^*$ is a homomorphism of exterior algebras (theorem 0.5.9): $\mathbb{R}$-linear in each degree and sending a wedge to the wedge of the images. Applying it at each point gives (1).
For (2), $g\omega$ scales $\omega_{f(p)}$ by $g(f(p))$ and $\Lambda^k(df_p)^*$ is linear, so $(f^*(g\omega))_p = g(f(p))(f^*\omega)_p$. For (3), on a coordinate 1-form the covector pullback gives $f^*(dx^i) = d(x^i \circ f)$ (proposition 12.1.7); the multiplicativity (1) and the C^∞-rule (2) extend this to $f^*(a_I\, dx^I) = (a_I \circ f)\, d(x^{i_1} \circ f) \wedge \cdots \wedge d(x^{i_k} \circ f)$, and summing over I yields the formula. Each $d(x^i \circ f)$ is a smooth 1-form and each $a_I \circ f$ is smooth, so $f^*\omega$ is a smooth k-form by proposition 12.4.5.
For (4), the chain rule $d(f \circ h)_p = df_{h(p)} \circ dh_p$ transposes to $(d(f \circ h)_p)^* = (dh_p)^* \circ (df_{h(p)})^*$, and the exterior power of a composite is the composite of the exterior powers (theorem 0.5.9); hence $(f \circ h)^*\omega = h^*(f^*\omega)$ at every point, and $\mathrm{id}^* = \mathrm{id}$ is immediate. This establishes the proposition.

Pullback transports forms against the direction of a map, making $\Omega^\bullet$ a contravariant functor from manifolds to graded algebras. Its multiplicativity lets a computation on M be transferred verbatim to N: a form written in coordinates on M is pulled back by substituting the component functions of f and differentiating. The factor produced when a top form is pulled back is the Jacobian determinant, which is why the pullback of an n-form is what integration (chapter 13) evaluates.

Example 12.4.12: Pullback in Polar Coordinates
Let $f\colon (0,\infty) \times \mathbb{R} \to \mathbb{R}^2$, $f(r,\theta) = (r\cos\theta,\ r\sin\theta)$, and write (x,y) for the target coordinates. By proposition 12.1.7,

$$f^*(dx) = d(r\cos\theta) = \cos\theta\, dr - r\sin\theta\, d\theta, \qquad f^*(dy) = d(r\sin\theta) = \sin\theta\, dr + r\cos\theta\, d\theta$$

so by multiplicativity

$$f^*(dx \wedge dy) = f^*(dx) \wedge f^*(dy) = \big(\cos\theta\, dr - r\sin\theta\, d\theta\big) \wedge \big(\sin\theta\, dr + r\cos\theta\, d\theta\big) = r\, dr \wedge d\theta$$

the surviving terms being $r\cos^2\theta\, dr \wedge d\theta - r\sin^2\theta\, d\theta \wedge dr = r\, dr \wedge d\theta$. The coefficient r is the Jacobian determinant of f: the pullback of the area form is the area form scaled by the change-of-variables factor, the mechanism by which $\int f^*\omega$ will compute $\int \omega$ in new coordinates (chapter 13).

Exercise 12.4.1

1. On $\mathbb{R}^3$ let $\alpha = dx - 2\,dz$ and $\beta = dx \wedge dy + dy \wedge dz$. Compute $\alpha \wedge \beta$ and express it as a multiple of $dx \wedge dy \wedge dz$.
2. For the vector field $X = x\,\partial_y - y\,\partial_x$ on $\mathbb{R}^3$ and $\mu = dx \wedge dy \wedge dz$, compute $\iota_X \mu$ and $\iota_X(dx \wedge dy)$.
3. Using that $\iota_Z^2 = 0$ for every vector field Z (proposition 12.4.8), prove that $\iota_X \iota_Y = -\iota_Y \iota_X$ for all $X, Y \in \mathfrak{X}(M)$, and explain how the case $Y = X$ recovers $\iota_X^2 = 0$.
4. Let $f\colon \mathbb{R}^2 \to \mathbb{R}^2$, $f(u, v) = (u^2 - v^2,\ 2uv)$. Compute $f^*(dx \wedge dy)$ and verify that its coefficient equals the Jacobian determinant of f.
5. Let $\omega \in \Omega^k(M)$. (a) Show that $\omega \wedge \omega = 0$ whenever k is odd. (b) Exhibit a 2-form ω on $\mathbb{R}^4$ with $\omega \wedge \omega \neq 0$.

12.5 The Exterior Derivative

The graded algebra $\Omega^\bullet(M)$ of section 12.4 carries the wedge product and the interior product, but not yet an operator that differentiates a form. In degree zero the differential of section 12.1 already does this, sending a function to the 1-form df. This section extends it to a single operator d acting on forms of every degree. Three properties pin d down: it raises degree by one, it is an antiderivation for the wedge, and it squares to zero. From these follow a coordinate-free description of d, its compatibility with pullback, and Cartan's formula tying it to the Lie derivative.

The differential $df = \sum_i (\partial f/\partial x^i)\,dx^i$ takes a 0-form to a 1-form and, as section 12.1 recorded, satisfies $df = 0$ exactly when f is locally constant. The task is to continue this to a map $\Omega^k(M) \to \Omega^{k+1}(M)$ for every k. The requirements that force the answer are algebraic: d should be $\mathbb{R}$-linear, should reduce to the differential on functions, should obey a Leibniz rule for the wedge product, and should compose to zero (a requirement made plausible by the equality of mixed partial derivatives, which already gives $d(df) = 0$). These conditions leave no freedom.

Theorem 12.5.1: The Exterior Derivative

On a smooth n-manifold M there is a unique family of $\mathbb{R}$-linear maps

$$d\colon \Omega^k(M) \to \Omega^{k+1}(M), \qquad k = 0, 1, \ldots, n$$

satisfying:

1. on $\Omega^0(M) = C^\infty(M)$, d is the differential $f \mapsto df$ of definition 12.1.3;
2. (antiderivation) $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$ for $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^\ell(M)$;
3. $d \circ d = 0$.

In any chart (U, x), writing a k-form as $\omega = \sum_I a_I \, dx^I$ over increasing multi-indices $I = (i_1 < \cdots < i_k)$ with $dx^I = dx^{i_1} \wedge \cdots \wedge dx^{i_k}$, the operator is given by

$$d\omega = \sum_I da_I \wedge dx^I = \sum_I \sum_j \frac{\partial a_I}{\partial x^j} \, dx^j \wedge dx^I \tag{12.6}$$

***Proof*:**
The argument has three parts: any operator with properties (1)-(3) is local; locality forces the coordinate formula (12.6), giving uniqueness; and the formula, verified to satisfy (1)-(3), glues across charts to give existence.

Step 1: Any such d is local. The claim is that if $\omega, \omega' \in \Omega^k(M)$ agree on an open set V, then $d\omega = d\omega'$ on V. By $\mathbb{R}$-linearity it suffices to show $\omega|_V = 0$ implies $d\omega|_V = 0$. Fix $p \in V$ and choose a bump function $\varphi \in C^\infty(M)$ with $\varphi \equiv 1$ on a neighborhood of p and $\operatorname{supp}\varphi \subseteq V$ (theorem 1.5.4). Then $\varphi\omega = 0$ on all of M: where $\varphi \neq 0$ the point lies in V, where ω vanishes. Applying (2) with the 0-form φ (so $(-1)^0 = 1$),

$$0 = d(\varphi\omega) = d\varphi \wedge \omega + \varphi \, d\omega$$

At p, the function φ is constant near p, so $d\varphi_p = 0$, while $\varphi(p) = 1$; evaluating the identity at p gives $(d\omega)_p = 0$. Since $p \in V$ was arbitrary, $d\omega|_V = 0$.

Step 2: Uniqueness. Work in a chart (U, x). By locality it suffices to compute $d\omega$ on U from $\omega|_U = \sum_I a_I \, dx^I$. First, $d(dx^i) = d(dx^i) = 0$ by (3), since $dx^i = d(x^i)$ is itself the differential of the coordinate function. Iterating the antiderivation rule (2) over the factors of dx^I and using $d(dx^{i_r}) = 0$ at each step gives $d(dx^I) = 0$. Then, again by (2) applied to a_I (a 0-form) and dx^I,

$$d(a_I \, dx^I) = da_I \wedge dx^I + a_I \, d(dx^I) = da_I \wedge dx^I$$

Summing over I and using $\mathbb{R}$-linearity yields (12.6). Thus any operator satisfying (1)-(3) agrees with the coordinate formula on every chart, hence is unique.

Step 3: Existence. Define d_U on each chart (U, x) by (12.6), and check (1)-(3). Property (1) is immediate, since for a 0-form f the formula reads $d_U f = \sum_j (\partial f / \partial x^j) \, dx^j = df$.

For (2), by bilinearity it suffices to treat $\alpha = a_I\, dx^I$ (degree k) and $\beta = b_J\, dx^J$ (degree ℓ). Then $\alpha \wedge \beta = a_I b_J\, dx^I \wedge dx^J$ and

$$d_U(\alpha \wedge \beta) = d(a_I b_J) \wedge dx^I \wedge dx^J = (b_J\, da_I + a_I\, db_J) \wedge dx^I \wedge dx^J$$

Here $d_U(\alpha) \wedge \beta = (da_I \wedge dx^I) \wedge (b_J\, dx^J) = b_J\, da_I \wedge dx^I \wedge dx^J$, moving the 0-form b_J freely. For the other piece, the 1-form db_J passes the degree-k form dx^I with a factor $(-1)^k$, so

$$(-1)^k \alpha \wedge d_U(\beta) = (-1)^k a_I\, dx^I \wedge db_J \wedge dx^J = a_I\, db_J \wedge dx^I \wedge dx^J$$

Adding the two matches $d_U(\alpha \wedge \beta)$, establishing (2).

For (3), first take a 0-form: $d_U(df) = \sum_j d(\partial_j f) \wedge dx^j = \sum_{i,j} \frac{\partial^2 f}{\partial x^i \partial x^j}\, dx^i \wedge dx^j$. The coefficient $\partial^2 f/\partial x^i \partial x^j$ is symmetric in i, j while $dx^i \wedge dx^j$ is antisymmetric, so the double sum vanishes and $d_U(df) = 0$. Consequently $d_U(dx^I) = 0$ as in Step 2. For a general term $a_I\, dx^I$, using the antiderivation rule (2) just proved for the 1-form da_I,

$$d_U\big(d_U(a_I\, dx^I)\big) = d_U(da_I \wedge dx^I) = d_U(da_I) \wedge dx^I - da_I \wedge d_U(dx^I) = 0 - 0 = 0$$

so $d_U \circ d_U = 0$.

Finally, the local operators glue. On an overlap $U \cap V$ of two charts, d_V is an operator on $\Omega^\bullet(U \cap V)$ satisfying (1)-(3), so by the uniqueness of Step 2 it equals the coordinate formula in the x-chart, namely d_U. Thus $d_U = d_V$ on $U \cap V$, and the d_U assemble into a single global operator d satisfying (1)-(3). This is the exterior derivative, completing the proof.

The three defining properties are exactly what makes d useful. Linearity and the coordinate formula make it computable; the antiderivation rule lets it interact with the wedge; and $d^2 = 0$ is the identity that turns $\Omega^\bullet(M)$ into a cochain complex, the object whose cohomology chapter 14 will study. In low dimensions d collects the differential operators of vector calculus into one operator.

Example 12.5.2: The Exterior Derivative on $\mathbb{R}^3$

On $\mathbb{R}^3$ with coordinates (x, y, z), identify vector fields and 1-forms and 2-forms with triples of functions in the usual way, a 1-form as $P\, dx + Q\, dy + R\, dz$ and a 2-form as $P\, dy \wedge dz + Q\, dz \wedge dx + R\, dx \wedge dy$. The exterior derivative in each degree is a classical operator.

On a 0-form f,

$$df = \frac{\partial f}{\partial x}\, dx + \frac{\partial f}{\partial y}\, dy + \frac{\partial f}{\partial z}\, dz$$

whose coefficient triple is the gradient ∇f.

On a 1-form $\omega = P\, dx + Q\, dy + R\, dz$, expanding (12.6) and collecting the basis 2-forms,

$$d\omega = \Big(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\Big) dy \wedge dz + \Big(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\Big) dz \wedge dx + \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big) dx \wedge dy$$

whose coefficient triple is the curl $\nabla \times (P, Q, R)$.

On a 2-form $\eta = P\, dy \wedge dz + Q\, dz \wedge dx + R\, dx \wedge dy$,

$$d\eta = \Big(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\Big) dx \wedge dy \wedge dz$$

the divergence $\nabla \cdot (P, Q, R)$ times the volume form.

The identity $d^2 = 0$ reads, across the two degrees, as the two staples $\nabla \times \nabla f = 0$ (curl of a gradient) and $\nabla \cdot (\nabla \times F) = 0$ (divergence of a curl). Both are now special cases of a single fact.

The coordinate formula (12.6) defines d but hides that the result is independent of the chart. Uniqueness already guarantees chart-independence, yet it is worth having a formula for $d\omega$ written entirely in terms of operations that make sense without coordinates: the action of vector fields on functions and the Lie bracket. Such a formula also reveals how d encodes the failure of vector fields to commute.

Proposition 12.5.3: The Invariant Formula

For $\omega \in \Omega^k(M)$ and smooth vector fields $X_0, \ldots, X_k$ on M,

$$d\omega(X_0, \ldots, X_k) = \sum_{i=0}^{k} (-1)^i\, X_i\big(\omega(X_0, \ldots, \widehat{X_i}, \ldots, X_k)\big) + \sum_{0 \le i < j \le k} (-1)^{i+j}\, \omega\big([X_i, X_j], X_0, \ldots, \widehat{X_i}, \ldots, \widehat{X_j}, \ldots, X_k\big)$$

where a hat denotes omission of an argument and $[\cdot, \cdot]$ is the Lie bracket (definition 5.5.1).

***Proof*:**
Write $D\omega(X_0, \ldots, X_k)$ for the right-hand side. Both $d\omega$ and $D\omega$ are $\mathbb{R}$-multilinear in the X_i; the claim is that they agree. The strategy is to show $D\omega$ is $C^\infty(M)$-multilinear and alternating, hence a $(k+1)$-form, and then to check the identity on coordinate vector fields, where both sides are determined by their pointwise values.

Step 1: $D\omega$ is $C^\infty(M)$-linear in X_0. Replace X_0 by fX_0 for $f \in C^\infty(M)$. In the first sum the $i = 0$ term becomes $fX_0(\omega(X_1, \ldots, X_k))$, already f times the original. For $i \ge 1$, the argument fX_0 sits inside ω, and the Leibniz rule for the derivation X_i gives

$$(-1)^i X_i\big(f\, \omega(X_0, \ldots, \widehat{X_i}, \ldots, X_k)\big) = f \cdot (-1)^i X_i\big(\omega(\ldots)\big) + (-1)^i (X_i f)\, \omega(X_0, \ldots, \widehat{X_i}, \ldots, X_k)$$

In the second sum the terms with $i = 0$ use $[fX_0, X_j] = f[X_0, X_j] - (X_j f)\, X_0$, so

$$(-1)^j\, \omega\big([fX_0, X_j], \ldots\big) = f \cdot (-1)^j\, \omega\big([X_0, X_j], \ldots\big) - (-1)^j (X_j f)\, \omega\big(X_0, X_1, \ldots, \widehat{X_j}, \ldots, X_k\big)$$

The f-multiples of the two sums reassemble $f \cdot D\omega(X_0, \ldots, X_k)$. The remaining correction terms cancel in pairs: for each $j \ge 1$ the extra term from the first sum (index $i = j$) is $(-1)^j (X_j f)\, \omega(X_0, \ldots, \widehat{X_j}, \ldots, X_k)$, and the extra term from the second sum (index $i = 0$) is its negative, since $-X_j(f)X_0$ occupies the first slot of ω. Hence $D\omega(fX_0, X_1, \ldots, X_k) = f\, D\omega(X_0, \ldots, X_k)$.

Step 2: $D\omega$ is alternating. It suffices to show that swapping two adjacent arguments X_p, X_{p+1} reverses the sign. In the first sum, every term with $i \notin \{p, p+1\}$ keeps its outer factor X_i and its sign $(-1)^i$, while the two swapped arguments both sit inside the alternating ω; so that term changes sign. The two terms with $i \in \{p, p+1\}$ exchange roles: the $i = p$ term of the swapped tuple equals $(-1)^p X_{p+1}(\omega(\ldots X_p \ldots))$, which is the negative of the $i = p+1$ term of the original, because the signs $(-1)^p$ and $(-1)^{p+1}$ differ while the argument lists coincide, and symmetrically. The second sum reverses by the same accounting together with the antisymmetry $[X_p, X_{p+1}] = -[X_{p+1}, X_p]$ and the alternation of ω. Thus $D\omega$ changes sign under an adjacent transposition, and since these generate all permutations, $D\omega$ is alternating. Combined with Step 1, $C^\infty(M)$-linearity holds in every argument (move any slot to the first by a transposition, apply Step 1, move it back), so $D\omega$ is $C^\infty(M)$-multilinear and alternating, hence a $(k+1)$-form.

Step 3: Agreement on a coordinate frame. Two $(k+1)$-forms are equal once they agree on every increasing tuple of coordinate vector fields. Fix a chart and an increasing multi-index $J = (m_0 < \cdots < m_k)$, and evaluate both sides on $(\partial_{m_0}, \ldots, \partial_{m_k})$, where $\partial_m = \partial/\partial x^m$. All brackets $[\partial_{m_i}, \partial_{m_j}]$ vanish, their components being constant (proposition 5.5.2), so only the first sum survives:

$$D\omega(\partial_{m_0}, \ldots, \partial_{m_k}) = \sum_{r=0}^{k} (-1)^r \, \partial_{m_r} \big(\omega(\partial_{m_0}, \ldots, \widehat{\partial_{m_r}}, \ldots, \partial_{m_k})\big)$$

Writing $\omega = \sum_I a_I \, dx^I$, the evaluation $\omega(\partial_{m_0}, \ldots, \widehat{\partial_{m_r}}, \ldots, \partial_{m_k})$ picks out the coefficient $a_{J\setminus m_r}$ of the increasing k-index $J \setminus m_r$ obtained by deleting m_r. On the other side, from (12.6),

$$d\omega(\partial_{m_0}, \ldots, \partial_{m_k}) = \sum_I \sum_j \frac{\partial a_I}{\partial x^j} \, (dx^j \wedge dx^I)(\partial_{m_0}, \ldots, \partial_{m_k})$$

and $(dx^j \wedge dx^I)(\partial_{m_0}, \ldots, \partial_{m_k})$ is nonzero only when $\{j\} \cup I = J$, that is $j = m_r$ and $I = J \setminus m_r$ for some r; moving dx^{m_r} into sorted position contributes $(-1)^r$. Hence $d\omega(\partial_{m_0}, \ldots, \partial_{m_k}) = \sum_r (-1)^r \, \partial_{m_r} a_{J\setminus m_r}$, the same expression. The two forms agree on every coordinate frame, so they are equal, as we sought to show.

The invariant formula is coordinate-free by construction: its right-hand side never mentions a chart. The most-used instance is $k = 1$, where it reads

$$d\omega(X, Y) = X\big(\omega(Y)\big) - Y\big(\omega(X)\big) - \omega([X, Y]) \tag{12.7}$$

The bracket term is what distinguishes $d\omega$ from a naive antisymmetrized derivative: if X and Y commute (for instance coordinate fields) it drops out, but in general it records their failure to do so. This is the formula through which the exterior derivative enters the theory of connections and curvature (the Riemannian geometry volume), where $d\omega(X, Y)$ measures a holonomy around an infinitesimal parallelogram.

The exterior derivative was built chart by chart, so it is a construction internal to a single manifold. It is also compatible with smooth maps between manifolds, in the strongest sense: it commutes with pullback. This is the property that will let d descend to a map on cohomology.

Proposition 12.5.4: Naturality of the Exterior Derivative

Let $f\colon M \to N$ be smooth. Then $f^*(d\omega) = d(f^*\omega)$ for every $\omega \in \Omega^k(N)$; that is, $f^* d = d f^*$.

***Proof*:**
For $k = 0$ the claim is $f^*(dg) = d(g \circ f)$ for $g \in C^\infty(N)$, which is proposition 12.1.7. For $k \geq 1$, both sides are forms, and the identity is local on M, so it may be checked near a point $p \in M$

inside a chart (V, y) of N containing $f(p)$. Write $\omega = \sum_I a_I \, dy^I$ on V. Pullback commutes with the wedge (proposition 12.4.11) and sends dy^i to $f^*(dy^i) = d(y^i \circ f)$ by the $k = 0$ case, so

$$f^*\omega = \sum_I (a_I \circ f) \, d(y^{i_1} \circ f) \wedge \cdots \wedge d(y^{i_k} \circ f)$$

Apply d and use the antiderivation rule together with $d\big(d(y^{i_r} \circ f)\big) = 0$, which kills every term except the one differentiating the leading coefficient:

$$d(f^*\omega) = \sum_I d(a_I \circ f) \wedge d(y^{i_1} \circ f) \wedge \cdots \wedge d(y^{i_k} \circ f)$$

On the other side, $d\omega = \sum_I da_I \wedge dy^I$, and applying f^* (again commuting with the wedge) gives

$$f^*(d\omega) = \sum_I f^*(da_I) \wedge f^*(dy^{i_1}) \wedge \cdots \wedge f^*(dy^{i_k}) = \sum_I d(a_I \circ f) \wedge d(y^{i_1} \circ f) \wedge \cdots \wedge d(y^{i_k} \circ f)$$

where $f^*(da_I) = d(a_I \circ f)$ is once more the $k = 0$ case. The two expressions coincide, so $f^*(d\omega) = d(f^*\omega)$, completing the proof.

Naturality says the exterior derivative is a natural transformation: it is the same operator on every manifold, transported correctly by every smooth map. Two operators already met, the interior product and the wedge, interact with d; the third is the Lie derivative, which measures how a form changes along a flow. Cartan's formula expresses that interaction in closed form.

The Lie derivative of a vector field was defined in section 5.10 by dragging the field back along the flow and differentiating. The same construction applies verbatim to forms, using the pullback in place of the pushforward.

Definition 12.5.5: Lie Derivative of a Form

Let X be a smooth vector field on M with flow θ (section 5.7). The *Lie derivative* of $\omega \in \Omega^k(M)$ along X is the k-form

$$\mathcal{L}_X\omega = \frac{d}{dt}\Big|_{t=0} \theta_t^*\omega$$

For each $p \in M$ the flow θ_t is defined for $|t|$ small on a neighborhood of p (section 5.7), so $\theta_t^*\omega$ is a k-form near p for small t, and the t-derivative at 0 defines $(\mathcal{L}_X\omega)_p$; the result depends smoothly on p.

On a 0-form f the definition gives $\mathcal{L}_X f = \frac{d}{dt}\big|_0 (f \circ \theta_t) = Xf$, the derivative of f along X, matching the vector-field Lie derivative $\mathcal{L}_X Y = [X, Y]$ of theorem 5.10.2. Computing $\mathcal{L}_X\omega$ from the definition requires the flow, which is rarely explicit. Cartan's formula removes the flow entirely, expressing $\mathcal{L}_X$ through d and the interior product ι_X of definition 12.4.7.

Theorem 12.5.6: Cartan's Magic Formula

For every smooth vector field X on M,

$$\mathcal{L}_X = d\iota_X + \iota_X d$$

as operators on $\Omega^\bullet(M)$.

***Proof*:**
Both operators are $\mathbb{R}$-linear and local: $\mathcal{L}_X$ because it is defined by the flow near each point, and $d\iota_X + \iota_X d$ because d and ι_X are. Set $L = d\iota_X + \iota_X d$. The proof shows both $\mathcal{L}_X$ and L are degree-0 derivations of $\Omega^\bullet(M)$ that commute with d and agree on $\Omega^0(M)$, and that these properties determine such an operator.

Step 1: Both are derivations. For $\mathcal{L}_X$, pullback commutes with the wedge (proposition 12.4.11), so $\theta_t^*(\alpha\wedge\beta) = \theta_t^*\alpha\wedge\theta_t^*\beta$; differentiating at $t=0$ with $\theta_0^* = \mathrm{id}$ gives $\mathcal{L}_X(\alpha\wedge\beta) = \mathcal{L}_X\alpha\wedge\beta+\alpha\wedge\mathcal{L}_X\beta$. For L, recall d and ι_X are antiderivations of degrees $+1$ and -1: $d(\alpha\wedge\beta) = d\alpha\wedge\beta+(-1)^k\alpha\wedge d\beta$ and $\iota_X(\alpha\wedge\beta) = \iota_X\alpha\wedge\beta+(-1)^k\alpha\wedge\iota_X\beta$ for $\alpha\in\Omega^k(M)$. Expanding $d\iota_X(\alpha\wedge\beta)$ and $\iota_X d(\alpha\wedge\beta)$ and adding, the four cross terms carry the coefficients $(-1)^{k-1}$ and $(-1)^k$ in the pair $d\iota_X$, and $(-1)^{k+1}$ and $(-1)^k$ in the pair $\iota_X d$; these cancel in the two matching pairs, leaving

$$L(\alpha\wedge\beta) = L\alpha\wedge\beta+\alpha\wedge L\beta$$

So L is a degree-0 derivation.

Step 2: Both commute with d. Naturality (proposition 12.5.4) gives $\theta_t^*(d\omega) = d(\theta_t^*\omega)$; differentiating at $t=0$ yields $\mathcal{L}_X(d\omega) = d(\mathcal{L}_X\omega)$. For L, using $d\circ d = 0$,

$$Ld = (d\iota_X+\iota_X d)d = d\iota_X d = d(d\iota_X+\iota_X d) = dL$$

Step 3: Both agree on functions. As noted, $\mathcal{L}_X f = Xf$. On the other side, $\iota_X f = 0$ for a 0-form, so $Lf = \iota_X(df) = df(X) = Xf$. Thus $\mathcal{L}_X$ and L agree on $\Omega^0(M)$.

Step 4: Such an operator is unique. Let D be any degree-0 derivation commuting with d. In a chart every form is a sum of terms $a\,dx^{i_1}\wedge\cdots\wedge dx^{i_k}$. On a coordinate differential, $D(dx^i) = D(dx^i) = d(Dx^i)$, so D on dx^i is determined by D on the function x^i; by the Leibniz rule D on the product $a\,dx^{i_1}\wedge\cdots$ is then determined by its values on the functions a and x^i. Hence a degree-0 derivation commuting with d is fixed on all forms, locally, by its restriction to $\Omega^0(M)$, and by locality globally. Since $\mathcal{L}_X$ and L are two such operators agreeing on $\Omega^0(M)$, they are equal, so $\mathcal{L}_X = d\iota_X+\iota_X d$, completing the proof.

Cartan's formula makes $\mathcal{L}_X$ computable without ever integrating the flow: it reduces to one application of d and one contraction. It also displays the three operators as a tightly linked system and, read as $\mathcal{L}_X - \iota_X d = d\iota_X$, foreshadows the homotopy arguments of chapter 14, where an operator of the form $dh+hd$ certifies that a closed form is exact.

Example 12.5.7: The Area Form under Scaling

On $\mathbb{R}^2$ take the radial field $X = x\,\partial/\partial x + y\,\partial/\partial y$ and the area form $\omega = dx \wedge dy$. Since ι_X is an antiderivation and $\iota_X dx = dx(X) = x$, $\iota_X dy = dy(X) = y$,

$$\iota_X\omega = (\iota_X dx) \wedge dy - dx \wedge (\iota_X dy) = x\,dy - y\,dx$$

Then $d(\iota_X\omega) = d(x\,dy - y\,dx) = dx \wedge dy - dy \wedge dx = 2\,dx \wedge dy$, while $\iota_X(d\omega) = 0$ because $d\omega = 0$. Cartan's formula gives

$$\mathcal{L}_X\omega = d\iota_X\omega + \iota_X d\omega = 2\,dx \wedge dy = 2\omega$$

This agrees with the flow: X generates $\theta_t(p) = e^t p$, which scales area by e^{2t}, so $\theta_t^*\omega = e^{2t}\omega$ and $\frac{d}{dt}\big|_0 \theta_t^*\omega = 2\omega$.

The identity $d^2 = 0$ splits the forms into two nested families whose comparison is the central question of the next chapter.

Definition 12.5.8: Closed and Exact Forms

A form $\omega \in \Omega^k(M)$ is *closed* if $d\omega = 0$, and *exact* if $\omega = d\eta$ for some $\eta \in \Omega^{k-1}(M)$.

Because $d \circ d = 0$, every exact form is closed: if $\omega = d\eta$ then $d\omega = d(d\eta) = 0$. The two families sit inside one another,

$$\{\text{exact } k\text{-forms}\} \subseteq \{\text{closed } k\text{-forms}\} \subseteq \Omega^k(M)$$

and the question of whether the first inclusion is an equality is the one the exterior derivative poses. For 1-forms the vocabulary is already familiar: by theorem 12.2.7 a 1-form is exact exactly when it is conservative, so an exact 1-form has path-independent line integrals. Closedness, $d\omega = 0$, is the local integrability condition, weaker than exactness in general. The gap is real: on $\mathbb{R}^2 \setminus \{0\}$ the angle form

$$\omega = \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

of section 12.2 is closed but not exact, its line integral around the origin being 2π rather than 0. Whether a given closed form is exact depends on the shape of M, not on ω alone; measuring this dependence, degree by degree, is the subject of de Rham cohomology (chapter 14), which turns the quotient of closed forms by exact ones into a topological invariant. That question is posed here and answered there.

Exercise 12.5.1

1. On $\mathbb{R}^3$ let $\omega = xy\,dx + yz\,dy + zx\,dz$. Compute $d\omega$ in the basis $\{dx \wedge dy,\ dy \wedge dz,\ dz \wedge dx\}$, and then compute $d(d\omega)$ directly to verify $d^2\omega = 0$.
2. On $\mathbb{R}^2$ let $\omega = x\,dy$, $X = \partial/\partial x$, and $Y = \partial/\partial y$. Compute $d\omega$ from the coordinate formula and evaluate $d\omega(X, Y)$. Then verify that the $k = 1$ invariant formula $d\omega(X,Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X,Y])$ gives the same value.
3. Using the antiderivation rule and $d^2 = 0$, prove: (a) if α and β are closed then $\alpha \wedge \beta$ is closed; (b) if α is exact and β is closed then $\alpha \wedge \beta$ is exact.
4. On $\mathbb{R}^2$ let $X = -y\,\partial/\partial x + x\,\partial/\partial y$ be the rotation field and $\omega = dx \wedge dy$ the area form. Use Cartan's formula to compute $\mathcal{L}_X\omega$, and interpret the answer.

5. Using $\iota_X \circ \iota_X = 0$ (definition 12.4.7) and Cartan's formula, show that $\mathcal{L}_X \circ \iota_X = \iota_X \circ \mathcal{L}_X$.
6. Verify directly that the angle form $\omega = (-y\,dx + x\,dy)/(x^2 + y^2)$ on $\mathbb{R}^2 \setminus \{0\}$ is closed, by computing $d\omega$.

12.6 Orientation and Volume Forms

The preceding sections built the exterior calculus on a fixed manifold; this one turns its top-degree forms back on the notion of orientability. An orientation of M has already been defined twice, by an atlas of charts with positive Jacobian transitions (definition 11.7.3) and by an orientation of the tangent bundle (section 11.7). A single n-form encodes the same data, and among such forms the *volume forms* are exactly what the integral of chapter 13 pairs against an oriented manifold. This section makes the identification and singles out the boundary orientation that Stokes' theorem uses.

The starting point is a dimension count. For an n-dimensional vector space V the top exterior power $\Lambda^n V^*$ is one-dimensional (theorem 0.5.9: $\dim \Lambda^k V^* = \binom{n}{k}$, so $\dim \Lambda^n V^* = 1$), and a nonzero element of it is an alternating n-form on V that vanishes on no basis. Such a form sorts the ordered bases of V into the two classes on which it is positive and negative, which is precisely a choice of orientation of V. Applied fiberwise to $V = T_pM$, this says a nowhere-zero section of $\Lambda^n T^*M$ orients every tangent space at once.

Definition 12.6.1: Orientation Form

An *orientation form* on a smooth n-manifold M is a nowhere-vanishing n-form $\omega \in \Omega^n(M)$. It *orients* M by declaring an ordered basis $(v_1, \dots, v_n)$ of T_pM to be *positively oriented* when $\omega_p(v_1, \dots, v_n) > 0$ and negatively oriented when $\omega_p(v_1, \dots, v_n) < 0$. Two orientation forms *determine the same orientation* if they induce the same class of positively oriented bases at every point.

Because $\Lambda^n T^*M$ is a line bundle, any two n-forms differ by a function: if ω, ω' are orientation forms then $\omega' = f\omega$ for a unique $f \in C^\infty(M)$, and f is nowhere zero since both forms are. On the fiber at p the sign of $\omega'_p(v_1, \dots, v_n) = f(p)\,\omega_p(v_1, \dots, v_n)$ agrees with that of $\omega_p(v_1, \dots, v_n)$ exactly when $f(p) > 0$. Thus ω and ω' determine the same orientation iff $f > 0$ everywhere. A nowhere-zero continuous function on a connected manifold has constant sign, so on a connected M there are exactly two orientations, carried by ω and $-\omega$.

Example 12.6.2: The Standard Orientation of $\mathbb{R}^n$

On $\mathbb{R}^n$ the form $\omega = dx^1 \wedge \cdots \wedge dx^n$ is nowhere zero: evaluated on the standard coordinate frame it gives

$$(dx^1 \wedge \cdots \wedge dx^n)\left(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\right) = \det\left[dx^i\left(\frac{\partial}{\partial x^j}\right)\right] = \det(I_n) = 1$$

so the standard basis is positively oriented, and ω carries the standard orientation of definition 11.7.3. Since $\Lambda^n T^*\mathbb{R}^n$ is trivialized by this single frame, every n-form on $\mathbb{R}^n$ is $a\,dx^1 \wedge \cdots \wedge dx^n$ for one smooth function a; it is an orientation form iff a vanishes nowhere, and $a\,dx^1 \wedge \cdots \wedge dx^n$ and $b\,dx^1 \wedge \cdots \wedge dx^n$ give the same orientation iff $a/b > 0$ throughout.

An orientation form need not exist. The manifolds that carry one are exactly the orientable ones, and the correspondence refines to a bijection between orientation forms up to positive scaling and orientations in the sense already fixed for the tangent bundle.

Theorem 12.6.3: Orientation Forms and Orientability

A smooth n-manifold M admits an orientation form iff M is orientable (definition 11.7.3). An orientation form induces an orientation of TM, every orientation of TM arises this way, and two orientation forms induce the same orientation of TM iff they differ by a positive function. In particular the orientations of M correspond bijectively to the equivalence classes of orientation forms under positive scaling.

***Proof*:**
Step 1: An orientation form orients TM. Let $\omega \in \Omega^n(M)$ be nowhere zero. At each p orient T_pM by the rule of definition 12.6.1, writing μ_p for that orientation. To see this choice is locally constant in the sense of definition 11.7.1, work in a connected chart (U, x): there $\omega = a\, dx^1 \wedge \cdots \wedge dx^n$ with $a = \omega(\partial/\partial x^1, \ldots, \partial/\partial x^n)$ smooth and nowhere zero, hence of constant sign on U. The coordinate frame is therefore positively oriented at every point of U (if $a > 0$) or negatively oriented at every point (if $a < 0$), so the fiber orientations are carried by the local frame of a trivialization and vary locally constantly. Thus the assignment $p \mapsto \mu_p$ is an orientation of TM, and M is orientable and oriented by definition 11.7.3.

Step 2: An orientation of TM produces an orientation form. Suppose M is oriented. Choose an oriented atlas $\{(U_\alpha, x_\alpha)\}$, that is, charts whose coordinate frames are positively oriented, so that the transition Jacobians $\det D(x_\alpha \circ x_\beta^{-1})$ are positive (definition 11.7.3). On each U_α set $\omega_\alpha = dx_\alpha^1 \wedge \cdots \wedge dx_\alpha^n$, a nowhere-zero n-form. On an overlap, writing the transition as $\tau = x_\alpha \circ x_\beta^{-1}$ gives $dx_\alpha^i = \sum_j (\partial\tau^i/\partial x_\beta^j)\, dx_\beta^j$, and wedging all n of these,

$$\omega_\alpha = \det\left(\frac{\partial \tau^i}{\partial x_\beta^j}\right) \omega_\beta$$

since Λ^n of a linear map is multiplication by its determinant (theorem 0.5.9). The scalar $\det(\partial\tau/\partial x_\beta)$ is the transition Jacobian, positive by the choice of atlas. Let $\{\rho_\alpha\}$ be a smooth partition of unity subordinate to $\{U_\alpha\}$ (theorem 1.5.5) and set

$$\omega = \sum_\alpha \rho_\alpha\, \omega_\alpha$$

a smooth global n-form. Fix p and a positively oriented basis $b = (b_1, \ldots, b_n)$ of T_pM. For every α with $p \in U_\alpha$ the chart is oriented, so $\omega_{\alpha,p}$ takes the value 1 on the positively oriented coordinate frame, whence $\omega_{\alpha,p}(b) > 0$ (a positively oriented basis differs from the coordinate frame by a positive-determinant change). Therefore

$$\omega_p(b) = \sum_\alpha \rho_\alpha(p)\, \omega_{\alpha,p}(b)$$

is a sum of nonnegative terms with $\sum_\alpha \rho_\alpha(p) = 1$, so at least one summand is strictly positive and $\omega_p(b) > 0$. Hence ω is nowhere zero and, by Step 1, induces the orientation started from.

Step 3: The correspondence. Steps 1 and 2 show orientation forms and orientations of TM map to one another, and the map from forms is surjective. Two orientation forms differ by a nowhere-zero function f, and by the discussion after definition 12.6.1 they induce the same orientation iff $f > 0$. So the induced-orientation map is constant exactly on the positive-scaling classes, giving the stated bijection, as we sought to show.

Orientability now has three equivalent faces: an atlas of charts with positive Jacobian transitions (definition 11.7.3), an orientation of the tangent bundle (section 11.7), and a nowhere-vanishing top form. The last is the computational package, a single global object rather than a compatible family of local choices, and it is exactly what the integral of the next chapter pairs against a manifold. The equivalence also reads directly off the bundle theory: a nowhere-zero n-form is a nowhere-zero section of the line bundle $\Lambda^n T^*M$ (proposition 11.8.2), which exists iff that line bundle is trivial (corollary 11.4.4), iff T^*M is orientable (proposition 11.7.2); and T^*M is orientable iff TM is, since their transition functions are inverse transposes of one another and $\det(\tau^{-1})^{\mathsf{T}}$ has the sign of $1/\det \tau$.

Example 12.6.4: A Volume Form on S^n

Let $\Omega = dx^1 \wedge \cdots \wedge dx^{n+1}$ be the standard orientation form on $\mathbb{R}^{n+1}$ and let $N = \sum_i x^i\, \partial/\partial x^i$ be the outward radial vector field, which along S^n is the outward unit normal. The interior product (definition 12.4.7) produces the n-form

$$\sigma = \iota_N \Omega\big|_{S^n}$$

on S^n, and σ is nowhere zero. Indeed at $p \in S^n$ the normal N_p is transverse to T_pS^n, so for any basis $(v_1, \ldots, v_n)$ of T_pS^n the tuple $(N_p, v_1, \ldots, v_n)$ is a basis of $\mathbb{R}^{n+1}$, and

$$\sigma_p(v_1, \ldots, v_n) = \Omega(N_p, v_1, \ldots, v_n) \neq 0$$

Thus σ is an orientation form; it is the outward-normal orientation of example 11.7.4. For $n = 1$, $\iota_N(dx \wedge dy) = x\, dy - y\, dx$, which restricts on S^1 to the angle form $d\theta$. For $n = 2$,

$$\sigma = x\, dy \wedge dz + y\, dz \wedge dx + z\, dx \wedge dy$$

and at the north pole $(0, 0, 1)$ it gives $\sigma(\partial/\partial x, \partial/\partial y) = z = 1 \neq 0$, confirming nonvanishing there.

Every orientation form assigns a signed volume to a positively oriented parallelepiped of tangent vectors, and integrating it will measure the total volume of the manifold. When a manifold is already oriented, the forms that respect the orientation deserve a name, since only those integrate to a positive volume.

Definition 12.6.5: Volume Form

A *volume form* on an oriented manifold M is an orientation form that is positively oriented, that is, $\omega_p(v_1, \ldots, v_n) > 0$ for every positively oriented basis $(v_1, \ldots, v_n)$ of T_pM. Equivalently, having fixed one orientation form ω_0 compatible with the orientation, the volume forms are exactly $f\omega_0$ with $f \in C^\infty(M)$, $f > 0$.

On an oriented manifold the volume forms are thus the nowhere-zero top forms inducing the chosen orientation, and they form a convex cone: any positive smooth multiple of a volume form is one, and any positive combination of two is one. The term *volume form* is often used loosely for any orientation form, the choice of orientation being understood. A metric refines the freedom to a canonical choice: a Riemannian metric together with an orientation singles out the *Riemannian* *volume form*, locally $\sqrt{\det(g_{ij})}\, dx^1 \wedge \cdots \wedge dx^n$, whose integral is the Riemannian volume. That construction and its geometry belong to the Riemannian geometry volume; here the point is only that the top forms exist and encode orientation.

Example 12.6.6: The Möbius Band Has No Volume Form

The open Möbius band B is non-orientable: its tangent bundle restricts along the core circle to the nontrivial Möbius line bundle (example 11.7.4). By theorem 12.6.3 it therefore admits no nowhere-vanishing 2-form, hence no volume form. The obstruction is visible directly. Traverse the core circle once and transport a local frame $(\partial/\partial s, \partial/\partial t)$ along it; the gluing that closes the band reverses the transverse direction, returning the frame as $(\partial/\partial s, -\partial/\partial t)$. A nowhere-zero 2-form ω has $\omega(\partial/\partial s, \partial/\partial t)$ a continuous nowhere-zero function along the loop, yet after one circuit the sign flip forces it to equal its own negative, so by the intermediate value theorem it vanishes somewhere, a contradiction.

Orientation and integration meet at the boundary. To integrate over a manifold with boundary and compare the result with an integral over the boundary, as Stokes' theorem will, the boundary must carry an orientation determined by that of M. A top form gives it through contraction with an outward vector.

Proposition 12.6.7: The Induced Boundary Orientation

Let M be an oriented n-manifold with boundary (definition 1.1.16), $n \geq 1$, with orientation form ω. For $p \in \partial M$ let $N_p \in T_pM$ be any *outward-pointing* vector, one not tangent to ∂M and pointing out of M. Then

$$\omega_\partial = (\iota_N \omega)\big|_{\partial M}, \qquad (\omega_\partial)_p(v_1, \ldots, v_{n-1}) = \omega_p(N_p, v_1, \ldots, v_{n-1})$$

is a nowhere-vanishing $(n-1)$-form on ∂M whose orientation is independent of the choice of outward N. Its class is the *induced boundary orientation*: a basis $(v_1, \ldots, v_{n-1})$ of $T_p\partial M$ is positively oriented iff $(N_p, v_1, \ldots, v_{n-1})$ is positively oriented in T_pM.

***Proof*:**

Step 1: ω_∂ is a nowhere-zero $(n-1)$-form. A smooth outward field along ∂M exists. In a boundary chart modeled on $\mathbb{H}^n = \{x_n \geq 0\}$ the field $-\partial/\partial x^n$ points outward; patching such fields with a partition of unity $\{\rho_\alpha\}$ subordinate to a boundary atlas (theorem 1.5.5) gives $N = -\sum_\alpha \rho_\alpha\, \partial/\partial x^n_\alpha$, still outward because a positive combination of vectors on the outward side of the boundary hyperplane stays on that side. Then $\iota_N\omega$ is a smooth $(n-1)$-form on M; restrict it to ∂M. At p the outward N_p is transverse to the hyperplane $T_p\partial M$, so $T_pM = \mathbb{R}N_p \oplus T_p\partial M$; for a basis $(v_1, \ldots, v_{n-1})$ of $T_p\partial M$ the tuple $(N_p, v_1, \ldots, v_{n-1})$ is a basis of T_pM, and $\omega_p(N_p, v_1, \ldots, v_{n-1}) \neq 0$. Hence $(\omega_\partial)_p \neq 0$.

Step 2: Independence of the outward vector. Let N'_p be another outward vector. Both lie strictly on the outward side of $T_p\partial M$, so $N'_p = aN_p + w$ with $a > 0$ and $w \in T_p\partial M$. For $v_i \in T_p\partial M$,

$$\omega_p(N'_p, v_1, \ldots, v_{n-1}) = a\, \omega_p(N_p, v_1, \ldots, v_{n-1}) + \omega_p(w, v_1, \ldots, v_{n-1})$$

The final term evaluates ω_p on the n vectors $w, v_1, \ldots, v_{n-1}$, all lying in the $(n-1)$-dimensional space $T_p\partial M$, hence linearly dependent, so it vanishes because ω_p is alternating. Thus $(\iota_{N'}\omega)_p = a\,(\iota_N\omega)_p$ with $a > 0$, the same orientation of $T_p\partial M$. The boundary orientation is therefore well defined, and since $(\omega_\partial)_p(v_1, \ldots, v_{n-1}) = \omega_p(N_p, v_1, \ldots, v_{n-1})$, a boundary basis is positively oriented exactly when $(N_p, v_1, \ldots, v_{n-1})$ is, completing the proof.

The outward-normal-first convention is fixed precisely so that Stokes' theorem $\int_M d\omega = \int_{\partial M} \omega$ holds with no compensating sign, which is why it, rather than an inward-first or last-vector convention, is the one carried forward. With orientation packaged as a top form and the boundary oriented from it, the objects integration needs are in place: a top form is what one integrates over an oriented manifold, and chapter 13 defines that integral and proves Stokes' theorem. The gap between closed and exact forms recorded in the previous section, and the volume forms built here, are the two threads that de Rham cohomology (chapter 14) will weave together.

Exercise 12.6.1

1. Show that every n-form on $\mathbb{R}^n$ equals $a\, dx^1 \wedge \cdots \wedge dx^n$ for a unique smooth function a, and deduce that the orientation forms on $\mathbb{R}^n$ are exactly those with a nowhere zero. Show that $a\, dx^1 \wedge \cdots \wedge dx^n$ and $b\, dx^1 \wedge \cdots \wedge dx^n$ induce the same orientation iff $a(x)b(x) > 0$ for all x.

2. Using that a nowhere-zero continuous function on a connected space has constant sign, prove that a connected orientable manifold has exactly two orientations, and that an orientable manifold with k connected components has 2^k.

3. For the 2-form $\sigma = x\, dy \wedge dz + y\, dz \wedge dx + z\, dx \wedge dy$ on $\mathbb{R}^3$, verify that $\sigma = \iota_N(dx \wedge dy \wedge dz)$ for the radial field $N = x\, \partial/\partial x + y\, \partial/\partial y + z\, \partial/\partial z$, and show that σ restricts to a nowhere-zero form on S^2. Conclude that S^2 is orientable.

4. Let (M, ω_M) and (N, ω_N) be oriented manifolds of dimensions m and n, with projections $\mathrm{pr}_M, \mathrm{pr}_N$ from $M \times N$. Show that $\mathrm{pr}_M^*\omega_M \wedge \mathrm{pr}_N^*\omega_N$ is an orientation form on $M \times N$, so that a product of orientable manifolds is orientable.

5. Give $\mathbb{H}^n = \{x_n \geq 0\}$ the orientation of $dx^1 \wedge \cdots \wedge dx^n$. Compute the induced boundary orientation on $\partial\mathbb{H}^n = \{x_n = 0\}$ and show it equals $(-1)^n\, dx^1 \wedge \cdots \wedge dx^{n-1}$. Interpret the answer for $n = 1$.

13

Integration on Manifolds

A top-degree differential form is built to be integrated. On an oriented n-manifold it assigns a number to the whole space, and the number is coordinate-free for a precise reason: under a change of chart a form transforms by the Jacobian determinant while the Euclidean integral transforms by its absolute value, and an orientation is exactly the choice that makes the two signs agree. The first task of this chapter is to make that cancellation into a definition and prove it well posed.

The integral is then bound to the boundary by a single identity. Stokes' theorem,

$$\int_M d\omega = \int_{\partial M} \omega,$$

makes the exterior derivative the adjoint of the boundary operator, and it contains, as the cases $n = 1, 2, 3$, the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical Stokes theorem of vector analysis. One proof, reduced to two half-space models, yields all of them.

What Stokes leaves behind is where topology enters. A closed form integrates to zero over any boundary, so its integral over a cycle depends only on the cycle's homology class; the resulting numbers, the periods, are the first measurable trace of a manifold's shape, and the angle form whose period around a puncture is 2π is the standing example. Whether the periods determine a closed form up to an exact one is the question de Rham cohomology answers, and it is posed here and handed to chapter 14. A final section frees integration from orientation altogether, by integrating densities rather than forms, so that even the Möbius band acquires an integral. The forms calculus of chapter 12 and the Euclidean change of variables are cited; the integral and Stokes' theorem are proved here.

13.1 Integration of Forms over Oriented Manifolds

The construction proceeds in three steps. A top form supported in a single oriented chart is integrated by reading it as an ordinary integral over Euclidean space; the reading is shown to be independent of the chart, the one place where orientation is used; and a partition of unity then patches the local integrals into a single number attached to the whole manifold. The section closes by recording how that number transforms under a diffeomorphism.

Throughout, M is a smooth oriented n-manifold with $n \geq 1$, and $\Omega^n_c(M)$ denotes the compactly supported n-forms on M, that is, the top forms ω whose support $\operatorname{supp}\omega = \overline{\{p \in M \mid \omega_p \neq 0\}}$ is compact. An *oriented chart* is a chart (U, x) whose coordinate frame $(\partial/\partial x^1, \ldots, \partial/\partial x^n)$ is positively oriented at every point (definition 11.7.3); equivalently, $x\colon U \to x(U) \subseteq \mathbb{R}^n$ carries the orientation of M to the standard orientation of $\mathbb{R}^n$. A boundary is allowed: a boundary chart maps into the half-space $\mathbb{H}^n = \{u \in \mathbb{R}^n \mid u_n \geq 0\}$, and the integral over $x(U) \subseteq \mathbb{H}^n$ is again an ordinary integral over a subset of $\mathbb{R}^n$, so nothing below changes; the boundary itself is a Lebesgue-null slice and does not affect any integral. The systematic use of the boundary is section 13.2.

Definition 13.1.1: The Integral over an Oriented Chart

Let $\omega \in \Omega^n_c(M)$ be supported in the domain of a single oriented chart (U, x), and write $\omega = f\, dx^1 \wedge \cdots \wedge dx^n$ on U with $f \in C^\infty(U)$ compactly supported. The *integral of ω over M* is

$$\int_M \omega \;=\; \int_{x(U)} (x^{-1})^*\omega \;=\; \int_{x(U)} (f \circ x^{-1})(u)\, du^1 \cdots du^n \tag{13.1}$$

the right-hand side being the ordinary (Riemann or Lebesgue) integral over the open set $x(U) \subseteq \mathbb{R}^n$ of the coefficient of the pulled-back form $(x^{-1})^*\omega = (f \circ x^{-1})\, du^1 \wedge \cdots \wedge du^n$, where $u^1, \ldots, u^n$ are the standard coordinates on $\mathbb{R}^n$.

The definition converts a form into a function to be integrated: the chart map x trades the abstract form ω for the concrete top form $(x^{-1})^*\omega$ on Euclidean space, whose single coefficient $f \circ x^{-1}$ is what the ordinary integral uses. The pullback under x^{-1} is the device that will make the recipe chart-independent (proposition 13.1.3), since $(x^{-1})^*(dx^i) = d(x^i \circ x^{-1}) = du^i$ carries the coordinate covectors to the standard ones. Only two properties of the right-hand side are used, both furnished by the Euclidean theory: the integral is finite (guaranteed by compact support and continuity) and it obeys the change-of-variables law. The compact-support hypothesis is one sufficient condition for the first; the formula (13.1) applies verbatim to any top form whose chart coefficient is absolutely integrable.

Example 13.1.2: A Gaussian Top Form on $\mathbb{R}^n$

Take $M = \mathbb{R}^n$ with its standard orientation, a single global oriented chart given by the identity $x = \mathrm{id}$. The top form

$$\omega = e^{-|x|^2}\, dx^1 \wedge \cdots \wedge dx^n, \qquad |x|^2 = (x^1)^2 + \cdots + (x^n)^2$$

is not compactly supported, but its coefficient $e^{-|x|^2}$ is absolutely integrable, so (13.1) applies with $x^{-1} = \mathrm{id}$:

$$\int_{\mathbb{R}^n} \omega = \int_{\mathbb{R}^n} e^{-|u|^2}\, du^1 \cdots du^n = \prod_{i=1}^{n} \int_{\mathbb{R}} e^{-(u^i)^2}\, du^i = (\sqrt{\pi})^n = \pi^{n/2}$$

the product factoring because the integrand and the domain both factor over the coordinates. Here no chart-matching is needed: $\mathbb{R}^n$ is one oriented chart, and the integral is the Euclidean integral of the coefficient with no further work.

For (13.1) to define anything, its value must not depend on which oriented chart carries the support of ω. Two oriented charts differ by a transition map with positive Jacobian determinant, and the pulled-back coefficient acquires exactly that Jacobian; the Euclidean change of variables absorbs its absolute value, and positivity of the Jacobian identifies the absolute value with the value itself. The sign that an unoriented change of variables would leave unaccounted is precisely the sign an orientation fixes.

Proposition 13.1.3: Chart-Independence of the Integral

Let $\omega \in \Omega^n_c(M)$ be supported in $U \cap V$, where (U, x) and (V, y) are oriented charts. Then

$$\int_{x(U)} (x^{-1})^*\omega = \int_{y(V)} (y^{-1})^*\omega$$

so $\int_M \omega$ of definition 13.1.1 is independent of the oriented chart used to compute it.

***Proof*:**
Set $\alpha = (x^{-1})^*\omega$ on $x(U)$ and $\beta = (y^{-1})^*\omega$ on $y(V)$, and let $\tau = y \circ x^{-1} \colon x(U \cap V) \to y(U \cap V)$ be the transition map, a diffeomorphism between open subsets of $\mathbb{R}^n$.

Step 1: The two Euclidean forms are related by the transition pullback. Functoriality of the pullback (proposition 12.4.11) and $y^{-1} \circ \tau = y^{-1} \circ y \circ x^{-1} = x^{-1}$ give

$$\tau^*\beta = \tau^*(y^{-1})^*\omega = (y^{-1} \circ \tau)^*\omega = (x^{-1})^*\omega = \alpha$$

on $x(U \cap V)$. Writing $\beta = b\, du^1 \wedge \cdots \wedge du^n$ for its single coefficient b on $y(V)$, the pullback of a top form under a map of open Euclidean sets is

$$\tau^*\beta = (b \circ \tau)\, \det(D\tau)\, du^1 \wedge \cdots \wedge du^n$$

since $\tau^*(du^i) = d\tau^i = \sum_j (\partial\tau^i/\partial u^j)\, du^j$ and wedging the n rows multiplies by the determinant (proposition 12.4.11). Thus the coefficient of α is $(b \circ \tau)\, \det(D\tau)$.

Step 2: Change of variables with the orientation sign. Because both charts are oriented, τ preserves orientation, so $\det(D\tau) > 0$ throughout $x(U \cap V)$ and $\det(D\tau) = |\det D\tau|$. The support of ω lies in $U \cap V$, so α and β vanish outside $x(U \cap V)$ and $y(U \cap V)$ respectively, and each integral is unchanged by restricting the domain to those sets. The Euclidean change-of-variables theorem [14, Change of Variables for C^1 Diffeomorphisms], applied to the diffeomorphism τ and the coefficient b, reads

$$\int_{y(U\cap V)} b\, dv = \int_{x(U\cap V)} (b \circ \tau)\, |\det D\tau|\, du$$

Combining the two steps,

$$\int_{x(U)} \alpha = \int_{x(U\cap V)} (b \circ \tau)\, \det(D\tau)\, du = \int_{x(U\cap V)} (b \circ \tau)\, |\det D\tau|\, du = \int_{y(U\cap V)} b\, dv = \int_{y(V)} \beta$$

which is the asserted equality, as we sought to show.

The proof isolates the role of the orientation to a single equality, $\det(D\tau) = |\det D\tau|$. Drop the orientation hypothesis and the transition Jacobian may be negative on part of the overlap; the change of variables still holds with the absolute value, but the naive chart integral would then depend on the chart, differing by the sign of the Jacobian. An orientation is exactly a coherent choice of charts on which all transition Jacobians are positive, and that coherence is what makes (13.1) well posed on the overlaps. This is why a top-form integral belongs to oriented manifolds and, on a manifold with no orientation, to no consistent recipe at all (section 13.5 recovers integration there by a different device).

A general $\omega \in \Omega_c^n(M)$ need not be supported in one chart. Its support is compact, however, so it meets only finitely many members of any locally finite cover, and a partition of unity subordinate to an oriented atlas splits ω into finitely many pieces, each supported in a single chart. Summing the chart integrals of the pieces defines the integral in general.

Definition 13.1.4: The Integral over an Oriented Manifold

Let $\omega \in \Omega_c^n(M)$. Choose an oriented atlas $\{(U_\alpha, x_\alpha)\}$ and a smooth partition of unity $\{\rho_\alpha\}$ subordinate to it (theorem 1.5.5). Each $\rho_\alpha\omega$ is a compactly supported top form supported in U_α, and $\rho_\alpha\omega = 0$ for all but finitely many α (those with $\mathrm{supp}\rho_\alpha \cap \mathrm{supp}\omega \neq \emptyset$). The *integral of ω over M* is

$$\int_M \omega = \sum_\alpha \int_M \rho_\alpha\omega$$

a finite sum, each term given by definition 13.1.1 in the chart (U_α, x_α).

The chart integral of definition 13.1.1 is $\mathbb{R}$-linear in the form (its coefficient depends linearly on ω, and the Euclidean integral is linear), a fact used repeatedly below and worth recording: for forms supported in a common chart and scalars a, b, $\int_M (a\omega + b\eta) = a\int_M \omega + b\int_M \eta$. Two choices enter definition 13.1.4, the atlas and the partition of unity, and the definition is only legitimate if the resulting number is insensitive to both. It is, and the proof turns entirely on chart-independence.

Proposition 13.1.5: Well-Definedness of the Integral

The value $\int_M \omega$ of definition 13.1.4 is independent of the choice of oriented atlas and subordinate partition of unity.

***Proof*:**
Let $\{(U_\alpha, x_\alpha), \rho_\alpha\}$ and $\{(V_\beta, y_\beta), \sigma_\beta\}$ be two oriented atlases with subordinate partitions of unity, and write

$$I = \sum_\alpha \int_M \rho_\alpha\omega, \qquad J = \sum_\beta \int_M \sigma_\beta\omega$$

for the two candidate values. Both are finite sums, since $\mathrm{supp}\omega$ is compact and meets only finitely many $\mathrm{supp}\rho_\alpha$ and finitely many $\mathrm{supp}\sigma_\beta$.

Fix α. The form $\rho_\alpha\omega$ is supported in U_α, and $\sum_\beta \sigma_\beta = 1$ gives $\rho_\alpha\omega = \sum_\beta \sigma_\beta\rho_\alpha\omega$, a finite sum of forms each supported in U_α. Linearity of the chart integral in (U_α, x_α) yields

$$\int_M \rho_\alpha\omega = \sum_\beta \int_M \rho_\alpha\sigma_\beta\omega$$

Each summand $\rho_\alpha \sigma_\beta \omega$ is supported in $U_\alpha \cap V_\beta$, so it lies in the domain of both oriented charts x_α and y_β; by chart-independence (proposition 13.1.3) its integral is the same computed in either. Summing over α,

$$I = \sum_\alpha \int_M \rho_\alpha \omega = \sum_{\alpha,\beta} \int_M \rho_\alpha \sigma_\beta \omega$$

The identical computation with the roles of the two families exchanged, using $\sum_\alpha \rho_\alpha = 1$ and linearity of the chart integral in (V_β, y_β), gives

$$J = \sum_\beta \int_M \sigma_\beta \omega = \sum_{\beta,\alpha} \int_M \sigma_\beta \rho_\alpha \omega$$

The two double sums are finite and have the same terms $\int_M \rho_\alpha \sigma_\beta \omega$, so $I = J$, completing the proof.

The integral is now a well-defined map $\int_M : \Omega^n_c(M) \to \mathbb{R}$, linear by the same patching argument, and agreeing with definition 13.1.1 whenever the form happens to be supported in one chart (take a partition of unity refining that chart). The machinery of atlases and partitions of unity has vanished from the answer: it enters the definition and leaves no trace in the number, which is what proposition 13.1.5 asserts. The two sphere integrals below are the first genuine computations, each requiring one chart that omits a negligible set.

Example 13.1.6: Integrating over S^1 and S^2

Both spheres carry the outward-normal orientation and the volume form $\sigma = \iota_N \Omega$ of example 12.6.4, where $\Omega = dx^1 \wedge \cdots \wedge dx^{n+1}$ and N is the outward radial field.

On S^1, that form is $\sigma = x\,dy - y\,dx$. The map $\gamma\colon (0, 2\pi) \to S^1$, $\gamma(\theta) = (\cos\theta, \sin\theta)$, is an oriented chart onto S^1 minus the single point $(1, 0)$, and

$$\gamma^*\sigma = \cos\theta\, d(\sin\theta) - \sin\theta\, d(\cos\theta) = (\cos^2\theta + \sin^2\theta)\, d\theta = d\theta$$

The omitted point is a single point, hence Lebesgue-null in the chart, so it contributes nothing to the integral, and

$$\int_{S^1} \sigma = \int_0^{2\pi} d\theta = 2\pi$$

the length of the circle.

On S^2, $\sigma = x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy$. Spherical coordinates give the oriented chart $\Phi\colon (0, \pi) \times (0, 2\pi) \to S^2$,

$$\Phi(\varphi, \theta) = (\sin\varphi\cos\theta, \sin\varphi\sin\theta, \cos\varphi)$$

onto S^2 minus a meridian half-circle (a Lebesgue-null set). Expanding the pullback with $dx = \cos\varphi\cos\theta\, d\varphi - \sin\varphi\sin\theta\, d\theta$ and its analogues for dy, dz, the three terms combine to

$$\Phi^*\sigma = \sin\varphi\, d\varphi \wedge d\theta$$

with $\sin\varphi > 0$ on $(0, \pi)$, confirming that (φ, θ) is positively oriented for the outward orientation. Hence

$$\int_{S^2} \sigma = \int_0^{2\pi} \int_0^{\pi} \sin\varphi\, d\varphi\, d\theta = 2\pi\big[-\cos\varphi\big]_0^{\pi} = 4\pi$$

the surface area of the sphere.

An orientation-preserving diffeomorphism relabels the manifold without disturbing any of the data the integral sees, so it should leave every integral unchanged once the form is carried along by pullback. Reversing the orientation, on the other hand, flips the sign that proposition 13.1.3 was built to control. Both facts follow from the chart definition.

Proposition 13.1.7: Naturality under Diffeomorphisms

Let $F\colon N \to M$ be a diffeomorphism of oriented n-manifolds and $\omega \in \Omega^n_c(M)$. Then $F^*\omega \in \Omega^n_c(N)$, and:

1. if F is orientation-preserving, $\displaystyle\int_N F^*\omega = \int_M \omega$;
2. writing $-M$ for M with the opposite orientation, $\displaystyle\int_{-M} \omega = -\int_M \omega$; consequently, if F is orientation-reversing, $\displaystyle\int_N F^*\omega = -\int_M \omega$.

***Proof*:**
Since F is a diffeomorphism, $\operatorname{supp} F^*\omega = F^{-1}(\operatorname{supp}\omega)$ is compact, so $F^*\omega \in \Omega^n_c(N)$. The two clauses are proved in turn.

1. *Orientation-preserving invariance.* Suppose first $\operatorname{supp}\omega \subseteq U$ for a single oriented chart (U, x) of M. Then $z = x \circ F$ is a chart of N on $F^{-1}(U)$, oriented because F preserves orientation and x is oriented, with $z(F^{-1}(U)) = x(U)$. From $z^{-1} = F^{-1} \circ x^{-1}$ and functoriality of the pullback (proposition 12.4.11),
$$(z^{-1})^*(F^*\omega) = (F \circ z^{-1})^*\omega = (F \circ F^{-1} \circ x^{-1})^*\omega = (x^{-1})^*\omega$$
so the two Euclidean forms coincide on $x(U)$, and definition 13.1.1 gives $\int_N F^*\omega = \int_{x(U)} (z^{-1})^*(F^*\omega) = \int_{x(U)} (x^{-1})^*\omega = \int_M \omega$.
For a general ω, take an oriented atlas $\{(U_\alpha, x_\alpha)\}$ of M with subordinate partition of unity $\{\rho_\alpha\}$; then $\{F^{-1}(U_\alpha)\}$ is an oriented atlas of N and $\{\rho_\alpha \circ F\}$ a subordinate partition of unity. Since $F^*(\rho_\alpha\omega) = (\rho_\alpha \circ F)\, F^*\omega$ and each $\rho_\alpha\omega$ is supported in U_α, the one-chart case applies to every piece:
$$\int_N F^*\omega = \sum_\alpha \int_N (\rho_\alpha \circ F)\, F^*\omega = \sum_\alpha \int_N F^*(\rho_\alpha\omega) = \sum_\alpha \int_M \rho_\alpha\omega = \int_M \omega$$
2. *Orientation reversal.* It suffices to prove $\int_{-M} \omega = -\int_M \omega$: an orientation-reversing $F\colon N \to M$ is orientation-preserving as a map $N \to -M$, so part (1) gives $\int_N F^*\omega = \int_{-M} \omega = -\int_M \omega$. By linearity and a partition of unity it is enough to treat ω supported in one oriented chart (U, x) of M. Let $r(u^1, u^2, \ldots, u^n) = (-u^1, u^2, \ldots, u^n)$ and $\tilde{x} = r \circ x$; the transition $\tilde{x} \circ x^{-1} = r$ has $\det Dr = -1 < 0$, so $\tilde{x}$ is an oriented chart for $-M$. Running the computation of proposition 13.1.3 with the orientation-reversing r in place of τ, the coefficient of $(x^{-1})^*\omega$ becomes $(b \circ r)\, \det Dr = -(b \circ r)\, |\det Dr|$, so the sign persists through the change of variables:
$$\int_{-M} \omega = \int_{\tilde{x}(U)} (\tilde{x}^{-1})^*\omega = -\int_{x(U)} (x^{-1})^*\omega = -\int_M \omega$$

This proves both clauses, completing the proof.

Naturality is the statement that $\int_M$ depends on the oriented manifold and the form, not on any presentation of either; a diffeomorphism is an equivalence of oriented manifolds precisely when it respects integration up to the orientation sign it carries. The two clauses will be used constantly: clause (1) is the substitution rule that lets a computation be moved to whichever model is convenient (the sphere integrals above are one instance, the transported chart being spherical coordinates), and clause (2) is the sign bookkeeping that makes $\int_{\partial M}$ well posed once the boundary is oriented, which is where Stokes' theorem (section 13.2) begins.

Exercise 13.1.1

1. Give the torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ the product orientation, with angle coordinates $\theta^1, \dots, \theta^n$ each valued in $[0, 2\pi)$. Compute $\int_{T^n} d\theta^1 \wedge \cdots \wedge d\theta^n$. (For $n = 2$ this is the standard flat area of the torus.)
2. Prove that the integral is $\mathbb{R}$-linear: for $\omega, \eta \in \Omega_c^n(M)$ and $a, b \in \mathbb{R}$, $\int_M (a\omega + b\eta) = a\int_M \omega + b\int_M \eta$. (Reduce to a common partition of unity and use linearity of the chart integral.)
3. Suppose $\omega \in \Omega_c^n(M)$ is supported in the domain of one oriented chart (U, x). Show that the partition-of-unity value of definition 13.1.4 equals the chart value $\int_{x(U)} (x^{-1})^*\omega$ of definition 13.1.1, so the two definitions agree where both apply.
4. Let M be a compact oriented n-manifold admitting an orientation-reversing diffeomorphism $F\colon M \to M$. Show that every $\omega \in \Omega^n(M)$ with $F^*\omega = \omega$ satisfies $\int_M \omega = 0$. Deduce that no volume form on M is F-invariant.
5. Recompute $\int_{S^2} \sigma = 4\pi$ for the volume form σ of example 13.1.6 using stereographic projection from the south pole,
$$\psi(u, v) = \frac{1}{1 + u^2 + v^2}\,(2u,\ 2v,\ 1 - u^2 - v^2)$$
an oriented chart onto S^2 minus the south pole, given that $\psi^*\sigma = \dfrac{4}{(1 + u^2 + v^2)^2}\, du \wedge dv$. Explain why the agreement with example 13.1.6 is guaranteed in advance.

13.2 Stokes' Theorem

The integral of section 13.1 pairs a top form with an oriented manifold. This section binds that pairing to the boundary. For a compactly supported $(n-1)$-form ω on an oriented n-manifold with boundary, the integral of $d\omega$ over the whole manifold equals the integral of ω over the boundary alone, and the entire content is a single application of the fundamental theorem of calculus once the problem is cut down, by a partition of unity, to two Euclidean models.

The setting is the manifold with boundary of definition 1.1.16: a space modeled on the closed half-space $\mathbb{H}^n = \{x \in \mathbb{R}^n \mid x_n \geq 0\}$, whose boundary ∂M is the $(n-1)$-manifold of points carried into $\partial\mathbb{H}^n = \{x \mid x_n = 0\}$. When M is oriented, its boundary inherits the orientation of proposition 12.6.7, fixed by the outward-normal-first rule: a basis $(v_1, \dots, v_{n-1})$ of $T_p\partial M$ is positively oriented exactly when $(N_p, v_1, \dots, v_{n-1})$ is positively oriented in T_pM for an outward-pointing N_p. That convention was chosen so that the identity below carries no compensating sign. On the model $\mathbb{H}^n$ oriented by $dx^1 \wedge \cdots \wedge dx^n$, the induced boundary orientation of $\partial\mathbb{H}^n$ is $(-1)^n\, dx^1 \wedge \cdots \wedge dx^{n-1}$, computed in section 12.6; this sign is the one that will surface in the model calculation.

A boundary $(n-1)$-form is integrated over ∂M after restriction along the inclusion $\iota\colon \partial M \hookrightarrow M$, that is, as the pullback $\iota^*\omega$ (proposition 12.4.11). With this understood, the statement is clean.

Theorem 13.2.1: Stokes' Theorem

Let M be an oriented smooth n-manifold with boundary, $n \geq 1$, and let $\omega \in \Omega^{n-1}(M)$ be compactly supported. Give ∂M the induced boundary orientation of proposition 12.6.7. Then

$$\int_M d\omega = \int_{\partial M} \omega$$

where the right-hand side denotes the integral over ∂M of the pullback $\iota^*\omega$ along the inclusion $\iota\colon \partial M \hookrightarrow M$. If $\partial M = \varnothing$ the right-hand side is zero, so $\int_M d\omega = 0$.

***Proof*:**
The proof has three steps: a partition of unity reduces the identity to a form supported in a single chart; each chart integral is transported to one of two Euclidean models; and the models are computed by the fundamental theorem of calculus.

Step 1: Reduction to a single chart. Cover M by an oriented atlas whose charts are of two kinds: *interior charts* (U, x) with $x(U)$ an open subset of the open half $\{x \mid x_n > 0\}$, hence of $\mathbb{R}^n$, meeting ∂M in the empty set; and *boundary charts* with $x(U)$ an open subset of $\mathbb{H}^n$ meeting $\partial\mathbb{H}^n$. Every point has a neighborhood of one kind (definition 1.1.16), and the atlas may be taken oriented, each chart carrying the orientation of M to the standard orientation of $\mathbb{H}^n$.

Since $\operatorname{supp}\omega$ is compact, finitely many of these charts $U_1, \ldots, U_m$ cover it. Choose a smooth partition of unity $\{\rho_i\}_{i=1}^m$ subordinate to $\{U_i\}$ with $\sum_i \rho_i = 1$ on a neighborhood of $\operatorname{supp}\omega$ (theorem 1.5.5). Then $\omega = \sum_i \rho_i\omega$, a finite sum in which each $\rho_i\omega$ is compactly supported inside the single chart U_i. The exterior derivative and the integral are $\mathbb{R}$-linear, so

$$\int_M d\omega = \sum_{i=1}^m \int_M d(\rho_i\omega), \qquad \int_{\partial M} \omega = \sum_{i=1}^m \int_{\partial M} \rho_i\omega$$

It therefore suffices to prove $\int_M d\eta = \int_{\partial M} \eta$ for a single compactly supported $\eta \in \Omega^{n-1}(M)$ whose support lies in one chart (U, x).

Step 2: Transport to the model. Fix such an η, supported in (U, x). By the chart definition of the integral (definition 13.1.1) and the naturality of the exterior derivative (proposition 12.5.4),

$$\int_M d\eta = \int_{x(U)} (x^{-1})^*(d\eta) = \int_{x(U)} d\big((x^{-1})^*\eta\big)$$

Set $\widetilde{\eta} = (x^{-1})^*\eta$, a compactly supported $(n-1)$-form on the open set $x(U)$; extend it by zero to a compactly supported smooth form on all of $\mathbb{R}^n$ (interior chart) or $\mathbb{H}^n$ (boundary chart), which is legitimate because $\operatorname{supp}\widetilde{\eta}$ is a compact subset of the open set $x(U)$. Then $\int_{x(U)} d\widetilde{\eta} = \int_{\mathbb{R}^n} d\widetilde{\eta}$ or $\int_{\mathbb{H}^n} d\widetilde{\eta}$ accordingly.

For the boundary side, if (U,x) is an interior chart then $U \cap \partial M = \varnothing$, so $\iota^*\eta = 0$ and $\int_{\partial M}\eta = 0$. If (U,x) is a boundary chart, x restricts to an oriented chart $\bar{x}\colon U \cap \partial M \to \partial\mathbb{H}^n$ of the $(n-1)$-manifold ∂M: being a diffeomorphism of manifolds with boundary, x carries ∂M to $\partial\mathbb{H}^n$ and outward vectors to outward vectors, hence the induced boundary orientation of ∂M to that of $\partial\mathbb{H}^n$ (the construction $\iota_N\omega$ of proposition 12.6.7 is natural under such maps). Thus

$$\int_{\partial M}\eta = \int_{\partial\mathbb{H}^n}\widetilde{\eta}\,|_{\partial\mathbb{H}^n}$$

again by definition 13.1.1, now applied to ∂M. The theorem is reduced to two computations on the models $\mathbb{R}^n$ and $\mathbb{H}^n$.

Step 3: The Euclidean models. Let $\widetilde{\eta}$ be a compactly supported $(n-1)$-form on $\mathbb{H}^n$. In the standard coordinates every $(n-1)$-form is a combination of the basis forms obtained by omitting one factor, so

$$\widetilde{\eta} = \sum_{i=1}^{n} f_i\,dx^1\wedge\cdots\wedge\widehat{dx^i}\wedge\cdots\wedge dx^n$$

with $f_1,\ldots,f_n$ compactly supported smooth functions and $\widehat{\ }$ denoting omission. Applying the exterior derivative (theorem 12.5.1), only the $j=i$ term of $df_i = \sum_j \partial_j f_i\,dx^j$ is nonzero in the wedge, since a repeated differential vanishes; moving dx^i into its sorted position past the $i-1$ earlier factors contributes $(-1)^{i-1}$, so

$$d\widetilde{\eta} = \sum_{i=1}^{n}(-1)^{i-1}\frac{\partial f_i}{\partial x^i}\,dx^1\wedge\cdots\wedge dx^n \tag{13.2}$$

Integrating (13.2) over $\mathbb{H}^n$ with its standard orientation turns each term into an ordinary iterated integral. For $i<n$, the variable x^i ranges over all of $\mathbb{R}$, and the fundamental theorem of calculus gives

$$\int_{-\infty}^{\infty}\frac{\partial f_i}{\partial x^i}\,dx^i = 0$$

because f_i has compact support. Only $i=n$ contributes, and there x^n ranges over $[0,\infty)$, so

$$\int_0^{\infty}\frac{\partial f_n}{\partial x^n}\,dx^n = f_n(x',\infty) - f_n(x',0) = -f_n(x',0)$$

writing $x' = (x^1,\ldots,x^{n-1})$ and using compact support at the upper limit. Hence

$$\int_{\mathbb{H}^n} d\widetilde{\eta} = (-1)^{n-1}\int_{\mathbb{R}^{n-1}}\big(-f_n(x',0)\big)\,dx' = (-1)^n\int_{\mathbb{R}^{n-1}} f_n(x',0)\,dx' \tag{13.3}$$

Now compute the boundary side. On $\partial\mathbb{H}^n = \{x \mid x_n = 0\}$ the coordinate x^n is constant, so dx^n restricts to zero; every basis form of $\widetilde{\eta}$ except the $i=n$ one contains a factor dx^n and therefore restricts to zero, leaving

$$\widetilde{\eta}\,|_{\partial\mathbb{H}^n} = f_n(x',0)\,dx^1\wedge\cdots\wedge dx^{n-1}$$

The boundary carries the induced orientation $(-1)^n\,dx^1\wedge\cdots\wedge dx^{n-1}$ (proposition 12.6.7, computed on $\mathbb{H}^n$ in section 12.6), so integrating against it multiplies the ordinary $\mathbb{R}^{n-1}$-integral by the sign $(-1)^n$:

$$\int_{\partial\mathbb{H}^n}\widetilde{\eta} = (-1)^n\int_{\mathbb{R}^{n-1}} f_n(x',0)\,dx' \tag{13.4}$$

Comparing (13.3) and (13.4) gives $\int_{\mathbb{H}^n} d\widetilde{\eta} = \int_{\partial\mathbb{H}^n} \widetilde{\eta}$, the boundary-chart identity. For an interior chart the same computation applies on $\mathbb{R}^n$, where x^n too ranges over all of $\mathbb{R}$; the $i = n$ term then also integrates to zero by the fundamental theorem, so $\int_{\mathbb{R}^n} d\widetilde{\eta} = 0$, matching the empty boundary. By Step 2 each chart term satisfies $\int_M d(\rho_i\omega) = \int_{\partial M} \rho_i\omega$, and summing over i as in Step 1 yields $\int_M d\omega = \int_{\partial M} \omega$, completing the proof.

The identity reads d as the adjoint of ∂: differentiating a form and then integrating over the interior gives the same number as restricting to the boundary and integrating there. The proof localizes everything to one computation, the fundamental theorem of calculus on a half-space, and the outward-normal-first orientation is exactly the bookkeeping that makes the two signs in (13.3) and (13.4) agree. The classical integral theorems of vector analysis are the cases $n = 1, 2, 3$ of this one statement, each recovered in section 13.3 by choosing the manifold and the form.

The most immediate consequence is the boundaryless case, worth isolating because it is the seed of the comparison between closed and exact forms.

Corollary 13.2.2: Exact Forms on Closed Manifolds

Let M be a compact oriented n-manifold without boundary. Then $\int_M d\omega = 0$ for every $\omega \in \Omega^{n-1}(M)$. In particular a volume form on M is never exact.

***Proof*:**
Since M is compact, every form is compactly supported, and $\partial M = \varnothing$, so theorem 13.2.1 gives $\int_M d\omega = \int_{\partial M} \omega = 0$. If μ were a volume form with $\mu = d\eta$, then $\int_M \mu = \int_M d\eta = 0$; but μ is positively oriented (definition 12.6.5), so its integral over the positively oriented M is strictly positive, a contradiction. Hence no volume form is exact, as we sought to show.

A compact oriented manifold without boundary is a *closed* manifold. On such a manifold the map $\omega \mapsto \int_M \omega$ annihilates every exact top form yet is nonzero on a volume form: integration sees the difference between a closed form and an exact one. Which closed forms this distinguishes, and how the resulting numbers encode the shape of M, is the question of periods taken up in section 13.4 and answered by de Rham cohomology in chapter 14.

Stokes' theorem was stated for a smooth boundary, but the regions met in practice, a rectangle, a spherical triangle, a solid cone, have edges and vertices where the boundary is only piecewise smooth. The identity survives, because the corner set is too small to contribute to either integral.

Definition 13.2.3: Region with Corners

A *region with corners* in a smooth n-manifold M is a compact subset D such that each $p \in D$ has a chart (U, x) of M with $x(p) = 0$ in which

$$x(D \cap U) = x(U) \cap \left\{ y \in \mathbb{R}^n \;\middle|\; y^i \geq 0 \text{ for } i \in I_p \right\}$$

for some subset $I_p \subseteq \{1, \ldots, n\}$. Points with $|I_p| = 0$ are interior. Those with $|I_p| = 1$ make up the smooth *faces*, the $(n-1)$-dimensional part of the boundary ∂D, and those with $|I_p| \geq 2$ form the *corner set*, of dimension at most $n - 2$.

Theorem 13.2.4: Stokes' Theorem on a Region with Corners

Let D be a region with corners in an oriented smooth n-manifold M, $n \geq 1$, and let $\omega \in \Omega^{n-1}(M)$. Orient each face of D by the outward-normal-first rule (proposition 12.6.7). Then

$$\int_D d\omega = \int_{\partial D} \omega$$

the right-hand side being the sum over the faces of the integrals of $\iota^*\omega$ along their inclusions, and the corner set contributes nothing.

***Proof*:**
The reduction of theorem 13.2.1 applies unchanged: a partition of unity subordinate to a cover by charts of three kinds (interior, one-face, and corner) writes $\omega = \sum_i \rho_i \omega$ with each $\rho_i\omega$ supported in a single chart, and $\mathbb{R}$-linearity of d and of the integral reduces the claim to one chart. Interior and one-face charts are the two models already computed in theorem 13.2.1. The new case is a corner chart, where $x(D \cap U)$ is the orthant $\mathbb{H}^n_I = \{y \mid y^i \geq 0,\ i \in I\}$ for a set I with $|I| \geq 2$.

Transport $\eta = \rho_i\omega$ to $\widetilde{\eta} = (x^{-1})^*\eta$, a compactly supported $(n-1)$-form on $\mathbb{H}^n_I$, written $\widetilde{\eta} = \sum_j f_j\, dx^1 \wedge \cdots \wedge \widehat{dx^j} \wedge \cdots \wedge dx^n$. As in (13.2), $d\widetilde{\eta} = \sum_j (-1)^{j-1} \partial_j f_j\, dx^1 \wedge \cdots \wedge dx^n$. Integrating over $\mathbb{H}^n_I$ term by term, the fundamental theorem of calculus kills the j-th term when $j \notin I$, since then x^j ranges over all of $\mathbb{R}$ and f_j has compact support. For $j \in I$ the variable x^j runs over $[0,\infty)$ and $\int_0^\infty \partial_j f_j\, dx^j = -f_j|_{x^j=0}$. Hence

$$\int_{\mathbb{H}^n_I} d\widetilde{\eta} = \sum_{j\in I} (-1)^j \int_{\{x^j=0\}} f_j|_{x^j=0}\, dx^1 \cdots \widehat{dx^j} \cdots dx^n$$

The boundary faces meeting the chart are $F_j = \{x^j = 0\} \cap \mathbb{H}^n_I$ for $j \in I$. On F_j the coordinate x^j is constant, so every basis form of $\widetilde{\eta}$ except the j-th restricts to zero, leaving $\widetilde{\eta}|_{F_j} = f_j\, dx^1 \wedge \cdots \wedge \widehat{dx^j} \wedge \cdots \wedge dx^n$. The outward normal along F_j is $-\partial_j$, so the induced orientation (proposition 12.6.7) is $(-1)^j\, dx^1 \wedge \cdots \wedge \widehat{dx^j} \wedge \cdots \wedge dx^n$, and integrating against it reproduces the j-th term above. Summing over $j \in I$ gives $\int_{\mathbb{H}^n_I} d\widetilde{\eta} = \sum_{j\in I} \int_{F_j} \widetilde{\eta}$. The lower strata $\{x^j = x^{j'} = 0\}$ have dimension $n-2$ and are null for the $(n-1)$-dimensional face integrals, so they contribute to neither side. Summing the partition of unity as in theorem 13.2.1 yields $\int_D d\omega = \int_{\partial D} \omega$, completing the proof.

Example 13.2.5: Stokes on the Annulus
Let $M = \{(x,y) \in \mathbb{R}^2 \mid 1 \leq x^2 + y^2 \leq 4\}$ be the closed annulus, an oriented 2-manifold with boundary, oriented by $dx \wedge dy$. Take the 1-form

$$\omega = -y\, dx + x\, dy$$

Its exterior derivative is $d\omega = dx \wedge dy - (-dx \wedge dy) = 2\, dx \wedge dy$, twice the area form, so

$$\int_M d\omega = 2\int_M dx \wedge dy = 2 \cdot \mathrm{Area}(M) = 2(\pi \cdot 2^2 - \pi \cdot 1^2) = 6\pi$$

The boundary is the outer circle C_2 of radius 2 and the inner circle C_1 of radius 1. The outward direction from M points away from the origin along C_2 and toward the origin along C_1, so by the outward-normal-first rule (proposition 12.6.7) C_2 acquires the counterclockwise orientation and C_1 the clockwise one. Parametrizing a circle of radius R by $t \mapsto (R\cos t, R\sin t)$ counterclockwise gives $\omega = R^2\,dt$, so $\int_{\text{ccw circle of radius } R} \omega = 2\pi R^2$. Therefore

$$\int_{\partial M} \omega = \int_{C_2} \omega - \int_{C_1} \omega = 2\pi \cdot 4 - 2\pi \cdot 1 = 6\pi$$

the minus sign recording the clockwise orientation of the inner circle. The two sides agree, as Stokes' theorem requires.

For the running example S^n, the boundaryless corollary already carries information. The sphere is a closed oriented manifold, so $\int_{S^2} d\omega = 0$ for every $\omega \in \Omega^1(S^2)$; yet the area form $\sigma = \iota_N(dx \wedge dy \wedge dz)$ of example 12.6.4 is a volume form, hence by corollary 13.2.2 it is not exact. A closed 2-form on S^2 can thus fail to be the derivative of any 1-form, and integration is what detects the failure. This is the first appearance of the gap between closed and exact forms as something measured by an integral.

Exercise 13.2.1

1. Verify Stokes' theorem for the 1-form $\omega = x\,dy$ on the closed unit disk $D^2 = \{(x, y) \mid x^2 + y^2 \leq 1\}$, oriented by $dx \wedge dy$: compute $\int_{D^2} d\omega$ directly and $\int_{\partial D^2} \omega$ over the boundary circle with its induced orientation, and check that they agree.

2. Let $f \in C^\infty(\mathbb{H}^1)$ be compactly supported on $\mathbb{H}^1 = [0, \infty)$. Using the fundamental theorem of calculus, show $\int_{\mathbb{H}^1} df = -f(0)$. Reconcile the sign with the induced boundary orientation of the point $\{0\} = \partial\mathbb{H}^1$, which is $(-1)^1 = -1$ (a negatively oriented point), so that Stokes' theorem $\int_{\mathbb{H}^1} df = \int_{\partial\mathbb{H}^1} f$ holds.

3. On $M = [0, \infty)$ take the constant function $f \equiv 1$. Compute $\int_M df$ and $f(0)$, and explain why $\int_M df = \int_{\partial M} f$ fails. Identify precisely which hypothesis of theorem 13.2.1 is violated.

4. Let M be a compact oriented n-manifold without boundary and let $\alpha, \beta \in \Omega^{n-1}(M)$ satisfy $d\alpha = d\beta$. Show that $\int_M d\alpha = 0$, and deduce that any two closed n-forms differing by an exact form have the same integral over M.

5. Let $M = S^1 \times [0, 1]$ be the cylinder, with the product orientation. Describe the induced boundary orientations on the two boundary circles $S^1 \times \{0\}$ and $S^1 \times \{1\}$, and explain why for a 1-form ω the boundary integral $\int_{\partial M} \omega$ is a difference of the two circle integrals rather than a sum.

13.3 The Classical Integral Theorems

Stokes' theorem is one identity in every dimension; the integral theorems of classical vector calculus are what it becomes on manifolds of dimension one, two, and three, once forms are translated back into functions and vector fields. This section carries out that translation. Each classical theorem is theorem 13.2.1 applied to a specific form on a specific low-dimensional region, and the bridge between the two languages is a single computation on $\mathbb{R}^3$: the exterior derivative is the gradient, the curl, and the divergence in turn, one operator named three times according to the degree it acts on.

The lowest-dimensional case fixes the pattern and, more importantly, the sign convention. On an oriented 0-manifold, a finite set of points each tagged $+$ or $-$, the integral of a 0-form (a function) is the signed sum of its values. With that reading, the fundamental theorem of calculus is Stokes' theorem on a closed interval.

Corollary 13.3.1: The Fundamental Theorem of Calculus

Let $f\colon [a,b] \to \mathbb{R}$ be smooth. Orient $[a,b]$ by dx, and give the boundary $\partial[a,b] = \{a,b\}$ its induced orientation (proposition 12.6.7). Then

$$\int_a^b f'(x)\,dx = f(b) - f(a)$$

***Proof*:**
The interval $[a,b]$ is a compact oriented 1-manifold with boundary (definition 1.1.16), and $f \in \Omega^0([a,b])$ is a compactly supported 0-form. Its exterior derivative is $df = f'(x)\,dx$, so theorem 13.2.1 gives

$$\int_{[a,b]} f'(x)\,dx = \int_{[a,b]} df = \int_{\partial[a,b]} f$$

the left side being the integral of the top form $f'(x)\,dx$ over the identity chart, that is the ordinary integral $\int_a^b f'(x)\,dx$ (definition 13.1.4). For the right side, the induced boundary orientation tags each endpoint by whether an outward vector is positively oriented in $T_p[a,b] = \mathbb{R}$. At b the outward direction is $+\partial_x$, positively oriented, so b carries the sign $+$; at a the outward direction is $-\partial_x$, negatively oriented, so a carries the sign $-$. The signed sum of f over the oriented boundary is therefore $f(b) - f(a)$, as we sought to show.

The minus sign in the fundamental theorem is thus a statement about orientation: the two ends of an interval receive opposite signs because "outward" points in opposite directions at the two ends. Every classical theorem below inherits its signs the same way, from the induced boundary orientation of proposition 12.6.7, and this is the one place a reader must attend to conventions.

To reach dimensions two and three in the language those theorems are usually stated in, a vector field on $\mathbb{R}^3$ must be turned into a form and back. Two identifications do this: one turns a vector field into a 1-form using the Euclidean metric, and the other into a 2-form by contraction with the volume form, which needs no metric at all. On $\mathbb{R}^3$ with its standard metric both are explicit relabelings of components, which is all that is needed here; the coordinate-free version of the first, on a general metric, belongs to Riemannian geometry. For a vector field $X = P\,\partial_x + Q\,\partial_y + R\,\partial_z$ and the standard volume form $\mu = dx \wedge dy \wedge dz$, set

$$X^\flat = P\,dx + Q\,dy + R\,dz \in \Omega^1(\mathbb{R}^3), \qquad \iota_X\mu = P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy \in \Omega^2(\mathbb{R}^3)$$

the first the *flat* of X (lower the index), the second its contraction into the volume form (definition 12.4.7); a function f is identified with the 3-form $f\mu$. These match Ω^0 with vector-field-valued data one way, Ω^1 and Ω^2 with vector fields, and Ω^3 with functions. Under them the exterior derivative becomes the three operators of vector analysis.

Proposition 13.3.2: Grad, Curl, and Divergence as the Exterior Derivative

On $\mathbb{R}^3$ with $\mu = dx \wedge dy \wedge dz$, let $f \in C^\infty(\mathbb{R}^3)$ and $X = P\,\partial_x + Q\,\partial_y + R\,\partial_z$. Then

1. $df = (\operatorname{grad} f)^\flat$, where $\operatorname{grad} f = \big(\partial_x f, \partial_y f, \partial_z f\big)$;
2. $d(X^\flat) = \iota_{\operatorname{curl} X}\,\mu$, where $\operatorname{curl} X = \big(\partial_y R - \partial_z Q,\ \partial_z P - \partial_x R,\ \partial_x Q - \partial_y P\big)$;
3. $d(\iota_X \mu) = (\operatorname{div} X)\,\mu$, where $\operatorname{div} X = \partial_x P + \partial_y Q + \partial_z R$.

***Proof*:**
Each part is a direct computation.

1. By definition of the exterior derivative on functions, $df = \partial_x f\,dx + \partial_y f\,dy + \partial_z f\,dz$, which is exactly $(\operatorname{grad} f)^\flat$.
2. Expanding $d(X^\flat) = dP \wedge dx + dQ \wedge dy + dR \wedge dz$ and keeping only the surviving wedges,
$$dP \wedge dx = -\partial_y P\,dx \wedge dy + \partial_z P\,dz \wedge dx, \quad dQ \wedge dy = \partial_x Q\,dx \wedge dy - \partial_z Q\,dy \wedge dz,$$
$$dR \wedge dz = -\partial_x R\,dz \wedge dx + \partial_y R\,dy \wedge dz$$
Collecting the coefficients of $dy \wedge dz$, $dz \wedge dx$, and $dx \wedge dy$ gives
$$d(X^\flat) = (\partial_y R - \partial_z Q)\,dy \wedge dz + (\partial_z P - \partial_x R)\,dz \wedge dx + (\partial_x Q - \partial_y P)\,dx \wedge dy$$
which is $\iota_{\operatorname{curl} X}\mu$ by the display defining the contraction into μ.
3. With $\iota_X \mu = P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy$, each term contributes only through the derivative in its missing direction:
$$d(\iota_X \mu) = \partial_x P\,dx \wedge dy \wedge dz + \partial_y Q\,dy \wedge dz \wedge dx + \partial_z R\,dz \wedge dx \wedge dy = (\partial_x P + \partial_y Q + \partial_z R)\,\mu$$
since each 3-form on the right equals μ after an even permutation of the factors, completing the proof.

One operator, three names: d on Ω^0 is the gradient, on Ω^1 the curl, on Ω^2 the divergence. The nilpotency $d^2 = 0$ is then two identities of vector calculus at once. Applying d twice from a function reads $\operatorname{curl}(\operatorname{grad} f) = 0$, and from a 1-form $\operatorname{div}(\operatorname{curl} X) = 0$: the two ways a field can be built by differentiation are exactly the two ways it is annihilated by the next derivative.

Example 13.3.3: The Rotation Field

Take $X = -y\,\partial_x + x\,\partial_y + z\,\partial_z$, so $P = -y$, $Q = x$, $R = z$. Its divergence is $\operatorname{div} X = \partial_x(-y) + \partial_y(x) + \partial_z(z) = 1$. Its curl is $\operatorname{curl} X = (\partial_y z - \partial_z x,\ \partial_z(-y) - \partial_x z,\ \partial_x x - \partial_y(-y)) = (0, 0, 2)$. Checking part (2) against the definitions, $X^\flat = -y\,dx + x\,dy + z\,dz$ has
$$d(X^\flat) = -dy \wedge dx + dx \wedge dy = 2\,dx \wedge dy = \iota_{(0,0,2)}\mu$$
and $\iota_{(0,0,2)}\mu = 2\,dx \wedge dy$ agrees, confirming $d(X^\flat) = \iota_{\operatorname{curl} X}\mu$.

In dimension two the region is a piece of the plane, and Stokes pairs a line integral around its boundary with a double integral over its interior. This is Green's theorem, and it is nothing more than theorem 13.2.1 for a 1-form on a plane region.

Theorem 13.3.4: Green's Theorem

Let $D \subseteq \mathbb{R}^2$ be a compact 2-manifold with boundary, oriented by $dx \wedge dy$, and give ∂D the induced boundary orientation. For smooth functions P, Q on a neighbourhood of D,

$$\oint_{\partial D} (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy$$

with ∂D traversed counterclockwise.

***Proof*:**
Let $\omega = P\,dx + Q\,dy \in \Omega^1(D)$; it is compactly supported because D is compact. Its exterior derivative is

$$d\omega = dP \wedge dx + dQ \wedge dy = \partial_y P\,dy \wedge dx + \partial_x Q\,dx \wedge dy = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \wedge dy$$

By theorem 13.2.1, $\int_D d\omega = \int_{\partial D} \omega$. The left side is the integral of the top form $(\partial_x Q - \partial_y P)\,dx \wedge dy$ over D with its standard orientation, which by definition 13.1.4 is the ordinary double integral $\iint_D (\partial_x Q - \partial_y P)\,dx\,dy$. It remains to identify the induced boundary orientation with the counterclockwise sense. At $p \in \partial D$ with outward vector N, a tangent vector v is positively oriented for the boundary iff (N, v) is a positively oriented basis of $\mathbb{R}^2$, that is iff $\det[\,N\ v\,] > 0$; this holds exactly when v is N rotated by a quarter turn counterclockwise, which is the counterclockwise direction along ∂D. Hence $\int_{\partial D} \omega = \oint_{\partial D} (P\,dx + Q\,dy)$ with the counterclockwise traversal, as we sought to show.

Green's theorem reads a line integral as an area integral of an infinitesimal circulation $\partial_x Q - \partial_y P$. Choosing the 1-form well turns it into a formula for area itself: with $P = -y/2$ and $Q = x/2$ the integrand becomes $\partial_x Q - \partial_y P = 1$, so

$$\operatorname{Area}(D) = \iint_D dx\,dy = \frac{1}{2} \oint_{\partial D} (x\,dy - y\,dx) \tag{13.5}$$

the enclosed area computed from the boundary alone. A planimeter traces this integral mechanically.

Example 13.3.5: The Area of an Ellipse
Let D be the region bounded by the ellipse $x = a\cos t$, $y = b\sin t$, $t \in [0, 2\pi]$, with $a, b > 0$. Along the boundary $dx = -a\sin t\,dt$ and $dy = b\cos t\,dt$, so

$$x\,dy - y\,dx = a\cos t \cdot b\cos t\,dt - b\sin t \cdot (-a\sin t)\,dt = ab(\cos^2 t + \sin^2 t)\,dt = ab\,dt$$

and (13.5) gives $\operatorname{Area}(D) = \frac{1}{2}\int_0^{2\pi} ab\,dt = \pi ab$, the expected value, recovered from the boundary curve without integrating over the interior.

In dimension n the natural pairing is between the spreading of a vector field throughout a region and the flux of the field out through the boundary. Stating it needs a notion of how fast a field spreads, and that notion is available without a metric: a field distorts any chosen volume form at a rate that measures its divergence.

Definition 13.3.6: Divergence

Let μ be a volume form on M (definition 12.6.5) and $X \in \mathfrak{X}(M)$. Since $d(\iota_X \mu)$ is an n-form and μ is nowhere zero, there is a unique function $\mathrm{div}_\mu X \in C^\infty(M)$ with

$$d(\iota_X \mu) = (\mathrm{div}_\mu X)\, \mu$$

called the *divergence* of X with respect to μ.

The definition is well posed because a nowhere-zero top form trivializes the line bundle of n-forms: every n-form is a unique smooth multiple of μ, and $\mathrm{div}_\mu X$ is that multiple. On $\mathbb{R}^n$ with $\mu = dx^1 \wedge \cdots \wedge dx^n$ and $X = \sum_i X^i\, \partial_{x^i}$, the same computation as part (3) of proposition 13.3.2 in each coordinate gives $d(\iota_X \mu) = \big(\sum_i \partial_{x^i} X^i\big)\mu$, so $\mathrm{div}_\mu X = \sum_i \partial_{x^i} X^i$ is the ordinary divergence. The name is justified by Cartan's formula (theorem 12.5.6): since μ is closed, $\mathcal{L}_X \mu = d(\iota_X \mu) + \iota_X(d\mu) = (\mathrm{div}_\mu X)\, \mu$, so the divergence is the rate at which the flow of X expands the volume μ, positive where the flow spreads volume out and negative where it draws it in.

Theorem 13.3.7: The Divergence Theorem

Let M be a compact oriented n-manifold with boundary, μ a volume form, and $X \in \mathfrak{X}(M)$; give ∂M the induced boundary orientation. Then

$$\int_M (\mathrm{div}_\mu X)\, \mu = \int_{\partial M} \iota_X \mu$$

***Proof*:**
The contraction $\iota_X \mu$ is an $(n-1)$-form on M, compactly supported because M is compact. Applying theorem 13.2.1 to it,

$$\int_{\partial M} \iota_X \mu = \int_M d(\iota_X \mu) = \int_M (\mathrm{div}_\mu X)\, \mu$$

the last equality by definition 13.3.6, completing the proof.

The right side is a flux. On $\mathbb{R}^3$, writing N for the outward unit normal along ∂M and dA for the area element of the boundary surface, the metric identifies $\iota_X \mu|_{\partial M} = (X \cdot N)\, dA$, so the theorem takes the familiar form $\iiint_M (\mathrm{div} X)\, dV = \iint_{\partial M} (X \cdot N)\, dA$: the total divergence inside equals the net outflow across the boundary. The identification of the flux 2-form with $(X \cdot N)\, dA$ uses the Euclidean metric and belongs to Riemannian geometry; the form-level statement above needs no metric at all, only the volume form.

Example 13.3.8: Flux of the Radial Field

Let $M = \overline{B}^3$ be the closed unit ball with $\mu = dx \wedge dy \wedge dz$, and $X = x\,\partial_x + y\,\partial_y + z\,\partial_z$ the outward radial field. Its divergence is $\operatorname{div} X = 1 + 1 + 1 = 3$, so the left side is

$$\int_{\overline{B}^3} 3\,\mu = 3 \cdot \operatorname{vol}(B^3) = 3 \cdot \frac{4}{3}\pi = 4\pi$$

For the right side, $\partial M = S^2$ and on the unit sphere the outward unit normal is the position vector, so $X \cdot N = |X|^2 = 1$ there; the flux is $\iint_{S^2} 1\,dA = \operatorname{Area}(S^2) = 4\pi$. The two agree, as the theorem requires.

The remaining classical theorem lives on a two-dimensional surface sitting inside $\mathbb{R}^3$. It equates the circulation of a field around the boundary curve of the surface with the flux of the field's curl through the surface, and it is theorem 13.2.1 applied to the 1-form $X^\flat$ on the surface, together with part (2) of the dictionary.

Theorem 13.3.9: The Classical (Kelvin-)Stokes Theorem

Let $S \subseteq \mathbb{R}^3$ be a compact oriented 2-manifold with boundary, with inclusion $j\colon S \hookrightarrow \mathbb{R}^3$, and give ∂S the induced boundary orientation. For a smooth vector field X on a neighbourhood of S,

$$\oint_{\partial S} j^*(X^\flat) = \int_S j^*\big(\iota_{\operatorname{curl} X}\,\mu\big)$$

Classically, with $d\mathbf{r}$ the oriented line element of ∂S and $d\mathbf{S}$ the oriented area element of S, this reads $\oint_{\partial S} X \cdot d\mathbf{r} = \iint_S (\operatorname{curl} X) \cdot d\mathbf{S}$.

***Proof*:**

Let $\alpha = X^\flat \in \Omega^1(\mathbb{R}^3)$, and let $j^*\alpha \in \Omega^1(S)$ be its pullback to S, compactly supported because S is compact. By theorem 13.2.1 on the oriented 2-manifold with boundary S,

$$\int_S d(j^*\alpha) = \int_{\partial S} j^*\alpha$$

Naturality of the exterior derivative (proposition 12.5.4) commutes d with the pullback, so $d(j^*\alpha) = j^*(d\alpha)$. Part (2) of proposition 13.3.2 identifies $d\alpha = d(X^\flat) = \iota_{\operatorname{curl} X}\mu$. Substituting,

$$\int_S j^*\big(\iota_{\operatorname{curl} X}\mu\big) = \int_S j^*(d\alpha) = \int_S d(j^*\alpha) = \int_{\partial S} j^*(X^\flat)$$

which is the claimed equality, as we sought to show.

The circulation of a field around a closed loop equals the flux of its curl through any surface the loop bounds. Two consequences are immediate from the dictionary. A gradient field $X = \operatorname{grad} f$ has $X^\flat = df$, so its circulation around any bounding loop is $\oint_{\partial S} df = \int_S d(df) = 0$: gradient fields are conservative, the vector-calculus face of $d^2 = 0$. And the flux of $\operatorname{curl} X$ through S depends only on the boundary ∂S, since two surfaces with the same boundary differ by a closed surface, over which the flux of a curl vanishes by the divergence theorem and $\operatorname{div}(\operatorname{curl} X) = 0$.

Example 13.3.10: Circulation of the Rotation Field

Let S be the upper unit hemisphere $\{x^2 + y^2 + z^2 = 1,\ z \geq 0\}$, oriented by the upward normal, so ∂S is the unit circle in the xy-plane traversed counterclockwise. Take $X = -y\,\partial_x + x\,\partial_y$, whose curl is $(0, 0, 2)$ by example 13.3.3. The circulation is

$$\oint_{\partial S} j^*(X^\flat) = \oint_{\partial S} (-y\,dx + x\,dy) = \int_0^{2\pi} \big((-\sin t)(-\sin t) + (\cos t)(\cos t)\big)\,dt = \int_0^{2\pi} dt = 2\pi$$

For the flux of the curl, $\iota_{(0,0,2)}\mu = 2\,dx \wedge dy$, and its pullback to the hemisphere integrates as $\int_S 2\,dx \wedge dy = 2\iint_D dx\,dy$, where D is the image of S under the orientation-preserving projection $(x, y, z) \mapsto (x, y)$ onto the unit disk. That integral is $2 \cdot \mathrm{Area}(D) = 2\pi$, matching the circulation.

The four theorems are one theorem. Each is theorem 13.2.1 on a compact oriented manifold with boundary, differing only in the dimension of the manifold and in which entry of the grad-curl-div dictionary translates the exterior derivative into classical notation.

Exercise 13.3.1

1. Use Green's theorem (theorem 13.3.4) to evaluate $\oint_C (-y^3\,dx + x^3\,dy)$, where C is the unit circle $x^2 + y^2 = 1$ traversed counterclockwise.
2. Using the area formula (13.5), compute the area enclosed by the astroid $x = \cos^3 t$, $y = \sin^3 t$, $t \in [0, 2\pi]$.
3. Use the divergence theorem (theorem 13.3.7) to compute the flux $\int_{\partial M} \iota_X \mu$ of the field $X = x\,\partial_x + 2y\,\partial_y + 3z\,\partial_z$ out of the unit cube $M = [0, 1]^3$.
4. Compute the outward flux of $X = x^3\,\partial_x + y^3\,\partial_y + z^3\,\partial_z$ across the unit sphere S^2 by applying theorem 13.3.7 on the closed unit ball.
5. Show that for every $f \in C^\infty(\mathbb{R}^3)$ the field $\operatorname{grad} f$ has zero circulation around the boundary of every compact oriented surface $S \subseteq \mathbb{R}^3$. Identify the entry of proposition 13.3.2 and the property of d that make this automatic.
6. Show that $\operatorname{div}(\operatorname{curl} X) = 0$ for every smooth vector field X on $\mathbb{R}^3$, arguing at the level of forms rather than by expanding components.

13.4 Closed Forms, Periods, and the de Rham Question

Stokes' theorem couples the exterior derivative to the boundary operator, and what the coupling leaves behind is a pairing. A closed form integrated over a compact oriented cycle produces a number that is insensitive to the two moves Stokes controls: adding an exact form to the integrand, and replacing the cycle by another that differs from it by a boundary. This section isolates that number, the period, proves the two invariances that make it well defined on classes rather than on representatives, and reads the angle form on the punctured plane as a closed form whose nonzero period certifies that it is not exact. The gap between closed and exact that the period detects is the object de Rham cohomology measures, and the question of whether the periods determine a closed form up to an exact one is posed here and handed to chapter 14.

The setting is a smooth k-form on M integrated over a k-dimensional piece of M. The piece is packaged as a smooth map from a model manifold, which is the flexible notion that homotopies and disjoint unions preserve.

Definition 13.4.1: Period of a Closed Form

A *smooth k-cycle* in M is a smooth map $c\colon Z \to M$ from a compact oriented k-manifold Z without boundary. For a k-form $\omega \in \Omega^k(M)$ the *period* of ω over c is

$$\int_c \omega = \int_Z c^*\omega$$

the integral over the oriented manifold Z (definition 13.1.4) of the pullback $c^*\omega \in \Omega^k(Z)$, a top-degree form on Z and hence compactly supported since Z is compact. The period is of principal interest when ω is *closed* (definition 12.5.8).

The pullback $c^*\omega$ is a k-form on a k-manifold, so it is a top form and the integral of section 13.1 applies with no further hypothesis than compactness of Z. When Z is an embedded compact oriented submanifold and c its inclusion, the period is the integral of the restricted form $\omega|_Z$; the map formulation only adds the freedom to let c be any smooth image, which is what the invariance results below require. A period is $\mathbb{R}$-linear in the form, and $\int_{c \sqcup c'} \omega = \int_c \omega + \int_{c'} \omega$ over a disjoint union of cycles, both directly from the corresponding properties of the integral; reversing the orientation of Z reverses the sign (proposition 13.1.7).

The first example is the one that recurs throughout: the angle form on the punctured plane, whose period around the puncture is where the topology of $\mathbb{R}^2 \setminus \{0\}$ becomes a number.

Example 13.4.2: The Angle Form

On $M = \mathbb{R}^2 \setminus \{0\}$ consider the 1-form

$$\omega = \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

of section 12.2, which is closed: a direct computation gives $d\omega = 0$ on the punctured plane. Let $c\colon [0, 2\pi] \to M$ trace the unit circle, $c(t) = (\cos t, \sin t)$, with the standard counterclockwise orientation, so c is a smooth 1-cycle (the circle S^1 is compact, oriented, and without boundary). Pulling ω back along c, with $x = \cos t$, $y = \sin t$, $x^2 + y^2 = 1$, and $dx = -\sin t\,dt$, $dy = \cos t\,dt$,

$$c^*\omega = \frac{-\sin t\,(-\sin t\,dt) + \cos t\,(\cos t\,dt)}{1} = (\sin^2 t + \cos^2 t)\,dt = dt$$

so the period is

$$\int_c \omega = \int_0^{2\pi} dt = 2\pi$$

The form is the angular coordinate: on any half-plane where a smooth branch of θ exists, $\omega = d\theta$, yet no such branch is global, and the period 2π measures exactly the jump accumulated on going once around the origin. That the period is nonzero is the fact that distinguishes ω from every exact form, as the next result makes precise.

Whether the period $\int_c \omega$ carries information depends on how much of it is unaffected by the two degrees of freedom in the data. The integrand ω may be changed by any exact form and the cycle c by any boundary; Stokes' theorem says the period ignores both. This is the single mechanism of the section, and both halves are one application of theorem 13.2.1.

Proposition 13.4.3: Exact Forms and Bounding Cycles

Let M be a smooth manifold and $c\colon Z \to M$ a smooth k-cycle, with $\omega \in \Omega^k(M)$.

1. If ω is exact, then $\int_c \omega = 0$.
2. If ω is closed and c *bounds* in M (that is, c extends to a smooth map $b\colon W \to M$ from a compact oriented $(k+1)$-manifold with boundary W whose oriented boundary is $\partial W = Z$ and whose restriction is $b|_{\partial W} = c$), then $\int_c \omega = 0$.

***Proof*:**
Both parts pull the form back and apply Stokes' theorem (theorem 13.2.1), once on Z and once on W; naturality of the exterior derivative (proposition 12.5.4) converts d before pullback into d after it.

Part (1). Write $\omega = d\eta$ with $\eta \in \Omega^{k-1}(M)$. Pullback commutes with the exterior derivative, so

$$c^*\omega = c^*(d\eta) = d(c^*\eta)$$

Thus $c^*\omega$ is an exact top form on Z. Since Z is compact and has no boundary, Stokes' theorem gives

$$\int_c \omega = \int_Z d(c^*\eta) = \int_{\partial Z} c^*\eta = 0$$

the last equality because $\partial Z = \emptyset$, so the boundary integral is over the empty set.

Part (2). Now $d\omega = 0$, and $b\colon W \to M$ extends c over the compact $(k+1)$-manifold W with $\partial W = Z$ and $b|_{\partial W} = c$. The pullback $b^*\omega \in \Omega^k(W)$ is compactly supported, so Stokes' theorem on W applies:

$$\int_{\partial W} b^*\omega = \int_W d(b^*\omega) = \int_W b^*(d\omega) = \int_W b^*0 = 0$$

using $d(b^*\omega) = b^*(d\omega)$ and $d\omega = 0$. On the boundary, $b|_{\partial W} = c$ carries the induced boundary orientation (proposition 12.6.7) of $\partial W = Z$, so $\int_{\partial W} b^*\omega = \int_Z c^*\omega = \int_c \omega$. Combining, $\int_c \omega = 0$, completing the proof.

The two parts are the two sides of a single duality. Part (1) says the period depends on ω only through its class modulo exact forms: replacing ω by $\omega + d\eta$ changes no period. Part (2) says the period depends on c only through its class modulo boundaries: replacing c by a cycle that differs from it by something that bounds changes no period. Together they make $(\omega, c) \mapsto \int_c \omega$ a bilinear pairing between closed forms modulo exact ones and cycles modulo boundaries. The first quotient is de Rham cohomology (chapter 14); the second is an oriented bordism group, which maps to but does not coincide with the singular homology of [11, Singular Homology Group] (the precise duality between de Rham cohomology and singular homology is de Rham's theorem, taken up in chapter 14). The period is the value of that pairing on a pair of classes, and it is the first invariant of this book that reads topology off the differential calculus of forms.

Part (1) already settles the angle form. Its period over the unit circle is $2\pi \neq 0$, so by the contrapositive of proposition 13.4.3(1) the closed form ω of example 13.4.2 is *not* exact: no globally defined function has ω as its differential, though every point has a local angular potential. The gap between closed and exact, recorded abstractly in definition 12.5.8, is here a measured quantity. Part (2) reads the same number as a statement about the cycle: were S^1 the boundary of a compact oriented surface inside $\mathbb{R}^2 \setminus \{0\}$, its period against the closed form ω would vanish; since it does not, the unit circle bounds no such surface in the punctured plane. Integration detects at once that the form is not exact and that the circle is not a boundary.

Part (2) has an immediate consequence for cycles that are deformed rather than replaced. A homotopy sweeps out a cylinder whose two ends are the deformed cycles, and that cylinder exhibits their difference as a boundary.

Corollary 13.4.4: Homotopy Invariance of Periods

Let $\omega \in \Omega^k(M)$ be closed and let $c_0, c_1 \colon Z \to M$ be smoothly homotopic k-cycles, meaning there is a smooth map $H \colon Z \times [0,1] \to M$ with $H(\cdot, 0) = c_0$ and $H(\cdot, 1) = c_1$. Then $\int_{c_0} \omega = \int_{c_1} \omega$. In particular an orientation-preserving diffeomorphism $F \colon M \to N$ satisfies $\int_{F \circ c} \omega = \int_c F^*\omega$ for every closed ω on N, so periods are diffeomorphism invariants.

***Proof*:**
The cylinder $W = Z \times [0,1]$ is a compact oriented $(k+1)$-manifold with boundary; orient it with the product orientation in which a positively oriented frame is $(\partial_t, e_1, \ldots, e_k)$ whenever $(e_1, \ldots, e_k)$ is positively oriented for Z, where t is the coordinate on $[0,1]$. At the top face $Z \times \{1\}$ the outward vector is $+\partial_t$, so by the outward-first convention (proposition 12.6.7) the boundary orientation of $Z \times \{1\}$ agrees with that of Z; at the bottom face $Z \times \{0\}$ the outward vector is $-\partial_t$, so the boundary orientation is the reverse of Z. Thus, as oriented manifolds, $\partial W = (Z \times \{1\}) \sqcup -(Z \times \{0\})$, with H restricting to c_1 on the first and c_0 on the second. Since ω is closed, $d(H^*\omega) = H^*(d\omega) = 0$ by proposition 12.5.4, and $H^*\omega$ is compactly supported on the compact W. Stokes' theorem (theorem 13.2.1) gives

$$0 = \int_W d(H^*\omega) = \int_{\partial W} H^*\omega = \int_Z c_1^*\omega - \int_Z c_0^*\omega$$

the sign on the second term coming from the reversed orientation of the bottom face. Hence $\int_{c_0} \omega = \int_{c_1} \omega$. For the diffeomorphism statement, functoriality of pullback (proposition 12.4.11) gives $(F \circ c)^*\omega = c^*(F^*\omega)$, so $\int_{F \circ c} \omega = \int_Z c^*(F^*\omega) = \int_c F^*\omega$, as we sought to show.

Homotopy invariance is the concrete reason periods deserve to be called invariants: a cycle may be slid and stretched freely, and as long as its endpoints of deformation stay closed cycles, the period against any closed form is unchanged. The proof used nothing beyond Stokes' theorem, because homotopic cycles cobound the homotopy cylinder and part (2) of proposition 13.4.3 already disposes of cobounding cycles. The deeper homotopy invariance, that homotopic *maps* $M \to N$ induce the same map on de Rham cohomology through a chain homotopy built from integration along the cylinder, belongs to chapter 14 and is not needed here.

The angle form shows one closed non-exact form on one space. The general problem it opens is whether the periods see everything: given a closed form all of whose periods vanish, must it be exact? The question is the exact converse of proposition 13.4.3(1), and its answer is the content of the next chapter.

> **13.4.5 Foreshadowing: The de Rham Question** proposition 13.4.3(1) shows that an exact form has every period zero. The converse is the *de Rham question*: on a smooth manifold M, is a closed k-form ω with $\int_c \omega = 0$ for every smooth k-cycle c necessarily exact? Equivalently, does the period pairing separate the closed forms modulo exact ones from zero, so that the classes are detected entirely by their periods against cycles? De Rham's theorem answers yes, and more: the pairing identifies the quotient $\{\text{closed}\}/\{\text{exact}\}$ in each degree (the de Rham cohomology $H^k_{\mathrm{dR}}(M)$) with the singular cohomology of M with real coefficients, so that a form-theoretic invariant computed by integration equals a purely topological one. The proof rests on the homotopy invariance of cohomology and the Poincaré lemma, and is the programme of chapter 14. The present chapter poses the question and provides its first evidence: the angle form, a closed form whose single nonzero period certifies a nontrivial class.

Exercise 13.4.1

1. Let $\omega = (-y\,dx + x\,dy)/(x^2+y^2)$ on $\mathbb{R}^2 \setminus \{0\}$, and for $R > 0$ let $C_R\colon [0, 2\pi] \to \mathbb{R}^2 \setminus \{0\}$, $C_R(t) = (R\cos t, R\sin t)$. Using the closed annulus $A = \{\, 1 \le x^2 + y^2 \le 4 \,\}$, whose oriented boundary is $\partial A = C_2 \sqcup -C_1$, show that $\int_{C_2} \omega = \int_{C_1} \omega$. Then compute $\int_{C_R} \omega$ for general R and confirm it is 2π independent of R. Give a second explanation of the R-independence using corollary 13.4.4.
2. For an integer m let $c_m\colon [0, 2\pi] \to \mathbb{R}^2 \setminus \{0\}$ be $c_m(t) = (\cos mt, \sin mt)$. Compute the period $\int_{c_m} \omega$ of the angle form, and deduce that c_m and $c_{m'}$ are not smoothly homotopic in $\mathbb{R}^2 \setminus \{0\}$ when $m \neq m'$.
3. Let $c\colon Z \to M$ be a smooth 1-cycle, so Z is a disjoint union of circles, and let $f \in C^\infty(M)$. Show directly, without invoking proposition 13.4.3, that $\int_c df = 0$. (Reduce to a single circle and use the fundamental theorem of calculus after pulling back.)
4. Let $c\colon Z \to M$ be a smooth k-cycle and $\omega, \eta \in \Omega^k(M)$. Prove that $\int_c (a\omega + b\eta) = a\int_c \omega + b\int_c \eta$ for $a, b \in \mathbb{R}$, that $\int_{-c} \omega = -\int_c \omega$ where $-c$ denotes c with the reversed orientation on Z, and that $\int_{c \sqcup c'} \omega = \int_c \omega + \int_{c'} \omega$.
5. Suppose a closed k-form $\omega \in \Omega^k(M)$ has $\int_c \omega \neq 0$ for some smooth k-cycle c. Using proposition 13.4.3, deduce that ω is not exact and that c does not bound in M. Identify the instance of this statement furnished by example 13.4.2.

13.5 Densities and Integration without Orientation

Every integral in this chapter has required an orientation. The chart formula of definition 13.1.1 reads a top form as an $\mathbb{R}^n$-integral, and proposition 13.1.5 glues the charts together only because an orientation forces each transition Jacobian to be positive, turning the $|\det|$ produced by the change-of-variables theorem into a bare det. A manifold that admits no orientation (the Möbius band, $\mathbb{RP}^2$, the Klein bottle) carries no nowhere-vanishing top form by theorem 12.6.3, hence nothing for the form-integral to act on. Such a manifold nonetheless has a serviceable notion of area or mass. This section builds the object that captures it. A *density* transforms by the *absolute value* of the Jacobian rather than by the Jacobian itself, and because $|\det|$ is always positive it needs no compatibility of signs across charts. A density therefore integrates over any manifold at all, orientable or not.

The transformation law of a density is exactly the one appearing in the Euclidean change-of-variables theorem, $\int f = \int (f \circ g)\,|\det Dg|$ ([14, Change of Variables for C^1 Diffeomorphisms]). Reading that law backwards, the natural object to integrate on a manifold is not a top form but whatever quantity scales by $|\det Dg|$ under a coordinate change. On the space Λ^n of top forms the scaling factor is det; taking absolute values throughout replaces the alternating line $\Lambda^n T^*M$ by a new line bundle, the density bundle, whose sections are the densities.

Definition 13.5.1: Density

Let V be a real vector space of dimension n, and write $\mathrm{Fr}(V)$ for the set of ordered bases (frames) of V. A *density* on V is a function $\mu\colon \mathrm{Fr}(V) \to \mathbb{R}$ such that for every frame $e = (e_1, \ldots, e_n)$ and every $A = (A_{ij}) \in \mathrm{GL}_n(\mathbb{R})$,

$$\mu\big(e \cdot A\big) \;=\; |\det A|\,\mu(e) \qquad \text{where } (e \cdot A)_j = \sum_{i=1}^{n} A_{ij}\, e_i$$

A density is determined by its value on a single frame and that value may be any real number, so the densities on V form a one-dimensional real vector space, written $|\Lambda^n V^*|$. A density μ is *positive* if $\mu(e) > 0$ for one, equivalently every, frame e (the sign of $\mu(e)$ is frame-independent because $|\det A| > 0$).
The *density bundle* of a smooth n-manifold M is the real line bundle

$$|\Lambda^n T^*M| \;\longrightarrow\; M$$

whose fiber over p is the space of densities on T_pM; its transition functions are the $|\det|$ of the tangent-space transition matrices. A *density on* M is a smooth section $\mu \in \Gamma\big(|\Lambda^n T^*M|\big)$, positive if μ_p is positive for every p.

The density bundle differs from the determinant line bundle $\Lambda^n T^*M$ of chapter 12 in one place: the transition functions carry an absolute value. That single change removes every sign, and with it the orientation obstruction. In a chart (U, x) the coordinate frame $\big(\partial/\partial x^1, \ldots, \partial/\partial x^n\big)$ trivializes the bundle over U, so a density restricts to

$$\mu = f_x\,|dx^1 \wedge \cdots \wedge dx^n|, \qquad f_x(p) = \mu_p\left(\frac{\partial}{\partial x^1}\bigg|_p, \ldots, \frac{\partial}{\partial x^n}\bigg|_p\right)$$

where $|dx^1 \wedge \cdots \wedge dx^n|$ is the coordinate density assigning 1 to the coordinate frame, and $f_x \in C^\infty(U)$ is the local representative. On the overlap of (U, x) with a second chart (V, y) the coordinate frames are related by $\partial/\partial y^j = \sum_i (\partial x^i/\partial y^j)\,\partial/\partial x^i$, so the y-frame is the x-frame acted on by the matrix $A = (\partial x^i/\partial y^j)$. Applying the density law gives $f_y = |\det(\partial x/\partial y)|\, f_x$, or, inverting,

$$f_x \;=\; \left|\det \frac{\partial y}{\partial x}\right| f_y \qquad \text{on } U \cap V \tag{13.6}$$

where $\partial y/\partial x = \big(\partial y^i/\partial x^j\big)$ is the Jacobian of the transition $\psi = y \circ x^{-1}$. This is the transformation law that makes the integral below chart-independent.

Example 13.5.2: Coordinate Densities and the Absolute Value of a Form

On $\mathbb{R}^n$ with the identity chart, the coordinate density $|dr^1 \wedge \cdots \wedge dr^n|$ assigns to the standard frame the value 1; a general density is $f\,|dr^1 \wedge \cdots \wedge dr^n|$ for $f \in C^\infty(\mathbb{R}^n)$, and it will integrate to the ordinary Lebesgue integral of f.

Any top form $\omega \in \Omega^n(M)$ produces a density $|\omega|$ by taking absolute values pointwise,

$$|\omega|_p(v_1, \dots, v_n) = \big|\omega_p(v_1, \dots, v_n)\big|$$

Since an alternating n-form on an n-dimensional space satisfies $\omega_p(Av_1, \dots, Av_n) = \det(A)\,\omega_p(v_1, \dots, v_n)$, the absolute value obeys $|\omega|_p(Av) = |\det A|\,|\omega|_p(v)$, which is the density law. In a chart $\omega = h\,dx^1 \wedge \cdots \wedge dx^n$ gives $|\omega| = |h|\,|dx^1 \wedge \cdots \wedge dx^n|$; the representative $|h|$ is continuous everywhere and smooth wherever $h \neq 0$. Thus $|\omega|$ is a continuous density, smooth away from the zero set of ω.

The two constructions in example 13.5.2 show densities come in two kinds relevant here: the manufactured coordinate density $|dx^1 \wedge \cdots \wedge dx^n|$, which exists in every chart, and the absolute value $|\omega|$ of a form, which lives globally as soon as ω does. The first will let densities be built on any manifold; the second will relate density integration back to form integration when an orientation happens to be present.

Definition 13.5.3: Integral of a Density

Let μ be a continuous density on M (in particular any smooth one). If μ is supported in a single chart (U, x) and $\mu = f_x\,|dx^1 \wedge \cdots \wedge dx^n|$ there, define

$$\int_U \mu \;=\; \int_{x(U)} \big(f_x \circ x^{-1}\big)\,d\lambda$$

the ordinary Lebesgue integral over $x(U) \subseteq \mathbb{R}^n$ of the compactly supported continuous function $f_x \circ x^{-1}$, with λ the Lebesgue measure. For a compactly supported density μ on all of M, choose a partition of unity $\{\rho_\alpha\}$ subordinate to an atlas $\{(U_\alpha, x_\alpha)\}$ (theorem 1.5.5) and set

$$\int_M \mu \;=\; \sum_\alpha \int_{U_\alpha} \rho_\alpha\,\mu$$

a finite sum, since $\mathrm{supp}\mu$ is compact and the cover is locally finite.

For $M = \mathbb{R}^n$ with the single identity chart this reads $\int_{\mathbb{R}^n} f\,|dr^1 \wedge \cdots \wedge dr^n| = \int_{\mathbb{R}^n} f\,d\lambda$: integrating a density on Euclidean space is integrating its coefficient against Lebesgue measure. The definition is identical in shape to the form integral definition 13.1.4, with one difference of substance. The chart formula has no orientation built in (there is no requirement that the charts be positively oriented, and indeed no orientation is chosen); the freedom is harmless, because the absolute value in (13.6) is exactly the factor the change-of-variables theorem gives.

Proposition 13.5.4: Well-Definedness of the Density Integral

The integral $\int_M \mu$ of definition 13.5.3 is independent of the chart used in the single-chart case, and independent of the atlas and the partition of unity in the general case. No orientation of M is required.

***Proof*:**
Step 1: Chart-independence. Suppose μ is supported in $U \cap V$ for two charts (U, x) and (V, y), with representatives f_x and f_y related by (13.6). Put $\Omega = x(U \cap V)$, $\Omega' = y(U \cap V)$, and let $\psi = y \circ x^{-1} \colon \Omega \to \Omega'$ be the transition diffeomorphism, so $D\psi = \partial y/\partial x$. The Euclidean change-of-variables theorem ([14, Change of Variables for C^1 Diffeomorphisms]), applied to the compactly supported continuous function $f_y \circ y^{-1}$ on Ω', gives

$$\int_{\Omega'} \big(f_y \circ y^{-1}\big)\, d\lambda \;=\; \int_{\Omega} \big(f_y \circ y^{-1} \circ \psi\big)\, |\det D\psi|\, d\lambda \;=\; \int_{\Omega} \big(f_y \circ x^{-1}\big)\, |\det D\psi|\, d\lambda$$

using $y^{-1} \circ \psi = x^{-1}$. By (13.6), $f_x = |\det(\partial y/\partial x)|\, f_y = |\det D\psi|\, f_y$ on $U \cap V$, so $\big(f_y \circ x^{-1}\big)\, |\det D\psi| = f_x \circ x^{-1}$ on Ω. Hence the right-hand side equals $\int_\Omega (f_x \circ x^{-1})\, d\lambda$, which is $\int_U \mu$; the left-hand side is $\int_V \mu$. The two chart definitions agree. The absolute value given by change of variables is precisely the factor the density law requires, and no positivity of $\det D\psi$ (that is, no orientation) was used.

Step 2: Independence of the atlas and partition. Let $\{\rho_\alpha\}$ be subordinate to $\{(U_\alpha, x_\alpha)\}$ and $\{\sigma_\beta\}$ subordinate to $\{(V_\beta, y_\beta)\}$. Each $\rho_\alpha \sigma_\beta \mu$ is a density supported in $U_\alpha \cap V_\beta$, a set contained in both charts, so its single-chart integral is unambiguous by Step 1. Because $\operatorname{supp}\mu$ is compact, only finitely many products $\rho_\alpha \sigma_\beta$ meet it and every interchange of summation is finite:

$$\sum_\alpha \int_{U_\alpha} \rho_\alpha \mu = \sum_\alpha \int_{U_\alpha} \Big(\sum_\beta \sigma_\beta\Big) \rho_\alpha \mu = \sum_{\alpha,\beta} \int \rho_\alpha \sigma_\beta \mu = \sum_\beta \int_{V_\beta} \Big(\sum_\alpha \rho_\alpha\Big) \sigma_\beta \mu = \sum_\beta \int_{V_\beta} \sigma_\beta \mu$$

where $\sum_\alpha \rho_\alpha = \sum_\beta \sigma_\beta = 1$ on $\operatorname{supp}\mu$. The two partition-of-unity sums coincide, completing the proof.

The argument mirrors the well-definedness of the form integral proposition 13.1.5 line for line, and the single divergence is the decisive one: there the Jacobian had to be positive, and orientability was the hypothesis that guaranteed it; here the absolute value makes the same computation go through with no hypothesis at all. A positive density is, in each chart, a positive continuous function integrated against Lebesgue measure, so $\int_M \mu \geq 0$ for a positive density and equals its total mass. In this sense a density is the coordinate-free version of a measure: a positive density on M determines a Borel measure on M by $A \mapsto \int_M \mathbf{1}_A\, \mu$, and integration of densities is measure theory ([14]) transported to a manifold without the crutch of a global coordinate system.

Densities exist in profusion, and their existence is the structural reason integration on a non-orientable manifold is possible at all.

Proposition 13.5.5: Existence of a Positive Density

Every smooth manifold M admits a smooth positive density. Consequently the density bundle $|\Lambda^n T^*M|$ is trivial.

***Proof*:**
Choose an atlas $\{(U_\alpha, x_\alpha)\}$ and a subordinate partition of unity $\{\rho_\alpha\}$ (theorem 1.5.5). On U_α the coordinate density $|dx_\alpha^1 \wedge \cdots \wedge dx_\alpha^n|$ is positive; extending $\rho_\alpha\,|dx_\alpha^1 \wedge \cdots \wedge dx_\alpha^n|$ by zero off U_α produces a globally defined nonnegative smooth density, and

$$\mu \;=\; \sum_\alpha \rho_\alpha\,|dx_\alpha^1 \wedge \cdots \wedge dx_\alpha^n|$$

is a locally finite sum, hence a smooth density. At any p some $\rho_\alpha(p) > 0$, and the corresponding summand is positive at p while the rest are nonnegative, so $\mu_p > 0$. Thus μ is a nowhere-vanishing section, which trivializes the line bundle $|\Lambda^n T^*M|$, as we sought to show.

The contrast with top forms is exact and is worth stating plainly. The same construction applied to forms, $\sum_\alpha \rho_\alpha\, dx_\alpha^1 \wedge \cdots \wedge dx_\alpha^n$, can vanish at points where the coordinate forms carry conflicting signs, and a nowhere-vanishing top form exists if and only if M is orientable (theorem 12.6.3). Positivity of a density, on the other hand, is a convex condition preserved under the sum because every coordinate density is positive in every chart: the transition factor $|\det|$ can never turn it negative. Orientability is an obstruction to a global volume *form*; there is no obstruction to a global volume *density*.

When an orientation is present, the two integrals are related by the absolute value that separates them.

Proposition 13.5.6: Densities from Forms

Let M be an oriented n-manifold and $\omega \in \Omega_c^n(M)$ a compactly supported top form. Then $|\omega|$ is a compactly supported continuous density and

1. $\displaystyle\int_M |\omega| \;\geq\; \left|\int_M \omega\right|$;
2. if ω vanishes nowhere and induces the given orientation (definition 12.6.1), then $\displaystyle\int_M |\omega| = \int_M \omega$.

If in addition M is connected, then for any nowhere-vanishing ω it follows that $\int_M |\omega| = \pm \int_M \omega$, the sign being $+$ when ω induces the given orientation and $-$ when it induces the opposite one.

***Proof*:**
That $|\omega|$ is a continuous density with the same compact support as ω was checked in example 13.5.2. Fix a partition of unity $\{\rho_\alpha\}$ subordinate to positively oriented charts (U_α, x_α), and write $\omega = h_\alpha\, dx_\alpha^1 \wedge \cdots \wedge dx_\alpha^n$ on U_α. Because the chart is positively oriented, definition 13.1.1

carries no absolute value, so

$$\int_{U_\alpha} \rho_\alpha\, \omega = \int_{x_\alpha(U_\alpha)} (\rho_\alpha h_\alpha) \circ x_\alpha^{-1}\, d\lambda, \qquad \int_{U_\alpha} \rho_\alpha\, |\omega| = \int_{x_\alpha(U_\alpha)} |\rho_\alpha h_\alpha| \circ x_\alpha^{-1}\, d\lambda$$

the second by definition 13.5.3, using $\rho_\alpha \geq 0$ so that $\rho_\alpha|\omega| = |\rho_\alpha \omega|$.

Step 1: The inequality. Summing and applying the triangle inequality twice (first inside each integral, then across the finite sum) gives

$$\int_M |\omega| = \sum_\alpha \int |\rho_\alpha h_\alpha| \circ x_\alpha^{-1}\, d\lambda \;\geq\; \sum_\alpha \left| \int (\rho_\alpha h_\alpha) \circ x_\alpha^{-1}\, d\lambda \right| \;\geq\; \left| \sum_\alpha \int (\rho_\alpha h_\alpha) \circ x_\alpha^{-1}\, d\lambda \right| = \left| \int_M \omega \right|$$

which is part (1).

Step 2: The equality. If ω vanishes nowhere and induces the given orientation, then in each positively oriented chart the coordinate frame is positively oriented, so $h_\alpha(p) = \omega_p(\partial/\partial x_\alpha^1, \ldots, \partial/\partial x_\alpha^n) > 0$. Hence $|\rho_\alpha h_\alpha| = \rho_\alpha h_\alpha$ everywhere, each triangle inequality in Step 1 is an equality, and $\int_M |\omega| = \int_M \omega$, which is part (2).

Finally, reversing the orientation of M negates $\int_M \omega$ by the sign rule of proposition 13.1.7, while $\int_M |\omega|$ is unchanged because a density has no orientation to reverse. Now suppose M is connected. Since ω vanishes nowhere, the sign of ω_p relative to the given orientation is locally constant, hence constant on the connected M, so ω induces either the given orientation at every point or its opposite at every point. Part (2) gives $\int_M |\omega| = \int_M \omega$ in the first case; applying it after reversing the orientation gives $\int_M |\omega| = -\int_M \omega$ in the second, completing the proof.

The proposition says a density forgets exactly the orientation-dependent sign that a form remembers. On an oriented manifold the two integration theories carry the same information about a positively oriented volume element, and $|\omega|$ is the bridge. Off an oriented manifold only the density survives, which is where the following example lives.

Example 13.5.7: The Möbius Band Carries a Density but No Volume Form

Realize the closed Möbius band as the quotient

$$\overline{E} = \big(\mathbb{R} \times [-1,1]\big)/\mathbb{Z}, \qquad n \cdot (u,v) = \big(u+n,\, (-1)^n v\big)$$

a compact 2-manifold with boundary (definition 1.1.16), with covering map $q\colon \mathbb{R} \times [-1,1] \to \overline{E}$ and deck generator $g(u,v) = (u+1, -v)$. It is non-orientable, so theorem 12.6.3 denies it any nowhere-vanishing top form and the form-integral $\int_{\overline{E}}(\,\cdot\,)$ is not even defined. The obstruction is visible on the cover: the area form $\omega_0 = du \wedge dv$ pulls back under the deck map to

$$g^*\omega_0 = d(u+1) \wedge d(-v) = -\,du \wedge dv = -\omega_0$$

so ω_0 is anti-invariant and descends to no form on $\overline{E}$. Its absolute value fares better:

$$g^*|\omega_0| = |g^*\omega_0| = |-\omega_0| = |\omega_0|$$

because g has Jacobian determinant -1 and $|-1| = 1$. Thus $|du \wedge dv|$ is $\mathbb{Z}$-invariant and descends to a smooth positive density μ on $\overline{E}$. The band is compact, so μ has compact support and $\int_{\overline{E}} \mu$ is defined by definition 13.5.3 (the boundary contributes measure zero in each half-space chart). A fundamental domain for the $\mathbb{Z}$-action is $[0,1] \times [-1,1]$, on which μ is $du \wedge dv$ in absolute value, so

$$\int_{\overline{E}} \mu = \int_0^1 \int_{-1}^1 du\, dv = 2$$

The Möbius band has area 2, computed with no orientation, precisely because the very step that fails for forms (descending ω_0 through the sign-reversing gluing) succeeds for the density $|\omega_0|$.

A positive density is the datum that turns a bare manifold into a measure space, and geometry gives one canonically. A Riemannian metric g furnishes a positive density with local representative $\sqrt{\det(g_{ij})}$ against $|dx^1 \wedge \cdots \wedge dx^n|$, the Riemannian density; on an oriented Riemannian manifold it is $|\mathrm{vol}_g|$ for the Riemannian volume form vol_g, but the density itself needs no orientation and so integrates functions over any Riemannian manifold, $\mathbb{RP}^2$ and the Klein bottle included. The metric geometry behind $\sqrt{\det g}$ is taken up in the companion volume on Riemannian geometry; here it is enough that a density is what an orientation-free manifold integrates, and that de Rham cohomology, which measures what oriented integration detects, is developed in chapter 14 for the oriented world where forms and densities agree.

Exercise 13.5.1

1. Using the single identity chart on $\mathbb{R}^2$, compute $\displaystyle\int_{\mathbb{R}^2} e^{-(x^2+y^2)}\,|dx \wedge dy|$ as a density integral, and confirm the answer does not depend on choosing an orientation of $\mathbb{R}^2$.
2. Fix a positive density μ_0 on a manifold M. Show that every density ν on M can be written uniquely as $\nu = h\,\mu_0$ with $h \in C^\infty(M)$, that ν is positive if and only if $h > 0$, and deduce that the smooth densities on M form a free rank-one module over $C^\infty(M)$.
3. Let M be a connected oriented n-manifold and $\omega \in \Omega^n_c(M)$. Prove that $\int_M |\omega| = \left|\int_M \omega\right|$ holds if and only if ω does not change sign relative to the orientation (that is, in every positively oriented chart the coefficient h has one and the same sign, or is zero).
4. For a diffeomorphism $F\colon N \to M$ and a density μ on M, define the pullback $F^*\mu$ by $(F^*\mu)_p(v_1, \dots, v_n) = \mu_{F(p)}(dF_p v_1, \dots, dF_p v_n)$. Show that $F^*\mu$ is a density on N, that in charts it has representative $(f \circ F)\,|\det DF|$, and that $\int_N F^*\mu = \int_M \mu$ for compactly supported μ, with no orientation hypothesis on F. Contrast this with proposition 13.1.7.
5. The round area form ω on S^2 has $\int_{S^2} |\omega| = 4\pi$. Show that the antipodal map $a(x) = -x$ reverses orientation on S^2, so that $\mathbb{RP}^2 = S^2/\{\pm 1\}$ admits no volume form, yet $|\omega|$ descends to a positive density μ on $\mathbb{RP}^2$; compute $\int_{\mathbb{RP}^2} \mu$.

14

De Rham Cohomology

Closed forms that fail to be exact have appeared throughout this book: the angle form with its period 2π around a puncture, the top form of a sphere that integrates to its volume and so cannot be a boundary. Each time, the failure of exactness signalled a hole. This chapter turns that signal into an invariant. The closed forms modulo the exact ones form, in each degree, a real vector space, and across all degrees a graded algebra under the wedge: the de Rham cohomology of the manifold. The first task is to establish its formal properties, that it is a functor and an algebra, and that in degree zero it counts connected components.

The invariant is then shown to be topological in three steps. The Poincaré lemma trivialises it wherever there are no holes, on any star-shaped region, so cohomology is a purely global phenomenon. Homotopy invariance follows, and with it the fact that de Rham cohomology sees a manifold only up to homotopy equivalence. The Mayer-Vietoris sequence then makes the invariant computable, cutting a manifold into pieces whose cohomologies determine the whole, and it delivers the cohomology of the spheres, the tori, and the punctured spaces by hand.

The culmination is the de Rham theorem. Integrating a closed form over a cycle, an operation well defined precisely because of Stokes' theorem, pairs de Rham cohomology with the singular homology that algebraic topology builds from continuous data, and the pairing is perfect: the analytic operator d measures exactly the topology that homology measures, and a closed form is exact if and only if all of its periods vanish. Poincaré duality and the cohomological degree then cash the identification back into geometry, closing the degree and period threads the book opened. The forms calculus of chapter 12 and the integration of chapter 13 are used throughout; the singular side is cited to the companion volume; and Hodge theory, sheaf cohomology, and characteristic classes are handed forward to chapter 15 and the books beyond.

14.1 De Rham Cohomology

definition 12.5.8 recorded two nested subspaces of $\Omega^k(M)$: the *closed* forms, on which d vanishes, and the *exact* forms, in the image of d, the first containing the second because $d \circ d = 0$. Their quotient, taken in each degree, is the de Rham cohomology, and this section constructs it together with the algebra it carries: the wedge descends to a graded-commutative product, a smooth map induces a homomorphism the other way, and the degree-zero part counts connected components. It thus enters as a functor from manifolds to graded $\mathbb{R}$-algebras, before a single form has been shown non-exact.

Definition 14.1.1: De Rham Cohomology

Let M be a smooth manifold. For each integer k write

$$Z^k(M) = \left\{\omega \in \Omega^k(M) \;\middle|\; d\omega = 0\right\}, \qquad B^k(M) = \left\{d\eta \;\middle|\; \eta \in \Omega^{k-1}(M)\right\}$$

for the spaces of *closed* and *exact* k-forms, with the convention $\Omega^k(M) = 0$ for $k < 0$, so that $B^0(M) = 0$. Because $d \circ d = 0$ (theorem 12.5.1), every exact form is closed, $B^k(M) \subseteq Z^k(M)$, and the *k-th de Rham cohomology* of M is the quotient vector space

$$H^k_{\mathrm{dR}}(M) = Z^k(M)/B^k(M)$$

The image of a closed form ω is its *cohomology class* $[\omega] = \omega + B^k(M)$; two closed forms with $[\omega] = [\omega']$, that is $\omega - \omega' = d\eta$ for some η, are called *cohomologous.* The *total de Rham cohomology* is the graded vector space

$$H^*_{\mathrm{dR}}(M) = \bigoplus_{k \geq 0} H^k_{\mathrm{dR}}(M)$$

A class $[\omega]$ is zero exactly when ω is exact, so $H^k_{\mathrm{dR}}(M)$ measures the closed k-forms up to the ambiguity of adding a differential. It vanishes precisely where every closed form has a global potential, and it is nonzero exactly where some closed form does not. The passage from ω to $[\omega]$ discards the choice of representative and keeps only the obstruction, which is what makes the quotient an invariant rather than a space of forms. The degree k ranges over $0 \leq k \leq \dim M$, since $\Omega^k(M) = 0$ above the dimension, so all but finitely many summands of $H^*_{\mathrm{dR}}(M)$ vanish.

Example 14.1.2: Cohomology of a Point

Let $M = \{p\}$ be a single point, a zero-dimensional manifold. Its tangent space is $T_pM = 0$, so $\Lambda^k T_p^* M = 0$ for $k \geq 1$ and hence $\Omega^k(\{p\}) = 0$ for $k \geq 1$; the only nonzero cochain group is $\Omega^0(\{p\}) = C^\infty(\{p\}) = \mathbb{R}$, a smooth function on a point being a single real number. The differential $d\colon \Omega^0 \to \Omega^1$ lands in the zero space, so $Z^0 = \mathbb{R}$ and $B^0 = 0$, while all higher Z^k and B^k vanish. Therefore

$$H^0_{\mathrm{dR}}(\{p\}) = \mathbb{R}, \qquad H^k_{\mathrm{dR}}(\{p\}) = 0 \quad (k \geq 1)$$

The cohomology is a single copy of $\mathbb{R}$ concentrated in degree 0. This is the cohomology that every contractible manifold shares, once homotopy invariance is available (section 14.3).

The wedge product multiplies forms; the first structural fact is that it descends to cohomology. Closedness is preserved by the product, and changing either factor by an exact form changes the product by an exact form, so the operation is well defined on classes.

Proposition 14.1.3: The Cup Product

Let M be a smooth manifold. The wedge product descends to a well-defined bilinear map

$$H^k_{\mathrm{dR}}(M) \times H^l_{\mathrm{dR}}(M) \to H^{k+l}_{\mathrm{dR}}(M), \qquad ([\alpha], [\beta]) \mapsto [\alpha][\beta] := [\alpha \wedge \beta]$$

the *cup product*. Under it $H^*_{\mathrm{dR}}(M)$ is an associative $\mathbb{R}$-algebra with unit $[1]$, the class of the constant function $1 \in \Omega^0(M)$, and it is graded-commutative: for $[\alpha] \in H^k_{\mathrm{dR}}(M)$ and $[\beta] \in H^l_{\mathrm{dR}}(M)$,

$$[\alpha][\beta] = (-1)^{kl}\,[\beta][\alpha]$$

***Proof*:**
Throughout, $\alpha \in Z^k(M)$ and $\beta \in Z^l(M)$ are closed, and the Leibniz rule $d(\mu \wedge \nu) = d\mu \wedge \nu + (-1)^{\deg \mu}\mu \wedge d\nu$ is that of the exterior derivative (theorem 12.5.1).

Step 1: The product of closed forms is closed. By the Leibniz rule,

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta = 0$$

since $d\alpha = 0$ and $d\beta = 0$. Thus $\alpha \wedge \beta \in Z^{k+l}(M)$ and $[\alpha \wedge \beta]$ is defined.

Step 2: Well-definedness. Suppose $\alpha' = \alpha + d\eta$ is cohomologous to α, with $\eta \in \Omega^{k-1}(M)$. Then $\alpha' \wedge \beta = \alpha \wedge \beta + d\eta \wedge \beta$, and the Leibniz rule gives

$$d(\eta \wedge \beta) = d\eta \wedge \beta + (-1)^{k-1}\eta \wedge d\beta = d\eta \wedge \beta$$

because $d\beta = 0$. Hence $d\eta \wedge \beta$ is exact and $\alpha' \wedge \beta$ is cohomologous to $\alpha \wedge \beta$, so the class $[\alpha \wedge \beta]$ does not depend on the representative of $[\alpha]$. Symmetrically, if $\beta' = \beta + d\zeta$ with $\zeta \in \Omega^{l-1}(M)$, then

$$d(\alpha \wedge \zeta) = d\alpha \wedge \zeta + (-1)^k \alpha \wedge d\zeta = (-1)^k \alpha \wedge d\zeta$$

so $\alpha \wedge d\zeta = (-1)^k d(\alpha \wedge \zeta)$ is exact and the class is independent of the representative of $[\beta]$. The product is therefore well defined, and it is bilinear because the wedge is.

Step 3: Algebra structure. Associativity and the unit are inherited from the wedge product on forms: $(\alpha \wedge \beta) \wedge \gamma = \alpha \wedge (\beta \wedge \gamma)$ passes to classes, and $1 \wedge \alpha = \alpha = \alpha \wedge 1$ with $d1 = 0$ gives $[1][\alpha] = [\alpha] = [\alpha][1]$. Graded-commutativity of the wedge, $\alpha \wedge \beta = (-1)^{kl}\beta \wedge \alpha$, yields on classes $[\alpha][\beta] = [\alpha \wedge \beta] = (-1)^{kl}[\beta \wedge \alpha] = (-1)^{kl}[\beta][\alpha]$, completing the proof.

De Rham cohomology is thus not only a graded vector space but a graded ring, and the ring structure is a finer invariant than the list of dimensions: two manifolds may have isomorphic cohomology in each degree yet distinct multiplication. The cup product records how forms of complementary degrees multiply to a top-degree form, and section 14.5 identifies it with the cup product of singular cohomology.

A smooth map cannot in general be inverted, so it does not push forms forward; but it pulls them back, and pullback is compatible with everything just built. The result is that cohomology is contravariant.

Proposition 14.1.4: Functoriality of de Rham Cohomology

A smooth map $f\colon M \to N$ induces, for each k, an $\mathbb{R}$-linear map

$$f^*\colon H^k_{\mathrm{dR}}(N) \to H^k_{\mathrm{dR}}(M), \qquad f^*[\omega] = [f^*\omega]$$

and together these form a homomorphism $f^*\colon H^*_{\mathrm{dR}}(N) \to H^*_{\mathrm{dR}}(M)$ of graded $\mathbb{R}$-algebras. Moreover $\mathrm{id}_M^* = \mathrm{id}$ on $H^*_{\mathrm{dR}}(M)$ and $(g \circ f)^* = f^* \circ g^*$ for smooth $M \xrightarrow{f} N \xrightarrow{g} P$, so H^*_{dR} is a contravariant functor from smooth manifolds to graded-commutative $\mathbb{R}$-algebras.

***Proof*:**
Pullback of forms $f^*\colon \Omega^k(N) \to \Omega^k(M)$ commutes with the exterior derivative, $f^*(d\omega) = d(f^*\omega)$, by naturality of d (proposition 12.5.4). Consequently f^* carries closed forms to closed forms, since $d\omega = 0$ gives $d(f^*\omega) = f^*(d\omega) = 0$, and exact forms to exact forms, since $\omega = d\eta$ gives $f^*\omega = f^*(d\eta) = d(f^*\eta)$. Thus f^* maps $Z^k(N)$ into $Z^k(M)$ and $B^k(N)$ into $B^k(M)$, so the assignment $[\omega] \mapsto [f^*\omega]$ is a well-defined linear map on the quotients.

That f^* respects the cup product and unit follows from the corresponding facts for forms (proposition 12.4.11): $f^*(\alpha \wedge \beta) = f^*\alpha \wedge f^*\beta$ passes to $f^*([\alpha][\beta]) = f^*[\alpha]\, f^*[\beta]$, and $f^*1 = 1$ gives $f^*[1] = [1]$. The functor identities $\mathrm{id}^* = \mathrm{id}$ and $(g \circ f)^* = f^* \circ g^*$ hold at the level of forms (proposition 12.4.11) and therefore descend to cohomology, as we sought to show.

Contravariance is the reason de Rham cohomology is a cohomology rather than a homology: arrows reverse. A map into N is probed by pulling back the cohomology of N, and a class on N that pulls back to zero on M witnesses that f cannot see that part of N. Functoriality already forces one qualitative fact: a map that factors through a point induces the zero map in each positive degree, because it factors through the cohomology of a point, which vanishes above degree 0 (example 14.1.2).

The lowest degree is computed with no machinery beyond the meaning of d on functions. A closed 0-form is a function whose differential vanishes, hence one that is constant on each piece of the manifold, and there are no exact 0-forms to quotient by. The zeroth cohomology therefore counts components.

Proposition 14.1.5: Zeroth de Rham Cohomology

Let M be a smooth manifold. Then $B^0(M) = 0$, and $Z^0(M)$ is the space of locally constant real-valued functions on M; hence $H^0_{\mathrm{dR}}(M)$ is canonically that space. If M has finitely many connected components, say c of them, then

$$H^0_{\mathrm{dR}}(M) \cong \mathbb{R}^c, \qquad \dim_{\mathbb{R}} H^0_{\mathrm{dR}}(M) = c$$

the number of connected components; for infinitely many components $H^0_{\mathrm{dR}}(M)$ is the space of all real functions on the set of components. In particular M is connected if and only if $H^0_{\mathrm{dR}}(M) \cong \mathbb{R}$, generated by the class of the constant function 1.

***Proof*:**
Since $\Omega^{-1}(M) = 0$ by convention, $B^0(M) = \mathrm{im}(d\colon \Omega^{-1} \to \Omega^0) = 0$, so $H^0_{\mathrm{dR}}(M) = Z^0(M) = \ker(d\colon C^\infty(M) \to \Omega^1(M))$.

Step 1: $df = 0$ iff f is locally constant. In a chart (U, x) one has $df = \sum_i (\partial f/\partial x^i)\, dx^i$, so $df = 0$ on U if and only if every partial derivative vanishes there. On a connected chart domain (a ball, say) the vanishing of all first partials forces f constant, by the mean value theorem applied along coordinate segments. Every point of M has such a chart neighborhood, so $df = 0$ if and only if f is constant near each point, that is, locally constant.

Step 2: A locally constant function is constant on each component. Let M_i be a connected component of M, fix $p \in M_i$, and put $c = f(p)$. The set $A = \{q \in M_i \mid f(q) = c\}$ is open because f is locally constant, and its complement in M_i is $\bigcup_{c' \neq c} \{q \in M_i \mid f(q) = c'\}$, a union of open sets and hence open; so A is closed in M_i. As A is nonempty and M_i is connected, $A = M_i$, and $f \equiv c$ on M_i.

Step 3: Identification. By Steps 1 and 2, a closed 0-form is exactly an assignment of a real constant to each connected component of M. Conversely any such assignment defines a locally constant, hence smooth, function, because the components are open (manifolds are locally connected). Therefore $Z^0(M)$ is isomorphic to the space of functions from the set of components to $\mathbb{R}$; when there are finitely many components, say c, this is $\mathbb{R}^c$, of dimension c. When M is connected, $c = 1$ and every closed 0-form is a global constant, so $H^0_{\mathrm{dR}}(M) = \mathbb{R} \cdot [1]$, completing the proof.

The zeroth cohomology is the crudest topological reading a form can give: it sees only whether the manifold is in one piece or several. Combined with functoriality, it already shows that a connected manifold admits no smooth retraction onto a two-point subset $\{a, b\}$: such a retraction would make $H^0_{\mathrm{dR}}(\{a, b\}) \cong \mathbb{R}^2$ inject into $H^0_{\mathrm{dR}}(M) \cong \mathbb{R}$, which is impossible. The higher cohomology, which requires closed forms that are not exact, is the subject of the rest of the chapter; the circle furnishes the first case where it does not vanish.

Example 14.1.6: Cohomology of the Circle
Present S^1 as the quotient $\mathbb{R}/2\pi\mathbb{Z}$ with covering map $q\colon \mathbb{R} \to S^1$, $q(t) = [t]$, oriented so that q is orientation-preserving. Since q is a surjective local diffeomorphism, the pullback $q^*\colon \Omega^*(S^1) \to \Omega^*(\mathbb{R})$ is injective, with image the 2π-periodic forms.

In degree 0 the circle is connected, so $H^0_{\mathrm{dR}}(S^1) = \mathbb{R}$ by proposition 14.1.5. In degree 1, the manifold is one-dimensional, so $\Omega^2(S^1) = 0$ and every 1-form is closed: $Z^1(S^1) = \Omega^1(S^1)$. For $\omega \in \Omega^1(S^1)$ write $q^*\omega = f(t)\, dt$ with $f \in C^\infty(\mathbb{R})$ of period 2π. The *angle form* $d\theta \in \Omega^1(S^1)$ is the 1-form determined by $q^*(d\theta) = dt$; it descends because dt is translation-invariant, and it is the restriction of the angle form of example 13.4.2 along the inclusion $S^1 \hookrightarrow \mathbb{R}^2 \setminus \{0\}$, $[t] \mapsto (\cos t, \sin t)$.

Define $\Phi\colon \Omega^1(S^1) \to \mathbb{R}$ by averaging the coefficient,

$$\Phi(\omega) = \frac{1}{2\pi} \int_{S^1} \omega = \frac{1}{2\pi} \int_0^{2\pi} f(t)\, dt$$

This is linear, and $\Phi(d\theta) = \frac{1}{2\pi} \int_0^{2\pi} dt = 1$, so Φ is surjective. Its kernel is exactly the exact forms.

If $\omega = d\bar{g}$ is exact, then $\int_{S^1} \omega = 0$ by proposition 13.4.3(1) applied to the fundamental cycle of S^1, so $\Phi(\omega) = 0$. Conversely, suppose $\Phi(\omega) = 0$, that is $\int_0^{2\pi} f = 0$. Set $g(t) = \int_0^t f(s)\,ds$; then $g \in C^\infty(\mathbb{R})$ and

$$g(t + 2\pi) - g(t) = \int_t^{t+2\pi} f(s)\,ds = \int_0^{2\pi} f(s)\,ds = 0$$

using the periodicity of f, so g is 2π-periodic and descends to $\bar{g} \in C^\infty(S^1)$ with $q^*\bar{g} = g$. Then $q^*(d\bar{g}) = d(q^*\bar{g}) = dg = f\,dt = q^*\omega$, and injectivity of q^* gives $\omega = d\bar{g}$, so ω is exact. Hence $\ker\Phi = B^1(S^1)$, and Φ induces an isomorphism

$$H^1_{\mathrm{dR}}(S^1) \xrightarrow{\cong} \mathbb{R}$$

with generator $[d\theta]$. In summary $H^0_{\mathrm{dR}}(S^1) = \mathbb{R}\cdot[1]$, $H^1_{\mathrm{dR}}(S^1) = \mathbb{R}\cdot[d\theta]$, and $H^k_{\mathrm{dR}}(S^1) = 0$ for $k \geq 2$. As a ring, $[d\theta][d\theta] = [d\theta \wedge d\theta] = 0$ because a 2-form on a 1-manifold vanishes, so $H^*_{\mathrm{dR}}(S^1) \cong \mathbb{R}[x]/(x^2)$ with $x = [d\theta]$ in degree 1.

The circle is the first manifold whose cohomology is not that of a point: the single generator $[d\theta]$ in degree 1 records the one loop that cannot be filled. Its period $\int_{S^1} d\theta = 2\pi$ is the number that certifies the class is nonzero, exactly the pairing between forms and cycles opened in chapter 13 and completed into the de Rham theorem later in this chapter. The computation was elementary because S^1 is one-dimensional; for higher spheres and tori the same quotient is out of reach by hand, and section 14.4 gives the tool that makes it computable.

Exercise 14.1.1

1. Let M be the disjoint union of m copies of $\mathbb{R}^n$. Using proposition 14.1.5, determine $H^0_{\mathrm{dR}}(M)$ and exhibit an explicit basis of $Z^0(M)$ by locally constant functions. What is $\dim_{\mathbb{R}} H^0_{\mathrm{dR}}(M)$?
2. Using graded-commutativity of the cup product (proposition 14.1.3), show that any class $[\alpha] \in H^k_{\mathrm{dR}}(M)$ of odd degree k satisfies $[\alpha][\alpha] = 0$. Confirm the statement directly for $[d\theta] \in H^1_{\mathrm{dR}}(S^1)$ from example 14.1.6.
3. Show that a diffeomorphism $f\colon M \to N$ induces an isomorphism of graded $\mathbb{R}$-algebras $f^*\colon H^*_{\mathrm{dR}}(N) \to H^*_{\mathrm{dR}}(M)$, with inverse $(f^{-1})^*$. Conclude that the graded ring H^*_{dR} is a diffeomorphism invariant.
4. Prove that $H^*_{\mathrm{dR}}(S^1) \cong \mathbb{R}[x]/(x^2)$ as graded rings, with x in degree 1. Then compute the cohomology class of $\omega = (2 + \cos\theta)\,d\theta$ on S^1 as a multiple of $[d\theta]$, where θ is the coordinate with $q^*\omega = (2 + \cos t)\,dt$.
5. Let $f\colon M \to N$ be smooth. Show that $f^*[1_N] = [1_M]$. Then let $\kappa\colon M \to N$ be a constant map, with image a single point. Using proposition 14.1.4 and example 14.1.2, show that $\kappa^*\colon H^k_{\mathrm{dR}}(N) \to H^k_{\mathrm{dR}}(M)$ is the zero map for every $k \geq 1$.

14.2 The Poincaré Lemma

The functor of section 14.1 is so far uncomputed: nothing yet shows that $H^k_{\mathrm{dR}}(M)$ is ever zero. The Poincaré lemma is the first vanishing theorem. On a star-shaped open subset of $\mathbb{R}^n$ every closed form of positive degree is exact, so such a set carries no de Rham cohomology above degree zero. The proof is constructive. It exhibits a linear operator that turns a closed form into a primitive, built by integrating the form along the rays that contract the set to its center. Because every point of every manifold has a star-shaped coordinate neighborhood, the lemma also shows that closedness is a purely local condition: a closed form is exact near each point, and whatever obstruction de Rham cohomology records is therefore global, not local.

The hypothesis names the class of sets on which the contraction is available.

Definition 14.2.1: Star-Shaped Set

A subset $U \subseteq \mathbb{R}^n$ is *star-shaped* about a point $x_0 \in U$ if for every $x \in U$ the closed segment from x_0 to x lies in U, that is,

$$(1-t)x_0 + tx \in U \qquad \text{for all } t \in [0,1]$$

The set U is *star-shaped* if it is star-shaped about at least one of its points.

Star-shapedness about x_0 says that U can be shrunk to x_0 by sliding each point straight in along its segment. It is weaker than convexity, which requires the segment between *every* pair of points, and it is exactly the property the radial contraction below requires.

Example 14.2.2: Star-Shaped and Non-Star-Shaped Regions

Every convex set is star-shaped about each of its points, since the defining segment lies in the set by convexity. In particular $\mathbb{R}^n$, every open ball, every open box, and every open half-space is star-shaped.

The slit plane $U = \mathbb{R}^2 \setminus \{(x,0) \mid x \le 0\}$, the plane with the nonpositive x-axis removed, is star-shaped about $(1,0)$ but is not convex. A segment from $(1,0)$ to a point $(x,y) \in U$ is $\big((1-t)+tx,\ ty\big)$; its second coordinate vanishes only at $t=0$ or $y=0$, and when $y=0$ the target has $x>0$, so the segment stays on the positive x-axis and never meets the slit. Convexity fails because two points straddling the slit cannot be joined within U. This set is where a single-valued branch of the argument, and hence a global primitive of the angle form, lives.

The punctured plane $\mathbb{R}^2 \setminus \{0\}$ is star-shaped about no point. For any candidate center $c \neq 0$ the point $-\frac{1}{2}c$ lies in the set, yet the segment from c to $-\frac{1}{2}c$ passes through the origin, which is absent; and 0 itself is not in the set. The same failure afflicts every annulus. On these sets the conclusion of the Poincaré lemma is false, as the angle form of example 13.4.2 shows.

To prove exactness constructively one needs a systematic way to manufacture a primitive η with $d\eta = \omega$ from a closed ω. Star-shapedness gives a canonical mechanism: contract U to its center along the rays $x \mapsto tx$, and integrate the form along the contraction. The bookkeeping of this integration is a single linear operator, the *homotopy operator*, whose defining identity does the whole work of the lemma. After translating the center to the origin, take U star-shaped about 0, so that $tx \in U$ whenever $x \in U$ and $t \in [0,1]$.

Lemma 14.2.3: The Homotopy Operator on a Star-Shaped Domain

Let $U \subseteq \mathbb{R}^n$ be open and star-shaped about 0. For each $k \geq 1$ there is an $\mathbb{R}$-linear operator

$$K\colon \Omega^k(U) \to \Omega^{k-1}(U)$$

given on $\omega = \sum_I a_I \, dx^I$, the sum over increasing multi-indices $I = (i_1 < \cdots < i_k)$, by

$$K\omega = \sum_I \sum_{\alpha=1}^{k} (-1)^{\alpha-1} \Big(\int_0^1 t^{k-1} a_I(tx)\, dt \Big)\, x^{i_\alpha}\, dx^{i_1} \wedge \cdots \wedge \widehat{dx^{i_\alpha}} \wedge \cdots \wedge dx^{i_k} \tag{14.1}$$

where the hat marks an omitted factor. It satisfies the *homotopy identity*

$$d(K\omega) + K(d\omega) = \omega \qquad \text{for every } \omega \in \Omega^k(U),\ k \geq 1 \tag{14.2}$$

***Proof*:**
The operator is assembled from the radial contraction and the two ends of the interval that parametrizes it. Introduce the smooth map

$$H\colon U \times [0,1] \to U, \qquad H(x,t) = tx$$

which is well defined precisely because U is star-shaped about 0; it has $H(x,1) = x$ and $H(x,0) = 0$. Write the two end inclusions $\iota_s\colon U \to U \times [0,1]$, $\iota_s(x) = (x,s)$, for $s \in \{0,1\}$, so that $H \circ \iota_1 = \mathrm{id}_U$ and $H \circ \iota_0$ is the constant map c_0 sending every point to 0.

Step 1: The dt-decomposition and the operator. Use coordinates $x^1, \ldots, x^n$ on U and t on $[0,1]$. Every form $\eta \in \Omega^m(U \times [0,1])$ splits uniquely as

$$\eta = \beta + dt \wedge \gamma$$

where β and γ contain no dt: collecting the coordinate basis forms dx^I that omit dt into β and those that contain it, each of the latter written $dt \wedge dx^J$, exhibits the split, and uniqueness is the linear independence of the basis. Define the *fiber integral*

$$Q\eta = \int_0^1 \gamma \, dt \in \Omega^{m-1}(U)$$

by integrating each coefficient of γ over $t \in [0,1]$; the result is a smooth form on U, since differentiation under the integral sign applies to the coefficient functions, which are smooth on $U \times [0,1]$. Set $K\omega = Q(H^*\omega)$.

To read off (14.1), compute $H^*\omega$ for a single term $\omega = a\, dx^I$. The component functions of H are $x^i \circ H = tx^i$, so $H^*(dx^i) = d(tx^i) = t\, dx^i + x^i\, dt$ and $H^*a = a(tx)$ by proposition 12.4.11. Hence

$$H^*\omega = a(tx)\, \big(t\, dx^{i_1} + x^{i_1} dt\big) \wedge \cdots \wedge \big(t\, dx^{i_k} + x^{i_k} dt\big)$$

Since $dt \wedge dt = 0$, at most one factor may contribute its dt summand. The term with no dt is $a(tx)\, t^k\, dx^I$; the term taking dt from the α-th factor is $a(tx)\, t^{k-1} x^{i_\alpha}$ times $dx^{i_1} \wedge \cdots \wedge dt \wedge \cdots \wedge dx^{i_k}$

with dt in slot α. Moving dt to the front past $\alpha - 1$ one-forms contributes the sign $(-1)^{\alpha-1}$, so

$$H^*\omega = a(tx)\, t^k\, dx^I + dt \wedge \sum_{\alpha=1}^{k} (-1)^{\alpha-1} a(tx)\, t^{k-1} x^{i_\alpha}\, dx^{i_1} \wedge \cdots \wedge \widehat{dx^{i_\alpha}} \wedge \cdots \wedge dx^{i_k}$$

Thus $\beta = a(tx)\, t^k\, dx^I$ carries no dt, and integrating the dt-part over t gives $K\omega = Q(H^*\omega) = \sum_\alpha (-1)^{\alpha-1} \big(\int_0^1 t^{k-1} a(tx)\, dt\big) x^{i_\alpha}\, dx^{i_1} \wedge \cdots \wedge \widehat{dx^{i_\alpha}} \wedge \cdots \wedge dx^{i_k}$. Summing over I by linearity yields (14.1).

Step 2: The fiber-integral identity. The claim is that for every $\eta \in \Omega^m(U \times [0,1])$,

$$d(Q\eta) + Q(d\eta) = \iota_1^*\eta - \iota_0^*\eta \tag{14.3}$$

Write $\eta = \beta + dt \wedge \gamma$ with $\beta = \sum_I b_I(x,t)\, dx^I$ and $\gamma = \sum_J c_J(x,t)\, dx^J$. The exterior derivative on $U \times [0,1]$ separates the t-derivative from the derivative in the x-directions: for a function g, $dg = d_U g + \frac{\partial g}{\partial t}\, dt$, where d_U differentiates only in x. Applying this to the coefficients,

$$d\beta = d_U\beta + dt \wedge \sum_I \frac{\partial b_I}{\partial t}\, dx^I, \qquad d(dt \wedge \gamma) = -\, dt \wedge d\gamma = -\, dt \wedge d_U\gamma$$

the last equality because the dt-summand of $d\gamma$ meets the leading dt and dies. Adding, the part of $d\eta$ that carries dt is $\sum_I (\partial b_I/\partial t)\, dx^I - d_U\gamma$, so

$$Q(d\eta) = \int_0^1 \Big(\sum_I \frac{\partial b_I}{\partial t}\, dx^I - d_U\gamma\Big)\, dt = \sum_I \Big(\int_0^1 \frac{\partial b_I}{\partial t}\, dt\Big) dx^I - \int_0^1 d_U\gamma\, dt$$

The fundamental theorem of calculus evaluates the first integral: $\int_0^1 (\partial b_I/\partial t)\, dt = b_I(x,1) - b_I(x,0)$, and since $\iota_s^*(dt \wedge \gamma) = 0$ (as $\iota_s^* dt = d(s) = 0$) the pullback $\iota_s^*\eta = \iota_s^*\beta = \sum_I b_I(x,s)\, dx^I$ picks out exactly the value at $t = s$. Hence the first sum is $\iota_1^*\eta - \iota_0^*\eta$. In the second, d_U acts only in x and so commutes with $\int_0^1 (\cdot)\, dt$, giving $\int_0^1 d_U\gamma\, dt = d_U \int_0^1 \gamma\, dt = d(Q\eta)$. Combining the two,

$$Q(d\eta) = \big(\iota_1^*\eta - \iota_0^*\eta\big) - d(Q\eta)$$

which rearranges to (14.3).

Step 3: Specialization to the radial homotopy. Apply (14.3) to $\eta = H^*\omega$. Naturality of the exterior derivative (proposition 12.5.4) gives $d(H^*\omega) = H^*(d\omega)$, so $Q(d\eta) = Q(H^*d\omega) = K(d\omega)$, while $d(Q\eta) = d(K\omega)$ by definition. The right side of (14.3) is

$$\iota_1^* H^*\omega - \iota_0^* H^*\omega = (H \circ \iota_1)^*\omega - (H \circ \iota_0)^*\omega = \mathrm{id}_U^*\omega - c_0^*\omega$$

using functoriality of pullback (proposition 12.4.11). The first term is ω. For the second, the constant map c_0 has $c_0^*(dx^i) = d(x^i \circ c_0) = d(0) = 0$, so by multiplicativity of pullback c_0^* annihilates every form of degree ≥ 1; as $k \geq 1$, $c_0^*\omega = 0$. Therefore $d(K\omega) + K(d\omega) = \omega$, which is (14.2), completing the proof.

The identity (14.2) is a chain homotopy between the identity and the zero map on the de Rham complex of U in positive degrees, with K the homotopy. Chain-homotopic maps agree on cohomology, and the zero map sends everything to zero, so the positive-degree cohomology of U collapses. The Poincaré lemma is the immediate consequence, and its proof is now a single line.

Theorem 14.2.4: The Poincaré Lemma

Let $U \subseteq \mathbb{R}^n$ be open and star-shaped. Then every closed k-form on U with $k \geq 1$ is exact. Explicitly, if $\omega \in \Omega^k(U)$ satisfies $d\omega = 0$ and $k \geq 1$, then $\omega = d(K\omega)$ with $K\omega$ the $(k-1)$-form of (14.1).

***Proof*:**
Translating coordinates, assume U is star-shaped about 0, so the operator K of lemma 14.2.3 is available. Let ω be closed of degree $k \geq 1$. The homotopy identity (14.2) reads $d(K\omega)+K(d\omega) = \omega$, and $K(d\omega) = K(0) = 0$ since ω is closed. Hence $\omega = d(K\omega)$, exhibiting $K\omega \in \Omega^{k-1}(U)$ as a primitive, as we sought to show.

The lemma converts the abstract quotient H^k_{dR} into a computation on the simplest spaces, and the formula (14.1) makes the primitive explicit rather than asserting it exists.

Example 14.2.5: A Primitive by the Homotopy Operator

On $\mathbb{R}^3$, star-shaped about 0, consider

$$\omega = (2xy + z)\,dx + (x^2 + z)\,dy + (x + y)\,dz$$

a 1-form. It is closed: the mixed partials match, $\partial_y(2xy + z) = 2x = \partial_x(x^2 + z)$, $\partial_z(2xy + z) = 1 = \partial_x(x + y)$, and $\partial_z(x^2 + z) = 1 = \partial_y(x + y)$, so $d\omega = 0$. For a 1-form ($k = 1$) the weight t^{k-1} is $t^0 = 1$, and (14.1) collapses to the single function

$$K\omega = \int_0^1 \big(x\,a_1(tx, ty, tz) + y\,a_2(tx, ty, tz) + z\,a_3(tx, ty, tz)\big)\,dt$$

with $a_1 = 2xy + z$, $a_2 = x^2 + z$, $a_3 = x + y$. Substituting the scaled arguments,

$$x\,a_1(t\cdot) = x\big(2t^2xy + tz\big), \quad y\,a_2(t\cdot) = y\big(t^2x^2 + tz\big), \quad z\,a_3(t\cdot) = z\big(tx + ty\big)$$

whose sum is $t^2\big(2x^2y+x^2y\big)+t\big(xz+yz+xz+yz\big) = 3t^2x^2y+2t(xz+yz)$. Integrating, $\int_0^1 3t^2\,dt = 1$ and $\int_0^1 2t\,dt = 1$, so

$$K\omega = x^2y + xz + yz$$

Direct differentiation confirms $d(x^2y + xz + yz) = (2xy + z)\,dx + (x^2 + z)\,dy + (x + y)\,dz = \omega$. The operator has produced the unique primitive vanishing at the origin.

For a star-shaped set the Poincaré lemma computes the entire de Rham cohomology, degree by degree.

Corollary 14.2.6: Cohomology of a Star-Shaped Set

Let $U \subseteq \mathbb{R}^n$ be open and star-shaped (for instance $U = \mathbb{R}^n$). Then

$$H^k_{\mathrm{dR}}(U) \cong \begin{cases} \mathbb{R}, & k = 0 \\ 0, & k \geq 1 \end{cases}$$

In particular $H^k_{\mathrm{dR}}(\mathbb{R}^n) = 0$ for all $k \geq 1$ and $H^0_{\mathrm{dR}}(\mathbb{R}^n) = \mathbb{R}$.

***Proof*:**
In degree $k \geq 1$ the Poincaré lemma (theorem 14.2.4) says every closed k-form is exact, so $Z^k(U) = B^k(U)$ and $H^k_{\mathrm{dR}}(U) = Z^k(U)/B^k(U) = 0$. In degree 0 there are no nonzero exact forms (nothing maps into Ω^0 under d), and $H^0_{\mathrm{dR}}(U) = \mathbb{R}^{\#\pi_0(U)}$ counts connected components by proposition 14.1.5. A star-shaped set is path-connected, since every point is joined to the center by the segment inside U, so it has a single component and $H^0_{\mathrm{dR}}(U) = \mathbb{R}$. This gives the stated values, and $\mathbb{R}^n$ is convex, hence star-shaped, completing the proof.

Euclidean space has no de Rham cohomology beyond the constants: its only closed-but-inexact obstruction is the trivial one in degree zero. Every interesting class in $H^*_{\mathrm{dR}}(M)$ must therefore come from the way M is glued together from Euclidean pieces, not from the pieces themselves. This is what makes the cohomology a genuine topological invariant, and it is the base case on which the inductive computations of section 14.4 rest.

Applied in a chart, the lemma gives a local statement that holds on every manifold, with no star-shapedness hypothesis on M itself.

Corollary 14.2.7: Local Exactness of Closed Forms

Let $\omega \in \Omega^k(M)$ be closed with $k \geq 1$. Then every point of M has an open neighborhood on which ω is exact.

***Proof*:**
Fix $p \in M$ and a chart $\varphi\colon V \to \varphi(V) \subseteq \mathbb{R}^n$ with $p \in V$. Shrinking V, take $\varphi(V)$ to be an open ball, which is convex and hence star-shaped. The pullback $(\varphi^{-1})^*\omega$ is a closed k-form on $\varphi(V)$ by naturality of d (proposition 12.5.4), so by theorem 14.2.4 it equals $d\eta$ for some $\eta \in \Omega^{k-1}(\varphi(V))$. Pulling back along φ and using naturality again, $\omega = d(\varphi^*\eta)$ on V, so ω is exact on the neighborhood V of p, as we sought to show.

Local exactness is the reason the passage from closed to exact is a global question at all: the failure of a closed form to be exact is never visible in any single chart, only in how the local primitives fail to agree on overlaps. The angle form of example 13.4.2 is the model, closed on $\mathbb{R}^2 \setminus \{0\}$ and locally the differential of a branch of the argument, yet with no global primitive because those branches differ by 2π around the puncture. Assembling the local primitives into a global obstruction is precisely what the next two sections mechanize, first through homotopy and then through the Mayer-Vietoris sequence.

14.2.8 Foreshadowing: The de Rham Complex as a Resolution Local exactness has a sheaf-theoretic reading. The exterior derivatives $\mathbb{R} \hookrightarrow \Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \cdots$ form, on small enough open sets, an exact sequence: the kernel of the first d is the locally constant functions $\mathbb{R}$ (proposition 14.1.5), and corollary 14.2.7 makes the sequence exact at every later stage locally. The de Rham complex is thus a resolution of the constant sheaf $\mathbb{R}$ by the sheaves of forms, and de Rham cohomology is the cohomology of that resolution. This is the entry point to sheaf cohomology, taken up in chapter 15 and its companion volumes.

Exercise 14.2.1

1. On $\mathbb{R}^3$ let $\omega = (y + 2xz)\,dx + x\,dy + x^2\,dz$. Verify that ω is closed, then apply the homotopy operator K of (14.1) to compute a primitive f with $df = \omega$, and check the answer by differentiating.
2. On $\mathbb{R}^3$ compute $K(dx \wedge dy)$ from (14.1), and verify directly that $d\big(K(dx \wedge dy)\big) = dx \wedge dy$.
3. Prove that a star-shaped set is path-connected, and deduce from proposition 14.1.5 that $H^0_{\mathrm{dR}}(U) = \mathbb{R}$ for every star-shaped open $U \subseteq \mathbb{R}^n$. Show by example that a disjoint union of two open balls in $\mathbb{R}^n$ is not star-shaped.
4. Show that the punctured plane $\mathbb{R}^2 \setminus \{0\}$ is star-shaped about no point. Using the angle form of example 13.4.2, explain why the conclusion of the Poincaré lemma fails on $\mathbb{R}^2 \setminus \{0\}$, and identify which hypothesis of theorem 14.2.4 is violated.
5. On $\mathbb{R}^2$ let $\omega = x\,dy$, which is *not* closed. Compute $K\omega$, $d(K\omega)$, $d\omega$, and $K(d\omega)$ separately, and confirm that $d(K\omega) + K(d\omega) = \omega$. Explain why this verifies more than the Poincaré lemma asserts.
6. A closed 1-form ω on a manifold M is exact on a neighborhood of every point by corollary 14.2.7. Explain why this does not force ω to be exact on all of M, and describe what the collection of local primitives fails to do in the case of the angle form on $\mathbb{R}^2 \setminus \{0\}$.

14.3 Homotopy Invariance

The Poincaré lemma (theorem 14.2.4) killed the cohomology of a star-shaped set by contracting it to its center along the straight-line homotopy, then integrating a closed form along that contraction to write it as d of something. The mechanism deserves to be isolated from the accident of a Euclidean domain. Any homotopy at all, not just a linear contraction, produces a comparison between the two maps it interpolates, and the comparison lives one degree down: a smooth homotopy from f to g manufactures an operator whose exterior derivative is $g^* - f^*$ on forms. On closed forms the difference becomes exact, so f and g agree on cohomology.

The consequence is that de Rham cohomology cannot distinguish two maps that are smoothly homotopic, nor two manifolds that are homotopy equivalent. It sees a manifold only up to homotopy type. This upgrades the period-level statement of corollary 13.4.4, that deforming a cycle leaves the integral of a closed form unchanged, into a statement about the cohomology classes themselves, and it delivers the converse the Poincaré lemma pointed at: cohomology is trivial wherever the manifold can be contracted.

Smooth homotopy of maps was defined in definition 7.5.1: a smooth map $F\colon M \times [0,1] \to N$ with $F(\cdot,0) = f$ and $F(\cdot,1) = g$, written $f \simeq g$. The relation between whole manifolds that homotopy invariance turns into an isomorphism is the following.

Definition 14.3.1: Smooth Homotopy Equivalence

Smooth manifolds M and N are *smoothly homotopy equivalent* if there are smooth maps $f\colon M \to N$ and $h\colon N \to M$ with

$$h \circ f \simeq \mathrm{id}_M \qquad \text{and} \qquad f \circ h \simeq \mathrm{id}_N$$

Such an h is a *homotopy inverse* of f. A manifold M is *contractible* if it is smoothly homotopy equivalent to a one-point space, equivalently if id_M is smoothly homotopic to a constant map. A *smooth deformation retraction* of M onto a subset $A \subseteq M$ is a smooth map $r\colon M \times [0,1] \to M$ with $r(\cdot,0) = \mathrm{id}_M$, with $r(\cdot,1)$ taking values in A, and with $r(a,t) = a$ for all $a \in A$ and t; the subset A is then a *deformation retract* of M.

Homotopy equivalence is looser than diffeomorphism: it allows the dimension to drop, collapsing a manifold onto a lower-dimensional core as long as the collapse can be run smoothly through a one-parameter family. A deformation retraction is the concrete way such a collapse usually presents itself; the map $\iota\colon A \hookrightarrow M$ and the time-one map $r(\cdot,1)\colon M \to A$ are then homotopy inverses, since $r(\cdot,1) \circ \iota = \mathrm{id}_A$ on the nose and $\iota \circ r(\cdot,1) = r(\cdot,1) \simeq r(\cdot,0) = \mathrm{id}_M$ through r itself.

Example 14.3.2: Convex Sets Are Contractible

Let $U \subseteq \mathbb{R}^n$ be convex, and fix a point $x_0 \in U$. The straight-line homotopy

$$F\colon U \times [0,1] \to U, \qquad F(x,t) = (1-t)\,x + t\,x_0$$

is smooth, stays in U by convexity, and satisfies $F(\cdot,0) = \mathrm{id}_U$ and $F(\cdot,1) \equiv x_0$. Thus id_U is smoothly homotopic to the constant map at x_0, so U is contractible. This is the very homotopy that ran the homotopy operator of the Poincaré lemma; read now as a statement about maps rather than about one form at a time, it says U has the homotopy type of a point. The same formula with x_0 the center shows a star-shaped open set is contractible, since the segment from x to the center lies in the set by definition of star-shaped.

The engine behind homotopy invariance is a single algebraic identity on the cylinder $M \times [0,1]$. Write t for the coordinate on the $[0,1]$ factor, $\partial_t = \partial/\partial t$ for the corresponding vector field, and

$$i_0, i_1\colon M \to M \times [0,1], \qquad i_s(x) = (x,s)$$

for the inclusions of the bottom and top slices. Every form on the cylinder splits, in a coordinate chart $(x^1,\dots,x^n)$ on M together with t, into a part containing no dt and a part that does. The operator below integrates the dt-part along the fiber $[0,1]$ and discards the rest, dropping the degree by one.

Lemma 14.3.3: The Homotopy Operator on a Cylinder

For each $k \geq 0$ (with the convention $\Omega^{-1}(M) = 0$) define the *homotopy operator* $K\colon \Omega^k(M \times [0,1]) \to \Omega^{k-1}(M)$ by

$$K\omega = \int_0^1 i_t^*\big(\iota_{\partial_t}\omega\big)\,dt$$

where ι_{∂_t} is the interior product (definition 12.4.7) and the integral is taken coefficientwise in t. Then, on $\Omega^k(M \times [0,1])$,

$$i_1^* - i_0^* = d\,K + K\,d \tag{14.4}$$

***Proof*:**
Both sides of (14.4) are $\mathbb{R}$-linear in ω and commute with restriction to open subsets of M, so it suffices to verify the identity on a chart $U \times [0,1]$, where U carries coordinates $(x^1, \ldots, x^n)$. There every k-form is a finite sum of monomials of two kinds,

$$\omega = f(x,t)\,dx^I \qquad \text{and} \qquad \omega = g(x,t)\,dt \wedge dx^J$$

with $|I| = k$ and $|J| = k-1$; the first contains no dt, the second does. The exterior derivative on the cylinder splits accordingly as $d = d_M + dt \wedge \partial_t$, where d_M differentiates only in the x-variables and ∂_t differentiates coefficients in t. It is enough to check (14.4) on each kind.

Step 1: The type $f\,dx^I$. Here $\iota_{\partial_t}\omega = 0$, so $K\omega = 0$ and $dK\omega = 0$; the whole left contribution comes through Kd. Its slice pullbacks are $i_s^*\omega = f(x,s)\,dx^I$, whence

$$i_1^*\omega - i_0^*\omega = \big(f(x,1) - f(x,0)\big)\,dx^I$$

On the other side,

$$d\omega = d_M f \wedge dx^I + \partial_t f\,dt \wedge dx^I$$

The operator K annihilates the first term (no dt) and, on the second, integrates the coefficient and drops dt:

$$Kd\omega = \left(\int_0^1 \partial_t f(x,t)\,dt\right) dx^I = \big(f(x,1) - f(x,0)\big)\,dx^I$$

by the fundamental theorem of calculus. Thus $dK\omega + Kd\omega = i_1^*\omega - i_0^*\omega$.

Step 2: The type $g\,dt \wedge dx^J$. Since $t \circ i_s \equiv s$ is constant, $i_s^*(dt) = 0$, so $i_s^*\omega = 0$ for both s, and the left side of (14.4) vanishes. It remains to see that $dK\omega + Kd\omega = 0$. Write $G(x) = \int_0^1 g(x,t)\,dt$. Contracting, $\iota_{\partial_t}\omega = g\,dx^J$, so $K\omega = G\,dx^J$ and

$$dK\omega = d_M G \wedge dx^J = \sum_i \left(\int_0^1 \partial_{x^i} g(x,t)\,dt\right) dx^i \wedge dx^J$$

using differentiation under the integral sign, valid because g is smooth and $[0,1]$ is compact. For the other term, $dg \wedge dt \wedge dx^J = d_M g \wedge dt \wedge dx^J$ because $dt \wedge dt = 0$, and moving dt past the single factor dx^i costs one sign,

$$d\omega = \sum_i \partial_{x^i} g\,dx^i \wedge dt \wedge dx^J = -\sum_i \partial_{x^i} g\,dt \wedge dx^i \wedge dx^J$$

Applying K integrates each coefficient and drops the leading dt:

$$K\,d\omega = -\sum_i \left(\int_0^1 \partial_{x^i} g(x,t)\,dt\right) dx^i \wedge dx^J$$

This is the negative of $dK\omega$, so $dK\omega + Kd\omega = 0$, matching the left side. Both types satisfy (14.4), and by linearity so does every form, completing the proof.

The identity (14.4) is the integrated form of Cartan's formula (theorem 12.5.6): where $\mathcal{L}_X = d\iota_X + \iota_X d$ measures the infinitesimal change of a form along the flow of X, the operator K integrates the interior product ι_{∂_t} over the whole fiber and produces the finite change between the two ends of the cylinder. The two slice inclusions i_0 and i_1 are what a homotopy pulls apart, and (14.4) says they are cochain-homotopic. Composing with the homotopy itself carries this from the cylinder down to the maps.

Theorem 14.3.4: Homotopy Invariance

Let $f, g\colon M \to N$ be smooth maps. If f and g are smoothly homotopic, then they induce the same map on de Rham cohomology:

$$f^* = g^*\colon H^k_{\mathrm{dR}}(N) \to H^k_{\mathrm{dR}}(M) \qquad \text{for every } k$$

***Proof*:**
Choose a smooth homotopy $F\colon M \times [0,1] \to N$ with $F \circ i_0 = f$ and $F \circ i_1 = g$. Define

$$\mathcal{K} = K \circ F^*\colon \Omega^k(N) \to \Omega^{k-1}(M)$$

with K the homotopy operator of lemma 14.3.3. For any $\omega \in \Omega^k(N)$, apply (14.4) to $F^*\omega \in \Omega^k(M \times [0,1])$:

$$i_1^*(F^*\omega) - i_0^*(F^*\omega) = d\,K(F^*\omega) + K\,d(F^*\omega)$$

Functoriality of pullback (proposition 12.4.11) turns the left side into $(F \circ i_1)^*\omega - (F \circ i_0)^*\omega = g^*\omega - f^*\omega$. On the right, $K(F^*\omega) = \mathcal{K}\omega$, and because F^* commutes with the exterior derivative (proposition 12.5.4) one has $Kd(F^*\omega) = K(F^*d\omega) = \mathcal{K}(d\omega)$. Therefore

$$g^*\omega - f^*\omega = d(\mathcal{K}\omega) + \mathcal{K}(d\omega) \tag{14.5}$$

for every form ω on N. Now let ω be closed, $d\omega = 0$, representing a class $[\omega] \in H^k_{\mathrm{dR}}(N)$. The second term of (14.5) drops out, leaving $g^*\omega - f^*\omega = d(\mathcal{K}\omega)$, an exact form. Hence $[g^*\omega] = [f^*\omega]$ in $H^k_{\mathrm{dR}}(M)$, and since f^* and g^* on cohomology are computed on representatives, they coincide, as we sought to show.

The operator $\mathcal{K}$ of (14.5) is a *cochain homotopy* between f^* and g^*: it witnesses their equality on cohomology without their being equal on forms. This is the standard homological device, and it says something sharp. Two manifolds that look different but are homotopy equivalent have, by the same computation applied twice, the same de Rham cohomology.

Corollary 14.3.5: Cohomology Is a Homotopy Invariant

If M and N are smoothly homotopy equivalent, then $H^*_{\mathrm{dR}}(M) \cong H^*_{\mathrm{dR}}(N)$ as graded $\mathbb{R}$-algebras. In particular, if A is a smooth deformation retract of M, then the inclusion $\iota\colon A \hookrightarrow M$ induces an isomorphism $\iota^*\colon H^*_{\mathrm{dR}}(M) \to H^*_{\mathrm{dR}}(A)$.

***Proof*:**
Let $f\colon M \to N$ and $h\colon N \to M$ be homotopy inverses, so $h \circ f \simeq \mathrm{id}_M$ and $f \circ h \simeq \mathrm{id}_N$. Pullback is contravariantly functorial (proposition 12.4.11), so $f^* \circ h^* = (h \circ f)^*$ and $h^* \circ f^* = (f \circ h)^*$ on cohomology. By theorem 14.3.4 the homotopic composites induce the same maps as the identities:

$$f^* \circ h^* = (h \circ f)^* = \mathrm{id}_M^* = \mathrm{id}, \qquad h^* \circ f^* = (f \circ h)^* = \mathrm{id}_N^* = \mathrm{id}$$

Thus f^* and h^* are mutually inverse. Each is a ring homomorphism (proposition 14.1.4), so $f^*\colon H^*_{\mathrm{dR}}(N) \to H^*_{\mathrm{dR}}(M)$ is an isomorphism of graded algebras. For a deformation retraction r, the maps ι and $r(\cdot, 1)$ are homotopy inverses as noted after definition 14.3.1, giving the stated isomorphism, completing the proof.

Contractibility is the extreme case: a manifold with the homotopy type of a point has the cohomology of a point.

Corollary 14.3.6: Cohomology of a Contractible Manifold

If M is contractible, then

$$H^k_{\mathrm{dR}}(M) \cong \begin{cases} \mathbb{R} & k = 0 \\ 0 & k \geq 1 \end{cases}$$

***Proof*:**
A one-point space $\{p\}$ has $\Omega^0(\{p\}) = \mathbb{R}$ and $\Omega^k(\{p\}) = 0$ for $k \geq 1$, so its only nonzero cohomology is $H^0_{\mathrm{dR}}(\{p\}) = \mathbb{R}$. A contractible M is smoothly homotopy equivalent to $\{p\}$ by definition, so corollary 14.3.5 transports these groups to M. The degree-zero value also reads off directly from proposition 14.1.5, since a contractible manifold is connected, completing the proof.

For a star-shaped open $U \subseteq \mathbb{R}^n$, example 14.3.2 makes U contractible, so corollary 14.3.6 returns $H^k_{\mathrm{dR}}(U) = 0$ for $k \geq 1$. This reproves the Poincaré lemma (theorem 14.2.4) and its corollary $H^k_{\mathrm{dR}}(\mathbb{R}^n) = 0$ (corollary 14.2.6), now as the special case of a general principle: the homotopy operator on the cylinder specializes, when the homotopy is the linear contraction, to the radial homotopy operator that proved the lemma by hand. The two arguments are the same argument at different levels of generality.

Everything above was carried out in the smooth category, with smooth maps and smooth homotopies. The restriction is only apparent.

14.3.7 Note: De Rham Cohomology Is a Topological Invariant A continuous map between smooth manifolds is homotopic (continuously) to a smooth one, and two smooth maps that are continuously homotopic are smoothly homotopic, both by the Whitney approximation theorem (theorem 7.9.1). Consequently smoothly homotopy equivalent and continuously homotopy equivalent mean the same thing for smooth manifolds, and H^*_{dR} depends only on the underlying topological space up to homotopy: it is a topological invariant, computed by a smooth apparatus. The de Rham theorem (section 14.5) will make this precise by identifying it with a cohomology built from continuous data alone.

The first genuine computation the theorem enables, before the machinery of the next section, is the cohomology of a punctured Euclidean space, which collapses onto a sphere.

Example 14.3.8: Punctured Euclidean Space

For $n \geq 1$ the punctured space $\mathbb{R}^n \setminus \{0\}$ deformation-retracts onto the unit sphere S^{n-1}. The retraction pushes each point radially inward or outward to the sphere:

$$r\colon \big(\mathbb{R}^n \setminus \{0\}\big) \times [0,1] \to \mathbb{R}^n \setminus \{0\}, \qquad r(x,s) = (1-s)\,x + s\,\frac{x}{|x|}$$

This is smooth on its domain because $|x|$ never vanishes there. It stays off the origin: writing $r(x,s) = \lambda(x,s)\,x$ with $\lambda(x,s) = (1-s) + s/|x|$, both summands are nonnegative and they are never simultaneously zero (for $s < 1$ the first is positive, and at $s = 1$ the second is $1/|x| > 0$), so $\lambda(x,s) > 0$ and $r(x,s) \neq 0$. At $s = 0$ it is the identity, at $s = 1$ it lands on S^{n-1} since $|x/|x|| = 1$, and for $x \in S^{n-1}$ one has $|x| = 1$ and $r(x,s) = x$ throughout. Thus r is a smooth deformation retraction, and corollary 14.3.5 gives

$$H^k_{\mathrm{dR}}\big(\mathbb{R}^n \setminus \{0\}\big) \cong H^k_{\mathrm{dR}}(S^{n-1}) \qquad \text{for every } k$$

The punctured space is not contractible for $n \geq 2$, so this cohomology is not that of a point. In the plane, $\mathbb{R}^2 \setminus \{0\} \simeq S^1$, and the class detecting the hole is that of the angle form $d\theta = (-y\,dx + x\,dy)/(x^2+y^2)$, whose period around the origin is 2π (example 13.4.2); it is closed, not exact, and generates $H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}) \cong \mathbb{R}$. The retraction has turned the two-dimensional puncture into a one-dimensional circle without changing what the closed forms detect.

The cohomology of the spheres $H^*_{\mathrm{dR}}(S^{n-1})$ on the right-hand side is computed in section 14.4 by the Mayer-Vietoris sequence; the example has reduced a family of open subsets of $\mathbb{R}^n$ to that single computation.

Exercise 14.3.1

1. On $M = \mathbb{R}$ with coordinate x, take the 1-form $\omega = xt^2\,dt$ on the cylinder $\mathbb{R} \times [0,1]$. Compute $K\omega$ from lemma 14.3.3, then compute both sides of the identity $i_1^*\omega - i_0^*\omega = dK\omega + Kd\omega$ directly and confirm they agree.
2. Call a smooth map $f\colon M \to N$ *null-homotopic* if it is smoothly homotopic to a constant map. Show that a null-homotopic f induces the zero map $f^* = 0$ on $H^k_{\mathrm{dR}}(N)$ for every $k \geq 1$, and deduce that any smooth map from a contractible manifold M to any N induces the zero map on positive-degree cohomology.

3. Verify in detail that the map r of example 14.3.8 is smooth, that $r(\cdot, 1)$ maps $\mathbb{R}^n \setminus \{0\}$ into S^{n-1}, and that $r(x, s) = x$ for $x \in S^{n-1}$. Conclude that the inclusion $S^{n-1} \hookrightarrow \mathbb{R}^n \setminus \{0\}$ induces an isomorphism on de Rham cohomology.

4. Show that smooth homotopy is an equivalence relation on the set of smooth maps $M \to N$. For transitivity, given $f \simeq g$ and $g \simeq h$, build a single smooth homotopy $f \simeq h$; you will need to reparametrize each given homotopy so that it is constant in t near the endpoints before concatenating.

5. Let N be a smooth manifold. Show that the projection $\pi \colon N \times \mathbb{R} \to N$ and the inclusion $j \colon N \to N \times \mathbb{R}$, $j(y) = (y, 0)$, are homotopy inverses, and conclude $H^*_{\mathrm{dR}}(N \times \mathbb{R}) \cong H^*_{\mathrm{dR}}(N)$.

14.4 The Mayer-Vietoris Sequence

The Poincaré lemma (theorem 14.2.4) and homotopy invariance (theorem 14.3.4) compute de Rham cohomology wherever a manifold is contractible, and there the answer is trivial. A manifold with holes is not contractible, but it can be cut into pieces that are, and the problem becomes one of reassembly: given a cover $M = U \cup V$ by open sets, how do the cohomologies of U, V, and $U \cap V$ determine that of M. This section produces the exact sequence that answers the question. It rests on a single short exact sequence of forms, built from restriction and a partition of unity, and on the passage from a short exact sequence of complexes to a long exact sequence in cohomology. With it, the cohomology of the spheres, the tori, and the punctured spaces is computed by hand, reducing each to the Poincaré lemma applied on contractible pieces.

Proposition 14.4.1: The Mayer-Vietoris Short Exact Sequence

Let M be a smooth manifold with an open cover $M = U \cup V$, and write the inclusions of $U \cap V$ into U and V and of U, V into M. For each k the sequence of restriction maps

$$0 \to \Omega^k(M) \xrightarrow{i} \Omega^k(U) \oplus \Omega^k(V) \xrightarrow{j} \Omega^k(U \cap V) \to 0$$

with $i(\omega) = (\omega|_U, \omega|_V)$ and $j(\alpha, \beta) = \beta|_{U\cap V} - \alpha|_{U\cap V}$ is exact. The maps i and j commute with d, so this is a short exact sequence of cochain complexes.

***Proof*:**
Restriction of a form to an open set is its pullback along the inclusion, so i and j commute with the exterior derivative by naturality (proposition 12.5.4); this gives the cochain-map claim once exactness is shown. Exactness is checked at each of the three positions.

Injectivity of i. If $i(\omega) = 0$ then $\omega|_U = 0$ and $\omega|_V = 0$. Since $\{U, V\}$ covers M, every point lies in U or V, so $\omega = 0$ on M.

Exactness at the middle. For $\omega \in \Omega^k(M)$,

$$j(i(\omega)) = \omega|_V|_{U\cap V} - \omega|_U|_{U\cap V} = \omega|_{U\cap V} - \omega|_{U\cap V} = 0$$

so im$i \subseteq \ker j$. Conversely, suppose $j(\alpha, \beta) = 0$, that is α and β agree on $U \cap V$. Define ω on M by $\omega = \alpha$ on U and $\omega = \beta$ on V. The two definitions agree on the overlap $U \cap V$, and U, V are open, so ω is a well-defined smooth k-form on M with $i(\omega) = (\alpha, \beta)$. Thus $\ker j \subseteq \mathrm{im} i$.

Surjectivity of j. Let $\{\rho_U, \rho_V\}$ be a smooth partition of unity subordinate to $\{U, V\}$ (theorem 1.5.5), so $\rho_U + \rho_V = 1$ on M, supp$\rho_U \subseteq U$, and supp$\rho_V \subseteq V$. Given $\omega \in \Omega^k(U \cap V)$, the form $\rho_U\omega$ is defined on $U \cap V$ and has support contained in supp$\rho_U \subseteq U$. It extends by zero to a smooth k-form on V: on the open set $V \setminus \mathrm{supp}\rho_U$ it is set to 0, on the open set $U \cap V$ it equals $\rho_U\omega$, and the two agree where they overlap since ρ_U vanishes there; these two open sets cover V, because a point of V outside $U \cap V$ lies outside $U \supseteq \mathrm{supp}\rho_U$. Write $\widetilde{\rho_U\omega} \in \Omega^k(V)$ for this extension, and symmetrically $\widetilde{\rho_V\omega} \in \Omega^k(U)$ for the extension by zero of $\rho_V\omega$ across U. Set

$$\alpha = -\widetilde{\rho_V\omega} \in \Omega^k(U), \qquad \beta = \widetilde{\rho_U\omega} \in \Omega^k(V)$$

On $U \cap V$ the extensions restrict to $\rho_V\omega$ and $\rho_U\omega$, so

$$j(\alpha, \beta) = \beta|_{U\cap V} - \alpha|_{U\cap V} = \rho_U\omega + \rho_V\omega = (\rho_U + \rho_V)\omega = \omega$$

Hence $\omega \in \mathrm{im}j$, and j is surjective, completing the proof.

The three maps have direct readings. Injectivity of i says a form is determined by its restrictions to the pieces of a cover. Exactness at the middle says a pair of forms, one on U and one on V, patch to a global form exactly when they agree on the overlap. Surjectivity is the only place the smooth structure does real work: a form on the overlap need not extend to either piece, but the partition of unity splits it into two summands, each of which does, and these no longer agree on $U \cap V$. That failure to agree is precisely what the connecting map of the long exact sequence will measure. The concrete case to keep in mind is the circle covered by two arcs: U and V are arcs slightly larger than the upper and lower semicircles, each diffeomorphic to an interval, and $U \cap V$ is a pair of small arcs near the two endpoints, so $U \cap V$ has two components while $M = S^1$ has one.

The exterior derivative turns proposition 14.4.1 into a short exact sequence of cochain complexes, and a short exact sequence of complexes always induces a long exact sequence on cohomology. That passage is the zig-zag lemma of homological algebra, developed on the singular side in [11, Algebraic Snake Lemma]; applied here it produces the sequence below. The one piece special to the de Rham setting is the connecting map, which the proof makes explicit, since it is what the computations use.

Theorem 14.4.2: The de Rham Mayer-Vietoris Sequence

Let $M = U \cup V$ be an open cover of a smooth manifold. There is a long exact sequence

$$\cdots \to H^{k-1}_{\mathrm{dR}}(U\cap V) \xrightarrow{\delta} H^k_{\mathrm{dR}}(M) \xrightarrow{i^*} H^k_{\mathrm{dR}}(U) \oplus H^k_{\mathrm{dR}}(V) \xrightarrow{j^*} H^k_{\mathrm{dR}}(U\cap V) \xrightarrow{\delta} H^{k+1}_{\mathrm{dR}}(M) \to \cdots$$

where $i^*[\omega] = ([\omega|_U], [\omega|_V])$ and $j^*([\alpha], [\beta]) = [\beta|_{U\cap V}] - [\alpha|_{U\cap V}]$ are induced by the maps of proposition 14.4.1. For a partition of unity $\{\rho_U, \rho_V\}$ subordinate to $\{U, V\}$, the connecting homomorphism is

$$\delta[\omega] = [\, d\rho_U \wedge \omega \,]$$

the class of the closed $(k+1)$-form on M that equals $d\rho_U \wedge \omega$ on $U \cap V$ and 0 off a closed subset of $U \cap V$.

***Proof*:**
The maps i, j of proposition 14.4.1 commute with d, so

$$0 \to \Omega^*(M) \xrightarrow{i} \Omega^*(U) \oplus \Omega^*(V) \xrightarrow{j} \Omega^*(U \cap V) \to 0$$

is a short exact sequence of cochain complexes. The zig-zag lemma ([11, Algebraic Snake Lemma]) then yields the displayed long exact sequence with induced maps i^*, j^* on cohomology and a connecting homomorphism δ. It remains to compute δ in the present setting.

Construction of the class. Let $[\omega] \in H^k_{\mathrm{dR}}(U \cap V)$ with ω closed. The surjectivity argument of proposition 14.4.1 produces the lift $(\alpha, \beta) = (-\widetilde{\rho_V \omega}, \widetilde{\rho_U \omega})$ with $j(\alpha, \beta) = \omega$. Its image under d satisfies

$$j(d\alpha, d\beta) = d\beta|_{U\cap V} - d\alpha|_{U\cap V} = d\big(\beta|_{U\cap V} - \alpha|_{U\cap V}\big) = d\omega = 0$$

so $(d\alpha, d\beta) \in \ker j = \operatorname{im} i$. There is thus a unique $\eta \in \Omega^{k+1}(M)$ with $\eta|_U = d\alpha$ and $\eta|_V = d\beta$. It is closed: $(d\eta)|_U = d(d\alpha) = 0$ and $(d\eta)|_V = d(d\beta) = 0$, and $\{U, V\}$ covers M, so $d\eta = 0$. Tracing the zig-zag lemma, $\delta[\omega] = [\eta]$.

Explicit form. On U the representative is $\eta|_U = d\alpha = -d(\widetilde{\rho_V \omega})$, which on $U \cap V$ equals $-d(\rho_V \omega) = -d\rho_V \wedge \omega - \rho_V\, d\omega = -d\rho_V \wedge \omega$, using $d\omega = 0$. On $U \cap V$ the relation $\rho_U + \rho_V = 1$ gives $d\rho_U = -d\rho_V$, so $\eta = d\rho_U \wedge \omega$ there. The same computation on V gives $\eta|_V = d(\rho_U \omega) = d\rho_U \wedge \omega$ on $U \cap V$, agreeing. Thus η is the global closed form equal to $d\rho_U \wedge \omega$ on $U \cap V$ and to 0 elsewhere, its support contained in $\operatorname{supp}\rho_U \cap \operatorname{supp}\rho_V$, a closed subset of $U \cap V$. Therefore $\delta[\omega] = [d\rho_U \wedge \omega]$, as we sought to show.

The connecting map is the mechanism of the whole sequence. A class on the overlap is exact on each piece separately, by the partition of unity, but the two local primitives disagree, and the disagreement $d\rho_U \wedge \omega$ is a global closed form on M whose class obstructs assembling the local data into a global primitive. Where that obstruction vanishes, i^* and j^* recover $H^*(M)$ from the pieces by linear algebra alone. The sequence is exact in every degree, so each unknown group $H^k_{\mathrm{dR}}(M)$ is squeezed between its neighbours; when the pieces are chosen small enough to be contractible, their cohomology is the Poincaré lemma, and the sequence becomes a finite computation. This is the tool that makes de Rham cohomology calculable. The first payoff is the running example of the book.

Example 14.4.3: Cohomology of the Spheres
For $n \geq 1$,

$$H^k_{\mathrm{dR}}(S^n) \cong \begin{cases} \mathbb{R} & k = 0 \text{ or } k = n \\ 0 & \text{otherwise} \end{cases}$$

Cover S^n by $U = S^n \setminus \{p_-\}$ and $V = S^n \setminus \{p_+\}$, the complements of the south and north poles. Stereographic projection makes each of U, V diffeomorphic to $\mathbb{R}^n$, so by the Euclidean computation (corollary 14.2.6) both have $H^0_{\mathrm{dR}} = \mathbb{R}$ and $H^k_{\mathrm{dR}} = 0$ for $k \geq 1$. The overlap $U \cap V = S^n \setminus \{p_+, p_-\}$ deformation retracts onto the equator S^{n-1} by sliding along meridians, so by homotopy invariance (corollary 14.3.5) $H^k_{\mathrm{dR}}(U \cap V) \cong H^k_{\mathrm{dR}}(S^{n-1})$.

Since S^n is connected for $n \geq 1$, $H^0_{\mathrm{dR}}(S^n) = \mathbb{R}$ (proposition 14.1.5). For $k \geq 2$ the Mayer-Vietoris sequence around degree k reads

$$\underbrace{H^{k-1}_{\mathrm{dR}}(U) \oplus H^{k-1}_{\mathrm{dR}}(V)}_{=0} \to H^{k-1}_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} H^k_{\mathrm{dR}}(S^n) \to \underbrace{H^k_{\mathrm{dR}}(U) \oplus H^k_{\mathrm{dR}}(V)}_{=0}$$

both flanking groups vanishing because U, V are contractible and $k-1 \geq 1$. Exactness makes δ an isomorphism, so

$$H^k_{\mathrm{dR}}(S^n) \cong H^{k-1}_{\mathrm{dR}}(U \cap V) \cong H^{k-1}_{\mathrm{dR}}(S^{n-1}) \qquad (k \geq 2)$$

The remaining degree is $k = 1$, where the low end of the sequence, with $H^1(U) \oplus H^1(V) = 0$, is

$$0 \to H^0_{\mathrm{dR}}(S^n) \xrightarrow{i^*} H^0_{\mathrm{dR}}(U) \oplus H^0_{\mathrm{dR}}(V) \xrightarrow{j^*} H^0_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} H^1_{\mathrm{dR}}(S^n) \to 0$$

For $n \geq 2$ the equator S^{n-1} is connected, so $H^0_{\mathrm{dR}}(U \cap V) = \mathbb{R}$, and the sequence is $0 \to \mathbb{R} \to \mathbb{R}^2 \to \mathbb{R} \to H^1_{\mathrm{dR}}(S^n) \to 0$. Here i^* is injective with one-dimensional image, so by exactness $\ker j^*$ is one-dimensional and $\mathrm{im} j^*$ is $(2-1)$-dimensional, filling $H^0_{\mathrm{dR}}(U \cap V) = \mathbb{R}$; thus j^* is surjective, $\delta = 0$, and $H^1_{\mathrm{dR}}(S^n) = \mathrm{im}\delta = 0$. For $n = 1$ the intersection retracts onto S^0, two points, so $H^0_{\mathrm{dR}}(U \cap V) = \mathbb{R}^2$, and the sequence is $0 \to \mathbb{R} \to \mathbb{R}^2 \to \mathbb{R}^2 \to H^1_{\mathrm{dR}}(S^1) \to 0$. Now $\mathrm{im} i^*$ is the diagonal of $\mathbb{R}^2$, of dimension 1, so $\mathrm{im} j^*$ has dimension $2 - 1 = 1$ and the cokernel $\mathbb{R}^2/\mathrm{im} j^*$ has dimension 1; exactness identifies it with $H^1_{\mathrm{dR}}(S^1)$, giving $H^1_{\mathrm{dR}}(S^1) = \mathbb{R}$.

The two computations combine into an induction on n. The base case S^1 has $H^0 = H^1 = \mathbb{R}$ and $H^k = 0$ for $k \geq 2$, the last because $H^k_{\mathrm{dR}}(S^1) \cong H^{k-1}_{\mathrm{dR}}(S^0) = 0$ for $k \geq 2$. Assume the claim for S^{n-1} with $n \geq 2$. Then $H^0(S^n) = \mathbb{R}$, $H^1(S^n) = 0$, and for $k \geq 2$

$$H^k_{\mathrm{dR}}(S^n) \cong H^{k-1}_{\mathrm{dR}}(S^{n-1}) = \begin{cases} \mathbb{R} & k-1 = n-1 \\ 0 & \text{else} \end{cases} = \begin{cases} \mathbb{R} & k = n \\ 0 & \text{else} \end{cases}$$

since $k - 1 = 0$ is excluded by $k \geq 2$. Collecting degrees gives $\mathbb{R}$ in degrees 0 and n and 0 otherwise, as claimed.

The sphere carries exactly two independent cohomology classes: the constants in degree 0, present on any connected manifold, and a class in the top degree n. The top class is the one detected by integration; a generator is any n-form on S^n with nonzero total integral, such as the volume form, which cannot be exact because Stokes' theorem would force its integral to vanish (proposition 13.4.3). The vanishing of every intermediate group is the statement that the sphere has no holes of intermediate dimension. The single class in the top degree is where the Brouwer degree of a self-map $S^n \to S^n$ is read off, matching the degree theory of section 8.4, a connection made precise by the cohomological degree at the end of the chapter.

The next computation shows a manifold whose cohomology is large in every degree. The torus is a product of circles, and Mayer-Vietoris turns the single-circle factor into a recursion.

Example 14.4.4: Cohomology of the Tori

For the n-torus $T^n = (S^1)^n$ there is an isomorphism of graded algebras

$$H^*_{\mathrm{dR}}(T^n) \cong \Lambda^* \mathbb{R}^n$$

In particular $\dim H^k_{\mathrm{dR}}(T^n) = \binom{n}{k}$, and the total dimension is 2^n.

The single-circle recursion. Let X be any smooth manifold with finite-dimensional cohomology, and cover S^1 by two open arcs A, B, each contractible, whose intersection $A \cap B$ is a disjoint union of two arcs. Then $X \times S^1$ is covered by $U = X \times A$ and $V = X \times B$, with $U \cap V = X \times (A \cap B)$ a disjoint union of two copies of $X \times$ (arc). By homotopy invariance (corollary 14.3.5) the projections give $H^k_{\mathrm{dR}}(U) \cong H^k_{\mathrm{dR}}(V) \cong H^k_{\mathrm{dR}}(X)$ and $H^k_{\mathrm{dR}}(U \cap V) \cong H^k_{\mathrm{dR}}(X) \oplus H^k_{\mathrm{dR}}(X)$. Under these identifications a class α on U and a class β on V restrict to each of the two components of $U \cap V$ as the same underlying class of X, so

$$j^*(a, b) = (b - a,\, b - a) \in H^k_{\mathrm{dR}}(X) \oplus H^k_{\mathrm{dR}}(X)$$

Hence $\ker j^* = \{(a, a)\} \cong H^k_{\mathrm{dR}}(X)$ and $\mathrm{im} j^*$ is the diagonal, so the cokernel $H^k_{\mathrm{dR}}(X)^2/\mathrm{im} j^* \cong H^k_{\mathrm{dR}}(X)$ as well. The Mayer-Vietoris sequence breaks at $H^k(X \times S^1)$ into the short exact sequence

$$0 \to \operatorname{coker} j^*_{k-1} \xrightarrow{\delta} H^k_{\mathrm{dR}}(X \times S^1) \xrightarrow{i^*} \ker j^*_k \to 0$$

reading off $\mathrm{im}\delta = \ker i^*$ and $\mathrm{im} i^* = \ker j^*_k$. Taking dimensions,

$$\dim H^k_{\mathrm{dR}}(X \times S^1) = \dim H^{k-1}_{\mathrm{dR}}(X) + \dim H^k_{\mathrm{dR}}(X)$$

The dimension count. Writing $b_k(Y) = \dim H^k_{\mathrm{dR}}(Y)$, the recursion is $b_k(T^n) = b_k(T^{n-1}) + b_{k-1}(T^{n-1})$, with base $b_k(T^1) = b_k(S^1)$ equal to 1 for $k \in \{0, 1\}$ and 0 otherwise, that is $b_k(T^1) = \binom{1}{k}$. This is Pascal's rule, so by induction $b_k(T^n) = \binom{n}{k}$, and $\sum_k \binom{n}{k} = 2^n$. Since $\dim \Lambda^k \mathbb{R}^n = \binom{n}{k}$, the graded dimensions of $H^*_{\mathrm{dR}}(T^n)$ and $\Lambda^* \mathbb{R}^n$ agree.

The algebra structure. Let $\theta^1, \ldots, \theta^n$ be the angular coordinates on the n circle factors. Each $d\theta^i$ is a globally defined closed 1-form on T^n (the coordinate θ^i is defined only locally, but its differential is not), and for a multi-index $I = (i_1 < \cdots < i_k)$ the wedge $\omega_I = d\theta^{i_1} \wedge \cdots \wedge d\theta^{i_k}$ is closed. The $\binom{n}{k}$ classes $[\omega_I]$ are linearly independent: integrating over the coordinate k-subtorus T_J spanned by the directions in a multi-index J of size k gives

$$\int_{T_J} \omega_I = \begin{cases} (2\pi)^k & J = I \\ 0 & J \neq I \end{cases}$$

since a factor $d\theta^i$ with $i \notin J$ restricts to 0 on T_J. The pairing matrix of the $[\omega_I]$ against the $[T_J]$ is thus a nonzero multiple of the identity, so no nontrivial linear combination of the $[\omega_I]$ can be exact (an exact form integrates to 0 over every closed subtorus, proposition 13.4.3). Together with $\dim H^k_{\mathrm{dR}}(T^n) = \binom{n}{k}$, the $[\omega_I]$ form a basis of $H^k_{\mathrm{dR}}(T^n)$. The map $\Lambda^* \mathbb{R}^n \to H^*_{\mathrm{dR}}(T^n)$ sending $e_{i_1} \wedge \cdots \wedge e_{i_k}$ to $[\omega_I]$ carries a basis to a basis and respects the wedge, so it is an isomorphism of graded algebras, the wedge on the left matching the cup product (proposition 14.1.3) on the right.

The contrast with the sphere is the point of the two computations. Both are compact and orientable, yet the sphere has cohomology only at the extremes while the torus has it in every degree, and the numbers $\binom{n}{k}$ record the number of independent k-dimensional cycles, the k-fold products of the circle factors. De Rham cohomology separates the two spaces by counting holes of each dimension, and it does so with the analytic data of closed forms modulo exact ones alone. The last example returns to the space that opened the chapter.

Example 14.4.5: Cohomology of Punctured Space

For $n \geq 2$ the punctured space $\mathbb{R}^n \setminus \{0\}$ deformation retracts onto the unit sphere S^{n-1} by the radial map $x \mapsto x/|x|$, a smooth deformation retraction. Homotopy invariance (corollary 14.3.5) and example 14.4.3 give

$$H^k_{\mathrm{dR}}(\mathbb{R}^n \setminus \{0\}) \cong H^k_{\mathrm{dR}}(S^{n-1}) = \begin{cases} \mathbb{R} & k = 0 \text{ or } k = n-1 \\ 0 & \text{otherwise} \end{cases}$$

For $n = 2$ this reads $H^0 = H^1 = \mathbb{R}$, and the generator of $H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\})$ is the class of the angle form $(-y\,dx + x\,dy)/(x^2 + y^2)$, whose period 2π around the origin is exactly the pairing computed in chapter 13 certifying that the class is nonzero. The single nonvanishing class in degree $n-1$ is the obstruction to filling in the puncture: it is the class of the solid-angle form, the pullback to $\mathbb{R}^n \setminus \{0\}$ of the generator of $H^{n-1}(S^{n-1})$.

Every computation of this section reduced a manifold to contractible pieces glued along overlaps that were themselves handled by an earlier case. This is the general pattern: a manifold covered by finitely many open sets, all of whose finite intersections are contractible, has its cohomology determined by the combinatorics of the cover through iterated Mayer-Vietoris, and the computation terminates in the Poincaré lemma. Such a cover exists on any compact manifold, so the cohomology of a compact manifold is finite-dimensional, a fact taken up with the de Rham theorem in section 14.5. The exactness that drives these computations is the same exactness that, matched against the singular Mayer-Vietoris sequence of [11, Mayer-Vietoris Sequence in Cohomology], will identify de Rham cohomology with singular cohomology.

Exercise 14.4.1

1. Compute $H^*_{\mathrm{dR}}(S^1 \times \mathbb{R})$ in two ways: first by homotopy invariance, using that $S^1 \times \mathbb{R}$ deformation retracts onto $S^1 \times \{0\}$; then by the Mayer-Vietoris sequence for the cover $U = A \times \mathbb{R}$, $V = B \times \mathbb{R}$, where A, B are two open arcs covering S^1 with $A \cap B$ a pair of arcs. Confirm the two answers agree.

2. Let $M = \mathbb{R}^2 \setminus \{p, q\}$ be the plane with the two points $p = (-1, 0)$ and $q = (1, 0)$ removed. Cover M by $U = \{x < 1\} \setminus \{p\}$ and $V = \{x > -1\} \setminus \{q\}$, so that U and V are each a convex half-plane with one puncture, hence homotopy equivalent to $\mathbb{R}^2 \setminus \{0\}$, while $U \cap V = \{-1 < x < 1\}$ is a convex strip, hence contractible. Use Mayer-Vietoris to show $H^0 = \mathbb{R}$, $H^1 = \mathbb{R}^2$, and $H^k = 0$ for $k \geq 2$.

3. With $M = U \cup V$ and $\omega \in \Omega^k(U \cap V)$ closed, let $\{\rho_U, \rho_V\}$ and $\{\rho'_U, \rho'_V\}$ be two partitions of unity subordinate to $\{U, V\}$. Show that the two representatives $d\rho_U \wedge \omega$ and $d\rho'_U \wedge \omega$ of $\delta[\omega]$ differ by an exact form on M, so that $\delta[\omega]$ does not depend on the choice of partition of unity. (Consider the global form $(\rho_U - \rho'_U)\omega$.)

4. For a manifold Y with finite-dimensional de Rham cohomology, define the Euler characteristic $\chi(Y) = \sum_k (-1)^k \dim H^k_{\mathrm{dR}}(Y)$. Using the exactness of the Mayer-Vietoris sequence and the fact that an alternating sum of dimensions along a finite exact sequence is zero, prove the inclusion-exclusion formula $\chi(M) = \chi(U) + \chi(V) - \chi(U \cap V)$ for a cover $M = U \cup V$ by sets

with finite-dimensional cohomology. Then compute $\chi(S^n)$ and $\chi(T^n)$ from example 14.4.3 and example 14.4.4.

5. Apply the single-circle recursion of example 14.4.4, namely $\dim H^k_{\mathrm{dR}}(X\times S^1) = \dim H^{k-1}_{\mathrm{dR}}(X) + \dim H^k_{\mathrm{dR}}(X)$, with $X = S^2$ to compute $\dim H^k_{\mathrm{dR}}(S^2 \times S^1)$ in each degree, and check that the total dimension is 4.

14.5 The de Rham Theorem

The Mayer-Vietoris computations of section 14.4 returned, for the spheres and the tori, exactly the Betti numbers a topologist assigns those spaces by counting the generators of their homology. This section shows that the agreement is not particular to those examples. Integration of a closed form over a cycle, the pairing whose invariances were settled in section 13.4, assembles into a map from de Rham cohomology to the singular cohomology that [11, Cohomology Group] builds from continuous data, and the de Rham theorem asserts that this map is an isomorphism. The differential operator d then computes the same invariant as the boundary operator of singular chains: the analytic and the topological measurements of the holes of M coincide.

The task is threefold. The first is to build the map, the de Rham map, and prove it well defined by Stokes' theorem. The second is the isomorphism, obtained by comparing the de Rham Mayer-Vietoris sequence with its singular counterpart through the five lemma, the base of the induction being the Poincaré lemma. The third is to read off two consequences: the period pairing between cohomology and homology is perfect, so a closed form is exact exactly when all of its periods vanish, which settles the question posed in section 13.4; and the de Rham cohomology of a compact manifold is finite-dimensional, its dimensions the Betti numbers.

Singular cohomology with real coefficients is the invariant of [11, Cohomology Group], assembled from continuous maps of simplices into M. Integrating a k-form requires a smooth domain, so the comparison is made through smooth simplices. The smooth singular cohomology this produces agrees with the continuous one: every continuous simplex is uniformly approximable, relative to its already smooth faces, by a smooth one, and any two such approximations are smoothly homotopic, both by the approximation theorem of chapter 7; the resulting chain-homotopy equivalence between the smooth and continuous singular complexes gives them the same cohomology ([11, chain homotopic chain maps, Homology Isomorphism, then Cohomology Isomorphism]), and the two are written alike.

Definition 14.5.1: Smooth Singular Chains and Cochains

A *smooth singular k-simplex* in M is a smooth map $\sigma\colon \Delta^k \to M$ from the standard k-simplex

$$\Delta^k = \left\{t \in \mathbb{R}^{k+1} \;\middle|\; t_i \geq 0 \text{ for all } i,\ \textstyle\sum_i t_i = 1\right\}$$

where smoothness means σ extends to a smooth map on a neighborhood of Δ^k in its affine span. The *smooth singular chain complex* $(C^\infty_\bullet(M), \partial)$ has $C^\infty_k(M)$ the real vector space with basis the smooth k-simplices, and boundary

$$\partial\sigma = \sum_{i=0}^{k} (-1)^i\, \sigma \circ F_i^k$$

where $F_i^k\colon \Delta^{k-1} \to \Delta^k$ is the affine inclusion onto the i-th face (the face opposite the i-th vertex). The dual complex $C^k_\infty(M;\mathbb{R}) = \operatorname{Hom}(C^\infty_k(M), \mathbb{R})$, with coboundary $(\delta\varphi)(\sigma) = \varphi(\partial\sigma)$, computes the *singular cohomology* $H^k(M;\mathbb{R})$.

Under the affine identification $\Delta^1 \cong [0,1]$ a smooth 1-simplex is a smooth path σ, and its boundary $\partial\sigma = \sigma(1) - \sigma(0)$ is the difference of its two endpoints, each a 0-simplex; this is the algebra of chains specializing to the endpoints of a curve. The relation $\partial^2 = 0$ holds because the faces of a face repeat with canceling signs ([11, Complex Condition]), so H^k is well defined.

The bridge between the two complexes is integration, and its single structural property is that it converts the exterior derivative into the singular coboundary. That conversion is Stokes' theorem, read on a chain rather than on a manifold with boundary.

Lemma 14.5.2: Stokes' Theorem for Smooth Chains

For $\omega \in \Omega^{k-1}(M)$ and a smooth singular k-chain c in M,

$$\int_{\partial c} \omega = \int_c d\omega$$

where $\int_c \omega = \int_{\Delta^k} \sigma^*\omega$ on a simplex σ and is extended linearly to chains.

***Proof*:**
Both sides are linear in c, so it suffices to treat a single smooth k-simplex $\sigma\colon \Delta^k \to M$. Naturality of the exterior derivative (proposition 12.5.4) gives $\sigma^*(d\omega) = d(\sigma^*\omega)$, so

$$\int_\sigma d\omega = \int_{\Delta^k} \sigma^*(d\omega) = \int_{\Delta^k} d(\sigma^*\omega)$$

an integral of the top form $d(\sigma^*\omega)$ over Δ^k, oriented by the standard orientation of its affine span. The simplex Δ^k is a compact manifold with corners; its corner set, where two or more faces meet, has codimension two and hence measure zero in Δ^k and in each face. Stokes' theorem for manifolds with boundary (theorem 13.2.1) therefore applies away from that negligible set and reads the boundary as the signed sum of the codimension-one faces, whose induced boundary orientations produce exactly the alternating signs $(-1)^i$:

$$\int_{\Delta^k} d(\sigma^*\omega) = \int_{\partial\Delta^k} \sigma^*\omega = \sum_{i=0}^{k}(-1)^i \int_{\Delta^{k-1}} (F_i^k)^*\sigma^*\omega$$

Functoriality of pullback (proposition 12.4.11) turns $(F_i^k)^*\sigma^*\omega$ into $(\sigma \circ F_i^k)^*\omega$, so each term is $\int_{\sigma\circ F_i^k}\omega$, and the sum is $\int_{\partial\sigma}\omega$ by the definition of $\partial\sigma$. Hence $\int_\sigma d\omega = \int_{\partial\sigma}\omega$; summing over the simplices of a general chain gives the stated identity, as we sought to show.

Integration over smooth simplices thus defines, for each $\omega \in \Omega^k(M)$, a cochain $I(\omega) \in C^k_\infty(M;\mathbb{R})$ with $I(\omega)(\sigma) = \int_\sigma \omega$, and lemma 14.5.2 is the statement $I(d\omega) = \delta\, I(\omega)$ that I is a cochain map. A cochain map descends to cohomology, and the result is the object of the section.

Definition 14.5.3: The de Rham Map

The *de Rham map* of M is the map on cohomology

$$\mathcal{I}\colon H^k_{\mathrm{dR}}(M) \to H^k(M;\mathbb{R}), \qquad \mathcal{I}[\omega] = \left[\,\sigma \mapsto \textstyle\int_\sigma \omega\,\right]$$

induced by the integration cochain map $I\colon \Omega^\bullet(M) \to C^\bullet_\infty(M;\mathbb{R})$; it sends the class of a closed k-form ω to the singular cohomology class of the cocycle $\sigma \mapsto \int_\sigma \omega$.

That $\mathcal{I}$ is well defined, and carries the structure of section 14.1 to the singular side, is the content of the next proposition. Its well-definedness is Stokes' theorem in two guises, one for each of the two quotients folded into a cohomology class.

Proposition 14.5.4: The de Rham Map is a Natural Ring Homomorphism

The de Rham map $\mathcal{I}\colon H^k_{\mathrm{dR}}(M) \to H^k(M;\mathbb{R})$ is well defined and $\mathbb{R}$-linear; it is natural, meaning $\mathcal{I}_M \circ f^* = f^* \circ \mathcal{I}_N$ for every smooth map $f\colon M \to N$; and it is a homomorphism of graded rings, carrying the wedge product of proposition 14.1.3 to the singular cup product of [11, Cohomology Ring and Algebra] and the unit to the unit.

***Proof*:**
Step 1: Well-definedness. By lemma 14.5.2, $I(d\omega) = \delta\, I(\omega)$. If ω is closed then $\delta\, I(\omega) = I(d\omega) = 0$, so $I(\omega)$ is a cocycle and has a class in $H^k(M;\mathbb{R})$. If $\omega = d\eta$ is exact then $I(\omega) = I(d\eta) = \delta\, I(\eta)$ is a coboundary, hence zero in cohomology. Thus $[\omega] \mapsto [I(\omega)]$ depends only on the de Rham class, and $\mathcal{I}$ is well defined.

Step 2: Linearity. The cochain $I(\omega)$ is $\mathbb{R}$-linear in ω, since $\int_\sigma (a\omega + b\omega') = a\int_\sigma \omega + b\int_\sigma \omega'$ for a fixed simplex, and passing to classes preserves the linear combinations. Hence $\mathcal{I}$ is $\mathbb{R}$-linear.

Step 3: Naturality. Let $f\colon M \to N$ be smooth and $\omega \in \Omega^k(N)$ closed. For a smooth k-simplex σ in M, functoriality of pullback (proposition 12.4.11) gives

$$I_M(f^*\omega)(\sigma) = \int_{\Delta^k} \sigma^* f^*\omega = \int_{\Delta^k} (f\circ\sigma)^*\omega = \int_{f\circ\sigma} \omega = I_N(\omega)(f\circ\sigma)$$

The singular map f^* on cochains is precomposition with the chain map $\sigma \mapsto f \circ \sigma$, so the right-hand side is $(f^* I_N(\omega))(\sigma)$. Thus $I_M \circ f^* = f^* \circ I_N$ at the cochain level, and passing to cohomology gives $\mathcal{I}_M \circ f^* = f^* \circ \mathcal{I}_N$, which is the naturality of $\mathcal{I}$ compatible with the functoriality of proposition 14.1.4.

Step 4: The unit and the product. The constant function $1 \in \Omega^0(M)$ has $I(1)(\sigma) = \int_\sigma 1 = 1$ for every 0-simplex σ, the augmentation cocycle that represents the unit of $H^0(M;\mathbb{R})$; so $\mathcal{I}$ takes the unit to the unit. Multiplicativity, $\mathcal{I}([\alpha] \wedge [\beta]) = \mathcal{I}[\alpha] \cup \mathcal{I}[\beta]$, is the one property whose proof lies outside the scope of this book: at the level of cochains, integration is not multiplicative on the nose, and the wedge of forms agrees with the cup product of cochains only up to a natural cochain homotopy. The comparison is the acyclic-models theorem, a homological-algebra tool that [11] itself cites externally (a recorded gap in the series). Granting it, $\mathcal{I}$ is a ring homomorphism. The additive statement, which the rest of the section proves is an isomorphism, needs none of this and is established in full above, completing the proof.

The de Rham map records each cohomology class by the numbers it produces against cycles, and proposition 14.5.4 says those numbers respect every operation in sight: sums, pullbacks, and products. What remains, and what the name "de Rham theorem" refers to, is that the record is complete and lossless. The circle is the first case where completeness can be checked by hand.

Example 14.5.5: The de Rham Map on the Circle

On S^1 the de Rham cohomology is $H^0_{\mathrm{dR}}(S^1) \cong \mathbb{R}$ and $H^1_{\mathrm{dR}}(S^1) \cong \mathbb{R}$, the latter generated by the angle form ω (example 14.1.6). Let $c\colon S^1 \to S^1$ be the identity, a smooth 1-cycle tracing the circle once; its class generates $H_1(S^1;\mathbb{R}) \cong \mathbb{R}$ ([11, The Hurewicz Map for the Circle]). The period of the angle form over c is $\int_c \omega = 2\pi$ (definition 13.4.1, computed in example 13.4.2), so the cocycle $I(\omega)$ takes the value $2\pi \neq 0$ on a generator of H_1. Since $H^1(S^1;\mathbb{R}) \cong \mathbb{R}$, the class $\mathcal{I}[\omega]$ is a nonzero element of a one-dimensional space, hence a generator, and $\mathcal{I}$ is an isomorphism in degree 1. In degree 0 the constant function 1 goes to the augmentation cocycle, a generator of $H^0(S^1;\mathbb{R}) \cong \mathbb{R}$, so $\mathcal{I}$ is an isomorphism in degree 0 as well. The de Rham map is an isomorphism for S^1, the smallest instance of the theorem below.

The proof for a general manifold cannot proceed form by form. The mechanism is instead the one that made the computations of section 14.4 possible: cut the manifold into pieces on which cohomology is already understood, and let two Mayer-Vietoris sequences, joined by $\mathcal{I}$, propagate the isomorphism from the pieces to the whole. The pieces are the sets of a good cover.

Definition 14.5.6: Good Cover

An open cover $\{U_\alpha\}$ of a smooth n-manifold M is a *good cover* if every nonempty finite intersection $U_{\alpha_0} \cap \cdots \cap U_{\alpha_p}$ is diffeomorphic to $\mathbb{R}^n$. The cover is *finite* if its index set is finite.

On each set of a good cover, and on each of their intersections, both cohomologies are already known: $\mathbb{R}^n$ is star-shaped, so its de Rham cohomology is that of a point by the Poincaré lemma (corollary 14.2.6), and $\mathbb{R}^n$ is contractible, so its singular cohomology is that of a point by homotopy invariance in [11, Homotopy Equivalence, Then Isomorphism, Homology Group of a Point, Homology Isomorphism, then Cohomology Isomorphism]. A good cover is thus a decomposition of M into de-Rham-acyclic and singular-acyclic pieces at once, which is what lets a single induction handle both sides.

14.5.7 Note: Existence of Finite Good Covers Every smooth manifold admits a good cover, and every compact smooth manifold admits a finite one. The construction takes an auxiliary Riemannian metric (which exists by a partition of unity, a tool already used in this book) and covers M by geodesically convex neighborhoods, whose nonempty finite intersections are geodesically convex and hence diffeomorphic to $\mathbb{R}^n$; compactness extracts a finite subcover. The convexity theory this rests on is the geometry of the exponential map, developed in the companion volume on Riemannian geometry and pointed to in chapter 15, and is not needed in what follows beyond the existence statement: the de Rham theorem below is proved for any manifold that admits a finite good cover, and this note records that compact manifolds are among them.

Example 14.5.8: A Good Cover of the Circle

Cover S^1 by three open arcs U_1, U_2, U_3, each slightly longer than a third of the circle, so that consecutive arcs overlap in a single arc and no point lies in all three. Each U_i is an open arc, diffeomorphic to $\mathbb{R}$; each nonempty pairwise intersection $U_i \cap U_j$ is a single open arc, diffeomorphic to $\mathbb{R}$; and $U_1 \cap U_2 \cap U_3 = \varnothing$. Every nonempty finite intersection is therefore diffeomorphic to $\mathbb{R}^1$, so this is a finite good cover. A cover by only two arcs fails: two arcs covering S^1 meet in two disjoint arcs, a space diffeomorphic to $\mathbb{R} \sqcup \mathbb{R}$ rather than to $\mathbb{R}$, so it is not a good cover.

The propagation step is the compatibility of $\mathcal{I}$ with the Mayer-Vietoris machinery. Integration is insensitive to which open set a simplex is regarded as lying in, and this is precisely what makes it a map of the two short exact sequences that generate the sequences.

Lemma 14.5.9: The de Rham Map is a Morphism of Mayer-Vietoris Sequences

Let $M = A \cup B$ with A, B open. The de Rham map commutes with the restriction maps and with the Mayer-Vietoris connecting homomorphisms, so that it carries the de Rham sequence of theorem 14.4.2 to the singular Mayer-Vietoris sequence of [11, Mayer-Vietoris Sequence in Cohomology], forming a commuting ladder in which every vertical map is an instance of $\mathcal{I}$.

***Proof*:**
The de Rham sequence of theorem 14.4.2 is the long exact sequence of the short exact sequence of complexes

$$0 \to \Omega^\bullet(M) \xrightarrow{r} \Omega^\bullet(A) \oplus \Omega^\bullet(B) \xrightarrow{s} \Omega^\bullet(A \cap B) \to 0$$

with $r(\omega) = (\omega|_A, \omega|_B)$ and $s(\alpha, \beta) = \beta|_{A\cap B} - \alpha|_{A\cap B}$. Its singular counterpart is the long exact sequence of

$$0 \to C^\bullet_\infty(M; \mathbb{R}) \to C^\bullet_\infty(A; \mathbb{R}) \oplus C^\bullet_\infty(B; \mathbb{R}) \to C^\bullet_\infty(A \cap B; \mathbb{R}) \to 0$$

the short exact sequence of smooth singular cochains whose exactness rests on the subdivision of a chain into small simplices, each lying in A or in B ([11, Homology Of Covered Space]). Integration maps the first sequence to the second: for a simplex σ whose image lies in A, the value $\int_\sigma \omega$ is unchanged by restricting ω to A, so I commutes with the restriction map r and, by the same token, with the difference map s. Both squares therefore commute on the nose, and I is a morphism of short exact sequences of complexes.

The connecting homomorphism of the long exact sequence attached to a short exact sequence of complexes is natural in that sequence, being the boundary map of the snake lemma ([11, Naturality of the Connecting Homomorphism]). Applied to the morphism I, naturality gives a commuting ladder relating the two long exact sequences, and on cohomology the vertical maps are exactly the maps $\mathcal{I}$ for M, A, B, and $A \cap B$. This is the assertion, completing the proof.

With the base case in hand and the ladder available, the isomorphism follows by an induction on the size of a good cover, the five lemma converting isomorphisms on the pieces into an isomorphism on the union.

Theorem 14.5.10: The de Rham Theorem

Let M be a smooth manifold admitting a finite good cover, in particular any compact smooth manifold (box 14.5.7). Then for every k the de Rham map

$$\mathcal{I}\colon H^k_{\mathrm{dR}}(M) \to H^k(M;\mathbb{R})$$

is an isomorphism, and taken over all k it is an isomorphism of graded rings (proposition 14.5.4).

***Proof*:**
The proof is by induction on the number p of sets in a finite good cover $\{U_1, \dots, U_p\}$ of M.

Step 1: Base case. For $p = 1$ the manifold $M = U_1$ is diffeomorphic to $\mathbb{R}^n$. Its de Rham cohomology is $H^0_{\mathrm{dR}} = \mathbb{R}$ and $H^k_{\mathrm{dR}} = 0$ for $k \geq 1$ (corollary 14.2.6); its singular cohomology, $\mathbb{R}^n$ being contractible, is $H^0(\mathbb{R}^n;\mathbb{R}) = \mathbb{R}$ and $H^k(\mathbb{R}^n;\mathbb{R}) = 0$ for $k \geq 1$ ([11, Homotopy Equivalence, Then Isomorphism, Homology Group of a Point, Homology Isomorphism, then Cohomology Isomorphism]). In every degree $k \geq 1$ both groups vanish, so $\mathcal{I}$ is an isomorphism there. In degree 0 a closed form is a constant a, and $\mathcal{I}(a)$ is the class of the cochain valued a on every 0-simplex, that is a times the generator of $H^0(\mathbb{R}^n;\mathbb{R}) \cong \mathbb{R}$; hence $\mathcal{I}\colon \mathbb{R} \to \mathbb{R}$ is the identity $a \mapsto a$, an isomorphism. The same computation applies to a point, which is the base of the induction matching the Poincaré lemma.

Step 2: Inductive step. Suppose the theorem holds for every manifold admitting a good cover with fewer than p sets, and let M have the good cover $\{U_1, \dots, U_p\}$ with $p \geq 2$. Put $A = U_1 \cup \cdots \cup U_{p-1}$ and $B = U_p$, so $M = A \cup B$. Then $\{U_1, \dots, U_{p-1}\}$ is a good cover of A with $p - 1$ sets; B is diffeomorphic to $\mathbb{R}^n$, a good cover with one set; and

$$A \cap B = \bigcup_{i=1}^{p-1} (U_i \cap U_p)$$

has the good cover $\{U_i \cap U_p\}_{i<p}$ with $p - 1$ sets, since every nonempty finite intersection $U_{i_1} \cap \cdots \cap U_{i_r} \cap U_p$ is a nonempty finite intersection of the original good cover and hence diffeomorphic to $\mathbb{R}^n$. By the inductive hypothesis $\mathcal{I}$ is an isomorphism, in every degree, for A, for B, and for $A \cap B$.

Apply lemma 14.5.9 to $M = A \cup B$. Around the term $H^k_{\mathrm{dR}}(M)$ the commuting ladder reads

$$\begin{array}{ccccccccc}
H^{k-1}_{\mathrm{dR}}(A) \oplus H^{k-1}_{\mathrm{dR}}(B) & \to & H^{k-1}_{\mathrm{dR}}(A \cap B) & \to & H^k_{\mathrm{dR}}(M) & \to & H^k_{\mathrm{dR}}(A) \oplus H^k_{\mathrm{dR}}(B) & \to & H^k_{\mathrm{dR}}(A \cap \\
\downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
H^{k-1}(A;\mathbb{R}) \oplus H^{k-1}(B;\mathbb{R}) & \to & H^{k-1}(A \cap B;\mathbb{R}) & \to & H^k(M;\mathbb{R}) & \to & H^k(A;\mathbb{R}) \oplus H^k(B;\mathbb{R}) & \to & H^k(A \cap
\end{array}$$

with both rows exact (Mayer-Vietoris) and every square commuting. The four vertical maps flanking the central one involve only A, B, and $A \cap B$, and are isomorphisms by the inductive hypothesis. The five lemma ([11, The Five Lemma]) then forces the central vertical map $\mathcal{I}\colon H^k_{\mathrm{dR}}(M) \to H^k(M;\mathbb{R})$ to be an isomorphism. As k was arbitrary, $\mathcal{I}$ is an isomorphism for M, and the induction closes.

Every manifold with a finite good cover, and by box 14.5.7 every compact manifold, therefore has $\mathcal{I}$ an isomorphism in all degrees. That it is moreover a ring isomorphism is the multiplicativity recorded in proposition 14.5.4, completing the proof.

The de Rham theorem identifies two invariants of different origin: the quotient of closed forms by exact forms, defined by the differential calculus of M, is the singular cohomology that [11, Cohomology Group] extracts from continuous maps of simplices, and the isomorphism is realized by the concrete operation of integration. The two Mayer-Vietoris sequences are not only abstractly isomorphic; they are the same sequence, one written with forms and one with cochains, and $\mathcal{I}$ is the dictionary. A remark on scope: the proof covers every compact manifold, which carries the results below; a non-compact manifold with only an infinite good cover is handled by writing it as an increasing union of finite-good-cover open sets and passing to the limit, an argument of the same homological character that [11] carries out on the singular side.

The first consequence answers the question left open at the end of chapter 13. There the period of a closed form over a cycle was shown to vanish whenever the form is exact (proposition 13.4.3); the converse, whether vanishing periods force exactness, was posed as the de Rham question in box 13.4.5 and deferred. The de Rham theorem gives it, by turning the isomorphism into a statement about the pairing itself.

Corollary 14.5.11: Perfection of the Period Pairing

Let M admit a finite good cover. The period pairing

$$H^k_{\mathrm{dR}}(M) \times H_k(M;\mathbb{R}) \to \mathbb{R}, \qquad ([\omega],[c]) \mapsto \int_c \omega$$

is well defined and perfect: it induces an isomorphism $H^k_{\mathrm{dR}}(M) \cong \mathrm{Hom}(H_k(M;\mathbb{R}),\mathbb{R})$, so the two spaces are dual. Consequently a closed k-form is exact if and only if all of its periods vanish, that is $\int_c \omega = 0$ for every smooth singular k-cycle c.

***Proof*:**
The pairing is well defined: by lemma 14.5.2, if $\omega = d\eta$ is exact then $\int_c \omega = \int_c d\eta = \int_{\partial c} \eta = 0$ since $\partial c = 0$, and if $c = \partial b$ is a boundary then $\int_c \omega = \int_{\partial b} \omega = \int_b d\omega = 0$ since ω is closed; so the value depends only on the two classes. Its adjoint sends $[\omega]$ to the functional $[c] \mapsto \int_c \omega$, and

this adjoint is precisely the de Rham map $\mathcal{I}$ followed by the universal-coefficient isomorphism

$$H^k(M;\mathbb{R}) \cong \operatorname{Hom}(H_k(M;\mathbb{R}),\mathbb{R})$$

which holds because the coefficients form a field ([11, Universal Coefficient Theorem For Cohomology]). By theorem 14.5.10 $\mathcal{I}$ is an isomorphism, so the composite $H^k_{\mathrm{dR}}(M) \to \operatorname{Hom}(H_k(M;\mathbb{R}),\mathbb{R})$ is an isomorphism, which is the perfection of the pairing (that $H_k(M;\mathbb{R})$ is in turn the dual of $H^k_{\mathrm{dR}}(M)$ follows since both are finite-dimensional).

For the criterion, let ω be closed. Then $[\omega] = 0$ in $H^k_{\mathrm{dR}}(M)$, meaning ω is exact, holds if and only if $\mathcal{I}[\omega] = 0$, since $\mathcal{I}$ is injective; and $\mathcal{I}[\omega] = 0$ says the functional $[c] \mapsto \int_c \omega$ is zero, that is $\int_c \omega = 0$ for every smooth singular k-cycle c. Hence ω is exact if and only if all of its periods vanish, as we sought to show.

This is the affirmative answer to the de Rham question, and in the strong form anticipated by box 13.4.5: the period pairing separates cohomology from zero, so a closed form is determined, up to an exact form, by the numbers it produces against cycles. The evidence collected earlier now reads as an instance. The angle form on $\mathbb{R}^2 \setminus \{0\}$ has period $2\pi \neq 0$ over the unit circle (example 13.4.2), so by the criterion it is not exact, recovering the conclusion drawn by hand in chapter 13. Every smooth k-cycle $c\colon Z \to M$ of definition 13.4.1, a map from a closed oriented manifold, determines a singular cycle by pushing forward the fundamental class of Z ([11, Existence and Uniqueness of the Fundamental Class]), so the periods of definition 13.4.1 are among those the criterion tests, and the pairing against singular homology extends the earlier one without changing its values.

The second consequence is finiteness. The Mayer-Vietoris induction that proved the isomorphism also bounds the size of the cohomology, and it does so without reference to the singular side.

Corollary 14.5.12: Finite-Dimensionality of de Rham Cohomology

If M admits a finite good cover, then $H^k_{\mathrm{dR}}(M)$ is finite-dimensional for every k, and it vanishes for $k > \dim M$.

***Proof*:**
Vanishing for $k > n = \dim M$ is immediate, as $\Omega^k(M) = 0$ there, so $H^k_{\mathrm{dR}}(M) = 0$. Finite-dimensionality is proved by induction on the size p of a finite good cover. For $p = 1$ the manifold is $\mathbb{R}^n$, with $H^0_{\mathrm{dR}} = \mathbb{R}$ and the rest zero (corollary 14.2.6), all finite-dimensional. For $p \geq 2$ write $M = A \cup B$ as in theorem 14.5.10, so that A, B, and $A \cap B$ have good covers with fewer sets and, by induction, finite-dimensional de Rham cohomology in every degree. The Mayer-Vietoris sequence (theorem 14.4.2) gives an exact segment

$$H^{k-1}_{\mathrm{dR}}(A \cap B) \to H^k_{\mathrm{dR}}(M) \to H^k_{\mathrm{dR}}(A) \oplus H^k_{\mathrm{dR}}(B)$$

in which the outer two spaces are finite-dimensional. The kernel of the right-hand map is the image of the left-hand map, a quotient of $H^{k-1}_{\mathrm{dR}}(A \cap B)$ and so finite-dimensional, while the image of $H^k_{\mathrm{dR}}(M)$ in $H^k_{\mathrm{dR}}(A) \oplus H^k_{\mathrm{dR}}(B)$ is finite-dimensional; hence $\dim H^k_{\mathrm{dR}}(M)$ is bounded by the sum of the two, and is finite. The induction closes, completing the proof.

Definition 14.5.13: Betti Numbers

For a smooth manifold M admitting a finite good cover, the *k-th Betti number* is $b_k(M) = \dim_{\mathbb{R}} H^k_{\mathrm{dR}}(M)$, a nonnegative integer by corollary 14.5.12.

The Betti numbers are defined here through the analysis of forms, yet by the de Rham theorem they equal the ranks of the singular homology groups of [11, Betti Number And Torsion Coefficients], the integers a topologist would compute from a cell structure. The spheres show this: from $H^*_{\mathrm{dR}}(S^n)$ (example 14.4.3) one reads $b_0(S^n) = b_n(S^n) = 1$ and $b_k(S^n) = 0$ otherwise, so the alternating sum $\sum_k (-1)^k b_k(S^n) = 1 + (-1)^n$ is 2 for even n and 0 for odd n. That alternating sum is the Euler characteristic $\chi(M)$ built from intersection theory in chapter 8, now expressed through the de Rham complex, and the Morse inequalities of chapter 9 bound each b_k by the number of critical points of index k of a Morse function. The pairing that remains to exploit is the wedge followed by the integral, $([\alpha], [\beta]) \mapsto \int_M \alpha \wedge \beta$, whose nondegeneracy on a compact oriented manifold is Poincaré duality, taken up in section 14.6.

Exercise 14.5.1

1. Let M be a smooth manifold with r connected components. Describe $H^0_{\mathrm{dR}}(M)$ (proposition 14.1.5) and $H^0(M;\mathbb{R})$, and show directly that the de Rham map $\mathcal{I}\colon H^0_{\mathrm{dR}}(M) \to H^0(M;\mathbb{R})$ is an isomorphism by identifying both sides with $\mathbb{R}^r$ and computing $\mathcal{I}$ on a locally constant function.
2. Let $\sigma\colon \Delta^2 \to \mathbb{R}^2$ be the affine 2-simplex sending the vertices to 0, $e_1 = (1, 0)$, and $e_2 = (0, 1)$, and let $\omega = x\,dy \in \Omega^1(\mathbb{R}^2)$. Compute $\int_\sigma d\omega$ and $\int_{\partial\sigma} \omega$ directly from the definitions, and verify that they agree, confirming lemma 14.5.2 in this case.
3. Show that a cover of S^2 by two open sets, each diffeomorphic to $\mathbb{R}^2$ (for instance the complements of the north and south poles), is not a good cover, and identify which intersection fails the defining condition. Explain in one sentence why this does not obstruct the de Rham theorem for S^2.
4. Let $f\colon M \to N$ be a smooth homotopy equivalence between smooth manifolds admitting finite good covers. Using the naturality of the de Rham map (proposition 14.5.4), the de Rham theorem, and the homotopy invariance of de Rham cohomology (corollary 14.3.5), prove that the induced map $f^*\colon H^k(N;\mathbb{R}) \to H^k(M;\mathbb{R})$ on singular cohomology is an isomorphism for every k.
5. For the 2-torus T^2, use example 14.4.4 to list the Betti numbers b_0, b_1, b_2 and compute the Euler characteristic $\chi(T^2) = \sum_k (-1)^k b_k$. Then, using corollary 14.5.11 and the fact that $H_1(T^2;\mathbb{R}) \cong \mathbb{R}^2$ is generated by the two coordinate circles, state the precise condition on its periods under which a closed 1-form on T^2 is exact.

14.6 Poincaré Duality and the Degree

A compact oriented manifold has, in its top degree, the simplest cohomology available: a single line, coordinatized by integration. This section proves that fact and then spends it twice. Paired against that line by the wedge product, cohomology in complementary degrees becomes a nondegenerate duality, Poincaré's, which forces the Betti numbers to be symmetric. And the pullback of a smooth map between two such manifolds acts on the top line by a single scalar, which is exactly the Brouwer degree of chapter 8, so the integer that counts preimages with sign reappears as the effect of f on a one-dimensional cohomology group. The two results together close the degree and period threads the book opened.

The top computation reduces to a local statement about Euclidean space, dual to the Poincaré lemma of theorem 14.2.4. That lemma trivialized closed forms; its compactly supported companion trivializes compactly supported top forms of total integral zero.

Lemma 14.6.1: Exactness of Zero-Integral Top Forms

Let $\omega \in \Omega^n_c(\mathbb{R}^n)$ be a compactly supported n-form with $\int_{\mathbb{R}^n} \omega = 0$. Then $\omega = d\eta$ for some compactly supported $\eta \in \Omega^{n-1}_c(\mathbb{R}^n)$.

***Proof*:**
The argument is induction on n, building a compactly supported primitive one variable at a time.

Step 1: The line. For $n = 1$ write $\omega = f\, dt$ with $f \in C^\infty_c(\mathbb{R})$ and $\int_{\mathbb{R}} f = 0$. Set $\varphi(t) = \int_{-\infty}^t f(s)\, ds$. Then φ is smooth with $\varphi'(t) = f(t)$, so $d\varphi = \omega$; and φ has compact support, being identically 0 to the left of supp$f$ and equal to $\int_{\mathbb{R}} f = 0$ to its right. Thus $\eta = \varphi \in \Omega^0_c(\mathbb{R})$ works.

Step 2: The inductive reduction. Assume the statement in dimension $n-1$. Write the coordinates on $\mathbb{R}^n$ as (y, t) with $y = (y^1, \dots, y^{n-1})$, and $\omega = f(y,t)\, dy^1 \wedge \cdots \wedge dy^{n-1} \wedge dt$ with $f \in C^\infty_c(\mathbb{R}^n)$. Fix a bump $e \in C^\infty_c(\mathbb{R})$ with $\int_{\mathbb{R}} e = 1$, and put $E(t) = \int_{-\infty}^t e(s)\, ds$. Define

$$g(y) = \int_{\mathbb{R}} f(y,t)\, dt \in C^\infty_c(\mathbb{R}^{n-1}), \qquad \varphi(y,t) = \int_{-\infty}^t f(y,s)\, ds - E(t)\, g(y)$$

Then $\varphi \in C^\infty_c(\mathbb{R}^n)$: it is compactly supported in y because f and g are, and in t because for t far to the left both terms vanish while for t far to the right the first term is $g(y)$ and $E(t) = 1$, so the two cancel. Let $\eta = \varphi\, dy^1 \wedge \cdots \wedge dy^{n-1} \in \Omega^{n-1}_c(\mathbb{R}^n)$. Only the dt-component of $d\varphi$ contributes to the wedge, so

$$d\eta = \partial_t \varphi\ dt \wedge dy^1 \wedge \cdots \wedge dy^{n-1} = (-1)^{n-1}\big(f - e(t)g(y)\big)\, dy^1 \wedge \cdots \wedge dy^{n-1} \wedge dt$$

using $\partial_t \varphi = f - e\, g$ and $dt \wedge dy^1 \wedge \cdots \wedge dy^{n-1} = (-1)^{n-1} dy^1 \wedge \cdots \wedge dy^{n-1} \wedge dt$. Writing $\omega' = e(t) g(y)\, dy^1 \wedge \cdots \wedge dy^{n-1} \wedge dt$, this reads $\omega = \omega' + (-1)^{n-1} d\eta$.

Step 3: Killing the remainder. The form $\omega' = \pi^*\alpha \wedge (e\, dt)$, where $\pi \colon \mathbb{R}^n \to \mathbb{R}^{n-1}$ is the projection and $\alpha = g\, dy^1 \wedge \cdots \wedge dy^{n-1} \in \Omega^{n-1}_c(\mathbb{R}^{n-1})$. Its total integral is $\int_{\mathbb{R}^{n-1}} \alpha = \int_{\mathbb{R}^{n-1}} g(y)\, dy = \int_{\mathbb{R}^n} f = 0$, so the inductive hypothesis gives $\alpha = d\beta$ with $\beta \in \Omega^{n-2}_c(\mathbb{R}^{n-1})$. Since $d(e\, dt) = e'(t)\, dt \wedge dt = 0$,

$$d\big(\pi^*\beta \wedge e\, dt\big) = \pi^*(d\beta) \wedge e\, dt = \pi^*\alpha \wedge e\, dt = \omega'$$

and $\pi^*\beta \wedge e\, dt \in \Omega^{n-1}_c(\mathbb{R}^n)$ is compactly supported. Therefore

$$\omega = d\big(\pi^*\beta \wedge e\, dt\big) + (-1)^{n-1} d\eta = d\big(\pi^*\beta \wedge e\, dt + (-1)^{n-1}\eta\big)$$

exhibits ω as d of a compactly supported $(n-1)$-form, completing the induction and the proof.

The zero-integral hypothesis is exactly what the lemma cannot dispense with: the total integral of an exact compactly supported top form is always zero by Stokes' theorem (theorem 13.2.1), so integration $\int_{\mathbb{R}^n} : \Omega_c^n(\mathbb{R}^n) \to \mathbb{R}$ descends to an isomorphism from the compactly supported top cohomology of $\mathbb{R}^n$ onto $\mathbb{R}$. The lemma is the injective half of that isomorphism; the surjective half is any bump form. On a compact manifold every form is automatically compactly supported, and this local fact globalizes to the top cohomology of the whole manifold.

Proposition 14.6.2: Top Cohomology of a Compact Oriented Manifold

Let M be a compact connected oriented n-manifold. Then integration

$$I\colon H^n_{\mathrm{dR}}(M) \to \mathbb{R}, \qquad I([\omega]) = \int_M \omega$$

is a well-defined isomorphism. In particular $H^n_{\mathrm{dR}}(M) \cong \mathbb{R}$, generated by the class of any n-form of nonzero total integral, for instance any orientation form.

***Proof*:**
Step 1: Well defined and surjective. If ω and $\omega + d\tau$ represent the same class, they have the same integral, since $\int_M d\tau = \int_{\partial M} \tau = 0$ by Stokes' theorem (theorem 13.2.1) and $\partial M = \emptyset$. So I is a well-defined linear map. For surjectivity, take an orientation form $\mu \in \Omega^n(M)$ (definition 12.6.1); in every positively oriented chart its integrand is positive, so $\int_M \mu > 0$, and the scalar multiples of $[\mu]$ already fill $\mathbb{R}$.

Step 2: A generating bump form. Fix once and for all a positively oriented chart diffeomorphic to $\mathbb{R}^n$ and a bump function (chapter 1) supported in it, producing an n-form ρ with $\rho \geq 0$, $\operatorname{supp}\rho$ inside that chart, and $\int_M \rho = 1$. The claim is that

$$[\lambda] = \Big(\int_M \lambda\Big)[\rho] \qquad \text{in } H^n_{\mathrm{dR}}(M)$$

for every n-form λ whose support lies in a single oriented chart. Granting this, Step 3 finishes the proof.

Suppose first $\operatorname{supp}\lambda$ and $\operatorname{supp}\rho$ lie in one common oriented chart U diffeomorphic to $\mathbb{R}^n$ by an orientation-preserving diffeomorphism; such a chart exists around every point, an open ball being diffeomorphic to $\mathbb{R}^n$. Transported to $\mathbb{R}^n$, the form $\lambda - (\int_M \lambda)\rho$ is compactly supported with total integral $\int_M \lambda - \int_M \lambda = 0$, so by lemma 14.6.1 it equals $d\eta$ for some η compactly supported in $\mathbb{R}^n$, that is in U. Its support being a compact subset of U, extension by zero carries η to a global $(n-1)$-form on M whose differential is $\lambda - (\int_M \lambda)\rho$. Hence $[\lambda] = (\int_M \lambda)[\rho]$ when both forms sit in one such chart.

In general $\operatorname{supp}\lambda$ lies in some oriented chart $U \cong \mathbb{R}^n$ and $\operatorname{supp}\rho$ in the chart U_0. Because M is connected it is covered by oriented charts diffeomorphic to $\mathbb{R}^n$ along a chain $U = W_0, W_1, \dots, W_m = U_0$ with $W_{j-1} \cap W_j \neq \emptyset$ for each j. Choose in each overlap $W_{j-1} \cap W_j$ a bump n-form σ_j of total integral $\int_M \lambda$. Comparing consecutive forms inside a single chart, the one-chart case gives $[\lambda] = [\sigma_1] = \cdots = [\sigma_m] = (\int_M \lambda)[\rho]$, the last equality because σ_m and $(\int_M \lambda)\rho$ share the chart U_0 and have equal integral. This proves the claim.

Step 3: Injectivity. Cover the compact M by finitely many oriented charts $\{U_j\}$ diffeomorphic to $\mathbb{R}^n$ with a subordinate partition of unity $\{\rho_j\}$ (theorem 1.5.5). For any $\omega \in \Omega^n(M)$, each $\rho_j\omega$ is supported in the single chart U_j, so Step 2 gives $[\rho_j\omega] = (\int_M \rho_j\omega)[\rho]$. Summing over the finite index set,

$$[\omega] = \sum_j [\rho_j\omega] = \Big(\sum_j \int_M \rho_j\omega\Big)[\rho] = \Big(\int_M \omega\Big)[\rho]$$

Thus $I([\omega]) = \int_M \omega = 0$ forces $[\omega] = 0$, so I is injective. Being also surjective, I is an isomorphism, and $H^n_{\mathrm{dR}}(M) = \mathbb{R}\,[\rho]$, as we sought to show.

The proposition is the analytic form of orientability. A compact connected manifold carries a top cohomology class detected by integration precisely when it is orientable; the generator $[\rho]$ is the de Rham *fundamental class*, and pairing a closed top form against it returns its integral. The connectedness hypothesis is what makes the line one-dimensional rather than one copy of $\mathbb{R}$ per component: for a compact oriented manifold with c components, the same argument on each component gives $H^n_{\mathrm{dR}}(M) \cong \mathbb{R}^c$, matching the count of H^0 in proposition 14.1.5 with degrees exchanged. Orientability is essential: a compact connected non-orientable M has $H^n_{\mathrm{dR}}(M) = 0$, since it admits no top form of nonzero integral, and every top form is then exact.

Example 14.6.3: The Top Cohomology of the Sphere

For $M = S^n$, the Mayer-Vietoris computation of example 14.4.3 already gives $H^n_{\mathrm{dR}}(S^n) \cong \mathbb{R}$, and proposition 14.6.2 identifies the isomorphism concretely: the class of a top form ω is $\int_{S^n}\omega$ times the class of a normalized bump. The round volume form μ of S^n, an orientation form, has $\int_{S^n}\mu = \mathrm{vol}(S^n) > 0$, so $[\mu]$ generates $H^n_{\mathrm{dR}}(S^n)$. A closed top form on S^n is exact if and only if it integrates to zero, which is why the sphere's volume form, integrating to its volume, cannot be exact, the fact first met as a period in chapter 13.

The one-dimensional line at the top pairs with the rest of the cohomology. Wedging a closed k-form with a closed $(n-k)$-form gives a closed n-form, and integrating it produces a number that depends only on the two classes. This bilinear pairing is the subject of Poincaré duality: on a compact oriented manifold it is as nondegenerate as a pairing can be, so each $H^k_{\mathrm{dR}}(M)$ is the dual space of $H^{n-k}_{\mathrm{dR}}(M)$. For the general (possibly noncompact) statement the correct partner of H^k_{dR} is the compactly supported cohomology, which is named here as the dual side.

Definition 14.6.4: Compactly Supported de Rham Cohomology

Let $\Omega^k_c(M) \subseteq \Omega^k(M)$ be the compactly supported k-forms. Since d preserves compact support, $(\Omega^*_c(M), d)$ is a subcomplex, and its cohomology

$$H^k_{\mathrm{dR},c}(M) = \frac{\ker\big(d\colon \Omega^k_c(M) \to \Omega^{k+1}_c(M)\big)}{\mathrm{im}\big(d\colon \Omega^{k-1}_c(M) \to \Omega^k_c(M)\big)}$$

is the *compactly supported de Rham cohomology* of M. When M is compact every form is compactly supported, so $H^k_{\mathrm{dR},c}(M) = H^k_{\mathrm{dR}}(M)$.

Compactly supported cohomology has its own Poincaré lemma and its own Mayer-Vietoris sequence, parallel to theorem 14.2.4 and theorem 14.4.2 but with the variance of the maps reversed: an open inclusion pushes compactly supported forms forward by extension by zero, rather than pulling them back. Its full development is not needed for the geometry of this book and is pointed forward to chapter 15 and the companion volumes; the proof of Poincaré duality below uses only its two computational inputs, recorded as they arise. The duality is cleanest to state for manifolds whose topology is captured by a finite combinatorial datum.

That datum is a finite good cover (definition 14.5.6), which every compact smooth manifold admits (box 14.5.7).

Theorem 14.6.5: Poincaré Duality

Let M be an oriented n-manifold admitting a finite good cover, for instance any compact oriented manifold. Then for each k the pairing

$$H^k_{\mathrm{dR}}(M) \times H^{n-k}_{\mathrm{dR},c}(M) \to \mathbb{R}, \qquad ([\alpha],[\beta]) \mapsto \int_M \alpha \wedge \beta$$

is nondegenerate, and both spaces are finite-dimensional; equivalently the induced map $\mathrm{PD}\colon H^k_{\mathrm{dR}}(M) \to \big(H^{n-k}_{\mathrm{dR},c}(M)\big)^*$ is an isomorphism. When M is compact, $H^*_{\mathrm{dR},c} = H^*_{\mathrm{dR}}$, so the pairing $H^k_{\mathrm{dR}}(M) \times H^{n-k}_{\mathrm{dR}}(M) \to \mathbb{R}$ is nondegenerate and

$$b_k(M) = b_{n-k}(M)$$

for every k, where $b_k(M)$ is the k-th Betti number (definition 14.5.13).

The proof takes as inputs two properties of the compactly supported side, deferred as above: its Poincaré lemma, that $H^k_{\mathrm{dR},c}(\mathbb{R}^n) = \mathbb{R}$ for $k = n$ (integration) and 0 otherwise; and its Mayer-Vietoris sequence, that for $M = U \cup V$ open there is a long exact sequence in $H^*_{\mathrm{dR},c}$ whose maps go the opposite way to those of theorem 14.4.2, and whose connecting homomorphism is compatible up to sign with the wedge-integration pairing. Granting these, the duality follows by induction on the size of a finite good cover.

***Proof*:**
Step 1: The pairing descends. If α is closed and $\alpha' = \alpha + d\gamma$, then for closed compactly supported β one has $\alpha' \wedge \beta - \alpha \wedge \beta = d\gamma \wedge \beta = d(\gamma \wedge \beta)$, using $d\beta = 0$; the form $\gamma \wedge \beta$ is compactly supported, so its integral vanishes by Stokes' theorem (theorem 13.2.1). The same computation in the second slot, with $\beta' = \beta + d\delta$ and δ compactly supported, shows the pairing depends only on the two cohomology classes. It therefore defines $\mathrm{PD}\colon H^k_{\mathrm{dR}}(M) \to (H^{n-k}_{\mathrm{dR},c}(M))^*$.

Step 2: The base case. Let M be diffeomorphic to $\mathbb{R}^n$, a single good-cover piece. By corollary 14.2.6 the ordinary cohomology $H^k_{\mathrm{dR}}(\mathbb{R}^n)$ is $\mathbb{R}$ for $k = 0$ and 0 otherwise, and the compactly supported input gives $H^{n-k}_{\mathrm{dR},c}(\mathbb{R}^n) = \mathbb{R}$ for $k = 0$ and 0 otherwise. Both sides thus vanish unless $k = 0$, where the pairing sends the constant function 1 and a compactly supported top form β to $\int_{\mathbb{R}^n} \beta$; on the generator with $\int_{\mathbb{R}^n} \beta \neq 0$ this is nonzero, so PD is an isomorphism $\mathbb{R} \to \mathbb{R}^*$. The same holds for every finite intersection of good-cover sets, each being diffeomorphic to $\mathbb{R}^n$.

Step 3: The inductive step. Suppose PD is an isomorphism for every oriented manifold with a good cover of at most p sets, and let M have a finite good cover $\{U_1, \ldots, U_{p+1}\}$. Put $U = U_1 \cup \cdots \cup U_p$ and $V = U_{p+1}$. Then U has a good cover of size p; V is diffeomorphic to $\mathbb{R}^n$;

and $U \cap V = \bigcup_{i \le p}(U_i \cap V)$ has a good cover of size at most p, since each $U_i \cap V$ and its further intersections are finite intersections of the original cover, hence diffeomorphic to $\mathbb{R}^n$. By the inductive hypothesis PD is an isomorphism for U, V, and $U \cap V$.

The ordinary Mayer-Vietoris sequence (theorem 14.4.2) for $\{U, V\}$ and the dual of the compactly supported Mayer-Vietoris sequence, obtained by applying $(-)^* = \mathrm{Hom}(-, \mathbb{R})$ (exact, since these are vector spaces over a field), sit in a commuting ladder whose vertical maps are PD on M, U, V, and $U \cap V$. The squares between the homomorphisms induced by inclusions commute on the nose; the square involving the connecting maps δ and δ_c^* commutes up to a fixed sign, which is the compatibility recorded as the second compactly supported input and verified by a Stokes computation with the partition of unity defining δ. Four of the five vertical maps at each stage are isomorphisms by the inductive hypothesis, so the five lemma ([11, The Five Lemma]) forces the middle one, PD for M, to be an isomorphism as well.

Step 4: Conclusion. Induction gives that PD is an isomorphism for every M with a finite good cover, which is the asserted nondegeneracy; finite-dimensionality of each $H^k_{\mathrm{dR}}(M)$ propagates up the same induction from the finite-dimensional Euclidean case through the exact Mayer-Vietoris sequences, and duality carries it to $H^*_{\mathrm{dR},c}$. For compact M the two cohomologies coincide (definition 14.6.4), so $H^k_{\mathrm{dR}}(M) \cong (H^{n-k}_{\mathrm{dR}}(M))^*$; taking dimensions gives $b_k(M) = b_{n-k}(M)$, completing the proof.

Poincaré duality is the cohomological form of the geometric duality between a submanifold and its transverse complement: the pairing $\int_M \alpha \wedge \beta$ is the de Rham incarnation of the oriented intersection number of chapter 8, and the symmetry $b_k = b_{n-k}$ is the statement that a k-cycle and an $(n-k)$-cycle meet in a number that determines each class from the other. The Betti symmetry is not automatic: it fails without orientability, and the failure is visible in low degree. The real projective plane $\mathbb{RP}^2$ is compact and connected but not orientable, so by the remark after proposition 14.6.2 its top cohomology $H^2_{\mathrm{dR}}(\mathbb{RP}^2) = 0$, whereas $H^0_{\mathrm{dR}}(\mathbb{RP}^2) = \mathbb{R}$; the numbers $b_0 = 1$ and $b_2 = 0$ are unequal, and no nondegenerate pairing between the two groups can exist. Orientability is exactly what the wedge-then-integrate pairing needs.

Example 14.6.6: Poincaré Duality on the Torus

The torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ is compact, connected, and orientable, with $H^k_{\mathrm{dR}}(T^n) \cong \Lambda^k \mathbb{R}^n$ from example 14.4.4: a basis of $H^k_{\mathrm{dR}}(T^n)$ is given by the classes $[dx^I] = [dx^{i_1} \wedge \cdots \wedge dx^{i_k}]$ for increasing multi-indices $I = (i_1 < \cdots < i_k)$, where $x^1, \ldots, x^n$ are the angular coordinates. Hence $b_k(T^n) = \binom{n}{k} = \binom{n}{n-k} = b_{n-k}(T^n)$, confirming the Betti symmetry. The duality pairing is explicit on these generators: for multi-indices I of size k and J of size $n-k$,

$$\int_{T^n} dx^I \wedge dx^J = \pm 1 \text{ if } J = I^c, \qquad \int_{T^n} dx^I \wedge dx^J = 0 \text{ otherwise}$$

since $dx^I \wedge dx^J$ is a nonzero multiple of the volume form $dx^1 \wedge \cdots \wedge dx^n$ exactly when J is the complementary multi-index I^c, and integrates to zero when I and J share an index. Thus $[dx^I]$ pairs nontrivially with $[dx^{I^c}]$ and with nothing else, so the pairing matrix, in these dual bases, is a signed permutation matrix, hence nondegenerate.

The second use of the top line reads a map through its effect on cohomology. A smooth map $f\colon M \to N$ of compact oriented n-manifolds pulls back the top cohomology of N to that of M; both are copies of $\mathbb{R}$ by proposition 14.6.2, so f^* on the top is multiplication by a scalar. That scalar is the degree, and identifying it recovers the Brouwer degree of chapter 8 from the analytic side.

Proposition 14.6.7: The Cohomological Degree

Let M and N be compact connected oriented n-manifolds and $f\colon M \to N$ smooth. Then

$$\int_M f^*\omega = (\deg f) \int_N \omega$$

for every $\omega \in \Omega^n(N)$, where $\deg f$ is the Brouwer degree (definition 8.4.1). Equivalently, under the identifications $H^n_{\mathrm{dR}}(N) \cong \mathbb{R} \cong H^n_{\mathrm{dR}}(M)$ by integration (proposition 14.6.2), the map $f^*\colon H^n_{\mathrm{dR}}(N) \to H^n_{\mathrm{dR}}(M)$ is multiplication by $\deg f$.

***Proof*:**
Every n-form on the n-manifold N is closed, since $\Omega^{n+1}(N) = 0$. The two linear functionals on $\Omega^n(N)$

$$L(\omega) = \int_M f^*\omega, \qquad R(\omega) = (\deg f) \int_N \omega$$

both vanish on exact forms: R because $\int_N d\tau = 0$ by Stokes' theorem (theorem 13.2.1), and L because $f^*(d\tau) = d(f^*\tau)$ so $\int_M f^*(d\tau) = 0$ likewise. Hence L and R descend to linear functionals on $H^n_{\mathrm{dR}}(N) \cong \mathbb{R}$, and it suffices to check $L = R$ on a single class of nonzero integral.

Choice of representative. By Sard's theorem (theorem 7.2.1) f has a regular value $y_0 \in N$, and $f^{-1}(y_0)$ is a finite set $\{x_1, \ldots, x_r\}$ (a compact 0-manifold, by corollary 3.2.10 and compactness of M). At each x_i the differential df_{x_i} is an isomorphism, so f restricts to a diffeomorphism of a neighbourhood of x_i onto a neighbourhood of y_0. Choose disjoint such neighbourhoods $V_i' \ni x_i$; the set $K = M \setminus \bigcup_i V_i'$ is compact and $y_0 \notin f(K)$, so there is a ball $B \ni y_0$ contained in $\left(\bigcap_i f(V_i')\right) \setminus f(K)$. Setting $V_i = V_i' \cap f^{-1}(B)$ gives $f^{-1}(B) = \bigsqcup_i V_i$ with each $f|_{V_i}\colon V_i \to B$ a diffeomorphism. Take ω_0 a bump n-form supported in B with $\int_N \omega_0 = 1$.

The computation. Since $\mathrm{supp}\omega_0 \subseteq B$, the pullback $f^*\omega_0$ is supported in $\bigsqcup_i V_i$, and by naturality of the integral under diffeomorphisms (proposition 13.1.7),

$$L(\omega_0) = \int_M f^*\omega_0 = \sum_{i=1}^{r} \int_{V_i} (f|_{V_i})^*\omega_0 = \sum_{i=1}^{r} \varepsilon_i \int_B \omega_0$$

where $\varepsilon_i = +1$ if $f|_{V_i}$ preserves orientation and -1 if it reverses it, that is $\varepsilon_i = \operatorname{sign}\det df_{x_i}$. Thus $L(\omega_0) = \left(\sum_i \operatorname{sign}\det df_{x_i}\right) \int_N \omega_0 = \deg(f, y_0)$, which equals $\deg f$ by theorem 8.4.4. On the other hand $R(\omega_0) = (\deg f) \int_N \omega_0 = \deg f$, so $L(\omega_0) = R(\omega_0)$. The two functionals therefore agree on all of $H^n_{\mathrm{dR}}(N)$, giving $\int_M f^*\omega = (\deg f) \int_N \omega$ for every ω. Reading this through the integration isomorphisms of proposition 14.6.2 says precisely that f^* on the top cohomology is multiplication by $\deg f$, completing the proof.

The cohomological degree returns the counting invariant of Part II as an analytic one, and each of the degree's properties becomes transparent from this side. Homotopy invariance is immediate: homotopic maps induce the same f^* on cohomology by theorem 14.3.4, hence the same scalar, reproving that $\deg f$ depends only on the homotopy class of f. Multiplicativity under composition is the functoriality

$(g \circ f)^* = f^* \circ g^*$ of proposition 14.1.4, so $\deg(g \circ f) = \deg g \cdot \deg f$. And that the degree is an integer, not an arbitrary real scalar, is the content of the counting formula $\deg(f, y_0) = \sum_i \operatorname{sign} \det df_{x_i}$: the analytic ratio $\int_M f^*\omega / \int_N \omega$ is forced to be a signed count of preimages. Combined with the Hopf theorem (theorem 8.4.10), which says the degree is a complete homotopy invariant of self-maps of the sphere, this identifies $[S^n, S^n]$ with $\mathbb{Z}$ through the action on $H^n_{\mathrm{dR}}(S^n)$.

Example 14.6.8: Degree of the Power Map on the Circle

Let $f_m : S^1 \to S^1$ be $z \mapsto z^m$ for an integer m, which in the angular coordinate is $\theta \mapsto m\theta$. The class $[d\theta]$ generates $H^1_{\mathrm{dR}}(S^1) \cong \mathbb{R}$, with $\int_{S^1} d\theta = 2\pi$. Pulling back, $f_m^*(d\theta) = d(m\theta) = m\, d\theta$, so $f_m^*[d\theta] = m[d\theta]$, and proposition 14.6.7 reads off $\deg f_m = m$. This agrees with the geometric count: for $m \neq 0$ every point of the target has exactly $|m|$ preimages, each contributing the sign of m, so $\deg f_m = m$; for $m = 0$ the map is constant and $f_0^*[d\theta] = 0$. The degree is the number of times f_m wraps the circle around itself, now computed as the scalar by which it multiplies the generator of the top cohomology.

The identification of the analytic invariant d with the topology that singular homology measures, and the two geometric payoffs above, complete the thesis this book set out to prove: smooth methods compute topological invariants, and the same integers appear whether counted geometrically or read off the de Rham complex. Three roads continue from here. Hodge theory selects a canonical harmonic representative in each cohomology class once a metric is fixed, making Poincaré duality an isometry; sheaf cohomology places de Rham cohomology inside a general framework that computes the cohomology of any coefficient system; and characteristic classes attach cohomology classes to vector bundles, refining the invariants of this book into a calculus for bundles. We take up these directions, and the companion volumes that carry them, in chapter 15.

Exercise 14.6.1

1. Let $M \xrightarrow{f} N \xrightarrow{g} P$ be smooth maps of compact connected oriented n-manifolds. Using functoriality of pullback on de Rham cohomology (proposition 14.1.4) and proposition 14.6.7, prove that $\deg(g \circ f) = \deg g \cdot \deg f$. Deduce that if f is a diffeomorphism then $\deg f = \pm 1$.

2. Let $f : M \to N$ be a smooth map of compact connected oriented n-manifolds, and suppose f is not surjective. Using the cohomological degree formula with a bump form supported in the open set $N \setminus f(M)$, show that $\deg f = 0$. Conclude that for $n \geq 1$ a constant map has degree 0.

3. Let M be a compact connected oriented n-manifold and μ a nowhere-vanishing n-form on M (an orientation form). Show that $[\mu] \neq 0$ in $H^n_{\mathrm{dR}}(M)$ and that $[\mu]$ generates it. Deduce that a nowhere-vanishing top form on a compact oriented manifold is never exact.

4. Let M be a compact orientable manifold of odd dimension n. Using the Betti symmetry $b_k = b_{n-k}$ of theorem 14.6.5, prove that the Euler characteristic $\chi(M) = \sum_{k=0}^n (-1)^k b_k$ vanishes. (You may assume each b_k is finite, which theorem 14.6.5 also provides.)

5. On the torus T^2 with angular coordinates x^1, x^2, the classes $[dx^1], [dx^2]$ span $H^1_{\mathrm{dR}}(T^2)$. Compute the four pairings $\int_{T^2} dx^i \wedge dx^j$ for $i, j \in \{1, 2\}$, and check directly that the resulting pairing $H^1_{\mathrm{dR}}(T^2) \times H^1_{\mathrm{dR}}(T^2) \to \mathbb{R}$ is nondegenerate, as theorem 14.6.5 predicts for the middle degree $k = n - k = 1$.

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What Next?

This book began with a claim about what a smooth manifold is not: it is not a geometric object, only the bare substrate on which a geometry may later be imposed. Everything since followed that substrate as far as it goes on its own. Part I built the manifold and its calculus, the tangent space, the rank theorems, and the flows of vector fields. Part II made "generic" precise with Sard's theorem and transversality (chapter 7) and read topology off differential data: the degree of a map (section 8.4), the cells a Morse function carves out (chapter 9), and the framed cobordism of the Pontryagin-Thom construction (chapter 10). Part III returned to the calculus of bundles and forms and closed with the de Rham theorem (theorem 14.5.10), where the analytic operator d was shown to compute the same invariant as the singular chains of algebraic topology. What remains is everything the substrate does not settle on its own, and it divides in two: the geometries one may impose on a manifold, and the topology one may draw out of it more deeply. This chapter maps both, pairing each direction with the companion volume of the series that develops it and with the standard texts a working reader keeps within reach.

Riemannian geometry. The first and most basic rigidification is a metric. It gives at a stroke what the smooth manifold lacked, lengths and angles on tangent vectors, and with them geodesics, curvature, and the exponential map, whose geodesically convex balls are exactly what gave the good covers of chapter 14 their existence (box 14.5.7). The metric also singles out, through the Levi-Civita connection, a canonical way to differentiate vector fields. This is the subject of the companion *Riemannian Geometry* [6]; the standard introductions are do Carmo's *Riemannian Geometry* and John M. Lee's *Introduction to Riemannian Manifolds.*

Connections, curvature, and characteristic classes. A geometry on a vector bundle is a connection, a rule for differentiating sections, and its curvature is the obstruction to that rule being flat. The reduction of a bundle's structure group (section 11.5), identified in Part III as the exact point where a geometry enters, is what a connection refines; and the Chern-Weil homomorphism turns the curvature of any connection into de Rham cohomology classes, the characteristic classes, invariants of the bundle independent of the connection used to compute them. They sharpen the bundle constructions of this book into a calculus that classifies bundles up to isomorphism, and they are the geometric half of the companion *Vector Bundles and K-Theory* [13]; Milnor and Stasheff's *Characteristic Classes* is the classic account.

Hodge theory. Fixing a metric does more than add geometry; it makes the topology of chapter 14 canonical. Each de Rham class then contains exactly one harmonic representative, the unique form of least norm, so cohomology becomes a subspace of forms rather than a quotient, and Poincaré duality (theorem 14.6.5) is realized as an isometry through the Hodge star. On a Kähler manifold, where a metric, a complex structure, and a symplectic form agree, this refines into the Hodge decomposition of cohomology by type. The real theory is the companion *Hodge Theory* [3]; Warner's *Foundations of Differentiable Manifolds and Lie Groups* proves the Hodge theorem in the setting of this book, and Griffiths and Harris's *Principles of Algebraic Geometry* and Voisin's *Hodge Theory and Complex Algebraic Geometry* carry it into the complex world.

Symplectic and complex structures. A metric is only one of the rigidifications named in the introduction. A symplectic form, a closed nondegenerate two-form, makes a manifold the phase space of a mechanical system and opens symplectic geometry, developed in McDuff and Salamon's *Introduction to Symplectic Topology.* A complex structure turns real calculus into holomorphic calculus and sits at the border of *Complex Analysis* [12] and *Algebraic Geometry* [1]; Huybrechts's *Complex Geometry* is the standard bridge.

Lie groups and symmetry. The structure groups whose reductions name the geometries above are Lie groups, and the manifolds most open to computation are those a Lie group acts on transitively, the homogeneous spaces met in chapter 4. The structure theory of Lie groups, their Lie algebras, and their actions is the companion *Lie Groups* [16], with the representation theory that accompanies it in *Algebraic Groups and Representations* [8]; Warner's book again and Knapp's *Lie Groups Beyond an Introduction* are the standard references.

Algebraic topology. The de Rham theorem (theorem 14.5.10) matched the analytic cohomology of this book against the singular cohomology a topologist builds from continuous chains, but only named the singular side. Its full development, the fundamental group and covering spaces, singular and cellular homology, the homotopy groups that the Pontryagin-Thom construction of chapter 10 computed in one family of cases, and the Poincaré-Lefschetz duality that completes the compactly supported cohomology of definition 14.6.4, is the companion *Algebraic Topology* [11]. Hatcher's *Algebraic Topology* is the standard modern text, and Bott and Tu's *Differential Forms in Algebraic Topology* is the one closest in spirit to the present book, developing the subject through the de Rham complex throughout.

Cobordism and stable homotopy. The cobordism groups of chapter 10 and the Pontryagin-Thom construction that computes them are the entrance to stable homotopy theory, where manifolds up to cobordism are organized by the Thom spectra and the cobordism rings are read off by characteristic numbers. Stong's *Notes on Cobordism Theory* is the classical reference, and the characteristic numbers that detect cobordism class are again those of Milnor and Stasheff.

Sheaf cohomology. Finally, de Rham cohomology is one instance of a single general construction. The de Rham complex resolves the constant sheaf $\mathbb{R}$, and its cohomology is the sheaf cohomology of that sheaf; replacing $\mathbb{R}$ by another sheaf of coefficients, the holomorphic functions, the sections of a bundle, a local system, yields the cohomology theory adapted to it. This is the framework of *Algebraic Geometry* [1] and of complex geometry, where the analytic, the topological, and the algebraic accounts of cohomology are unified. Bott and Tu again, and the opening chapters of Griffiths and Harris, are the gentlest entries; Hartshorne's *Algebraic Geometry* is the systematic one.

Each of these directions begins where this book ends, at the boundary between a smooth manifold and the extra structure that would make it geometric. The substrate has been mapped; what one imposes on it, and what one draws out of it, fills the volumes named above.

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